

RESEARCH METHODOLOGY



**Prof. Prathap Haridoss, Prof. G. Phanikumar, Prof. Arun K. Tangirala,
Prof. M.S. Ananth, Prof. C. Balaji, Prof. Abhijit P. Deshpande,
Prof. Edamana Prasad**

Department of Metallurgical and Materials Engineering
IIT Madras

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Lecture - 01

Group Discussion on Reserch

Prof. Prathap Haridoss: Hello, welcome to this Group Discussion on Research. This is the first class, an introductory class on this course which we have on Introduction to Research. I am Prathap, I am a professor in the Department of Metallurgical and Materials Engineering at IIT Madras, and these are our panelist.

Prof. Arun K. Tangirala: Hi, I am Arun Tangirala, I am a professor in the Department of Chemical Engineering at IIT Madras.

Prof. G. Phanikumar: Hi, I am Phanikumar Gandham; professor in Department of Metallurgical and Materials engineering.

Prof. Abhijit P. Deshpande: Hi, I am Abhijit Deshpande, I am a faculty member in Department of Chemical Engineering.

Prof. Andrew Thangaraj: Hello, I am Andrew Thangaraj, I am a faculty in the Electrical Engineering Department, IIT Madras.

Prof. Prathap Haridoss: Right. So, when we discuss about Research I think the aspects that we need to look at are that many students aspire to get degrees in which are basically research based degrees. And often based on our background a particular person may not be completely aware of what is involved in getting such a degree and how is it perhaps different from other degrees. So the first thing, first topic that we will discuss today is, what does a research degree such as MPhil, MS or PhD imply or represent.



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A dark blue rectangular slide with white text centered in the middle. The text asks: "What does a Research degree such as MPhil, MS, or PhD imply or represent?"

What does a Research degree
such as MPhil, MS, or PhD imply or
represent?

Prof. Arun K. Tangirala: I think what I would say is that it is really different from the under graduate degree that normally one pursue you say B.Tech or a B.Sc and so on. Because, normally in such under graduate degrees you do a lot of courses, whereas in research degrees what we normally think of is doing something new, contributing, first of all knowing what is there out in that area, picking out favorite area and most importantly contributing something. It may be a small contribution but drops make an ocean. So, that really involves a lot of effort from our side; whereas in a course based program like B.Tech and so on we take courses we learn. There is only a small part of discovering something; it is more of learning. Whereas, in research based degrees you are trying to really discover something, think and then postulate, validate, perform certain experiments and so. So, there is lot of self-contribution in this research based degrees.

Prof. Abhijit P. Deshpande: Yeah adding on to that, so that is the difference as Arun pointed out that there are various things which depend on oneself during a research degree. So, a course based degrees is where you know the exams, the assignments everything is sort of catered and everything is well designed. But when a research degree lot of it depends on the researcher himself or herself. And those are the aspects we will be discussing further on as to how what is meant by these aspects of research.

Prof. Andrew Thangaraja: So, one crude way of comparing these two things is, if you conduct a final exam for a course at the end of a subject of a class you write a final exam,



everybody is supposed to turn in a very similar kind of answer script right. So that is the key and you write similar answers and you all get good marks. In fact, at the end of MS or PhD, you cannot submit a thesis which is similar to anybody else's thesis. Suppose to produce something which is unique and different and only you have done the work. So, that kind of represents the big shift between a normal degree course and PhD.

Prof. G. Phanikumar: Yes.

Prof. Andrew Thangaraja: So, have that and mind, it is not just one more degree after your MPhil, but or after your Masters it is a really a different sort of a degree when you pursue a PhD.

Prof. G. Phanikumar: Yes, so that brings to the point about the qualitative difference in PhD with other degrees. One may have a misconception that the amount of work, the quantitative work at a bachelor's degree, you do little bit more may be then you can get a Master's degree. But then you do twice of that, that does not mean that you can get a PhD degree. PhD is a qualitatively a very different degree where you are training yourselves to become a researcher and it is one different in also a sense of the ability of a PhD. Once you have finished the degree you are also expected to be at a level that you could guide another person to do a PhD, it is very different that from other disciplines. So, qualitatively there is a big difference.

Prof. Prathap Haridoss: Yeah and so, I would also like to add that you know when you do an undergraduate degree the boundaries are well defined. There is a start point and you can tell exactly when you are half way through the degree and when you have completed the degree. You can tell on the day you joined which is your graduation day, which is the day you pick up your degree all those things are well defined. In a research based degree, it is open ended. So, there is a lot of discovery in the process; you have to figure out when you have learnt enough and you are able to contribute enough and you have become a master or somebody who is well known in that field who has contributed a lot in to that field, and therefore you are in a position to pick up a PhD degree. So, that's very different from typical undergraduate degree.



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So, we will now look at this question; What is research? So, I will may be start with Andrew and he can tell you something.

Prof. Andrew Thangaraja: Ok. So, I keep coming back to this contrast between a course and research. So, let us take again very concrete differences. You have a text book and a research paper, I think these are the two things you can compare. Text book is accumulated knowledge over several years. Somebody has understood and formalized that knowledge in his head and he is putting it out in a sequence of subject so that somebody else can learn, that is a text book and that is why you learn and that is what you to do. When you do research, it is completely unclear what will happen to that work after that. You know you are going to the real cutting edge of current day development round of the subject and you trying to push the boundaries in a direction in which nobody has thought of before. So, and nobody can be sure what will happen to that work after that, maybe it is very interesting, maybe it is not very exciting later on, you cannot predict anything. And when you do it will never be organized in a very clean systematic way like a text book is. You will have to organize in your head and take time over it. So to me the text book versus research paper difference of, may be a cutting as research paper differences what comes to mind when I answer the question what is research.



Prof. G Phanikumar: Yes, in often when students ask, is there an any text book I can refer to so that I could do my research. May be if there were a text book them may be it is too late to do research in that.

Prof. Andrew Thangaraja: Yes, Exactly

Prof. Arun K. Tangirala: I would also like to add that, in fact reiterate what Andrew just said that the beginnings of research are quite hazy. In fact, it is so foggy and hazy that you may think that is nothing or nothing is clear to you in the beginning. It is only the passion that drives you. So, what is research? Big integral part of research is learning; that is a very important part in fact you see the term search in research so you have to search and discover.

So the searching is for both learning and for discovering. And that of course we will expand shortly on that as to what you do to carry out this searching part in research. But I think the important part is the beginnings are hazy, but there is a joy to the discovery; provided you actually persistently pursue what you want to really find out and so on. In fact, it is like going into a city where you do not know what the destination is, you know that this city is good you heard, it is your favorite city, it is like choosing your favorite field of research but you do not know which monument you want to visit where you want to spend most of your time.

Prof. Andrew Thangaraja: Yes.

Prof. Arun K. Tangirala: And then in the beginning you take, that is where you do a literature survey, find out what are all interesting monuments or sightseeing places in the city; and then you pick your favorite monument and or your sightseeing place and spend more time trying to know what is there and may be contribute to that place and so on. It is, yeah, there is a lot of trial and error in research and you have to be prepared for it.

Prof. Prathap Haridoss: Yes, I think that is a very great analogy that Arun just gave you and in fact we would like to just share this quote from Albert Einstein, which basically highlights this fact that the research that we do involves you know exploring unknown territory, so to speak in, in a particular area.



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*"If we knew what we were doing,
it would not be called research,
would it?"*

- Albert Einstein

He said, if we knew what we were doing, it would not be called research, would it? So, that pretty much highlights the point that you know there something new, it is not something that you can just go look up and then say now I have done the research. There is something new and you are going to be the person who searches for this new stuff, finds it and then shares it with all the people around you.

Prof. Abhijit P. Deshpande: Yes. The other aspect of research even though we are trying to give you in one hour some sort of gist of what might be research is, from so far discussion it is already clear that it is very, very open ended.

Prof. Arun K. Tangirala: Yes, in fact you need to research to find out what is research also.

Prof. Andrew Thangaraj: Your answer will be different from (Refer Time: 09:43) our answer.

Prof. G Phanikumar: There is another point about that there are some prefixes used for research for example, incremental research and path breaking research. Generally, people don't start of saying that I am going to now do path breaking research that is not how we go about. We do have a structured way of going about performing our routine activities for research and then we are trying to incrementally add knowledge to what is already there and in the process by serendipity or because our mind is prepared, we are looking



for clues and then path breaking research happens. It usually happens rather than you go out starting to do that.

Prof. Arun K. Tangirala: And I just want to add that, in research you are the one who questions and you are also the one who finds the answers.

Prof. Prathap Haridoss: Right.

Prof. Arun K. Tangirala: When compared to text book thing where the questions are then and you have to answer.

Prof. Andrew Thangaraj: (Refer Time: 10:31) everyone knows the answer.

Prof. Arun K. Tangirala: Right.

Prof. Andrew Thangaraj: Just wants you to write it down.

Prof. Arun K. Tangirala: But in research you ask questions and you find answers. The only thing that is important to keep in mind is you ask relevant questions.

Prof. Andrew Thangaraj: Yes.

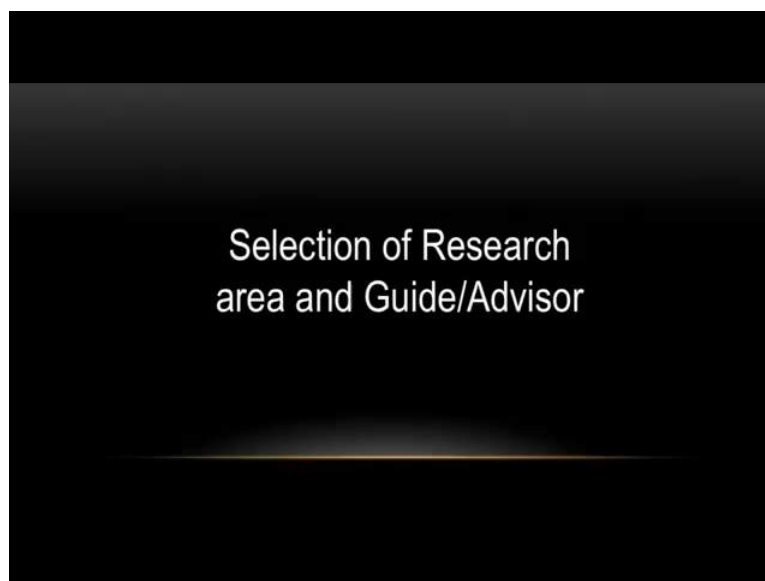
Prof. Arun K. Tangirala: Not absolute questions. And you give answers that make a difference.

Prof. Andrew Thangaraj: Yes this is true.

Prof. Prathap Haridoss: Through this discussion some of the points that we are going to look at are you know philosophical points like, what we have been looking at discussing so far which are you know the grand scheme of things within which we you know do carry out a research or we look at a research and so on. There are also lot of mundane activities that happen in that actually result in the process through which we do a research in universities and so on. We will also touch up on those because those are the day to day things that we get involved in as we try to carry out a research.



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So, one of those things is selection of a research area and selection of your guide or advisor. So, these are the first things that you would do when you try formally get into research. Of course, you could do research on your own as an independent person but most of us are not doing that you know in our own lab hidden away from everybody, we do it as a part of a community, a scientific community which you often you know is thriving in university setting also in research labs and in industry, R and D setting and so on. So at least in the university setting, one of the key things is you have to find the area that you would like to work on and the guide that you would like to work with. And there is no, I mean like in many of these things there is no single way in which you do it, but the area you are interested in is something that you have to understand on your own. I mean this could be due to a lot of general reading you have done in over the years as a student at various levels and you have found some particular area that appeals to you, that interests you in terms of kinds of details that they are looking at, the manner in which the area is being explored and so on.

And so, that is one way in which you can get a sense of you know this is an area that you would interested in. You also have through your general reading you will also have a sense of how much impact that area has to the neighborhood you are in or to the scientific community or the vault over all. So, that may also influence your interest in a particular research area. So, there is multiple factors that influence what you may get interested in and you should take all of those into account. But it is I would say



fundamentally it is very important that you should be interested in that area. If you end up trying to do research, if you just say that I want a degree and I want a PhD degree and you just go join a university and you just pick up whichever advisor is willing to pick you up. The biggest problem that will come is, the day to day; you know the exploration that you do which is part of your research will not appeal to you. And if it does not appeal to you it is difficult for you to do good work in that area. So, you have to look at it in the inverse manner. First of all, it should appeal to you; only then you should be getting into it. So, that is something that I wanted to share with you.

Prof. Abhijit P. Deshpande: One important aspect of this selection of both research areas and guide and advisor, is the central element in your thought should not be worry. In the sense, we always worry about you know; Will I get a job afterwards? Is this area really relevant? So, there are lot of things associated with worry. Actually, one should look at it as a challenge. And so, if you look at it from that point of view because we emphasize enough that it will be you who is the focus of this. In general, one can say that in most research areas and with many guides and advisor, if you drive you will be able to actually make a very good thesis and in the end actually use that for to furthering your overall career goals. So, central element should not be worry but central element should be challenge and interest. And that is how you should try to make this decision.

Prof. Andrew Thangaraj: So maybe I should touch up on the more mundane aspect of selecting area and advisor. So typically, I mean this is what happened to me I think for most people this is what happened. May be in your undergraduate or postgraduate studies there was a course that you picked up and there was something you did which really appeal to you maybe it was thought to you in a very well, very nice fashion and that instructor also appeal to you. So, this is typically how it happens. So, it can even happen in the undergraduate stage. You do a substitute really like it, then may be as a post graduate student entering university you take a course with someone you really like how the course went, how the person handled the class, the subject matter and also how they interacted with you.

So, all these things together can appeal to you and then you might want to think, ok; I can do a PhD in this area, I like this area and do it. This is the more mundane way in which it happens, but I have seen many students do this. I think this is something it shows that the student is not very prepared I think. They would take an undergraduate



core area. Like in Electrical engineering there is an area called signals and systems they would take that area and they will say I want to do research in signals and systems. It is fine, but except that you know all research in Electrical engineering is in signals and systems. So that is the difficulty you know. So, while you like a subject you should also know that you know just because you learnt it in undergraduate you cannot do research directly in that area, it is very highly specialized beyond that. The area will get more and more and more specialized as you go deeper and deeper into it and you have to kind of anticipate that, but like everybody said the guiding principles are yes you should be interested in the area. And also, say will be motivated by your advisor both as a technically capable person and also as a person. All these things go into having a good fit between you and your advisor.

Prof. G. Phanikumar: So, I would like to add that among all the degrees that student normally under goes, PhD perhaps is the longest duration on a single aspect with a single person.

Prof. Andrew Thangaraj: With a single person.

Prof. G. Phanikumar: It draws for about 4, 5 years, sometimes may be even 6 or 7 in some sciences. It is very important to pick the right advisor who has both the energy to drive you to take up the challenging aspects of the work and also the experience to guide you when you are into the difficulties coming over some of these topics. So, I think choosing an advisor is very important.

Prof. Abhijit P. Deshpande: Yes, there is also a style of your own and the advisor's. And I think there is good bit of thought that you can give to that also. Because sometimes those can also influence because as we have said it is day to day interaction over such a long period.

Prof. Arun K. Tangirala: I think it is a very important point as you must have probably now guessed, there is no definitive formula that we can give to choose an advisor or an area. I think as for as choosing an advisor is concerned as Abhijit just mentioned also mental compatibility. I think the style of thinking whether it matches, some students are really inclined towards doing carrying out theoretical work. While some students are made for experimental work. So, I think it goes back again to the same point that we have mentioned earlier you need to introspect a bit in and then figure out what suits you,



what you are interested in; whether a theoretical work or an experimental work or a mixed work and so on or you want to do really the cutting-edge stuff and so on.

So then of course, today there is no dearth of information you can really go to website, find out the profiles, check out the publication record even probably read a couple of publications and so on. To figure out whether the work really excites you, and remember in all of these there is no correct answer. There is always a discovery process, there are cases in which students have switched advisors, of course that is a very small proportion. But even if that happens you should not be really getting a depressed about that. Ok.

So, what is important is upfront to be very frank and honest with your advisor, express what your interests are and also listen to what your potential advisor has got to say, and do your own little bit of research about the advisor. And the same goes with the area as well you have to really introspect as Andrew said there are subjects that interest you, but that is just a beginning it is just giving you an idea of what to get into and you have to really probe a bit more into a broad field to figure out where you are. And I think the guide is a very appropriate term going by the previous analogy we said; research is about going into an unknown territory or visiting an unknown city and you are visiting monuments remember you have guides telling you giving to the history about it. So, here also the guides will tell you, when you pick a certain area and when you are talking to a potential advisor you can find out from the potential advisor, whether it is worth carrying out research? How crowded this area is? How crowded this bus that you are going to get in to is? Whether there are any empty seats or you have to stand for long time? Right so keep that in mind there is a lot of trial and error; but your gut feeling will tell you whether this is a right area and this is a right advisor right. It is also discovery process.

Prof. G. Phanikumar: And I would like to stress also, perhaps per our country the importance of homework. Generally, it is known that we have been used to spoon feeding; you know most of our schools, schooling system and later on. So, it is very important do adequate homework before we decide upon whether to do PhD and then also on what topic and with whom.

Prof. Arun k. Tangirala: So, that groundwork is very essential.

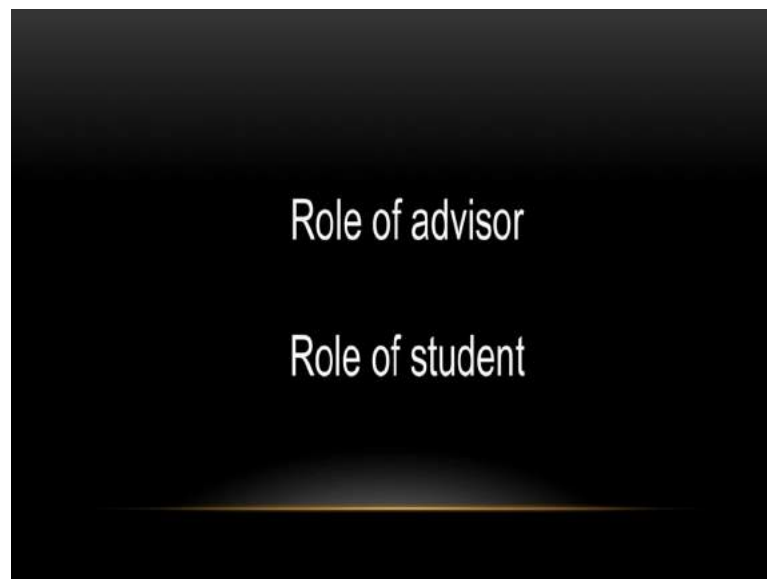


Introduction to Research
Prof. Prathap Haridoss
Prof. Arun K. Tangirala
Prof. G. Phanikumar
Prof. Abhijit P. Deshpande
Prof. Andrew Thangaraj
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture - 02
Role of advisor and Role of student

Prof. Prathap Haridoss: So, maybe we can have some more discussion on what we feel is are now specific aspects that a student should consider as role of a student in the PhD process, and may be what he or she can expect from a guide, and therefore, what would be a reasonable expectation as a role of a guide. So, these are some things that maybe looked at.

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Prof. G. Phanikumar: So, I will prompt a question and then see. So, do you expect an advisor to give a day-to-day guidance on what to do? Do you expect the student to turn up and say, Sir, I have done this today and what should I do next? Is that how it works?

Prof. Arun K. Tangirala: It is not a mission mode thing, where you have set of instructions.



Prof. Andrew Thangaraj: Yes.

Prof. Arun K. Tangirala: And then, you keep executing those instructions by the day, there is a calendar of instructions, there is nothing like that. Remember, the term that we are using is advisor or a guide, which means, that you keep exploring and when you think you are getting lost or when the advisor sees that you are getting lost, the advisor changes the direction of your research slightly, and it is like you are travelling in an unknown sea, you are steering the boat, but the advisor is just watching where you are steering. And also, the role of the advisor is to help you interpret the results, put the results in the right perspective and so on. So, I think that is a very important point which is probably missed out by many students. They expect the guides to actually give out day-to-day instructions or, you know, or weekly instructions, I have done this, now what next? That is not at all research. Research is all about trying to find out by yourself, but with the support of an advisor who will make sure that you are not lost in the unknown territories forever.

Prof. Andrew Thangaraj: So, I mean going back to my own experiences and maybe similar experiences, if you look at the more like mundane aspects. So, typically once you fix, once you are fixed under an advisor and you start working, you go to the advisor and start talking about what work is may be interesting to you? What is happening currently in the area? So, the advisor probably may tell you, may be this area is very interesting, you go look at papers. There are some advisors, who would go to the extent of giving you a list of 10 papers, you read these 10 papers and read this 11th paper, and then try to reproduce; they would do that. Some advisors, would say, you know, go read, this is a new area, exciting area. And now, again, it is again a style issue and a compatibility issue of which student kind of likes this first kind of advisor, who just says this area you try it and then see what happens; and the other advisor, who gives you a series of papers and says, you reproduce the result in the 11th paper. You know, I think some students, in fact, do not like the second type also; they would like to explore more, so that roles change a lot, but once... but that is usually common. Your advisor starts you off. I think that is more or less common in 99 percent of the cases. And then, after that, the student plays a huge role in what happens.

Prof. Arun K. Tangirala: I think that it is a very important point.



Prof. Andrew Thangaraj: And it is completely up to the student, there are students who just go out and read, left right centre and everything they read, and then do a lot of work, and come back to the advisor, and say, I have got this; I have got this; I have got this; and there are students who will slack off, and then, you have to chase them, and it just changes from case to case.

Prof. G. Phanikumar: So, now we are coming close to the role of the students; I will just prompt some question. Abhijith, just tell me, is it necessary or important for a student to be super intelligent before they take up research or they should be getting into adventures before they can get into research? Should they have that kind of a mind set or should they be very, very hard working to be able to do research? What characteristics do you think, students should have so that their role can be fulfilled as a researcher?

Prof. Abhijit P. Deshpande: Yes, I think to begin with like what we said from the beginning is, you know, interest, spirit of learning; I mean those are really the basic traits that you start out with. You do not start out by saying that I will do the... as we said it may not be path breaking immediately and things. So, the initial part of things is this spark and excitement, and these are the words which actually describe our spirits initially. And going back to my experience during my Masters' degree, for example, I would go and discuss with the advisor, and while we are discussing the advisor will say, why don't we try this for fun? And, you know, that really excited me, and sometimes, the reverse would probably happen, that actually without my advisor telling me, I actually did data acquisition for my experimental setup; then advisor says, oh! this is very exciting. So, actually both of us have to really spur excitement in each other about that technical topic. So, that is a very important role that both actually compliment, proceed further.

Prof. G. Phanikumar: Passion.

Prof. Andrew Thangaraj: And also, there is a routine to it. I keep coming back to routine; I do not know why! You know, for instance, most advisors and students meet once a week right, that happens, and they meet regularly, I think that helps, it helps. I think it helps in the long run to meet regularly even if your advisor is not particular about meeting, it's good to force a meeting once in a while because there the exchange of ideas, just you voicing out the idea, just saying it out loud in front of an advisor helps a



lot in clearing things in your head. When regular meetings happen and what happens in those meetings who pushes where, like Abhijith says, it changes from time to time. Might be times when student is down and the advisor steps in, and says maybe you look at it this way. Maybe times when the advisor is not interested and student has to jump in. So, it changes over a period of time. Some advisors are very hands on, they work with you on the board, write down, derive, etcetera. Some advisors will say, I am not going to be hands on, you read the paper, it's your problem. It changes, there is no universal rule. There are benefits and negative points about both, but I would say regular meetings, yes; very important.

Prof. Arun K. Tangirala: If I can just make three points, which are probably consolidation of what Abhijith and Andrew have been saying. One, I think is the attitude, attitude - the student's attitude makes a big difference. When a student comes to me and says, well, I want to work, I want to really work towards a PhD under your guidance, I say that I don't expect you to have knowledge, going back to the question, I don't expect you to have the requisite knowledge at all because knowledge is typically acquired through course work and learning. What is most important is attitude. If you have the inclination to learn and carry out research, you are ready. And secondly, I think what Andrew was pointing out is some advisors would define the problem for you upfront, where as some advisors would like you to define the problem, and in that context, they will give you a few papers and so on. Some students enjoy the defining their own problems, some students don't; they want the problem to be predefined and that plays an important role in choosing your advisor. These days, students as he said, homework, do a lot of groundwork before, and then, you know, find out how depending on the reputation the advisor has and so on, and choose the advisor.

And, I think the third thing is sharing your results with the advisor makes a big difference. Some advisors are extremely busy and some students don't like that. Some students would like to see their advisors every week and share. And simple discussion Like Abhijith said, a simple discussion will even before the advisor gives you ideas, just a listening ear will actually give you ideas. So, I think there are several factors that go in to choosing a advisor and there are several factors to watch out for as a role - as what your role is - and yes, I think we probably outline this.

Prof. Abhijit P. Deshpande: And the basic, again going back to some of the routine, note



taking, noting down things, making systematic consolidation of what you have done, what you want to do, time management; there is lot of this that is actually very important because again it is a long-term process, and a student must keep on reminding about many of these things, and actually implementing some of these things, as the time goes on.

Prof. Andrew Thangaraj: Just to close out one final thing. So, initially you might think that the advisor has more knowledge or more awareness of the area than a student. But at the end of the day, once the thesis is ready, the student is the expert in that topic, in the thesis, that's what is expected.

Prof. Arun K. Tangirala: And typically, the advisor looks up to the student to give inputs on what is a next problem to work on.

Prof. Andrew Thangaraj: Yes, exactly.

Prof. Andrew Thangaraj: So, that, it is kind of a role reversal.

Prof. Arun K. Tangirala: That means you are ready to get a degree.

Prof. Andrew Thangaraj: That point you are ready to leave.

Prof. Arun K. Tangirala: The green signal is on.

Prof. Prathap Haridoss: I would also like to add to something that Arun was saying earlier, which is about know, advisor's role in helping you interpret your results. So, basically, the advisor brings experience; experience in that area, and in that relationship as an advisor and a student relationship. So, at least in earlier days of your PhD, I mean as Andrew said, by the end you have more experience and knowledge in that particular aspect of what you are examining. But, in general, and the earlier days of a PhD, your advisor has a lot of experience in that area, and so, when you present results to him or her, and you discuss the results with your advisor, often the advisor's role is to help you have perspective on what your work is. So, sometimes as a student because you do not yet have enough experience in the area, you may tend to give lot of importance to a particular aspect of your result. Whereas, the advisor maybe in a better position to tell you that, that is not the most critical aspect of that result that you are presenting whereas, something else maybe more interesting. So, of course, this gives you an opportunity to



discuss, to consider your ideas again, and maybe, you know, bounce your ideas off with your advisor and so on, and that may help you gain that experience. The kind of experience that your advisor has had over the years is something that is getting transferred to you when you have that interaction. So, that is I think where the role of the both the student and the advisor sort of come together.

Prof. G. Phanikumar: Yes, one word about attitude also. I want to say that when it comes to the attitude, we must have a very healthy attitude on keeping the long-term objective. But then, doing things on a day-to-day basis in a very steady fashion, keeping the journal, and then moving forward in an incremental manner. There is something called Obsessive Compulsive Disorder, where a person wants something, and they want it immediately, urgently, and then smallest obstacle, then the person is totally away from it. Definitely, you know one should be under that kind of an attitude. You must be able to keep the long-term goal in mind.

Prof. Arun K. Tangirala: Be patient.

Prof. G. Phanikumar: Be patient, then keep chipping away, and then you finally, make your thesis.

Prof. Arun K. Tangirala: So, be patient; do not become a patient.

Prof. Prathap Haridoss: So, maybe now, maybe you talk about it.

Prof. Abhijit P. Deshpande: That brings to actually this, what Phani was just saying that you know, you keep the long term in mind, and with that long term in mind is where, enquiries come up, postulates come up. I mean you start thinking, you know, what can I do? And that basic spirit of enquiry needs to be at the basis of all of your research that you are doing.

Prof. Andrew Thangaraj: Yes, that is the key difference between a course-based program and research-based program. You learn about something, but you do not stop there when you do research. You have to ask questions now. You have to ask: what else can I do in this field? For what else are people doing in this field currently? How else, what difference can I make to what they are doing? And then, you say maybe if I try this, this will happen. So, that requires first of all, some sort of an intuitive understanding of the



blocks of, I mean, intuitive understanding of the different ideas that make makeup that area and you will say, if you change this idea a little bit that might happen. So, you should be able to analytically reason out, ask the right question, postulate some answer, and then, check it out and then.

Prof. Arun K. Tangirala: I think an integral element is persistence; and in fact, this key thing called persistence appears even in data driven modeling. Of course, I am talking more technical stuff now, but we say that if you want to really build a model - mathematical model of the process - the inputs have to be persistently exciting. So, I keep saying this to my students that the inputs that you are giving into a research are the questions, and you have to really keep persistently thinking, thinking just as the way you are trying... you are basically hunting for a treasure which is buried underneath the soil and you cannot really say, well I will dig a few feet today, and I will come back a few months later, and then you have to again restart all over again, collect your thoughts and so on. So, I think persistence and patience, and it is a complicated thinking that is happening in your mind. It is a wonderful dish, but you know, when, before the dish comes out to you it is all boiling and it is all chaos, but when, you know, when the stove is switched off, and when the dish settles down, then it comes out to a very nice taste. So, there is going to be chaos before clarity comes in, for sure in the session, you have to be really able to persist though that chaos and sustain that with the hope of finding the clarity.

Prof. Andrew Thangaraj: Yes.

Prof. Arun K. Tangirala: Then, when you get that clarity, the joy is unbounded and that is what I think makes research, drives research.

Prof. Andrew Thangaraj: That moment when you...

Prof. Arun K. Tangirala: Yes. Yes.

Prof. Andrew Thangaraj: Something clicks in.

Prof. Arun K. Tangirala: You will forget, it is like climbing Mount Everest and people have forgotten all the pain that have gone by.

Prof. Andrew Thangaraj: Everybody, every researcher works for that moment; this is



little bit of a moment. But, I have been saying the difference between the regular course work and research work, but let me point out one thing which is often over looked by people which is actually very similar what you should do in research, which you do regularly for course work. For most courses, good students have a notebook, and they have notebooks, and that notebook ends, they take another note book, that is very important in research. So, do not think that just because the research is different from course work I will not have a notebook. You will find very few serious researchers, who do not have a notebook, everybody has a notebook. They keep taking notes, they keep looking at something. It may not be a physical note these days, it might be electronic for some people, but there is those notebook and it is irreplaceable. You have to have a notebook with you, keep noting down, work on something, very regularly, every single day, more time is spent on something, more results you will get, absolutely no difference.

Prof. Arun K. Tangirala: I think in research, you have to be prepared to ask even silly questions. There is nothing like silly questions, but even at some point you may question your own knowledge of what you thought you understood for years and suddenly you come to a point and say did I really know this. Or sometimes, I even say that maybe you sit down and ask is really $1 + 2 = 3$ and so on. I mean there are points, you go through some real moments like that, and you just be prepared. The other integral part of research, I would say attitude is to be able to take feedback from what you can. I think feedback is a very, very integral part. Feedback from your meetings with the advisor, feedback from your own results, feedback from the mistakes that you make; everything, even your success gives you a feedback, everything.

Prof. Andrew Thangaraj: So, there is one student of mine I remember he said, no I really enjoy working in this area, I thought it will be exciting and everything, but after doing research I realized that about 80 percent of my time I am not doing the exciting stuff, doing the exciting stuff only 20 percent of the time. 80 percent of the time I am doing either coding up a simulation or doing something.

Speaker 3: It is 80-20 rule.

Prof. Andrew Thangaraj: It is a lot of drudgery and routine, and that is actually common for every field; you look at Sachin Tendulkar on the cricket pitch that is 20 percent of his time.



Prof. Arun K. Tangirala: True.

Prof. Andrew Thangaraj: So, 80 percent of his time, he is practicing some nets.

Prof. Arun K. Tangirala: Yes.

Prof. Andrew Thangaraj: Doing in the nets. So, that is what happens in every single area. You might enjoy it, I am sure; Sachin Tendulkar enjoyed his cricket, there is no doubt about it, but it involves an equal amount of routine, and you have to be ready for that even if you really are passionate and you enjoy your area, you have to be ready for the routine.

Prof. Arun K. Tangirala: He has perhaps done a lot of research on how the ball comes and so on.



Introduction to Research
Prof. Prathap Haridoss
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Prof. Abhijit P. Deshpande
Prof. Andrew Thangaraj
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture - 03
Art of inquiry - Postulating

Prof. Prathap Haridoss: I think, many of the points that you heard so far do convey this idea that first of all, you have to be interested in the area, and that is where we feel, you know, that you should have... when we keep saying passion is something that comes because over period of time you have been very interested in that area and you have recognized that you are interested in that area. So, that itself takes some while for you to figure that out.

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There many things that you might feel that you are interested in, but there is something that really strikes a chord with you, so then that is something that you pickup. I think through this discussion also indicated that there is a general acknowledgment - both in



the student community and you know people who have already been researchers and so on - that research is challenging. I mean, there are a lot of aspects of it that are challenging. For I think the most important reason for that is also the fact that we are learning about research as we do research; so especially the first-time researchers. When you come in as a student, you actually still may not fully be, you know, aware of what all is involved here, and as you keep going through the process, you are learning the process. Whereas, when you do course work, it is something that you have done since first grade or first standard onwards you have been doing course work, you do classes, you write exams, you get marks; that is a pattern that you know. This is a completely new pattern that you are handling, which comes, you know, all of a sudden after you do your bachelor's degree or your first masters degree and so on. It becomes challenging for that reason because you are both trying to work to, you know, go through the process, and you know complete the process, and if in the back of your mind, the intention was to get a degree, then that is also there ahead of you. And the whole process of doing it, is also something that you are discovering. So, that is why, that is one of the reasons why it tends to be challenging. And as we also pointed out, there is lot of routine work that is involved here, which you should not over look. It is not like, you know, you come in here and suddenly you come in to a research program somewhere, and then, you know, one fine morning, you suddenly get a bright idea and that's it, your work is done; it never works that way. It is a lot of routine work that goes on, that you have to keep working on, and building on all of that routine work, you know, your experience goes up; your experience goes up, your confidence goes up, your ability to, you know, identify insights into that area starts going up and that is when you actually start making those in-roads, which you can look back and say, you know, I was now beginning to do research; so that is the point that you need to understand. So I would say, you know, even when you look at research, and when you say from the perspective of how you discuss it in front of others, when you do experimental work, and you collect data, in a sense you are doing that drudgery; all those hard work that, you know, is involved in collecting the data. When you discuss it, when you try to put perspective into it, when you try to say that, you know, this is the reason why the experiments gave that particular answer, that is when you are sort of actually doing research, beginning to analyze that data, and convey something out of it, which is more than just simply saying that you know particular parameter increases when you an increase some other parameter. So, that is just data, but why it is increasing is what research is all about.



Prof. Abhijit P. Deshpande: An aspect of why we have said also research is challenging is, is its long term and cyclic nature of it. This I have experienced with second year, third year, undergraduate students when they come to try to just see, you know, I want do a research project, what happens is when you start off, there is of course learning and you are doing new things, and you might get some good results also, but sooner or later three weeks or four weeks down the line, you will... may be either something may not work or may be what you thought actually was opposite of what you are getting, and all those things - so that's the cyclic process. So, research, somewhere you will find that there are things which actually are not according to your expectations and that is where the what I found is, some of the students who can actually pass that, during those phases to fall back on routine, to fall back on your knowledge, and try to think of something different, and then pass through that negative and then come up again, so that's the big spirit, that is where it is very, very challenging. You have to persist as Arun said earlier.

Prof. Andrew Thangaraj: It is all in the mind.

Prof. Abhijit P. Deshpande: Yes.

Prof. Andrew Thangaraj: This quote.

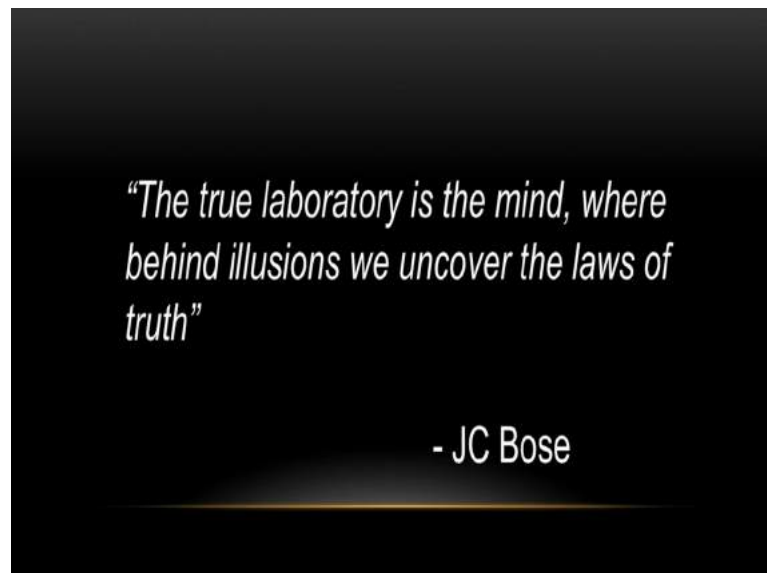
Prof. Arun K. Tangirala: I think also, may be to reiterate what Phani mentioned earlier, as you are working through, it is important to keep focus, to stay focused on the short-term part, but not to lose sight of the big picture that you are working towards, and it always helps projecting what you have in the context of the big picture. Where does your result fit in? Is it really fitting into the big picture nicely or is it really taking you away? And the other thing that I want to mention is in research you may start out with the problem definition in mind and as you are working towards a problem, solving the problem, you may find some other problem which requires or probably or which is of equally important proportion or magnitude, that is probably worthwhile carrying out research in itself, which means that your problem definition is subject to change along the pathway. Of course at some point in time, for practical reasons, you have to freeze a problem that you are working on, but however, in the first year of your working, the problem definition can change, and some things that you thought were trivial have really not been



addressed by anyone, and you find that there is a lot to work in there, and that can become your own problem of research and so on. So there is a quite a bit of flexibility in the initial years, which has to slowly freeze. It's like you start off making ice cream, and finally, you have a frozen product as an ice cream, you cannot really have a liquid state also. There is a molten state, then there is liquid state, and then there is a solid state. It goes that way, right.

Prof. Prathap Haridoss: So, some of the things that we have been discussing so far are nicely captured in this quote which is from JC Bose.

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And it simply says, "The true laboratory is the mind, where behind illusions we uncover the laws of truth". So, that is the quote attributed to JC Bose and it sort of brings out this idea that, you know, we are applying our mind to look at new things, in new ways, and trying to come up with learning, which we can then convey to others.



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Sequence in which a research work is communicated is usually not the sequence in which the work was conceived or carried out

As we also discussed through this process about the mundane aspects of research; we also had this comment from Abijith, which was that it is a cyclic process. So, often you start off, and you feel very happy you know that your initial experiment started working, and then I know two, three weeks down the road something does not work; and I would say that is the biggest difference between what I say an undergraduate student trying experiments for the first time goes through and say a more seasoned researcher goes through. Typical undergraduate student going through research or attending the research for first time gets totally disheartened, when the first set of experiments fail, and it really looks like the whole process was pointless, the whole exercise was pointless. Whereas a true researcher, who has had experience in the process appreciates that, you know, this is likely to happen, is quite comfortable with the idea that, you know, his first set of experiment did not work out, and sits back, and then tries to analyze, ok we got this and it did not go the way we wanted, so where have we gone wrong or where is it that we need to reassess our approach. That is something that we would like to now discuss, which is dealing with failure both when you do the work, when you try to present the work, and may be, how other people have had experience with it.

Prof. Andrew Thangaraj: Yes. Actually, if you look at it, most research methodologies course, layout like a path by which you do your research. You do literature survey, then



you identify a problem, and work on the problem, you solve it, then you publish it, you get out. At least in my experience, I have seen very few problems work that way. In case you kind of identify any area, start working, you do something, and then you invariably may not make progress, then you stumble on something else, then you try something else, then you see some other area, you listen to some talks, somebody is talking about that or maybe I try this; so it kind of goes randomly. There is colleague of mine, who put it very nicely, he said, research is a random walk. So, when you are walking in that fashion there will be failure, and of course, you should expect that; I mean, this is not something unknown to people but dealing with that becomes hard. Maybe in the initial phases you can deal with it little bit because you have not really published it, but when you try to publish the work, even there you will see things will get rejected. First time you submit your work, very high probability, it will get rejected because you are trying to convey something to somebody else, very first time you are writing something, it will happen that way, and you have to be expecting it. The idea is there are two ways to react to it, when something gets rejected or something does not work, you can get either totally dejected or you can throw it away, or you will get totally combative, say, no this guy is talking nonsense, etcetera. So, you have to hit a nice middle path; you take what is possible. And one great advantage of research, which you can use to get over your failure, in many cases, you should change your initial assumptions. You are allowed to do that, this is not a text book problem, where conditions are given and you have to get the answer right; you can keep changing the questions. So, quite often people miss that point; you change some assumptions, you change the application, you can change so many things to make even your failure become a success.

Prof. Arun K. Tangirala: That was a keyword; assumption was a keyword ringing in my mind just a few minutes ago. Every research work is based on certain assumption; there is no research work which is devoid of any assumptions, and what I tell my students is, first be aware of the assumptions that you have made. That is the frame work in which you are solving the problem.

Prof. Andrew Thangaraj: Yes.

Prof. Arun K. Tangirala: And also ask yourself whether those assumptions are too



restrictive; I mean it only applies to your mind and it only that you have conceived of those assumptions or do you really find many problems falling into this frame work. So, your assumption should not be too restrictive; as long as that's not the case, as long as that's the case that your assumptions are not too restrictive and as long as your results are consistent with the assumptions, then you are absolutely fine. And I think failures definitely are bound to happen, and they can happen at different stages, like Andrew said, it could be at the publication stage or it could be just in your own discovery as Abijith said, something completely contrary to your expectation, to your intuition can happen. But remember there is nothing like success or failure in research, what is important is, what you expect and what you have discovered, and taking that into your account and coming up with the solid discovery, with the solid answer to a question. Even saying that there exists no proof for some theorem or for some result is also a great result. That should not be taken in a negative stride. Again, failures are very important, but most importantly what I would say is when you get a result, to avoid may be rejections by reviewers and so on which are bound to happen, I think you should be convinced of your result first, that's a most important thing. Many-a-times, we have come across different students, some students are not sure of the results and the advisor has to really put in some effort to convince that yes, this is right. And there are situations where the student, the PhD student, the researcher is convinced that this has to be the result, even if the advisor says no. So, that conviction of what your result has is very important and that requires a combination of your intuition, and of course, a procedure that you have followed, and of course, with validation. There are three things: intuition, and then discovery, and validation. If you have done all of these in a proper way, then you can be convinced about your result and it is all a matter of convincing the other person. But, of course, dealing with failures is important, and failures occurring towards... what you call as a failure, you should have to first define a failure, but what people call as failure and if it occurs towards the end of your degree or so-called expected duration, then yeah, it is more difficult to say and that is where I think advisors play a very critical role in cushioning, and in giving comfort, and so on, and giving advice in the right direction. In fact, sometimes with just change of idea, you can convert your failure to a success.

Prof. Abhijit P. Deshpande: Yeah, and failure can be in various ways, I mean, one example where I had with my PhD advisor was I was writing some two page summary



report, and it went through iterations and the first, second, third, fourth iteration, and then we both again met and again he pointed out something, and so that is where I had a sense of failing because I was getting irritated, and that is where, he then told me that look, this is not something to get irritated about. I mean, So, there, then if you again persist with it and say ok let me learn from whatever is being the process I am going through, then seventh and eighth iteration, finally I was able to write that report. Based on what is expected of me for being a researcher. So there again a sense of failing was there, but somehow I was able to come out of it by persisting.

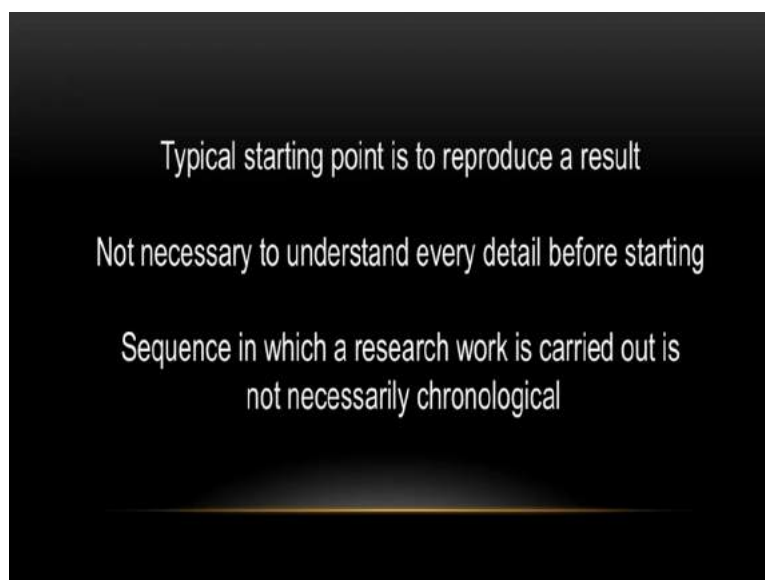
Prof. Prathap Haridoss: I also want to add that, you know, when we talk of you send in a paper for publication, and then it comes back rejected, and I think there are already few comments on these aspects. So one of things you should be open to, I mean, I am not suggesting that you should be over confident, but one of things you should be open to is the possibility that the expert who looked at the paper may also not be right. So, they send, they look at your paper from some perspective, their own knowledge of the field and so on, their idea of what is important and what is not important in that area, their idea of what is possible, not possible, etcetera. Lot of things they are bringing into the process when they look and evaluated your paper. So, when you get a review from a journal, which comes from, you know, few different reviewers whom the journal has approached, it is not necessary that they are always right. So, feel free to openly look at the review. When you get the review even if it says, no, we do not accept this paper, it is not up to the standards or something is not correct about it, you read it, don't just say that ok two out of three people said it is not correct, so that is means it is not correct. So you make an independent assessment of the report that you have got ok. So, you look at it, you try to assess it, and then see if that makes sense to you, whether you agree with what they have said or you feel that they have actually missed the point. In which case, you need to, you know, revisit, how you present your work; you may need to give the right kind of background so that when you send it to another journal, the reviewer does not go into the same tangent. You are preparing them appropriately so that they are directed into your line of thought. So, that is the way you need to look at it.

Prof. Abhijit P. Deshpande: Important aspect of dealing with failure also is to be able to talk about it and I think there are multiples sort of places, where such help can be there.



So, your own friends first you can talk, your own lab mates or your advisor; and so, there is a one important point, which we will discuss the later on, is also to recognize a peer community, so we will come to that little later in our talk.

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Prof. Prathap Haridoss: So, one point that I want to bring out here is again when you read research papers, and you also you know as part of this thing that you were told to, you should do some literature survey and so on. Often those papers are, in fact, not often, always those papers are presented in a very systematic and sequential manner, ok so and as a first-time researcher when you read those papers, and you think of the work that you are doing in the lab, you really feel bad because you are not doing anything in that level of sequence at which they are doing it. Now, very systematically they have done some three, four things; they have systematically identified a specific set of materials which they need to work on, and they have done only you know 25 experiments, from that they were able to get a very nice pair of graphs, from that they could you know clearly tell you that one particularly region is a maximum, they give you the answer. So, that's very nicely presented in a paper. The truth is, often in most cases, that is not how the work actually happened. So, they would have also gone through the same kind of frustration that you are currently going through, they are trying out various different samples, and many of them are, you know, completely giving them tangential results or not at all



showing any kind of relevance to that particular parameter that they are trying to explore. They learn something from it, maybe they have chosen a system that doesn't work and so on, they change the system and go on. So, in the paper they publish, they do not often tell you that we looked at 45 systems out of which 39 failed, only 6 of them were interesting and those 6... they do not show you all those graphs of failed research, all of that they ignore; they just show you the 6 that work and in the order in which it logically leads to the conclusion that they have eventually reached. But behind each paper there is a lot of failure in terms of samples that didn't work, in terms of analysis that was wrong, may be experiments that were wrong, experimental setups that were wrong, all of which is not getting reported. So what you experience in the lab is not new, everybody experiences similar stuff in the lab; I mean equipment fail, experiments go wrong and so on. We just have to keep working with it; so, that is something that I want to.

Prof. Arun K. Tangirala: Having the analogy that I wanted to give is, when you go to some one's place, and you know, that person is presenting you with a new dish, I am fond of analogies, and you really like that dish and it is been presented very nicely, ornated, and when you ask for the recipe, there is a list of instructions given in a very detailed and a sequential manner, and you think really that the person followed that. That has come out of experience. That cook will not tell you that there were ten dishes of this types which were burnt earlier; this is the eleventh one that is being presented; and this eleventh one is success because of the experience with those ten ones. And obviously, nobody wants to present that because sometimes it may not be relevant and sometimes probably it will put you in bad light. But most importantly for the benefit of readership and the audience always the paper is presented in a coherent manner, because you don't want the reader to go through the same torture that you have gone through in discovering this. So, I think it is a very important point that Prathap makes, and you should only read the paper to know what it is saying, but rather not to mimic the sequence in which they actually arrived at, that they used to arrive at the result.

Prof. Andrew Thangaraj: Yes.

Prof. G. Phanikumar: Which also means that when we read a research paper, we also do not read from the first line to the last line in a sequential manner. We can actually look at



what is new there, and what is that I need to do to validate what that person has done, and then what is it that I can extend. So, even the reading of paper is not in the same sequence as it is written, and very often, actually, we now need to also think whether we should go through every single detail that is there in that paper to be able to conform that they are up to something good that I can build up on. So, I believe that we take some kind of a philological approach, that is given these conditions what is it that I need to learn, pick up, so that I can go and validate my hypothesis, and then, go for the idea that I want to prove. So in other words, there is something like a black box approach, we do not need to know every single detail, but at the same time we do not need to also brush away the importance of details. So we should come to the details where is necessary, but we should be able to move forward with the assumptions.

Prof. Andrew Thangaraj: That is true. So, I think again we have been focusing on a new researcher who is beginning to do research, and one very good way to start your research, of course, is to go out and once you fix the area and area is decided, go out and see what is latest happening in that area.

Prof Arun Tangirala: Not reinvent the wheel.

Prof. Andrew Thangaraj: Go, look at recent conferences, the most recent conference what is been published, what is being talked about, and then maybe you pick a paper which you like for some reason, and the first thing to focus on, I believe is to be result oriented when you read the paper. How can I reproduce the same result that this person has, is it even possible? If your conditions are not allowing you to do it, maybe you should not focus on that, you should move to other.

Prof. Arun K. Tangirala: There is always one advantage to that; you are trying to reproduce a result.

Prof. Andrew Thangaraj: Yes.

Prof. Arun K. Tangirala: One, of course is, it helps you get familiar with all the nit bits, and the other thing is may be that researcher has made a mistake.



Prof. Andrew Thangaraj: Yes.

Prof. Arun K. Tangirala: Not deliberately of course; hopefully not, but has over looked and you probably have the chance to do that. Always remember, the creator has enough made sure that there are enough complexities in this world. That every researcher forever can actually keep discovering something new; that is what I believe in.

Prof. Andrew Thangaraj: Yes. Yes. That is true. And actually, when you are result oriented, you will also feel much better about doing your work. You are doing something; I mean not just reading and reading and reading. So, many initial researchers fall into this trap that they need to know all the basics of every single tool or technique that is being used in that paper before they can reproduce that result; that is not really needed.

Prof. Arun K. Tangirala: That is why I tell my students that, look at the baby, when all of us were babies, we really did not say that I learnt how to crawl and walk first, and then only learnt how to eat or vice versa. I was trying to learn how to pick an object, how to eat, how to crawl, how to pinch, how to scream everything at the same time and this not necessarily happening in a sequence. I think beyond a certain point, reading should happen as a parallel action rather than as necessary.

Prof. Andrew Thangaraj: It is true and when your result oriented, once you figure out how to get that result you will also know what tweaks you can make.

Prof. Arun K. Tangirala: True and also what material to read up further.

Prof. Andrew Thangaraj: Yes. Exactly. So, that does not mean that you can ignore large chunks of the paper and then hope to get anywhere, you know.

Prof. Arun K. Tangirala: Right, Right.

Prof. Andrew Thangaraj: To be able to reproduce, you should know enough about that



paper, the important aspects of the paper, which help in getting that result, you should know, and you should know them very deeply; only then you can innovate and make those little tweaks.

Prof. Arun K. Tangirala: True. Actually, that clears the lot of haziness that exists in initial stages and kind of brings clarity.

Prof. Andrew Thangaraj: Yes.

Prof. Arun K. Tangirala: You start seeing, where what your direction is, what material to read up, and your learning becomes more contextual.

Prof. Andrew Thangaraj: Yes.

Prof. Arun K. Tangirala: So, you really know which text book to pick up or which research paper to read, and in fact, how to read future papers.

Prof. Andrew Thangaraj: Yes, correct.



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Indian Institute of Technology, Madras

Lecture - 04
Group Discussion on Research Part 4 of 4

Prof. Prathap Haridoss: So, I think one of the things which comes out of the discussion is that you know, this trying to reproduce a work as the starting point, as Arun said and as Andrew also pointed out.

First of all, it makes you very familiar with the area and it also helps you built your sort of the kinds of experiments that you are trying to try out. So, and in that process one of the ideas that you have to become comfortable with is that, eventually when you are I mean further down the road your results may add up to something significant. But, on a day to day basis, most of the time your experiments or your results are going to be small steps, each individual step will add to something.

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So, just making a small experimental setup may take you like 2-3 weeks. To get an



experimental setup, that is working correctly, where you have been able you know when we said you know reproduce some original, some already published data. That experimental setup is able to actually consistently reproduce some data. So, that takes some time. So, if you have managed to you know, taken, if you used about 2-3 weeks to setup that experimental process that is not a waste of time and you should feel happy, that you should feel happy that you have actually progressed from having nothing you had a clean state, you had a clean board ahead of you and then, now you have a setup that consistently gives you data that is believable, defensible and believable.

Now, you can try some new sample. So, this idea that there are small steps that you have to work on is something that you have to get used to and for those small steps to occur you really need to be a regular researcher. So, you have to put in those hours in the lab. It is not something that you can you know, suddenly one fine day everything will be in the lab and your first experimental work, it does not work that way. You have to keep working on it slowly and steadily and therefore, those regular hours in the lab really make a difference. I would also like to add that you know, sometimes there is this misconception that only if you are working late night hours that you are actually doing great work. It is not like that, if you even if you work very regularly from 8 in the morning to 5 in the evening and then take your evening off and be free and comfortable, you could still be doing great work. You could still be doing research work, you may be thinking about your research later in the evening without necessarily running the experiment at that point and time, but still that would be a lot of good work. So, just keeping up those regular hours, helps you add those small, small steps which then add up to something that is significant.

Prof. Abhijit P. Deshpande: All of this, actually the notebook that Andrew mentioned earlier is a central aspect. So, noting down let say, you suddenly while you are looking at some equations and you notice some dependence. So, it depends on the cube root, note it down, observation. So, it is a simple small fact, but it is an observation that should come into your note book or let say after 1 week of things, you plot it something, something, plot it graphic, put it in your notebook, that is a simple result that you got.

Prof. Arun K. Tangirala: It is lot easier to do this electronically also.

Prof. Abhijit P. Deshpande: Yes, electronically. So, all that should go into your notebook.



So, these are all these simple observations, results which then eventually will be helpful to build later on.

Prof. Andrew Thangaraj: Yes. So, one for the point you want make about all these when you get a small result, you have to be first of all excited by it, you have to believe in it and you have to be confident about it and you have to proudly talk about it to others; your students, your advisor, everybody and you have to show that you know, simply it might look like, why should I be so proud about something so small? I am doing something so small, other people are doing great. Yes, they might be doing something great in your opinion, but you have to be also proud of the small thing you are doing.

The reason is, there is lots of reasons for it because when you are proud of that you keep thinking about it a lot, you keep going back and back about thinking about every single way in which it can be extended and all that builds up on this small result of yours and you get a bigger result. So, I mean Prathap was mentioning, it is experiments a lot and it does not necessarily apply only for experimental work, even for theoretical work all these things are true. The final big result that you derive, a theorem that you derive, does not come just like that. You know, you might show it for one special case one day and then one more special case but, if you don't believe in it you are not excited by it, you do not have confidence in it, you are not going to keep exhausting all the cases and build the general idea that gives you the big result. So, it is very important.

Prof. Arun K. Tangirala: I just wanted to add, the continuation of what Andrew just said. Somehow there is a misconception that experiments have to be performed carefully and simulations don't have to be. The same rules, when we teach courses on design of experiments we make it very clear, that all the rules that you learn or the theory that you learn in design of experiments equally applies to simulations. The only difference in simulation is you can play around a lot more. But, research is not just about playing around it, its playing around with a purpose in mind. So, even experiments are done with a purpose, simulations are done with a purpose. So, you have to choose your simulation settings. You have to know, how your simulator works? What kinds of solvers are being used? And what is the engine that's running underneath and so on?

To make sure that, you have chosen the right settings and tool that you are using is being used for the purpose because, simulator is a very deadly weapon and you can actually



end up producing so much data and only a small fraction of it is going to be useful to you and you may be lost in the deep of data. So, please remember that the rules that apply, the theory that applies to experiments or equally applies to simulation as well and like Andrew said, even in to theoretical work.

Prof. G. Phanikumar: And then, there is a small point about the discipline also. When we see one small result that is promising and we are proud of it. To grow into a big topic, worth making a publication or a presentation and may be becoming a whole thesis by itself, that process requires lot of discipline. So, putting in those hours every day and the being very meticulous about the day to day work and then subjecting every extension to the logical process is very important and, if you go around the labs where research is being done, whether it is universities or national labs or a research divisions of companies, you actually do not see too many people in dirty clothes, long hair, incompetent, working in wee hours, actually you see normal people, well dressed and coming to their work for a significant number of hours every day.

Maybe more than an average employer because you see that people do put in more number of hours of work in research than outside, that is mainly because of the passion. But, you need to put on a regular basis and in a disciplined manner, so that you can grow your small idea to a big one.

Prof. Andrew Thangaraj: But Phani, there are some badly dressed researchers also.

Prof. Arun K. Tangirala: And having said that, I think I just want to re-iterate what Prathap just said that. See, in research your thinking does not end with the time you close your lab or your computer or your notebook and so on. The thinking process is a 24 hour process; it even appears in your dreams. I do not know if you know Kekule discovered the Benzene structure in his dream, right. Which means, he was really thinking into it, may be thinking about it while he was eating, in his shower, where he was walking, running, everywhere so I think research that thinking process, the thought process that goes into it, is a never ending one. You cannot close that lab of the mind that JC Bose mentioned, that lab cannot be closed. All the physical labs you may close, but not the mental lab that you have. So, make sure that you are thinking about the problem and you will hit the treasure very soon.



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"Science is simply common sense at its best, that is, rigidly accurate in observation, and merciless to fallacy in logic."

- Thomas Henry Huxley

Prof. Prathap Haridoss: So, another quote in this context of whatever we have been discussing so far, is attributed to Thomas Huxley and it simply says "Science is simply common sense at it is best, that is, rigidly accurate in observation, and merciless to fallacy in logic." So, I think that sums up some of the ideas that we have been try to convey so far. We will now look at, how do you as you start collecting all those small results and you know, building up your depository of results.

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How do you know your results are right and significant?

Peer community of your area

Identify your peer community

How do you know that your results are right? How do you know they are significant?



And what is the process involved in this? So, maybe Andrew can start with that.

Prof. G. Phanikumar: Yes, I want to say something.

Prof. Andrew Thangaraj: Yes, yes.

Prof. G. Phanikumar: So, basically I want to say that, the research enterprise that we have been having so far several hundreds of years is not a democratic process. That is, if everybody agrees to it, it is it. No, it is not like that. The one expert can prove everybody before him was wrong and then his or her idea can then prevail forever after that. So, it is not a democratic process. It is also not a process where, somebody highly recognized is able to tell and then everybody agrees. So, it is a process where actually the peers are willing to review each other and then hold something as true so long that it does not been disproved. So, that peer part is very important and if you notice the way we go about disseminating the research results, it is basically by peer review and then peers criticizing and validating each of this work and going about. So, identifying what is your peer community is very important and I think professional societies play a big role in that aspect and you have something to say Andrew.

Prof. Andrew Thangaraj: Yes, I think that is important quite a few, we all work like Phani was mentioning it, it is a global area of work today. I mean you cannot say my work applies only for India or some in a local way it does not work, research is global today, your competitors are all over the globe definitely in China, Japan, USA, Europe, everywhere and all of those places are connected together by the peer community and peer community is international and there is also a significant national peer community and you have to identify the peer community. For instance, in Electrical Engineering many of these areas IEEE is a big peer group and the technical societies of the IEEE cover almost all the areas of Engineering and I am sure every area has a similar things that are local chapters, etcetera.

The important thing to do is to identify the journals that are run by these societies of reputed peer communities and then identify conferences that are run by them. There is lot of things there, they will give something called Technical course sponsorship etcetera, etcetera those are different from the conferences that the society itself runs. So, you have to identify those which are critical and then try to see, how high you can go and there are different levels there I mean it is there a different gradations for conference, at different



gradation for the journals, even when it is run by the same society and it is ok, to identify the best you can do and then be wherever you are. But, identify a solid reputed research to peer communities both, nationally and internationally and try to publish your results there and that is the number one way of figuring out how relevant your work is, how good you are.

Prof. Arun K. Tangirala: So, I just wanted to add. Going back to the question of, how do you know your results are right and significant? Now, as I think I have made this remark earlier as well, that you need to have an intuition. Sometimes you do not, sometimes you do. But, you do develop the intuition with experience and somewhere you know, that this result make sense or does not make sense which means, that when you get a result it is important to validate it qualitatively and quantitatively, that is the first important step. When, if you want to say the result is right, the conviction should begin from you, you should not depend on someone to say it is right right and then sharing with your, discussing with your advisor that is a second step because, there is a discussion and may be you are right and you can convince or maybe you are wrong then sharing it with your own research group.

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So, run your research through you know, make a small group presentation. Even if your advisor does not insist on it, you say that I would like to discuss this results in the open with my group mates because and hope that you will get a critical feedback and very



often you will get a critical feedback. Simple discussion like they say, when you are upset, when you are sad, simply pouring out your grief, listening here will really help you. Likewise, when you are excited about your results also, sharing it with your group members really helps you because a lot of perceptive come in.

I think, the key word in research is perceptive and sometimes insignificant results can become very significant. Just with a change of perceptive and that is a very important part and then comes sharing it, communicating it to the scientific community, through conferences, through papers and so on and that is why the peer community plays a very important role. But, the starting process is with you, you are the owner of the problem, you are the owner of the result, you take complete ownership of the credibility of the result, right and that is, for that to happen you should have done enough validation either theoretically, experimentally, through simulations, whatever. You should have a second way of showing that your result makes sense.

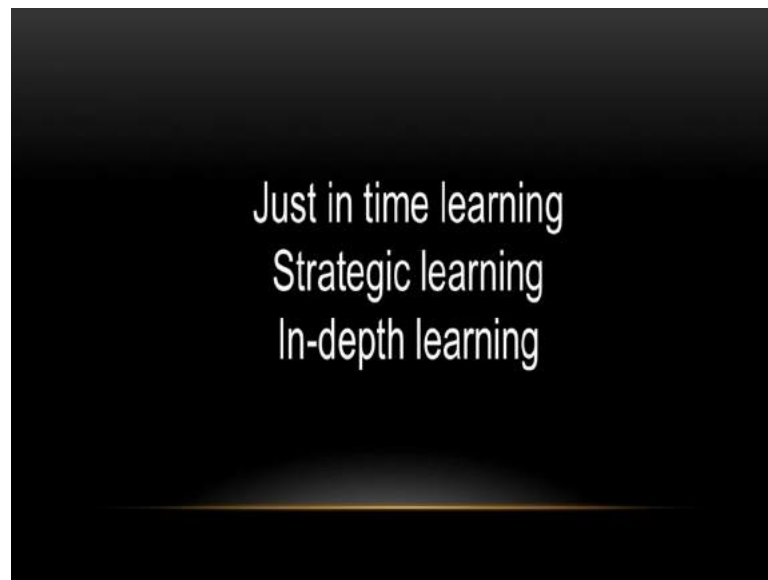
Prof. G. Phanikumar: There is one point that we want to make about learning. Do we go into every specific detail of every tool and pick up all the things before we embark on research.

Prof. Arun K. Tangirala: For research.

Prof. G. Phanikumar: So, in research I think there is a combination of multiple techniques of learning.



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So, we normally talk about, what is called Just in time learning, which is applicable to for example, when you are making a dish out of a recipe you know you need to only know about the detail which is necessary to make the dish. You are not going to the theory of what can mix with what, for example;

Prof. Arun K. Tangirala: And you do not want the science of material.

Prof. G. Phanikumar: Do not want to go to the science of material. Then, there is also another learning paradigm called Strategic learning. Where, you basically have a goal and up to achieving that goal only you are interested, you are not interested in anything else and then, there is another paradigm called In-depth learning, where you are really interested to know go to the bottom of the knowledge and then get that knowledge to your long-term memory. I think research encompasses all these three in a way. For example, let us say you want quickly make a plot in a way that conveys the idea, you want to pick up a mat lab, the script for that. So, you just learn to that extent, you do not need to go into see, how mat lab actually is doing all these things?

And then, Strategic learning you need to basically solve a problem. You understand only to the extent that is necessary to solve but then, when it comes to your own core area of research, I think it is very important to achieve in depth learning. So, it is very important to be little agile in the way you learn about in research, the way you have been doing in this high school or when you are preparing for this entrance exams, doesn't apply to



research. So, your attitude of learning has to change the moment you want to start on research.

Prof. Arun K. Tangirala: And all three have different time scales, obviously.

Prof. G. Phanikumar: All of these are at different time scales.

Prof. Arun K. Tangirala: Different time scales and PhD typically involves all these three and while converges to in-depth. Starts with Just in time and ends up with In-depth.

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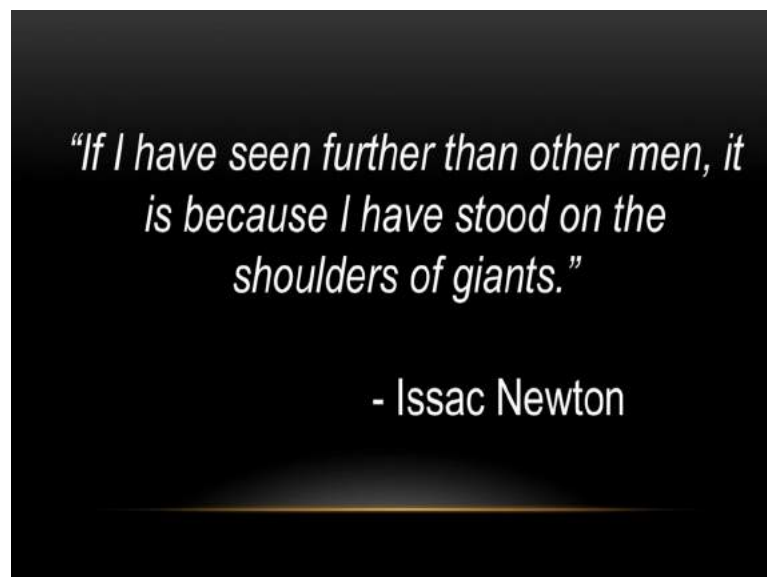
Prof. Prathap Haridoss: Right. So, in research one of the aspects, we have not touched upon so far is Teamwork. You actually consciously or unconsciously participate in Teamwork. Rarely have we, I mean it is not that it is impossible but, in general we don't have people who are completely isolated and then doing research on their own. There are some, but most in most of cases we are working with the group. You are typically student in a group or a Post Doc in a group and so, there are people with a wide range of different experience who are working along with you some you are learning from, some you are teaching and sometimes the roles are reversed and so that is a process that you have to get comfortable with and you have to understand.

So, it is always going to be Teamwork and you have to be you know willing and happy participant in it. You will invariably find groups and you will invariably find the occasional researcher who does not like to participate in Teamwork. I think in the long



run that's not a great idea, maybe in the short run you do see 1 or 2 of them being successful, they try to hide the resources, they do not share the resources and so on. But, in the long run that is not a good idea. World does get along, we spoke about peer community and so on. It is a same peer community, that they will see that you will see and just a matter of time before people accept that you know, this is not a person that is easy to work with. So, Teamwork is very important. You should know how to work with the people around you and you should know how to share credit. So, when you write publications you should know whose work was this? Was the core of that particular publication and therefore, that person should get priority and so that something that you should become comfortable with and just to share a quote here it is from Issac Newton.

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And is something I am sure many of you heard of it says "If I have seen further than other men, it is because I have stood on the shoulder of giants." And, I think that very beautifully captures this idea of teamwork. In fact, he is talking of teamwork in a grander scale, where he is talking of you know not just his group but, you know other people who worked before him, in various parts of the world. Who have contributed to that area and he has understood those contributions and built on those contributions. So, that's teamwork in a grander scale, but there is team work even it at a local scale.

Prof. Abhijit P. Deshpande: Even much local scale for example, you may be using the same computer, you may be using the same balance to weigh things and so there is



teamwork involved in everything because it is coordination, inter working with each other. So, all of these are very important aspects of.

Prof. Arun K. Tangirala: You know, interacting with team really gives you good breadth of the research. Which is also important in a research not just your own depth getting lost and so on. Just not going, single one sighted.

Prof. Andrew Thangaraj: So, having a active group for instance, going into an active group which is publishing in conferences regularly, publishing in journals regularly.

Prof. Arun K. Tangirala: It is positively contiguous.

Prof. Andrew Thangaraj: Yes. It is huge. Just gives you the setting in the atmosphere and completely changes your attitude to everything and you will be surprised how that can make a big difference to starting out in research, going in to a active groupers, very crucial and that it is you might say it is, some of them you might work with and they might be part of a team, some of them you may not even work with. That they are talking about things and they are exciting you and you are motivated by them. It works in a great fashion.

Prof. Prathap Haridoss: So, yes. We are sort of a getting closer to the conclusion of this discussion, just a couple more points, before we look at some general aspects about you know, Human characteristics, Research characteristics and then we will close this discussion.



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So, one of the points that you should get accustomed to is that, you have to be the most critical examiner of your work. So, while you know, you send your work out and you get you know, peer review and you are likely to most likely to get some kind of critical comment. You should be the one who is most ready to critically look at your work and to examine whether you really have done something to the level of detail that your discussion is trying to convey and so on and that requires a lot of honesty with yourself with your immediate group.

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Because, when you convey these things in you know, very blunt and clear manner to your immediate co-workers, you also get feedback which is very, very useful. So, that you have actually covered much of the ground that you need to cover when you send a publication out, so that when a peer who is not working with you, looks at your paper. He or she also understands that you know, you have really covered a lot of a detail, you have looked at all possibilities and therefore, your result is much more believable. So, I think honesty is very fundamental in research because ultimately you are trying to discover something. If you are not being honest about it, there is no meaning in saying you discovered something in it. So, that's it.

Prof. Arun K. Tangirala: In that context, I just want to say that there is nothing like a final result and that's it. Research in that field is over, right. People typically use mature and so on and infancy the researcher infancy and so on. So, what is more important is when you critically evaluate your results, to ask first of all if the assumptions that you have made are not too restricted as I remarked earlier. And secondly, whether the result qualitatively makes sense and most and thirdly whether people are able to reproduce what you have done. In fact, there is a theory of reproducible research in which the researchers are encouraged, if they are doing computational work. These days, they are encouraged to a publish quotes. So, that they can be reproduced or you know if there is an experimental data out of which you have drawn some inferences, post that data, if it is not confidential and so on.

So, you should ask yourself if this result is a onetime affair or you can actually repeatedly do it and in many times, on many occasions when you are performing experiments there is a repeatability issue associated with it, you should have done enough repeatability analysis. So, the critical part is that where you get your result, look at the credibility of the result and ask if it is, if it make sense, if it is reproducible and then of course, talk to your advisor to find out if it is significant.

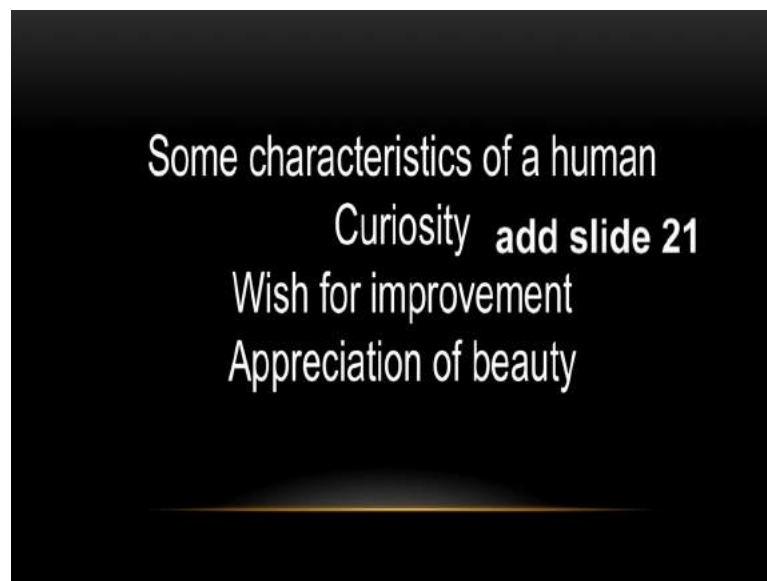
Prof. Prathap Haridoss: So, I think in this discussion we started off with certain philosophical aspect about, what is Research? Then, we have through the past half hour or so looked at lot of you know mundane day to day aspects of research, which is what you will actually experience on the ground, when you are researcher in a group and right now, we will again step back and look at a few more philosophical aspects which will help us put a larger perspective on the whole discussion and the process and with that we



will close. So, may be Abhijith can.

Prof. Abhijit P. Deshpande: Yeah, I mean we started out by posing the question, right. That, what PhD and M Phil? And, it is we want to get these degrees, we want to make good livings. So, that is really the underlying feature in some sense. But, all throughout our discussion you also heard that you being the centre of all of this. So, researcher sometimes wants to step back and try to look at you know, what does a researcher do? And how can how is it? Sort of all this hard work and all these thought and all these labs, lab mind laboratory being mind. So, how do we sort of place ourselves? How does a researcher place ourselves and so in all normal things we are just humans? So, just like humans are curious, all the humans want to actually improve things around us and we of course, all have very appreciation for beautiful things. So, similarly for motivation for research you know, we many times want to justify our work that we are doing in terms of we are trying to do something which because nobody knows about it. So, therefore, I want to know about it. Therefore, I am a researcher.

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Some other times we may step back and say that no, some of the thing that I am trying to do is actually to improve, may be make less effort for some manual work, may be to take something in 2 seconds, what used to take longer time? Of course, these days with the world around us also we are equally, not just humans that we are concerned about it is the earth as a whole. So, therefore, betterment of society and world around us, is also an



important aspect of why we do research and of course, one very important aspect in all of this is also we find, I mean we have talked about joy, we have talked about all these emotions, we have talked about being passionate. So, therefore, there is a duty and essay text in science. There is something when we look at a nice equation, we feel very nice about it. When we see a very elegant experiment we say, oh wow! How could it be shown so elegantly? So, therefore, there is for being a researcher all of these could it be, could be also involved.

Prof. Arun K. Tangirala: So, that is an art in researches

Prof. Abhijit P. Deshpande: Yes.

Prof. Arun K. Tangirala: We have talked about the science part of it.

Prof. Abhijit P. Deshpande: Yes, yes.

Prof. Arun K. Tangirala: But, I think research is both science as well as an art and of course, you know the degree of art varies from a research problem to the other. But, I think it is a very important to ask ourselves and be honestly with the purpose of carrying out research. I think that's it.

Prof. Andrew Thangaraj: Various, there are various ways of looking at it. I think one other way of asking this question is we spoke a quite a bit about, how to do a research? And there is also why aspect and Why, we said in terms of passion extra but, where does the passion come from. So, ultimately the why is answered individually by different people and I am sure you will have your own answer in you when it comes to it.

Prof. Arun K. Tangirala: Yes, yes.

Prof. Andrew Thangaraj: But, you have to answer that question. Otherwise, you do not survive for a long time in research, if you don't know or don't appreciate the why inside a few very clearly and that is actually a life philosophy. You should know why you are doing something.

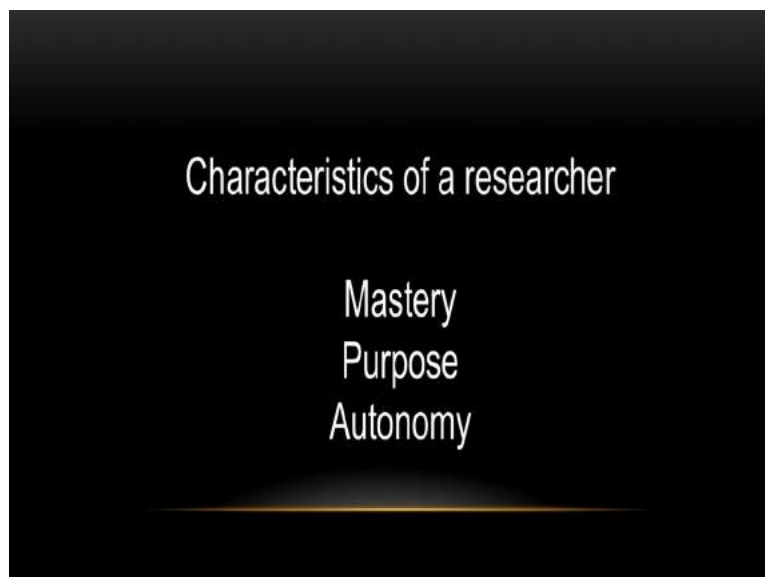


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Prof. G. Phanikumar: So, here I want to relate to what Daniel Pink had summarized from research on motivation. Why do people work? And why people are driven to work? This is something that he has done very nicely in his book, Title Drive and there he brings three aspects.

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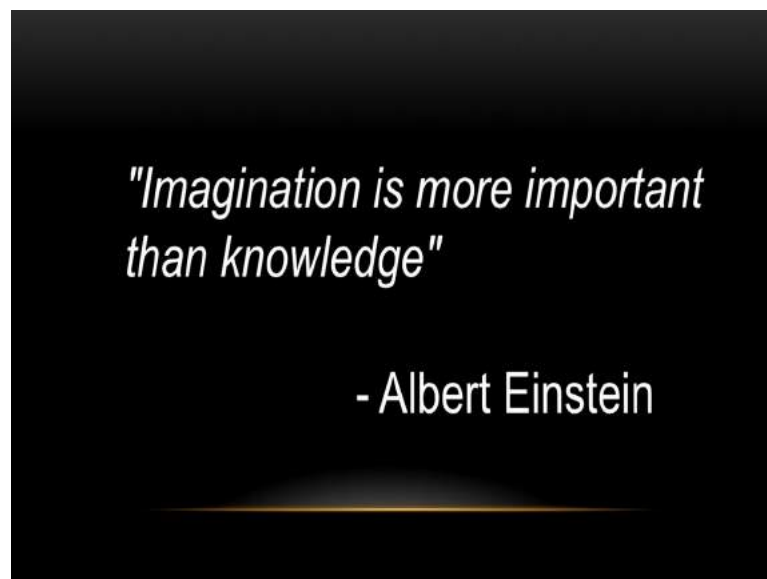
One is Mastery, Purpose and Autonomy. So, very interestingly all these three very nicely relate to research you know, Mastery for example, is where you see a sense of getting better and better as you go along. Researchers are like that you know, as you keep on



doing research over a period of time you are very good at the equipment, the techniques, the methodologies, etcetera. So, you can see that you are gaining Mastery. Purpose, we have already discussed, Abhijith has said that. So, we see a larger purpose for the work that we are doing and then Autonomy, researchers want to solve their problems and the problems for the betterment of people around them in their own way, by choosing the techniques that they wish to use, by choosing the specific methodologies that they wish to adopt, etcetera. So, there is lot of autonomy by which people work. So, naturally if you put all these things together, researchers are intrinsically motivated.

Prof. Prathap Haridoss: So, I think we have now reached the conclusion of this discussion. We sort of started with a quote from Albert Einstein and I think we will it is kind of appropriate to close with the quote from him since, many of us recognized him to be a very significant contributor to the scientific community over the years and it simply says this, "Imagination is more important than knowledge" and that's very philosophical statement.

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So, you need to think about it and interpret it in a manner that make sense to you and then possibly utilize it in the manner that you feel is appropriate. So, I think with that we will close. I hope you had an overview of what research is and I don't know if you add some closing comments.

Prof. Arun K. Tangirala: I think, what is important is this, if you look at if someone is



looking for a work flow for a research, the disappointing answer is there is no sequential work flow. However, there are at each stage there is a systematic way of doing it. Whatever you are doing, you have to do it in a systematic way and learn to and also, make it a habit to analyze what you get out of and then it is a lot of a trial and error. There is a lot of iterative process, but at every stage there is a feedback you take it and then you find out where the problem is and fix it there and come back and so on. So, there is lot of back and forth. However, at each stage you have to do things systematically and analyze the results that come out of it. But, if you look at the grand scale of things yeah, I know people go back and forth and research and every researcher goes through that, not necessarily that the researchers shares those things.

Prof. G. Phanikumar: So, I would summarize saying that, if you want to do research be at it.



=Introduction to Research
Prof. M. S. Ananth
Distinguished Visiting Professor
Former Director of IIT Madras
Department of Chemical Engineering
Indian Institute of Technology, Bombay
Lec-05

Introduction to Research, this is the outline of my talk. I am going to say a few words about the Idea of a University.

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About the nature of science, nature of research, just briefly what drives research, then what do research scholars work on. This is a summary from when I was doing research here, I did a summary of what people work on try to classify the problems, it comes under 6 or 7 classifications. And then I am going to talk about a little bit about learning and creativity, because while course work is mostly about logic; research is a lot about intuition. While you use logic, you make your leads only through intuition. And we have some idea understanding of it from the work of Sperry and coworkers, it is called the Split brain experiments. Sperry got his noble price in 1981 in neuro research. Then I will give some advice that I cannot resist for a research scholars and (Refer Time: 01:06) that's also based on my experience here with students and a last line on ethics.



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A presentation slide titled 'THE IDEA OF A UNIVERSITY' in blue. Below the title is a horizontal line. The slide lists 'The three basic assumptions [Renaissance thinkers - E.O. Wilson]' followed by three bullet points: 'the material world is lawful', 'there is an underlying unity in knowledge - can be discovered by a combined study of the natural and social sciences', and 'education can lead to indefinite human progress'. Below these are three more bullet points: 'Research and teaching go hand-in-hand [von Humboldt]', 'Arguably the most civilised of human undertakings [Oakshott]', and 'Purpose is refinement of the mind not employment'. At the bottom, there is a footer with '01/12/15', 'MSA', and '3'.

Basically, the university itself is conceded somehow we forget to mention this, and I think we ourselves forget it. University is based on, it is a Renaissance concept; essentially the Current University. In the Renaissance thinkers made three assumptions. The first assumption is that the material world is lawful. That is nonliving world is governed by laws; that's what it means. The second is that there is an underlying unity in knowledge and this can be discovered only by a combined study of the natural and social sciences.

We have divorced the two essentially natural sciences includes engineering, social sciences includes humanities. We think they don't have an effect; I will just give one example for example, when in my own area in molecular thermodynamic, Gibbs made the assumption that all the microscopic states of an isolated system are equally probable. This idea, a priori that everything should be equal unless there is a good reason for it not to be, came from the then social ideas of Karl Marx. If you had been at any other time, the Gibbs may have made some other assumption, but we would have finally come to this assumption because this is what leads to results that compare well with the experiment. But the point is that the ideas in society influence your thinking in terms of science also.



Then the third is that education can lead to indefinite human progress. These are assumptions these are the postulates. Then it was Von Humboldt in the 19th century who said research and teaching have to go hand in hand. He said research brings passion to teaching and teaching rejuvenates the researcher. So both are important. And the ever since then all the research universities have been based on this four premises. There is a nice article by Oakshott, Oakshott was used to be a professor of political science in Oxford and it is called the idea of a university. In fact, you can Google it and find the article.

He says it is arguably the most civilized of human undertakings. The idea is simply of course that education passes on accumulated knowledge to the next generation. And no other animal does that, all other animals have to only learn by copying, so by watching and copying. In fact, that's probably the only the reason we are on top of the food chain. Then but the purpose of education is refinement of mind not employment; I think that is important to realize many people come to education thinking this will lead to employment afterwards; it does but that is a corollary. A refined mind will more opportunities and therefore find, I mean more jobs available for it. But what has happened is this has been twisted out of sing, because employment is certainly important for individuals. So, everybody thinks will I get a better job after this? And therefore, they come for it that is not the purpose of education at all.



(Refer Slide Time: 04:07)

A presentation slide titled "THE HOUSE OF EDUCATION" in blue text. Below the title is a horizontal line. The slide content is organized into two sections: "Like an ancient monument" and "Yet very different", each with a bulleted list of points. At the bottom, there is a footer with the date "01/12/15", the text "MSA", and a small number "4".

THE HOUSE OF EDUCATION

Like an ancient monument

- A metaphor like Oppenheimer's House of Science
- A vast house; some wings complete & perfect
- Some parts are still receiving their finishing touches
- Scaffolding being built for new rooms, new wings

Yet very different

- No preconceived design; wonderful randomness
- No shut doors, open to all comers

01/12/15 MSA 4

In fact, the universities are now burdened with not only educating students but also placing them. Which should never have been the case. Anyway, I would like to draw on a metaphor, like this metaphor this is Oppenheimer has a book called Science and the Common understanding. It's a beautiful book! should read it. He draws the metaphor he calls it the House of Science; I have extended it to the house of education. It is like an ancient monument, it is a vast house, it is in some wings have complete and perfect and some parts are still receiving finishing touches. And there are also parts for which the scaffolding is being built. And there are new wings being created. But it is also very different from a monument.

First it is not done to your preconceived design and it has a wonderful randomness suggestive of unending growth. And if it has no shut doors it is open to all comers. Of course, this seems a little ironical conceiving the way people struggled to get it into institution, good institutions in India. But that is an artificial result of our not having enough seats for the number of students who seek education. Basically, it is an open house; it is open to all comers. I mean if you want a degree of course, you have to register and so on and there are limitations.



(Refer Slide Time: 05:30)

A presentation slide with a white background and a blue title. The title is "THE NATURE OF SCIENCE" in blue, uppercase letters. Below the title is a horizontal line. There are three bullet points, each starting with a blue dot. The first bullet point says "Science seeks unity in the wild variety around us". The second bullet point says "It is a pyramid built by research traditions; the pyramid is robust but needs constant modification; " take Copernicus out and you have to rethink Einstein". The third bullet point says "Its theories are produced by intuitive insight, validated by logic and experiment and improved by iteration". Below the bullet points is another horizontal line. The slide is flanked by two black vertical bars on the left and right sides.

THE NATURE OF SCIENCE

- Science seeks unity in the wild variety around us
- It is a pyramid built by research traditions; the pyramid is robust but needs constant modification; " take Copernicus out and you have to rethink Einstein"
- Its theories are produced by intuitive insight, validated by logic and experiment and improved by iteration
- Its theories can only be corroborated not proved

Let me say a few words about science, nature of science. Science seeks unity in the wild variety around us, when I say science it include engineering. And I contrast this in particular with the industry, which thrives on differences. If you go to the industry, you have to first tell everybody what your product is and how it's different from everybody else's, otherwise you cannot sell your product. Whereas, the university seeks unity, it asks what is the minimum number of loss on the basis of which I can describe the entire diversity around us. And you will never seek unity after you leave the university you have to come back to the university to seek unity. It is basically a pyramid built by research traditions; the pyramid is robust, but needs constant modification. Then a saying is that take Copernicus out and you have to re think Einstein. You have to rethink, it does'nt mean you have to completely change everything; normally it's a fairly robust structure, but at the same time, there will be minor changes that occur.

For example, when Einstein introduced the limitations of lights movement, and Newton's idea of instantaneous reaction to a force was lost, it did'nt make any difference to the macroscopic world; it only made a difference when you started talking about speeds comparable to light. But on the other hand, you did have to make that modification. The theories of science are produced by intuitive insight and validated by logic and experiment and improved by iteration. Please note that no theory can be proved; it can



only be corroborated; you say that this is the theory current theory and as results come in you may have to modify your thinking.

(Refer Slide Time: 07:09)

A presentation slide with a light blue background and a dark blue header. The title 'SCIENCE IS AN EVOLUTIONARY PROCESS' is in blue. Below the title is a horizontal line. The slide contains two bullet points. The first bullet point is 'Science evolves by deliberate idea mutation subject to the following selection rules:' followed by three sub-points: 'Ignore ideas that lack testable consequences', 'Reject ideas that fail the tests', and 'Seek ideas that make the widest possible range of exact predictions'. The second bullet point is 'Science uses a double negative process (disproof of incorrect theories) to create a growing store of useful theories'. The name '[Popper and Kuhn]' is written in blue next to the first bullet point.

SCIENCE IS AN
EVOLUTIONARY PROCESS

- Science evolves by deliberate idea mutation subject to the following selection rules:
 - Ignore ideas that lack testable consequences
 - Reject ideas that fail the tests
 - Seek ideas that make the widest possible range of exact predictions [Popper and Kuhn]
- Science uses a double negative process (disproof of incorrect theories) to create a growing store of useful theories

You should read the other book you should read is Popper and Kuhn, they both write, I mean it's not a book they were articles by both of these people. They have also written books, but it is primarily about science as an evolutionary process. What they say is science evolves by deliberate idea mutation, subject to the following selection rules. First, it ignores all ideas that lack testable consequences. You must realize that eliminates about two-thirds of life; because there are lot of things in life do not have testable consequences. In fact, in all of science almost all of science and engineering we deal primarily with materials that don't have a memory and therefore, you are able to deal with them the way we deal with them. If they have a memory then it's very difficult especially if they have a long memory, it's very difficult to have testable consequences for such substances.

Secondly, you reject ideas that fail the tests. If you compare with experiment doesn't agree then the idea has to be rejected. Thirdly you seek ideas that make the widest possible range of predictions. If you can predict two things with one idea and three things with the second idea, you keep the second idea and throw out the first. Because



you are seeking unity, you want to know what is the minimum number of basic laws on the basis of which you can explain everything. So, science uses a double negative process. That is, it disproves incorrect theories, it uses double negative process to create a growing store of useful theories.

(Refer Slide Time: 08:39)

A presentation slide with a light blue background and a dark blue header. The title "HOW DO SCIENTISTS WORK?" is in dark blue. Below the title is a horizontal line. There are two bullet points, each with a small blue square icon. The first bullet point is a quote from Carl Sagan about balancing contradictory attitudes. The second bullet point is a quote from Isaac Newton about standing on the shoulders of giants.

HOW DO SCIENTISTS WORK?

- Balance two seemingly contradictory attitudes: remain open to new ideas no matter how bizarre or counterintuitive they may be and scrutinize ruthlessly and skeptically all ideas – Carl Sagan
- "If I have seen further than others, it is by standing upon the shoulders of giants" – Isaac Newton

So, how do scientists work? Basically, I have got two quotations that tell you quite graphically. First you balance two seemingly contradictory attitudes; this is from Carl Sagan. Carl Sagan was an astrophysicist who taught in Cornell, he passed away about 10 years ago. And he said the one attitude is to remain open to new ideas, no matter how bizarre or counter intuitive they may be. And the other attitude is, scrutinize ruthlessly and skeptically all ideas that come to you. So, this apparently conflicting ideas are what you have to work with. And Newton said if I have seen further than others it is by standing on the shoulders of giants, it is a very famous quote. What this tells you is please do your literature search very well, because you should know what others have done before you are start doing your work.



(Refer Slide Time: 09:31)

A presentation slide with a white background and a blue title "NATURE OF RESEARCH". The slide is flanked by two vertical black bars. It contains a bulleted list of points about research. A horizontal blue line is positioned above the first bullet point.

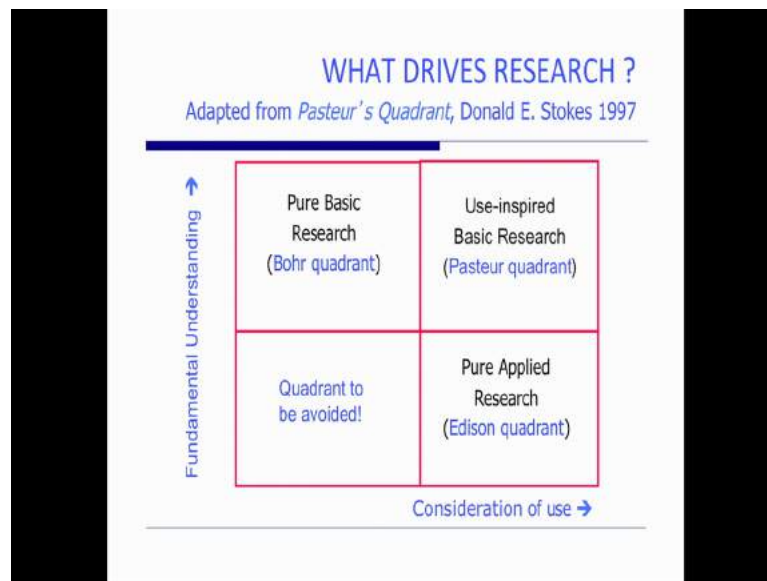
- Research is a **creative process**; it is complex and iterative
- It is the **search for truth** whether mundane or profound
- It is about going up blind alleys to see if they are really blind
- Two kinds: **academic and developmental**
 - **Academic**: curiosity driven, 99% routine 1% inspired
 - **Developmental**: application driven, team effort; needs leadership and considerable funding

Let me say a few words about the nature of research. Research is basically a creative process; it's complex and it's iterative. It is the search for truth, whether the truth is mundane or profound. It's about going up blind alleys to see if they are really blind. Sometimes you try out a method after four years, you discover it is not a useful method it is still publishable you will still get your PhD, it doesn't mean, you would not get your PhD. You are seeing does not have to of course, normally for a negative result, you should have done experiments too. A pure theory that gives a negative result usually doesn't lead to a PhD. So, you have to be careful, if you are working on pure theory you have to be little more sure of your ground.

Then there are two kinds of research in general the university; one is academic, the other is developmental. Academic is basically curiosity driven, but you must remember even here 99 percent is routine, only one percent is inspired. It is routine because people have said things before you keep verifying those things, you keep testing them out and so on. And one percent inspired is of course the great people if you have seen in science and in engineering, various fields. The developmental research is application driven; it's a team effort it is needs leadership and considerable funding.



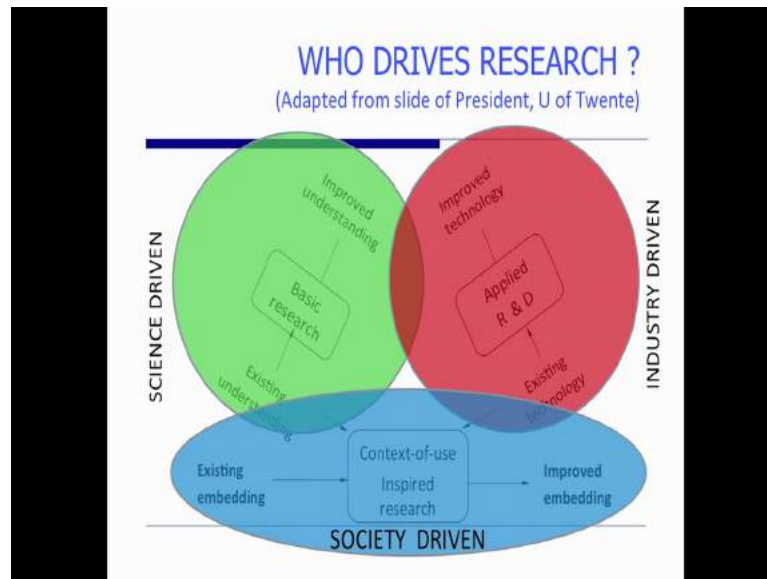
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So, what drives research? This is from Donald Stokes book, but it is basically a well-known diagram. On the one hand, what drives research is fundamental understanding, you are trying to understand things in a fundamental way. The other is the consideration of use. So, you divide the areas space into four quadrants and there is a quadrant to be avoided. The Pure basic research quadrant is called the Bohr quadrant, Use-inspired basic research is called the Pasteur quadrant, Pure applied research is called the Edison quadrant. Incidentally, this is a famous quotation of Edison that said, I mean they asked him I believe he succeeded in making the light bulb only out in his 10,000th trial. So, they asked him how, they went and said you failed 9,999 times. He said no, I didn't fail; I found out how not to make a bulb in 9,999 ways, so that's also a contribution to research. So, these are what you have to do. This quadrant is to be avoided, I will say a few more about few words about that later.



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This is a slide that I borrowed and adapted from President of the University of Twente; we met in a meeting in Korea, it was about research in what how university should pursue it. And the idea is the following, primarily you have basic research which has takes existing understanding to an improved understanding, then you have existing technology which goes to improved technology. So, we have industry driven science and research.

The reason I put this in is, because I think we should realize increasingly what's happening is we are not able the control the applications of science and technology. So, you have some results that come out, turns out that you have made an improvement to understanding, you have made an improvement to technology. But there is a side effect in typically for example, it may be an environmental side effect of a technology. And then there are lots of objections to it and many scientists and technologies have complained that these people who object to it are eco-terrorists. But actually they are not, the point is they didn't know about this technology till you told them about it. So, the idea is increasingly in universities that you should have a society driven research, where you should inform people about what you are working on, where it will go, so that social scientists are also involved right at the beginning and they warn you that this might lead to consequences.



So, the idea is if you take them in the team right at the beginning then you are likely to produce technologies that will be actually useful. Now a lot of your technologies may not be useful. So, there is always a risk of your running into problems, because you did not take these considerations into account. So, it's called context of use inspired research; it is you must know, where it is going to be used, and you must discuss it with people in humanities and social sciences. Sociologist will tell you the society will not take kindly to this or they may say with the existing structure this technology can lead to problems and so on.

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A presentation slide with a white background and a blue title. The title is "WHAT DRIVES THE RESEARCH SCHOLAR?". Below the title is a list of seven bullet points, each preceded by a blue square. The bullet points are: "Interest in understanding an aspect of our universe; making things the 'good God forgot to make'", "PhD, a pre-requisite for some jobs/ career advancement", "A good route to change career path", "Hopefully not want of current employment!", "Not money! At most 10% more in life-time earnings!", "Value beyond monetary returns; good for your ego!", and "Research is its own reward; rewards don't come later!". The slide is flanked by two black vertical bars.

WHAT DRIVES THE RESEARCH SCHOLAR?

- Interest in understanding an aspect of our universe; making things the "good God forgot to make"
- PhD, a pre-requisite for some jobs/ career advancement
- A good route to change career path
- Hopefully not want of current employment!
- Not money! At most 10% more in life-time earnings!
- Value beyond monetary returns; good for your ego!
- Research is its own reward; rewards don't come later!

What drives the research scholar? I am hoping it is interest in an understanding of some aspect of the universe or making things the "good God forgot to make" that is a quote from it's an old quote now, I think some 2001 or 2002, the Arthur Miller was the president of the US Institution of Engineers, American Institutional Engineers. And they have a one week celebration in February of not an Engineer's day, they have Engineer's week. They will have lots of posters, and Miller was apparently sitting in a bus in New York. When a little girl was sitting next to him with her mother, and this girl kept asking her mother, what do engineers do? Then she kept looking around these posters, and finally, she said oh now I know what engineers do, they make all the things that the good God forgot to make. He says that is a very good description of what engineers do. So that



could be one reason why you are a research scholar in this institute, which is a good thing.

But PhD is also a prerequisite for some jobs in career advancement, if you want to be a faculty member, if you want to be a research scientist, you are going to get a PhD. It's also a good route to change career path. If you are in one area and you want to change to another area, it's good to come back to graduate school do a PhD in that area. Then hopefully it is not for want of current employment; unfortunately, a sizable fraction of our research scholars do come because of this reason. And it is also not money, this result is actually for chemical engineers but I suspect it's true of all engineering. They say in chemical engineering there is a huge survey that the American institute of chemical engineers did. They found a maximum of ten percent more in lifetime earnings, I mean you have to leave out the outliers some people make it to the top no matter what their discipline is those people you leave out. By and large you take people who have ordinary career paths and 10 percent more in terms of lifetime earnings. So, it is not because you are going to make more money. Hopefully for the love of it because you are curious and so on.

So, the value is actually beyond monetary returns of course, it's good for your ego. If somebody has to call you doctor so and so. It is a little harder to curse you. And then research is its own reward; this is something that I want to emphasize. Very often research scholars seem to think the reward comes later, that PhD is only a piece of paper, it is something that recognizes what you have done, but more than that if the research doing the research itself is the reward that's when you are suddenly understand something then I think you have.



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A presentation slide with a light blue background and a dark blue header. The title 'WHAT DO RESEARCH SCHOLARS WORK ON ?' is in dark blue. Below the title is a horizontal line. A bulleted list follows, with each item preceded by a blue dot. The items are: 'Intuitive, imaginative research leading to unifying laws', 'Systematic gathering of empirical knowledge for applications/ to check validity of existing theories', 'Extensions of existing theories', 'Synthesising Materials and characterising them', 'Mathematical modeling and simulation', and 'Empirical correlations for design'.

WHAT DO RESEARCH SCHOLARS
WORK ON ?

- Intuitive, imaginative research leading to unifying laws
- Systematic gathering of empirical knowledge for applications/ to check validity of existing theories
- Extensions of existing theories
- Synthesising Materials and characterising them
- Mathematical modeling and simulation
- Empirical correlations for design

Then what do research scholars work on? This is a sort of summary of the kinds of problems people work on. First intuitive imaginative research leading to unifying laws; this is rare. If you are in that category then you do not need this lecture, you do not need any of it. The second is systematic gathering of empirical knowledge for applications this is measuring properties. For example, you may measure the viscosity of many substances. Of course, lots have been measured, but there are a new substance coming up, you will have to measure their viscosity for the sake of applications. Or you may be doing it in order to check the validity of existing theories. I think these constitute about 60 percent or it may be little less, may be about 40 percent of the entire work. If you look at journals and look at all the papers that come you will find this is a very large fraction. Then extensions of existing theories, this is also quite common. A theory may be applicable to Newtonian fluids, you may want to extend it to non-Newtonian fluids.

Then synthesizing materials and characterizing them. This is always been a big area and now it is much more so because of so called nano science and technology. 1959, I think when Feynman gave a lecture to the American physical society. The lecture was titled There is a Plenty of Room at The Bottom; it was the first lecture on nano science. And essentially Feynman said that we have reached the point when we can manipulate individual molecules or sets of molecules, and therefore, you can change the structure of



material, so that you can get the properties you want. See, when I was a student this was unthinkable and I mean that tells you partly how old I am, but basically it was unthinkable. They said this is the property, you substitute it that into the different transport equations, you predicted what would happen, and you say if you have this material, this is what will happen.

Now you know what you want and you ask, can you make a material that would respond to shear-stress in this fashion. So, you can ask such questions, you can ask for materials for example, we have materials that are conducting one way and non-conducting the other way. And they are used often in applications where heat is conducted one way, but you can touch the material from the other side very practically touch it. So, I think there are now what you have is a great potential to synthesize materials, you can change molecular additives here and there, and create substances that have properties that you want, but you have to characterize them of course. That is a very large fraction of the research.

Now at the moment, I think nano science and technology probably covers sixty percent of all current. Mathematical modeling and simulation, this is an important part although I must warn you that you should never get into mathematical modeling, if you don't have a physical understanding of the system. I mean doing mathematical modeling for its own sake is meaningless, although there are some useful occasions where you do that. For example, you do neural networks and this is all this is done primarily for control. It is not so much for understanding the science behind it, but to understand the relationship between output and input of a system; and use that to control the system. In chemical engineering, it is used, presumably in metallurgy also because in large number of systems you have a lag in measurement.

Ideally what you would like is find a product, set a set point, find the difference in properties between the product, and what you wanted, use that difference to drive the process to correct it that is feedback control. But in most systems in chemical engineering, particularly the feedback comes too late. For example, if you are saponifying oil to produce soap, you are adding sodium hydroxide to oil to produce soap. If you have added too much sodium hydroxide, you will know only after the results



come back from the lab; by that time that batch is gone. So, what people do is now do a mathematical model of the process, they give the disturbances that come in the feed to the model and ask what is the consequence. Use that consequence to control the process. So, basically mathematical modeling is done in order to do anticipated control.

Then second is simulation. Modeling gives you conceptual understanding and simulation supposed to give you actual understanding. So, mathematical modeling and simulation is now a large fraction of the work. Finally, empirical correlations for design should realize that the lot of our... the science is still inadequate. A lot of systems are not sufficiently well understood. And you must also realize very often industry is ahead of us; that is if there is a profit to be made people will find a technology to make that material or to produce that service even if you do not understand the science behind the technology. So, very often science comes after the technology is successful. You will always need empirical correlations for design. Partly also because we do not understand turbulence. So because you don't understand turbulence, you cannot do the correlations for heat transfer for all kinds of processes. So, what you have to do is do it empirically.

And if you look at analysis basically in all of research, we divide a large problem that is to be solved into parts. we divide the small parts sometimes as a conceptual difficulty, you leave the half that contains the conceptual difficulty and solve all the problems related to the other half. And once all of those are solved people take the other half again you divide it into half. So, analysis always proceeds that way. We do not know how to proceed otherwise; occasionally a person has an intuitive flash and he produces an understanding that completes the design process. But 99 percent of the cases we still have to cross the final design using empiricism. So, empirical correlations for design are very much a part of engineering design.



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A presentation slide with a white background and a blue title. The title is "WHAT ARE THE CHARACTERISTICS OF A GOOD RESEARCHER" in blue, all-caps, sans-serif font. Below the title is a horizontal blue line. Underneath the line is a bulleted list of seven characteristics, each preceded by a blue dot. The list items are: "A prepared and open mind", "Broad interests in several areas", "A capacity for hard work", "A desire to know the truth / an ability to challenge prevailing paradigms", "Discrimination and aesthetics", "Ability to learn from mistakes of the past/ of others", and "A positive attitude and faith in the scientific method". The slide is flanked by two vertical black bars on the left and right sides.

WHAT ARE THE CHARACTERISTICS OF A GOOD RESEARCHER

- A prepared and open mind
- Broad interests in several areas
- A capacity for hard work
- A desire to know the truth / an ability to challenge prevailing paradigms
- Discrimination and aesthetics
- Ability to learn from mistakes of the past/ of others
- A positive attitude and faith in the scientific method

So, now what are the characteristics of a good researcher. The first thing you need is a prepared and an open mind. when I say open mind, I don't mean a empty mind. I mean a mind that is full, but is still receptive to new ideas. Then you must have a broad interest in several areas, I must say a lot of you although you have chosen your field still don't know where your interest is. So, I think it is important for you to discover your interest and to keep your interest broad in several areas partly because an idea in one field may help progress in another field.

So, I would recommend for example, that you attend a large number of seminars. In fact, in my own case my PhD problem, I solved the problem, actually I had to deal with molecules with angle dependent forces, and the theory was already known for molecules mono atomic substances. And in order to deal with this I have to deal with rotation, and I had to deal with four dimensional harmonics in the mathematics. And I was quite ignorant what is already in the field at that that time; in chemical engineering there was nothing. So, I ended up deriving a lot of properties of four dimensional harmonics. And I very uncomfortable with them because the algebra was very complex. I had results that looked somewhat clumsy. Then finally, I went to a physics department seminars simply because they have gave very good eats. But the seminar was by a guy called Rose and still remember, I didn't understand 90 percent of what he said, but he was talking about



angular momenta and quantum mechanics. And he suddenly showed me the four-dimensional harmonics results and that were already been done and I just had to go back and verify, I was very happy that my results were right, but they had very elegant notation and a beautiful way of treating them. The idea is that it gave me such confidence then I could proceed further; I think sometimes the idea comes from various fields, you never know where it will come from.

Then third property you need is capacity for hard work. I think there is no substitute for hard work. Anything you get without hard work will always leave you a little insecure. So, I mean it's always... our students have a favorite way of saying their fundaaes are weak da, so I don't know. The main reason they are weak is because you haven't put on the hard work to understand them. And then a desire to another characteristic is desire to know the truth; you must have a desire to know the truth. In this context, nowadays it is particularly important that you don't pretend to the truth. I mean there are people who are good research workers who are fallen for this, who have done plagiarism or have manipulated their experiments, it never pays in the long run. So, you must have a desire to know the truth and an ability to challenge the prevailing paradigms.

Then Discrimination and aesthetics. I think this is very important. Ultimately all the research proceeds based on your sense of aesthetics. You think this is a beautiful way of doing it and not that. And I think that's very, very important you have to develop a sensitive aesthetics. When you read a paper just reading the abstract you should be able to say whether the treatment is beautiful or not. I think you should have an opinion on that. You may change it, but you should cultivate tastes.

Discrimination is very important, because if I teach you thermodynamics and you have a problem in thermodynamics, you also read transport phenomena which is irreversible processes. If you arrive at an explanation of a thermodynamic problem at equilibrium problem through a non-equilibrium process, it is a very bad way of dealing with it. I mean ultimately an equilibrium process should not require that you know how the system changes when it's not at equilibrium.



I mean it's basically you must have the discrimination to say I will use only tools that belong to the equilibrium case and not use tools that belong there. So, and the other thing I keep telling undergraduates is and it applies to you, when you do course work in graduate school; discrimination also means know the amusing story is that I show an apple and ask you what is it, and you shout orange. Then you have told me that you don't know an apple, you don't know an orange; you don't know both. And a person like me, I grade by negative marks. I give you a 100 and then subtract whenever you make a mistake. So, if you kept quiet you would have lost only 5 marks, if you shout the wrong answer you get minus 5 twice.

So, I think it's important that you be able to discriminate between in this case apples and oranges. Then ability to learn from mistakes of the past and mistakes of others; this is a peculiar ability that human beings have and I think it's important that you learn that. And finally, a positive attitude and faith in the scientific method. Basically, you can challenge the method. But basically you have to have faith that this is a process by which I can do things in a reliable manner.

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A presentation slide with a white background and a blue title. The title is "THE SPLIT-BRAIN EXPERIMENTS" in blue, uppercase letters. Below the title is a horizontal line. The text on the slide reads: "Roger Sperry (1981 NL in Physiology and Medicine) and co-workers have elucidated the learning process:". This is followed by a bulleted list with three items: "• The two sides of the brain are fundamentally different", "• The left brain is logical, thinks in words and is good at step-by-step reasoning", and "• The right brain is intuitive and musical and uses visual images to draw conclusions". The slide is flanked by two vertical black bars on the left and right sides.

THE SPLIT-BRAIN EXPERIMENTS

Roger Sperry (1981 NL in Physiology and Medicine) and co-workers have elucidated the learning process:

- The two sides of the brain are fundamentally different
- The left brain is logical, thinks in words and is good at step-by-step reasoning
- The right brain is intuitive and musical and uses visual images to draw conclusions

Let me say a few words about learning and creativity. The split-brain experiments are experiments that were conducted by Roger Sperry. He got the 1981 Nobel Prize in



Physiology and Medicine. He and his coworkers have elucidated the learning process. There's a beautiful book by his student, Gazzaniga, on the brain but I think a very nice summary is given by a person called Blakesley. It is called the right brain, and this is the summary. It says, essentially two sides of the brain are fundamentally different. The left brain is logical, it thinks in words and is good at step-by-step reasoning. And the right brain is intuitive and musical and uses visual images to draw conclusions.

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The creative process itself this is a over simplified summary but gives you a pretty good idea. There's a preparation stage where information is gathered by the left-brain. In fact, I keep telling my colleagues, if students don't listen in class, don't worry, their left-brain is anyway collecting the data, it's actually true you will know when you read for an exam. And even when you are not attentive in class when you are reading for the exam you suddenly tell yourself oh! this is why the guy was going on and on in class about this. You suddenly understand something and you know why the teacher went on and on and on about it even though you are not listening, because your left-brain was listening. But that's the preparation stage where you gather data, it's information gathering stage.

The second and the third stage belong to the right brain. It is called the incubation stage when the right brain tries to see the whole picture. And the illumination stage when the



right brains insight and intuition generate possible solutions and you think you have hit the right solution. But this is often this happens to you at night when you are reading just before the exam. Suddenly things fall into place and everything seems to be right, but when you enter the hall, it breaks down and you become nervous again. And this is typical I mean the right brain is intuitive, it generates solutions but very often these solutions are wrong. So, you need a verification stage where the left brain logically tests the solution. In fact, we take this for granted now. But it is only Galileo, who said experimentation is very important, verification before that there was a peculiar situation.

(Refer Slide Time: 29:49)

A presentation slide titled "THE SYNERGY" in blue text. Below the title is a horizontal line, with the left portion being blue and the right portion white. The slide contains six bullet points in blue text. At the bottom, there is a footer with the date "01/12/15", the name "msa", and the number "19".

- Freedom from logic and structure makes right brain so effective in generating ideas!
- Since most of such ideas fail when tested logically, the left brain is equally important!
- Synergy between the two parts is the real basis of creativity
- Left brain can be that of one and right brain of another!
- Creativity often results from a 'meeting of unlike minds'
- University trains only the left brain; nurture your right brain

First let me say this. Freedom from logic and structure is what makes the right brain so effective in generating ideas. Since most such ideas fail when tested logically, the left brain is equally important. It is a synergy between the two parts that is the real basis of creativity, you need both. And basically, the course work is usually based on logic. All of university education is mostly based on logic, because I cannot really teach you intuition, I can only tell you what I understand logically I can explain to you. So, you have to nurture your right brain, you have it and you should nurture it. And you should realize that in our tradition and in the Greek tradition, the left brain was usually that of one person in the right brain was that of an another. Typically, we collected data and went to a sage who used his right brain his or her right brain and explained to you how all the



confusing things that you had in your mind fit together, but because the sage said it we couldn't verify it.

I think my favorite anecdote is that of Aristotle you know, he said women have fewer teeth than men. So, some 260 AD and it was 1450 AD before somebody finally, said no, no Aristotle was wrong, I have actually counted the teeth of men and women they have the same number of teeth. And mind you, Aristotle had two wives and he never counted that teeth apparently. So, but I think this kind of thing happened in both and it's Galileo who said you have to question everything. And now if you have a good new theory of science, you have to propose an experiment that may lead to its downfall. For example, Einstein proposed that you measure the bending of light during a solar eclipse from a place in Africa, where it was a cloud you know cloudless weather and there was new moon. So, you could actually see the light and bending. And Edington went and measured it, it was exactly as Einstein predicted. If it wasn't Einstein's theory, would have been thrown out, but Einstein had to propose that experiment.

So, basically the creativity results from a meeting of unlike minds that's somebody's strong right brain, somebody has a strong left brain. So, you need not be in the same person and you should have a meeting of unlike minds which is why in the university I recommend very strongly you tend to one tends to group into like minds. You know you make friends of people who are like minded. But I think in the university you should make friends with people who have an unlike mind. Even if they make for a little unpleasantness, the chances of joint creativity are higher. The university basically trains the left brain you have to nurture your right brain.

I will tell you incidentally in my thermodynamics course, I have told students often that even if it's a complex problem you have an intuitive understanding of it. So, write the answer on the top. So and then I tell them, then do the problem logically and verify if logic confirms your intuition. Typical of IIT students, they write that intuitive answer in pencil, work out the problem; if it does not agree, they erase it and write this answer there. I tell them there are no marks for it but they still play it safe, it's not for me, it's for your own sake. If your logic confirms your intuition, you will get self-confidence and very often, you have a feel for what's right, what's likely to be this?



(Refer Slide Time: 33:22)

A presentation slide titled "MANAGING CREATIVITY" in blue text. Below the title is a horizontal line. The slide contains three bullet points: 1. The 'Magic Garden' approach: hire brilliant minds, create the right atmosphere & leave them alone! 2. The 'idea factory' approach: bring unlike minds together, create the right atmosphere but structure interactions 3. Minds can be 'unlike' in, followed by three sub-bullets: • Cultural background (Eg. US Graduate schools) • Disciplinary training (Eg. Bell Labs) • Attitude (University Research Parks). At the bottom left is the date "01/12/15", in the center is "mca", and at the bottom right is "21".

MANAGING CREATIVITY

- The 'Magic Garden' approach: hire brilliant minds, create the right atmosphere & leave them alone!
- The 'idea factory' approach: bring unlike minds together, create the right atmosphere but structure interactions
- Minds can be 'unlike' in
 - Cultural background (Eg. US Graduate schools)
 - Disciplinary training (Eg. Bell Labs)
 - Attitude (University Research Parks)

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So, then I want to talk a little bit about managing. This is special to IIT madras because we have a research park. The university is the source of almost all creativity in history. And question is how can university manage its creativity. There are two methods that are described usually. A guy called Shrager describes these. And I think he has a book on creativity I forget the name of the book. Basically, there is a magic garden approach where you hire brilliant minds create the right atmosphere and leave them alone. If you are director of the institute you pray because they may sometimes retire without doing anything. This is the risk you have to take, because the brain is very good you must have it in the university. But the second approach which is called the idea factory approach was actually evolved in research labs in the US in the thirties and forties, Bell labs is the most famous for it. The idea was to bring unlike minds together from different disciplines together, and allow them to create the right atmosphere, give them a lot of freedom, but structure interactions.

And anyway, minds can be unlike in different ways. First unlike success of meeting of unlike minds is the US graduate school. Typically, where they take people from different cultural backgrounds. You must realize that science is universal, but the scientist is not; the scientists has a cultural background therefore, a set of prejudices. So, if you want to overcome at least some prejudices, you must mix people from different cultures. And the



US did it by accident, because the country immigrants, they have been very successful. Second is disciplinary training, that is one I have told you about Bell labs. They brought together mathematicians, physicists, chemical engineers, electrical engineers all kinds of people in one group and that group produced most of the results in solid state physics, even Bardeen was in that group. But afterwards they called it the solid state physics group, then they lost a lot of their sink, because now you needed a visa. You should have done solid state physics. Either you are undergraduate or postgraduate to get in. I think you need to have a mix.

(Refer Slide Time: 35:42)

A presentation slide with a white background and a blue header. The header reads 'THE UNIVERSITY RESEARCH PARK' in blue, uppercase letters. Below the header is a horizontal blue line. A bulleted list follows, with each item preceded by a small blue square. The list items are: 'Property-based venture near a university campus', 'Creates a local concentration of skill and technology', 'Promotes innovation, competitiveness & entrepreneurship', 'Helps convert research ideas into innovative technologies', 'Houses R&D of companies', 'Creates and nurtures start-ups', and 'Drives technology-led regional economic development'. At the bottom left of the slide, the date '01/12/15' is written in small black text.

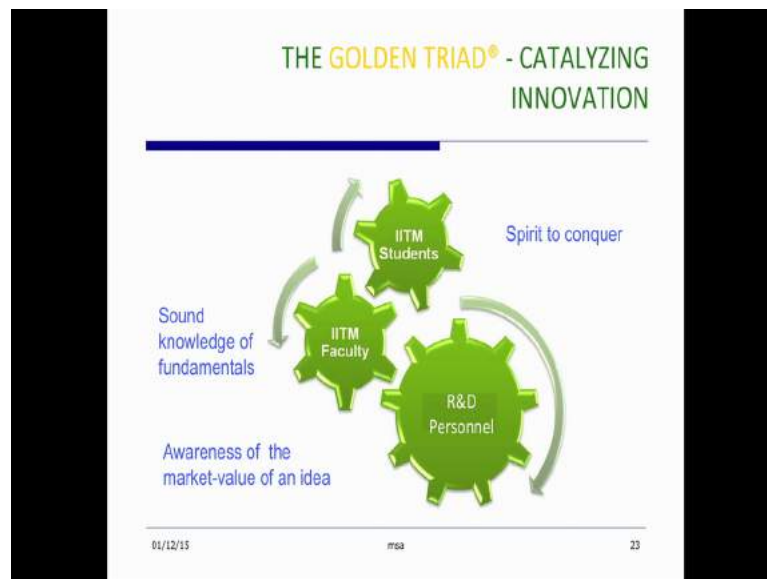
- Property-based venture near a university campus
- Creates a local concentration of skill and technology
- Promotes innovation, competitiveness & entrepreneurship
- Helps convert research ideas into innovative technologies
- Houses R&D of companies
- Creates and nurtures start-ups
- Drives technology-led regional economic development

01/12/15

And the third is attitude. University research parks, where you have the... I think I have; Yes, the university research park is basically a property based venture near a campus like we have done here. It creates a local concentration of skill and technology. It promotes innovation competitiveness and entrepreneurship. It helps convert research ideas into innovative technologies. It houses R and D of companies. Creates and nurtures start-ups and drives technology-led regional development.



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The idea in this is this it's called the GOLDEN TRIAD. You have three types of people: the students who have a spirit to conquer and IIT faculty who have a sound knowledge of fundamentals, and R and D personal from the industry who have an awareness of the market value of an idea. So, if you have these three minds together then... see students are likely to come up with a large number of ideas, but they have the advantage that they can come up with all kinds of wrong ideas, nothing happens to them. A proff for example, is considered an expert once you finished your PhD, you are considered an expert. Then if you say some rotten ideas, they will say don't you even know this. Whereas when you are a student you have the freedom, you can make 99 mistakes in the 100th idea may be great hit and that is all that is required. So, you need this combination.



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A presentation slide with a white background and a blue title. The title is "WHY IITM RESEARCH PARK?". Below the title is a horizontal line. Underneath the line are three bullet points. At the bottom of the slide, there is a footer with the date "01/12/15", the word "IITM", and the number "24".

WHY IITM RESEARCH PARK?

- Significant fraction of IPRs in Silicon Valley in the 90s have names of IIT alumni
- Louis Pasteur: "Discovery is the result of chance meeting a prepared mind"
- IITs have been preparing the minds and chance has been meeting them in Silicon Valley!

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And why IITM research park? We did an analysis of IPRs in Silicon Valley which is probably created the largest number of IPRs per unit time. In the 90s, and we have found a very large number of them have the names of alumni, IIT alumni almost 50 percent. So, I mean they have many names one of them is an IIT alumnus. And Louis Pasteur said Discovery is the result of chance meeting a prepared mind. In particular, I think he was talking about the discovery of penicillin. I mean for example, had an idea of what was when he saw this fungus. When he saw this thing grow, he realized that there was a cure for diseases.

Anyway, so he said this, he said discovery is the result of chance meeting a prepared mind in experimental research. I omitted that, and I quoted this to MHRD and I said IITs have been preparing minds and chance has been meeting them in Silicon Valley. So, I said they need to meet chance in our backyard and MHRD agreed.



(Refer Slide Time: 38:03)

A presentation slide titled 'THE IITM RESEARCH PARK' in blue text. The slide lists four bullet points: 'IITMPRP - an independent section 25 company that can hold shares in start-ups', '11.5 acres land from State Govt; Rs 100 Cr from MHRD', '1.2 m sq.ft in two phases (85% R&D, 15% incubation)', and 'Student Opportunities'. Under 'Student Opportunities', there are two sub-points: 'Summer Internships' and 'Incubation'. At the bottom, there is a footer with '01/12/15', 'msa', and '25'.

So, we started the IITM research park; it is an independent section 25 company, I think now the section has been changed, called section 8. You can hold shares in start-ups, IIT cannot. We got 11.5 acres from the State Government. I wanted it just outside the campus and luckily for us the MGR film city was closed down. So, they had 40 acres in out of which we got about nearly 12 acres, 11.5 acres. And the state government was very generous, they gave us that land; it is adjoining us. The idea is that the values in an academic institution are different from values in a market place. So, you shouldn't mix them if possible. So, I preserve in fact, I told the chief secretary. He said, you have 630 acres why are you asking me for 12. I said 630 acres of pure academic land where only Saraswati will be worshiped, and I want a place where I will worship Lakshmi along with you and with the industry. And just on that basis, he said yes, he was very enlightened chief secretary that time. And he gave me seven acres then the government changed they gave me another five and half acres. So, we got about eleven and half acres total.

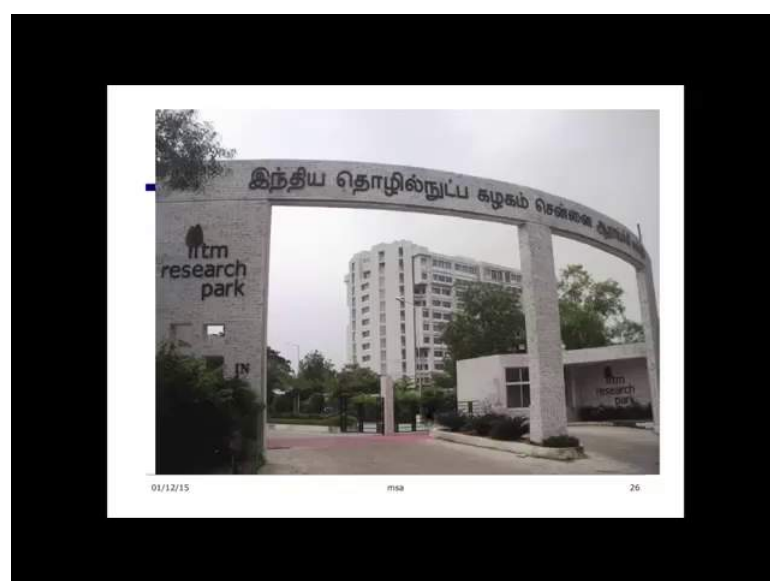
And MHRD gave us 100 Crores, it took about 7 years, but finally, they gave us a 100 crore loan. They going to make it a grant. And we decided to put 1.2 million square feet in two phases in this; 85 percent is for R and D, 15 percent was incubation. Research Park, University Research Parks exist everywhere else in the World. In India, this is the



first research park. In fact, when I started talking about it in 2001, when I became director 2005 is when Dr. Chidambaram was then our chairman he wanted me to go to the US to attend meeting of the association of University Research Parks. He said you go represent India. I said what India? what research park? there is none in India. He said you are the only one talking about it, so you go. And I went there and the first speaker was a Chinese, this girl lady, she is a quite young lady; she got up and said china has a very modest program only 100 research parks, only 100 acres per research park, only 10,000 companies in each research park and only 1.2 billion dollars of support from the government.

So, I put up my hand and said can I go last. I was next journalist and she said nothing doing you have to speak in this order. Then I went up and said, one research park, 10 acres site, you know even our budget was 300 cores at that time. But any way, actually china has over done it, a lot of our research parks are empty. But the point is, I think research parks are an important source of innovative technologies. Now research parks, as far as Research College is concerned also you give a chance for summer internships, because you get ideas from research there. And incubation, if you want start companies, a lot of IIT people have started companies already.

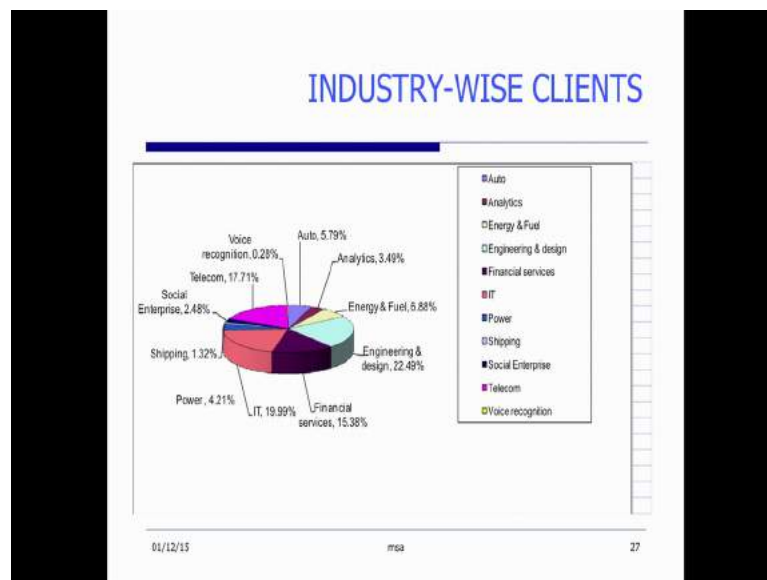
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So, this is the research park you must have seen it. I don't know if you have visited it there, ok good.

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Then this is our clients; we have got clients in all sectors. And there is also a lot of interaction between the clients that is also generated a lot of research. I think we have total of some 75 patents per year now. We used to be five and this is a very amusing because in 2005, IIT Madras got the award for the University that produce the largest number of patents. So, the commerce secretary called me and said I want to congratulate you. I said are you sure we got only 5, he said others have less. So, its not that we are not creative, but this patenting the idea of intellectual property right was not there, and we created a cell here, but after the research park came this is increased tremendously.



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A presentation slide with a white background and a blue header. The title "SOME INCUBATEES" is in blue. Below it is a list of seven bullet points, each preceded by a green square. At the bottom, there is a footer with the date "01/12/15", a small logo, and the number "28".

SOME INCUBATEES

- RTBI - Rural Inclusive Technology
- Uniphore - Voice Net and Biometric Systems
- ROPE - Rural Products to Global Market
- Villgro - Rural Innovations
- Dhvani Research - Non-Destructive Testing
- Aaum Research and Analytics – Analytics
- Gyan Data

01/12/15  28

And these are some incubates.

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A presentation slide with a white background and a blue header. The title "GENERAL ADVICE" is in blue. Below it is a list of five bullet points, each preceded by a blue square. The last bullet point has three sub-bullets, each preceded by a small circle. At the bottom, there is a horizontal line.

GENERAL ADVICE

- Problem is yours; Help choose it! the thesis is also yours
- Research is iterative; results often come only in spurts
- Clear intermediate steps are rare
- Avoid the temptation to attribute failure to external causes:
 - Advisor (indifferent/ incompetent)
 - Problem (too difficult/ ill-defined)
 - Environment (not conducive for good work)

So, let me now get to some advice. General advice, the PhD problem is yours. So, you have to help choose it. By and large, we find, I had 18 PhD students only two of them helped me choose, I may help choose their own problems. We give a general area



because most of us faculty have an area of expertise. So, we see within this area, if you want to you pick up a problem. Usually I used to give students about a year, year and half and they never came back with the problem; and one or two cases in two cases they came up with and. So, the thesis is also yours, when you write your thesis it should be in your style. Of course, if you make mistakes your guide will correct it, but by and large, you should have a style of writing. It's a story after all.

And research is iterative, you must remember that results very often come in spurts. There are whole times year, year and half when no result comes and you can feel very frustrated. And there are no clear intermediate steps, they very rare and you have to avoid the temptation to attribute the failure to external causes. Normally the adviser is what takes that. Every student thinks that advisor is indifferent, incompetent, they will come and tell you when you deal research, sir I have a problem and very seriously they will look at you they will say you should not be angry with me. But you know my advisor is indifferent and things like that. Actually, it is not indifferent, usually you take a PhD student, because you have run out of ideas you have got some basic idea but you don't know, how to tackle a particular problem. You are hoping this new mind will come up with an idea. And then you put it together. That's how... I mean if I could solve the problem myself why would I take a student at all. So, it is not occasionally this is true; 1 in a 100 may be true or 1 in a 1000. But by and large adviser is not indifferent because he also does not have an idea, he is waiting for the nucleus have an idea to come. His incompetence is not something that a student can usually judge; it's very rare.

Then the other thing is he say problem is too difficult it's ill defined. These are things that... first of all problem too difficult is common assumption, but very often after they finish they are very happy that they solved a difficult problem. It's ill defined very often you have to define the problem carefully. You have to know what's possible, what's not possible. The environment is not conducive for good work. Normally I find if a student is really good, they get into these conflicts right at the beginning, I mean not, they do not say the adviser indifferent, they go argue with the adviser has to why I should work on this problem. But that's out of interest. If you have goofed for three years and then you go and complain the adviser is indifferent or the problem is difficult; that means, you are just trying to escape from a disciplinary action that might follow. So, I think the



important thing as that the problem is yours, you have to define it in such a manner that's meaningful. You must know after you finish the problem, what's the direction of your research will be if you continue as a faculty.

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A presentation slide with a white background and a blue header. The title "DOs AND DON'Ts" is in blue. Below the title is a horizontal blue line. A list of six bullet points follows, each starting with a blue dot. The text in the bullet points is in blue. The slide is flanked by two vertical black bars.

DOs AND DON'Ts

- Acquire scholarship; most of us are not so original!
- Acquire a sense of aesthetics and discrimination
- Acquire a feel for the subject, interest in the problem and increasing obsession with it
- Anticipate results but retain your capacity to be surprised
- Do not investigate a problem without being sure that it is worth while
- Understand the physics before attempting 'optimisation'

There is some dos and don'ts. Acquire scholarship. When I say most of us are not so original, I did not being you know pessimistic. It is a fact that we have... there are very few people who are really original. I am taking about Newton, Einstein people of the highest caliber. But most of us are not that original. And what we are talking about is solving a problem for the first time but solving it has a follow up of some theories that has already been formulated. But you should acquire scholarship. I think that's very, very important. So, people very often don't take courses every student and the guide also comes to me and say: Sir we want to finish these four courses in the first year, as if it something we got rid off. So, you take whatever course is available actually you should take courses that are meaningful to your research, and if you take more courses it doesn't matter. In fact, I probably have a record for the largest number of graduate courses in US in Florida, because the Chairman told me why are you taking, so many courses I told him this: I told him I am not original, so I want scholarship.



And he was a very nice gentleman. He called me in the evening, he said this was in a coffee room. He said come and see me young man at 4.30 in the evening. And the secretary Karen Walker told me, Ananth you are in trouble! When he says come and see me young man, he is going to give you a lecture that you will be sorry to hear. So, I went in at 4.30 and I told him before you say anything let me say this: I think I am very clever in my class of thirty students, I think only Charlie the bros may be cleverer than me. I am certainly cleverer than all others, but on the other hand I do not think I am original enough to impress myself. Then he surprisingly he smiled he called for coffee and he said you are quite mature, you have to make this decision, but don't underestimate yourself, you should work very hard. The point is of course, you shouldn't underestimate yourself, but on the other hand, it is important to realize that many people are not so original. It is good to acquire scholarship, because when you learn many areas then ideas come in the area that you are really interested in.

I have already said this aesthetics and discrimination. Acquire a feel for this subject and interest in the problem and increasing obsession with it. I am afraid that is not something that seems to happen. I wish people would do that, I mean I have before I became director at least not many students many students did not know me. So, I would walk to the canteen, and try to over hear what students were discussing. Of course, they discuss a lot of politics, lot of cinema. But I was hoping at least 15 percent of the conversation will be about their research; I am afraid it wasn't! I think it was about ten percent I am hoping it will increase with time.

Then anticipate results, but retain your capacity to be surprised. When a synopsis is presented in dean researches office, people present their results as if it's routine. Some of it couldn't have been routine they must have been (Refer Time: 47:55) then I asked them did you really expect this? Then he says yes. Then I asked them what is your intuition about this; then they say the opposite. Then I tell them then why aren't you surprised, somehow you have lost that ability to be surprised. I think that shouldn't happen. Then do not investigate a problem without being sure it's worthwhile but this must begin right at the beginning. You must discuss with your guide, because the guide is also can also be mistaken about the value of our problem. There I mean at the PhD level you are more



like equals. Then understand physics before attempting optimization people try it to do modeling and optimization without understanding the physics.

(Refer Slide Time: 48:35)

A presentation slide with a light blue background and a dark blue header. The header text is "MORE DOs AND DON'Ts" in a bold, dark blue font. Below the header is a horizontal line. The main content is a list of five bullet points, each preceded by a blue dot. The text is in a dark blue font. The slide is flanked by two vertical black bars on the left and right sides.

- Discuss problems in the area constantly
- Do not be afraid of making mistakes
- Avoid sophistry - be jargon literate but don't hide behind it
- Explain your research to a school kid till he/ she understands you – compensate him/ her with eats!
- Communicating research findings quickly and clearly is both a privilege and a responsibility

So, more dos and don'ts; discuss problems in the area constantly. I would recommend that very strongly. Because you never know where ideas will come from; people who don't know the subject can give you very good ideas. Then do not be afraid of making mistakes. Avoid sophistry, you must know the jargon in your field but you shouldn't hide behind it. Explain your research to your school kid till he or she understands you. And you have to buy them some chocolates or something ice cream. But I have recommended this only two of my students took this suggestion. And they ended up with I mean some of students they explained to where their own cousins or something in school. But the point was that they when they wrote their thesis, I did not have to correct it at all. Because when you explain it you know the story fully. Otherwise, you end up having Ravana kidnapped Sita before Sita marries Rama. And there is no Ramayana at all. You have to have the right order. Anyway, communicating research findings quickly and clearly is both the privilege and a responsibility. It is a privilege, but it is also a responsibility. Because all research is a cumulative effort, everybody pitches inputs in epsilon and all of it together is what solves problems.



(Refer Slide Time: 49:45)

A presentation slide with a white background and a blue header. The header text is "COMMUNICATING RESEARCH FINDINGS" in blue, with "FINDINGS" on a new line. Below the header is a horizontal blue line. A list of six bullet points follows, each starting with a blue square. The text in the bullet points is in a mix of black and blue. The slide is flanked by two vertical black bars.

COMMUNICATING RESEARCH
FINDINGS

- Make many (preferably mental) drafts
- Be brief but complete; practice précis writing!
- Value of the work should be obvious from abstract
- Time spent on an aspect of the problem is seldom proportional to the number of pages it occupies in print!
- Give credit where it is due; don't plagiarise!
- Take reviewers comments seriously

Here I would say make many mental drafts. I think this is important, I prefer mental. Because when you write it takes a long time, but if you can envisage the whole thing and make the draft it would be wonderful. You have to be brief, but complete you have practice preci writing. I do not know if you have done preci writing in school. Our teachers used to make us write you write an essay they will say make it one third the size that is the préci; without losing content. And one student came back came with some 500 pages and then came back with one-third then I said make it one-third further he was about to hit me, but finally he was very happy with his thesis, because it is brief to the point into the excellent, very well written. I think it's not the volume, once you have the ideas right, you have write you have to learn to write it briefly.

Value of the work should be obvious from the abstract; you should write an abstract that tells everybody exactly what you have done in the field. Then second is time spent on the any aspect of the problem is seldom proportional to the number of pages it occupies in print. Sometimes the whole year's work may see only one line in the thesis, this can happen because you have now you tried different ways of doing it, didn't work out and then finally, you arrive at it. So, you will write a paragraph about how some of the approaches you used and how it didn't work, and the actual way you did it will take all the pages. Then give credit where it is to you. I think that's very important and don't



plagiarize. I think that's becoming increasingly important. I was just coming from Bombay by plane and one Civil Engineering Professor was traveling with me, he said he had just admitted a student to his group and throw him out on the first week itself. At the end of one week, he was met he met him and he said what are you going to do? He said, I don't know somebody must have done this problem, so I will take this that and put it together in submit a thesis. The man said leave the this thing, you are not going to stay. And also he said the student said if the experiment doesn't work, I know how to adjust things. I think those are things that you have to avoid at all times.

Take reviewers comment seriously. When you have to write your paper up and send it. Reviewers comments can be very damaging sometimes, sometimes very puzzling. I had my most interesting experience was with industrial engineering chemistry, we had done a work on deactivation of catalysts. There is carbon deposition and the catalyst gets deactivated. We had done a very complicated modeling of the system, and the coupled partial differential equations and we solved it had beautiful plots. But by the time we finished all that we have got so tired our results and discussion was one paragraph and we sent it off. And then the reviewer said very good problem, very good formulation, very good solution, so what? That is all, that is all he said. He said, so what? Then a pickford who was the editor of the journal said I could not agree more with the reviewer, please revise and send it. So, we sat down for one month we took to write the results and discussion.

The idea is, if you have done so much effort, you must be able to explain the results so that others don't have to do the computation and all that effort to understand what the physics is. So, when we finally, wrote the physics the discussion alone, we only changed that section. And then pickford said I don't have to send it to the reviewer I am accepting it, it appeared afterwards. I think in India particularly; we don't write results and discussions with enough care. Very often, you get exhausted by the time you solve the problem therefore, you don't, but I think that's very, very important.



(Refer Slide Time: 53:23)

A presentation slide titled 'ETHICS' in blue. It contains three bullet points: 'Conducting research is an exercise in ethics and a test of character', 'The results of research are ethically neutral; however the researcher cannot remain neutral in respect of its applications', and 'Be constantly aware of the 'value of values':'. Below the bullet points is a quote: 'Non-violence >> power, wealth, fulfillment of desires'. The quote is attributed to '[Swami Dayananda Saraswati]'. At the bottom of the slide, there is a date '01/12/15', the text 'msa', and the number '34'.

The last slide is about ethics. Conducting research is an exercise in ethics and a test of character. I think that's very important to realize. The results of research are ethically neutral, but the researcher cannot remain neutral in respect of its applications. The most dramatic example, I can think of is the atom bomb. All the research that was done to develop nuclear power was done after quantum mechanics was born twenties, thirties, forties. And when they finally, discovered a understood everything about fission then mean while due to historical accident people were afraid Germany will make the bombs or America rush to make the bomb, and they made it they dropped it. At that time the Japanese scientist were participated in the quantum mechanics and this thing was terribly devastated. He said he never thought his colleagues will do this. But eventually what happened was a lot of scientists went into a depression. Because they felt suddenly they had pursued truth, they had pursued the truth about nuclear fission, and it had resulted in a bomb that killed hundreds of thousands of people and maimed many more for many years.

So, you cannot control this and even after that the hydrogen bomb was worked on and Edward Taylor made a famous statement saying, I cannot help working on the hydrogen bomb, my curiosity is over whelming. How it is used is not my business! I think that's no longer true. The only way to control that it is not really you it is not in your hands but the



way to control it is, I think for scientist to write to about the research in common journals in you know popular journals not giving technical details, but telling them what the consequences is the research may be. A well-informed public is the best protection in a democracy against misuse of science. So, I think that's important.

Finally, I will close with a statement on value of values. And this was Swami Dayananda Saraswati who spoke here in IIT madras about 10 years ago. He passed away recently, he said don't try to tell students about values, they already know. If you tell for example, if you take a thief and tell him don't rob its not a good thing, he already knows its not a good thing. So, he said instead talk about the value of values, they are many values. One could be acquisition of power, another value is acquisition of wealth, another could be acquisition to just fulfillment of desires and so on. All of these are values. But there is one universal value which is called non-violence. Because you don't want to be hurt you shouldn't hurt others. But Swami Dayananda Saraswati like Gandhiji also he explained that non-violence is not just absence of physical violence, it could be mental violence, it could be any form, but you know when it is violence. So, you know when when somebody has used violence against you.

So, he said teach them that the value of non-violence is greater than the value of power, is greater than the value of wealth and greater than the value of fulfillment of desires. And then tell them that if they acquire wealth, power or fulfill their desires through non-violence means they welcome to it. And Lord Krishna says that's the path of the dharma; so if you walk the path of the dhrama, I will walk with you; otherwise, you walk alone; there is no punishment, you walk alone. Then he looked, there were 450 students in the hall he looked at them and said, do you think walking alone is easy, and most twenty year olds plus or minus three they all said of course, we will walk alone.

Then he said it was the day after the exam, you know this Taramani temple next to Taramani, I don't know a few you people still have that story. My students used to say if you come around that temple you are supposed to get five more marks. When you come to the exam you just cycle around it and come to the exam. He says may or may not be true but why take a chance if you get five marks why not. So, then he asked them he somehow knew about it he said and 400 hands went up in the CLT. Then he said that is



what Krishna means by walking with you. I mean it's not something dramatic, it is simply that human beings need something to hold on to and they create various forms of help for themselves. And I think the point he made was very valid that if you know that non-violent ways of getting anything is acceptable, then you welcome to it. If you use violent means, it is just not worth it and I think that's important remember right through and being ethical is rather important. So, I think that's the last slide I have.

Thank you for your patience.

Thank you.

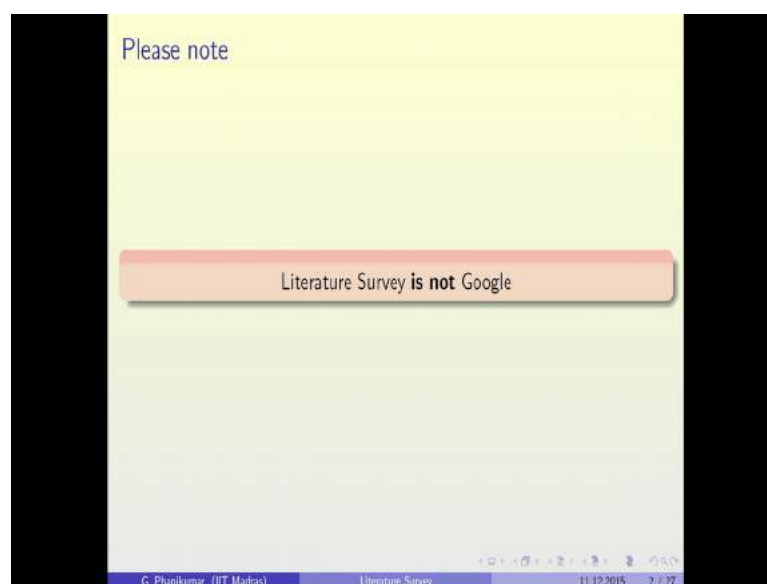


Introduction to Research
Prof. G. Phanikumar
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture - 06
Overview of Literature Survey

We are going to look at literature survey in the next three modules, each compressing roughly about twenty minutes.

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The first twenty minutes is going to be slides on the background of literature survey. And then, the next twenty minutes roughly will be on actual demonstration of the various literature sources. I am going to show you how you can search for literature, organize them in set of tools that you can use. And then, in the last twenty minutes, I will see how you can use these literature data in a document that you want to write up.

So, we will start off the literature survey by first making a disclaimer. Literature survey is not just googling for some information. The reason is Google search engine is going to look at private websites of various individuals. Whereas, by literature survey, what we basically mean is a survey of technical literature that is of authoritative in nature, pertaining to the scientific enterprise that is going on across the world in universities, research labs and so on. So, what comes closest to available, what is available in Google



would be perhaps Google scholar. So, please note that literature survey is not the same as googling out the information.

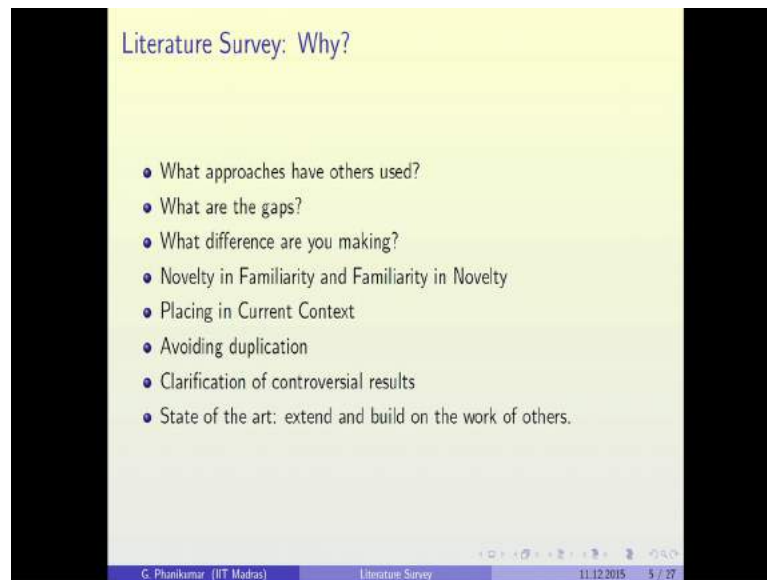
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So, we are going to look at the literature survey in this manner. First, a very brief background on why should we get into this literature survey at all; and then, where are the sources of literature that we could look up; what are the ways by which we can go about searching; and then, who are all the publishers who host the scientific archival literature that is being used by scientists all over the world. And then, towards the end, how to collect and store it in a manner that we can use it to write up a document.



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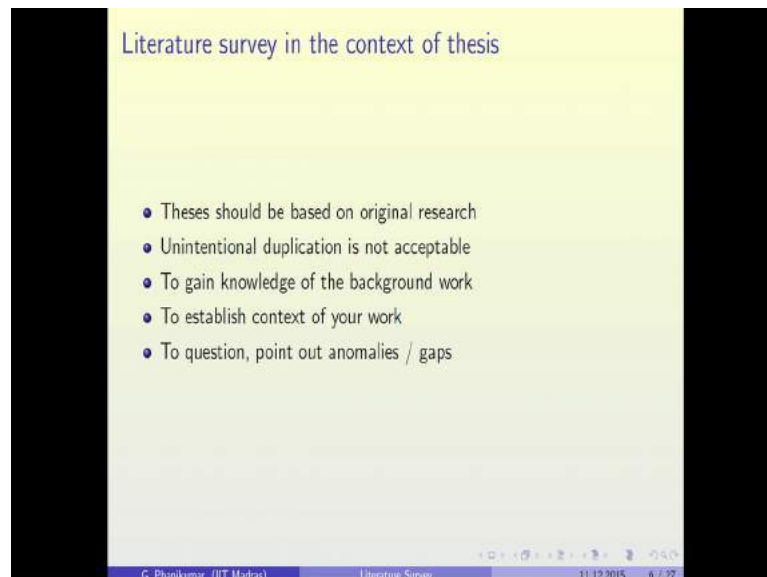
Okay So, why do we want to do literature survey at all? So, essentially the first requirement that we need to do is every scientist cannot replicate the entire history of how a particular topic has evolved. So, we must use approaches that are going beyond what others have done till now; which means that we must know what others have used; what kind of approaches people have used; where are the gaps that have come up; and, where is it that we can contribute. And then, what is the difference that we are making to the existing knowledge. And, very often a very novel technique used in a very familiarized area would be a good addition to the literature. And, sometimes a very familiar technique used in a totally novel area of research would also be valuable. So, novelty in familiarity as well as familiarity in novelty require you to know what is familiar and what is novel. And therefore, literature survey is necessary for you to know that.

And then, very often a study is valuable or seen as valuable by our peers when we are able to place that study in a current context, where people look at it as valuables. So, it is important to avoid duplication; and, it is also important to clarify certain controversial results that may be coming up. And, if you want to know what is the state of the art in the literature that is currently there and we want to extend and build on the work of others, definitely we need to know what is done till now. And, literature survey should help us in getting that status quo of the subject we are a trying to do research on. So, literature survey that way is a very important aspect. It should be done not only at the beginning of



the research but also throughout. But, at the beginning, it definitely makes a very important point.

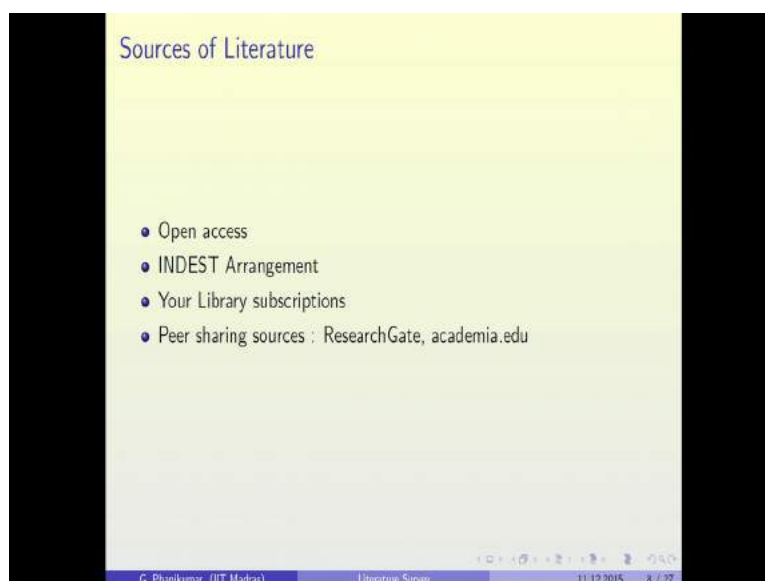
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Literature survey in the context of making a Ph.D. thesis or a research-based degree is important, because very often the research degrees are given based upon original work that is done by the student. And, duplication is not acceptable even if it is not intentional. So, it is important to know what is out there, what have people done elsewhere in the world on the research area that we are working on. And, sometimes it's also important to establish the context for your work, and its also important to have a background knowledge of the topic that we are working on okay. And, to also point out anomalies and gaps, we must know what has been done till now. So, in the context of a research thesis also research analysis and literature survey is very important.



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So, where do we look up this literature? So, very often we have various ways and I am going to look at them in one by one. The first way is obviously what is available for everyone in the world – open access. Unfortunately, the cost of keeping a journal publication or a conference proceeding in an archival manner, that is available in a reproducible manner over a long period of time is going to make the respective publishers spend money for that; so, which means that there is a subscription cost associated with every journal access. And therefore, open access - that is publicly available information from scientific archival data - is very limited. But, definitely, they are increasing in number as we go along. So, open access is very much available and we will see what kind of journals are available under open access.

And, in India, there is an arrangement made by the Ministry of Human Resources Development under the so-called INDEST program, which is being coordinated by IIT, Delhi, which provides access to a large number of literature sources to various education institutes in the countries. They are categorized into various groups: tier 1, tier 2, tier 3, etcetera. And, which means, that if your research organization or university is funded by government, then most probably, quite a few of literature survey can be done using the sources that are paid for by INDEST program. We will see what are all available on INDEST for our institute. And then, we also can see what is available in other institutes, where we could perhaps go visit, take permission, and do the literature survey there. Very often the literature sources that are subscribed by INDEST are not adequate. So, some



libraries do spend additional amount of money to make subscriptions beyond that set. And, this is something that will be available from your library. So, you definitely must get in touch with your librarian and find out what are all the journals that are subscribed by your library. And, of late, there has been also an effort to use some of the possibilities of authors being allowed to share their own publications on a portal, where upon request, it is possible to have the literature shared. So, these are called as peer sharing sources. So, Research Gate and academia.edu are such peer sharing sources, which also may be able to provide some amount of literature.

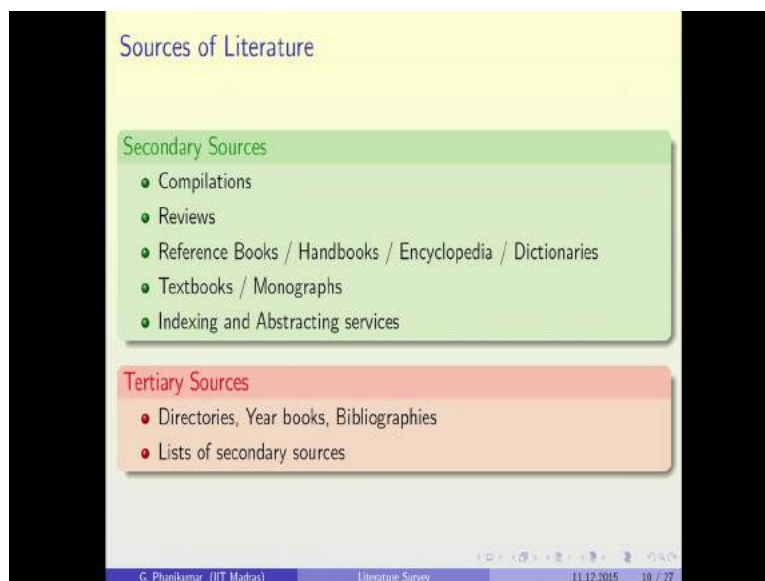
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So, what are these literature sources that we are talking about? We are basically going to look at periodicals, research reports, conference proceedings, official publications, standards, theses, dissertations, etcetera. Sometimes even publications that are going to come up in future can also act as a source of literature. We must, of course, have a pointer to refer to them, so that whenever they are complete, bibliographic details are available, we will be able to update them in our document.

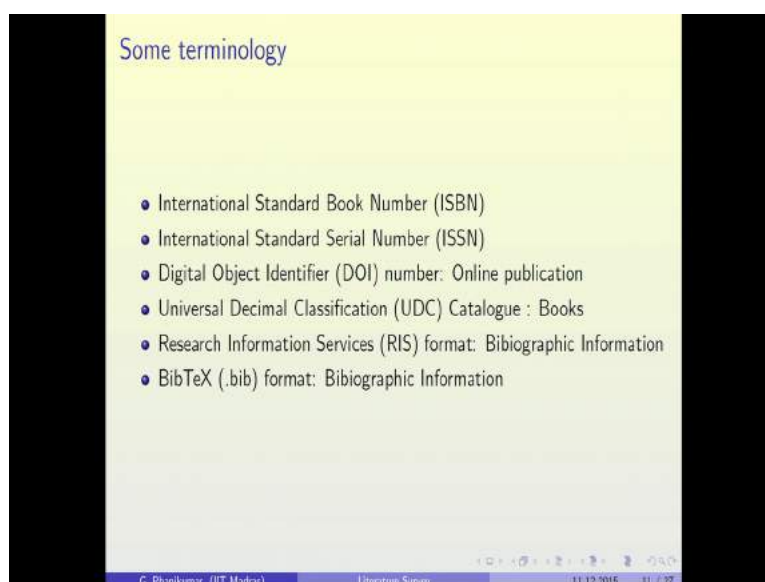


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And, there are also secondary sources of literature such as compilations of works done, and reviews, reference books, hand books, etcetera; text books, monographs, and abstracting services - all these can be also acting as literature surveys. And tertiary sources like dictionaries, year books, bibliographies, lists of secondary sources also can be used as sources of literature.

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So, when we go ahead with literature survey, we must be familiar with some terminology. Every book that we want to refer to must contain what is called an ISBN



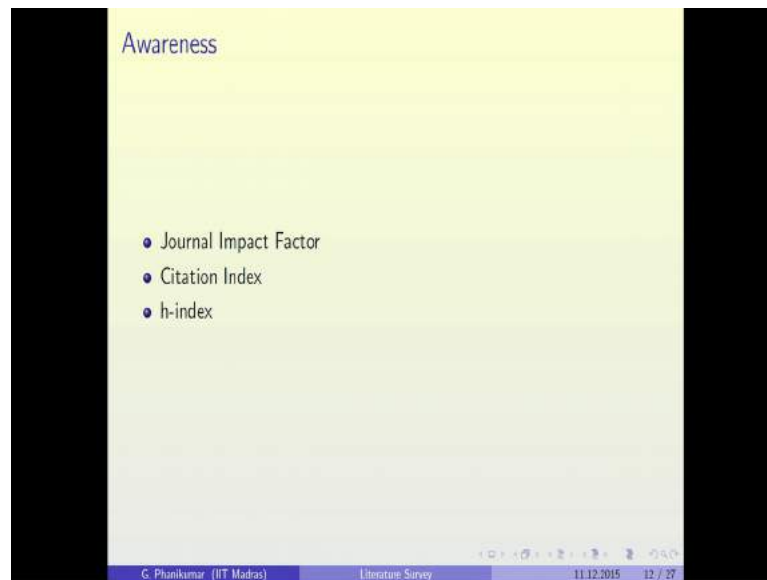
number – International Standard Book Number. This number is basically going to allow anybody in the world to identify the book and also possibly source it if it is available in print. Any book that does not have an ISBN, is essentially not traceable after sometime. And, it is important to note it down whenever we want to refer a book. Similarly, any journal that we want to refer to should have an ISSN number. And, these are also applicable for newspapers. And, ISSN number is going to be in ranges. We must pay attention to this when we are also referring to conference proceedings, which will usually have an ISSN number.

And, today in the digital world, we would like to have an identity for each document such that we will be able to access them directly. And, such an initiative is now available and the number is going by a name called Digital Object Identifier, namely, the DOI number. So, every document that is available online in the literature - open literature - will have a DOI number. And, there is an agency which translates the DOI number into a URL, where that particular full text resource is available. And, whenever the full text resource moves, the mapping will be modified by the agency – DOI agency. And therefore, we will be able to access the resource as the resource keeps moving its actual location; and that means that if you are going to refer to an online resource, make sure that you have a DOI number for the particular resource, so that you can refer to it by that number.

And of course, there are also other things that we must be familiar. They are something called UDC catalog. This is basically to physically be able to refer to that book in a library. And, the format of a bibliographic information comes in several methods. And, two such formats are very popular: RIS format and BIB format. We will look into the format of these two types of specification and what kind of fields are available etcetera as we collect literature survey through the demonstration that I will do briefly later on.



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And, there also some other things that we must be aware of while doing literature survey. One such thing is journal impact factor. What we mean by journal impact factor is that to see how many times the articles in a particular journal are being referred by other journals and other articles. It basically gives you a sense of how often these articles are being read and being used by other scientists. So, if you want to compare two journals in a particular area, then the one with the higher impact factor is being accessed, read, and referred more often. And, this is a number which is controversial; one cannot compare these impact factors across research areas. Very often it is seen that a research area in which the research is very actively being pursued, there are a large number of researchers that who are working. Then, normally, it will have journal impact factors, a little bit on the higher side. And, in communities - research communities - where the number of journals are limited, the number of scientists are limited, then the journal impact factor tends to be on the lower side. So, it is not an absolute method by which we can say a journal is less or more important. But it gives you a relative sense to compare two journals in a particular area of research.

There is also something called citation index. This is about a journal article; how many times has it been cited. We do call some articles as citation classics. What we mean by that is there is an article that has been referred by so many people, maybe hundreds, that you could now call it as a classic. And, citation is basically how many times it has been referred by other successful journal publications. h-index is something again a



controversial index; it basically refers to how many times a particular author's publications have been cited and how many of them exceed the number of publications; that is, basically when we say that an author has an h-index of say n , then he has n papers that are cited n times or more. So, which also means that an author, who has higher h-index, is an author whose papers are being read widely and are being referred widely. And, most probably, he is a more successful scientist.

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So, what are the various strategies to search literature? Its very important for a newbie in research to be very effective in searching the literature in his or her area, because the amount of time available to come to the focus area is very limited. And, we must identify the gaps very quickly, so that we can focus on the research that we want to pursue.

So, there are three major strategies that we can use to search literature. The first strategy is basically using a keyword. What we mean by a keyword search is using a word to identify the topic of the research result that we are looking for, or to use the author's name. We could also use a keyword that is mentioned by the article itself. So, essentially, it is like coming up with words that would describe and identify a particular research paper; and, which means that there are variants that may be available that point to the same article. We will see how we can do that shortly.

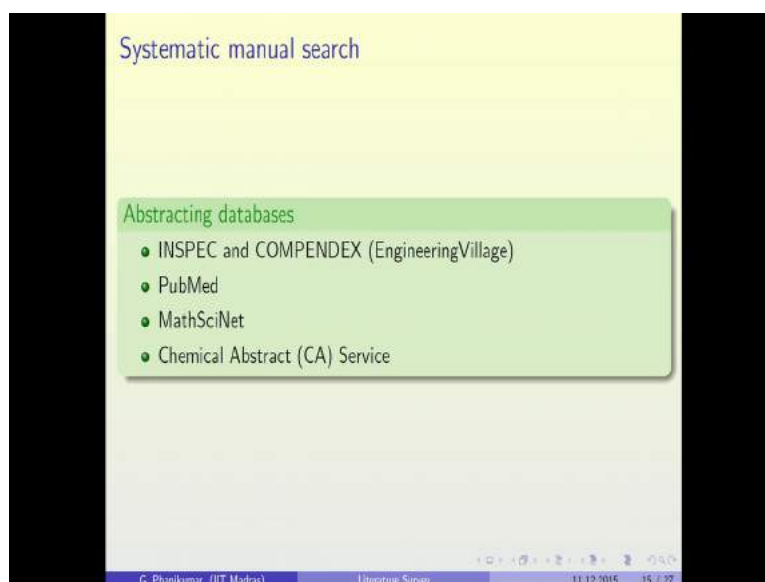


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There is also two other ways of searching, what is called the backward chronological search and a forward chronological search. What we mean by these two will be evident as we do the searching when we go ahead with a demonstration. Citation is something that is going to be playing a role in the forward chronological search.

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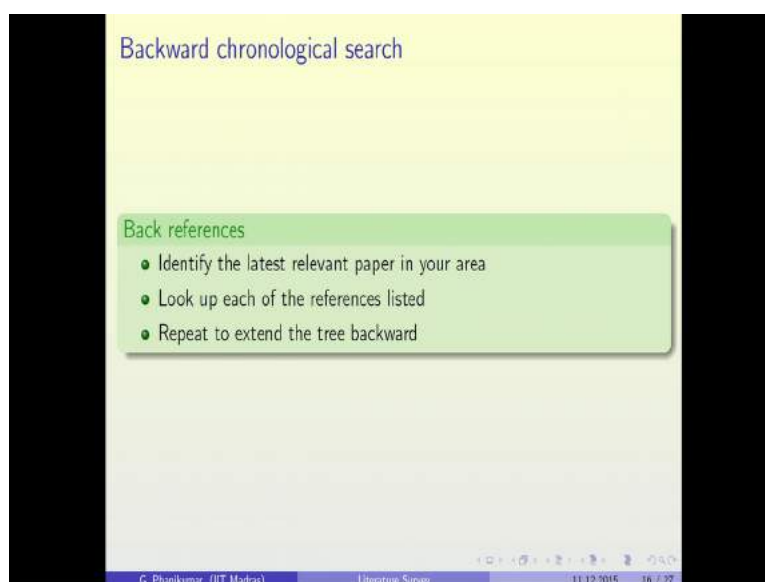


And, where do we do these kinds of searches. So, there are certain databases that are available, which contain the entire collection of journals and conference proceedings that will be collected together. And, one needs to have a subscription for these services to



have access - INSPEC and COMPENDEX are some such databases. PubMed is relevant for the medical community. And we have, of course, MathSciNet and Chemical Abstract services. So, if our libraries have access to these databases, then we can have an ability to search a large number of articles and conference proceedings in one place.

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The backward chronological search basically what it means is that we are going to look at an article, and see what are all the papers that it is referring to, which will be basically older than the article. And then, look at among those references, what are relevant for that topic, and then, going recursively back in time. So, this way actually we can go and discover the entire background of a particular research area.



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Forward chronological search

Citations

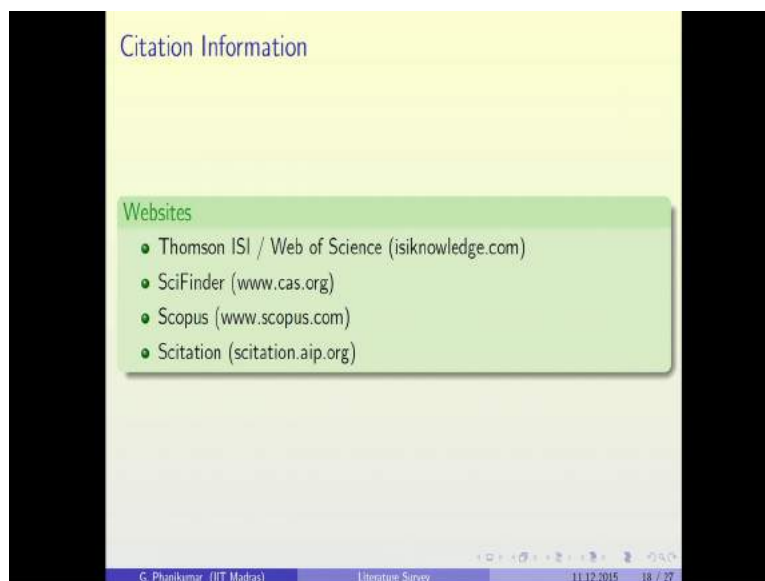
- Identify the citation classic in your area
- Look up which all papers refer to it
- Find out which of these are well cited
- Repeat to extend the tree forward

G. Phanikumar (IIT Madras) Literature Survey 11.12.2015 37 / 37

Forward chronological search is something that works in a different manner. We basically first identify a very important paper in our area of research, and then, we will see what all those publications that are citing this particular paper, which means that they are going to come in the future, in the sense, let us say we have identified a very good paper in 1994, then all those papers that are published after '94, which are referring to this particular paper. And, among those there will be some which will be very relevant to us. And then, we are going to pick them, and then, we will see which of the papers after those have cited those particular papers. And, that way we can go forward in time. Obviously, we can come only up to the year that we are searching, because the paper that will be published in future are not yet available to us at this point. Maybe by the volume number wise may be one year ahead, we may be able to see the papers that are coming in, but not beyond.



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So, what are the kind of information sources that we can use, where the citation information will be available. So, here again subscription is required, and two such sources I am going to show you: one is the Thomson ISI company's Web of Science and the other is Scopus. And, these two are seen as the most popular resources of information for citation. And I am going to show you a demonstration of how to look up the important research results in a particular area using these sources.

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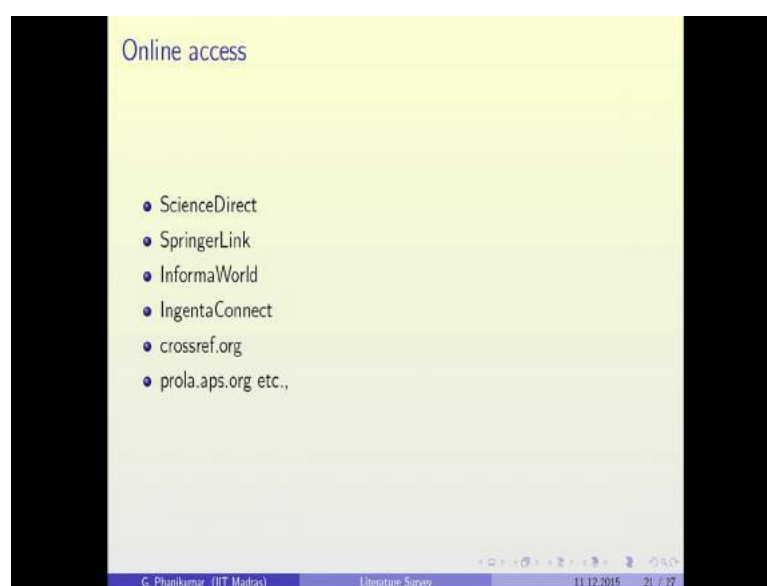


A word about publishers; roughly about eighty percent of the publications in the journal



forms are done by a very limited number of companies that I have listed here. Elsevier and Springer together they take up majority of the journal publications in the world. And then, comes the Taylor and Francis group, Nature Publications group and Maney Publishing. And then, there are a large number of journals that are published by societies: American Physical Society, American Institute of Physics, etcetera. So, there are journals that are published by societies and they are also publishing houses. So, we must know where to find the websites of online resources for these publishers, so that we can search for our articles there.

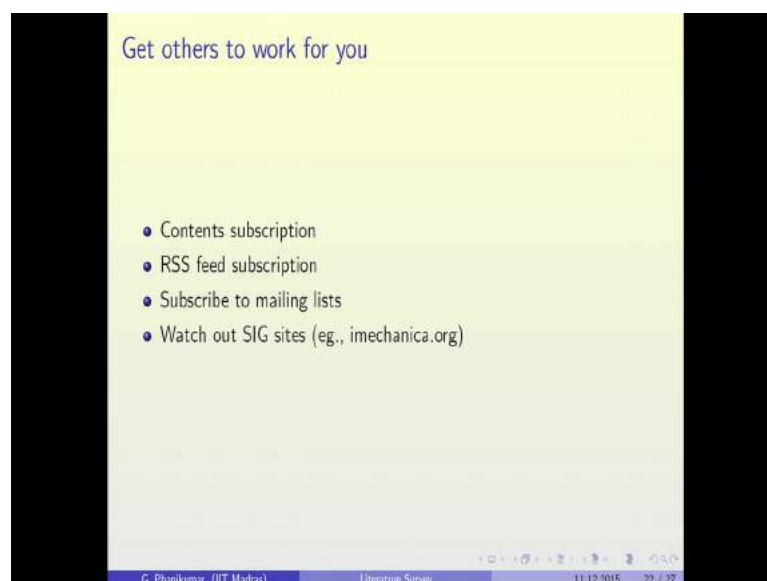
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Online articles are also available in a full text form from portals that are run by the respective publishers. So, ScienceDirect is one such portal that is run by Elsevier. SpringerLink is a portal run by the Springer publishing group. And, we have many such online sources that I am listing here. We will open up some of those sites, and then, see how we can capture those full texts that are available against subscription.



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So, sometimes it is also possible that the articles that are being published can come to our desk without our intervention. So, what we mean by that is we don't have to go to the journal website every month to check what is new. There is something called RSS feed subscription or content subscription. So, we can enroll our email with those respective publishers, and then, they will actually update us if there are new articles or if there are new issues every month. And, that way we can see that the literature survey is being pushed to our inboxes rather than we going and searching.

Naturally, accessing the articles that come with links in our emails through the subscription will require a subscription from our library; however, the basic information such as the title, authors, and the article details such as volume issue number and page number, etcetera will be available in our inbox through these subscriptions. Sometimes it is also very useful to visit special interest group websites. One example is imechanica.org which is an interest group for mechanical engineering community. These groups or these group websites will basically discuss about ongoing work in that particular domain. So, it is a very good idea to check out some such sites that may be active for your research area.



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A presentation slide with a yellow background and a black border. The title 'Thesis / Paper' is at the top. Below it, there are three bullet points: 'Numerical / Chronological + Alphabetical', 'Be aware of style guides for each purpose', and 'Use a reference manager'. At the bottom, there is a footer with the text 'G. Phankumar (IIT Madras)', 'Literature Survey', and '11.12.2015 24 / 37'.

So, when we want to then write up, we need to basically organize these references in a numerical manner, one after other in the sequence of referencing the document, or maybe chronological order or alphabetical order. So, we must be aware of what are the different styles in which the references are required for a particular document. This may be varying from university to university and also from a publisher to publisher when we want to send our manuscript for publications. So, its very important, for example, to use a reference manager to be able to change the style depending upon the source of information and the target where we want to send our document.

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A presentation slide with a yellow background and a black border. The title 'Reference Managers' is at the top. Below it, there is a list of software tools: 'JabRef', 'EndNote', 'ReferenceManager', 'ProCite', 'Mendeley', 'Zotero', and 'Comparison of reference management software on Wikipedia'. At the bottom, there is a footer with the text 'G. Phankumar (IIT Madras)', 'Literature Survey', and '11.12.2015 25 / 37'.



I will be introducing some of those reference managers. My favorite one is JabRef, which is an open source and freely available software. I will be actually doing a brief demonstration of this software. There are many paid ones also which I have listed here. And, some of them may be subscribed by your library for your university; you must check with your library what are available for your university as far as reference managers are concerned. There is also a very nice comparison of all the reference management softwares on wikipedia. You may want to have a look at it before you decide which ones to use.

And, I would like to thank Professor K. Ramamurthy and my colleagues for helping me with some of the pointers for this slides till now.



Introduction to Research
Prof. G. Phanikumar
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture - 07
Literature Survey Using Web of Science™

We will be discussing how to do literature survey using two tools: one is Web of Science and the other one is Scopus. And the first module will be on Web of Science.

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The slide has a dark teal background with a red vertical bar on the right side. The title "Disclaimer" is in white at the top left. Below it, there are four bullet points, each preceded by a white right-pointing triangle.

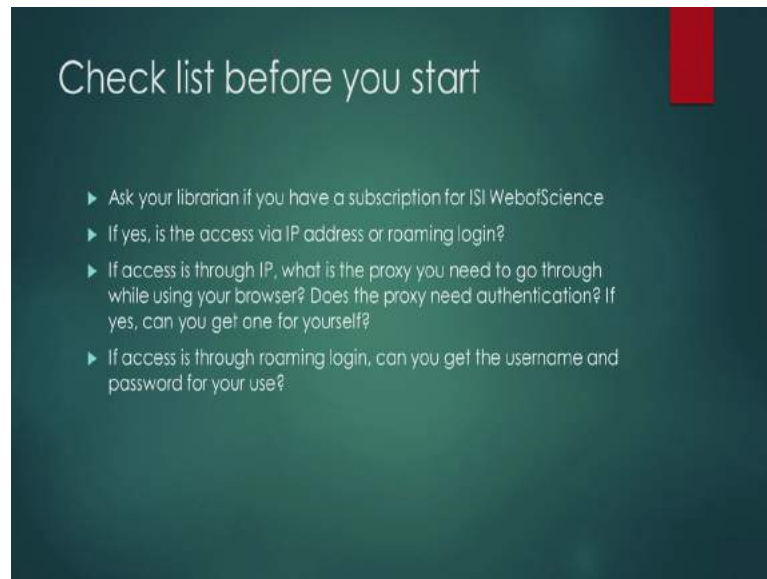
Disclaimer

- ▶ This module is one of the many to illustrate usage of different online tools for literature survey. In this module, WebofScience™ is being demonstrated.
- ▶ This module is not authorized by Thomson Reuters™ and viewers are advised to check with their librarian for guidance on applicable restrictions, licenses and copyrights.
- ▶ WebofScience™, EndNote™, WebofKnowledge™ are trademarks of THOMSON REUTERS™
- ▶ For simplicity, the™ symbol is being omitted for the rest of the presentation.

See there is a disclaimer that we need to provide before we proceed further.



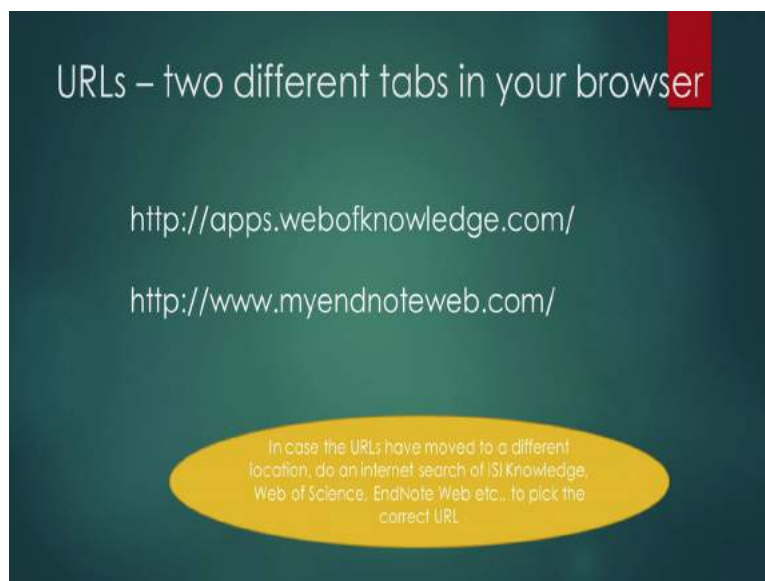
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So, before we start the literature survey, it is very important for us to know from our librarian whether we have subscription for these tools. So, please do ask your librarian whether your institute has a subscription for ISI Web of Science, and if they do have, then what kind of a subscription they have. The subscription comes often in a two modes - either by an IP address from where all connections will be honored for content or through a roaming login where the user name and password will be provided by the librarian. And, in case the access is through IP address, then we also need to know from the institute whether we need to use a proxy, and which means that whether you have a login and password for the proxy server from your institute. So, this information must be available before your start with literature survey using this tool.



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The two URLs that we will require to access while we do the literature survey using Web of Science are given here: Web of Knowledge and End Note Web. And these two URLs should be opened simultaneously in two different tabs in your browser, because we will be logging into both the portals and we will need the information to be seen simultaneously in both the portals. And if the URLs are changed by the vendor at any point of time from now, then you need to identify the correct URLs using a web search engine and you could that yourself at a later point.



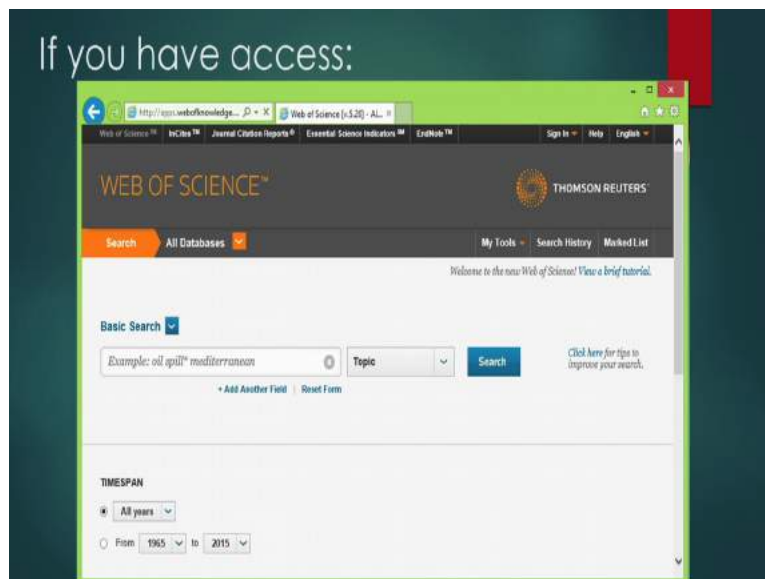
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In case you do not have a access for any reason, then when you open the Web of Science portal, then this is how it looks like. This portal is going to ask you to login and there will be no option for you to search anything on this portal. When you see this kind of a screen, please note that you need to approach your librarian and ask whether you have a subscription, and if yes, then what is the methodology by which you can access that particular site.

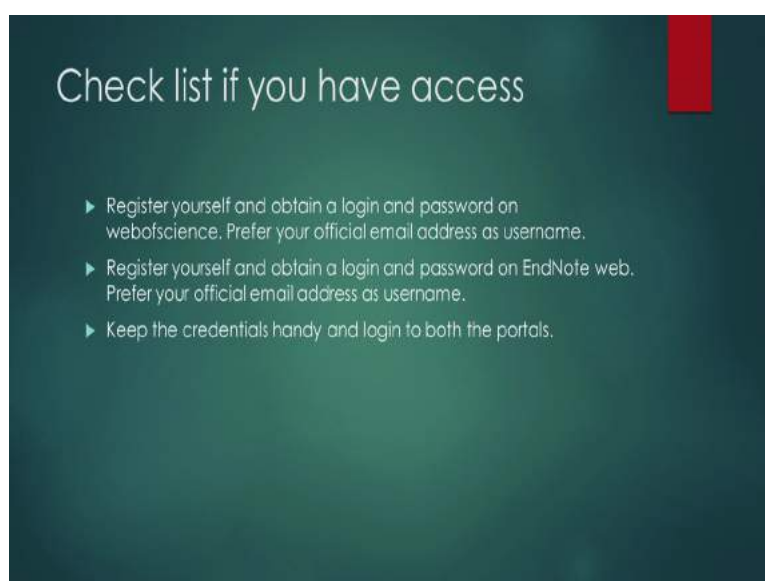


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In case you do have access, then when you open this portal on your browser, in your desktop, then this is how it looks like; you do have immediately a search possibility, a box that will let you type in the keywords, and then a search button there. So, which means that you are ready to start your literature survey.

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And there is a check list that we must clear before we proceed further. It's important for us to have a login access to both these portals. The reason being that the literature survey that we are doing should be stored in a profile that only we are able to access, so that we can come back to the work where we left earlier; and therefore, you need to register your user name and password with both these portals. It is often recommended to use your official e-mail address, because the authentication sometimes will honor you to use that address for access to the content, sometimes even when you are accessing from out side your campus location. So, you may want to check about these possibilities from the vendor, but its very important for us to register and create a login on both the portals. And we must be able to login at the time we are doing the literature survey, which means that we must have the user name and password handy before we start the literature survey.

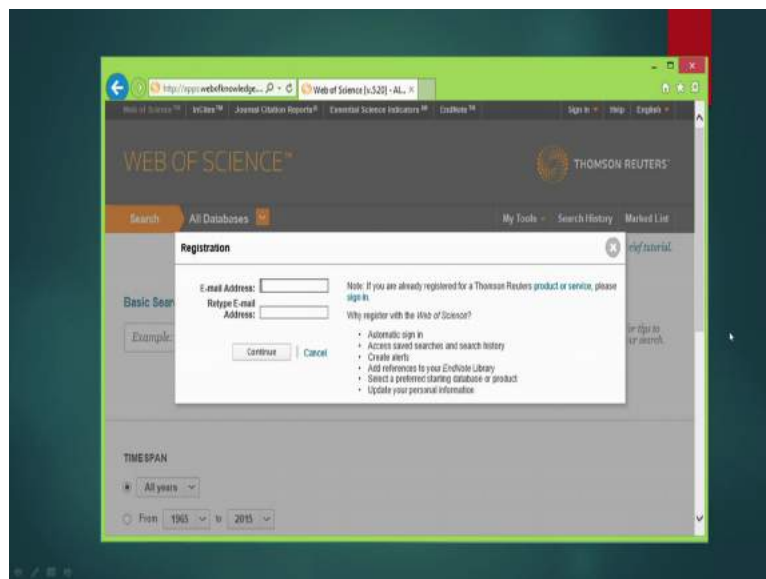
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So, the registration is very simple, when you open the Web of Science portal on the right hand side top there is a menu, under Sign In, and when you click on that there are three tabs that will open. The middle one - Register - is what you must click to register your details.



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And the registration form is very simple. Initially, you will be asked to provide your e-mail address, and once give your e-mail address, then there will be an e-mail confirmation, following which you will be able to set your password. The user name is your e-mail itself and the password is what you can set. And why is it that we need to have registration? You are already listed by Web of Science here under several possibilities that we can do with our login account.



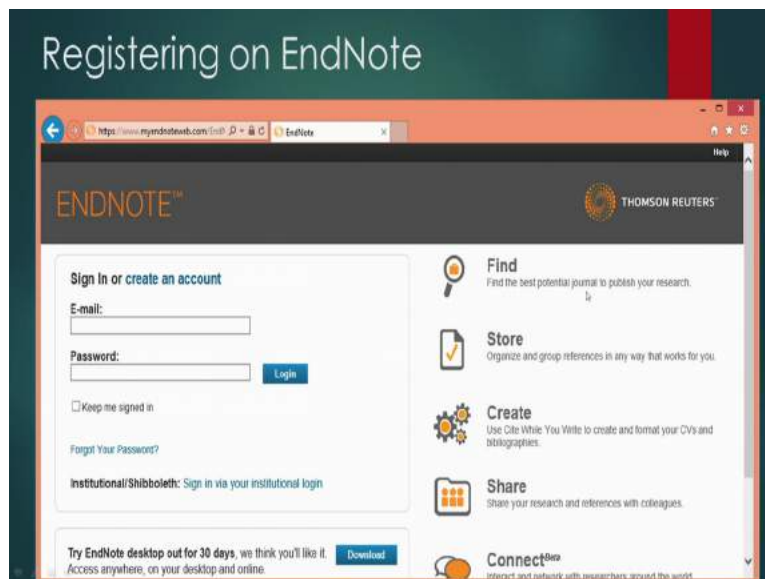
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But for our reference what we need to do is listed here; we want to save the searches that we are performing and we also want to save the search history so that we can continue from where we left off last time we did the literature survey. And most important, we need the login to Web of Science because we want to be able to add the references which we collect from Web of Science into End Note Library - which is also one of the products of Thomson Reuters - and its important for us to have a login because that is where the references will be entered, and only after that we can take them out in the format called BibTeX which I will be illustrating shortly.

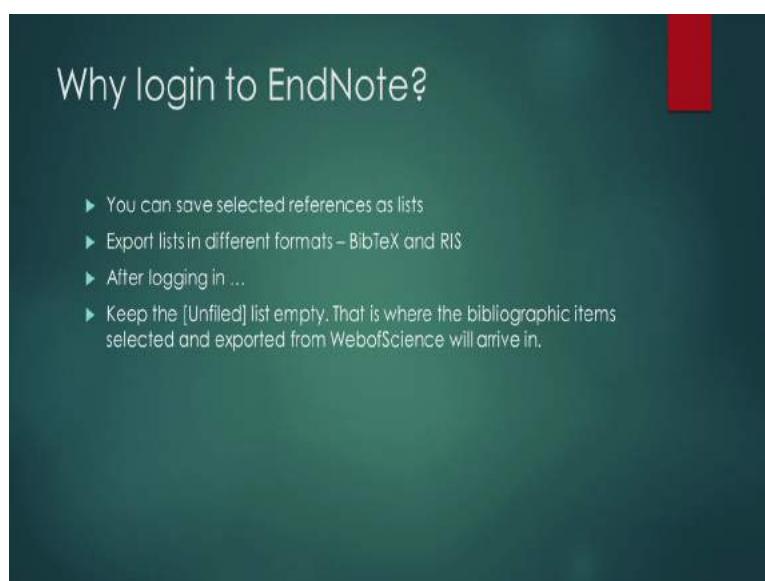


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Registering at the End Note portal is also quite straight forward. You initially login to endnote.com, and the website will then get redirected itself to myendnoteweb.com where you will be able to create your account or if you already have created then you can sign in. It is advisable that you need to sign into both portals in two different browser tabs and keep it ready.

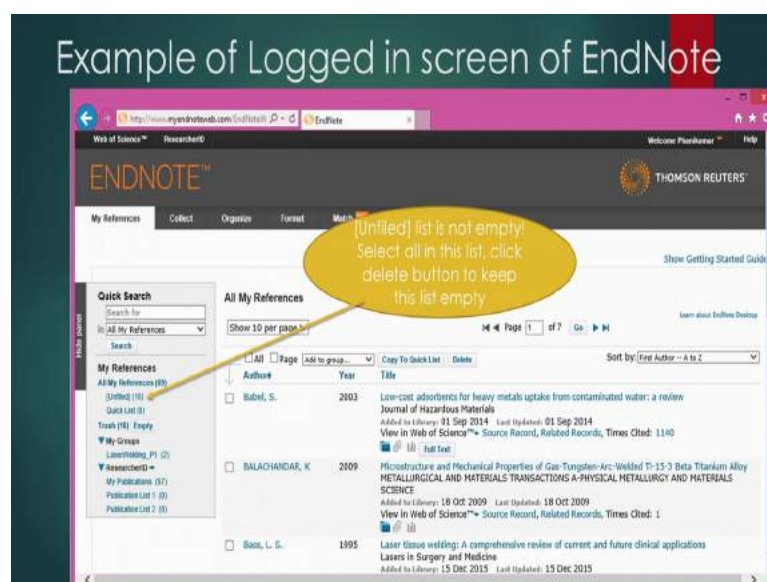
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And the reason why we want to login to End Note is because we want to save the selected references as lists and we want to export those lists in different formats particularly BibTeX and RIS formats. And then, after logging in, we must also pay attention to one aspect - the lists are all named with some heading which we can pick to be meaningful for the set of references that we collect. And what those lists have is also an item - an empty container - called unfilled, which when we first time login it will be empty, but as you keep doing literature survey this list will kept populated by various items. It is very important to keep it empty whenever we want to collect references, so that we can avoid mixing up two different literature surveys that we are doing simultaneously at two different topics.

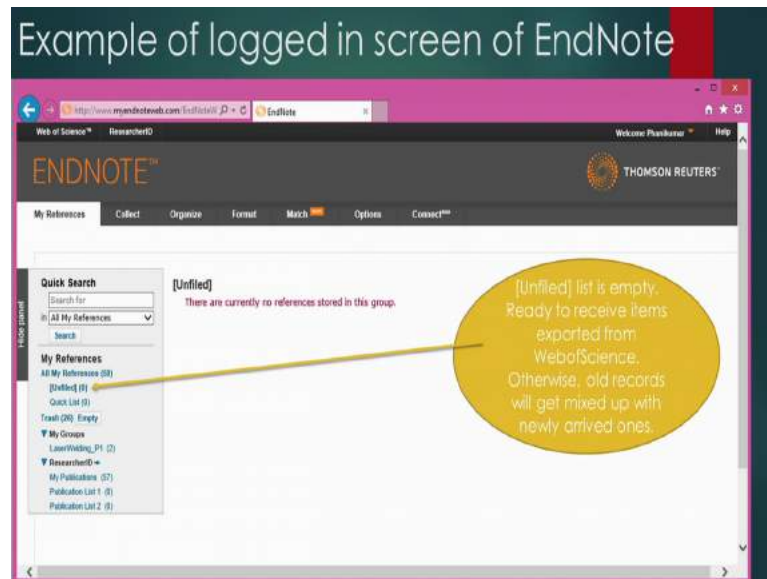
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This is how the logged in screen of End Note would look like. This is what I was alerting it to you - on the left hand side bar there is an item called unfilled with a bracket; it just shows you how many bibliographic items are there which are not categorized as any list and those must be emptied. So, what we need to do is select all of them using the All button here, and click on the button Delete, so that to that particular list becomes empty.



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So when it is empty it looks like this. Beside unfiled you have 0 and the list is looking empty. Which means that we are now ready with our folder to pick the literature survey items that are coming from Web of Science into End Note, following which we can then export in the format that we like.

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Keyword based search

- ▶ Choose alternate keywords with "OR" combination
- ▶ Choose the fields carefully : Topic / Title / Author / Author Identifiers / Editor / Group Author / Publication Name / DOI / Year Published / Address
- ▶ Choose the timespan :
 - ▶ All years
 - ▶ Specific year range (From YYYY to YYYY)
- ▶ Use advanced search features to construct a search string combining different fields, values and Boolean operators / parentheses.



So, first demonstration will be on a Keyword based search. What we mean by a Keyword based search is that we choose a set of words which describe the research area that we want to search for. As an example, in my case, I am going to do the literature survey on a topic called Melt Spinning. A very often Melt Spinning is also referred to as Rapid Solidification, because during Melt Spinning, Rapid Solidification phenomenon takes place. Therefore, its very important to find out what are all the alternative keywords, so that we can use them with OR combination, so that we can get a union of all the results that will match the set of keywords that we have combined.

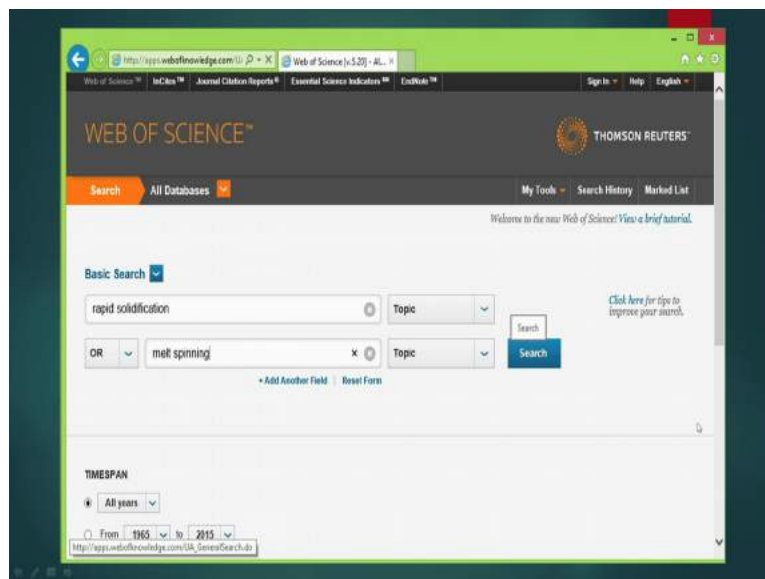
Its very important to also choose the fields. It is possible for us to choose the fields like Topic, Title, Author, Author Identifiers, Editor, Group Author, Publication Name, DOI link, Year of Publication, Address etcetera, so that we can search for articles pertaining to the respective field. Some times when you pick the wrong field but the right keyword, then you may not get the number of publication hits that you are expecting.

Also important to look at the time span over which the search has to be performed. For a first time search, it is important to have the search done for all the years where the data is available, but then as you move on, if you have done rigorous search, for example, this year, then next year you can do the search only for one year and continue from there on. So, its very important to pay attention to the time span over which you are doing the literature survey.

And if you are already an experienced programmer or user of data bases, then you will be able to configure search queries using different fields, and Boolean operators, and parentheses that will combine, so that you can perhaps get the right literature item within just one query.



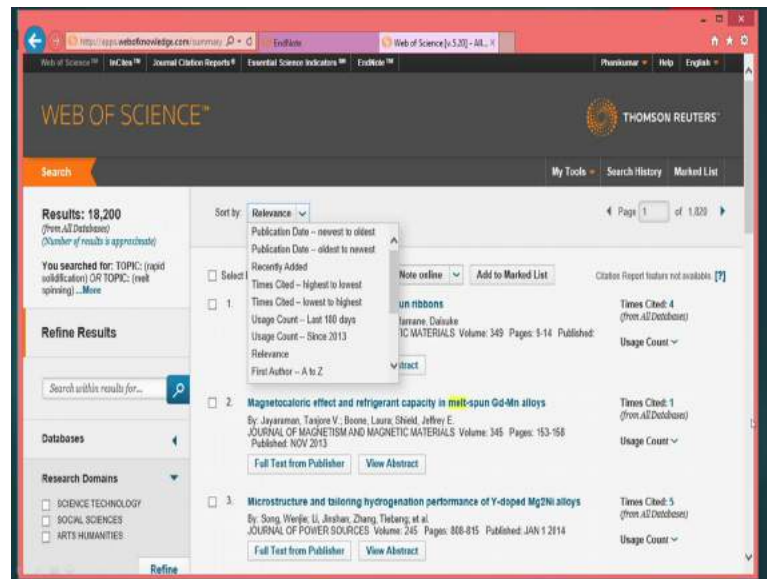
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We are just now going to try out this kind of a keyword based search, and as you can see from the screen, I have typed in two sets of words; one is rapid solidification and the other is melt spinning, and I am combining them here with, you know, OR Boolean operator, which means that we will get a union of the results that are combined from both these searches. You can actually add large number of such phrases by clicking on this item - Add Another Field, and you can mix and match OR, and to make a combination. You can change the type of field here on the right-hand side, click on this button to change it from Topic to, say, Author Name or Year of Publication etcetera. Once you are done with writing here, you can then click on the button Search, and then the search will be performed.



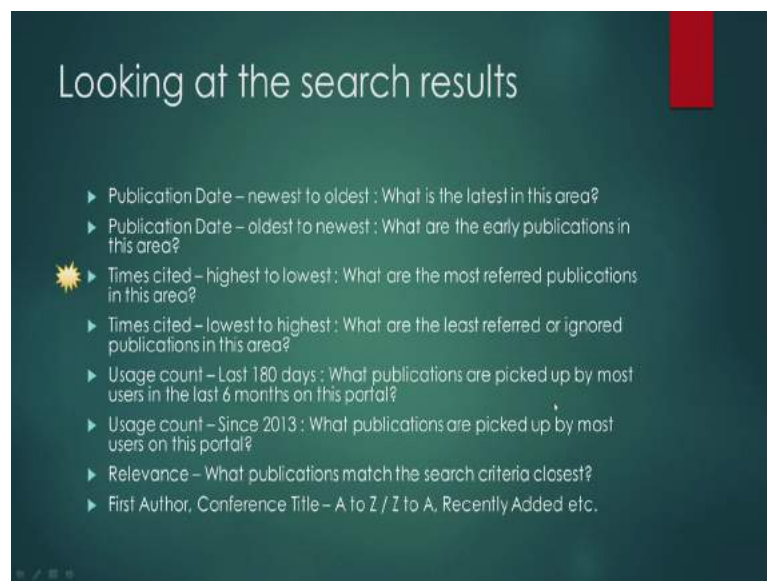
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Usually, a an established area is going to give a lot of search results. In this case, for example, melt spinning or rapid solidification being fairly established area of research in the metallurgical and materials engineering domain, you do have a large number of hits - namely about 18000 hits. So, naturally, for a beginner among the researchers, its very difficult to go through all of these to get an over view of the subject. Its very important to pick only few that are very important. We must sort these results in a meaningful fashion, so that we can identify those results that we want to go over. There is a ability here, by the portal here, to sort by different types of fields. So you can see that you can sort by relevance, which means that those results that are very closely matching will be picked up and there are various other types of sorting also. So, just briefly, to summarize how these different sortings are important is given here.



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It's very important that all the sorting types must be explored by a novice in research. Publication date, starting from the newest at the top, is important because that is where you will see what is a latest publication that is coming in this area. And sorting by publication date, where the oldest is at the top, is important to see what are all the early or very old publications in this area, so that you can see how this particular research area has started. And you can also sort the publications by the number of citations, and you can cite them from the highest level to the lowest level; highest being at the top. This is important because you may want to know what are all the most refereed publications in this area, what are the so-called citation classics in this particular research area.

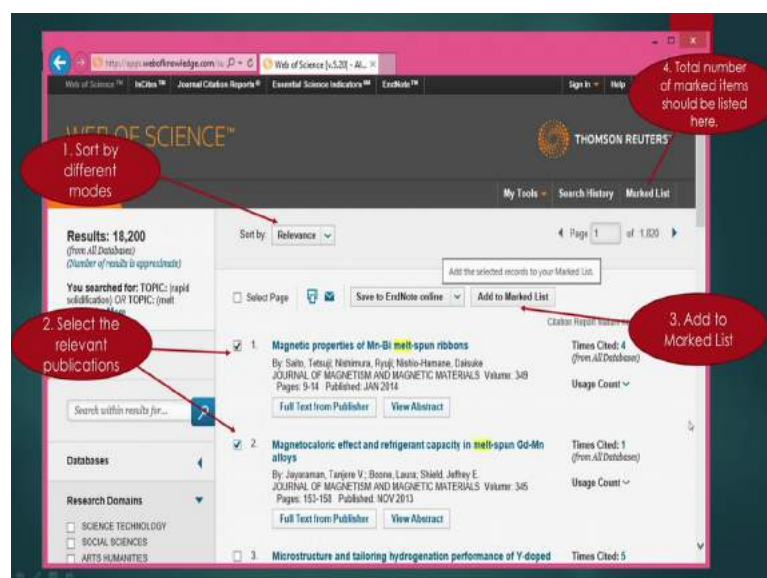
And you can also flip this sorting by making the lowest citations, namely zero, to be on the top, because that shows you what are the least referred or ignored publications. It's not necessary that all the ignored publications are important, but then, may be sometimes there are very important and useful publications that have somehow evaded the attention of a lot of researchers across the world. You can also use the search engine capabilities to see whether our peers who are also using this portal have something to convey to us. And the Web of Science portal keeps track of the number of users who have picked up your particular literature item by using the usage count. So you can look at what happens to the usage count of a particular item in the last six months and look at that number to



guide you to identify the kind of publication that you want to pick up from the portal.

You can also use it across the entire period over which this particular portal was running, so that you can actually see what are the publications that are picked up by most of the users on this portal. You can also pick them up by the First Author Name, Conference Title, A to Z or Z to A, and you know, recently added articles etcetera. So, there are lot of ways by which you can sort the results that come by searching for a set of keywords, and its important to explore all these types of sorting, so that we can pick the top few of each of these types, so that it gives us a overall picture of this particular research area.

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And here, I am just showing you now what's the next step. What we need to do is once we have sorted by the way that we want to sort, for each of the sorting types we need to first pick that type, and the moment you change the type here the page will refresh to sort the items by that particular type. And then, we use these check boxes to select items that we want to pick up ok. And then, after that, we can click on this button Add to Marked List, what it does is basically like a e-shopping website or like an electronic cart. It basically puts all these checked items into a basket and those items will be visible in this corner under the Marked List. And the number will keep going up as you perform more and more sorting and selection; and at some point of time you may say that you have had



enough of useful publications picked up, so that you can collect them on to your desktop.

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A presentation slide with a dark teal background and a red vertical bar on the right side. The title "Steps to collecting the reference items" is at the top in white. Below it is a bulleted list of instructions in white text.

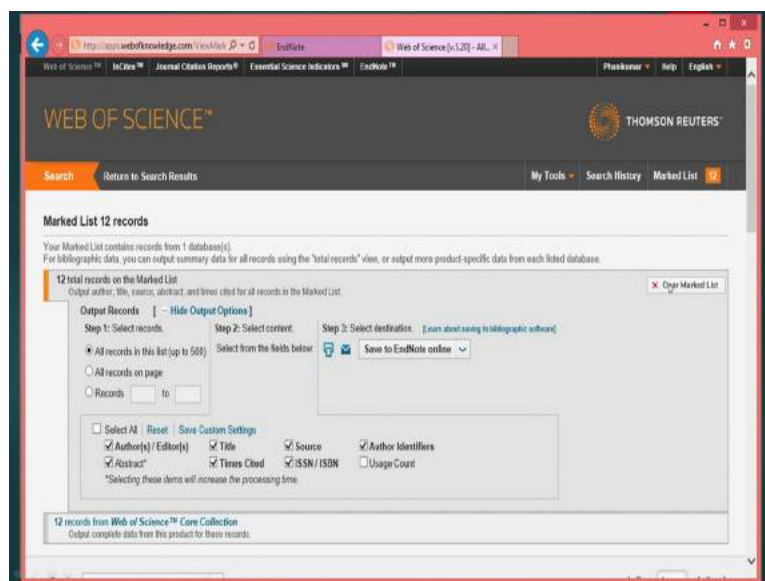
Steps to collecting the reference items

- ▶ Under each method of sorting the search results, "select" those you feel are important for your literature survey by "checking the box" against those items.
- ▶ Click on the button "Add to Marked List" to add these to your cart.
- ▶ The "Marked List" tab shows how many items are in the list currently.
- ▶ Once you are done collecting, click on the "Marked List tab" to proceed to the 3 step process of collecting the data.
 - ▶ Step 1: Select records (within this Marked List – usually all)
 - ▶ Step 2: Select content (choose abstract also, helps in identifying which publications to find the full text of)
 - ▶ Step 3: Select destination (choose Save to EndNote online)

So, these are the steps that are to be performed while collecting the reference items or the journal publication items. Under each method of sorting the search results, select those that you feel are important for your literature survey by checking the box against the items. Click on the button Add to Marked List to add those to your cart. The Marked List cart then should show how many items are there in the list already, and then, once you are done collecting, click on the Marked List tab to proceed to the three step process of collecting the data. The three step process is illustrated in this screen shot.



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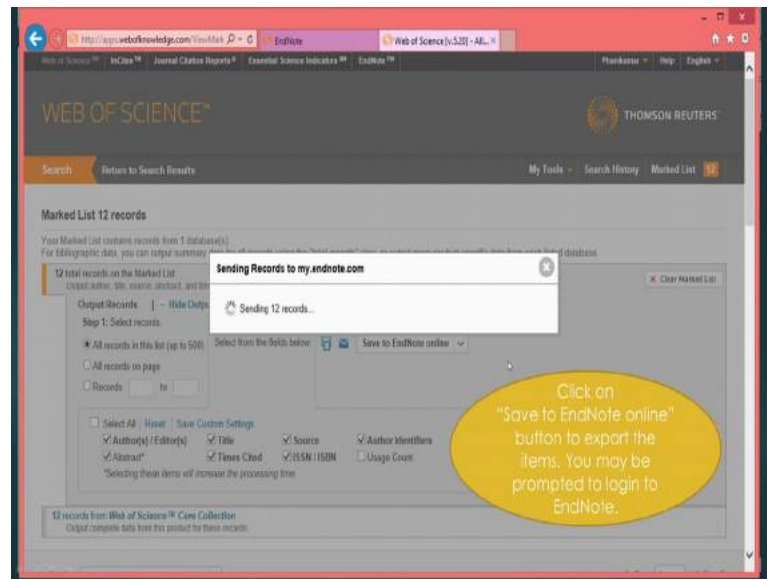
So, you should see this with the number of items that you have selected; in my case, I have selected only 12 items for illustration. The step one is to select the records among these 12; we are not selecting from the whole of search but among these 12; so, naturally, you have carefully selected 12, so you can select all of them.

And the step two is to select what is a content that we want to export. So, very often only the Author Titles, Source and Author Identifiers will be selected, and its very important to also select Abstract, because you may want to read the abstract to identify which ones among these 12 publications requires you to go through the full text of that particular article. It may not be necessary, for example, to select the time cited or usage count, because they are only of statistical importance after you have identified the important publications.

Step three is to select the destination - where do you want to take these 12 items to. So, unfortunately here, in this list, you don't have a methodology by which you can select the data of these 12 records as BibTeX format or RIS format directly. They will be available only after you go through the End Note portal which is again owned by Thomson Reuters. You have already performed the registration process for this portal, so you are now ready to save it to the End Note portal.



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So, the moment you click on the End Note portal, then all these 12 items will be sent across, you would see a progress circle here rotating for few seconds while the data is being transferred from the Web of Science portal to the myendnote.com portal. Once that is done, then the search screen will come back, and then, you are ready to then explore the items in the End Note portal.

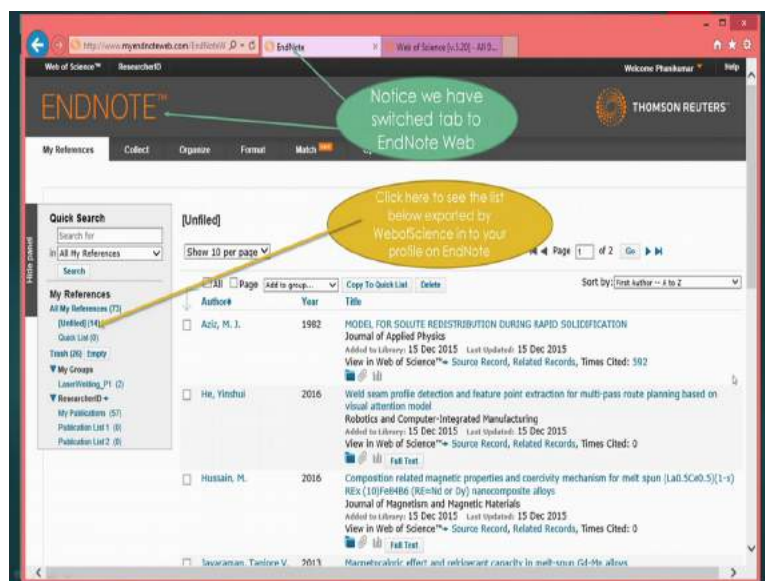
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Ok So, this is a summary. Once a dialog to export items to the End Note online has completed, login to the endnote.com portal in a separate tab, and then click on the link unfiled, under all my references in the left-hand side bar, and then you should you seeing the same set of references here as you have exported from the Web of Science portal.

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And I am now opening up the End Note portal here. You can note already that we have opened the second tab and the website name has changed here to End Note ok. And this is the unfilled link with a number of items that we have added and the items that we have selected are all listed here. Ok



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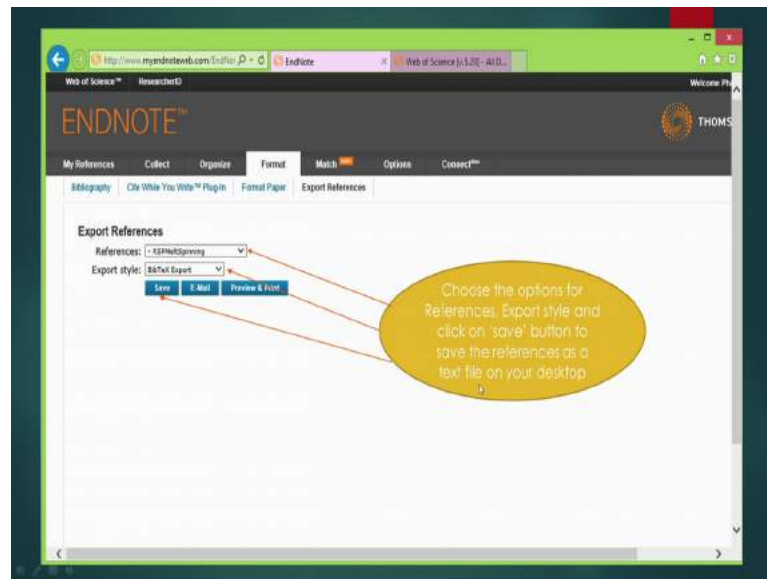
Save the list and export

- ▶ Click on "[Unfiled]" link to view all the items
- ▶ Click on the checkbox "All" to select all the items
- ▶ Choose "New group" under the drop down menu "Add to Group" and give a nice name to this list
- ▶ This new list will now appear under "My Groups".
- ▶ Click on the tab "Format" and choose "Export references"
- ▶ Under Export References, choose the new group of references you just made, choose the Export style to "BibTeX Export" and click on the "save" button to have the data reach your desktop as a text file "exportlist.txt"
- ▶ Rename the text file preferably same as the name of the group you created on EndNote web and keep the extension as ".bib".

And what do we do these list of items? So this is a way to export the data. Click on the link unfiled to view all the items; click on the check box All to select all the items; and choose a New Group under the drop-down menu Add to Group, and give an appropriate name to this list. And this list will now appear under My Groups; click on the tab format to choose Export References; under Export References choose the new group of references you have just made; choose Export Style to BibTeX export, click on the Save button to have the data reach your desktop as a text file called exportlist.text. And because the format that you have chosen is BibTeX format, you should actually rename the file to be .bib file.

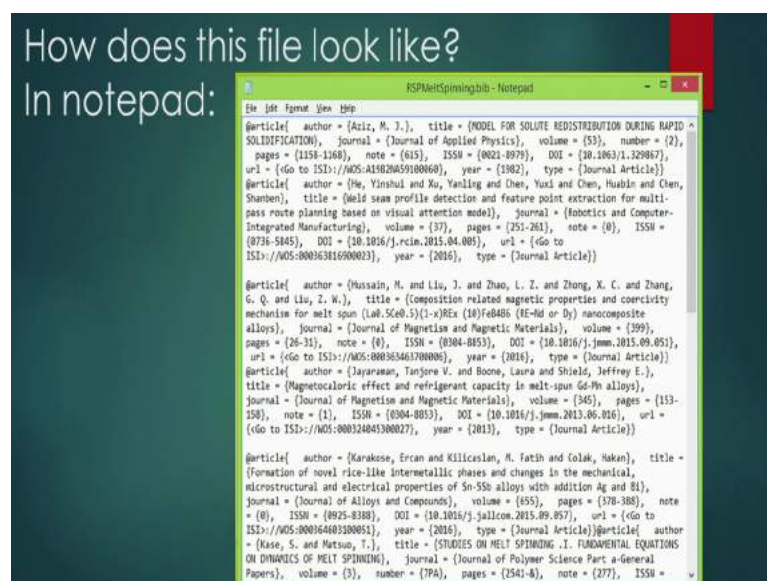


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And here is a screen that does it. After we select all the items, click on the tab Format, and then, click on the References, choose the group that we have collected all these items under, choose the Export Style, and then, click on the Save button. The file will come on to your desktop as a bib file.

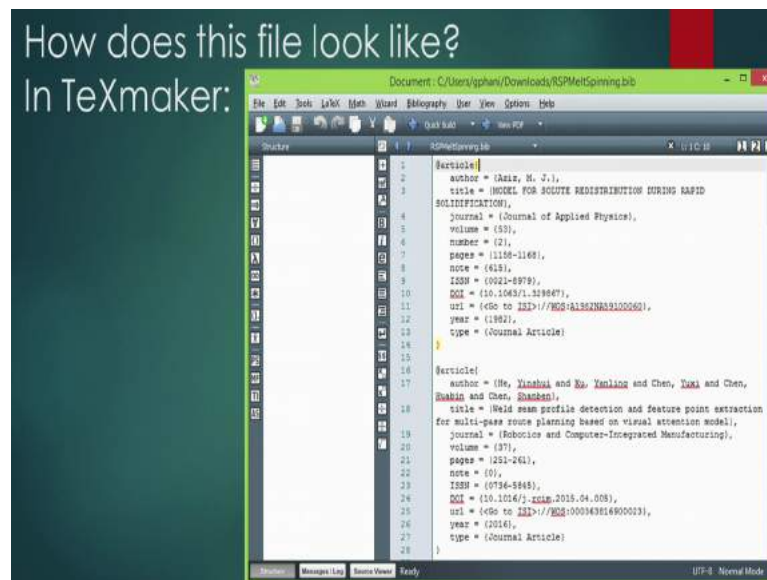
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Now, how does this bib file look like? So here I have shown you some screen shots in Notepad. The bib file is actually a pure text file, that is human readable file, and it tells you what is the type of the article, and what are all the various fields that are bounded by the braces, and every new item is starting with an @ symbol followed by an article. And this format is such that you can mix and match the different reference items yourself; you can cut and paste to the items into a Notepad editor and join your literature survey across the different periods or you can split one literature survey into multiple parts.

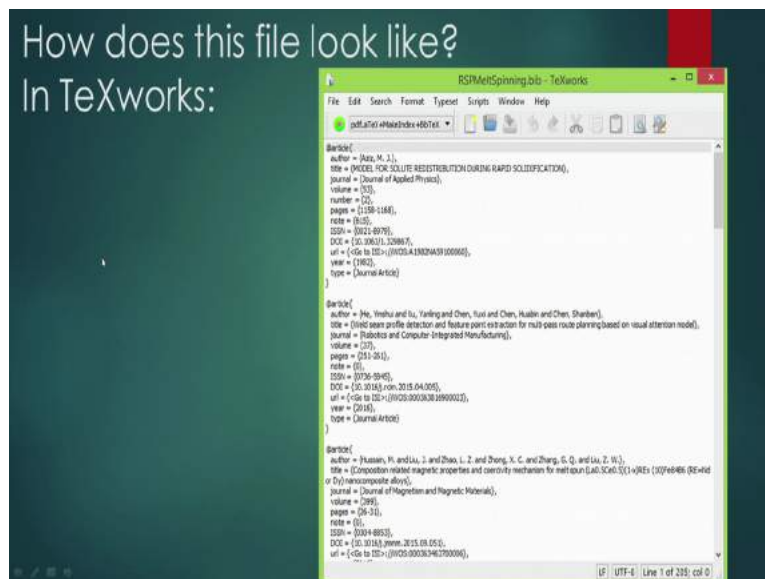
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In an editor called TeXmaker, the same file would look like this, where it is neatly formatted and you will be able to read it much better, because this editor freely available from the Internet called TeXmaker, is able to understand the format of LaTeX, and as I will illustrate later to you, you can use this bib file directly in your LaTeX documents to make references.



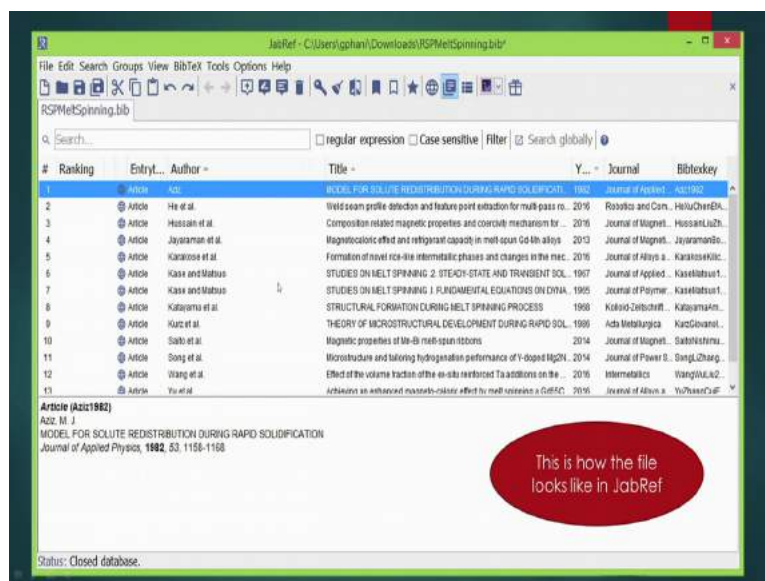
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And if you install a software called MikTeX , which is a freely available software for LaTeX on windows, then it also installs a LaTeX works which is an editor for the Tex files and the bib file would look like this in the TeXworks. So, you can see that essentially our literature data is a very well organized data with difference fields and it can be combined because its a pure text format. However, we may need a Reference Manager to be able to mix and match these data, analyze them, have our notes written, etcetera. And that's were I would recommend you to use the software called JabRef.



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And I am using a screen shot here to illustrate to you how the same data would appear in JabRef software, which is freely available from the Internet Open Source software, java based which will run on any platform. So, if you open the same bib file in JabRef software, then it looks like something familiar to you like an Excel sheet or Open Office Calc spread sheet software, where each of the bibliographic item is appearing as rows and different fields as columns. You can search for data within this software, you can also sort them out, and you can write notes. You can double click on any of the items and add notes for your own reference later on.



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What are the other advantages of a using reference manager to manage your literature survey data? Essentially, because they normally give you a spread sheet like appearance. So, that it is very easy to work with different data. You should be able to add notes, you should be able to split and merge different files, should be able to export different formats. And most importantly, these files reside on your desktop, which means, that you can go through these items at your own pace, offline, and identify those who you want to read the full text of, and so on.



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And why do we need to export the data as a bib file? This is very important. The objective of this module is to pick up the literature survey items from some citation data bases such as Web of Science and then have them help you in writing an article. And as you can see, Web of Science is providing that data, but then, you need to go through the End Note Web, so that the data can come as a bib file. Now, once the data is available as a bib file, then you can use LaTeX directly, and then, make those references in your article or you can use JabRef software to export the bib file to a XML format, and then, use that XML data in Microsoft Office, for example, to write your article. In other words, if you are planning to write an article or thesis, and you want to your literature data to come as a data file, then there is a procedure to adopt, and here I am showing you the data flow for that procedure.

Thank you

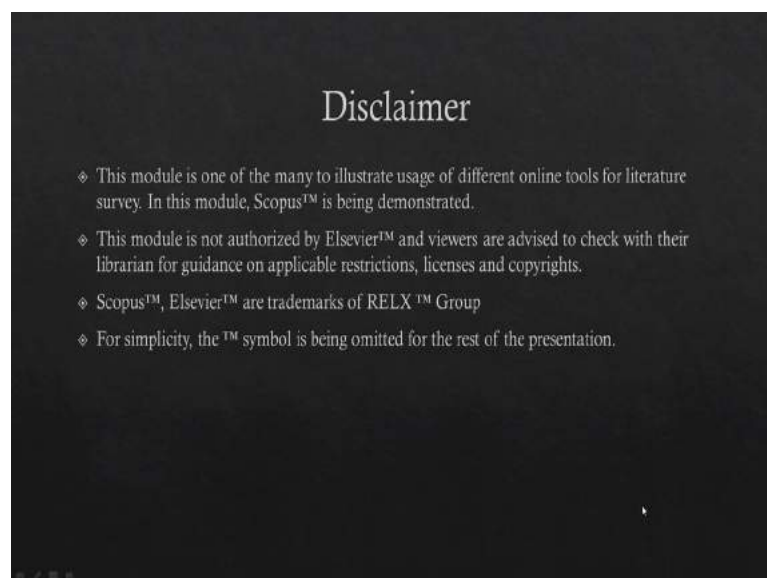


Introduction to Research
Prof. G. Phanikumar
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture – 08
Literature Survey Using scopus TM

Yeah. This is the second part of the Literature Survey module, on how to use tools such as Web of Science and Scopus to collect bibliographic data on to your desktop. The objective being that these data can then help you write an article or your thesis and populate the bibliographic fields in your thesis without having to type out every different bibliographic data field that you need and this second part of the module I am going to use Scopus as a source of bibliographic data.

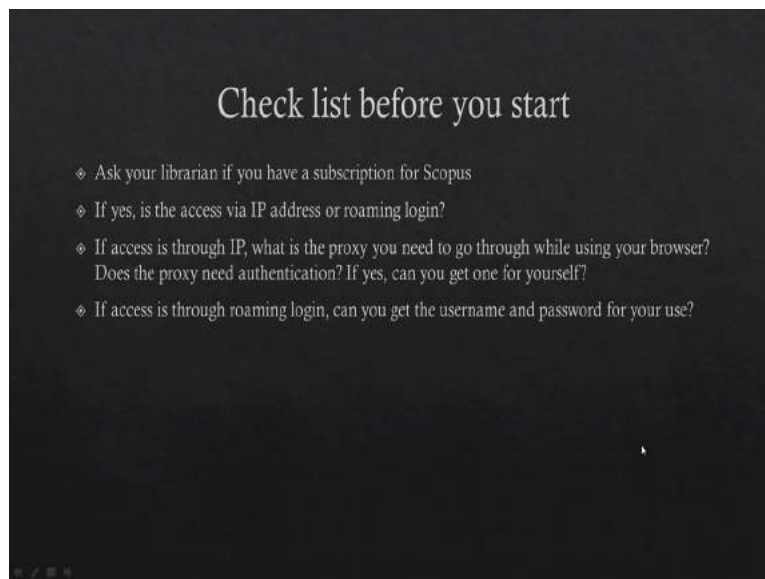
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As usual again we must make a disclaimer.



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This is a Check list that we must have before we start a literature survey using the Scopus as a tool. We must obviously ask whether we have access to the Scopus tool. So, please do ask your librarian if you have subscription for Scopus and if yes, then is the access via an IP address or a roaming login and if the access is through IP address, whether do you need to access it through a proxy, and if it is through proxy then whether it needs a user name and password and whether you have such a user name and password with you to be able to start the literature survey using this particular portal. If it is through a roaming login then you must get the user name and password for you to login through the portal.

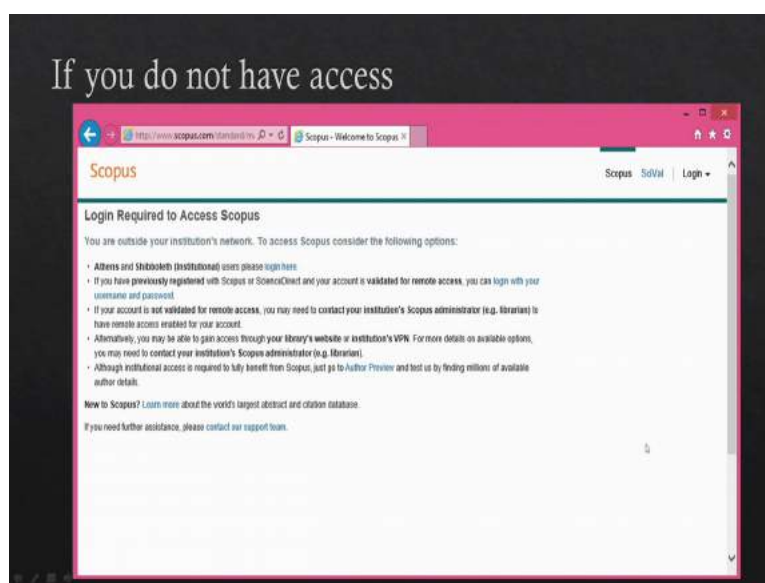


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So, assuming that you have the information then we can go on to now start the literature survey on this portal and I have given the URL of this portal here and in case the URL of this portal happens to change because the vendor or the company has changed then please do search the internet for the words like Elsevier and Scopus and identify the correct URL.

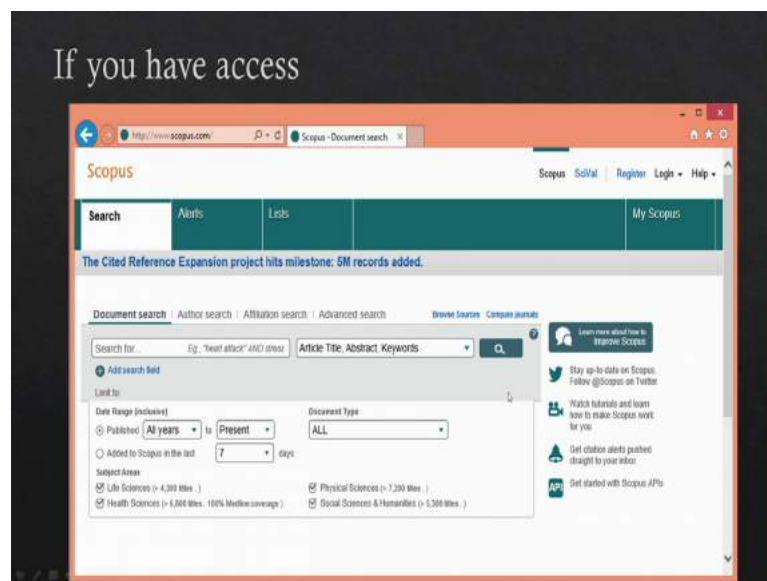
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In case, you do not have access and you want to open this portal then this is the screen that you will see. Namely the screen will not let you search for anything and it will ask you to login because it detects that you are logging in or trying to access from outside your institutions network. If you have already registered for access through your institute at least once then it may detect that you have logged in and provide you a partial access. However, you need to actually login before you start any process.

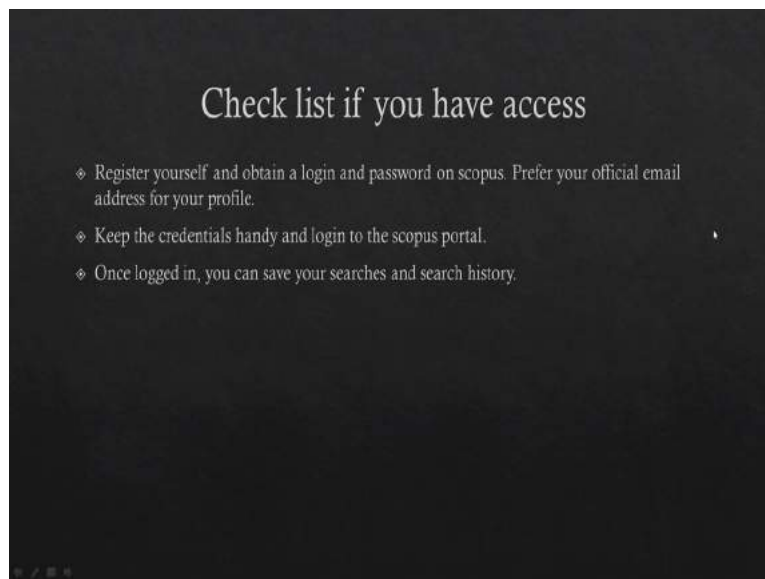
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And this is how the front page would look like, in case you have access, which means that you are ready now to search because the search form is readily available for you and I advice that you can register and login before you start searching because you can actually keep the history of your literature survey and all the search results in organized fashion in your own profile and that will be helpful for you to start off your work from where you left last time you did the literature survey.



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So some check list before you can start the process, register yourself, obtain a login and password on Scopus and again for the profile you must use official email, so that you will be able to show your affiliation to an institute while you do your literature survey and have these credentials handy, so that you can login to the portal whenever you are required to do the literature survey and once you are logged in you can save your searches and search history.



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Register yourself on the portal

Scopus

Search Alerts Lists My Scopus

Register

Registration is quick and free. It allows you to personalize the Elsevier products to which you have access.

[Privacy policy](#)

Your details

First name:

Family name:

Email and password

Enter a password between 5 and 200 characters. Your e-mail address will be your username.

E-mail address:

Password:

[Password guidelines](#)

☐ Add profile details

☐ I wish to receive information from Elsevier B.V. and its affiliates concerning their products and services

☐ I have read and understood the [Registered user agreement](#) and agree to be bound by all of its terms.

This is how it looks like; when you click on the link register there is a small form that will be shown to you. You can fill all these fields, naturally you are expected to agree to the license terms and then register.

(Refer Slide Time: 03:40)

Login to the site

Scopus

Search Alerts Lists My Scopus

Login using your Elsevier credentials

Username:

Password:

☒ Remember me

[Forgot your username or password?](#)

Document search

Search for:

[Add search filter](#)

Limit to:

☒ Published: to

☐ Added to Scopus in the last: days

Document Type

Subject Areas

☒ Life Sciences (> 4,300 titles)

☒ Health Sciences (> 6,000 titles, 100% Medline coverage)

☒ Physical Sciences (> 7,200 titles)

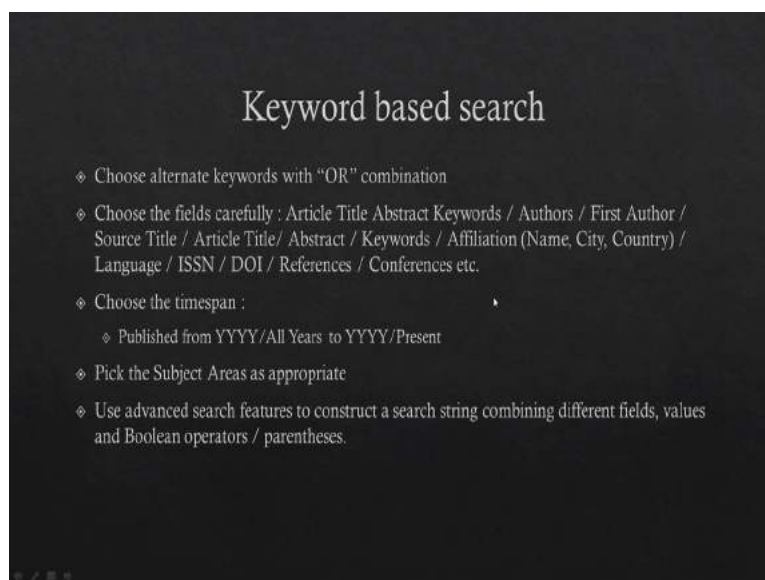
☒ Social Sciences & Humanities (> 5,100 titles)

Once, you register then you can login by using this login form and you can ask the portal



to remember your login so that you can perform the literature survey in different sessions without having to login a second time.

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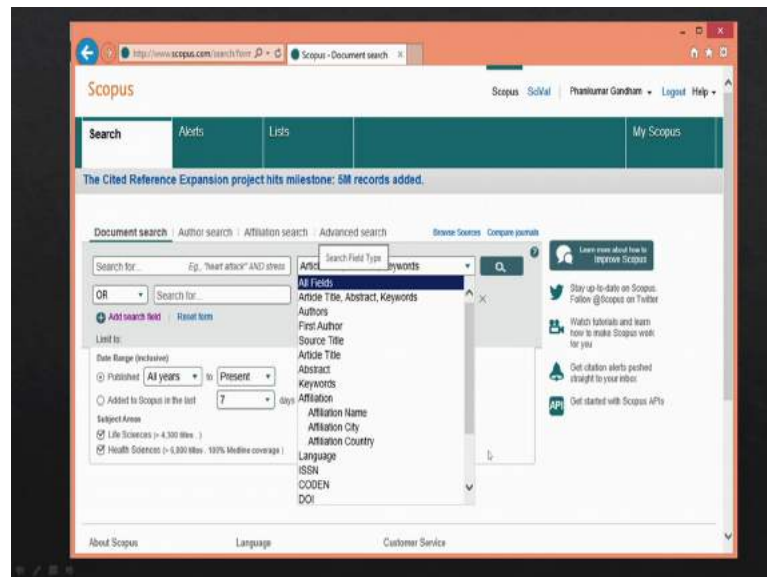
So, we are going to show a demonstration using a Keyword based searched and we are going to use a same set of keyword as we have seen in the other module on Web of Science. We must choose fields very carefully. The set of fields that are available in Scopus are slightly different from those that are available in Web of Science. You can actually choose Article, Title, Abstract, Keywords all of them together and search for a set of keywords in that or you can choose them in Authors, First Authors, Source Title, Article Title, Abstract, Keywords, Affiliation Name, etcetera.

You should also pay attention to the timespan and you can choose the timespan either from a particular year to another year or you can choose all years till the present, so that the search will be done across the entire database. For a person who is doing the literature survey on a topic for the first time, it is advised that the entire timespan is used for literature survey. The subject areas also have to be chosen up for appropriate links to show up. An engineering topic need not be searched, for example in the medicine area and vice versa. However, it is again advisable to keep all the subject areas open, so that you may find interdisciplinary results that will be showing up.



You can actually also perform advance search features using a search string that can combine different fields, values, Boolean operators and parentheses, if you are already familiar with advanced programming or SQL syntax.

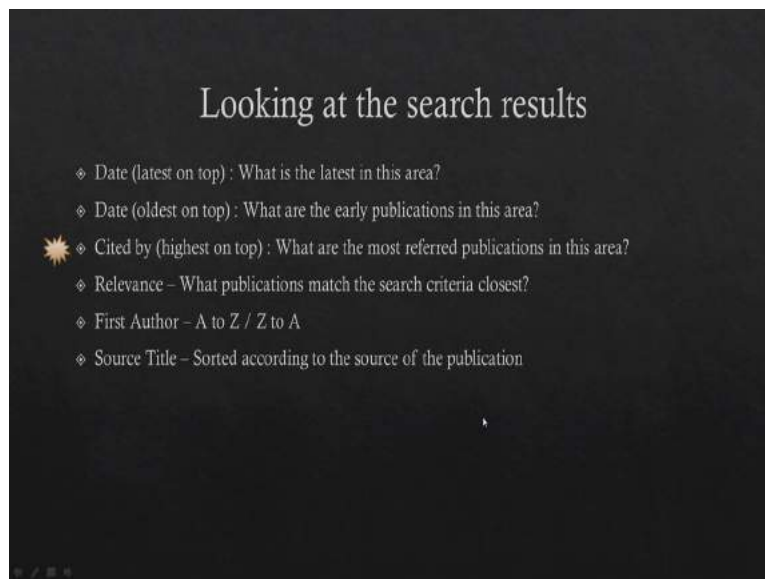
(Refer Slide Time: 05:21)



And this is the form that is used to search for the literature. This is at the front page, I have already logged in as you can see from my name here and the search history then will be stored in my profile. You can actually have multiple fields and you can add additional fields by using the plus button here and you can combine your searches across multiply fields using either OR or AND, I would advise OR because you can get a union of all the different results. In what fields should you search this keywords for is evident here, you can press this dropped down menu and you would see a list of fields that will be appearing.



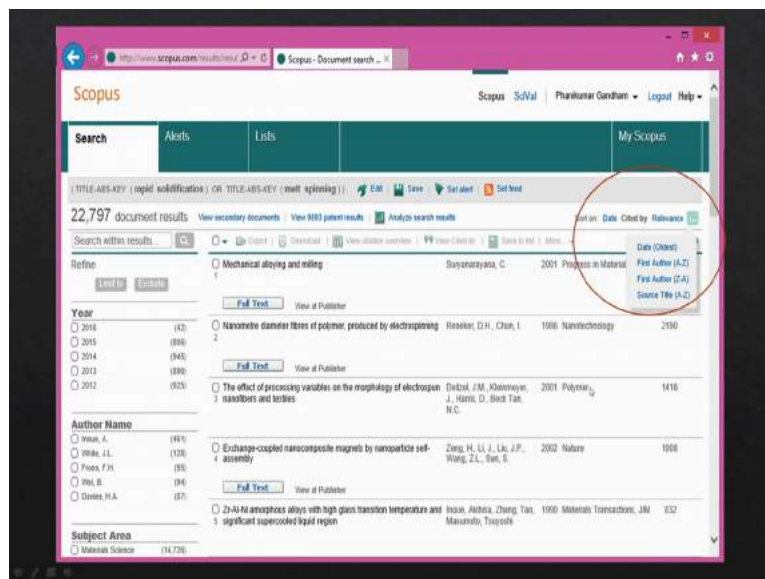
(Refer Slide Time: 06:02)



And what are the various fields, you can actually see that the fields are available in such a way that you can choose from the dates, the citations, the relevance, the first author and both forward and backward sorted order as well as from the title and here I have put a star again is the cited because it's very important for a novice researcher to identify, what are the publications that are referred by large number of researcher in this area, so that you identify the important pieces of work in this area immediately.



(Refer Slide Time: 06:34)



And I have searched, as an example for 2 phrases, rapid solidification and melt spinning and combined them with the Boolean operator OR, so that I am getting a union of these two and I have obtained number of hits 22000, which means that this is a fairly mature area of research. On the right hand side you can see that these results can be sorted by date, citations and relevance and in this 3 dots, if you click, you get 4 more ways of sorting, namely the inverse of the date search. Basically, here the oldest paper will be coming on the top and the first author searching both backwards and the forwards sorting order as well as the source title.



(Refer Slide Time: 07:15)

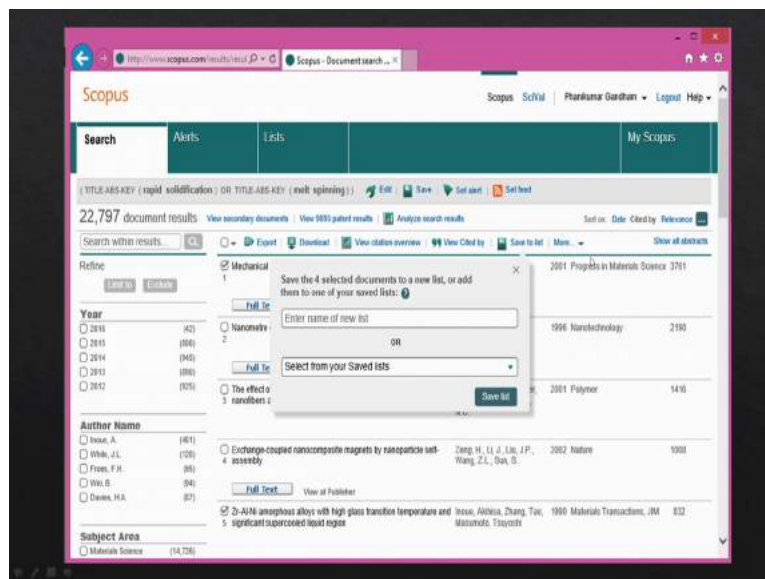
A presentation slide with a dark blue background and white text. The title "Collecting the reference items" is centered at the top in a large, white, serif font. Below the title is a bulleted list of instructions in a smaller, white, sans-serif font. The list contains seven items, each preceded by a diamond symbol. The first item is a general instruction about selecting items. The second is about clicking "Save to list". The third is about entering a new list name or selecting an existing one. The fourth is about viewing the number of items added. The fifth is about clicking "View or manage your saved lists". The sixth and seventh items are part of a three-step process for collecting data, starting with "Step 1: Select records".

Collecting the reference items

- ◆ Under each method of sorting the search results, "select" those you feel are important for your literature survey by "checking the box" against those items.
- ◆ Click on the button "Save to list" to add these to your list.
- ◆ "Enter name of a new list" if you wish to create a new name of this list. Else, add them to an existing list by picking it from the drop down menu below.
- ◆ The number of items added to your list are shown in a box along with a link to "View or manage your saved lists".
- ◆ Once you are done collecting, click on the link "View or manage your saved lists" to proceed to the 3 step process of collecting the data.
 - ◆ Step 1: Select records (within this Marked List – usually all)
 - ◆ Step 2: Select content (choose abstract also, helps in identifying which publications to find the full text of)
 - ◆ Step 3: Select destination (choose Save to EndNote online)

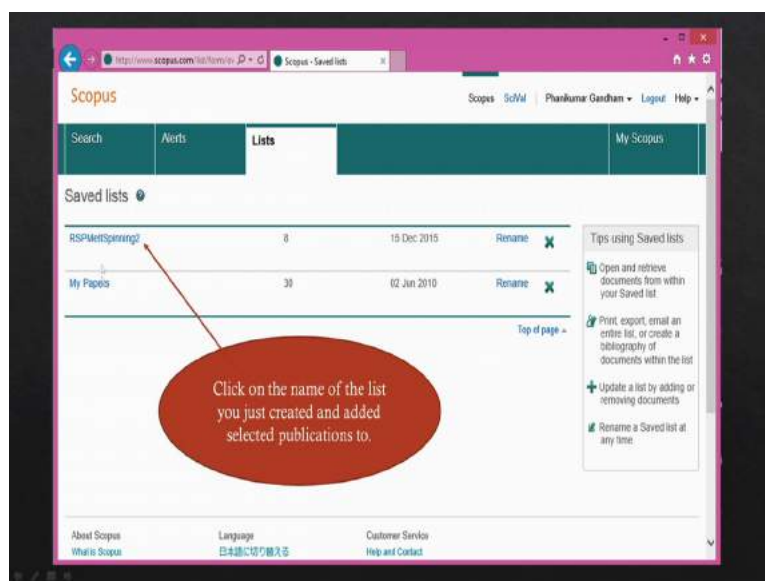
And how do we then collect the reference items that we need from this list. What we need to do is, for each method of sorting the results, we should select those we feel are important for our literature survey by checking the box against those items and then when we can click on the button save to list to add these to our list and then we can click on the item enter the list name of a new list if you want to create a new name for this list or we can also add them to an existing list that we have created just now. The number of items added to the list will be shown and we must verify that against what we are searching. And once we are done with collecting then we can click on the link view or manage your saved list, so that we can go to a three step process in collecting the data.

(Refer Slide Time: 08:01)



I am illustrating them here, you can use this click buttons to identify those articles that you want to collect and then click on the saved list button here and it will open up a dialogue, asking you whether you want to save these items as a new list or an existing list. You can choose as appropriate and click on the saved list and those items will be in your list.

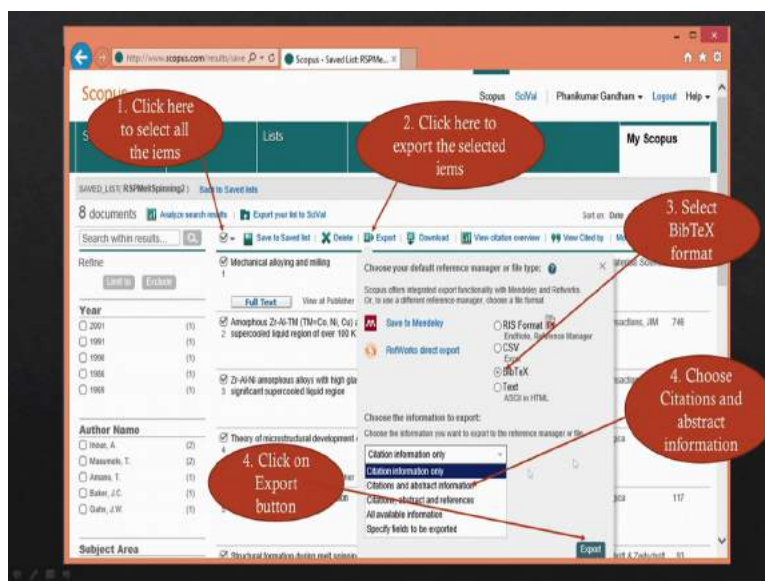
(Refer Slide Time: 08:25)





And those lists are then shown to you here and you can see that I have just given a name called RSP Melt spinning 2 for this literature and I have only identified 8 publications right now to save. And if you click on this link then you can see those items that we have just now identified and they are listed here.

(Refer Slide Time: 08:43)



What is the process now to export them? This is quite straight forward, we have taken efforts to sort in a different manner and identify most important publications. Therefore, all these we can actually pick up for our literature survey. So we can click on this button here, where you can select all the items that you have identified. This will not pick all the items for the search that you have originally made but for only those which you have picked and added to your list.

So first we select all of them, the second is to click on the export button and that will open up a dialogue here. The dialogue is fairly elaborate. In the dialogue, we must choose a format of the data that we want to take. The format can be chosen to be RIS format, CSV format for excel, BibTeX or text format. I would advise you to choose the BibTeX format because that is the native format for many reference management software and also it can be converted later to other formats. What information do you want to take along to this BibTeX format is again available here, you can click on this



dropped down menu and you can choose as I would advise citations and abstract information because very often it is important to read the abstract, so that we can identify those among these publications that require you to go to the full text. Once, you have chosen the options here then you can click on the button export here, to export these data as a BibTeX file on to your desktop and it will come to your desktop as a file that can be saved in your downloads folder.

(Refer Slide Time: 10:15)

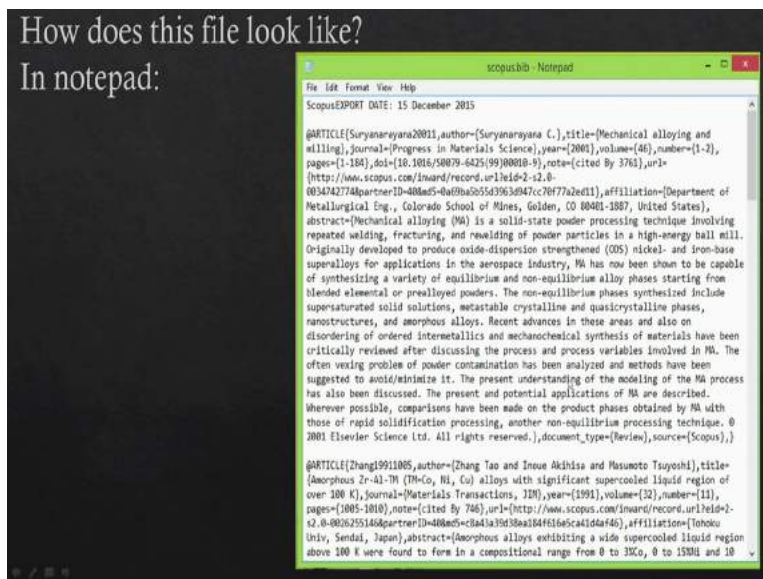
Save the list and export

- ❖ Click on the name of the list to view all the items you selected and saved.
- ❖ Click on the checkbox at the top of the first column of all the items to select all the items
- ❖ Click on the link "Export" to open the pop-up dialogue.
- ❖ Choose "BibTeX" format, "Citations and Abstracts" and click on the button "export" to save the results as a file named "scopus.bib".
- ❖ Go to your Downloads folder to rename the file to something meaningful, say, after the name of the list you had in your profile on scopus itself.

Ok, here is the summary that we have done, click on the name of the list to view all the items that you selected and saved. Click on the check box at the top of the first column of all the items to select all the items. Click on the link export to open up the pop up dialogue. Choose BibTeX format citations and abstracts and click on the button export to save the results as a file named Scopus dot bib. Go to your downloads folder to rename the file to something meaningful. For example, you can give the same name as the list that you have written and you can store it in a safe place that you can look up later on.



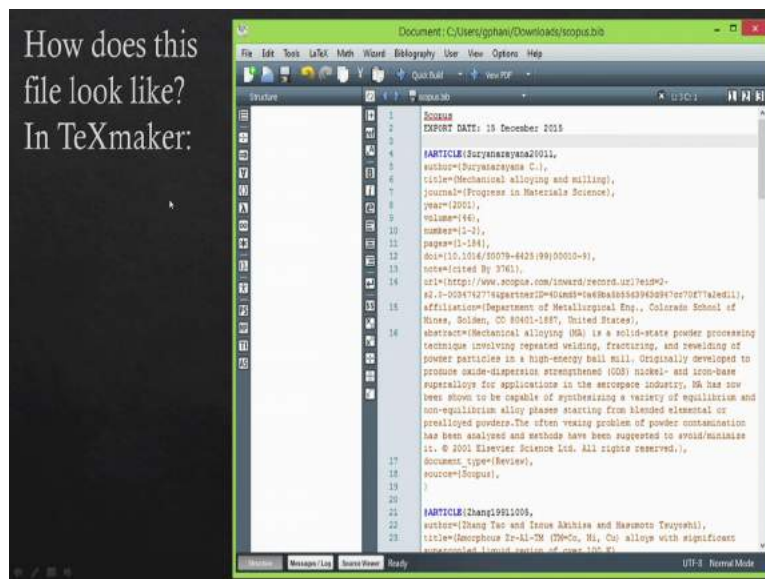
(Refer Slide Time: 10:48)



How does this file look? It is very much similar to the other demonstration we did on Web of Science. In a note pad it would like this, every item would have, at symbol and an article type followed by the data which is human readable. The full text is available in ASCII format, you can read it here. You can also use the notepad to cut and paste different bibliographic items from different bib files that you may want to combine or split.

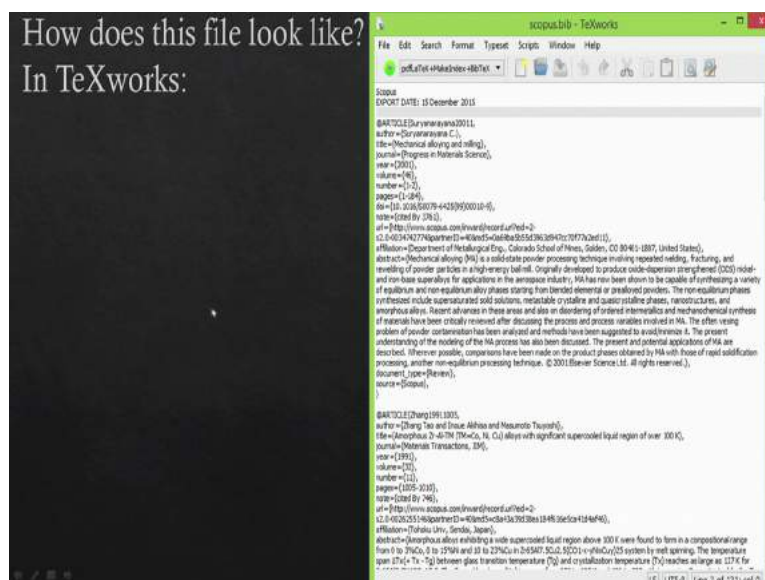


(Refer Slide Time: 11:14)



In TeXmaker, it would look like this neatly organized by different fields.

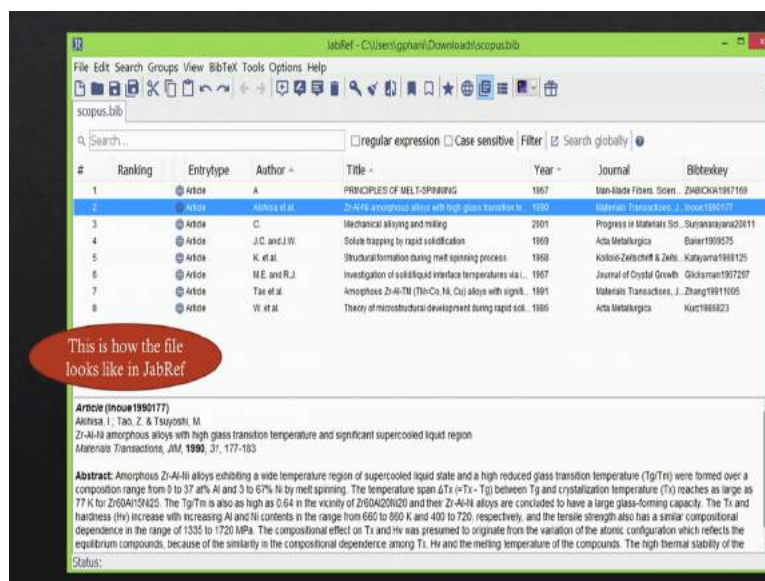
(Refer Slide Time: 11:20)



In TeXworks, it would look like this again neatly organized. These two tools, TeXworks and TeXmaker are freely available for all the platforms for you to install.



(Refer Slide Time: 11:30)



And my favorite would be then JabRef, which is a software that will let you show the literature data as a spreadsheet and here we can see and we can identify any of these items, double click to edit and add notes and we can read the abstract within the software itself and you can also export the data in multiple formats.

(Refer Slide Time: 11:51)

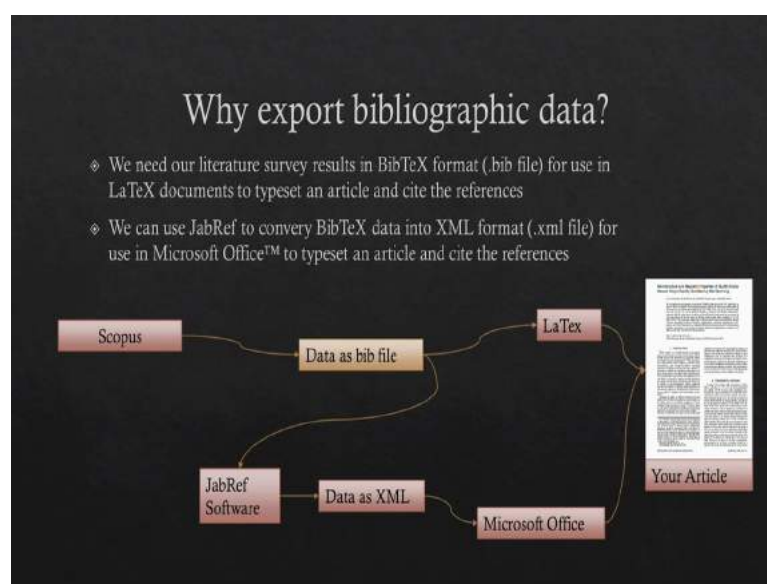
How to edit this .bib file?

- ◆ Use a Reference Manager like **JabRef** (opensource and free)
- ◆ Spreadsheet like appearance
- ◆ Possibility to add notes, split and merge similar files, export to different formats
- ◆ The file resides on your desktop – go through the items at your pace, offline, identify those you need to have the full text of and so on.



And how do we edit this bib files so that we can mix and match, combine, etcetera? We always need a reference manager and I would suggest JabRef because it is open source and free and it has a spreadsheet like appearance. And JabRef gives you the possibility to add notes, split and merge different files, export different formats. What is more important is that these files resides on your desktop, which means that you can go through these items at your own pace offline and identify those among these papers where you want to the read the full text.

(Refer Slide Time: 12:22)



And why do we need to export this bibliographic data, again the logic is similar to what we have shown on Web of Science demonstration. Scopus is giving you the data as a bib file directly and you can use this bib file, LaTeX software to create your article. You can also use JabRef software to convert the bib file to an XML file, which can be used in Microsoft Office to create your article. In other words, you can perform a literature survey in a citation database such as Scopus, export the data and use it directly using typesetting software such as LaTeX or Microsoft Office to generate your article, references at the end of the article and you can also use it for your thesis.

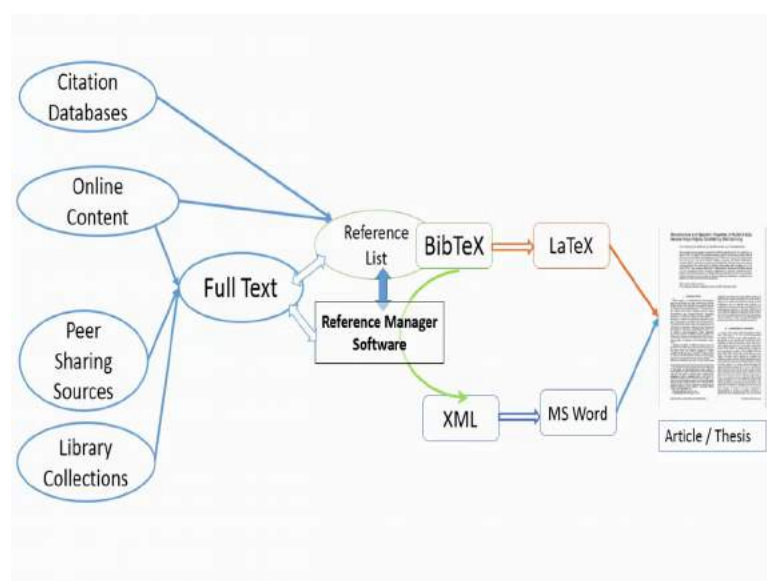
Thank you.



Introduction to Research
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture – 09
Literature Survey: Wrapping up

(Refer Slide Time: 00:01)



So this is the outline of how we went about this module. We have looked at Citation Databases such as Web of Science and Scopus from where we will be getting the list of references. We also have seen several links, where we can find the full text online such as Science Direct and Springer link. We also have given links where peer sharing of journal articles is available such as in Research Gate and academia.edu; and most of the universities also have libraries where hard copy collections are available for full text of articles. So we will be collecting full texts relevant to our research area - both online and hard copy, and we will also be collecting list of references. It's very important to link these two, so that we know which articles we need for our research area.

The reference management software is very useful in getting this done. JabRef is a very useful reference manager software. The native format for JabRef is BibTeX. And if you have your data on the list of references in BibTeX format, then you can use them directly in LaTeX, so that you can add citations to the article or the thesis that you are writing.



And if you prefer to use Microsoft Word to write your article or thesis, then you can use JabRef to convert the literature data from BibTeX to XML, and then use XML format to provide the bibliographic list of references as a bibliographic source in Microsoft Office.

(Refer Slide Time: 01:33)

Importance of References

- Typographical errors due to manual entries to be avoided
- All publications cited in the document are to be listed at the end of the document
- Style of referencing to be customized as per the standard format of University / Journal
- **Sequencing** to be taken care of automatically while editing the document

	Article	Thesis
Chapters	X	✓
(Sub)Sections	✓	✓
Referring Figures	✓	✓
List of Figures	X	✓
Referring Tables	✓	✓
List of Tables	X	✓
Cited Reference List	✓	✓
Size of Document	2 – 20 pages	50 – 200 pages
No. of References	10-30	100-300

LaTeX and Microsoft Word
can take care of these

Adding captions to Figures and
Tables and using them to cross-
refer

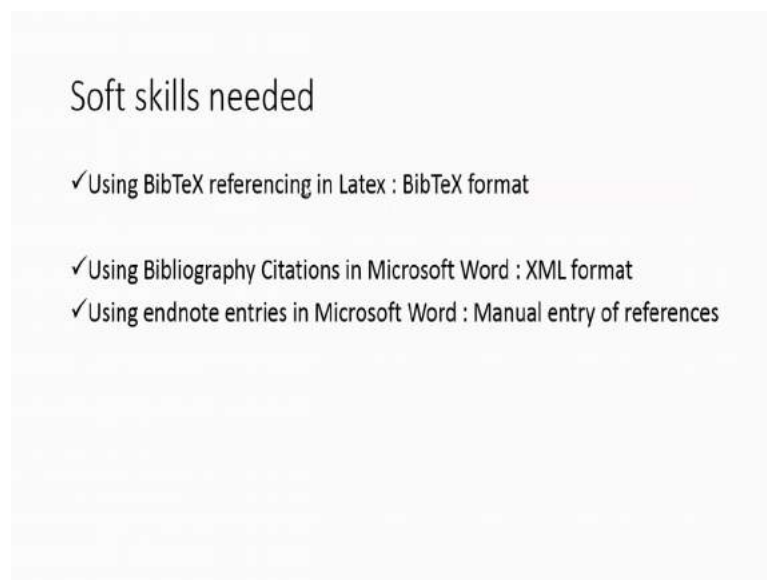
The reason why reference management is very important is evident from this summary. We can see that a journal article that you are going to write or the thesis that you will have to write at the end of your research degree is going to contain large number of sections which will be falling in figures, tables, references, etcetera. And the size of references is going to vary by the size of the document you are going to write. So, from 10 to 30 references may be made in an article, while about 100 or more references will be made in a thesis. So it's very important that when we write these references we do not make any typographical errors due to manual entry. And it's also important to see that every publication that is cited in the document is also listed at the end of the document.

The style of referencing will be customized as per the standard format recommended by the university or the journal where you are going to submit your article. The sequencing of these references is also very important, so that it can be taken care automatically by the software that you are using to write your article. In this sense, LaTeX and Microsoft Word can take care of these aspects very well. The way we manage references can also



be applicable to both figures and tables. However, we will not be looking at them here.

(Refer Slide Time: 02:52)



So you will need to learn couple of soft skills so that you can proceed to write up your document very neatly using the referencing automatically. If you are going to use LaTeX, then you should know about BibTeX, and if you are going to use MS Word then you should know how to use bibliography citations or how to make endnote entries.



(Refer Slide Time: 03:10)

Referencing using BibTeX in LaTeX

- Keep the "*.bib" file containing your reference data in the same folder as your "*.tex" file that contains your article. Ensure the Bibtexkey field is unique for all entries.
- Use the command `\bibliography{}` to point to the reference data.
- Use the command `\bibliographystyle{}` to choose the style of referencing.
- Cite a reference using the command `\cite{bibtexkey}` where appropriate in your tex file.
- Use the sequence "LaTeX → BibTeX → LaTeX → LaTeX" to compile your tex file.

Bibliographic citations using LaTeX is quite straight forward. A summary is given here: you have to keep the bib file containing your reference data in the same folder as your tex file and ensure that the bib file contains the BibTeX field unique for every entry. You can use the command `bibliography` to point the file name containing the bibliographic references. And you can use the command `bibliography style` to choose a style of referencing - alpha numeric, number wise or name plus year etcetera. And you can cite a given reference using the command `cite`, and you can place it at any point in your document; you may have to compile multiple times so that the references are available while compiling. So you will start off by compiling with LaTeX, and then BibTeX, and then again LaTeX couple of times, so that the bibliographic references are all set in the sequence that you have given.

And here, we see a brief demonstration of how to do it in BibTeX plus LaTeX.



(Refer Slide Time: 04:05)

Referencing using Bibliographic Sources in Microsoft Word

- Convert your "*.bib" file to "*.xml" file using JabRef
- Using the menu "References → Manage Sources", open the Source Manager Dialogue. Use the "Browse..." button to select the "*.xml" file you just created.
- Select and copy all the references from left side to the current list by clicking "Copy →" button. Close the dialogue.
- You can now cite a reference at any location in your article by clicking "References → Insert Citation" and choosing the reference.
- At the bottom of the text, use "References → Bibliography → References" to insert a list of cited references.
- Once you are done writing the article, do not forget to open the Source Manager Dialogue again and remove the unused reference from the current list by deleting them.
- Update the Reference List and you are done.

Referencing using bibliographic sources in Word is also quite straight forward. However, it requires that your bibliographic data is available in an XML format. You can use JabRef to convert the data from bib format to XML format. Using the menu on Microsoft Word, References and Manage Sources you can open the Source Manager dialogue and use the Browse button to select the XML file. Select and copy all the references from the left-hand side pane to the right-hand side in the current list by clicking the Copy button, and then, you can close the dialogue. You can now cite a reference at any location in your article by clicking References, Insert Citation, and then, choosing the reference that you want to cite. At the bottom of the text use References, Bibliography, References to insert a list of cited references. Once you are done writing an article do not forget to open the Source Manage dialogue again and remove the unused references from the current list by deleting them. Update the references list and then you are done.

We will see a brief demo of Microsoft Word using bibliographic sources now.



(Refer Slide Time: 05:18)

Referencing using manual endnote entries in Microsoft Word

- At any location in your article, use the menu "References → Insert Endnote" to insert reference details at the end of the article manually.
- Use "References → Cross-reference" to refer to same document again at a later point in your article.
- Use the "Footnote and Endnote" dialogue to change the numbering Format from Roman to Arabic.
- Use the menu "View → Draft", "References → Show Notes", Pick "Endnote Separator" from the drop down menu at the bottom and delete it.
- Use the "Home → Styles" to open the Styles dialogue and click on Style Inspector button to identify the style used for endnote reference. In the Style Inspector dialogue, use the Modify link to change the Font appearance of endnote reference.

Referencing using manual endnote entries in Microsoft Word is quite straight forward. At any location in your article use the menu item References, Insert Endnote to insert a reference detail at the end of the article manually. You have to copy paste the reference detail from some other source, you can do it from JabRef also. Use References, Cross reference to refer to the same document again at a later point in your article. Use a Footnote and Endnote dialogue to change the numbering format from Roman to Arabic. Use a menu View Draft, References, Show Notes; pick Endnote Separator from the drop-down menu at the bottom and delete it, if you do not want to see that line separating your document and the list of references. Use Home, Styles to open the Styles dialogue; click on the Style Inspector button and identify the style used for the endnote reference, and then, in the same dialogue you can use the Modify Link to change the font appearance of the Endnote Reference.

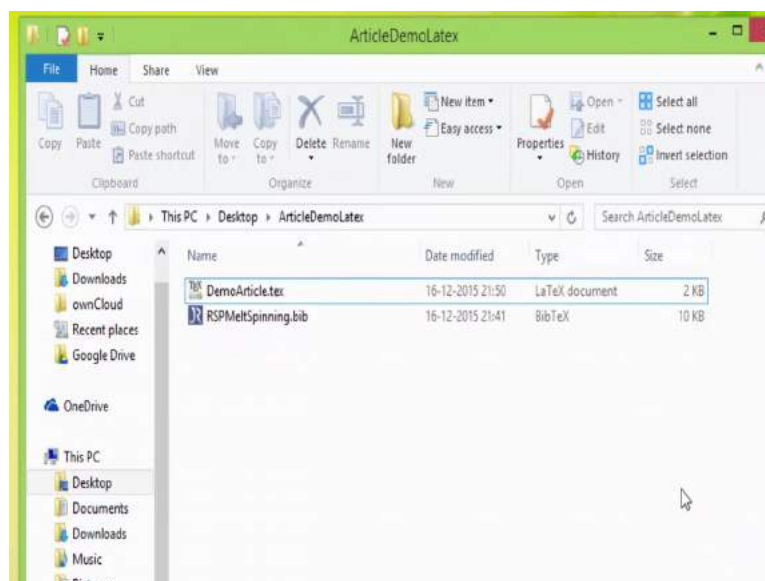
And this is all that is required to list the endnotes at that end of your document in the sequence that you are going to use in your document.



Introduction to Research
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture – 10
Tutorial on using BibTeX with LaTeX
to add Reference to a Document

(Refer Slide Time: 00:01)



Here is a brief demo on how to use referencing in LaTeX. I have now in my folder, ArticleDemoLaTeX, two documents - DemoArticle.tex which contains the article and then RSPMeltSpinning.bib which contains the reference data we have picked up from the literature survey.



(Refer Slide Time: 00:20)

#	Entrytype	Author	Title	Year	Journal	Bibtextkey
1	Article	Aziz	MODEL FOR SOLUTE REDISTRIBUTION DURING RAPID SOLIDIFICATION	1982	Journal of Applied Ph...	Aziz1982
2	Article	He et al.	Weld seam profile detection and feature point extraction for multi-pass rout...	2016	Robotics and Compu...	He2016
3	Article	Hussain et al.	Composition related magnetic properties and coercivity mechanism for ma...	2016	Journal of Magnetism...	Hussain2016
4	Article	Jayaraman et al.	Magnetocaloric effect and refrigerant capacity in melt-spun Gd-Mn alloys	2013	Journal of Magnetism...	Jayaraman2013
5	Article	Karakose et al.	Formation of novel rice-like intermetallic phases and changes in the mech...	2016	Journal of Alloys and ...	Karakose2016
6	Article	Kase and Matsuo	STUDIES ON MELT SPINNING. I. FUNDAMENTAL EQUATIONS ON DYNAM...	1965	Journal of Polymer S...	Kase1965
7	Article	Kase and Matsuo	STUDIES ON MELT SPINNING. 2. STEADY-STATE AND TRANSIENT SOLU...	1967	Journal of Applied Po...	Kase1967
8	Article	Katayama et al.	STRUCTURAL FORMATION DURING MELT SPINNING PROCESS	1968	Kolloid-Zeitschrift an...	Katayama1968
9	Article	Kurz et al.	THEORY OF MICROSTRUCTURAL DEVELOPMENT DURING RAPID SOLU...	1986	Acta Metallurgica	Kurz1986
10	Article	Sato et al.	Magnetic properties of Mn-Bi melt-spun ribbons	2014	Journal of Magnetism...	Sato2014
11	Article	Song et al.	Microstructure and taking hydrogenation performance of Y-doped Mg2Ni...	2014	Journal of Power Sou...	SongLiZhangEAL...
12	Article	Wang et al.	Effect of the volume fraction of the ex-situ reinforced Ta additions on the mic...	2016	Intermetallics	Wang2016

Article (Aziz1982)
 Aziz, M. J.
 MODEL FOR SOLUTE REDISTRIBUTION DURING RAPID SOLIDIFICATION
 Journal of Applied Physics, **1982**, 53, 1158-1168

Let me open jabRef and show you how this bib file would look like. It looks like a spread sheet with the Article Type, Author, Title, Year, Journal and a unique BibTex key that is made available here. In case if BibTex key field is empty, you can use the Generate BibTex Keys button that is here to generate them automatically. You can also edit those keys later on if you want to make them brief for the use. Let us see how the tex file would look like.



(Refer Slide Time: 00:51)

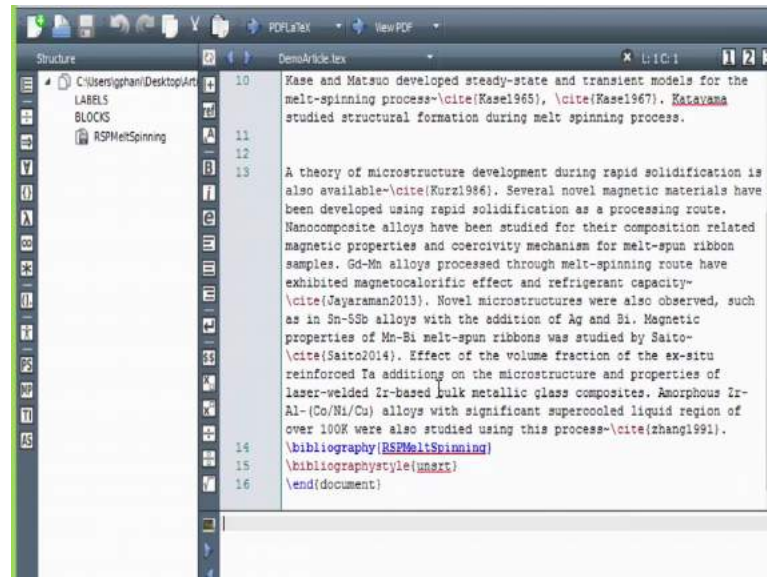
```

1 \documentclass[12pt,a4paper]{article}
2 \usepackage[utf8]{inputenc}
3 \author{G. Phanikumar}
4 \title{A brief review of rapid solidification literature}
5 \begin{document}
6 \maketitle
7
8 Aziz proposed a model for solute redistribution during rapid
9 solidification~\cite{Aziz1982}.
10
11 Hase and Matsuo developed steady-state and transient models for the
12 melt-spinning process~\cite{Hase1965}, \cite{Hase1967}. Karavama
13 studied structural formation during melt spinning process.
14
15 A theory of microstructure development during rapid solidification is
16 also available~\cite{Kurzi1986}. Several novel magnetic materials have
17 been developed using rapid solidification as a processing route.
18 Nanocomposite alloys have been studied for their composition related
19 magnetic properties and coercivity mechanism for melt-spun ribbon
20 samples. Gd-Mn alloys processed through melt-spinning route have
21 exhibited magnetocaloric effect and refrigerant capacity~
22 \cite{Jayaraman2013}. Novel microstructures were also observed, such
23 as in Sn-5Sb alloys with the addition of Ag and Bi. Magnetic
  
```

Here is a sample LaTeX file that I have generated to make a small article based upon this literature that we have picked up from the Internet. It's a very small article; only for demonstration purpose. You can see that the article will have at the top document class, 12 point, A4 paper, Article. Then, we are using a package that would allow a utf8 type of an input, author name is given, title is given, and then, we start the document. Command `\make title` will make the top header of the article.



(Refer Slide Time: 01:25)



And then we have the write up here; and at the bottom, we close the document with `\end document`. Here are the two commands that are most important: `\bibliography` and then the name of the bib file without the extension, and then, the `\bibliography style`, and then, a keyword that tells you in which way you want to cite the references. `unsrt` is a keyword which will tell you that you can sort these by numbers. The way you refer to the article is by inserting `\cite` and then the BibTeX key that is highlighted here, which is then going to be used for referencing.



(Refer Slide Time: 01:57)

```
\title{A brief review of rapid solidification literature}
\begin{document}
\maketitle

Aziz proposed a model for solute redistribution during rapid
solidification~\cite{Aziz1982}.

Kase and Matsuo developed steady-state and transient models for the
melt-spinning process~\cite{Kase1965}, \cite{Kase1967}. Katayama
studied structural formation during melt spinning process.

A theory of microstructure development during rapid solidification is
also available~\cite{Kurzi1986}. Several novel magnetic materials have
been developed using rapid solidification as a processing route.
Nanocomposite alloys have been studied for their composition related
magnetic properties and coercivity mechanism for melt-spun ribbon
samples. Gd-Mn alloys processed through melt-spinning route have
exhibited magnetocaloric effect and refrigerant capacity~
\cite{Jayaraman2013}. Novel microstructures were also observed, such
as in Sn-Sb alloys with the addition of Ag and Bi. Magnetic
```

You can insert these at any location that you like; only make sure that this key is available in the bib file.

(Refer Slide Time: 02:11)

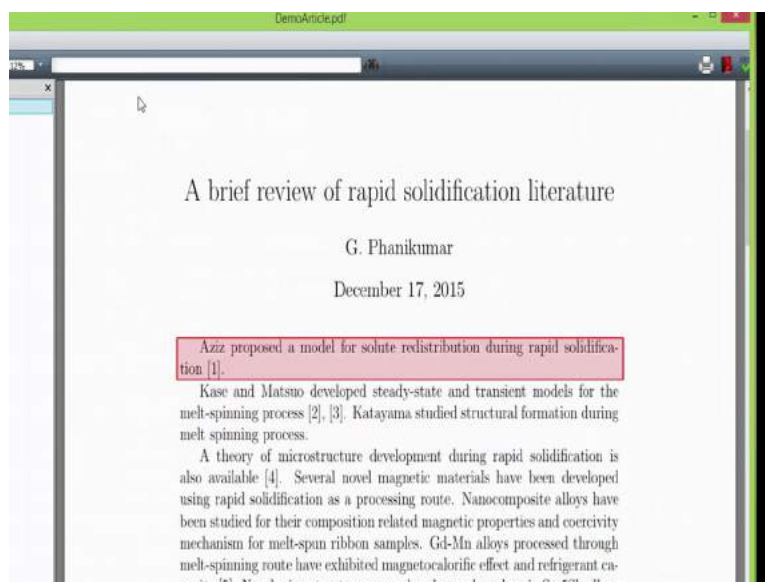
```
1 \documentclass[12pt,a4paper]{article}
2 \usepackage[utf8]{inputenc}
3 \author{G. Phanikumar}
4 \title{A brief review of rapid solidification literature}
5 \begin{document}
6 \maketitle
7
8 Aziz proposed a model for solute redistribution during rapid
9 solidification~\cite{Aziz1982}.
10
11 Kase and Matsuo developed steady-state and transient models for the
12 melt-spinning process~\cite{Kase1965}, \cite{Kase1967}. Katayama
13 studied structural formation during melt spinning process.
```

And then the way we compile is, first we should do one round of LaTeX, and then, it will complain perhaps that the citations are not available; and then, we can run BibTeX, so



that those citations are available. Then we will run LaTeX couple of times, and then, you can now see that there is no more complaint. And then, we will run PDF LaTeX to generate a PDF directly.

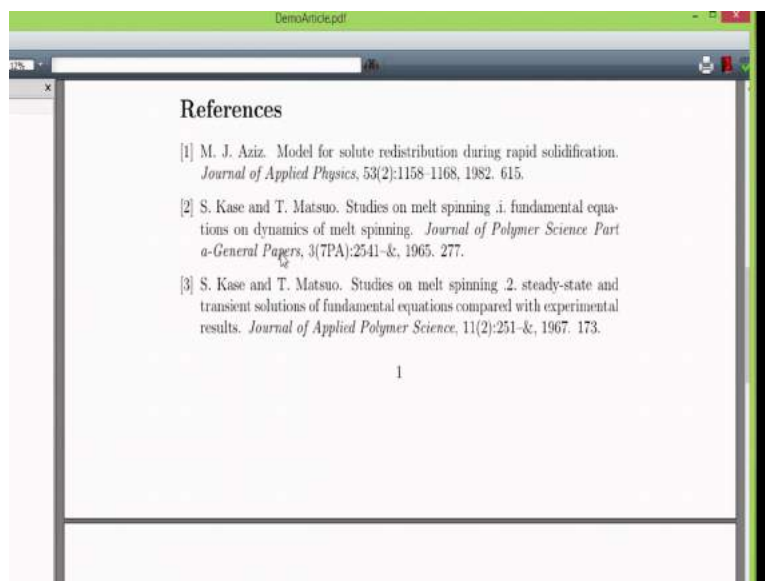
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Then we can click on View PDF. Now we have this article here. You can see that the references are here 1, 2, 3, 4 etcetera.

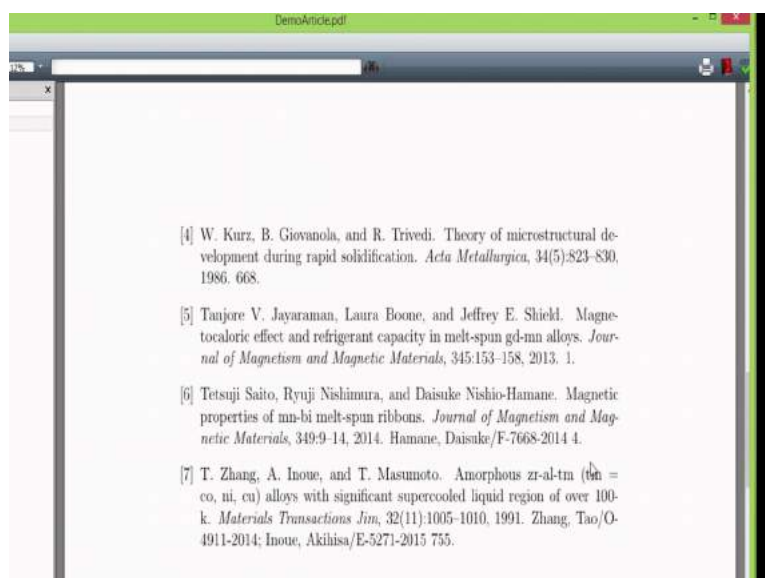


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In that, in the bottom automatically the references are listed with complete details in the same sequence as they have appeared in the way we have written.

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Let me just modify this slightly to see how it will affect the sequencing.



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```

5 \begin{document}
6 \maketitle
7
8 Aziz proposed a model for solute redistribution during rapid
solidification-\cite{Aziz1982}.
9
10 Kase and Matsuo developed steady-state and transient models for the
melt-spinning process-\cite{Kase1965}, \cite{Kase1967}. Katayama
studied structural formation during melt spinning process.
11
12
13 A theory of microstructure development during rapid solidification is
also available-\cite{Kurz1986}. Several novel magnetic materials have
been developed using rapid solidification as a processing route.
Nanocomposite alloys have been studied for their composition related
magnetic properties and coercivity mechanism for melt-spun ribbon
samples. Gd-Mn alloys processed through melt-spinning route have
exhibited magnetocaloric effect and refrigerant capacity-
\cite{Jayaraman2013}. Novel microstructures were also observed, such
as in Sn-5Sb alloys with the addition of Sn and Bi. Magnetic

```

I am just removing one line here and locating it somewhere else; then I am going to compile that again. Once through LaTeX, and then BibTeX, and then LaTeX - couple of times, and then PDF LaTeX. Now, if I view, you can see that the reference for Aziz paper has now renamed as 3 as per the sequence that we have referred and the same will be reflected also at the bottom of the text.

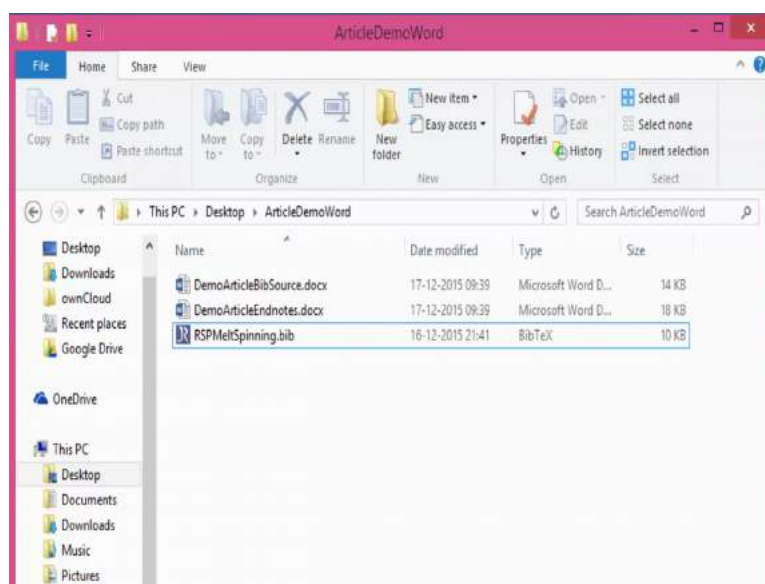
So referencing this way using bib file in LaTeX is quite straight forward. At any point if you want add more references, you could add them manually here in JabRef or you can do a literature survey, and then, merge those items into the bib file. You can use a software like MiKTeX, that is freely available in the Internet to compile LaTeX files on your desktop, and you can install TexMaker which is a very good editor that will let you write these documents, and compile on the fly, and then show you how the document would look like. It is as simple as that.



Introduction to Research
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture – 11
Tutorial on using Microsoft Word
with Bibliographic Sources

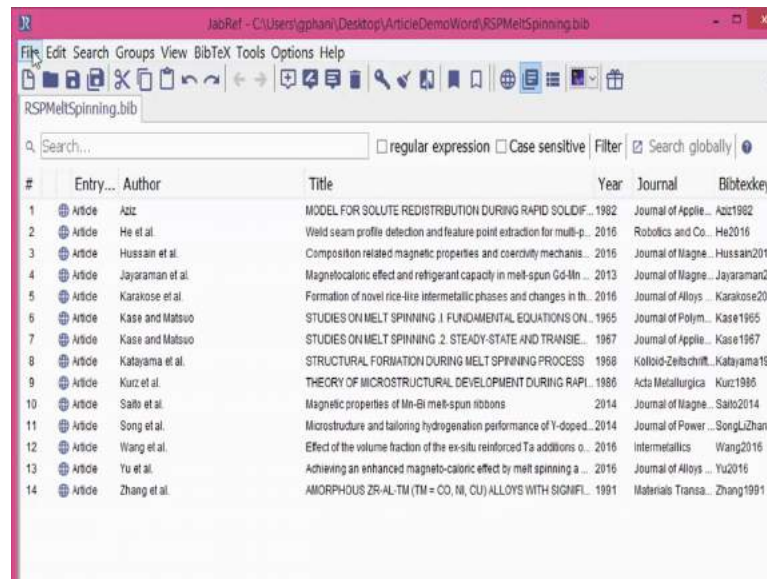
(Refer Slide Time: 00:01)



Here is a brief demo on how to use Microsoft Word with bibliographic sources to add references to your text. In my folder, I have a doc file which contains an article which I have written to demonstrate this purpose, and I have a bib file which contains references we have collected using the literature survey. Let us open the bib file to see how the references look like.

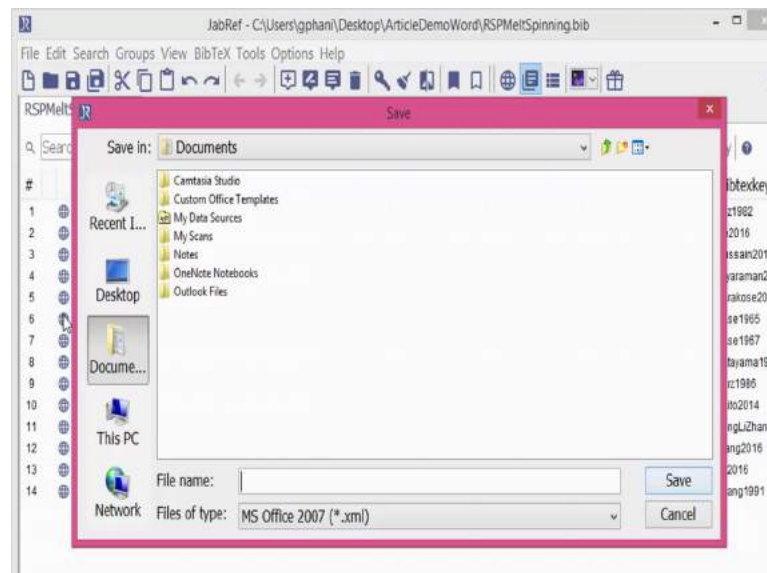


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And the bib file is here. We can now convert the format of this file from a BibTeX to XML by going to File, Export.

(Refer Slide Time: 00:40)

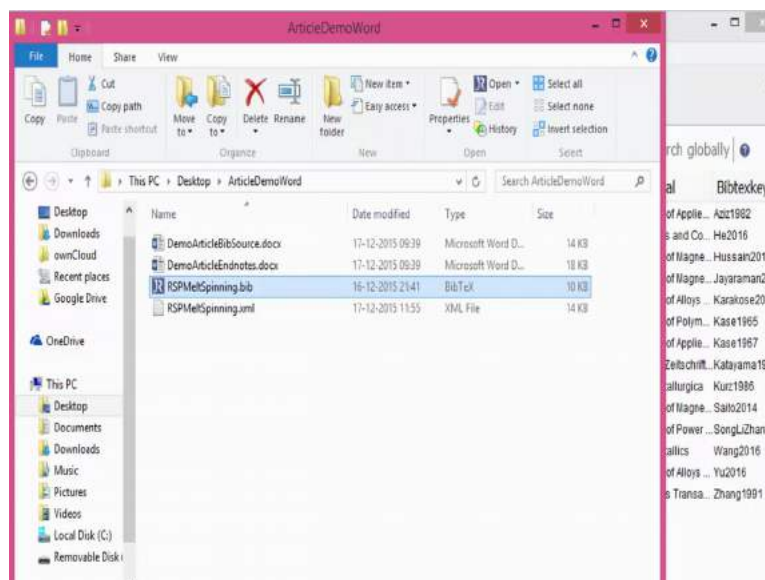


And then choose Microsoft 2007 XML as a format which is compatible with the later versions of Microsoft Office also. And then, where do we write that file? You should



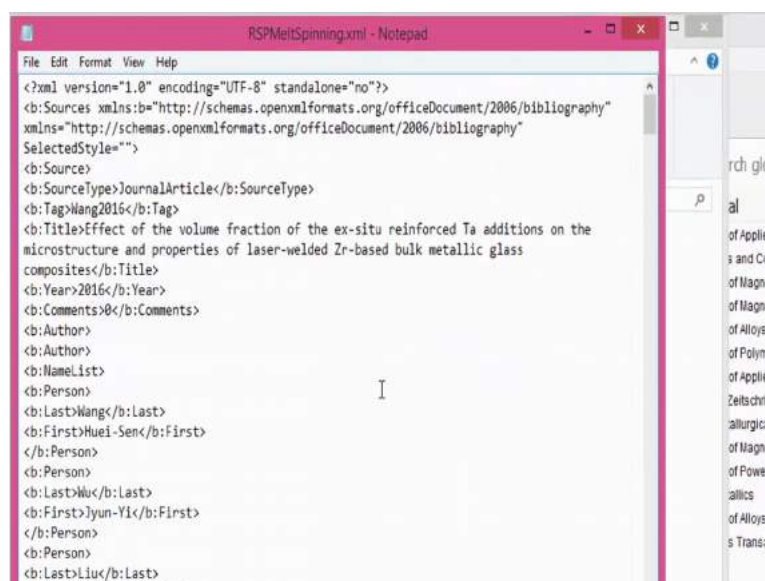
write it at a location that we are able to identify, which I will choose in the same file as I have picked up my document.

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So, I can now see that this is an XML file that has come in the same folder. You can open it with Notepad to see how it looks like.

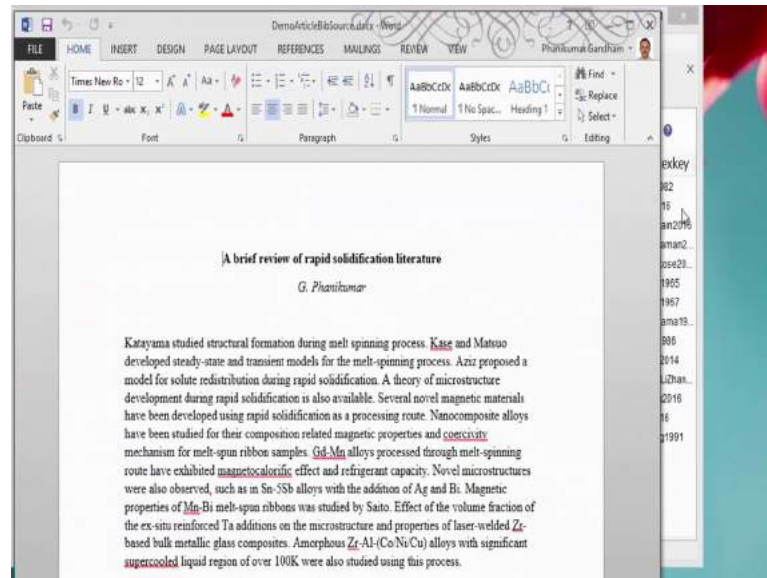
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It is basically pure text file, this may be not amenable for editing directly, but it can be opened in Microsoft Office as a bibliographic source.

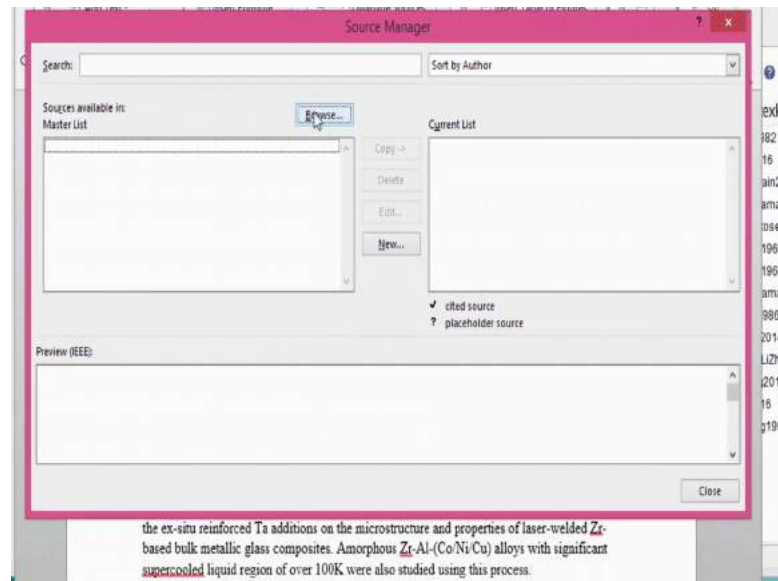
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Let us now open the Word file that we have written. Here is an article; the title is here, the author name, and then some write up. We now need to add the bibliographic resources in this. So what we now need to do is, initially we go to the References tab, Manage Sources, and use the Browse button to pick up the file.

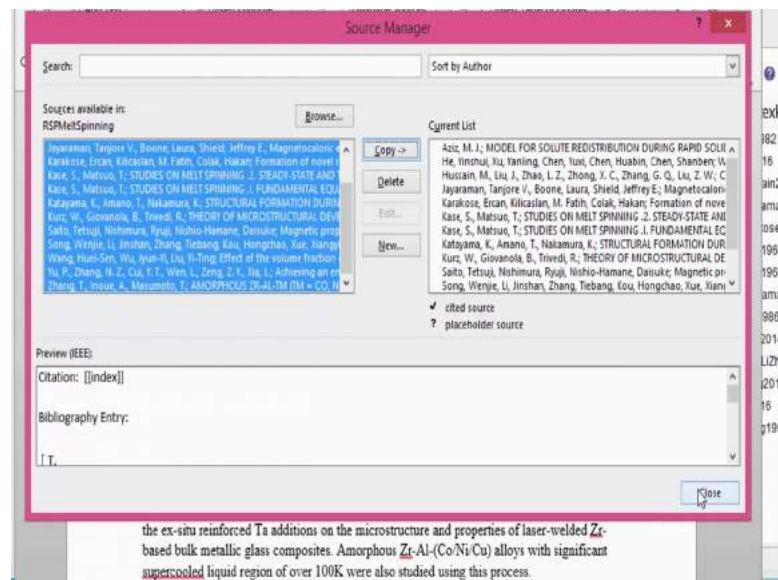


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And the file is to be located here, and then, select the XML file, click OK.

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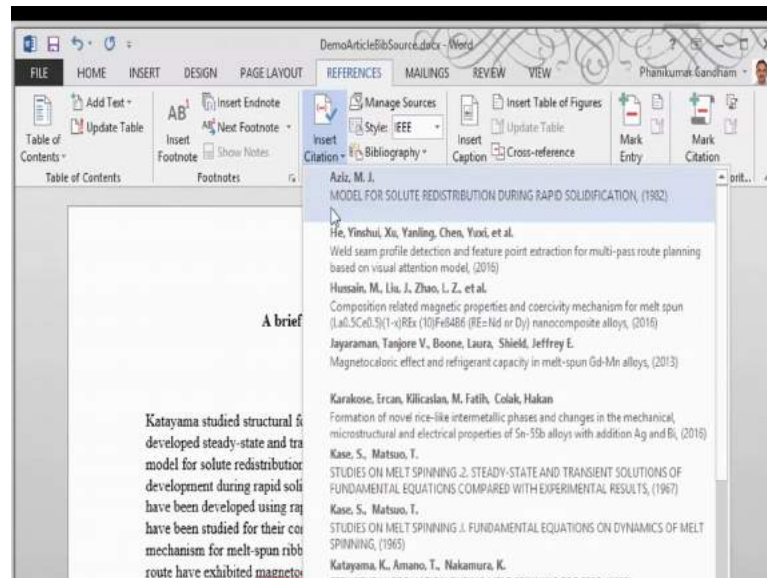


And then, we now see all the references have come on the left-hand side pane. We use Control + Arrow key to select all of them, and then copy them to the current list, and then close the dialog. At this point, we can now start inserting the references, we can just



locate where we want to insert, and then Insert Citation.

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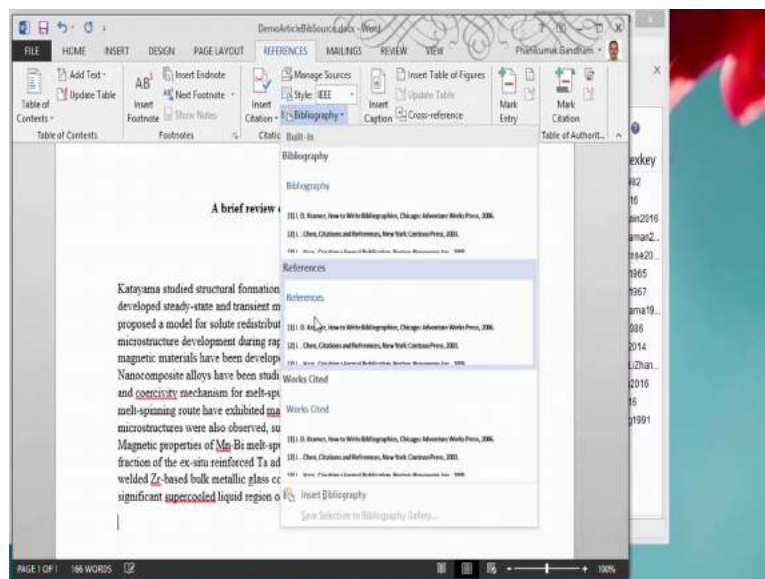


And we can do that at any location we wish. So, we will do that one comma, ok and then, as is reference also we would insert here.

Ok, so we can do that as we go along, as many references as we have in our source. And how do we now get these references listed at the bottom? It is quite simple; go to the end of the document, and then, insert the bibliography and references. And we see how the document references have come.



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Now, before we close, what we need to ensure is that unused references are removed. So, we can do that by going to the sources and on the right-hand pan you see that there's a tick mark for all the references that are used. Unused references can then be deleted. And I would just do that to illustrate, and come back, and click on the Title and Update, and you would see that only four references are present which are been referred here. And we can now copy-paste the text around, and you would see that the reference numbers would change appropriately, and to update if you just click on this Update Citations. And you would see that the numbers have changed here to reflect the sequence of referencing. And the document is now ready to be wound up. It is as simple as this.

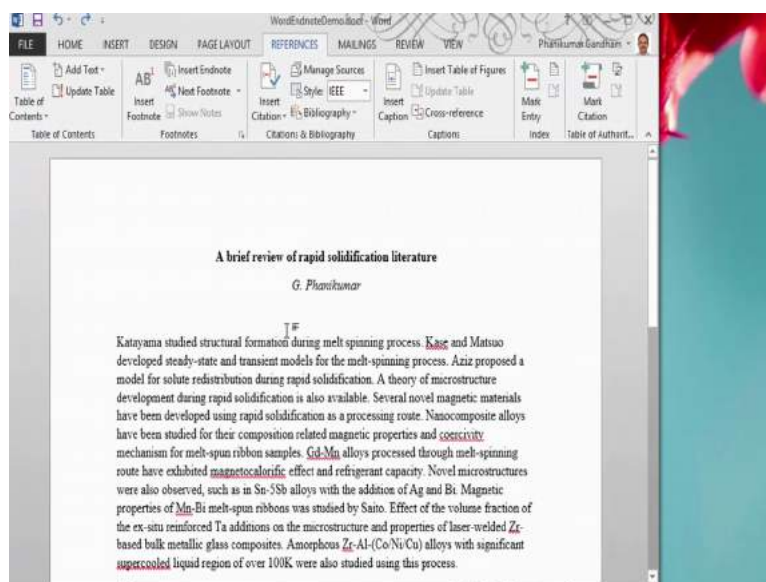


Introduction to Research
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture – 12
Tutorial on using End Note Entries to
Reference Data to a Microsoft Word Document

Here is a brief demo on how to use Endnote Entries to add reference data to Microsoft Word document.

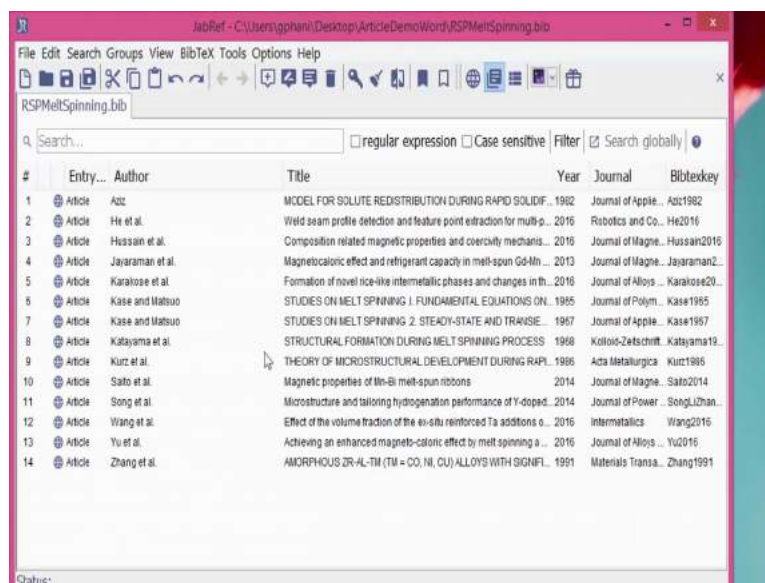
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I have a sample Word document here; the title, an author name, and a brief write up only for illustration purpose. We can add the reference details at the end of the sentence where we want to make the citation; we can use square bracket to enclose the endnote reference, and we can go to the Reference tab and Insert, Endnote. At the bottom of the page we have an ability to type out the reference. We can either type it out manually or we could copy-paste it from some other location where we have the reference information; in our case, we have them in JabRef.

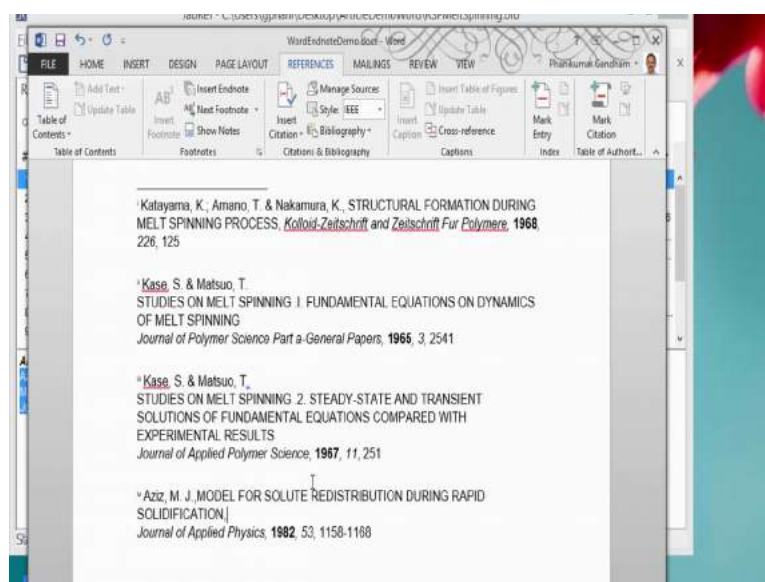


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We will do that by copying it from JabRef by copy-paste and I will do that now.

(Refer Slide Time: 00:54)

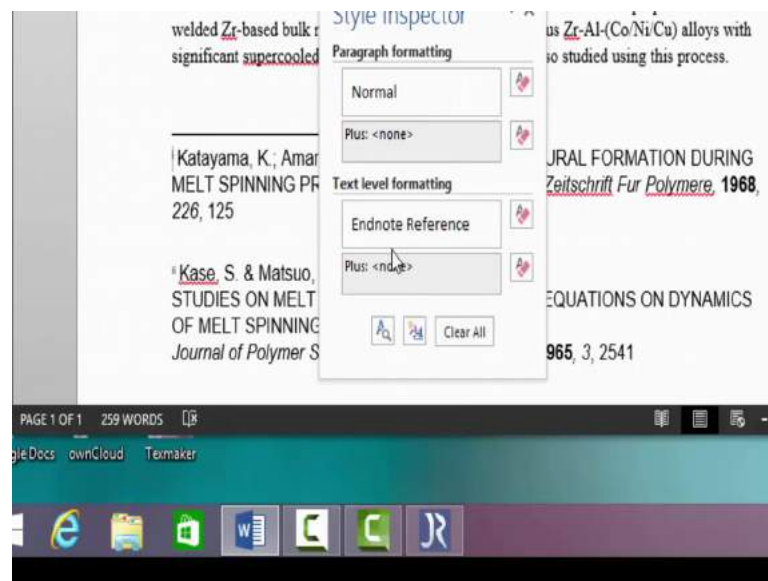


We could format it to remove any empty spaces as we wish to have. And you could have the same process to add multiple references. So, we add one more endnote, and at the bottom we have the text, and we would add the second reference, and again, may be one



more. Ok so we remove some page breaks to make the references to look a little neat. And now, you notice that the references have all come nicely at the bottom of the document, but then, the Endnote Reference is coming in the Roman letter but we could change it to the Arabic format by using what is called as the Style Inspector. For that part, what we need do is go to Home, and go to Styles, and then, in the center of the bottom you have Style Inspector - click on that and highlight the Endnote Reference.

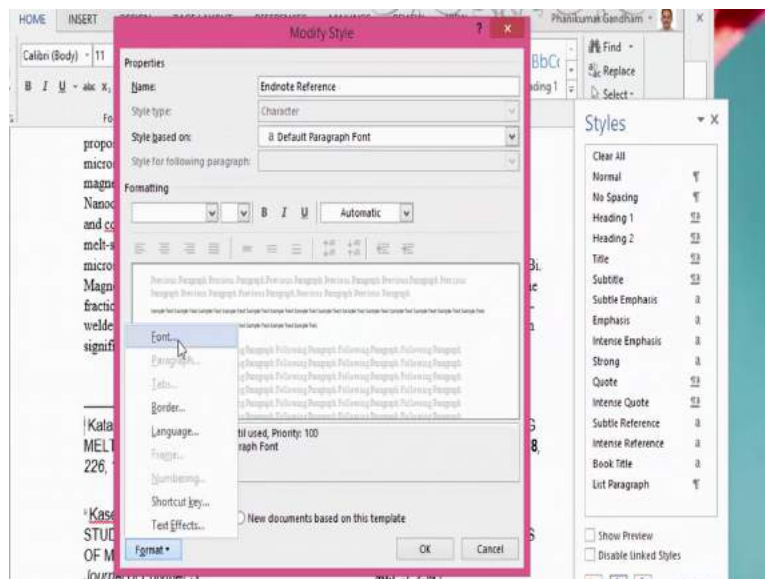
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And you would see that it has detected the Endnote Reference format. You open the dialog here and modify.

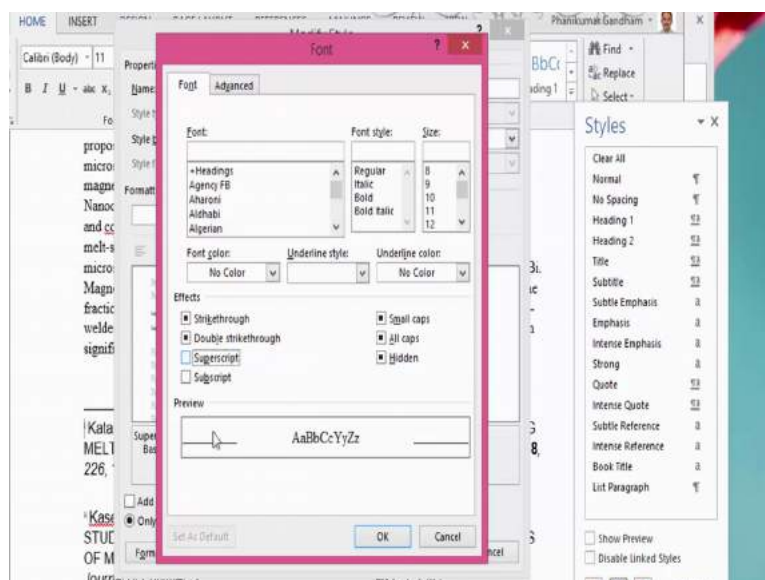


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And we can modify now the Font.

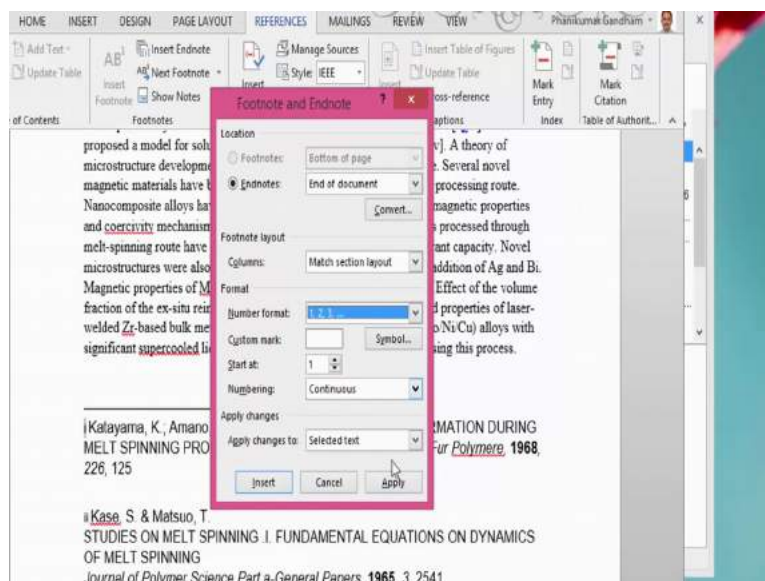
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As you can see there is a Super script written here, we remove that here, and say OK, and then OK here, and we then also go to References and Footnotes.



(Refer Slide Time: 02:40)



The format here and change it to 1, 2, 3, 4 and Apply. Now, you see that the references are coming quite nice in sequential number, as we would normally expect in most of the documents.

Now, there is also a line that comes here which you may not like, so we could change that also as follows. For that what we now need to do is go to View, and click on Draft; then go to References tab, then Show Notes; at the bottom, we then select Endnotes Separator and now we have the line. We simply remove that line and then Close. Now, you can see that we have got the references lined up in the same sequence as we have referred, and the line is gone, and we have seen that the four references are also appearing as hyperlinks; you can see the Endnote Reference appearing whenever we hover the mouse over the Endnote Reference.

We can now see that as we copy paste the text around while editing, then the reference numbers also should change. I have just now moved them around and you can see that the numbers have already changed.

Ok so, adding references in a Word document, so that the sequencing is taken care is as simple as that. However, using Endnotes would mean that you would be adding the



bibliographic references manually by typing it out or copy pasting from some other source. It is as simple as that.



Introduction to Research
Prof. Prathap Haridoss
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture – 13
Experimental Skills

Hello, today's class, we will look at experimental skills as part of our short course on Introduction to Research.

As you are probably aware from, you know, discussion with people around you, in general, research could be, you know, either theoretical - broadly speaking - could be theoretical or it could be experimental. Most often, it gets, you know, separated into these two categories. So, if you want to see, we have basically - you have theory and you have experiments. So, broadly your work depending on wherever you are working on, the kind of area that you are working on, could be of one of these two kinds. There are people who do fair bit of, you know, the combination of these and so on. But, still your activity could be one or the other, most of the cases.

It is important to understand what is the relationship between these two and also to see, as part of today's class, we will look at skills that are associated with the second aspect show here, experiments. But, I would like to point out what is the relationship between them, so that you are aware of what is the broader picture.

So, basically, we have, you know, phenomena in nature and we are trying to investigate this phenomena. So, we are trying to understand why those phenomena are the way they are? What options you have? Can you push the phenomena in one direction or the other direction and so on. In general, to understand the phenomenon we typically run experiments, we run controlled experiments, which means you make sure that certain set of conditions are constant, and then you change one quantity, and then you try to figure out what is the impact of that change on the other quantities that are present in the system. So, those are experiments that we do and we collect a lot of data.



In theory we are writing equations, we are trying to simulate what nature is trying to do. We are trying to say that, you know, maybe nature follows some set rules, and if it does follow those set of rules, what can you expect when you change a particular quantity, what can you expect in terms of that - the impact of that change on other quantities that are present in the system right. So, here you are doing some calculation to figure out how one change - change in one quantity - impacts another quantity. Here you are running some experiments to understand what is the change - impact of change of one quantity on another quantity. So, in both these cases, you are trying to do may be something similar.

Now, in general, the idea is that the experiments are supreme. Meaning, in general, in theory you are trying to cut out various aspects, and then you say that, you know, maybe the system follows certain set of rules. So, on that basis you make a prediction. So, what theory does is often it makes a prediction. It is the experiment which shows whether or not that prediction is correct. So, therefore, an experiment is very important. Experiments are very important because they are the ones that really validate something that the theory says right. So, it does not matter, how sophisticated the theory looks, how interesting the theory looks, how fascinating the equations are that have come together and so on. At the end of it all, if it makes predictions that are not borne out by experiments, then the theory is not considered good; it is considered, at the very least, it is considered incomplete. It means, either some major aspect that needs to be further incorporated into the theory before it can be considered correct. So, therefore, experiments in the hierarchy of, you know, science it is reasonably fair to say that experiments are supreme. You have to show by experiment that something has actually happening and that is when it is believed.

Now, so that's the basic, you know, relationship between these two. Now, therefore, it is also very important that you run your experiment correctly okay. So, if you run the experiment incorrectly or you have done it in an uncontrolled way, then, naturally, your results will not be appropriate, will not represent the actual phenomenon right. So, it's important to, when you say that experiments are supreme and all that, the underlying statement they have, that the underlying assumption they have, is that you have run the experiment correctly. So, that is a very important aspect associated with experiments.



So, today, that is the reason why we have going to spend some time on this concept or this idea of experimental skills. Because, experimental skills are the kinds of skills that we should have to ensure that our experiments are running correctly, for us to feel confident that the data that we have obtained from the experiments are correct; and therefore, we can now, you know, with great confidence we can say that if the theory does not match that experimental data, then there is some issue with the theory. If you have run the experiment incorrectly, then obviously, this is not going to hold true. So, in that case you cannot confidently say that the theory is not correct or the theory is incomplete. So, it is very important for you to run your experiments correctly and if you are an experimentalist, that is something that will have to put up with all the time. You should feel ready to indicate in what ways you have run your experiment correctly, and your experiment should be open to scrutiny - people should be able to ask you lot of questions on how you ran the experiment. You should be able to defend how you ran your experiments. And learning experimental skills is a very important aspect associated with that right.

So, we have to run our experiments carefully and correctly. So, today that is what we are going to look at - how to do this and, or at least, what I am going to do is, you know, if you take the idea of experiments it is a very vast world out there. There are lots of different experiments you can do to probe different phenomena, to demonstrate different phenomena, and so on. In most cases, there are specific experimental quantities or experimental parameters that we tend to control or we tend to measure. And then, based on that we investigate wide range of phenomena. So, I am going to take a few common parameters that we tend to control or measure and so on. And then, highlight to you, you know, what are the places where you can make an error, and therefore, you need to be cautious about, when you... you know... or double check or cross check what measurement you are making and how you would go about doing that. So, this is I will small... I mean process that we will go through with few different quantities that I feel, we commonly encounter.

And naturally, if there are other, they are bound to be lot of other quantities which you will measure in your experiments, which you should follow a similar process. So, it is not the intent of this class to, you know, exhaustively look at all possible quantities that



you can measure and, you know, figure out what all ways you can get it right or what all ways you can get it wrong. But, representatively, I am going to pick a few, I am going to highlight, where you can go wrong, and then, I am also going to highlight, how you can, you know, correct for it; how you can cross examine it; how you can make it better. And the same philosophy is what you should carry with you to any other experiment that you do where you may be dealing with quantities that are not necessarily shown to you today right.

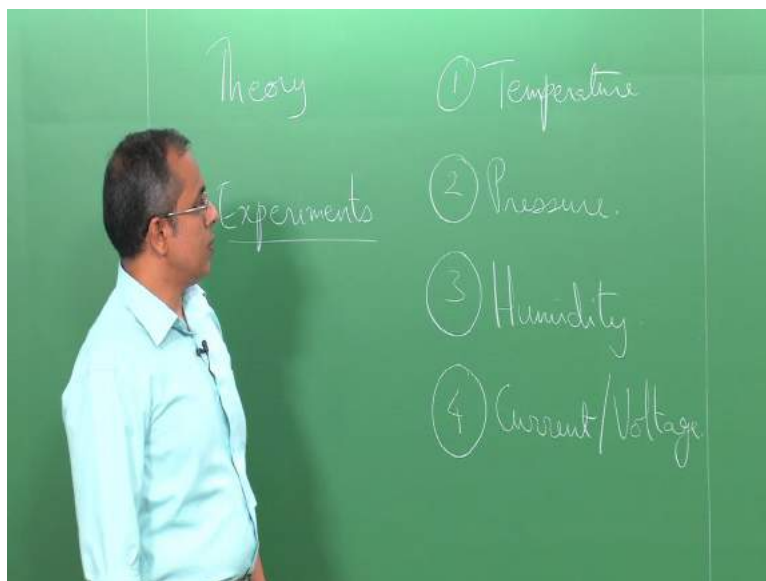
So, we will take some common quantities and we will start with that. Ones that we often encounter are temperature, there is pressure, humidity, and then, electrical quantities such as current or voltage. So, these are common quantities that you are likely to see in many experiments. You are measuring current, you are measuring voltage, you are applying a current, measuring a voltage, applying a voltage, measuring current, lot of things we do.

There are experiments where we worry about the humidity in the system and we want to quantify it or we want to ensure that it is a certain value or we want to claim that it is a certain value, so that's something we will look at. There is pressure; you have pressure gauges and so on to tell you what is the pressure in the system. I will look a little less on pressure, I will spend more time on temperature, humidity, and current and voltage.

Some of the basic approaches we use will be similar to what you will have to, you know, utilize when you look at pressure, what we will look at for temperature. So, conceptually similar; of course, the gauge used and so on going to be very different right.



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So, we will start with temperature. So now, many experiments that we do, commonly we will start by saying that, you know, the experimental setup was set some temperature. So, this is common statement that we make right. So, how do we know this right ? So, this is something we have to be very careful about it. So, for example, we may be running temperature experiments at say, 200 degrees C; somebody else may run an experiment at 500 degrees C. Or you may have a series of experiments, one of which runs at 200 C, another at 500 degrees C, another at 800 degrees C; at every 300 C increments you may have an experiment. So, you can have experiments over a set of temperatures right.

So, it is important that...so, when you present the data, you are going to say that, this data I have obtained at 200 degrees C; then I changed the temperature of the system to 500 degrees C and I obtained some other data; then I changed it to 800 degrees C, 1100 degrees C; and then you do a trend based on temperature. So, and then, you may get some activation energy values, lot of different things you may get out of this kind of data. So, it very important that your experiment actually when you are claiming it as a 200 degrees C, the experiment should actually be at 200 degrees C. So, that is the important aspect that I want to highlight.

So, one of the common sources of error in many our experiments, where we are dealing



with temperature is that we simply look at a display, which will be there on your furnace or your oven - and hot air oven - and it will simply say 200 degrees C, and we assume on faith that is 200 C, and we proceed; many, many first time experimenters do this.

Now, there is a lot of scope of error, scope for error, when you just look at that display, take that value and proceed. And that is something that I want highlight here, because this is... temperature is one of the most common quantities that we encounter in experimental setups. So, therefore, we need to be careful. So, the point that we have to keep in mind is that any experimental setup you buy, if it's a hot air oven or it is a furnace - let us say, it is a tube furnace, could be a box furnace or a tube furnace, in all these cases they have a sensor, which measures temperature. And that is how that sensor, that is how you see a display somewhere outside that says that it is 200 degrees C or whatever. It is very important for you to understand, what is the location of that sensor? That is very important; because you can have a situation where the sensor is located at one location, your sample is located at another location, and therefore, the temperature that the sensor is reporting to you is not that temperature at which your sample is sitting. And, this is an experimental thing that you have to be aware of right.

So, for example, let me just say that I have a box furnace, I am sorry, a tube furnace. So, let's say there is a tube here, and this is inside some, you know, structure that is there which envelopes it. So, this is a tube and inside this you have a boat - a ceramic boat right. So, you have a structure which has all the heating rods, heating elements are there on this, all are around this tube; in the middle there is this tube, it is a ceramic tube; and in that there is a ceramic boat. In this ceramic boat, you have some sample; some sample is sitting here; the sample that you are testing is sitting in this ceramic boat okay. So, now, let's say you are trying to run this at 800 degrees C. You want to say that your sample was at 800 degrees C for 1 hour; that is, let's say, that's the experiment that you are trying to do; that you are going to take the sample, keep it at 800 degrees C for 1 hour, under some atmosphere. So, those are two things you will specify: the temperature and the atmosphere.

So, let's say it is air, in which case you can keep it open or if it is some other gas, let's say it is under nitrogen atmosphere or argon atmosphere, then you have to ensure that



there is an inlet that, you know, air cannot freely get in here, there is a controlled way to introduce argon or nitrogen, and there is a control exit for argon or nitrogen, and that you will have a bubbler which will then check. So, for example, you may have something like this, this would be then closed except for this inlet, and you will have a tube coming out which will then go to a bubbler, and you will see bubbles of gas coming out. So, this is the way you can ensure that there is actually gas coming in, there is gas going out, and since there is an inlet and there is an outlet, there is no build up of pressure, just free flow of gas. Even with air you can ensure that you have an air bottle, which is just supplying air, so that it is not... it's some controlled flow of air, and then, you have a gas coming out right. So, this is the typical set up. So, you have heating elements.

So, what normally happens is that the manufacturer of this furnace will now provide you with a thermocouple which would be somewhere here, which will enter from somewhere here, and it will come in contact with the tube here. And from this you will have a display. And let us say, this is now showing you 800 degrees C. So, experimentally, now you have got a set up where you have got a tube, a tube furnace, you have got a boat, you have got a sample, and you have a display, you have got some controller here, you set the temperature to be 800 degrees C, and it shows your display of 800 degrees C right. So, it looks like your experiment is all running fine. Now, the issue is this - when you have... especially when you have gas flowing, some small flow rate of gas and so on, this area is actually being cooled. So, and also, it may be... this region is getting slowly cooled by the flow of gas plus the heaters are at specific locations. So, there are some heating elements here, there are some heating elements here, and so on. And therefore, you may even have a coil around the furnace, around the tube, and so on.

So, the point is not every region in this tube is at 800 degrees C. So, that's a very important piece of information you should be aware of. You are seeing a display of 800 degrees C, but you should be very cautious in understanding what it means, because the entire furnace is not at 800 degrees C. What it specifically implies is that the region that the thermocouple is accessing, that location is at 800 degrees C okay. And that location, if you look at this diagram, if this is the layout of the set up, that location is here, your sample is sitting here, and this could easily be of the order of say 2 or 3 inches, may be about 5 to 10 centimeters; 5 to 10 centimeters can be there between the location of your



thermocouple and the location of your sample.

And, in this region you could have gas flowing, you could have lot of different things happening, and therefore, even though this region is marked as 800 degrees C - that your thermocouple is registering 800 degrees C, your actual sample may be sitting at 750-760 degrees C; that's not a big difference in temperature in terms of, you know, that the cooling rate might have caused that, could easily caused that 50 degrees difference. And even...so, vertically there can be a difference in temperature; horizontally also there is a difference in temperature. So, your boat may also not be positioned exactly below this thermocouple. It may be off a little bit; it may be a couple of centimeters ahead or couple centimeters below, behind, because it's a long tube.

Based on the location of your boat, which then contains the sample, the location of your sample may not be at 800 degrees C; it could be easily be at 750 degrees C and that is a big difference, because you have an experiment were you are looking at 200, 500, 800 and 1100, and if each them is off by about 50 degrees C, 70 degrees C, 80 degrees C or even 100 degrees C, then the data is completely wrong; I mean or at least it is not the data that you think it is; it is correct data but it is not what you think it is. It is not data at 800 degrees C, that is data at 750 degrees C; what you thought was 500 degrees C, may not be 500, it may be 475 degrees C. And, the difference may not even be the same, it may be 25 degrees difference here, it may be 50 degrees difference here, it may be 100 degrees difference here. So, you are getting data at 475, 750 and 1000, when you think you are actually getting at 500, 800 and 1100.

So, as part of experimental skills you should be aware of this. That any experiment you set up, where you are trying to measure something, the manner of measuring it you have to really think about it, you have to examine how is that instrument making that measurement, and satisfy yourself that what it is displaying - because all these electronic devices, these days, display something. So, you should satisfy yourself that what it is displaying is correct, is what you think it is or what you feel it should be at the location where your sample is located right. So, in this case, a very common thing that we do, which is measuring temperature, in so many experiments we measured temperature or we try to control temperature, measure temperature, set temperature - it's a very



common experimental thing that we do. There is so much room for error. There is so much room for error. So, you have to be aware of this.

Now, this is not an insurmountable problem. This is a problem that you should just be aware of because you can easily handle it, if you know that it is... this is something that you have to address, something that you have to carefully note down, then you can easily address this. The simplest way to address this is to add another thermocouple. So, for example, this is a thermocouple. It is just a thermocouple wire that you see here, it is a K-type thermocouple. So, you can get this in all your labs, in experimental facilities you can always obtain it; most likely it is there in your lab; you can always obtain it. Its a long wire, thermocouple wire. So, all you have to do is, you have to insert it. You have to insert it; you have to make a provision such that it gets inserted and it comes to a halt just above your sample. So, this is something you can visually verify before you start the experiment. You can bring it such that, it is now, you know, barely half a centimeter to or even less above your sample, actual sample. It will come on top of the boat and it will come directly sit there right.

So, this is how you are now looking at the temperature. And this can then be connected to a different display right. So, what we have done now is in addition to whatever display was there on your experimental set up, we have added one more thermocouple, which I am now drawing here. So, we will just say that it is like this and you make sure that it is positioned here. So, this is another thermocouple. I have just drawn it like this because I have written something here, but basically it will be a straight wire, it will come out there, and this then you take to another display. So, as long as your thermocouple is good and there is no problem with it, etcetera, and that you calibrate again some standard setup, if your thermocouple is working right.

Now, this is now giving you the temperature barely millimeters above your sample. So, barely millimeters above your sample. So, this may now show you 760 degrees C. So, now, you know, that your sample is actually sitting at 760 degrees C even though the furnace is showing 800 degrees C. So, you simply have to... So, this is a problem; this is now solved, all you have got to do is you have to raise the set point of that furnace there, you may have to go to... it may not be exactly linear. You may have to go say let just



assume it is linear for the moment, you may have to go to 840 degrees C there. So, that your sample start seeing 800 degrees C.

So, now you bump this up to 840 degrees C. So, this will show 840, and that means, the furnace is trying to set it at 840 degrees C and the furnace is measuring 840 degrees C at that location. But, your actual sample this would have reached 800 degrees C and you now know that your sample is sitting at 800 degrees C right. So, this is a simple parameter, simple experimental quantity that we commonly... when I say simple it means very common, rather than simple I should say very common experimental quantity that we all measure, that we all encounter in so many experiments, and there is so much room for error in it, and there is also simple ways to make sure, there are simple ways to make sure that the temperature you are measuring is the temperature that you think it is. Or the temperature you are measuring is a correct measure of the temperature at that location of your sample right.

So, now if I have this additional thing thrown in here, I can actually run - confidently run - my experiment at 200 degrees C, 500 degrees C, 800 degrees C, 1100 degrees C. The margin for error is very minimal now right, and with much greater confidence I can actually see the trend in temperature, a trend in the behavior of some property of that sample as a function of temperature okay. So, I just want to highlight that therefore, like you have just seen now, I am going to show you couple more instances where these are, I mean things that you can think of.

When you see an experiment, when you run an experiment, you start running so when you go to a lab you have learnt some experiments, you learn from your colleagues, you learn from your, you know, seniors in the lab, etcetera, you start running some experiments to familiarize yourself with that experimental setup.

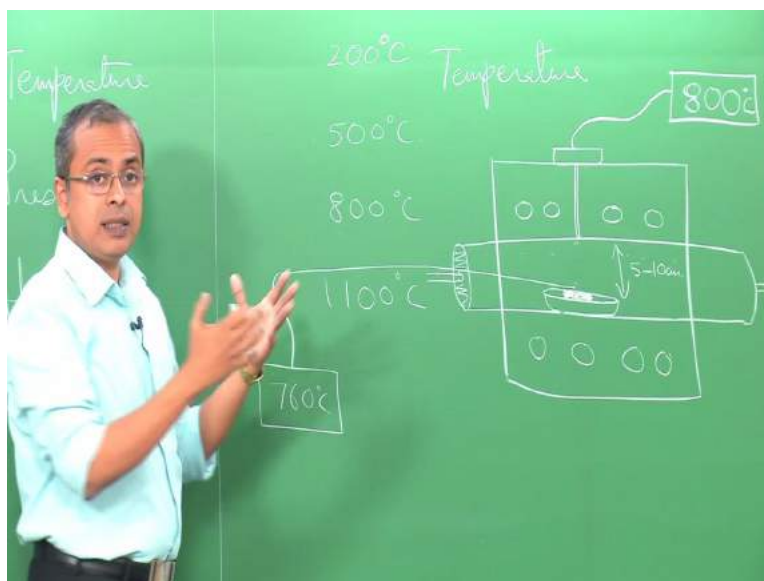
As part of the familiarization process, you should spend some time carefully understanding what are each of the quantities that you are measuring, what are each of the quantities that you are controlling, and try to understand - what is the process, what is that technique that is used to do that measurement, what is the technique that is used to do the control of that value, etcetera. And, when you understand that, you can figure out



if there is a scope for error. So, here we figured out that the location of the thermocouple can therefore give you a wrong idea of what is a temperature, because your sample is sitting somewhere else.

So, this idea may not be exactly the same for some other measurement; you may have to think of what is concept which creates the possibility of an error and then compensate for it. So, you compensate for it such that the actual measurement, you know, what is that actual measurement. So that, you have eliminated the error, you have adjusted for the error, compensated for the error. So, in using some standard sample, using some standard test conditions, you figure out what is that error that occurs in your instrument. So, instrument related error can be corrected, and therefore, you are now making a good measurement okay.

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So, this is one example of what can happen and this philosophy you should take with you for any other measurement you make, because the type of errors that may creep into those measurements may be different from what I am just showing you. So, that is something I wanted to highlight. This is with respect to this first quantity here - temperature. And as I said it's a very common experimental quantity that we all measure.



I won't spend too much time on pressure except, because we do not, I mean we do use it in many places, we do measure pressure in many places except to say that, you know, in similar to temperature you have to be cautious about - where is the sensor located? Where is the sensor located in your experimental setup and does it actually measure the pressure that you think it is measuring okay? And, in both these cases, you should also - both temperature and pressure - you should also make sure that the sensor that is doing the measurement is not, you know, polluted in some way, it is not clogged in some way, it is not, you know, covered by something else that it is getting an error in the measurement and so on. So, you should always check against some standard, so that your sensor is correct, and then, once it is correct, you should make sure the position of the sensor in your experiment is appropriate for the experiment that you are doing okay. So, those are two things.

Now, I will now spend a little bit of time on humidity and try to show you, you know, what is humidity measurement and, you know, what is the possibility of error in it, and so on. So, and I have picked these, because these are all relatively common quantities and they also give you some variety in terms of, you know, what is it that we need to be cautious about. So, in terms of humidity, I will just clear this drawing here. So, there are many experiments where we want to know - what is the relative humidity of the gas that is present in the experiment okay. So, often what is done is, you have something called a humidifier and this could be of various designs. So, in principle, what will happen is, there is a gas inlet, and then, there is some system here in which the gas mixes, in this case with water. So, there is water inside here, gas mixes with water, and then, there is a gas exit. So, the gas outlet. So, the gas that's coming out of this humidifier... this system would then be called a humidifier. The gas that comes out of the humidifier carries with it moisture and then that gets used to various experiments.

Now, the issue is that the amount of moisture that the gas can carry, varies. So, there is always an upper limit and that upper limit is based on temperature. So, at room temperature, for example, air can carry so much of moisture with it; some number of grams per some... so many grams of water - water in the form vapor - it can carry per litre of air. And, if you got to 100 degrees C, it will carry... usually it goes... the amount of carrying capacity of the gas goes up with of moisture, goes up with the temperature.



So, at 100 degrees C, same one liter of air can carry much more moisture. So, if you go from room temperature to 50 degrees C, 70, 80, 90, and 100, as you keep raising the temperature of the gas, it can carry more and more moisture. But, the point is that sets... that can be thermodynamically figured out. So, thermodynamically there is an upper limit for how much it is going to carry. But, the important point is that is only the upper limit. So, there are steam tables which give us these upper limits, but that is the upper limit. If you don't do this process correctly it may be carrying a lot less water than what it is capable of carrying okay.

So, therefore, you have to, if you are running an experiment where the humidity of your gas is an important quantity, that you want to say that humidity is 100 percent, meaning it is carrying the maximum amount of moisture that it can possibly carry, then you need to be able to measure it and satisfy yourself if that is the case. There may be experiments where you want the humidity to be only 50 percent, we call that relative humidity, relative humidity of 50 percent, which means if the gas can carry x grams of water in the vapor form per litre of gas, you are only running an experiment where it is only carrying half as much, x by 2. So, you can choose to do that; you can control that; you can say that I want 25 percent relative humidity is typically indicated as RH, you can say you want 25 percent, you want 50 percent, you want 75 percent, 100 percent, whatever, you can specify that and your experimental setup should enable you to do that okay.

So, therefore, it is very important for you to know what is the humidity of the gas that is coming out. Again, if you didn't know better, you would simply have a humidifier, and you will think that you are humidifying, and therefore, it should all be coming out at 100 percent RH. That is not the case. Based on the design of the humidifier, the efficiency with which the moisture is picked up varies, and therefore, often what is coming out is not at 100 percent relative humidity; it is typically lot less then that okay. So, therefore, you need to check, what is the humidity coming out. In other words, you need to calibrate your humidifier. So, what's the best way to do it? You can actually, you can use sensors to check your humidity, but invariably the sensors also face the same problem because you can have moisture condensing on the sensor, and in which case once again the sensor will give an erroneous reading, and therefore, you need to independently verify what you have got.



So, the simplest thing to do is to setup a system, where in you actually collect the water that is coming out with the gas, and then measure, how much water is coming out per, you know, volume of gas. So, the simple thing to do is therefore, to connect.

So, this is a basically a box, it is a very straight forward experimental setup that you can make and it enables you to check the validity of your humidifying system. So, it is simply a box, it has some holes on the top and a tube. So, the gas, you typically send in humidified gas that you want to check, the gas that you want to, you know, analyze is sent in through a tube, it goes inside this chamber, and then it exits out through the holes that are on top. And, inside this there is a desiccant; the desiccant simply absorbs moisture right. So, what happens is, the gas goes in, it goes through the desiccant, and all the moisture in the gas is absorbed by the desiccant, and then dry gas comes out of the box right. So, you can flow gas for some amount of time, before that basically you weigh this set up, you weigh this setup. So, you know, what is the weight of this box plus the desiccant plus this tube, this entire setup weight you know; you attach this to your humidifier. So, whatever is exiting out of the humidifier will go into this box, it goes inside this box, it releases all the water it has inside this box, and then, exits out dry. And then, when you continue this experiment for sometime at some fixed flow rate, you can know how much volume of gas went through this, and then, you take this bottle and weigh it; you know what is the gain in weight, that gain in weight is all the extra water that got collected in this box which was captured by desiccant, and so you know that many grams of water was carried by that much volume of gas; and therefore, you know, the humidity of that gas; because you know theoretically that same volume should have had how much of, how many grams of water.

So, this is a simple experimental setup, but it's a very good setup, because it helps you analyze and get a good idea of what is the actual humidity that is there in the gas that you are using right. You can make smaller versions of it and these are all homemade, that is why it looks, you know, simple boxes that you can get; based on the flow rate you can take smaller versions of it which essentially do the same thing. You only see one inlet, but there holes on top which are not visible to you through which the gas exits out. And of course, that you can do if it is air; if it is some other gas which needs to be handled carefully let us say, hydrogen, you have to make sure that the exit is handled properly,



you cannot just exhaust into air okay, into the ambient atmosphere.

So, this is a way in which you would check humidity. So, you would take this, exit it out into this bottle that we just had, send it into another bottle, and all the desiccants are here, the desiccant... and then the gas exits out. So, this is one way in which you would quantify what is the humidity that is present in the gas. And, this is very important, because as I said you need to know, you simply said something on the...

Usually these humidifier bottles also come with a temperature control. So, that will show you something; say 70 degrees C it will show. The general assumption is that you are seeing a humidifier set at 70 degrees C, you will assume that the gas is coming out at 100 percent RH, at 70 degrees C. What I am telling you is based on the flow rate that you have, based on the design of your humidifier, that may be completely incorrect assumption to make. You may, instead of getting 100 percent RH at 70 degrees C, you may only be getting 50 percent RH at 70 degrees C or 70 percent RH at 70 degree C; some, some number; it may be 100 percent RH. And, it's again a process that you can easily correct, because now once you know this information, once you know that at 70 degrees C it only captures, carries so many grams of water, you check the same humidifier at 80 degrees C, 90 degrees C and so on. So, you will find out that may be at once... and you do some interpolation between, so you will find out that may be if you set the bottle at 77 degrees C, it carries the correct amount of water which is what would be required for 100 percent RH at 70 degrees C right.

So, I am just saying that if this requires, you know, x grams of water to be carried per litre, x grams of water to be carried per litre of air, it may end up that at 77 degrees C, with a set point of 77 degrees C, it is actually carrying x grams of water per litre of air. So, it is still inefficient; at 77 degrees C, you again, I mean normally you would think that it should be 100 RH for 77 degrees C. It is actually giving 100 percent RH for 70 degrees C with set point at 77 degrees C. But, if your experimental setup is at 70 degrees C, you can always run your bottle at 77, and then, ensure that it cools down to 70 degrees C before it enters the bottle, enters your experimental setup. And therefore, when it goes into your experimental setup, the gas is now at 70 degrees C and it also contains x grams per litre, and therefore, it is 100 percent RH at 70 degrees C okay.



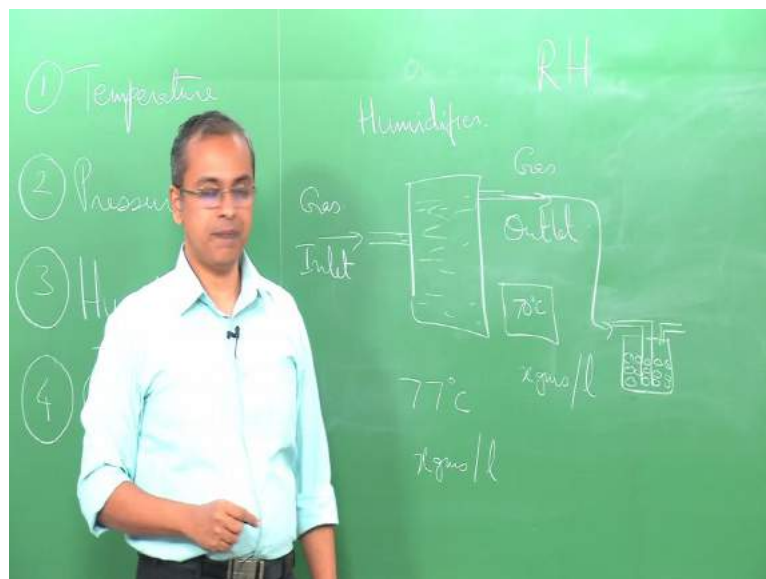
So, you have to worry about what is your humidifier doing; what is this path doing heading towards your experiment and then your experiment. And, so you can always change this setting here, so that the inlet for your experiment is correct right. So, and also I will add one additional detail. I said, I told you, you have worry about the path, because if this path drops to room temperature, if you don't maintain temperature of this path, if this is not heat traced as they call it, if this drops to room temperature, then the water will completely drop out as liquid water, and then, what you will have is gas entering the experimental setup, where gas will be an independent phase as gas, it will be close to, you know, room temperature relative humidity and independently you will have liquid water going in as liquid water not in vapor phase.

Whereas, in many of these experiments when you are talking of humidity, you don't want water in liquid phase, you want water in vapor phase right. So, in vapor phase, so therefore, it has to stay in vapor phase as it goes in. It does not help if you simply through in x grams of water and one litre of air. You cannot say that it is 100 percent RH at 70 degrees C. You have to have x grams of water in vapor vapor phase, along with one litre of air combined, sitting at 70 degrees C entering your experimental setup. So, that is important. So, therefore, it is necessary to keep this line heated, so that the temperature does not drop below 70 degrees C, and therefore, the correct value of water, correct amount of gas, at the right temperature, in the correct form which is vapor phase, enters the system okay.

So, there is a fair amount of detail here. You will often find many experimental setups are not addressing this level of detail. So, that's an experimental skill, to just look at your experiment carefully, to understand what is the possible sources of error in making a measurement, and then making sure that you have, you know, accommodated for all of them, so that your experiment actually runs correctly, is the experimental skill. So, here I have shown you there is a possible source of error in the bottle, possible source of error in the line, and then, what goes into your setup. And, I have shown you how you can correct for the bottle, you adjust the temperature here. I have shown you how you correct for the line, to adjust temperature here, and how to measure that you have got it right, all of them are possible. All these steps are possible, you just have to be alert to them, think about them and then implement them in your experimental setup.



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So, that's the idea. So, we have now seen temperature. I only just said I glossed over pressure, but I just told you that, you know, similar concepts are there. And then, we have discussed humidity because this is significantly different from what you would normally do right. So, these quantities we have looked at.

Finally, I will look at current and voltage and then with that we will, I mean, sum up this discussion. And again, many experimental setups we have, we are measuring current, we are measuring voltage and so on. I would say there are only two major aspects that you have to measure, be alert to in an experimental sense with respect to current and voltage.

The first is that it is always good to cross check your value, because you usually, again, I think the major mistake that most of us make with our experiments is that we simply trust the numbers that the display shows and that is fine, I mean many times the displays are working fine; I am not saying that the display is automatically wrong but it is very important for you to cross check the value that the display shows. So, many times you are drawing current, you may drawing, let us say, you are testing a battery, and you are drawing 50 amps from the battery; 50 amps from the battery; you are drawing 50 amps from the battery; and you simply have a display that says 50 amps. Sometimes the electronics of the system may not be correct, and so, it may be actually be drawing 47



amps or 45 amps or something like that or may be erroneously showing you 50 amps. There are many ways. So, you have a battery here, and then you have two leads coming out of it, and then you have some electronic setup here, which shows you 50 amps right.

What I would suggest to you, is that I mean, of course, you can always use different different meters to run it and so on. There are many facilities that are available these days, many electronic devices that are available these days, and they are all available with varying levels of sensitivity. So, you can buy electronic devices which have a highly sensitive devices, which can help you cross check this, and one of the simple devices that is available is something that are, I will just show you here, this is called a clamp on meter. So, you can always... this... it basically opens like a clock. So, you can put at any, any experimental setup where you have a wire through which current is flowing. So, you have basically a wire that is flowing, you simply have to put this around that wire; when you put this around the wire like that, so there is wire which perpendicular, which is headed towards you, and then you take this clamp on meter or since the circuit is complete you cannot just get in here, you open it, do this, close it, and then you will see a display of current.

So, now you can, of course, get this clamp on meter with varying levels of sensitivity and so on, and it may not be the best kind of technique to look at for, you know, getting very accurate values, but still, it's a good technique. What happens is... it's a good instrument; it is a very convenient instrument used because many experimental setups you do not want to disturb the experiment, the experiment is already running, you just want to cross check. Once in a way you want to cross check that the value is correct. So, you simply put a clamp on meter, on any... around anyone of these wires, and it should show you 50 amps. It's a good way for you to cross check.

So, one of the experimental skills you should have and you should pay attention to, is to have instruments available in your lab, and you should, you know, build up such instruments available with you so that you can cross check values, without interrupting the experiment. So, you know, from your, you know, activities in your lab, what is the kind of experiment you do. It is nice for you to build up a few extra techniques available with you, so that you can cross check at your convenience or as per your necessity you



can cross check each of those values. And, I will just show you one of them which a clamp on meter.

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The other thing, I will also tell you about current and voltage, is that at least... So, this is with respect to current; what I just showed you is with respect to current. With respect to voltage, the other major point that you have to remember is the location where you measure the voltage. So, for example, you have a sample; I am just showing you the top view of a sample. So, you looking down on a sample; let's say you simply want to measure the... you want to measure the resistance of the sample okay. So, let's say you want measure resistance of this sample. So, you can simply put, you know, current from this location, draw it out of this location and you also measure the voltage here. So, you measure the voltage drop here and you have a meter here. So, you can send current in, current comes out. So, this is just a lead through which you contact the sample. So, between these two leads current is following coming in here, coming out that way and you are measuring the voltage. And you simply say V equals IR , and therefore, the resistance is simply V by I , if it's a DC measurement and you have got the resistance.

Now, what you often don't realize is that there is a contact resistance. So, there is a resistance associated with the contact of this - of the leads - with this sample and that can



be significant. So, actually what you are measuring, this R that you are measuring, is not simply the resistance between this point and this point, that can be attributed entirely to the sample. I will say that there is a contact resistance R , which I will put $R_{\text{subscript C}}$, and we have one here and we have one here. So, actually what you are measuring and let's say the R for the sample is R_s , sample s . So, what you are actually measuring is R_C plus R_s plus R_c . So, you are measuring two contact resistance, associated with these two and you are measuring the resistance for the sample.

And, generally you cannot automatically assume that these are very small values; these could be significant values. It could upset your overall measurement, you are measuring this whole thing and that's what you are seeing as R here, and this could be easily, you know, off by 20 percent or 30 percent or something from your... you are assuming that the R is the belonging to the sample, but I am just saying that it could be off by 20, 30 percent from what is actually attributed to the sample.

So, therefore, so, this is called a two probe measurement, wherein both the entire process of connecting to your sample, some power source you are applying here, so through which you are connecting to the sample. The entire process of connecting to the sample occurs only at two locations - this one and this one. The same two locations you are checking the voltage and the current follows to the same two locations; this is called a two probe measurement. And a very simple way to correct for this, to avoid this contact resistance problem is to simply move the voltage lead away from here to two other locations in the sample, and therefore, now you will have voltage from here to here. So, volt meter connects to two locations which are towards the interior of the sample, the current goes from two outward locations. So, you have this situation and when you have situation, you basically avoid this contact resistance problem because the contact resistance is the... actually the amount of current going through the volt meter circuit is negligible, and therefore, technically you can say there is a contact resistance here also, but that contact resistance is extremely tiny relative to the contact resistance that you get here.

But, the way we are running... the current that follows to the circuit is the same. So, the same current goes through the entire circuit. It's just that we no longer bother about the



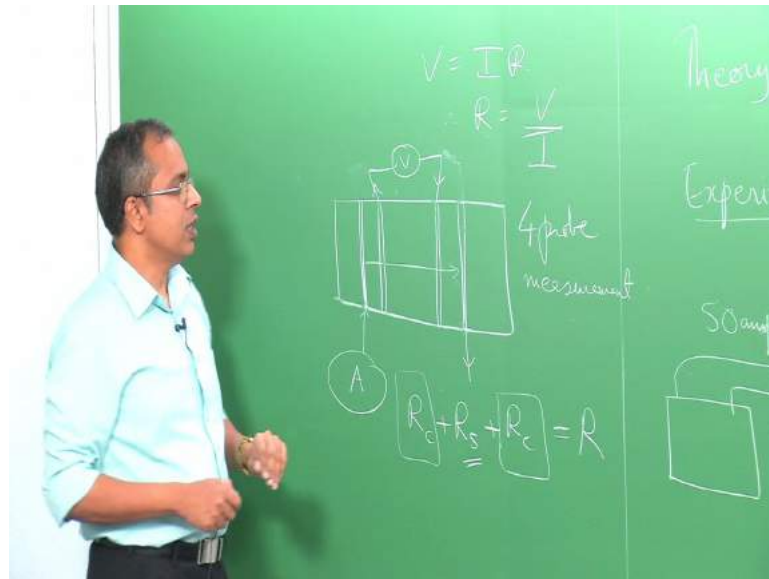
behavior of the sample on this side and the behavior of the sample on this side. We only look at the behavior of the sample here. And in this location, we know the current that is going through and the contact resistances involved with this measurement process is extremely tiny okay. And therefore, you eliminate the contact resistance problem. You still have two contact resistances, but they are now extremely tiny there are, you know, 3, 4 orders of magnitude smaller than what they were, when you were, when you had two probe measurement; this is called a four probe measurement.

And, mainly you are eliminating the IR drop associated with this contact and the IR drop associated with this contact. You are picking up two other IR drops, but these I 's are very small. The I going in this circuit is extremely small, relative to the I that is going through this circuit; and therefore, the IR drop associated with this measurement is extremely tiny. So, there's a way in which when you measure voltage, based on where you will measure the voltage, you can eliminate errors in the overall measurement. If your overall purpose was to get a resistance of the sample or more specifically you want to get resistivity of the sample, because that would then account for the sample dimensions.

I just showed you how the resistivity could be erroneously measured, if you had two probe measurement and can be much more correctly measured if you did a four probe measurement. And therefore, again, the way you run your experiment can give you either a wrong value or a correct value. Always, there are numbers on display and always you can note down those numbers, draw ratios, write up ratios, draw graphs and so on. I am just pointing out that, you know, in straight forward experiments that we commonly do, there are so many sources of error right.



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So, to sum up we have seen temperature, not a whole lot on pressure, but humidity, and current and voltage. And, I have just shown you that these are all common quantities that you measure. I just pointed out that, you know, there is so much room for error, based on where you may put your sensor; similarly, where you would put your sensor here for pressure, what kind of, you know, impurity or dirt, for example, covers the sensor; the location of the sensor or any other impurity that is on this temperature sensor also.

The humidity value based on the actual temperature of the system that is humidifying it, the design of the humidifier, the design of the path way between the humidifier and the actual experiment, the humidity value could be wrong.

Based on how you measure your current, you at least need to cross check the value of current, and location where you measure the voltage, could change the voltage and ... could change... may not change the voltage, but it will change the significance of the voltage, because that is different.

So, I just looked at some common quantities here, that we would typically measure. And I think the main thing that I want to convey is this idea that in all experiments there is room for error okay, there is scope for error, and we have to as experimentalists, pay



attention to it; first of all, look for it - that is the thing that when we say experimental skills, that is the experimental skill. The experimental skill is to take an experiment, and in your mind you should just, you know, rip it apart, you should look at every stage of that experiment, every step in that experiment and ask yourself - so, what can go wrong in this step okay? If you do a thorough analysis like this, you will feel very confident of your experiment. You go and make a presentation in a conference, you make a presentation in your college, in your university or some international conference or you send a paper for publication, you will feel very confident of the values that you are reporting. If somebody challenges your value, you can defend it; you can say, you know, I measured it like this, I have taken the following steps to account for errors, and therefore, I am confident that the value I am measuring is correct. And therefore, the significance of what I am measuring is also appropriate. I am highlighting it in the correct way.

So, this is the basic idea that I want to highlight and this whole process is something that I would call as, you know, being an experimentalist and as something that we should develop in our process and so that we can say that we have good experimental skills.

So, with this I will conclude and I wish you good luck with all the experiments you do. And, just take this philosophies with you and look at the quantities that you are measuring. I mean again as I said, you know, that many other quantities that I have not even spoken about here - you could be looking at magnetic properties, you could be looking at optical properties, so many other properties you could be looking at. In each measurement, look for that source of error and compensate for it, correct for it and feel confident with your results okay. With that I conclude the class.

Thank you.



Introduction to Research
Prof. Arun K. Tangirala
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture – 14 (2A)
Data Analysis

Hello. Welcome to the course on Introduction to Research; in particular, the lecture on Data Analysis. In the previous lectures, you have learnt what is meant by research. We have tried to give you an overview of research, and of course, we have also gone through a lecture on literature review. In this lecture, we will specifically look at what is data analysis, the types of analysis, and the different types of data that one encounters in data analysis, and of course, a systematic procedure.

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A presentation slide titled "Data Analysis" in the top left corner. The main heading is "Objectives" in a large, bold, blue font. Below it, the text "To learn the following:" is followed by a bulleted list of four items: "What is data analysis?", "Types of analyses", "Different types of data in analysis", and "Systematic procedure". Below the list, the text "With two hands-on examples in R® (a popular software for data analysis) ..." is displayed. At the bottom left, there is a small NPTEL logo and the text "Arun K. Tangirala, IIT Madras". At the bottom center, it says "Introduction to Research". At the bottom right, there is a small number "2".

Data Analysis

Objectives

To learn the following:

- ▶ What is data analysis?
- ▶ Types of analyses
- ▶ Different types of data in analysis
- ▶ Systematic procedure

With two hands-on examples in R® (a popular software for data analysis) ...

 NPTEL

Arun K. Tangirala, IIT Madras

Introduction to Research

2

We will end the lecture with two hands-on examples in a language or in a software platform called R which is an open source free software for statistical data analysis. Now, before we proceed further, let me tell you that data analysis is an integral part of any research work. In every type of research that one carries out, invariably, one collects some data or the other. Even if it's theoretical research, at some point in time, there is a bit of experimental work involved to corroborate the theory that has been developed. By



and large, if you look at today, the scenario is about data analytics. A lot of data has been generated not just in engineering arena, but every where, and therefore, there is a need to understand how to analyze data in a systematic manner. There are many, many software packages that can carry out analysis of different types of data. And one should remember that analyzing data is not just about throwing data into the software packages, and then, reporting some of the fancy graphs that are generated and some nice results that you get on your computer screen - that is not data analysis; it's just a very tiny part of that exercise.

As much as we are analyzing data, there is a theory to it, and of course, we are not going go over the theory of data analysis, but the procedure that one needs to follow to get meaningful results. First of all, we need to ask - what is data analysis?

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A presentation slide titled "What is data analysis?". The slide has a light blue background. At the top left, it says "Data Analysis". The main title "What is data analysis?" is in a large, bold, dark blue font. Below the title, there is a pink rounded rectangle containing the text "Extracting useful, relevant and meaningful information from observations in a systematic manner". Below this, the text "For the purposes of" is followed by a bulleted list of five items: "Parameter estimation (inferring the unknowns)", "Model development and Prediction (forecasting)", "Feature extraction (identifying patterns) and classification", "Hypothesis testing (verification of postulates)", and "Fault detection (process monitoring)". At the bottom left, there is a small NPTEL logo and the text "Arun K. Tangirala, IIT Madras". At the bottom right, it says "Introduction to Research" and a small number "3".

Analysis of data is about extracting useful, relevant, and meaningful information from observations. As I said, we collect data in different spheres and different forms, whether it's econometrics, whether it's engineering or medicine or any other field, we have different types of data collected and the purpose of collecting data is basically to know something about the process; but, of course, that's not the only purpose. As I have listed here, there are several purposes to data analysis. It could be commonly.. common



purposes, for example, parameter estimation where we are inferring certain unknowns, and when I say parameters here, it is not just model parameters; these parameters could also refer to statistical properties that we are commonly interested in, such as averages - that is mean, variability, variants or standard deviation, median and and so on. So, the term parameter here should be taken in a very generic manner. Essentially it's an unknown that we are trying to estimate from data. Obviously, that's one of the common exercises in data analysis.

Then, of course, we have the ubiquitous model development. Many of us collect data to develop models between two or more variables. And why do we develop models? Largely for predicting the variable of interest. Technical term for prediction is also called is also forecasting in econometrics and social sciences; it's a very common term and that's a huge field in itself. And then, we have feature extraction. So, if you take, for example, biomedical data, such as ECG or EEG data. In ECG data whether we have seen or not, but ECG is something that we all get to see whether it's a part of our research work or not. It's related to our human health, and if you see, if you recall ECG is a train, is a pulse, train of pulses which has regular peaks in it and we would like to extract those features. A doctor who looks at ECG essentially searches for those peaks, and also searches for anomalies in those peaks, and those anomalies could be indication of an unhealthy heart, for example. So, feature extraction is very common, not just of course in biomedical, but everywhere.

And classification. Once we extract features, we would like to classify image. In image analysis, this is a very very common exercise. We try to classify people with similar colors, hair colors, similar eye color, and facial skin color and so on.

And then, central to all of this is hypothesis testing. What is hypothesis testing? It's a very, very common exercise in data analysis, where the researcher postulates the truth being something. As an example, suppose I collect ambient temperature data. That is outside my room, I take a sensor and collect ambient temperature for about may be an hour or so. And I come back and postulate that the average temperature during that one hour is 25 degrees Celsius. Now that's a truth that I postulate for the average temperature. And hypothesis testing is about testing that claim that I am making whether



it holds or not, based on the data that I collected. So, the data I have collected serves as an evidence and I search for evidence in the data to support my hypothesis, and that has to be done in statistically sound manner. And hypothesis testing is an integral part of every statistical inferencing exercise. That involves, of course, a lot of theory, probability theory, estimation theory and so on. We don't, we will not go into depth in hypothesis testing and so on.

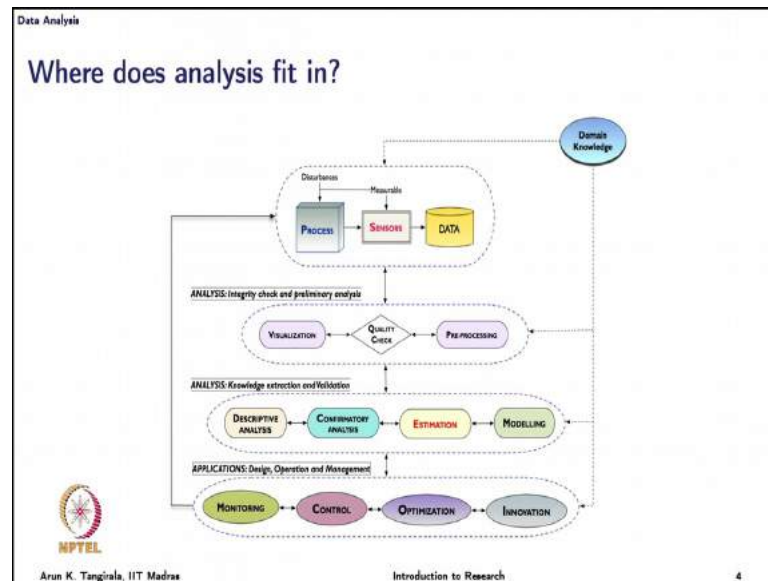
There is a MOOC course that we learn on introduction to hypothesis testing; that will begin perhaps in the second or third week of March. And if you are interested, you should sit through it because it is an integral part of every statistical data analysis exercise.

Then, we have data analysis being carried out probably for fault detection. As I explained earlier, a doctor, when the doctor looks at the ECG, is looking for anomalies or when a radiologist is looking at the x-ray, looking for anomalies based on some model or some template of normal conditions in mind, keeps projecting the present data on to the normal template. And that is how typically fault detection is carried out everywhere; even in process industry that is the situation. So, this is also known as Process Monitoring.

And there are several purposes that one can list in data analysis, but these primarily cover most of the applications of data analysis. As you can see the application are quite diverse and varied. Therefore, the procedure that we follow for an analysis, the tools that we deploy, and the interpretations that we draw have to definitely vary broadly.



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So, it's useful to look at data analysis and ask where does it fit in, in the broad scheme of data driven analysis. So, this pictorial representation here tries to give you some idea of where analysis fits in. At the top of the schematic, you have process, and this process is a very generic process; it need not be an engineering process; it could be a social process or a biological process and so on. And we have sensors collecting data from the process and these sensors need not be just the classical hardware or instrumentation sensors. It could be also human sensors. When we collect survey data, we are the sensors, in fact. In the end, we have data with us and that's where the journey begins in data analysis.

In this module, of course, we are not going to talk about how to collect data, the techniques of sampling and so on; we are more worried about analyzing data. So, once I have data, typically I put it through certain steps and we will talk about this a bit later. The purpose of this schematic is to tell you where analysis fits in. The two stages that you see titled analysis here, it is essentially the representation of what goes on in data analysis.

For example, visualization is a part of analysis. When you plot something, and try to observe something from the data, that is also a part of the data analysis. It's a very important part or you may carry out parameter estimation as I just discussed or you may



build models and so on. Ultimately, the results from this data analysis go into some of these very broad applications as I just explained; it could be for process monitoring; that is fault detection or it could be for process control, where I am taking feedback action based on measurements. So, the controller is taking in the data, performing some kind of analysis, and then, deciding how the input should move. That's also a part of the applications of the data analysis or optimization, I am trying to come up with an optimized way of operating a process, or making innovations or changes to the process.

Then, of course, there are few other applications that I have not listed here, but ultimately you should understand that data analysis is not the end of the road, but it is really a very, very critical art of data driven process, analysis process operations and so on; and when I say process here, as I mentioned earlier, its fairly general. And in all of this, of course, the domain knowledge is very, very important as I highlight here, and we will talk about this a bit later when we return to the schematic and discuss the systematic procedure that's involved in data analysis. So, hopefully, you understand where data analysis fits in. It actually connects the world of observations to the world of information.

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A presentation slide titled 'Data Analysis' with the subtitle 'Types of analyses'. It lists five types of analyses: 1. Exploratory vs. Confirmatory, 2. Quantitative vs. Qualitative, 3. Descriptive vs. Inferential, 4. Predictive vs. Prescriptive, and 5. ... Below the list, it states 'Exploratory and descriptive analyses are typically good starting points'. A callout box at the bottom says 'The objective of analysis determines the type of analysis and the tools that are used.' The slide includes the NPTEL logo, the name 'Arun K. Tangirala, IIT Madras', the title 'Introduction to Research', and the slide number '5'.

There are many different types of analysis, obviously, now, depending on the need and



the end use of your analysis, and there are several classifications that you can find in the literature. What I listed here is by no means exhaustive.

One of the common classifications of analysis that you see is exploratory verses confirmatory. In an exploratory type of analysis, you do not postulate anything upfront, but you are just exploring the data and trying to understand your process. For example, it could be looking at the mean or the standard deviation or looking at the graphs, trying to understand whether the process is linear or stationery and so on; trying to infer whether the process is oscillatory. There is a lot of visual plus some descriptive statistical analysis is involved; whereas in confirmatory analysis, you have a postulate upfront, and you would use the data to confirm whether what you have in mind is correct or not. So, obviously, there is a bit of contrast there.

And then, you have quantitative verses qualitative data analysis. In qualitative data analysis, for example, you look at trends; you are searching for certain features and oscillatory features, for example. Whereas, in quantitative you are being... you are looking at numerical results from your analysis.

Then, you have something called descriptive verses inferential. And in descriptive analysis, again, it is more or less like exploratory; whereas in inferential you are trying to infer something very specific from data and you are subjecting it to hypothesis testing. So, there is a lot of systematic, numerical, and statistical analysis involved in inferential analysis, and this could be a part of a larger analysis that you are doing such as building a model and so on. Whereas, descriptive analysis is not necessarily going to feed directly into a larger analysis; it's typically the first step in data analysis.

Then, you have predictive verses prescriptive. Predictive analysis is about asking where the process is heading based on data. And this is very useful in many, many applications; whereas prescriptive is about asking why something has happened. So, you have looked at the data either as a doctor or as an engineer or even as an econometrist and you looked at market data, and suddenly there has been a crash, and you are asking well – why did this crash occur or why did a particular fault occur in a process and so on. And then, you are, of course, going to suggest as to how to prevent this. So, there is going to be a



diagnosis and remedial thing involved in prescriptive analysis. So, obviously, there is a whole host of different types of analysis that are involved and there are many other classifications that I have not listed.

Typically, good starting points are exploratory and descriptive analysis and then one proceeds from there; even if you have decided that you are going to build a model for prediction and so on, it is recommended that you explore the data, you make friends with the data, and you familiarize yourself with the data, before you proceed to developing a model, for example.

Now, in general, the objective of analysis determines the specific types of analysis that we have listed here, that would you carry out. That means, essentially what is the end use of an analysis, how are you going to use the information that you extract.

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Data Analysis

Types of data

Several classifications (e.g., numerical vs. categorical, steady-state vs. dynamic, etc.) exist. In data analysis, the primary distinction is w.r.t the **data generating process (DGP)** or mechanism;

Deterministic (non-random) and **Stochastic** (non-deterministic) data

1. **Deterministic:** The DGP is perfectly known OR a mathematical function can perfectly explain the data OR there is no uncertainty in the observed values OR process is accurately predictable.
2. **Stochastic:** Observations are only a subset of many (countably finite or infinite) possibilities OR no mathematical function can explain the observed data OR observations have uncertainties in them. Stochasticity does NOT necessarily mean zero predictability

 In practice, no process is deterministic since perfect knowledge is never or rarely available.

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And before we proceed to data analysis, it may be very useful to know what are the different types of data that are available, because primarily as I will mention later, the tools, the interpretations, and so many other steps depend on the nature of the data. Now, once again, when you turn to data analysis literature, you will find different types of classification, a number of classifications exist. For example, you may find numerical



verses categorical classification or steady state verses dynamic and so on, but I am not going to really talk about those, because that's definitely important, but what is more important which people often ignore, at least beginners, in data analysis often ignore is this classification of deterministic verses random or stochastic data. That's very, very important because theoretically itself or the theory itself largely differs the moment you say the data has come out of a deterministic verses stochastic process.

Now, of course, some people use the term random. There is a subtle difference between random and stochastic; however, we will not observe that distinction here. Sometimes, the term non-deterministic is used. Before we ask why this classification or distinction is important, let's look at the definitions and the definitions are based on the data generating process. So, the deterministic and stochastic qualifiers are not for the data per se, it is actually, they are qualifiers for the process that is generating data. In data analysis, the data generating process is a very important entity. What we mean by data generating process is the process that you are collecting data from as well as your experimental process. That is how you are conducted the experiment. So, put together, everything is the data generating process. It's not just the physical process alone that you are analyzing. That's very important to remember because both contribute to the type of data that you generate.

As an example, let us say, I am observing some reactor temperature which is being controlled by coolant flow. This is very common in nuclear reactors and so on. Now, the reactor that I am observing is a physical process. There is a controller there trying to control the temperature and so on. Let's say it's doing a fantastic job of controlling the temperature. So, the temperature is held at its set point, but the sensor that I am using can be quite noisy. Ultimately, the data that comes to me is not directly from the process between the process, the physical process and the data I have a sensor and that sensor characteristics. The characteristics of that sensor actually goes into the data. If the sensor is very noisy, you are going to end up with a noisy data. If the sensor is a biased, your data will show a bias.

Now, the data generating process consists of both the reactor and the sensor. The DGP -as we shall use as an abbreviation that you will find even in literature - is set to be



deterministic, if you can find a mathematical function that can accurately explain the data; that is, it can predict the data, you can fit in perfectly, every data point lies on the curve of this mathematical function. That's one way of looking at it or you can say that you have exactly understood the data generating process; there is nothing that you cannot explain or that you can say that there is no uncertainty in the observed values.

You may have... many a times what happens is you may have very good knowledge of the physical process. For example, the reactor in the example that we discussed earlier, I may understand the dynamics, the kinetics, everything in the reactor, the thermodynamics and so on, but I may have poor knowledge of the sensor characteristics. Now, that makes it a very tricky situation. It puts in neither in the deterministic nor the stochastic thing and we will talk about it, but essentially that is not deterministic anymore. The data is not coming out of a deterministic process because a part of the DGP, which is a sensor, has not been understood properly. In general, this is the situation; sensors are never perfectly understood. You can design a very, very nice sensor that has very less noise, but no sensor is perfect and no sensor is really perfectly understood. So, in reality, there is nothing like a deterministic process. It's an idealization, that is very useful in data analysis and in developing the theory.

Now, obviously, stochastic is the counter part for the deterministic, where we say that the DGP is not accurately predictable or its not known or you can say there is uncertainty in the data and so on. And as I said, in practice, no process is deterministic since perfect knowledge is not known. Now, does it mean that every data generating process has to be treated as stochastic? Not really. There is a certain distinction still observed between deterministic and stochastic processes, even in practice even though no process is perfectly predictable. Typically, we say that a process is predominantly deterministic or predominantly stochastic and we will talk about it very soon; soon after this slide here.



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Data Analysis

Why is this assumption important?

The **theory**, tools for analysis and **interpretations of the results** heavily depend on the assumptions of determinism or otherwise.

- ▶ Example: definitions and computation of periodicities, mean, variance, stationarity

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So, going back to the question that we raised earlier - why should I worry about whether data has come from a deterministic or a stochastic process? Why is this assumption so important? Now, the answer is something that I have said earlier - that theory, the tool for analysis, and the interpretations of the results significantly depend on these assumptions. And there are many, many examples that one can give. For example, if I am looking at periodicity detection, then first I have to define what is a periodic deterministic signal is. If I am looking at deterministic data, and if I am looking at stochastic data, then I have to define what is meant by periodic random process. So, the definitions vary, and therefore right, the interpretations will also change. Likewise, when I am referring to the average of a stochastic signal, normally misconception is the sample mean, which is the mean that is computed from observations, is considered as the mean of the process itself; that is, it is not true at all. The sample mean is only an estimate of the true mean which is defined in a different way and we will quickly talk about it.

So, the bottom line here is, whether you are looking at mean variants or any other characteristic of the signal, you have to really go back to the books and ask how are these defined for the signal that I am looking at. If I am looking at a deterministic signal - what is the definition of periodicity? What's the definition of mean? If I am looking at a stochastic signal - what is the definition? And then only, you can make meaningful



interpretations. And a very, very common situation is that of power spectral density which is used for periodicity detection and so on. Anyway, we will not go so much in detail, but you should remember this fact, and every time you collect data and you sit do analyse data to ask certain important questions, and one of those questions is - what assumptions am I making about the data generating process.

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Data Analysis

In practice: deterministic or stochastic?

Q: When do we assume the DGP to be deterministic or stochastic? Can we know a priori?

- ▶ Determinism or stochasticity (random) rests with the knowledge of the user (analyst)
- ▶ Whenever the causes are unknown or the underlying mechanism is too complex to be understood, it is reasonable to assume the data to be stochastic (e.g. sensor noise, econometric data).
- ▶ Where the physical process is well-understood and/or the causal variables are known, and the errors in observations are "negligible", data may be assumed to be (predominantly) deterministic.

Q: Is it possible to have a mix of both situations?

- ▶ Yes. Several processes fall into this category (e.g. engineering data).

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So, before we move on to learning what is a procedure that's typically followed in data analysis, it's again good to close this discussion on deterministic or stochastic with this question - when do we assume the DGP to be deterministic or stochastic? Does some one comes and tell me that? Obviously not. Can I find in the literature? Some times yes, but most of the times no for the process that I am looking at. So, is there a way to know a priori? Well, unfortunately, there is no perfect formula for you to figure out whether the data is coming out of a deterministic or stochastic process; but you have to go by how much you know vis-a-vis how much you do not know right. That's about the process.

So, as an example, suppose I hold a ball, in a room, where there is not much breeze. The air is fairly still and I am going to drop the ball. We can pretty much with a help of physics calculate how long the ball takes to hit the floor. That's fairly a deterministic process. There is not so much uncertainty there. On the other hand, if I were to be asked



whether its going to may be rain tomorrow or in 6 months it's going to rain or if I look at any other atmosphere process characteristics or even econometric process and so on or a sensor noise and so on, it's very hard to predict what is the next value of the error in the sensor, for example.

In such situations, we take asylum in probability theory and we say that look I do not have a perfect knowledge of the process. Therefore, I shall assume that there are many possibilities, and I assign chances to each of those possibilities, and that's where probability theory takes birth, and that's where time series analysis comes in. There is also another situation in which you may have to assume the process to be stochastic and that situation is where the cause of the data that you are looking at is unknown. So, as a simple example, suppose, I am looking at the prediction of stock market index. There are so many causes that actually affect; there are so many factors responsible for stock market index. Now, unfortunately, even if I know the causes, I may not be able to measure them and sometimes I don't even know the factors exhaustively enough.

So, what do I do? I depend on the history, and hope that history has some reputation in it, and I exploit the historical patterns and correlations and so on, and make a prediction, but that prediction is going to be, obviously, not accurate. The point is here we apart from looking at the history, the main assumption that we make is that this stock market process – that which is some fictitious process that generating the index - is random in nature. It throws out, it can throw out any value of the stock market index out of which I am observing one. There could have been many other possibilities out of which one has occurred. In reality that may not be the case. The true process may not be behaving that way; it is an assumption that I am making, so as to move forward in data analysis; that's very, very important. In other words, whenever I assume a process to be stochastic, it is a formal way of saying I do not have sufficient understanding of the process.

But there are situations where I have a mixed case - where I have a fairly good knowledge of a part of the process and the other part of the process is beyond me. That's where we have a mix of deterministic and stochastic process. And that is very common in engineering arena because a physical process may be well understood, like in the reactor example that we discussed earlier, but the sensor or some effects of unmeasured



disturbances, something that's beyond my control, those have to be treated as stochastic simply because I can not predict them accurately.

So, there we may have to deal with both the deterministic and stochastic comments, and there is a theory that allows you to do that. So, the bottom line is please do spend a few moments asking - first of all whether you want to treat the data as a deterministic, coming out of a deterministic or stochastic process or a mixed process, and you should have the answer as to why you have made that assumption, because the course of analysis really depends upon that.

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A presentation slide titled "Further" under the heading "Data Analysis". It contains four bullet points with red arrowheads. The third bullet point is highlighted in red. At the bottom of the slide, there is a small NPTEL logo, the text "Arun K. Tangirala, IIT Madras", and "Introduction to Research". A small number "9" is in the bottom right corner.

Data Analysis

Further

- ▶ Be clear on why the data is assumed to be non-deterministic.
- ▶ Suppress the sources of randomness, if possible, during experimentation.
- ▶ **IMPORTANT: Any estimate inferred from data with uncertainties also has uncertainty in it!**
Therefore, always attempt to **compute the size of the error in the estimate.**
- ▶ Choose an estimation method that delivers estimates with the lowest possible error and improves with the sample size N , especially error goes to zero as $N \rightarrow \infty$.

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Okay so, just to emphasise, be clear on why the data is assumed to be non-deterministic, particularly this is a very common assumption; in fact, many many researchers carry out data analysis assuming the data to be stochastic without probably even pausing for a moment to ask why and what kind of stochastic. Just because I assumed data to be coming out of a stochastic process, I cannot simply go ahead and carry out my data analysis. Within the stochastic processes, there are several classifications and I have to be clear in my assumption. It is after all an assumption, but still I have to be clear as to whether it's coming out of a stationery stochastic process, non-stationery stochastic process or a periodic stochastic process and so on. So, there are different categories and



we should be very clear. This is.. all of this is apart from the regular classifications that we have to think about - which is whether the data is coming out of a steady state or dynamic process, linear or non-linear. All those still hold good, but this is a very important classification that you have to make upfront. And, in general, if you have access to the experiment, this is, of course, outside the purview of data analysis, but it's related to suppress the sources of randomness if possible during your experiment. That is a part, that is a topic that is dealt within detail in design of experiments and we don't go into that here.

The most important thing that people often miss out on - that is beginners - is that any estimate or any inference that I draw from data - and typically lot of data is assumed to come out of random or a stochastic process; those estimates are inferences are also random in nature, because you are putting through data which has uncertainty in it. Through some mechanism, some formula, some mathematical process and you are getting out, for example, I am computing the sample mean. I go and collect data, ambient temperature data, with a sensor.

Come back to my desk and feed in the data into my computer or my calculator and I actually compute the sample mean. The sample mean that I have calculated is simply the average; average of what? Data that is corrupted with uncertainties. So, those uncertainties have also made their way through to your sample mean. So, the value of sample mean that you have has also some uncertainties. The uncertainty here is not with respect to your calculation error okay; that should be kept aside. The uncertainty in it has got to do with a fact that you are not sure if this is the true mean of the process; that is the uncertainty. That's again because the data that you collected is has ..is one of the many possible data sets that you could have collected. So, you have a sample mean which is one of the many possible sample means that you could have obtained. And this is true for any parameter estimate whether you are estimating slope, intercept or any other correlation, any other complicated parameter that you may be estimating from stochastic data, you will have to remember that there is an error in the data, and as an analyst, it's the responsibility of the analyst to compute the size of the error or the uncertainty in the estimate or the inference that you are drawing. That's a very important part of data analysis. And in this context, you should choose an estimation method that



gives you an estimate with as small error as possible and we will talk about it a bit more in detail.

More importantly that as you collect more and more observations, the error in your estimate should ideally start shrinking and go to zero asymptotically as n goes to infinity. So, these are some criteria and very often many of the researchers may not be aware of this.

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A slide titled "Terminology in data analysis / statistical inferencing" with a list of three definitions. The slide includes the NPTEL logo and footer text: "Arun K. Tangirala, IIT Madras" and "Introduction to Research".

Terminology in data analysis / statistical inferencing

- 1. Population and Sample:**
Population refers to the collection of all events or possibilities, typically also known as the ensemble.
Sample refers to the subset of population (observations) that has been obtained through sampling.
- 2. Truth and Estimate:**
True value refers to the population characteristics in descriptive statistics or to that of true parameters / signals in parameter / signal estimation. Estimate is the inferred value of the unknown parameter / signal from observations.
- 3. Statistic and Estimator:** A statistic is a mathematical function of the given observations. An estimator is also a mathematical function of the observations, but devised for the purpose of inferring or estimating an unknown quantity / variable.

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So, before we move on to the data analysis procedure which is the last part of this lecture, it's important to at least gain familiarity with some terminology that's commonly found in the data analysis literature. And here, we will assume primarily that we are going to deal with data that comes from a non-deterministic process, because that's the prevalent situation everywhere. Even if you have a mix of deterministic and stochastic process, these terminologies apply.

So, the first terminology that we have to be familiar with is the term - population and sample. We know that in stochastic process, as we just mentioned earlier, we observe when we get a data record, the record of data that we have is one of the many, many possibilities - sometime countably finite, sometimes countably infinite or infinite - many



possible data records exist, out of which we have observed one; that is the theory, the entire theory is built on that premise. The population refers to all possibilities that you could have seen, although we do not observe that; it is not possible to really have or observe all the possibilities in most of the situations. So, population refers to the collection of all events or possibilities.

When with respect to the temperature measurement that we talked about earlier of the ambience. If I collect hundred observations with one sensor over a period of time, another sensor would have viewed that in a different manner; so that, another data record could have been impossible or the same sensor itself could have recorded, everything held constant could have recorded completely different values. So, it's not possible obviously to realize all those possible values, but theoretically it is important to imagine this all possible space and we call it as population of which we obtain one sample. When I say sample here it is not one observation, a sample in statistics typically refers to collection of observations. It basically refers to one data record, but in signal processing, a sample could be one value at an instant and that is in the deterministic world. Corresponding to this population and sample we have a truth and an estimate. We just said population is a collection of all possibilities, but that's characterized by the truth.

So, this entire population has, for example, certain mean. Let us say, I am looking at the average salary of a group - of an age group - in a very large country, let us say or in very large firm. It's not possible for me to go and ask every one. For practical reasons, I may only select a few people to ask to collect this information about their salary and from which I infer the true average salary. So, the true average salary is a truth, that is what characterizes the population - that age group - and estimate is what I obtain from data. Obviously, estimate will always be different from truth because I have not seen all possibilities. And the estimation theory tells you how to estimate the parameter of interest, in an efficient consistent way, from the data recorded. Associated with this, we have this estimation. You have this terms called statistical and estimator. A statistic is nothing but a mathematical function of your observations. Again, the sample mean is a statistic; sample variance is a statistic. It's simply a mathematical operation that you are performing on the observations. An estimator is also a mathematical function. You can say it's a soft device or virtual device that operates on the data in a mathematical way



and gets you an estimate of what you want. The difference between statistic and estimator is fairly subtle. Prima facie they appear to be the same, but the difference is, statistic is merely a mathematical function not necessarily devised for any purpose; whereas, an estimator is a mathematical operation that is carried out with a specific purpose of estimating something in mind okay.

So, if you go down further in the estimation theory, you will see more discussion on this difference between statistic and estimator, but as far as the mathematical operation, both are the mathematical functions of the data. So, in estimation theory, we use the statistics and estimators quite often. So, sample mean is both a statistic and an estimator of the true mean alright, because sample mean is devised for estimating the true mean, and again associated; as you can see it's all about the population and sample; you have ensemble and sample averages.

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Data Analysis

Terminology

4. **Ensemble and sample averages:**
Ensemble average is the average computed across all set of possibilities (population), at a single point in time (or another domain). Technically this is known as **Expectation**. Sample average is the average that is computed over all observations for one data record.
5. **Bias:** It is the systematic error in an estimate $\hat{\theta}$ of a parameter θ introduced by virtue of the estimation method. Bias is a measure of **accuracy**. Mathematically, $\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta_0$, where $E(\hat{\theta})$ is the ensemble average of estimates across all data records.
6. **Variability:** It is a measure of the spread of estimates obtained from all experimental records. Variability is a measure of the **precision**. Mathematically, $\text{var}(\hat{\theta}) = E(\hat{\theta} - E(\hat{\theta}))^2$, i.e., the ensemble average of the squared errors in the estimate across all records.

Ideally one would like to obtain estimates with zero bias and as low variability as possible!

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Ensemble average is the average that you would compute from all possible values that you could have ever seen, which, of course, we never get to see; in most of the situations we never get to collect all the possible data records or in the salary example, it may not be possible to collect the data from all possible people in meaningful time, Sample average, of course, is simply the average is computed from these samples. And these



averages are used to define what is known as an ergodic process and so on; we will not go into that.

Finally, the two terms bias and variability are very important in data analysis because they both talk about the error that one makes in estimating some parameter or a set of parameters. Bias is a systematic error that one ends up making; every estimation exercise will give you an estimate that's in error; there's no doubt about it. The question is whether the difference between estimate and truth is by chance or due to some deficiency or shortcoming of your estimator.

How do you know if there is bias in your estimation exercise or in your estimate? Imagine this thought process; I have collected let's say hundred data records or thousand data records or thousand data records for one process, and I compute the sample mean for each of the data records. If I average, let us say, not even thousand, about a million data records, per data record I have one sample mean; and for million data records, therefore I have million sample means. If I average the sample means, the million sample means that I have, I should be very, very, very close to the truth. If there are only million possibilities, I should get the truth, but if there is a difference that still exists between the average of this million sample averages or sample means and the truth, then there is a systematic error. This ensemble average that we talked about, as I have written above, is also known as expectation. So in other words, if coming back to bias, if the expected value of the estimate is different from the truth, then we say that there is a bias in the estimate.

Now, coming to variability, it's a measure of the spread of estimates across different data records. It's very, very important because in practice I only work with one data record and I make all my inferences from a single data record. I need to know, I need to know the answer to the question - had I collected another data record, and had I drawn the inferences from the data record, would those inferences be significantly different from what I have seen now. They will be different, obviously, numerically, but will the difference be too large? And if that is too large, then we say there is a large variability in the estimation or in the estimate. For this reason, variability is also referred to as precision. The lower the variability in the estimate, the more precise is the estimator. In



contrast, bias measures accuracy. Lower the bias, lesser the inaccuracy. In fact, zero bias would mean, it's an accurate; the estimator is an accurate one gives you an accurate estimate of the truth. Typically, we are more worried about variability. Yes, of course we want an accurate estimate; ideally, we want an accurate and a precise estimate. It's not possible to obtain both an accurate and precise estimate from finite samples, but what definitely we would like to have is low bias or zero bias if possible, and as small or low error as possible. That is why you see on many instruments that are being supplied to laboratories, you have these instruments being qualified as high precision instruments.

Remember, estimation is not just about we using a mathematical formula on data. Even a sensor actually, the reading that it gives you is an estimate of the truth. So, one should have a very broad view. So, what the company or the manufacturer means by high procession instrument is, if you were to repeatedly measure the same variable with the sensor, you would not see too much variability. Definitely there will be some variability. So, take a thermometer, and place it on your body, and take the reading. Note down the reading. Then again, take the thermometer, then place it on the same part of the body and take the reading. Of course, you will probably see the same reading there, because thermometers don't report beyond first decimal, but you may see an actual variability if you have thermometers reporting may be the fourth or fifth decimal and so on. That is called resolution of the instrument. But, anyway, the point is repeated measurement of the same variable definitely will give you different readings, but you don't want wide variations in those readings. So, you want a high precision instrument.

So, with that we come to the close terminology or the major terminology. Of course, there are many other terms that one would want to be familiar with, but those are the basics.

We will conclude the lecture with the procedure that one needs to follow for data analysis, and then, move on to a couple of illustrative examples in R.



(Refer Slide Time: 42:26)

A presentation slide titled "Data analysis procedure" in a blue font. The slide contains two main text blocks: "Almost all data analysis exercises are **iterative**, not necessarily sequential." and "The success of any data analysis exercise depends on two factors, **quality of data** and **methodology of data analysis**." The slide also features the NPTEL logo in the bottom left corner and footer text: "Arun K. Tangirala, IIT Madras", "Introduction to Research", and the slide number "12".

Data Analysis

Data analysis procedure

Almost all data analysis exercises are **iterative**, not necessarily sequential.

The success of any data analysis exercise depends on two factors,
quality of data and **methodology of data analysis**.

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The foremost thing to remember is that almost all data analysis procedures are iterative in nature. Don't expect step 1, step 2, step 3, step 4, done. It's never going to be that way, very rarely. So, they are not necessarily sequential, but there is a certain order in it. It doesn't mean that you go about doing in a haphazard manner. And also, remember that the success of any data analysis exercise depends on two things: the quality of data that you collect and the systematic procedure or the methods that you have, the tools that you have used in data analysis or the entire method methodology of data analysis that you followed. If the data quality is bad, you cannot do much. Even if you follow systematic procedure, you may not get marvellous results; and even if the quality of data is good, you follow a very unsystematic procedure and you use a very poor estimator, you may end up with a poor estimator. So, both are important.

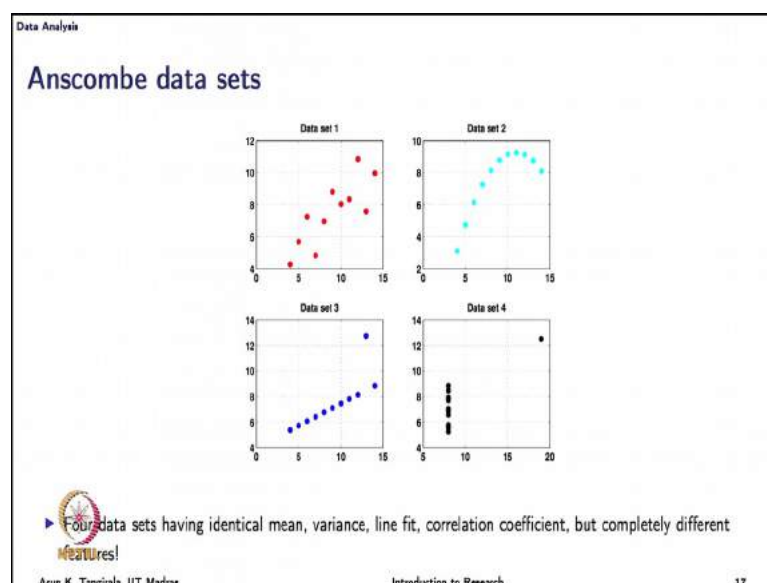
Now, of course, quality of data is many a times is not in our hands, but you should still be aware of the theory to be able to say when you look at a data and say that well this data quality is not good, so I should not expect good results. Suppose, you are the consultant, and you have been given some data, you are hired for the data analysis by a process industry and you are given some data, the first thing that you should be doing is look at a quality of a data and make an intuitive assessment of how good are the nature of the results that you can expect. If the data quality is too bad, you should return the



data and say well don't expect miracles. The data has to be collected in a much better manner, so that is so-called informative alright. So, please keep that in mind.

Now, we revisit the schematic that we saw at the beginning of this lecture which gives us some idea of what is the systematic procedure of the data analysis. So, at the top, we have, of course, the data coming out of the process - well, out of the sensors - and the first step in data analysis is going through some preliminary steps which involves three things: visualization, quality check, and pre-processing. Now, again within this, it's not necessarily sequential. Generally, the first step is to plot our data because the numbers may not tell you much, but pictures tell you a lot more and to emphasize that let me actually take you to the slide here.

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This is a very classical example based on what is known as an Anscombe data set. And this data set is also available in R as many data sets that come with it when you install. Now, the beauty of this Anscombe data set is... let me explain what I show you here. I have four data sets here, of pairs of variables; think of $x_1 y_1$, $x_2 y_2$, $x_3 y_3$ and $x_4 y_4$. They all look visually very different; however, if you had not looked at the data, and you had computed, for example, mean or variance or you had fit a model between the respective pairs or compute at a correlation, they would all be identical, they would all be



identical. So, based on the descriptive statistics or even your regression, you would not be able to make any distinction between these four data sets. However, as you can see, there is such a wide difference in the features of each of this data. For example, if you look at the data set 2, there is a parabolic kind of relationship, whereas data set 4, all the points fall on a straight line. Practically, there is no change in x , only y is changing apart from one outlier. So, this data set is also very useful to show you what outliers can do to your data analysis.

Outliers are those data points that fall out of the normal thing or that fall out of the region where the rest of the data. In fact, if you look at data set 3, all the data points lie on a straight line except for one data point that's an outlier right. So, two data sets can look completely identical when you look at descriptive statistics alone, whereas the moment you plot them, you know there is a huge difference. And then, of course, we will take corrective action and the necessary steps to be able to extract the features correctly. So, that hopefully this drives home the need for visualizing data prior to any data analysis.

And then, of course, we just did a quality check. For example, we observed the presence of the outliers. That's very important. Many a times data come with missing values. You have to decide how to deal with missing values. There is a whole lot of theory available whether you should replace it or live with it, interpolate, and so on. So obviously, it's not possible to talk about it here.

And then, other kinds of pre-processing that you may have to do. For example, you may have to sync the time stamps in multivariable processes; where you collect data from different variables, you may have to sync the time stamps or sometimes one variable is collected at one sampling rate, another variable is collected at an another sampling rate, then you may have to determine how to pre-process.

Sometimes, in fact, lot of times data comes with noise and we may have to remove the noise prior to any other step that you we may take as analysis, because noise really can hijack the inferences that we draw. It can significantly affect the course of data analysis. So, we may want to filter out noise.



So, there are many, many reasons why you want to spend time in this preliminary step which is perhaps the most important. Some bit of ... there is a bit of drudgery involved in it. It can really get on to your nerves, and there is a golden 80/20 rule which says of the 100 percent of the time that it takes for overall data analysis, you may have to spend 80 percent of the time on data pre-processing. That's the kind of effort that you may have to put in sometimes. If you are lucky, of course, you can just go pass through this step in a breeze.

Then comes, of course, the core part of your analysis which is modeling or estimation or exploratory or may be predictive kind or prescriptive kind of analysis diagnosis and so on, the results of which feed into many applications, such as monitoring, control, optimization, innovations as we have discussed earlier.

Now, very importantly, in all of these steps, you should keep in mind two things. One as the arrows indicate, they are not necessarily iterative. That means, sometimes you may have to go back. So, if I find that the data quality is poor, I may have to go back to my experiment or the person who gave me the data to give me better data or that I find that the features that I want are not present in the data, I may have to redo the experiment or after estimation I figure out that the data quality is not good enough or that I had not done sufficient pre-processing, I may have to go back and so on. That is the first point.

And the second point is if you have any domain knowledge, and that's very, very important; obviously, you should know where the data is coming from, whether it is coming from a biological process, an econometric process or an engineering process or some social media process and so on... or a chemical process and so on. That knowledge should be factored in at every step even in choosing what should be the sampling rate for your data - that is how often you should collect the data; that's where your journey of domain knowledge begins.

And if I know, for example, I am looking at a PH system, I know it's a non-linear system right. Then when at the modeling stage, I should not even think of building linear models, but if I don't know where the data is coming from, it just remains a data set and its of limited use to me, its just some kind of a toy that I am playing around with. So, the



domain knowledge has to be used at every stage in the best possible way and that completely depends, of course, on how much domain knowledge you have. It may help you in choosing the structure of the model, the kind of features that you should search for or the level of noise that you can expect - whether it is a sensor noise or a process noise and so on, or even in your applications. So, that should be kept in mind. All-in-all iterative in nature, but there is a certain systematizm to it - choose your models properly and make sure that you go through the data quality check; visualize the data; pre-process; then choose your models properly, if you are building models and so on. We will talk about that in the next lecture on modeling skills or estimate properly and so on.

So, we will now expand on this a bit more. So, before we proceed to data analysis, the first step, as we learned, is to ask certain preliminary questions.

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Data Analysis

Preliminary questions in analysis

- ▶ **What type of analysis is required?** Objectives or purpose of data analysis (e.g., exploratory or confirmatory, descriptive or predictive)
- ▶ **Where does the data come from?** Sources of data (domain)
- ▶ **How was the data acquired?** Method of data acquisition / collection (sampling mechanism, hardware used, etc.)
- ▶ **How informative is the data?** Quality of data (e.g., outliers, missing data, anomalies, noise levels)
- ▶ Lastly, but perhaps the most important, **what assumptions are being made on the data generating process?** (e.g., deterministic / stochastic, linear / non-linear, stationary / non-stationary)

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What type of analysis is required? Whether it is descriptive, predictive and so on, because that changes entire course of analysis. Where does the data come from? Sources of data - we have just asked, we have time domain, data frequency domain, data biological process data and so on. How was the data acquired? What kind of hardware was used? Was it a human who collected the data? Was it a sensor? If it was a sensor, what was the sampling rate? So many questions may have to be known for us to be able



to say how good a quality of analysis that I can expect, and what should be the kind of analysis tools that I should be using, and what should I be searching for. And then, of course, how informative is the data – that's a bit qualitative, but there are also certain quantitative measures of information. We will not talk about it, but as I said, it basically pertains to quality check whether there are missing data outliers and so on.

And lastly, but perhaps the most important - what assumptions are being made on the data generating process should be clear. You may be wrong. In the beginning, we may make wrong assumptions, we may assume a stochastic process to be deterministic, we may assume a non-linear process to be linear, a steady state process to be dynamic and so on; it's possible. But we should stick to the assumptions that we have made throughout the data analysis, and choose and be the consistent and commensurate - that is we should choose right tools and so on, so that if you have made a mistake, in the end, we do know that we have made a mistake. And then, we go back, and make corrections. And I am not going to really go into this detail. We have already talked about it. So, once you are through the preliminary questions, we prepare our data, through many data pre-processing steps; depends on what kind of problems the data has.

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The slide is titled "Data Analysis" and "Preparing your data". It contains three main sections with bullet points:

- ▶ **Data pre-processing**
 - ▶ Cleaning up the outliers or faulty data, re-scaling, filling in missing data
 - ▶ Robust methods can handle outliers or faulty data without an explicit treatment
 - ▶ Data reconciliation - adjusting data to satisfy fundamental conservation laws
 - ▶ Synchronization of time-stamps in multivariate data
- ▶ **Partitioning the data**
 - ▶ Partition into "training" and "test" data sets.
 - ▶ In certain applications, several data sets may have to be merged.
- ▶ **Assessing quantization or compression errors** for data from historians.

At the bottom left is the NPTEL logo, and at the bottom right is the text "Introduction to Research".

For example, if the data is outliers, I mean, I have to clean up, but there are also robust



methods. We say let the outliers stay in the data and still going to get you decent estimates and so on, or we may have to filter and so on.

The one thing that I didn't mention earlier is partitioning the data. If you are building a model or if you are performing classification by extracting features and so on, always you should have a part of the data in hands for cross validation. We should take the model that you trained on the data and test it on a fresh data which has come out of similar experimental conditions, or if we have built a classifier based on the training data, then we should test the classifier on the part of the data set that I have updated. So, always keep portion of the data set for cross validation.

And sometimes it may be necessary to assess quantization or compression errors that may arise in data that come from historians. These days it may not be so bad, but if you are looking at data may be 10 year ago, and so on, they be in highly quantized or compressed, and that can significantly change the course of your data analysis. They may introduce unnecessary non-linearities which are not a part of the physical process. It's just a part of your data acquisition process.

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A presentation slide titled "Data Analysis Visualization". It features a light blue header with the title. Below the title, a purple box contains the text: "Always convert numbers to pictorial representations before, during and after mathematical operations - visual analysis reveals a wealth of information". A bulleted list follows, listing various visualization methods: "Simple trend to high-dimensional plots", "Bar charts, Box plots, histograms", "Color maps, contour plots", "Specialized plots", and "...". At the bottom left is the NPTEL logo, and at the bottom right is the text "Introduction to Research". The slide number "16" is in the bottom right corner.

Data Analysis

Visualization

Always convert numbers to pictorial representations before, during and after mathematical operations - visual analysis reveals a wealth of information

- ▶ Simple trend to high-dimensional plots
- ▶ Bar charts, Box plots, histograms
- ▶ Color maps, contour plots
- ▶ Specialized plots
- ▶ ...

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And of course, we talked about visualization where in we could look at simple trend to



high dimensional plots, bar charts and so on. The list is obviously not exhaustive, but always convert numbers to pictorial representations before, during, and after analysis. So visualization is a part - integral part - of every stage of the data analysis. So, we have talked about the Enscombe data sets that basically emphasise a need for visualization.

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Data Analysis

Core steps

- ▶ **Select the domain of analysis** (e.g, time, frequency).
 - ▶ Transformation of data does wonders (e.g., Fourier transform for detecting periodicities)
- ▶ **Dimensionality reduction** for large-dimensional data
 - ▶ Multivariate data analysis tools (e.g., PCA, NMF)
- ▶ **Choose the right mathematical / statistical tool** appropriate for the purpose of analysis and commensurate with the assumptions made on the data.
 - ▶ In hypothesis tests, the appropriate statistic and significance level should be chosen
 - ▶ In analysis of variance (ANOVA) problems, sources of variability should be clearly identified.
 - ▶ In modelling, appropriate regressor set and model structure must be chosen.
- ▶ **Always factor in prior and/or domain knowledge** at each step.

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The core steps in analysis is what we will talk about and then conclude the lecture. Select the domain of analysis. What we mean by domain here is whether we want to analyze the data in the same domain that it has been obtained. Normally the time is a domain -independent domain - in which we collect data, but we may not want to analyze the data in the time domain. Sometimes we may want to move to frequency domain. In fact, that 's a very common procedure that's followed in periodicity detector. If you look at atmospheric data or even data coming from vibration machinery, ECG or EEG data, for all such data, the frequency domain analysis is a very convenient domain of analysis because it reveals the features that can be directly related to physical characteristics of the process. And of course, many a times you have to do joint time frequency analysis. Wavelets is one of them. So, select the domain of analysis and time need not be, it could be on space also. Many a times data can come to you as a function of frequency straight away, like observance spectroscopy data comes as a function of frequency from the chromatograph.



Then the second one that we may have to take a decision on is dimensionality reduction and particularly when we are looking at large dimensional data. This is now increasingly becoming common. It's not possible to analyse multivariable data in large dimensions, particularly when it goes to hundreds and thousands and so on, and they are very common today. And machine learning algorithms come very handy in this respect. Principal component analysis, non-negative matrix factorization, and many other dimensional reduction techniques are very powerful in analyzing large dimensionality data, large dimensional data. So, we have to take a decision whether to analyze in the dimensions that in which I have obtained or in a reduced order or a lower order dimension.

And most importantly is to choose a right mathematical or statistical tool. For example, if I want to estimate the mean, I can choose mean, sample mean or sample median as an estimator. Now, when should I choose sample mean as an estimator, when should I choose sample median - there is no perfect answer for it, but there are some very good guidelines. If I know that the data is free of outliers and it's coming out of a Gaussian distributed process, sample mean gives you the most efficient estimator - meaning estimate to the lowest possible error among all the estimators; but if I note that the data could be corrupted with outliers and I don't want my estimator to be affected by it, then sample mean is not robust. That means it's very sensitive outliers, then I should use the sample median. This depends on prior knowledge that I have and the knowledge that I have acquired from my preliminary data analysis step. And these kinds of decisions have to be made at every stage and there may not be unique answer.

Remember that; in data analysis, there is no unique answer necessarily at every stage of your analysis and even towards the end, even when you build a model, there may be many models that can explain the given data and you may have to actually pick the model based on some criteria. And likewise, in hypothesis test, we have to choose an appropriate statistic and analysis of variance. We may have to first know what the sources of variability are; at least guess fairly well enough or in modeling as I said you may have to choose the right regressor set and model structure and so on, and always factor in prior and/or domain knowledge. You may know a priori something, not necessarily related to the physics or the process, but may be from the sensor



characteristics and so on, and of course, you may know something about the domain or the physics or the chemistry or the biology of the process that you can easily factor in and save a lot of headaches.

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Data Analysis

Assessing and reporting your results

- ▶ Compute confidence regions and errors in estimates.
- ▶ Conduct hypothesis tests on estimates.
- ▶ Cross-validate models on fresh data sets (did the model over learn?).
- ▶ If multiple models are identified, use a mix of information-theoretic measures (e.g., Akaike Information Criterion) and end-use criteria to **select an appropriate model structure**. Remember we seek the **best working model**, not necessarily the right model!
- ▶ Where forecasts are computed, confidence bounds should be provided.

Any report from a data analysis exercise should include estimates, their error bounds and results from statistical tests of validation.

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And finally, once you are done with all your analysis, this is the most important part - assessing what results you have obtained and reporting them is a very important step. Always subject any parameter, estimate to a hypothesis' test - whether it is significant enough, whether you have just obtained a non-zero value by chance, whether truly you should have obtained zero. So, you should subject hypothesis test of the parameter being zero – that's null hypothesis, and then, carry on in a statistically sound manner.

And secondly, report confidence regions and errors in estimates. In fact, computing confidence regions and errors are an integral part of your hypothesis test. And then, if you are building models, then cross validate models. When I say models here, need not be regression models, it could be classifiers also. And if multiple models are identified, use a mix of criteria; as I just mentioned earlier, there may not be unique answer; typically, that is the case. My students always ask me, well I have two or three models explaining the data, which one should I pick? Well, the first thing is to make sure that you followed a very systematic procedure to minimize the number of solutions that you



have and even despite the best of efforts you may end up with multiple solutions, very common. Then you apply what are known as information theoretic measures such as Akaike information criteria or Bayesian information criteria. And also keep the end use of the model; may be your model has to be implemented online; so, simple model is preferred to a complicated one with a bit of sacrifice on the accuracy of the prediction and so on.

So, again there is no unique answer, but in the end, the final results that you report should have two things: a statistical backing to it and a sound argument to why you have decided to report these results and why the results are the way they are.

So, therefore, you should remember data analysis is not just science alone. Yes, a big part of it is science, but an important part of it is also an art. Data analysis is, therefore, both a science and an art. And never report estimates without reporting confidence regions and errors; as much as possible, you should definitely do that. Of course, in very complicated parameter estimation problems, it may be very difficult to obtain confidence regions, but you should try despite a best of your... try to put in the best of your efforts to report the errors and the confidence intervals.

(Refer Slide Time: 63:07)

Data Analysis

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These are some of the references. Of course, no way it's exhaustive. I have taken some of the examples and some material from these books; these are quite common books of which you may find the first book on Random Data Analysis very, very useful to you because it has been written by people who have practiced for 30-40 years, seen different kinds of data, and they have examples, very good examples in their text books. And the remaining three, at least the book by Johnson and Montgomery and Runger, they they are classical books in probability and statistics - applied probability and statistics. The book by Ogunnaike gives you the fundamentals of probability and statistics for engineers. If you are an engineer, you may be really interested in looking at the book. Of course, there are hundreds of other books that you can refer.

So, hopefully you enjoyed the lecture and you understood at least what is data analysis all about, what precautionary steps that you have to take, and how careful you have to be in data analysis, what is the systematic procedure, and some terminology. We will quickly look at a couple of illustrative examples in R and that will hopefully supplement this lecture.

Thank you.

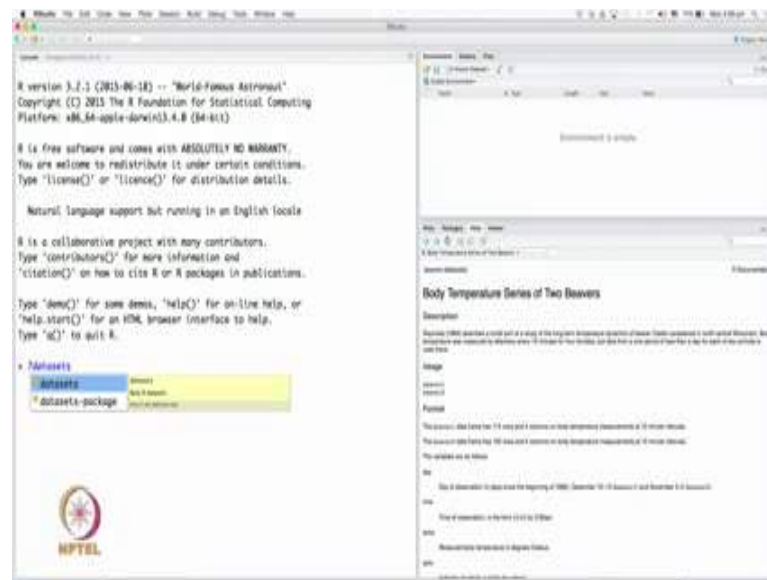


Introduction to Research
Prof. Arun K. Tangirala
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture – 15 (2 B)
Data Analysis

Welcome to the demonstration part of the Data Analysis lecture, where I am going to show you how to just analyze a couple of data sets in a package or software package called R, and as I had said R is an open source and free software, very popular. It's available on all platforms. I am not going to show you how to install R. You can just go to the website, download R; and for R, there is a very nice graphical user interface - GUI - called the R studio, and I strongly recommend you download that and install. All of that should be done fairly easily. I have done that on my computer. So, if you have a computer handy, you may want pause this video, install, and then, come back, and then, carry on with the instructions that I go through in this lecture.

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So, I have R studio installed. Of course, before installing R studio, I have installed R as well and I am going to open this R studio interface. This is how the interface should look like. Of course, it can look a bit different depending on how you configure it. The



purpose of this lecture is not to show you what R is about and so on. There are many, a number of, tutorials available on the web for that purpose. The purpose is to just show you what careful data analysis is about, just with a couple of data sets. Now, as I had mentioned during my lecture, the R package comes with a lot of data sets that the user can clear on with. That's one of the nice features of R.

Now, to know what data sets are available, one could use this syntax in R. The question marks syntax in R basically brings up help, and the nice thing about R studio is, it completes the keyword or whatever you are typing if it is a part of the R system and data sets is a package in R. Therefore, as I typed data sets, you can see it's trying to complete that for me and its also saying that is a package. So, what I am typing, therefore, is right; what I am looking for, it makes sense.

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Let's ask for help on data sets and it says, R data sets is a package. To know what data sets are available in R, go to the index at the bottom here, and that brings up the documentation for the package data set. I have the version 3.2.1 and they are listed in alphabetical order. Of course, it's impossible for me to go over all these data sets.

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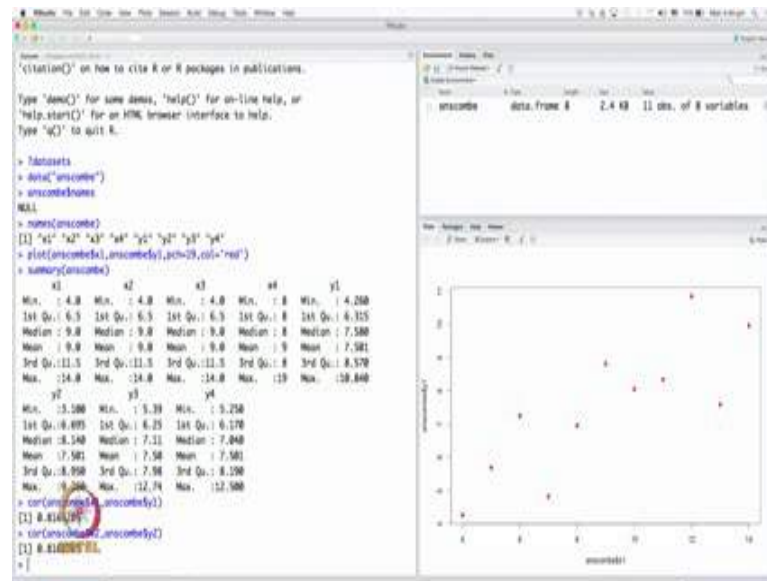


Let's look at this particular data set called Beavers, but before I do that, I want to draw your attention to this Anscombe data set that we talked about in the lecture. If you click on the Anscombe data set, it gives you a description of the data set. In fact, we can load that to begin with. To load a data set that comes with the R package, all you have to do is, type `data`, and then type the name of the data set, and once again R studio shows the list of the completions that one can have, and Anscombe is the one that I am looking for, and I choose that. Once I do that, as you can see on the top right, Anscombe data set has been loaded. Don't worry about the promise part; it's just not exactly loaded. The moment you talk about Anscombe or in the sense in the command prompt, the moment you start typing the name of the data set, it shows you the true nature of the data - whether it is a time series data or regular data and so on. It shows that is a data frame. A data frame in R is nothing but a matrix of data, but with the columns labeled, and the labels of Anscombe or any other data frame can be easily found.

For example, we could ask what are the labels for Anscombe data set. This is the syntax. So, these are the labels that go with the columns. Anscombe data set has 8 columns. Remember we had four pairs 4 x(s) and 4 y(s) and one could plot the x 1 verses y 1 or x 2 y 2 and regenerate the plots that I showed you in the lecture. For example, if I would, if I want to plot the Anscombe x 1 y 1 pair, then I could, I am just recalling the commands that I have in my history and that's a nice feature of R studio. So, here I am plotting x 1, sorry y 1 verses x 1 of the Anscombe data set, and the syntax here tells me PCH equals 19; these are all optional syntaxes. If I omit those, then it will just simply produce a scatter plot, but I don't, I want something more. I want solid circles that are colored red.



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So, PCH19 is basically telling which character it should use for plotting the markers, and that, and the color tells the plot to use a red color. So, if I do this, then it produces as you see on the right bottom screen, I have here the y_1 versus x_1 . As you can see on the y and x axis labels, you can also, of course, enhance and decorate these plots by providing x y labels. I am not going to go into that, but main thing that you should observe is this plot is exactly identical to what you have seen in the lecture and I welcome you to reproduce the other plots that I have shown you in the lecture.

And of course, what one could ask for is the summary statistics. Summary statistics are nothing but mean, median, and so on. So, we could ask for the summary statistics for the Anscombe data set. The nice thing about the summary command in R is that for a data frame, it looks through every column and reports the mean, minimum, maximum, median and so on. Particular attention that I want to draw your attention to is the mean. So, mean of x 1, x 2, y 2 or if you can look at mean of y 1 and mean of y 2 and y 3, y 4 are identical; may be mean of y 3 is just differing by one-third decimal. They all have the same average.

In fact, you can also check if they have the same correlation; that is x_1 and y_1 have the same correlation, x_2 and y_2 have the same correlation and so on. So, how do you



compute correlation? Let's say that we want to compute the correlation between x_1 and y_1 of the Anscombe data set. This dollar operator is an operator in R which allows you to access the columns of a data frame. Data frame is different from matrix but it looks like a matrix, but the dollar operator applies to data frames, and lists, and so on. So, this is the correlation between x_1 and y_1 . Let's see if they have the same - x_2 and y_2 have the same correlation; so exactly the same correlation. So, every pair in the Anscombe data set has the same correlation despite looking strikingly different.

Now, let's move on to the next data set that I want to show you. In fact, the nice thing about R is that it comes, although it comes with a few installed packages by default, one can go to the R website and look for packages of interest from the thousands of user contributed packages meant for additional purposes and there are many, many such packages, play around with them and the other nice thing is each package comes with a data set that you can test the routines in that package. Again, if you go to the R tutorial, you can realize how to do that.

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```

> library(ANSI)
> data(anscombe)
> summary(anscombe)
      x1      x2      x3      x4      y1      y2      y3      y4
Min.   4.0 Min.   4.0 Min.   4.0 Min.   4.0 Min.   4.200 Min.   4.200 Min.   4.200 Min.   4.200
1st Qu: 6.5 1st Qu: 6.5 1st Qu: 6.5 1st Qu: 6.5 1st Qu: 6.315 1st Qu: 6.315 1st Qu: 6.315 1st Qu: 6.315
Median: 9.0 Median: 9.0 Median: 9.0 Median: 9.0 Median: 7.580 Median: 7.580 Median: 7.580 Median: 7.580
Mean   9.0 Mean   9.0 Mean   9.0 Mean   9.0 Mean   7.581 Mean   7.581 Mean   7.581 Mean   7.581
3rd Qu:11.5 3rd Qu:11.5 3rd Qu:11.5 3rd Qu:11.5 3rd Qu: 8.570 3rd Qu: 8.570 3rd Qu: 8.570 3rd Qu: 8.570
Max.   14.0 Max.   14.0 Max.   14.0 Max.   14.0 Max.  10.840 Max.  10.840 Max.  10.840 Max.  10.840

      y1      y2      y3      y4
Min.   4.200 Min.   4.200 Min.   4.200 Min.   4.200
1st Qu: 6.315 1st Qu: 6.315 1st Qu: 6.315 1st Qu: 6.315
Median: 7.580 Median: 7.580 Median: 7.580 Median: 7.580
Mean   7.581 Mean   7.581 Mean   7.581 Mean   7.581
3rd Qu: 8.570 3rd Qu: 8.570 3rd Qu: 8.570 3rd Qu: 8.570
Max.  10.840 Max.  10.840 Max.  10.840 Max.  10.840

> cor(anscombe[,1:8])
      x1      x2      x3      x4      y1      y2      y3      y4
x1 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
x2 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
x3 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
x4 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
y1 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
y2 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
y3 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
y4 1.000 1.000 1.000 1.000 1.000 1.000 1.000 1.000
> data(beavers)
  
```

So, let's load another package called Beavers, and to know what these Beavers data set is about, we will go back to the help, and pick the Beaver's package. It says it's a body temperature series of two beavers. Beavers are this huge rodents that are found in North



America and so on, and they are very well known for building dams, small dams, and walls, and so on. So, they are kind of a beaver in that sense.

So, these Beavers has two data sets in it, beaver1 and beaver2 corresponding to the body temperature of two different beavers, and we look at the first data set - the beaver1; it has about 114 data points in it, collected at 10 minute intervals using a technique called telemetry and there is more information on this in the documentation.

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```

> #load datasets
> data("beaver")
> as.data.frame(beaver)
NULL
> names(beaver)
[1] "x1" "x2" "x3" "x4" "y1" "y2" "y3" "y4"
> plot(as.data.frame(beaver), pch=19, col="red")
> summary(as.data.frame(beaver))
      x1      x2      x3      x4      y1      y2      y3      y4
Min.   4.0 Min.   4.0 Min.   4.0 Min.   0 Min.   4.200
1st Qu. 6.5 1st Qu. 6.5 1st Qu. 6.5 1st Qu. 0 1st Qu. 4.315
Median  9.0 Median  9.0 Median  9.0 Median  0 Median  7.580
Mean    9.0 Mean    9.0 Mean    9.0 Mean    0 Mean    7.582
3rd Qu.11.5 3rd Qu.11.5 3rd Qu.11.5 3rd Qu. 0 3rd Qu. 8.570
Max.   14.0 Max.   14.0 Max.   14.0 Max.   0 Max.   12.840

      y1      y2      y3      y4
Min.   3.250 Min.   3.30 Min.   3.250
1st Qu. 6.095 1st Qu. 6.25 1st Qu. 6.170
Median  7.140 Median  7.11 Median  7.040
Mean    7.581 Mean    7.58 Mean    7.581
3rd Qu. 8.570 3rd Qu. 8.58 3rd Qu. 8.570
Max.   12.840 Max.   12.74 Max.   12.500

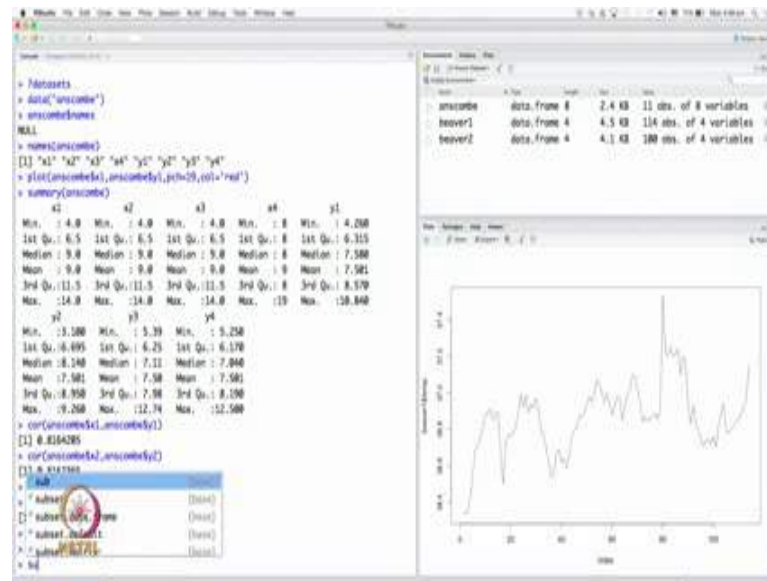
> cor(as.data.frame(beaver))
[1] 0.8164205
> cor(as.data.frame(beaver)[,1])
[1] 0.8164205
> data(beaver)
> names(beaver)
[1] "x1" "x2" "x3" "x4" "y1" "y2" "y3" "y4"
> plot(beaver)

```

So, to load the Beavers data set, as usual type data beaver, and as you can see in the environment here, the work space beaver1 and beaver2 have been loaded. Once again, the type of data will be shown the moment you access the beaver1. So, for example, if I want to know what type it is, just start typing the name and R studio will show you that it is a data frame as well. So how many columns does the data frame beaver1 have? It has 4 columns. To know what are the names of those, once again we type... we look at the attributes or the names, sorry, of the columns, and it shows that the first two columns give me the day and time at which this record measurements were taken, and the third one is a temperature that's of interest to us.



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Let's plot the temperature. So, we say plot beaver1 dollar temperature. And when I do this, it produces a scatter plot, but I want ideally also line plot, and I can ask for a line plot if I want, and then this is a line plot. I can also ask for both lines and points and so on; the story is endless; but this is how the temperature series looks like over two days of recording. Now, as you can see, the temperature hovers around a 36-37 and so on and there is some variability to it. Now the question is - should I ask whether this data comes out of a deterministic or a stochastic process? This is univariate series. Suppose, I want to predict what will be the temperature of the beaver the following day; I want build a model, should I treat this as a deterministic process? Now, I can treat this as a deterministic process if I see, at least from a visual analysis, that a mathematical function can explain this nicely.

Now, of course, I can always fit a polynomial of 113th degree to explain the data perfectly, but unfortunately the 113th degree polynomial will fail miserably in predicting the next point. That also is an indication of lack of determinism in the data, and also, it do not have any other factors that affect the beaver's temperature to help me predict this. So, it may be a wise thing to assume that this temperature series is stochastic, but on the other hand I also find some trends, and we will talk about it in the next and final data set that will take up. So, for now we can treat this data set to be stochastic and ask for mean



of beaver. For example, I can ask what is the mean of, sorry, beaver1-dollar temperature alright; we can do that; and that is the average. We can also ask for the standard deviation. You can recall the previous commands in the history by using the up and down arrows and this is the standard deviation and so on.

Now, what we want to do is also compare the means; that is suppose my hypothesis is that these two beavers have the same average temperature right. Definitely the temperature observations that I have are just one of the many possible readings that I could have obtained. So, I am imagining that there is a population of readings for beaver1, and population of readings for beaver2, and I am assuming that both these populations have same averages. Let us say I want test that. Now, statistically that's a hypothesis test. If μ_1 is a true mean for beaver1, and μ_2 is a true mean for beaver2, and here conducting a hypothesis test that μ_1 equals μ_2 or μ_1 minus μ_2 is 0.

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The screenshot shows the RStudio interface. The console on the left displays the following R commands and their outputs:

```

> cor(arscombe$arscombe1)
[1] 0.8164895
> cor(arscombe$arscombe2)
[1] 0.8162365
> data(beavers)
> names(beavers)
[1] "day" "time" "temp" "sex"
> plot(beaver$temp)
> plot(beaver$temp, type='l')
> mean(beaver1)
[1] NA
Warning message:
In mean.default(beaver1) : argument is not numeric or logical: returning NA
> mean(beaver1$temp)
[1] 36.46219
> sd(beaver1$temp)
[1] 0.1935172
> fit_lm1 <- lm(temp ~ day, data = beaver1)
> plot(fit_lm1)
Error in plot.lm(fit_lm1) : 'x' must be numeric
  
```

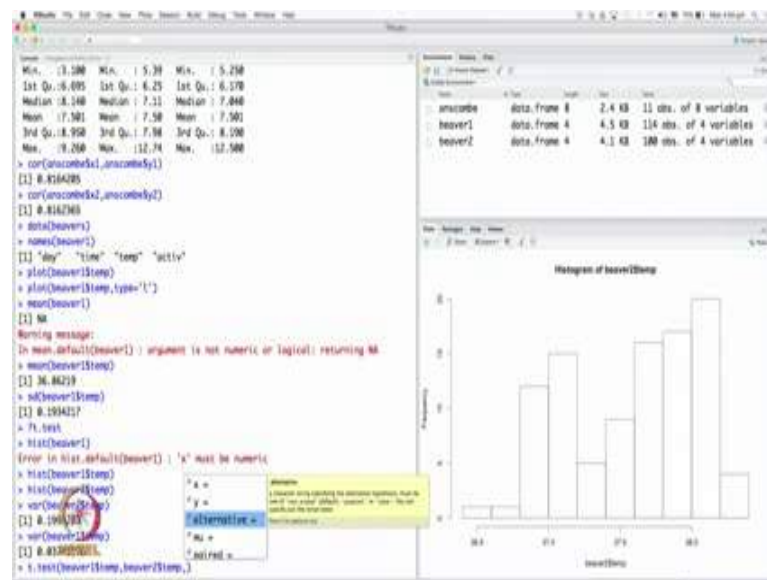
The right pane shows the 'Student's t-Test' plot, which is a boxplot comparing the temperatures of beaver1 and beaver2. The plot shows that the distributions are very similar, supporting the hypothesis that the means are equal.

Now, we are not going to go into the theory, but let me quickly tell you the routine or the command that helps you compare averages of two different populations, and that's the t dot test command, which is also known as a student's t test. The t refers to the distribution of the statistic that we shall use to carry out the hypothesis test. Remember statistic is some mathematical function of the data. So, what is done, at least



theoretically, just to give you an idea, is that sample means of both the beaver1 and beaver2 series are computed, and statistically we look at the distance - you can say so crudely - between the sample means, in presence of the fact that they have come from two different populations. What is a difference between them? I am hypothesizing that the means are the same, then, what is a difference? The difference is in the variability, in the variances; that is, they come out of different spreads. To know that, for example, whether my assumption that different variabilities hold good, we can look at a histogram and look at the spread also. So, let's look at the histogram of beaver1 which gives me an idea of the distribution of the data.

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So, now on the plot you see the histogram of data, it is beaver1 - temperature histogram - indicating some kind of a Gaussian distribution with the mean being the value that we have calculated here. What about beaver2? Does it look like a Gaussian distribution or some other distribution? Now, it doesn't look exactly like a Gaussian; I am not even so close, but what we can see strikingly is that the variability; if I look at the variance of beaver2 verses variance of beaver1, the variabilities - these are sample variabilities; that is, these have been computed from data - they are quite different from each other. So, there is some justification to the assumption that we are making that is the temperature readings of these two beavers have come from different variability. In other words, the



variability of temperature in both these beavers are quite different from each other. And it's possible even among human beings temperatures can vary quite significantly, but the means can be the same. So, we can now test the hypothesis and to know the syntax of any routine, for example, I can press the tab, and it tells me what are the default ones. I have to supply the first series, and then, supply the second series. And then also, there are other options that you want - what is the alternative hypothesis? Whenever I am testing an hypothesis or any null hypothesis there is an alternative hypothesis, then I have to specify. Here the alternative hypothesis is that the means are different from each other.

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```

[1] NA
Warning message:
In mean.default(beaver1) : argument is not numeric or logical: returning NA
> mean(beaver1$temp)
[1] NA
> sd(beaver1$temp)
[1] 0.1846217
> ?t.test
> t.test(beaver1)
Error in t.test.default(beaver1) : 'x' must be numeric
> t.test(beaver1$temp)
> t.test(beaver2$temp)
> var(beaver1$temp)
[1] 0.1996203
> var(beaver2$temp)
[1] 0.83741296
> t.test(beaver1$temp, beaver2$temp)

Welch Two Sample t-test

data: beaver1$temp and beaver2$temp
t = -15.235, df = 135.12, p-value = 2.2e-18
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.6204806 -0.6381334
sample estimates:
mean of a mean of y
 36.86219  37.09679

> length(beaver1$temp)
[1] 139
> data()
> plot(x)
```

In other situations, I may test the mean of beaver1 or the true average of a temperature of beaver1 is greater than that of beaver2 and so on, but we are not interested in that here. So, the default alternative if you look up the t test - t dot test - see documentation it says... it gives you the default values which is for the alternative one it says they are not different and there are many other options. For example, variance being equal - am I assuming the variability to be equal, no, and that's a default option as well. It says false. So, I don't have to do anything with that. And there are a few other options which I am not going to explain now, it requires a bit of theory, but let's do this.

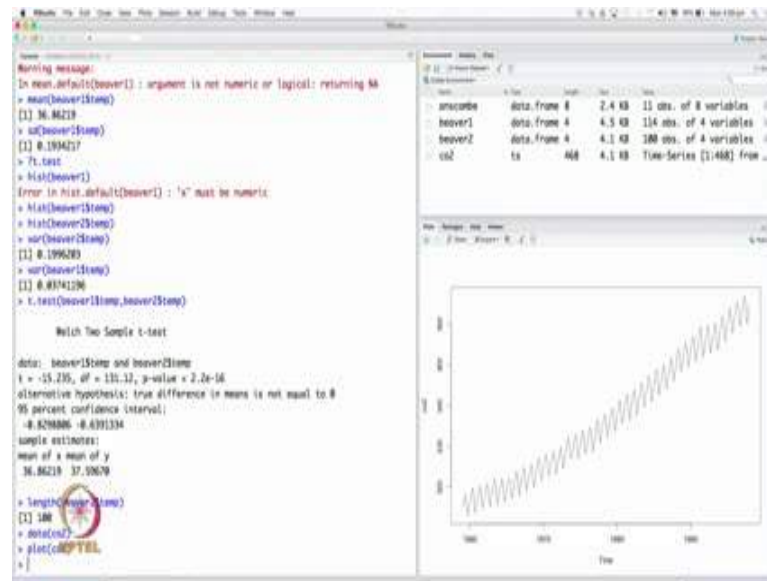


Let's see what the t dot test tells me. It reports a bunch of things, but what is of importance is this p value alright. And of course, it gives you the means of x and y. When you look at the sample means they look pretty close. I mean Of course, in the sense you may say no, I may say yes, and that's the reason why we are performing a statistical analysis; but the statistical analysis with its very, very low P value is indicating that the hypothesis is... null hypothesis is the means are equal does not hold good; it has to be rejected. Whenever the p value is low, the null hypothesis must go; so, that is a nice phrase that you will find in Ogunnaike's book. It is also telling you the alternative hypothesis against which the null hypothesis has been tested, and it also gives you the 95 percent confidence interval. The 95 confidence interval is on something. What is that? It is on the difference in the means. So, what it is testing is that the means are not different from each other, that's null hypothesis. And if you turn to statistics books, they will tell you that a hypothesis test of the form $\mu_1 - \mu_2 = 0$ can be also conducted by looking at the confidence interval for the difference in means.

So, the confidence interval for the difference in means is reported here, and it does not include the postulated value which is 0. If the confidence interval had included the postulated value for the difference in true means which is 0, then we cannot reject the null hypothesis. There is not enough evidence in the data to reject the null hypothesis. In this case, we have sufficient evidence in the data to come to a conclusion that the temperature - average temperature - of beaver1 and average temperature of beaver2 are different from each other. And we have done this despite the fact that the readings in beaver1 and beaver2 are of different sizes. Beaver1 has 114 data points; beaver2 has 100 data points. How do I know that? I can ask for the length or I can look the help and you can see here 100 data points; where as we had 114 data points. So, the theory allows you to collect different sample sizes, but the question is whether this test itself has been conducted correctly. Theoretically does it satisfy theoretical assumptions? Probably not because this t test actually assumes that both the samples have comes from Gaussian distributions, and we have seen that the beaver2 temperature series is not really conforming to the Gaussian distribution, but may be had we collected larger and larger samples, then the assumptions might have been met more strictly alright. So, the point to keep in mind is not just reporting the analysis, but also the assumptions that we have made and whether the data has met those assumptions.



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The final thing that I want to show you is another series, which is the carbon dioxide emissions in a certain region, during a certain period. Again, I welcome you to look up the help on this data set. Just want to show you the data set. And here, I have loaded the carbon dioxide data set and we shall plot this series. And you can see in this series there is a trend, the x axis is time - the year in which it was the carbon dioxide emission was collected. There is a linear trend or probably a slightly parabolic trend, we do not know, but vividly there are two things - a linear trend and then there is an oscillatory nature to it. Now, once again the question - should I treat this as a deterministic process or a stochastic process? On the face of it, it appears deterministic, because it's so dominant, there is a mathematical function that I can fit to explain the trend. I can also explain the oscillations using sinusoids. I can determine the frequency.

How do I determine the frequency? By looking at the periodogram or the power spectral density. You can fit a linear trend. I will show you how to fit linear trends when we go to the lecture on modeling skills. Basically, it is a matter of a regression, but this shows you that there is a possibility that you can think of the series as deterministic. However, once you have removed the deterministic part, you should ask if there is anything left in the series to be explained, and if that has some irregularities, some stochastic nature to it, then you come to the conclusion that the series is a mix of deterministic and stochastic



processes. Here, the determinism is purely as a function of time. In many processes, the deterministic nature can come about by a certain cause that you already know right. In the reactor temperature example that we discussed, if I give you the coolant flow and the reactor temperature series, using the coolant flow you can explain the variations in the reactor temperature significantly; yet there will be something left in the temperature that you may not be able to explain using a coolant flow which will be due to the sensor noise; there you have a mix of deterministic and stochastic process. We are not talking about linear or non-linear here.

Anyway, so hopefully I have introduced to you through these examples first of all the great free-open source software package called R, because some of the students have seen in the forum have requested for introduction to, an exposure to some programming language, but what is needed in data analysis is not necessary programming language, but a nice software tool that confirms to the theory, and R has been written by many of the pioneers in the world of statistical data analysis. Therefore, there is more credibility to it and there is a nice user forum. You can ask any questions and it's continuously developed. R studio is a fantastic piece of GUI for R. So, enjoy and yeah write to us if you have any questions. Hopefully you enjoyed the lecture.

Thank you.

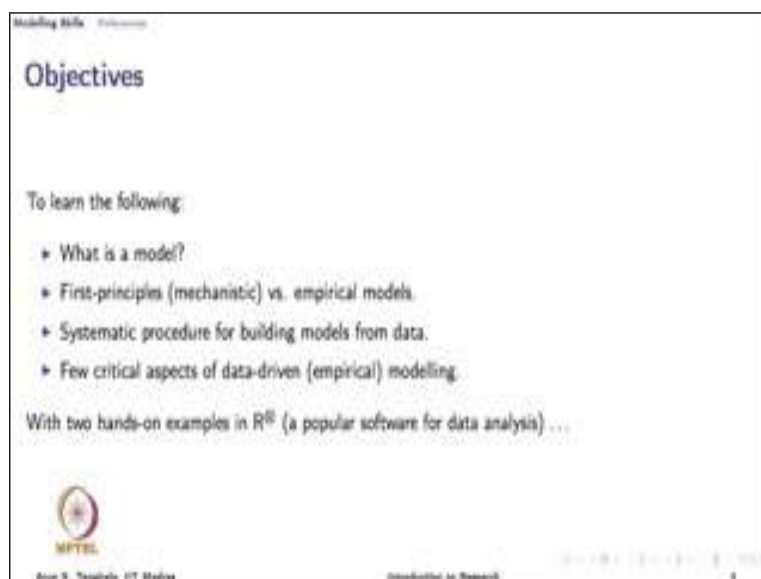


Introduction to Research
Prof. Arun K. Tangirala
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture – 3 A
Modelling Skills

Welcome to the course, once again, on Introduction to Research. In this lecture, we are going to learn certain aspects of modelling; in particular, data driven modelling. So, the lecture is titled as Modelling Skills, but it's not just about skills that we will discuss; we will also discuss what types of models we have, and what are the different ways of developing a model, and so on.

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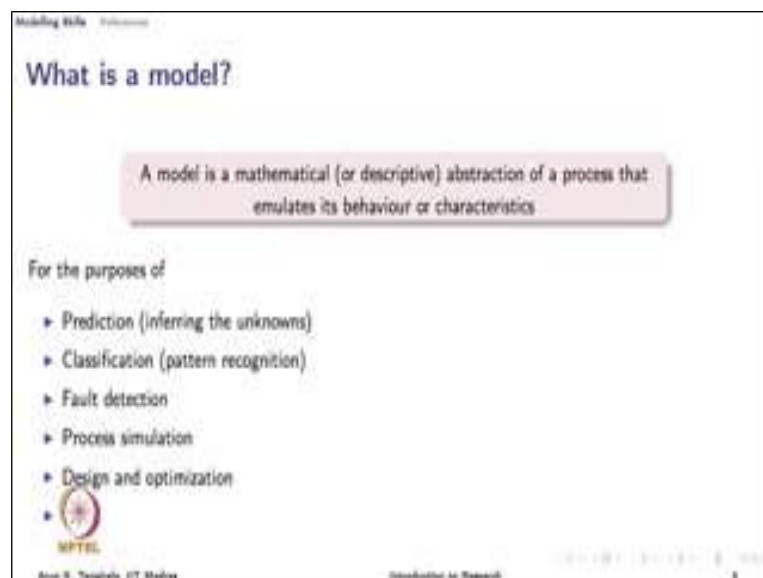
So, in particular, we will learn first what is a model? Of course, all of us know what is a model, and this term - model - is used in many fields with almost similar connotations, but it's good to really clarify it upfront, so that we are all on the same platform, and then, move on to learn what are first principles and empirical models. Now, essentially, these are not just models, these are also modelling approaches. The first principles approach is a fundamental approach, whereas empirical approach is a data driven approach. And then, we shall quickly go through a systematic procedure for building data driven models. As I said, a few moments earlier, we will largely focus on data driven modelling



because that's where a lot of expertise is not available or lot of beginners find it very difficult. And therefore, I would like to concentrate on the same.

And also, talk about a few critical aspects of data driven modelling - what are the things to watch out for, how to go about handling certain aspects such as noise in data, and over fitting or doing the right kind of experiment, and so on. We will also embellish this lecture with a couple of hands on examples like we did in the last lecture on data analysis in R. I hope now, with the previous lecture, after the previous lecture you are familiar with R.

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So, let's begin our discussion by asking - what is a model? In general, we know, we can answer qualitatively what's a model. It's some entity that allows us to emulate a process, the process behavior, process characteristics and so on; this term - model - is not only used in data analysis or research and so on, you can see this term being used even in fashion industry and elsewhere, or even models of houses and so on. So, we have a good feel of what is a model, but let's try and define what a model is. It is a mathematical or a descriptive that is quantitative or quantitative abstraction of a process. It allows us to describe a process in mathematical terms, so that we can emulate or simulate the process behavior.

And why would we need this model? There are several reasons. We know, again, but it's good to list some of the prime end uses of a model. One of the most popular uses of a



model is in prediction. That is as we call as the forecasting, like we mentioned in the last lecture. Once I have a model, and I know the inputs to the model - remember that a model consists of certain inputs from the user - and then, the model makes a prediction of how the process would respond, and we will talk more about it shortly. So, models are heavily used in prediction or inferring certain unknowns and also classification - we discussed this last time pattern recognition.

There we talked about models, not necessarily mathematical form; they are more of models in the form of classes. So, we say when the data falls into certain class, then the process belongs to a certain set of operating conditions and so on. So, models are heavily used in classification as well.

And the third application of modelling is in fault detection, where I build a model of the process under normal operating conditions. For example, I know how a friend of mine would talk, and sit, and so on when he or she is normal, but when something is wrong, may be something is mentally troubling my friend, then I know that something is definitely troubling him by observing the behavior. Now, what's happening underneath is I am projecting my friend's behavior against a normal behavior that I have observed over a period of time, and then, comparing both, and seeing well there is a huge difference, and then, finally, coming to a conclusion that something is abnormal. This is pretty much the same idea in fault detection as well. We built models of process under normal operating conditions from historical data or through first principles approaches and then, keep comparing the measurements that come out of the process against what the model is predicting. If there is a significant difference between the prediction and the measurement, then we raise an alarm, and probably conclude that there is a fault, and then, take it up for further diagnosis.

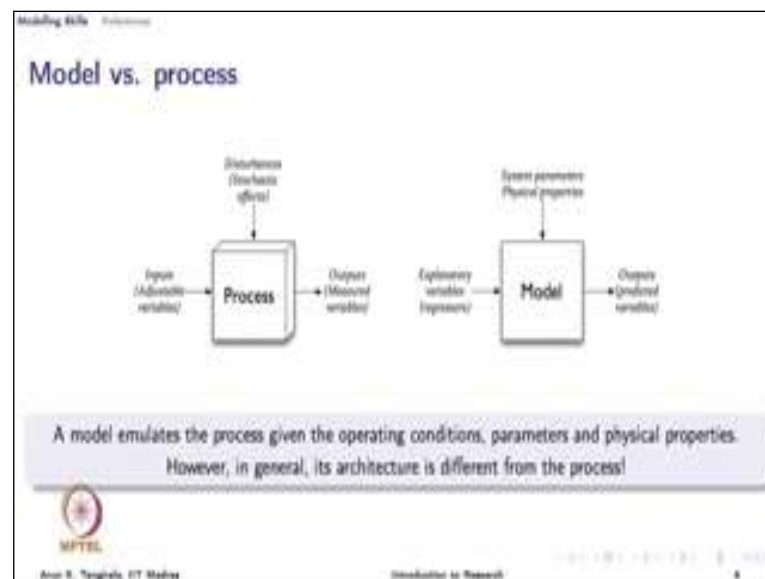
One of the prime uses of models is also in simulations. We have heard of simulators. Some of you must have worked with different simulators in chemical engineering, mechanical engineering, aerospace, and so on. You have heard of air craft or flight simulators. There, the primary role of the model again is in predictions. So, we give certain inputs to the model, the same inputs that we would see when we operate the process, and ask how the process would respond. So, the model would make a prediction. We need high fidelity models in such applications, whereas when we use models in control, although I don't list that here, models are heavily used in control,



where the model makes a prediction of where the process is heading, and then, a controller takes an action to keep the response of the process close to the set point. There, in such applications, that's in control, we may not need high fidelity models. We can work with fairly approximate models. And finally, we do find uses of modelling in design and optimization and so on.

So, obviously, the list is a bit more than what I have given here, but what should be remembered is the end uses of models vary a lot, and therefore, the type of model, the kind of accuracy, the nature of the model whether you want to build a steady state or dynamic model or you want to build a time domain or a frequency domain model and so on, really it depends on the end use. So, you have to work backwards, and then, make a decision on what kind of model you want to build.

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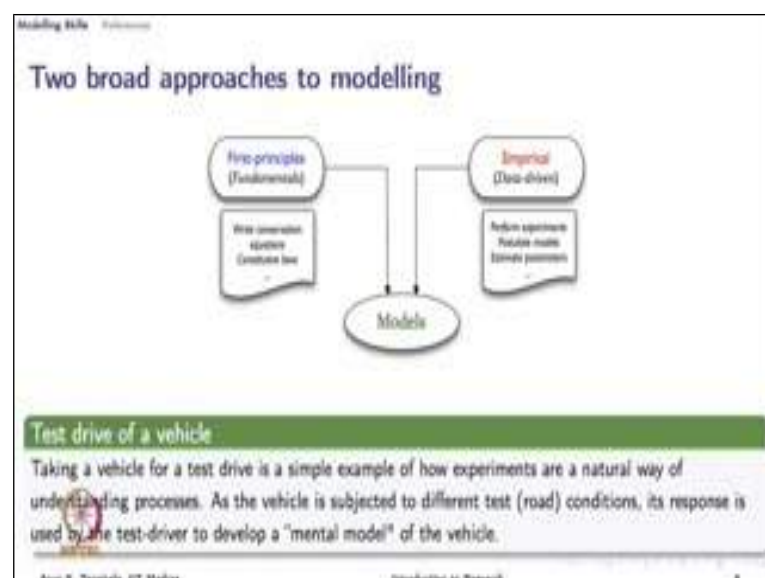
Okay so, just to even make this point very clear - the point on distinction between a model and a process and also a similarity - I am showing you a schematic here of the process and the model. So, if you carefully look at the process architecture, there are inputs which are causal and physical inputs going into the process, and then, there are disturbances acting on the process. You can think of... you can take any process and actually cast it into this architecture. And then, the process responds, which we call as outputs in the engineering terminology.



Now, models also give you the outputs that are of interest to you. In fact, typically, the output of a model is same as the output of the process, but typically the output of a model is nothing but the variable that you want to predict. And the inputs to the model need not be necessarily the physical inputs that go into the process. The inputs to the model are generally more or the same as the inputs that are going to the process, but for example, if you looking at a dynamic model, in a dynamic model, the output is modeled as a function of the present and the past, because transients are important to us. So, the inputs to the model are not only the present input but also the past; whereas the process is operating in real time, and it keeps receiving only the present input at any instant; the past has occurred but at that time. On the contrary, in the model, you do feed the past inputs and so on, and then make a prediction, because model is after all a mathematical abstraction.

And then, there are also certain user defined parameters and or user specific inputs that you will have to provide to the model along with the system parameters. So, the architectures are different, but the final use of the model is in prediction – basically, predicting the variable of interest to you. So, that's very important. So, do not get confused with the inputs that go into the process and the inputs that go into the model.

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Alright so, let's look at how models are developed because we want to really gain some insight into how to develop and build models right. There are two broad approaches to modelling. One approach is to start from fundamentals, where you invoke the laws of



physics, chemistry, biology, and so on; essentially science based or mechanistic models. And here, we invoke the laws of conservation primarily mass, energy, momentum, and use a few constitutive relationships may be from thermodynamics and fluid mechanics and so on; and finally come up with the model; the set of equations essentially. which we call as a model. Now, that's one approach.

The other approach, which is quite contrasting, where we don't rely on science as much as we rely on data. There is a data science, but we don't rely on the science of the physics to begin with; at some point in time maybe we can incorporate, but to begin with, we rely on data. And this approach is called an empirical modelling approach. And it's also called a data driven approach, where I will use the data to identify the relationship between the variables of interest; typically, the input and output and so on or sometimes only to build a model for the output which we call as a time series model. And here data is the primary food for identification. Without data, there is no empirical approach at all. And the kind of models that come out of empirical approaches are called black box models, typically, where you don't incorporate necessarily any physics of the process, explicitly. You work with a minimal understanding of the process, but that does not mean that there is no provision for incorporating the physics of the process or whatever you know about the process a priori; you can. And as you keep incorporating the prior knowledge into your empirical model, the black shade turns into gray, and there is some transparency that sets in and such models are known as gray box models.

So, in that respect, the first principles models are actually called white box models because they are very transparent. If I look at a model, first principles model, I will be able to associate every term in that model with some physical characteristics of the process; whereas, that's not necessarily the case with an empirical model. An empirical model is some mathematical fit between the input and output. So, to give you a simple example, when we go out on a test drive, let say to purchase a vehicle, the common sense thing that all of us do is take the vehicle, sit in the car and apply certain inputs to rotate the steering wheel and pedal, apply breaks, supply fuel, and so on, and you know, give all different kinds of inputs that we want to really test the vehicle on; and then, collect the response of the vehicle. So, what we are doing there is, we are applying inputs, and observing the response of the vehicle, putting it all together in our brain, and building a mental model. We may not be able to write an equation there. We are building an



empirical model. We are not really building a model of the car from first principles. I am sure that would be a very deadly approach indeed, but we never do it and we have not seen anyone doing it.

On the other hand, when we really sit in courses on in automobile engineering, mechanical engineering, and so on or in engineering design, we do learn what are the mechanics of a vehicle through equations, through first principles understanding and so on. So, that has its place, while empirical modelling has its place; where increasingly a lot of people are turning to empirical modelling, primarily because many processes that we are looking at, trying to understand, are quite complicated, quite complex, for us to write a first principles model. So, the experimental or the empirical approach is a natural recourse and that will continue; it's here to stay; it's been there since times immemorial; it has been there from the time man has started to build models, try to understand processes from observations.

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Modeling Skills

First-principles vs. Empirical modelling

First-principles	Empirical
Causal, continuous, non-linear differential-algebraic equations	Models are usually black-box and discrete-time
Model structures are quite rigid - the structure is decided by the equations describing fundamental laws.	Model structures are extremely flexible - implies they have to be assumed/known a priori.
Can be quite challenging and time-consuming to develop	Relatively much easier and are less time-consuming to develop
Require good numerical ODE and algebraic solvers.	Require good estimators (plenty of them available)
Very effective and reliable models - can be used for wide range of operating conditions	Model quality is strongly dependent on data quality - usually have poor extrapolation capabilities
Transparent and can be easily related to physical parameters/characteristics of the process	Difficult to relate to physical parameters of the process (black-box) and the underlying phenomena

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And for the rest of the lecture, we will focus on empirical approaches per se, but before we do that, I just want to draw your attention to a few salient differences between first principles and empirical modelling approaches. There are people who really vouch for first principles models, and say, that's the best and so on; and then, there are people who argue in favor of empirical models and so on, but there is no hard and definitive rule as to which modelling approach is the best. It is a situation that really determines the



answer to that question, and one has to take a very common sense approach through it, and a neutral approach. But in order to do that we have to understand what benefits or pros and cons that each of these approaches have got to offer.

With the first principles approach, we do get causal, physically meaningful models and so on, and also, its ability to predict the process behavior is good over a wide range of operating conditions. However, they are difficult to solve analytically. Typically, we end up with non-linear differential equations ODEs and PDEs and so on, and those models may take a lot of time and computational effort to solve; whereas, the empirical approaches offer a lot of flexibility in choosing the models.

In a first principles approach, I don't have any say on the structure of the model that I have. Whatever the laws that I am applying, give me, I have to live with those kinds of models. Of course, I can choose to build some approximations, but even those approximations can be quite complicated. Whereas with the empirical model, I am trying to find a mathematical fit that helps me understand or map the relation between the variables of interest, and also a map that helps me make good predictions; that's it. So, there may be many solutions of which I may pick the most simple one, and not really a very simple one, but simple enough model, and that flexibility really gives the empirical model a big, what do you say, you know, vote of favor. A lot of people really prefer that, particularly in control and so on. And therefore, relatively much easier to may be simulate, and less time consuming to develop, and so on. However, they require good estimation algorithms, because you are going to estimate parameters of the model and so on, and there can be considerable computational effort there. So, there is no escape there as well.

The first principles models give us effective and reliable models, whereas the empirical models are as good as your data. Now, that is a very common criticism of an empirical approach, but you have to be careful when you take that criticism into account. Clearly, training a model is pretty much like training a student in a subject. Whatever questions that you are going to ask later on to the student or whatever you are going to really test the student, depends on what you have taught. If you are going to ask a question to a student of a course, on a completely different topic that you have not taught, then obviously, the student may not be able to answer in most probability. And that's the same with an empirical model. You are going to develop the model from data. So, you have



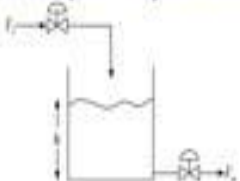
chosen a model structure and data, bring them together with the help of an estimation algorithm, and then, force the model to understand the data. If you have chosen the right model structure, and right estimation algorithm, things will work out well, and it would have captured the essence of the process for the operating conditions that are present in a data, that is under which the data has been generated.

Now, if you are going to present a completely new data set from a different operating condition, then it's very likely that the model may not work well, unless the process is linear and so on. So, to speak, the extrapolation capabilities of an empirical model are always questionable, but it depends on how you have trained the model. If you have shown the model a wide range of operating conditions, then, yes. So, the model will do a good job for you, but in any case, if the model is too complex, then, there is no other choice for us. We have to choose, we have to resort an empirical model.

The other aspect that is not reflected here is, when a process is random, that is stochastic - and we had discussed this part in the last lecture - in such situations, first principles models hardly come to your rescue. There is no choice, but to build models from data, which we call as time series models. So, empirical models have a lot of points in favor of them as long as you are aware of the limitations. The prime limitation being model quality is strongly dependent on the data quality. If you remember that, then, you will not really give excessive importance to the empirical modelling approach, really tread with a lot of wisdom, so to say.

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Example: Liquid level system



Case 1: Steady-state model between F_i and h :

$$F_o = C_v \sqrt{h}$$

(C_v is estimated from data)

Case 2: Dynamic, non-linear model of $h(t)$ for changes in $F_i(t)$:

$$A_t \frac{dh}{dt} = F_i - C_v \sqrt{h}$$

(requires numerical solvers)

Case 3: Approximate linear, dynamic model about an operating point

$$A_t \frac{dh}{dt} = \hat{F}_i - \beta \hat{h}, \quad \beta = \frac{C_v}{2\sqrt{h_o}}$$

$$\hat{F}_i = F_i - F_{i,o}, \quad \hat{h}_o = h_i - h_o$$

Typically, $F_{i,o}$ and $F_{o,o}$ are steady-state values.

Case 4: Approximate empirical, linear, grey-box, dynamic, discrete-time model,

$$h[k] = a_1 h[k-1] + b_1 F_i[k-1] + c[k]$$

(Parameters a_1, b_1 estimated from experimental data)

Case 5: Black-box, dynamic, discrete-time model

1. Model structure based on ease of estimation and end-use
2. Model may not be physically interpretable, but designed to give good predictions.

From S. Targui, IT Madras



So, let's move on now and just get a feel of what are the different types of model that one can develop for a liquid level system. Now, this is a very simple system that we see in our households everywhere, and also in industries, where there is a flow coming into a vessel or a tank, if you like it, and there is some storage of the liquid in a tank, and then, there is an outlet flow. Both the in and outlet flows are regulated by valves. And let us say, I am interested in knowing how a change in inlet flow affects the liquid level. This is very important, because look at the flush systems in our toilets and so on; they are based on an understanding of how the flow affects the liquid level. There, of course, you have a mechanical device controlling the liquid level, but in industries there are automated controllers controlling the liquid level. In all cases, we need to know how the liquid level changes, when there is a change in the inlet flow.

Now, the first scenario is probably that I want to understand its steady state. How the outlet flow and liquid level are related, because that actually becomes a part of the model eventually between the inlet flow and a liquid level. So, at steady state, we know from Bernoulli's principle, fluid mechanics, and so on, we can derive a relation between outlet flow and the liquid level of the head that people talk about. And we know that the outlet flow is linearly proportional to the square root of the liquid level. This is called a steady state model. This is true at steady state. All you have to do is perform an experiment at different steady states, measure the outlet flow rate and the liquid level, plot the outlet flow rate versus the square root of the liquid level; you should see more or less a straight line. Well, if you have access to an experiment, you should try it out and you will see that this is not perfectly right, but this is a fairly good model. Now, we just now said eventually I want a dynamic model; that is that tells me how the liquid level changes when not only at steady state but between two steady states, when there is a change in the inlet flow.

Now, for this we will have to write the material balance, that has apply the mass balance for the law of conservation of mass, and we assume incompressible flow, and we end up with a differential equation - first order differential equation – unfortunately, non-linear in nature. So, the equation that you see here has come about by applying two things. One, the conservation law of mass, and two, the steady state model that we just developed in the case one. Now, I can simulate this. Obviously as I mentioned earlier, this is a first principles model. There is no way to analytically solve this. I will have to



use a numerical solver; essentially use numerical integration techniques to determine the liquid level profile - for a profile, input profile - which is the inlet flow.

Now, what I could also do in order to have an analytical solution, I could approximate this non-linear model with a linear one, assuming that the changes in the inlet flow are not going to be too wild, and therefore, I can think of a linear relationship between the liquid level and the inlet flow. A fairly reasonable assumption under many conditions in many situations.

So, what we do is, we approximate the non-linearity, the source of the non-linearity in the ODE in case two is a square root. So, we can approximate that with a linear one using Taylor Series Expansion. So, what we have done is, we have approximated with a first order approximation of the square root, and whenever we approximate, typically approximations are within the vicinity of an operating condition of a nominal point, and therefore, we rewrite this ODE as a linear approximate model. It is still a first principles model, but it is an approximate model, but now the model is in terms of what are known as deviation variables. That is how far you are away from your reference operating point, typically chosen to be as a steady state. On the other hand, I may say, well, you know, I don't know any of this laws of conservation, and the valve equation and so on. I will choose to fit an empirical model, and then, there are two possibilities: a gray box and a black box model. Suddenly, I discover that in this empirical approach, no, no, no it's conservation mass is not so difficult to write. So, I do write, and I have some idea of how the model should look like as in case three; but now the problem is we are no longer in continuous time, because we are going to build models from data, and data is available only at sampled, in a sampled instance in time; it's not available at every point in time. So, we move from time t , which is a continuous time variable, to k - discrete time k . This k here, in case four, stands for the sampling instance, the k sampling instant.

That is I am observing now. I have observed the process in order when I was collecting data, at a regular intervals in time which we call as a sampling instance, and now from some prior knowledge of the process, I know that there is a first order linear relationship between the liquid level and the inlet flow rate. So, I write an approximate linear equation, but in discrete time, and it's a first order difference equation essentially. So, because I moved from continuous time to discrete time, I moved from a differential equation to a difference equation. And now, the goal in this kind of a modelling approach



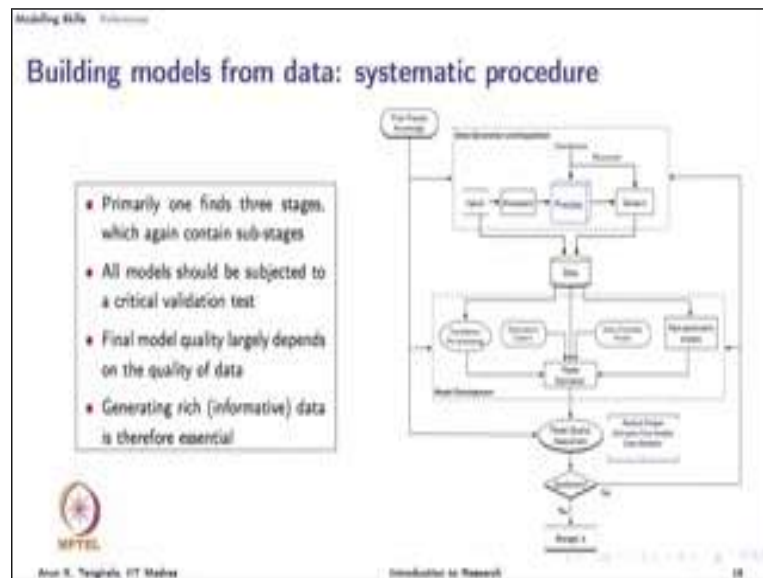
is to estimate these parameters a and b , and the epsilon that you see is a error that you have made in approximation. Of course, we also would like to know what the magnitude of this error is and so on, but the primary goal is to estimate these parameters a and b . So, here, we say this is a gray box empirical linear discrete time model, a lot of qualifiers, of course, but to be very clear, the gray box nature comes about because I have incorporated prior knowledge of a linear model and first order dynamics.

Suppose, I don't know any of that, and I just want to build a dynamic model; I can do that as well. Then, we end up with a black box model and I don't have any equation written up here because the choice is yours. Of course, through a systematic study you will be able to discover that eventually the relationship between the liquid level and the inlet flow rate is that of a linear one and a first order one. Typically, when we start off in black box modelling, we start off with simple models. So, we would start off with a linear model, and because this is a dynamic process, we have to choose the order of the dynamics first order, second order, and so on. And therefore, through a careful study we can converge to a first order linear model. And this case study is discussed in detail in the book - one of the books that have given as references at the end of this lecture alright.

So, this hopefully gives you a feel of the different kinds of models that one can develop for a process, and as you have seen here, the nature of the model that you develop depends on what you want to know, and what kind of rigor that you are looking for, and what you have with you - what kind of knowledge is with you. In an empirical approach, you have data and minimal process knowledge. In a first principles approach, you don't have data to begin with, but you have a pen and paper, and you have, you are equipped with good understanding of the physics of the process right, but a good solution - working solution - is always a gray box approach, which is a marriage of these two, so that you don't have to really take sides whether on the first principles or empirical.



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So, I just mentioned a few minutes ago, that a systematic procedure will help you get a good model, and in particular, I am talking about an empirical approach here. So, I give you a flow chart for a systematic way of identifying the model, and let me quickly go over this. Primarily, you have three stages. One is data acquisition and we don't talk about that here; we assume that data is available to us, and so, the first stage we really don't discuss much. And then, the second stage is, of course, model development, which is at the heart of this procedure. And the third stage is model assessment; that's very important and that applies to all models that we develop, whether it's a first principles or an empirical model, it's important to assess the goodness of the model; is the model capable of predicting the process over a good range of operating conditions? Is it doing well on a fresh data set? Even a first principles approach has to be validated. So, please don't be under that impression that this is not required for a fundamental model. And maybe we want to ask, if in an empirical approach, if the parameter estimates that I have obtained in a model, do they have large errors in them. That is a something of interest.

And the third thing that we have to watch out for an empirical modelling is over fitting. Remember I said building a model from data is pretty much similar to a student learning a subject. The student is presented with the text book, and the course material, and then, the student tries to understand the concepts of the subject, eventually, through a proper interaction with the course material and the instructor. Now, in the end, you have to ask if the student has over learnt; that may seem very strange - what is meant by over learning?



Now, over learning is, let us say, I as a student I am trying to solve an assignment problem and the assignment problem is based on a certain concept. If I have understood the concept, that is, if I at the end of the problem solving exercise, if I have gained mastery of the concept on which the question has been based, then the goal of solving the assignment problem is more or less achieved, but if I start paying attention to the numbers, the very fine details that are very specific to that question, but has very little to do with subject itself, and I am trying to really memorize all of that, then I am over learning right. So, that is probably a simple analogy of over fitting in modelling as well. And that occurs primarily because of presence of noise in data and I have an example to show you later on. So, remember that there are three stages: data acquisition, model development, and model validation.

Let me quickly talk about the model development part. You see that there is a pre-processing - data pre-processing - we talked about at in the last lecture. We have to watch out for missing values, outliers or any other anomalies, get rid of them and so on. And a big part of that is visualization; involves visualization of data. And we had an example in the last lecture highlighting the importance of visualization. Once we have understood the data well and it's ready for modelling, we should not straight jump away necessarily to build a model unless I know the model structure very well; that is, I know it's a first order or I know that it's a linear model, non-linear model, and so on of this type, and so on. So, an intermediate step involves what is known as a non-parametric analysis, where I make minimal assumptions on the process, and try to gather as much information as possible from the data, so as to make a good guess of the model structure; that's called a non-parametric approach. And this step, can be skipped if I already know the structure of the model that I am going to fit okay. So, in many situations the non-parametric analysis may not be even present.

Finally, I have some good decent guess of the model structure, and then, I have to estimate the parametric of that model, where I apply estimation algorithms which are essentially optimization algorithms to estimate the parameters, and then, I have a model with me. So, that's pretty much the model development in a very, very simple way that can be possible, but that can be possibly explained, but remember there is so much there, that's a huge ocean in itself, and in any of these, at any of these stages, there are no definitive answers or formulae for you to really go through, but there are very good



guidelines based on theory of data sciences. It's very important to know at least the basics of those principles, and have certain guidelines in place, so as to minimize the effort.

Now, one very important thing that you would see is that we are able to incorporate prior knowledge; that is, there is a provision for incorporating domain or prior knowledge at each of this stage, and I have also emphasized this aspect in the previous lecture as well. In any data analysis exercise, we should incorporate the domain and prior knowledge as much as possible.

And the last, but not the least important aspect of data driven modelling is that it is an iterative exercise. It's very unlikely that you will be able to get the best model in just one round of this iteration, one pass of this flow chart here. It is very likely that the model that we develop is not satisfactory in many respects, maybe it's not predicting well on a fresh data, maybe the parameter estimates have large errors in them and so on.

Now, why would this occur? Perhaps because the data quality itself is bad, that's very likely, which means we may have to go and repeat the experiment or the data quality is good and I have chosen a wrong model structure, which means I have to re-examine the models that have assumed, or that I have chosen a wrong estimation algorithm, in which case, I have to go back and choose a better estimation algorithm and so on. So a systematic study will really help us minimize this back and forth steps, and also, be able to pin point, with a fair degree of accuracy, as to what the source of problem is when the model is not satisfactory. So, please keep that in mind.



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A presentation slide titled "Two broad classes of models". It features a blue header bar with the text "Time-series models". Below this, there is a list of four bullet points, each preceded by a red arrow. The slide also includes the NPTEL logo in the bottom left corner and a footer with the text "Arjun K. Singh, IIT Madras" and "Introduction to Research".

Modeling Skills

Two broad classes of models

Time-series models

- ▶ Suited for modelling stochastic or random processes (e.g., stock market index, rainfall, sensor noise)
- ▶ Causes are either unknown or also random
- ▶ Usually dynamic models (in a few applications, steady-state as well)
- ▶ Challenges: choosing model structure, making the right assumptions on process characteristics, non-linearities, etc.

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Arjun K. Singh, IIT Madras

Introduction to Research

Now, within the empirical models, there are two broad classes of models that one encounters and I just want to briefly discuss those. I am not referring to the linear, and non-linear, and so on. This classification is based on more or less the same lines as we discussed in the data analysis lecture. The deterministic versus stochastic models and so on. So, the first class of models that one would see prevalently in the literature, predominantly in the literature is a time series models, which cater or which are suited for stochastic processes, where the causes are either unknown or known with error; that is, they are actually random themselves.

And a simple example for that would be that of an atmospheric process. Suppose I want to build a model that predicts the atmospheric temperature. Now, what do I do? Yes, I probably know that there are several factors affecting the temperature, but I have not measured them or probably I don't know the complete list of causes that influence the atmospheric temperature. So, a natural recourse is to take the historical data and hope that there is something in the history that will repeat itself; not exactly, but there are patterns, and correlations, and so on, and then, build what is known as a time series or a dynamic regressive kind of model. These are very common in many, many fields. On the other hand, let us say, I want to model the relative humidity of air. Now, I know that the relative humidity of air is significantly dependent on the temperature - atmospheric temperature. So, I have the relative humidity measurements, I have the temperature measurements and I can build a model between these two; that still counts as a time



series model in the general literature, primarily because both the temperature and the relative humidity are random in nature. So, we can build what are known as multivariate time series models, but then, you know, it's a matter of perspective. You can also call it as an input-output model. So, the time series model, when you look at the terminology, typically, it means you are building model based only on the response of the data, but many people use it with a larger connotation.

Now, some of the challenges in building time series models are choosing the right model structure; that is how much in the past, for example, affects the present, and that, of course, again, guidelines and some mathematical as well as statistical methods are available to help us, but again there is no definitive answer. And making the right assumptions. For example, in stochastic processes, do I assume the process is stationary or non-stationary or it is stationary with a deterministic trend like the one that we saw last time, the carbon dioxide emissions, we saw a trend. There's a linear trend, and then, on top of it there were oscillations and so on. So, we don't know. Or if the underlying random process is linear, non-linear and so on. And, of course, the unwritten challenge is estimating the parameters; that's anyway challenging in empirical modelling.

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Modeling Skills - Introduction

Two broad classes of models ... contd.

Input-output or causal models

- Suited for modelling relationship between a variable (or more) and other **explanatory** variables (a.k.a. **regressors** or **factors**) (e.g., power and current in a wire, temperature and coolant flow)
- Regressors may be known accurately or with some error
- Models could be steady-state or dynamic.
- Challenges: sufficient variability in factors, selection of regressors, measurements and regressors available at different sampling rates, choosing the order of dynamics, etc.

Remember

In practice, all modelling exercises demand a careful handling of uncertainties both at the experimental and modelling stages!

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Anur B. Targiere, IIT Madras

Now, on the contrary, we have what are known as input-output models, which are causal models, where I try to explain one variable using other variables that I think are causing or influencing the variable of interest. You may like the temperature relative humidity example or may be if I am measuring the power verses current in a wire or the



temperature of a reactor verses a coolant flow and so on. There I can think of an input-output relationship between these variables and will regressive models. Of course, multivariate models are also possible.

Now, one of the challenges here, in this input-output models, is this so-called Regressors or explanatory variables as we call them, may be known accurately or may not be known accurately. In the temperature, relative humidity example which we called as a multivariate time series models can also be considered as an input-output model, but the difference is that the temperature which is the regressor for relative humidity, is a measured variable, is not something that I am able to adjust. Unfortunately, I am not able to, I don't have the ability to adjust the atmospheric temperature. If people had that ability, then it could be a chaos right. So, the temperature is a measured variable and every measurement has error in it. So, the regressor, in that kind of a situation is known only with in error; whereas in a coolant flow verses temperature of a reactor, I probably have the provision to induce changes in a coolant flow and measure the temperature.

So, there I know exactly what kind of changes I am inducing in the coolant flow and only temperature is a measured variable. So, the coolant flow is considered a deterministic signal or a variable in which case the regressor is known accurately.

So, you have to look at a situation and determine how to treat each variable. So, it takes us back to the same thing that we learnt in last lecture, deterministic or stochastic. And of course, you can have different models and there are several challenges again here. For example, in cases where I have the privilege of performing an experiment where I change the factors or the explanatory variables - how should I change them? what should be the level of excitation and so on? Or which regressors? In multivariate regression there may be different factors, several factors affecting the variable of interest, then which variable should I factor in or which set of variables should I factor in? And there are also situations where the measurements and regressors are available at different sampling rates and so on.

Now, in practice, all modelling exercises involve a mix of both input-output and time series modelling, because even in an input-output modelling exercise, we may not be able to explain all the variations in the variable of interest. So, for instance, in the temperature and coolant flow reactor example, I may be able to explain most of the



changes in the reactor temperature using the coolant flow, but there is something in the measurement that I cannot explain using the changes in coolant flow and that something is probably sensor noise. So, how do I model that sensor noise? There, I have to take a time series approach. So, sometimes there may be no predictability in that sensor noise, then I only have to estimate the statistics, but in any case, I have to address both the deterministic and the stochastic portions of any any model. There is no escape to it. So, we will just quickly go through some of the critical aspects of empirical modelling.

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Excite the process sufficiently!

Example

Consider a steady-state model:

$$y[k] = b_0 + b_1 u[k] + b_2 u^2[k]$$

- Three unknowns
- Therefore, data corresponding to three steady-states is required
- A more general statement: The regressor matrix

$$U = \begin{bmatrix} 1 & u[k_1] & u^2[k_1] \\ 1 & u[k_2] & u^2[k_2] \\ 1 & u[k_3] & u^2[k_3] \end{bmatrix} \quad (1)$$

should be non-singular, i.e., of full rank.

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And the first aspect in empirical modelling where we perform experiments or even if you don't is that of the excitation in the regressor or in the factor or input, whatever you want to call it. So, let's look at a simple example where I have a steady state model. The y is the output and u is the input and y is a quadratic function of the input. So, I have three unknowns, and therefore, I need data corresponding to three steady states, clearly. In other words, if I write the equation for estimating the unknowns - three unknowns - in a matrix form, at these three instants - different instances - in time k_1, k_2, k_3 , remember we only have sample data. Therefore, we use this notation. When I write the equation of the model in a matrix form, then the big U matrix that we see here comes into picture, which maps basically y to the parameters b_0, b_1 , and b_2 . This big U matrix has to be non-singular or a full rank. Of course, in discussing this example, we have kept noise out of the picture. So, it's a noise free condition. We are not making any noise here.



When it comes to noise, there are other things to worry about, but even in the noise free condition, it's important to remember that we have sufficiently excited data.

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Modeling slide

For dynamic systems

Example

Consider a (deterministic) process

$$y[k] = b_1 u[k-1] + b_2 u[k-2] + b_3 u[k-3]$$

Suppose that the process is excited with $u[k] = \sin(\omega_0 k)$ (sine of single frequency). With this sine wave input,

$$\begin{aligned} y[k] &= b_1 \sin(\omega_0 k - \phi) + b_2 \sin(\omega_0 k - 2\phi) + b_3 \sin(\omega_0 k - 3\phi) \\ &= \left(b_1 + \frac{b_2}{2 \cos \omega_0}\right) \sin(\omega_0 k - \phi) + \left(b_3 + \frac{b_2}{2 \cos \omega_0}\right) \sin(\omega_0 k - 3\phi) \\ &= \left(b_1 + \frac{b_2}{2 \cos \omega_0}\right) u[k-1] + \left(b_3 + \frac{b_2}{2 \cos \omega_0}\right) u[k-3] \end{aligned}$$

Only two of the three parameters can be identified! Why?

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In fact, for a dynamic system, very interestingly, let us look at this example, where the output of a system is dependent on the past input, the one input beyond a past and the second input beyond the past. So, $u[k-1]$, $u[k-2]$ and $u[k-3]$ are essentially lagged inputs, but that 1 there, means essentially one sampling instant. This is the hallmark of a dynamic system. It has a memory essentially. So, I want to once again identify these three parameters.

Suppose I perform an experiment where the input is sinusoidal. This is very common in mechanical systems or all systems which have a kind an oscillatory characteristic and so on or which are characterized only at specific modes or frequencies. There, if suppose I perform an experiment with a single sinusoid, then, what i mean by single sinusoid is a single frequency sine wave, and then, it turns out that the process manifests as a two-parameter model. You just have to work through the trigonometry here. So, what I have done is, I have taken this input and plugged it into the equation here and asked how the response would look like. It turns out that the response appears as a two-parameter model to the user, to the experimentalist vis-a-vis the reality which is three parameter models. So, what has happened here? What is the consequence now? The consequence is only two of the three parameters can be identified.



In other words, going back to the same story that we had in the steady state case, now the big U which would now consist of $u(k-1)$, $u(k-2)$ and $u(k-3)$ in the first row and so on for the remaining two rows that becomes singular when input is a sine wave of single frequency. Therefore, I do not have enough information in the data to estimate all the parameters and that is the key. Always data should contain sufficient information to estimate the parameters. On the other hand, if I have an input which is made up of two frequencies - sinusoids of two frequencies - then I have sufficient information. You can show that, it's very easy, check for yourself. Construct this matrix - big matrix U - that I showed in the last slide, but now with the first row being $u(k-1)$, $u(k-2)$ and $u(k-3)$ likewise at $k-1$, $k-2$, $k-3$, for two different situations, and you will find that the matrix is singular in the first case, that is when you use a single frequency sine wave, and it is non-singular or a full rank when you use a multiple or two frequencies at least in the sine wave alright.

So, for dynamic systems the story looks different, but the bottom line is the same. Have sufficient information in the data. We will conclude with two things. One with understanding, and understanding of how noise effects our modelling, and then, finally is a bunch of questions that we want to answer in any empirical modelling exercise.

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Effects of randomness (noise) in data

The second challenging aspect of empirical modelling is the **presence of random or stochastic effects**. Randomness (uncertainties) in the process influences identification in several ways:

- Accuracy of predictions
- Errors in parameter estimates
- Goodness of the deterministic model

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So, how does noise affect our data? Well, of course, in many different ways. It affects the accuracy of predictions; we will not be able to predict the output accurately. It brings



about errors in parameter estimates. And three, it affects the goodness of the deterministic part of the process that we want to estimate.

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Signal-to-Noise Ratio (SNR)

SNR

A key measure that quantifies the effects of noise is the **Signal-to-Noise Ratio (SNR)**, which is defined as the ratio of variance of signal to the variance of noise in a measurement.

$$SNR = \frac{\sigma^2_{\text{signal}}}{\sigma^2_{\text{noise}}}$$

SNR can be interpreted as a measure of the degree of certainty (deterministic portion) to uncertainty

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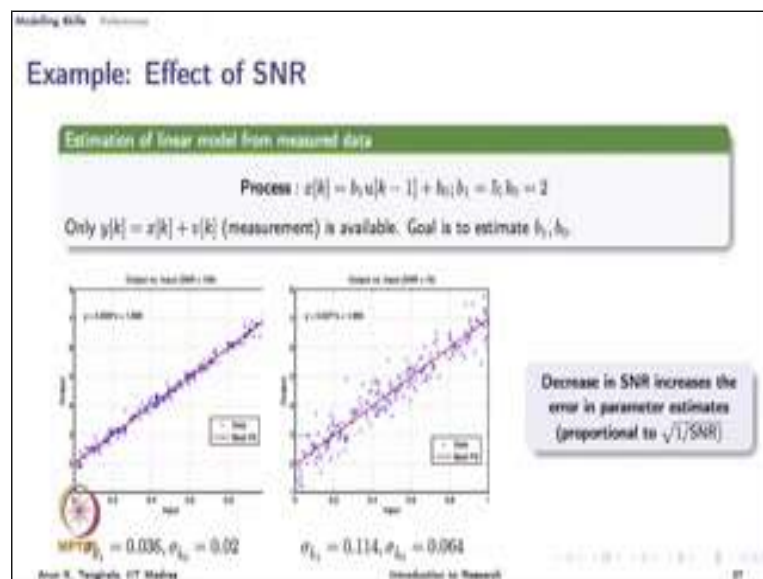
So, normally, in quantifying the effects of randomness or the noise in data, there is a quantity called a signal to noise ratio that is used widely. It is not just in electrical engineering, but in every data analysis exercise, this signal to noise ratio is a very nice quantity that helps us explain or understand the impact of randomness on the quality of the model or the parameter estimates. And a signal to noise ratio is defined as the variability in a signal. Think of it as the level of amplitude or the power in the signal. The signal, what we mean by signal, here is a deterministic part.

So, imagine that I am measuring, going back to the flow reactor and, sorry, the reactor coolant flow temperature, for example. I am performing an experiment, where I am introducing changes in coolant flow and I am measuring the temperature. There, I induce changes in coolant flow and I measure temperature. And when I measure temperature, the measurement contains two effects. One effect due to the changes in coolant flow and the other comes from the sensor noise or any other disturbance. The signal in that situation would be the true response, that is a response contained in the measurement due to changes in coolant flow only and the rest is all noise. So, obviously, if I want to get good estimates, that is of the model, of the coolant, of the reactor, then the level of response due to the coolant flow should be way higher than the noise right. And this is true for anything. If you are listening to a speaker in a classroom, the speaker has to



speak loud enough - which is hopefully the signal of interest to you - compared to the sources of noise in the classroom, which could be due to a fan or an air conditioner and so on. So, there the signal to noise ratio is the amount of power or the amplitude - you can say squared amplitude - in the speakers' speech signal **vicav (there is some term , not understandable)** or divided by the amplitude square of the noise contributions coming from the ceiling fans and air conditioners and so on.

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So, higher the SNR, better the parameter estimate is; that is lower the error. So, to give you a simple example, I have a process here, which I am going to simulate. So, I am simulating a simple process here. u is the input and b_1 and b_0 are parameters that I choose, and x is a true response of the process, but I don't have access to the true response. I have access only to the measurement which is y . So, the way I am simulating is I simulate I generate x first, the true response; and then, I add some random signal to, some kind of noise to x to generate my measurement. Now, I pretend that I do not really have access to x . I only have access to y and u , and that I know that the underlying relationship between x and u is this. So, when I fit a model between y and u using the data, what do I get?

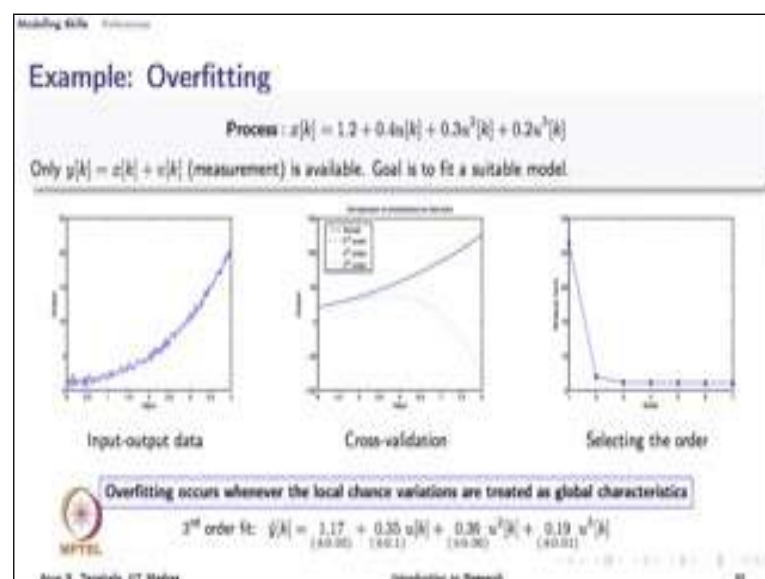
Let's look at it quickly here and the two different situations; on the left, you have the situation where the signal to noise ratio is 100. So, I have adjusted the noise level in v , such that the signal to noise ratio is 100 and the codes for this are available on my website. What you see on the right is the case or the situation for SNR 10. Clearly, on the



right-hand side, there is more noise as you can see in the scatter plot. In both cases, I am plotting y versus u . u is on the... input is on the x-axis, and what you see on the left top of the plot is the equation of fit of straight line. So, I have done this in matlab, and it gives me the equation of line, and it does a fairly good job of estimating b_1 and b_0 . The true values of b_1 are 5 and b_0 are 2; b_1 and b_0 are 5 and 2 respectively, and the estimates are pretty close in both cases, but the difference comes about when we look at the so-called the standard error in these estimates.

Now, what I have obtained here are $\hat{b}_1$ and $\hat{b}_0$. The hat denotes the estimates. So, $\sigma_{\hat{b}_1}$ and $\sigma_{\hat{b}_0}$ are so-called standard errors, which I compute through some theory. Let's not talk about that right now, but what is more important for us to notice is the errors that I am reporting for SNR100 are much lower than the errors that I report for SNR10 right. In fact, they share a relationship which is essentially that the errors in parameter estimates are proportional to square root of 1 over SNR. What this means is, I have a fall in the SNR by factor of 10 which means the errors should increase by factor of square root of 10, which is roughly about 3.6. So, you can see here in $\sigma_{\hat{b}_1}$ when SNR is 100 is 0.036, whereas when SNR is 10 $\sigma_{\hat{b}_1}$ is 0.114, which is roughly about three and half times the error in the case of SNR 100. So, there is theory to tell us how the errors are dependent on SNR, but what's more important to understand is wherever possible we should perform an experiment to make sure that the SNR is high enough; of course, respecting other constraints in the experiment.

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And the other aspect, which is a final aspect, of the noise that I want to talk about, which I have mentioned earlier, is over fitting that occurs primarily when you have noise. Once again what we do to simulate the data, we first generate the response of the true process, the equation for which is given at the top. It's a third order polynomial, and I add some noise to the true response maintaining a certain SNR, decent SNR like 10 and so on, and I generate my y ; that is the measurement. So, once again we pretend that we do not have access to the true response which is a reality. I have access only to the measurement y and the input that I have given to the process. Now, the input and output plot is shown here. I can see a polynomial kind of relationship, but I do not know whether it is a quadratic or cubic and so on. So, what I do is, I try out third order, fourth order, fifth order polynomials, and obviously, as I increase the order of the polynomial, here I have generated about 200 observations. Sorry 100 observations. I can fit a 99th degree polynomial to it. It exactly explains the relationship that I see in the left plot, but if I do that the danger is that on a fresh data, the 99th degree polynomial will fail miserably.

As you can see here in the center plot, the third order polynomial... what we see here in the center plot is the prediction of these polynomials of different orders that I have fit on a fresh data. So, I have reserved certain data for training and another data set for testing. The third order polynomial performs the best, the fourth order performs reasonably better and the fifth order polynomial goes for a toss. It predicts completely different. In fact, it's unstable and I could have avoided this over fitting. So, this is what we call as over fitting.

If I had looked at the improvement that I have obtained by fitting models of successive orders. So, the plot on the extreme right shows us that. What have done is with each model I have... Remember no model perfectly explains. So, there is going to be some residual. I have taken the variance in the residual or you can say the squared, sum squares of the residuals and plotted it verses the order that I have fit. So, when I start with the lowest order, obviously, the sum square is very high, and as I get closer to the true order, the sum square comes to a minimum, and thereafter, the improvement in the sum squares is very, very marginal. So, by increasing the order of the polynomial I have not benefited much. In fact, what I have lost out on is the ability to predict very well on a fresh data cell. So, a plot like this of how much improvement I am obtaining verses the



order or whatever model complexity that I am fitting is always helpful in avoiding over fitting okay.

So, why does over fitting occur? Essentially when I start confusing the local chance variation. So, if you see in the plot here, there is a global trend which is a polynomial trend on the left-hand side plot, but also there are some local fluctuations which are due to noise. So, if I start confusing those local fluctuations with a global trend, then I am over fitting, and that is what should be avoided in all situations. And that can be done with a careful study. So, just to give you a feel of what are the kind of errors that we obtained in the estimates of the third order polynomial, I report the estimates here, along with standard errors reported in the brackets underneath. They are called one sigma standard errors, and you can see that the standard errors in this parameter estimates are quite low compared to the estimates themselves, making us accept the third order model to be a satisfactory one. It has done a good job of predicting well on the fresh data and also the parameter estimates are low in errors. So, this should be the typical approach to empirical modelling.

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A presentation slide titled "Questions for reflection" in a blue serif font. It contains a bulleted list of seven questions in a black sans-serif font. At the bottom left is the NPTEL logo, and at the bottom right is the text "Introduction to Research".

Modeling Skills

Questions for reflection

- What type of models are possible? Which one(s) to choose?
- How do we "fit" a model that "explains" the variations observed in experimental data?
- How to "correctly" account for the deterministic and stochastic effects?
- Will the experiment influence the model that we fit? If yes, in what way?
- How do we set up and solve the problem of estimating the unknown model parameters?
- What kind of experiments should we design to obtain a good quality model?
- How much data do we collect? (what should be the sample size?)

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Dr. R. Venkatesh, IIT Madras

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Let's conclude this lecture with a few questions for reflection, and obviously, we are not going to discuss them in detail, but in any empirical modelling approach, in fact, also to a large extent first principles models, these questions apply. One is always faced with the question what type of models to choose? Again, there is no formula there. We have to go based on the end use, how easy it is to estimate, prior knowledge of the process and so



on. And the general guideline is keep the model as simple as possible. Not simpler, then simple, but as simple as possible. Good enough to explain, easy enough to estimate and so on, and of course, also convenient enough to implement, if you are going to implement the model online. And also, we will have to worry about two different models. One model that explains the deterministic portion and the other that explains the stochastic part. How do we correctly account for the deterministic and stochastic? That's a big challenge and one has to go through a careful procedure. There is no time to discuss those, but there are certain systematic procedures in place, and the key is the model assessment stage where you check for over fitting of the deterministic and stochastic model; whether you have nicely segregated the deterministic and the random effects.

Will the experiment influence the model that we fit? Of course, yes. There is no doubt about it. The data quality is highly influential on the model quality. Your model is as good as the data. So, perform experiments with care, think of the class of, range of models that you want to build. We have already seen excitation matters. If I poorly excite the process in the experiment, then I will have limited information, and therefore, I have not interrogated the process or interviewed the process enough to make a decision on a good model.

And how do I, what kind of experiment should be obtained or designed? As I said, we should perform experiments taking into account all the factors, suppressing the sources of disturbances as much as possible, choosing a nice instrumentation which limits the noise levels and so on. And then, there is a design of experiments subject which tells you how you should go about designing the experiments, not only to enhance a signal to noise ratio, but also making sure you have all the factors that affect the variable are excited sufficiently, how to minimize the time, perform in an optimal way, and so on. So, you should refer to the design of experiments subject.

How do we set up the problem of estimating? There are different estimation algorithms, least squares, maximum likelihood, Bayesian methods, so many different methods of estimation - which one should I pick? Again, you will have to understand how these estimation algorithms perform, but the general guideline is - choose the one that is efficient. That means, that gives you estimates with low errors and also computationally less burdensome; and usually these are conflicting factors. An algorithm that gives you efficient estimates need not be the computationally most friendly one.



And of course, how much data to be collected – that's a big question. It has a huge impact on the errors in the parameter estimates. So, general guideline is the errors fall down as a function of $1/\sqrt{n}$, where n is number of data points that we collect, if you have chosen the right estimation algorithms. So, obviously, more the data, better the estimate that one should expect.

Another question that doesn't crop of here is domain of modelling. I may collect data in time, but I may build a model in frequency domain. It's very, very likely, particularly, in periodic... in detection of periodicities and so on. So, that's another decision one has to make. Again, that completely depends on the application that you are looking at.

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So, with those words I would like to close this lecture and here are a few references. Again, not exhaustive. The case study that I was talking about earlier, the book that I referred to earlier in the lecture is the one that's given at the bottom - Principles of System Identification. There is a book, there is a website for this book on my web page and you can download some of the mat lab course. For example, for the over fitting and the signal to noise ratio examples that I illustrated. Please feel free to write to me if you have any questions. So, hopefully you enjoyed the lecture and that you have a good modelling session whenever it is.

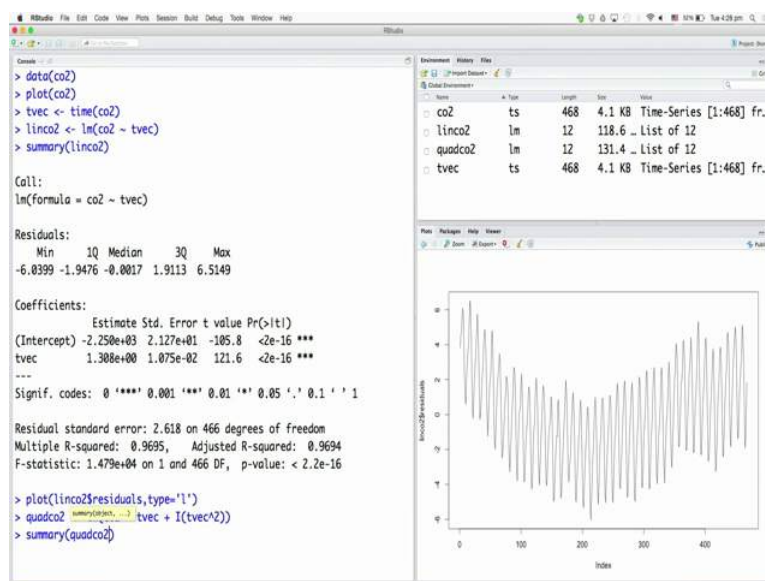
Thanks.



Introduction to Research
Prof. Arun K. Tangirala
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture – 17 (3 B)
Modelling Skills

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Welcome to the R session that supplements the lecture on Modelling Skills. In this session, we will work through two examples; in the first example, we will take up the CO 2 data set that we briefly looked at in the data analysis session. Once again, we start with a clean slate from R and load the CO 2 data set that comes with the base package. Always remember the CO 2 data set need not be just with one package, they are other packages when you load, you can also have the same data set. I am referring to the CO 2 data set that comes with the base package called data sets. So, it has loaded the CO 2 data. And to conform that's indeed the same data that we worked with last time. Just look at the plot and you must quickly notice that it is indeed the same one. It has a linear trend, atleast prima facie, but it could be a quadratic trend or a cubic one, you do not know, but definitely it has a trend function of time. On top of it, we also have an oscillatory trend which we would like to model. And then, there could be a stochastic component to it which also has to be modelled. So, the goal here is to model the series,



so that I can use that model for forecasting the CO₂ levels.

We shall not go through the complete modelling exercise, because a complete modelling exercise may also call for some theory of random processes, where we learn how to model the stochastic components. I will only show you how to model the trend and the oscillatory part, and then the rest is reserved for some other course or may be if you are already familiar with it, you can go ahead and do it.

So, here the first thing that we would do is we would model the trend, where the trend is a function of time. So, the time is a regressor and the CO₂ is the variable of interest. Or to extract the time vector we could use the time command. As you must notice here, in the work space panel, CO₂ is a time series object, and therefore, its always going to have time stamps as one of the attributes. We extract the time stamps and collect them in the tvec – it's just a variable name that I have chosen. Now, we could extract the trend using what a routine known as lm in R, lm stands for linear modelling or linear models. The estimation algorithm underneath this routine is a least square - standard least squares algorithm - with a lot of other features as well. You could, for example, model only using a subset of data, you could supply weights and so on, but we are going to use the most plain or vanilla version of lm. And let's call this model as lin CO₂ – that's just a variable name that I am choosing.

And let me show you what I am doing here. So, I say here I supply the formula that I want. Now by formula what we mean is the symbolic relationship between the predicted variable which is CO₂ and the regressor which is tvec. I do not have to tell lm that there is an intercept term as well. So, it is understood implicitly that there is an intercept term. On the other hand, if I do not want to the intercept term - for some reason I believe there is no intercept term - then I could use this syntax, but at the moment we do not know if there is an intercept or not; of course, we can look at things visually, but let us rely on the algorithm first to tell us if there is an intercept.

So, this is the model now we have. Now, we can examine this model and we should do it to proceed further. The summary command is a very multipurpose command which applies to different types of objects; lin CO₂ is an lm type object, basically it's a model



type object and it has several components to it. But first let's look at the summary and see what it brings out. So, on the top, it gives you the formula which essentially is the symbolic relationship that we are modelling. And then, it gives you some statistics on residuals, what is the median and so on. You see some idea of whether the residuals are of zero mean, what is the range and so on, but we will come back to that. The primary interest for us is the estimates of the intercept and the slope. We have just fit a linear model.

And as you can see here, there are four columns under the coefficients heading. The first column gives us the estimates; the second column gives us the standard error as we call one sigma error or the average error in the respective estimates. And then, the last two pertain to the statistics on these estimates and there are also stars here. So, let's quickly understand what all of this is about. So, the estimates, of course, we can read off; by themselves the estimates do not carry a lot of information, they have to be interpreted in the context of the standard error. So, for what I mean by that is, suppose the estimate here was a very small value, you cannot come to the conclusion that the estimate can be neglected. Suppose slope turned out to be 10^{-4} ; it has certain units; so, it is going to be sensitive to the units. So, the value of something of the order of 10^{-4} by itself does not make any meaning or carry any significant meaning to it, unless it is interpreted in the context of the standard error.

So, standard error here is much smaller relative to the estimates themselves. Conveying the fact that these estimates are to be treated as significant, which means you cannot really treat them to be negligible or theoretically zero. Now the t value column is something that I will not go over, it's pertaining to hypothesis testing. So, maybe in the course on hypothesis testing, you will find a lot of details on this. The last column, reports what is known as the p value; and the p value is again the probability of estimating this parameter, finding, obtaining an estimate larger than what we have observed, but without going too much into the details, the simple interpretation is, if the p value is extremely low, then the null hypothesis that the parameters are truly zero value. What are the parameters here? The intercept and slope, and that the entire purpose of this analysis is to test the hypothesis - that the individual parameters are truly zero.



So, when the p value is extremely low, the null hypothesis that the parameters should be zero value must be rejected. And if you recall, we use a phrase - if the p value is low the null hypothesis must go. So, it is just a catchy phrase to remember. And the three stars here are kind of telling you that the parameter estimates are significant. They all telling you the same thing; you can look at it from a hypothesis testing view point or you can look at it from a significance test for the parameters; either way what it's trying to tell us we cannot ignore the parameters estimates practically. Therefore, this linear model should have both slope and intercept.

Now the question is - if this linear model is sufficient? We have discovered that this linear model has significant parameters on it. Now, to do that, we can look at the residuals; for example, we could plot the residuals; the nice thing about R studio is it tells you whether the field that I am typing is a valid field in the `lm` model. And by default, it may plot a scatter plot. So, we will plot a line plot. So, you can see here the linear trend as been taken off, but it seems to be some quadratic trends as well, because a pure sine wave of that frequency cannot have a shape like that; maybe we can fit a quadratic model as well, and check if the residuals are much better behaved and so on. So, to build a quadratic model, you can go through this syntax here, and this is something that you may want to pay closer attention to. Again, the same story; we use `lm`, specify the relationship between the predicted variable and the regressor; the regressor is as usual our time vector. This time around we want to build a quadratic model.

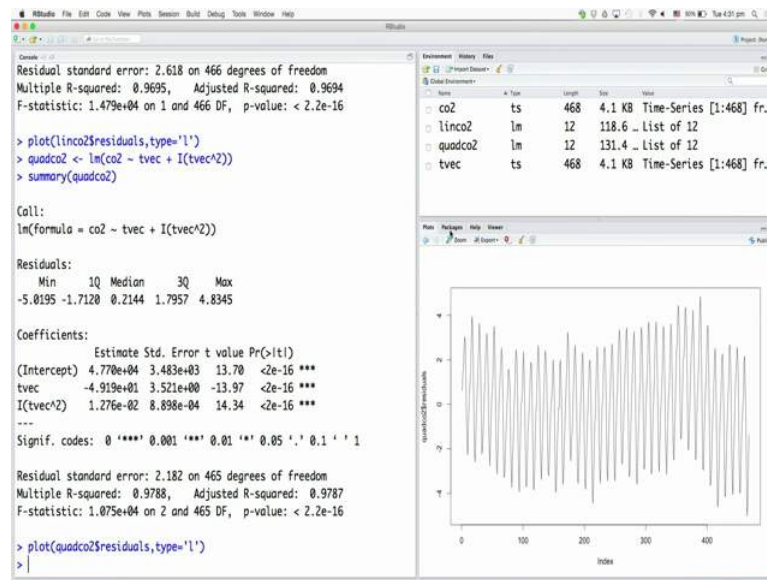
So, our model is $y = a_0 + a_1 t + a_2 t^2$. a_0 is implicitly understood to be present, so we present, we are supplying `tvec` here in the formula forcing `lm` to fit an a_1 , and then we used this syntax `I(tvec^2)`. This essentially tells R that `tvec^2` is another regressor; if we do not do this, and instead we say `tvec + tvec^2` what R would do is it would add up `tvec` plus `tvec^2`, and treat that as a single regressor, and that's not what we want alright.

So, always remember, when you are creating regressors out of a single explanatory variable - here the time vector - then we should follow the syntax. Tomorrow you may have `tvec` plus maybe cubic term or a logarithm term something like that; all of that has to be - each regressor - has to be encased in a similar fashion like this alright. So, we



have a model.

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And now, we can once again look at the quadratic model, and look at the parameter estimates. Of course, technically speaking, we should look at the residuals first, and then the parameter estimates, but we are taking a slight deviation from that. So, again, here the stars for me indicate or the p values as well indicate that the parameter estimates are significant. That is, we can reject the null hypothesis that each of the terms in my model a 0 intercept, a 1 - the first coefficient corresponding to t; a 2 - the second coefficient corresponding t square, are all not zero or the estimates themselves are significant. We do not have, unfortunately, in this case a true model. I do not know or we do not know how this CO₂ was generated. The actual process is far more complicated perhaps then the models that we are trying to fit; remember that. So, there is no truth to compare and that's why we go through this hypothesis testing; of course, if we know the truth then all this modelling exercise is futile alright.

So, there are a bunch of other pieces of information here that are reported by summary, but to go over that requires a good set of lectures on linear regression, and therefore, I am avoiding that. But if you are already familiar, you will enjoy these pieces of information and make more meaning out of this analysis.



So, once again let's look at the residuals and see if this time the trend - the quadratic trend - has vanished right. So, this time, we look at the residuals; yeah the trend has vanished. There could be, of course, a cubic one, we could do that, and then go on and check if there was a fourth order polynomial trend and so on. I leave that to you; now that I will shown you how to do this.

Let me move on to the next step where we analyse the residuals. For now, we shall assume that the quadratic trend is the only trend, but you should not do that, go further and fit a cubic trend. We will assume that the quadratic trend is only one and analyse the residuals. Now, when we look at these residuals here, we can clearly see an oscillatory behavior, which means we can now try to extract the frequency of this oscillation; that means, the CO₂ level has a periodicity to it. It repeats itself after a certain time. Now, how do we extract the frequency? I can do this visually, but that's going to be too rudimentary an analysis. Let's instead use the Fourier Transform rule. I don't know how many you are familiar, but if you are not, then Fourier Transforms allow us to detect the periodicity in a simple way by constructing what is known as a power spectrum or a power spectral density, where you analyse the contributions of different frequency components within the signal to the total power of the signal. And the simple rule of thumb, which has a theoretical basis to it, is in a power spectral plot if I see a peak at a certain frequency then that frequency component is significantly present in the signal.



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```

RStudio File Edit Code View Plots Session Build Debug Tools Window Help

Console
Residual standard error: 2.618 on 466 degrees of freedom
Multiple R-squared: 0.9695, Adjusted R-squared: 0.9694
F-statistic: 1.479e+04 on 1 and 466 DF, p-value: < 2.2e-16

> plot(linco2$residuals,type='l')
> quadco2 <- lm(co2 ~ tvec + I(tvec^2))
> summary(quadco2)

Call:
lm(formula = co2 ~ tvec + I(tvec^2))

Residuals:
    Min       1Q   Median       3Q      Max
-5.0195 -1.7120  0.2144  1.7957  4.8345

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.770e+04  3.483e+03  13.70  <2e-16 ***
tvec        -4.919e+01  3.521e+00  -13.97  <2e-16 ***
I(tvec^2)    1.276e-02  8.898e-04  14.34  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.182 on 465 degrees of freedom
Multiple R-squared: 0.9788, Adjusted R-squared: 0.9787
F-statistic: 1.075e+04 on 2 and 465 DF, p-value: < 2.2e-16

> plot(quadco2$residuals,type='l')
>

```

Now the base package in R does come with tools to perform spectral analysis or Fourier analysis; there's a command called spectrum, but I prefer the Time Series Analysis package and it's called the TSA. And I am going to load that, because it has a nicely wrapper or a nicely wrapped routine which is built on the base package to do this spectral analysis.

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```

RStudio File Edit Code View Plots Session Build Debug Tools Window Help

Console
> library("TSA", lib.loc="/Library/Frameworks/R.framework/Versions/3.2/Resources/library")
Loading required package: leaps
Loading required package: locfit
locfit 1.5-9.1 2013-03-22
Loading required package: mgcv
Loading required package: nlme
This is mgcv 1.8-7. For overview type 'help("mgcv-package")'.
Loading required package: tseries

'tseries' version: 0.10-34

'tseries' is a package for time series analysis and
computational finance.

See 'library(help="tseries")' for details.

Attaching package: 'TSA'

The following objects are masked from 'package:stats':

    acf, arima

The following object is masked from 'package:utils':

    tar

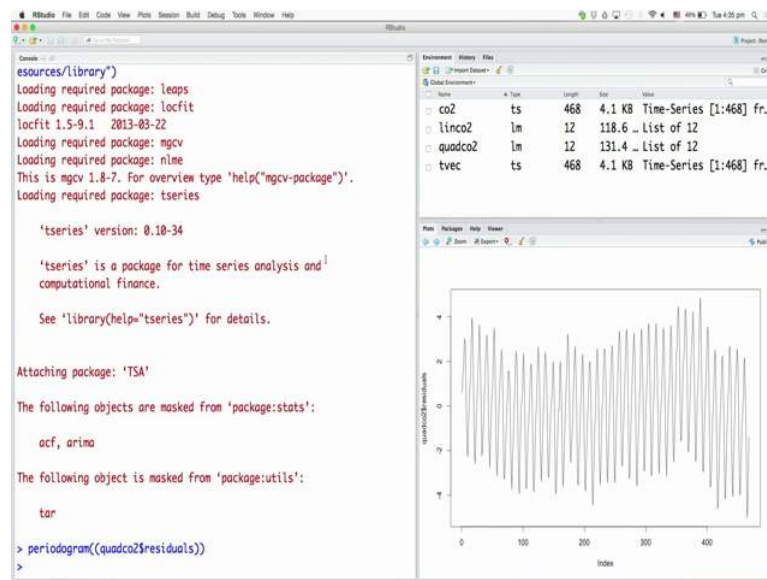
> periodogram((quadco2$residuals))

```



So, I have loaded the TSA package, and you should be... you should see this kind of a display. Now, when I load a new package in R, always there is a possibility that some of the routines in the new package share the same name with the routines in the existing or the base package. And R gives you a warning that the following objects are now marked from this package and so on. So, basically, they are telling you that acf, arima, tar and so on, if they are of interest to you, there is an overlapping between these two packages. So, you have to be careful and so on, but, anyway, we will not go into that. But one thing that I want to tell you, the TSA package also has a CO₂ data set, which looks quite a bit different from the CO₂ data that we have been working with.

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Anyway, so, let's quickly look at the periodogram, the command is `periodogram`. And I am going to really directly pull the residuals here, `periodogram` of `quad CO2 dollar residual`; you can see a dollar operator is being used to extract the component. And beautifully it shows me the frequency content of the signal. So, the way to interpret this plot here is on the x-axis I have power, and on the... sorry, on the y-axis I have power, and the x-axis I have frequency. And what it's basically telling me is that at this frequency, roughly, I have the maximum power; and this obviously, there is don't need to guess, is the main frequency component that we are interested in. But there is something else also this spectral analysis reveals which is a presence of a harmonic



alright. And this is, perhaps, well to conform whether there is harmonic or not - harmonics are integer multiples of fundamentals.

So, if you think of this big one - the frequency at which there is a big peak as a fundamental - then approximately the tinier one to the right of that, can be thought of as a harmonic. And typically, presence of harmonics means some kind of non-linearities in the generating phenomenon and so on, but it's very interesting, because visually we could not see this peak. So, you see, mathematical analysis brings out what we cannot really visually see, but at the same time visual examination is very important.

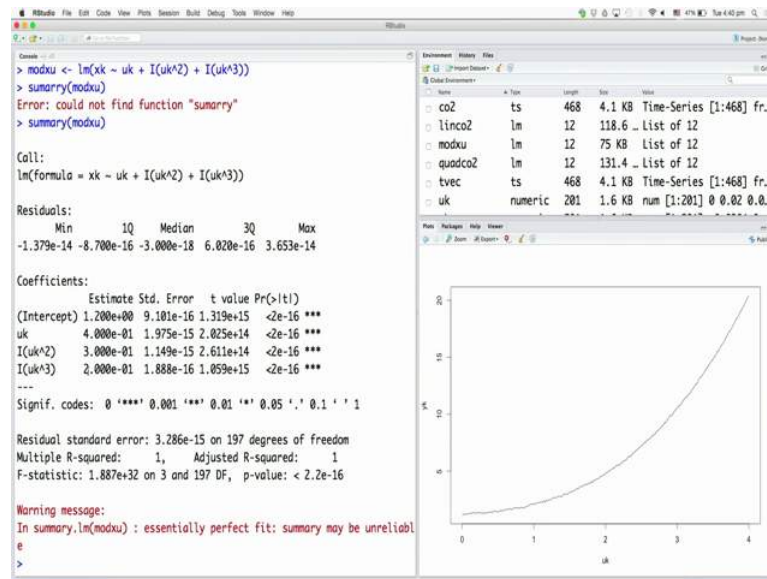
Now, if you look at the power contributions from very low frequencies - almost near zero frequencies - there seems to be some significant contribution that is perhaps indicative of a very low frequency trend, which probably is indicating that we should have also modelled the cubic one; even our visual analysis, in fact, reveal that if you go back to the plot, there seems to be some kind of a cubic trend. In fact, if you plot the cubic one, I should tell you that you will find the a three - that is a third coefficient, the forth coefficient - also being significant.

So, try it out, try fitting the cubic one and the fourth order one, and see what the analysis tells you whether the trend is indeed a third order or a fourth order polynomial, and then we have already learnt how to analyse the residuals, extract. So, now, you use these models to remove the trend and the sinusoid, and then you can see the residuals, which of course, I won't go through at this moment, but you can always write to us if you want know how to do it in more detail. We can send you the script.

Alright then, so, let's move on to the second example. In the first example, we have learnt, essentially, how to perform linear regression in a systematic manner in R. We are also going to do one more linear regression here, but this example pertains to that of over fitting example that we went through in the lecture.



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So, let's clear the screen here, and generate the data required for the over fitting example. What I am going to do is, I am going to generate the input as a sequence here, from 0 to 4 in steps of 0.02. So, I have 201 values of uk , and correspondingly I will generate the noise free response. If you recall, go and refer to the slide which gives you the relationship or the data generating expression for the noise free part, and this is what we had, and then let us call that as xk . Now, as we have always said, we never get access to the true response, we only have access to the measured value of it. So, to make the situation realistic, now we add noise. And we add noise in this fashion right; we add Gaussian distributed noise of the same length as xk . And we adjust the amplitude of the noise such that the signal to noise ratio is fairly high, and that's what I am doing here.

By default, if I don't have the scaling factor there, the variance of vk would be one; it's a unit variance random noise that I have been generating. So, let us generate the measurement; here I have yk equals xk plus vk and that's the noise part I have, noise or the measured value. So, let's simply plot here uk and yk , and see if this is what indeed we wanted yeah. So, it looks pretty similar to what we we wanted; of course, it looks a bit noisy, not so noisy. You can add... if you want, you can increase the levels of noise and repeat whatever we are going to do.



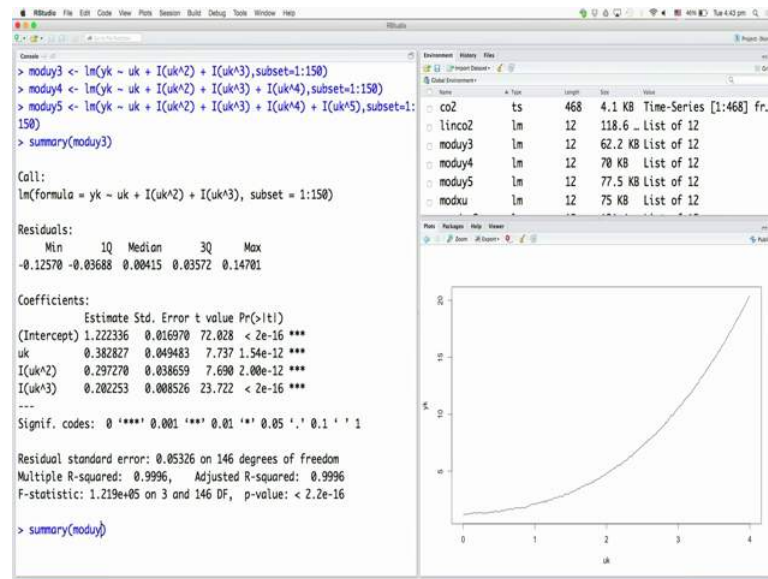
Now, let me go through two or three different models and then will conclude the session.

So, the first model I just want to show you, if you had access to the noise free part, and you had and you knew that the relationship is a third order polynomial, then you can... let's do that here okay. Let's call this as x_k , and call the l_m , and supply the symbolic relationship. So, we have here x_k tilda u_k plus $I u_k$ square plus I of u_k cube alright; so that's it alright. So, that brings up the summary for the model that we just fit between x and u . And I quickly want to draw your attention to the warning message at the bottom of this display. It says that essentially a perfect fit. It has to be, so what it says is, the relationship is perfect, there is nothing left unexplained, there is no residual from this model so to speak, and that's also reflected in the median of the residuals are extremely small. There is zero up to the numerical precision. And also, the parameters estimates have been identified in a perfect manner and so on, but this is all a utopian world, we don't have access to the noise free path. So, we move now to reality, am just showing you, that if you had everything in hand, the l_m will perfectly identified that for you.

So, let's move on to the over fitting demonstration or the over fitting example. In the lecture, I pointed out that when we over fit, that is, in this example if I fit a fourth order polynomial or a fifth order polynomial I am over fitting; if I fit a second order or a first order, I am under fitting. So, in practice - do we know what is a order of the polynomial? No, we do not. So, how do we go about determining it. Let's assume that we are given there is a polynomial relationship, even if you are not given, the plot of y versus u tells me or strongly indicates the polynomial kind of relationship. Now, I start guessing; we necessarily, obviously, rule out the first order, that is a linear one. We can begin with the second order, third order, fourth, and fifth, but I will skip the second, because the primary purpose is to show over fitting not under fitting okay.



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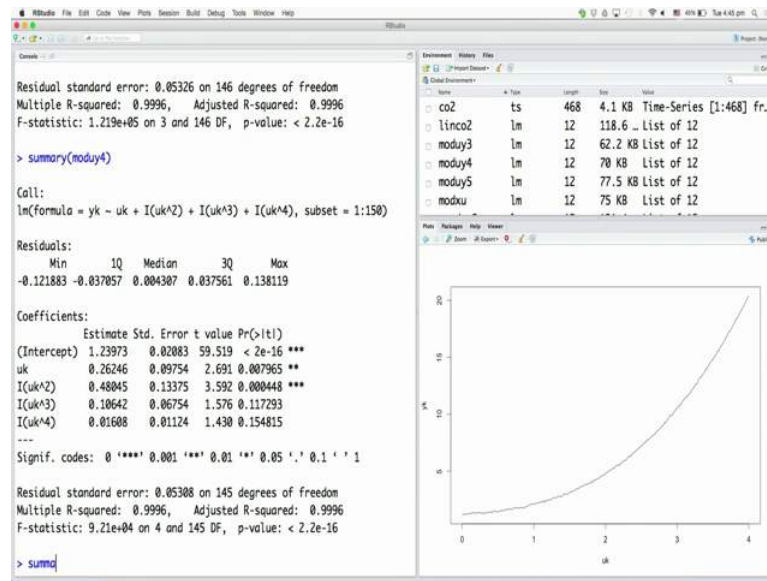


We will first identify the cubic polynomial relationship between y and u , and then build the other two models, and let me do that here. `moduy3`. So, I have here. And also, what I am going to do, is I am going to pick the first hundred and fifty points for building the models and use the remaining fifty-one for prediction. This is something that's probably going to be useful for you. And that exercise is also necessary, as I have mentioned the lecture, for cross validation. So, in `lm`, I can exercise this option by saying `subset` use a first one fifty points; you can use any one fifty points or even change the length and so on. Okay so, I have this `moduy3`; let's also build the other models right here in a similar fashion and the fifth order polynomial as well. So, I am going to build all of them at once. So, these are the three different polynomial models.

Now, we want to ask how good these models are right. A good test for these models, of course, is to look at the individual model - the parameters estimates in these individual models. So, here is a summary of the third order polynomial fit, and you can see that the parameter estimates are close to the truth. In this case, I know the truth because I have simulated the data and the estimates are quite close to the true values; you can never expect them to be exactly equal for obvious reasons. And the stars here also indicate along which are coming out of the low p values that the estimates are significant, good. So, third order polynomial is good.



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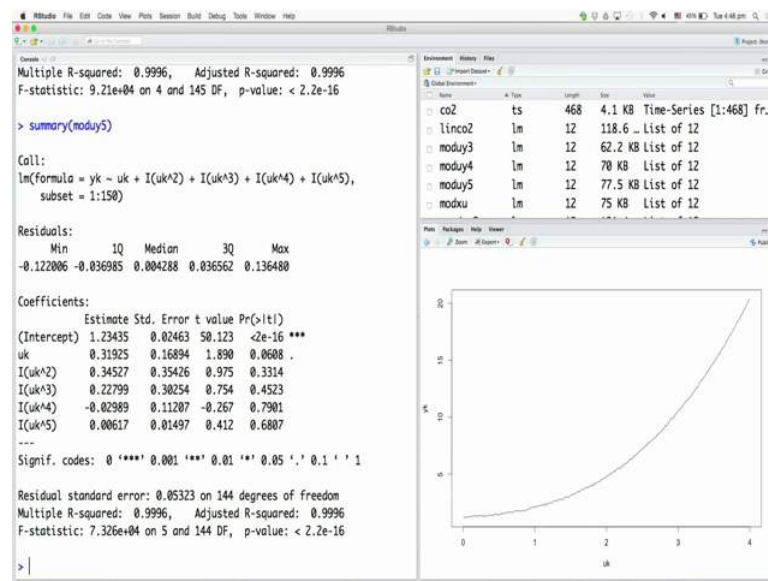
We can also look at in a similar way the fourth order. Now, you see something different. I have over fit, because this is the fourth order polynomial look at what is happen. The intercept term is pretty close to the truth, and then starts the jumble tumble here, that we have the estimates deviating from the truth significantly to the extent that the coefficient on the third order term is not significant; you can look at the error. The error in the estimate of the coefficient for u cube is quite significant compared to the estimate itself; and like wise for the fourth order coefficient as well. And that's why there are no stars here telling you that the null hypothesis, that the coefficients on u cube and u to the power four are zero, but we know that that is partially correct, but atleast we know for sure the coefficient on u to the four should be zero. What we should take home from this simple exercise is, when we over fit, we can have a situation where the parameter estimates, even though the true parameters are not zero, can turn out to be insignificant.

So, we do not know how this over fitting is going to hurt, and I always like to give this example, that data is the food for identification and parameters are guests. So, when you do not have enough food for your guests, then some guests may go satisfied and some others may go hungry. We do not know who will go hungry and who will go satisfied, but the bottom line is they will always be some guests who will go hungry and the guests here are parameters. What we mean by hungry is here large errors; we do not want that;



when we invite guests, we want all guests to actually go satisfied. So, always remember, data is food for identification; we don't want to over invite, we may under invite, but that is also not good in modelling. We want to invite exactly the number of guests that are meant for the food because you don't want any wastage of food either you have remember. And that analogy, hopefully, will stay with you whenever you are modelling.

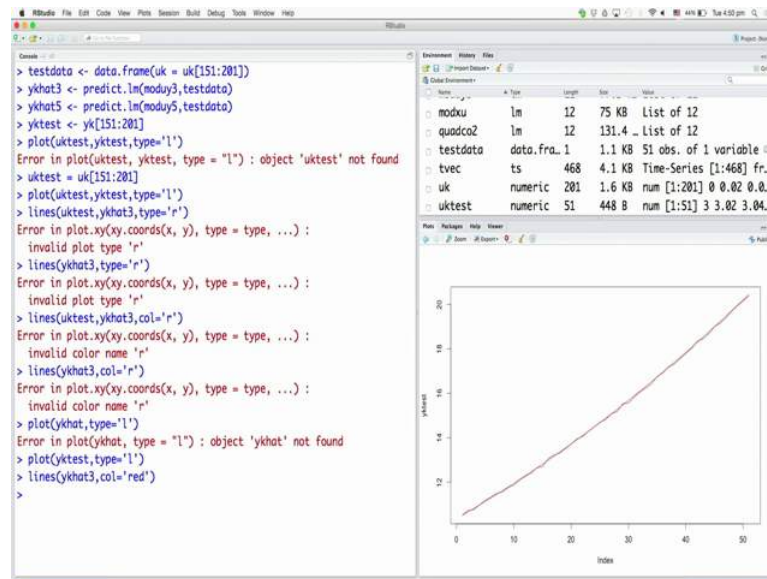
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And we should expect a similar behavior for the other one as well. So, for the fifth order polynomial as well, you can see none of the parameter estimates except the intercept term are significant, which means they have large errors in them. So, this has, obviously, gone for a full toss; we can compare, I will just compare the predictions. I will show you how to compute the predictions of the models. And we can quickly compare the predictions of the third order and fifth order polynomial on the remaining fifty-one points or we can choose the full data sets. We will just choose the remaining fifty-one points.



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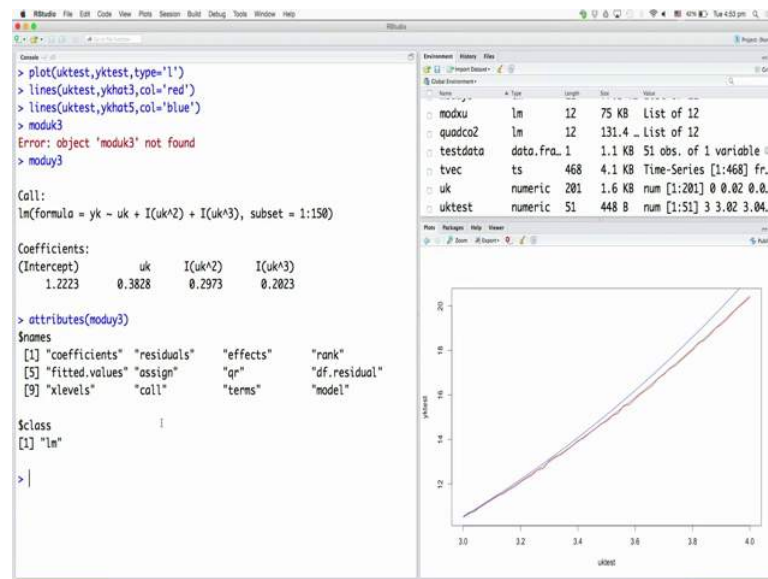
Let me do that. So, we call this as yk hat. So, let me... first the procedure to compute a prediction is to first create a data frame. Remember, that R accepts data frames for modelling and predictions and so on. So, let's create a data frame, where... I can simply say here uk 151 to 201. So, those are my... that's my test data uk. And now, I can make the prediction here alright. So, mod u y three dot test data predict dot lm or simply predict; predict knows by looking at the object that you feed, which is modu y 3, that's a model; it understands that it has to work on a linear model object. So, predict dot lm explicitly is telling R that the object or the model that I am supplying is a linear model type, and then, I am supplying the test data on which it has to compute predictions. So, I have done that. And similarly, let's do this for the fifth order as well. I leave it to you to compare the fourth order; something to work on for yourself.

Okay so, we call this as yk hat, hat is typically used in estimation for a prediction approximation and so on, and I have used the same convention. Now, let's compare the predictions here. So, I have a plot. Let's plot the yk first. In fact, let us call also yk test; create a variable which corresponds to the yk for the test data alright. So, now plot uk test or yk and test and; this is not uk test; we have not created, sorry. So, here is the plot for the 151 to 201 points between yk and uk. This is a true, the measurements. Let's ask what are the predictions of the third order and fifth order polynomial models. We can use



here lines and say uk test, yk hat 3, and we can use a red color to see how the plot is. So sorry; this is color. So, let's take a different approach here. Let's plot simply the yk hat, yk test, and then plot yk hat 3. So, we can compare the predictions here. I have plotted yk as the function of time.

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But we might as well plot as we did earlier here. We can plot this as a function of u as well okay. And then, here lines. Alright great ! And then we can also plot, maybe, on blue - in color blue; the prediction from the fifth order polynomial and see where it has gone. So, you can see, that in this case the predictions of the fifth order has gone apart. In the lecture, we have seen that the predictions had gone unstable as well. Now, how these predictions behave completely depend on the realization of the noise that we add, but the fact is, first of all, the parameter estimates are not reliable in the fifth order case. And therefore, we should not even trust the model; even if you were to trust that the predictions are not so great. I can tell you, if you were to repeat this exercise, these plots here will look different and the estimates will look different, because you will generate a different realization of noise.

But the bottom line is, if you were to over fit, there are two symptoms that will be clearly visible. One, that you would see the large errors in parameter estimates, which means



you have over estimated, you have over parameterised your model; and two that the predictions will fail somewhere between miserably to not so miserably, but they will be poor on a fresh data set. Always, in modelling therefore, if where ever possible have a cross validation data set; of course, some times may have two smaller data, but it should always be a practice to have a test data set, where you can test the trained model on the test data pretty much like we test students, trained on assignments, on exam papers right.

So, go through this exercise by yourself and get a feel of what it means to model. I just also want to show you whether it is a lin CO 2 or the models that we have been fitting for example, mod y k 3 sorry uk uy 3. If you just type, this is what it would give you, but if you want the full list of components in mod uy 3, attributes will tell you what are all the other things that are contained in each of these models that are returned by lm. You have coefficients, the fitted values, you can compare the fits; the fitted values are the predictions of the model on the training data; you can extract the residuals like we did; then you can also get other details as to how it was called and so on, depending on how well versed you are with linear equations.

There is also a non-linear version of the lm routine and there is a generalized version called glm. The non-linear version would use the non-linear least squares method; hopefully, some day we will have a chance to go over those as well, but we will have to conclude.

And let me do that by saying, by hoping that you have enjoyed this modelling session and write back to us, as always, with any questions that you may have pertaining to modelling or how to fit models in R and we will definitely be glad to help you.

Have a good day.



Introduction to Research
Prof. Parthap Haridoss
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture - 18
Safety in Laboratory

Hello. In the section, as part of a discussion on research activities, and specially a part of discussion on experimental research activities, we will look at safety in our research lab setting. So, now, safety is something that is a very broad subject. It's a subject that is dealt with in great detail in many occupational working condition kind of situations. And so, what we are going to do here, is briefly just highlight specific aspects of safety associated with some of the common things that people encounter in their labs. If you want a much more elaborate discussion on safety, you really have to attend courses which are focused on safety. And some of the points that I am highlighting, in this very brief section here, are actually points that are elaborated upon in great detail in the many of the safety courses that are there. Many of them are there internationally, and you should certainly, I mean, take recourse to actually looking those courses to fully understand the range of concerns that exist with respect to safety.

Now, safety is there in industrial setting; it is also there in research lab setting. There is one major difference, I feel, that exists between industrial setting and research lab setting. In the industrial setting, typically, the activities that each person carries out in the industry is actually well defined. And so, there is very little variability in that activity, little variations in that activity over a period of time. Therefore, the procedures that they follow, the safety equipment that they have, and the aspects of safety that they focus on are actually well defined for each individual. In a research lab setting, actually, quite the opposite is true. Most of the time we are trying to do new experiments. We are making new experimental set ups and we are trying out some new feature in our experiment and so on. Therefore, we are often pushing the boundaries in various different directions. And so, we have to be even more actively involved in figuring out the aspects of that experiment that require us to be more careful, and you know, where we have to put in more effort and ensure that the experiment is carried out safely. Now safety is important



for all the people who are present in lab, for yourself, and also for the equipment, and even for the experiments itself.

Now, I would also like to point out that, you know, safety is an aspect of culture. So, it is something that you have to build into you; it doesn't happen overnight; it's not something that, you know, it's just not just a matter following just a few procedures; it is being involved in the process of safety, and only then, you can actually, you know, make your work place a very safe work place.

Now, in a typical lab, some of the things that you would see in a typical lab, which require, you know, formal ways of, you know, addressing, so that you do in a safe manner are: first of all, in most labs you would see chemicals. So, chemicals is something that is very common in most labs; you will have, you know, acids; you are going to have bases; you are going to have, may be, powder samples; may be biological samples and so on; so, the variety of different samples which all are essentially chemicals. So, you could have a variety of chemicals which are present there in a lab. So, that is something which requires some level of formal attention for you to handle safely.

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The other thing that is commonly present in a lab is electricity. I mean any lab you go to, at least to power the basic infrastructure that is there in the lab - it could be lights, it could be fans, it could be some kind of, you know, temperature control system, whatever it is or to even just power equipment that you use for your experiment, you are going to



need electricity. So, electricity is another thing that is commonly present in the labs. And it is one of the things that we most often tend to take for granted, because it is something we see on daily basis, but still actually there are very great many things that we have to be aware of with respect to electricity, to ensure that we handle it safely.

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So, then, the third thing that is commonly present in labs is water. We always have, you know, some kind of a sink or a washbasin where you are going to, you know, clean may be glass utensils, may be your hands, few different things that you might do there. So, therefore, invariably, labs have some source of water, some arrangement for water in the lab. So, that's the third thing that is very common in the lab.



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The fourth thing that is very common in the labs is the gas bottles. Of course it depends on what experiments you are running, but many experiments will often require a some kind of a controlled atmosphere. So, maybe you have a nitrogen bottle maybe you have argon bottle or maybe even an air bottle, but most important thing is the bottle is a pressurized unit. There is gas present inside it under high pressure. So, there is certain way in which it has to be handled. So it has to be handled safely. So, that's the other common thing that we see in labs.

And finally, the other thing that I would highlight at least from the perspective of what is commonly present in labs is a heat source. In almost all our labs, we have at least a hot plate, where we are heating our samples or maybe you have an oven where you, again, you know, bake some sample to remove moisture from it or something like that or you have furnaces. And Furnaces, you know, could range in temperature, going from room temperature to something like 1000 degree C, 1500 degree C. So those are very high temperatures and you need to be careful when you handle furnaces. There are also furnaces which are vertical, there are furnaces which are horizontal, there are box type furnaces, there are tube type furnaces; so lot of variety in the type of furnaces that you have there, but principally they bring into your work place a source of heat which could be controlled, which would be running at a very high temperature, and therefore, you need be careful when you handle that source of heat.



So, these are the major things that you would see in a lab. And in next several minutes, we will look at at least some of the major safety aspects associated with each of these activities that you may do in a lab sometime.

I will begin by discussing with you the general apparel that we wear in a lab. So, common things that are good to have and good to wear on a regular basis, which makes automatically build safety into many of the things that you do are first of all to wear a lab coat. So, the moment you come to a lab, you should have a lab coat and you should put it on. It is a simple thing that you put on which immediately adds some safety to you. So, many things that if you have, you know, a few drops spilling on you, they do not directly fall on you, they fall on your lab coat. So, that provides you some protection. And most of the things that I am going to show you today are things that are very cheap, very easy to buy, and therefore, they don't cost you much, but they greatly enhance the safety for you in your work place.

So, the first one as I said is your lab coat. You should definitely have one which you use on a regular basis.

The other most common thing that you should have in a lab is, you know, a set of gloves. These are nitrile gloves. I will show you other forms of gloves that are around. So, this is a standard thing that you can wear. Our first protection is our skin, and then, you know if you are handling samples, you are actually going to put, you know, samples in your hands, which would then fill pores of your fingers if you are not having a set of gloves. So, whatever it is, whether it is, you know, carbon sample or any other sample that you are using, you don't know what is the thing that is going to be toxic. As a matter of routine, so you may be handling something, and you do not, you may not be immediately always alert to exactly what is the level of toxicity of the things. So, it is best to have a pair of gloves on your hands when you are in the lab as you start operating various experiments. It also provides automatically provides you with some insulation. These are thin enough so that you can have enough of feel of what it is that you are handling. So, they do not get in the way of your experiment, but they provide you with a lot of protection. May be the only thing you have to be concerned about when you wear a pair of gloves is that some of them may or may not handle a particular solvent or particular form of liquid. So you should be aware of what is the capability of the glove and then use it accordingly.



The third most important thing, I won't even say third most important thing, it is the most important thing that is there, but the third one that I am going to talk about is safety glasses okay. As you can see, I am already wearing a pair of glasses okay; so a lot of people assume that once you wear a pair of glasses, you don't need to wear anything else. In fact, that is a very wrong notion. So, safety glasses are distinctly different from just a pair of normal glasses. So, normal glasses these have power, they have some power. So, basically, they have been designed for my eyesight and that is why I am wearing them, but the most important... there are two aspects of this glass which makes it inadequate as a safety device. The first is that it does not provide you complete coverage. As you see that the side of my eyes are open. This side is open, top is open, bottom is open. So, if I am doing any experiment, if something spills, there is fair chance that it can reach my eyes and that is something that we need to avoid. So, therefore, we have safety glasses which do a much better coverage of your eyes. So, that something I am going to show you.

So, one aspect that this glass is not good at is the coverage that it provides, but there is another aspect in which it is not sufficient as a safety device, and that is that this is not shatter proof. So, this is just glass. So, if something strikes it, depending on the velocity with which it strikes it the momentum it has, the glass can shatter. So, if the glass shatters, actually, it is even more dangerous for your eye, than simply some object hitting your eye, because that glass now has lot of glass pieces which can enter your eye. So, therefore, the two things that the eye safety equipment has to do is to cover your eye and also be reasonably shatter proof. So, here is an example of a safety glass. You can see that it is fairly well covered. You can see the sides are covered okay. This side is covered, this side is also covered. The top is covered and the bottom is covered. So, if I were to... and usually these are sized slightly large, so that you can wear them on top of your prescription glasses or your standard glasses - that spectacles that you may be using. So, they are specifically designed that way. So, I can comfortably wear this and it would cover my eye. It would also cover the glasses that I am using. So, it doesn't affect my eye sight. I can still see perfectly fine whatever it is that I wish to see, but it provides me with a lot of protection.

May be the only thing that may cause discomfort to you, but for which there are a lot of variety of glasses that you can look at is the fact that it actually may make it little hot for



you inside here. So, based on where you are working, if it's a humid environment you are working in, it may tend to become slightly uncomfortable for that reason, but the safety that it provides to you is indispensable. And in fact, this is necessary not just for working with chemicals, I highlighted the aspects of it that would help you in the case of a chemical spill, but really even if you are, you know, putting a nail on a wall, the common thing that happens is as you are nailing, hitting the nail on wall, the nail slips and it flies off. You hit it and it just flies off. And when it flies, you have no control on where it flies. It could easily strike the person's eye. So, many accidents happen where they are just striking a nail, and it just strikes the person's eye. And this a simple device which just cost a few hundred rupees or may be just a few dollars, if you buy it in some foreign location, that provides your eye with great level of safety. And these glasses are available in a variety of different styling. So, some of them actually have well defined and well directed porosity that ensures that some amount of, you know, cross flow of air is there without compromising the protection that it provides you from the perceptive of a spill.

So, these are some safety devices that are prevalent. I will show you some more.

Right we will now look at specific safety equipment and I will highlight some of those features of those equipment. I started by showing you the goggles that I was wearing. I just want to point out that we have a lot of variety in those goggles. So, this was the one that I was showing you, but there are several other versions which are similar, which may serve different purposes. So, this has some additional shielding in case you are seeing something that has a strong glare associated with it. And you also have something which is a shield; basically, a shield that covers your face, which would cover a big broader region of your face, which you can then use to do certain other types of experiments. So, these are a few different types of safety equipment that you can use, which would primarily protect your eyes, but would also cover a fair fraction of your face. So, this is some equipment; as I said a lot of variety is there. If you look for it, you will find several versions of these, which may be targeted and you know specified for a particular purpose. So, you should really look at it, and find it, and buy it.

Based on the experiments that you are doing, you may also need to protect yourself from dust. So, dust is something that again depends on the kind of sample you are using and it also depends on the specification of the sample. So, this is a simple dust mask which is a surgical dust mask kind of thing that you can get, which I am pointing out here, and this



is easily available in many, you know, medical stores or drug stores. Then, you can use them, but it depends on your experiments. For some particular purposes, it may be sufficient; it may be highly ineffective with specific types of samples. So, you need to be very careful whether this is adequate for you or not. So, you need to find out whether it is adequate, but if it is adequate, you should be in a position to use it. What I am now showing you is a much more elaborate mask which deals with dust. Here there are some filters in the front, which you can select what kind of a filter you want to use, and then, use it. It would completely cover your... If you put it like that, it would then completely cover your nose, and then, you would... whatever you breathe would then be filtered through this device. So, it is a much more elaborate and, you know, designed device to protect you from dust of any particular chemical that may be present as part of your experiment.

As part of safety associated with thermal things that we do in a lab, perhaps the most important thing is handling hot items, equipment that may be hot or samples that may be hot. Straight forward thermal capability gloves are there which can handle fair bit of heat and protect your hands from the heat. So, these are slightly large gloves, which if you wear are usually a little cumbersome to handle samples, but you have to get used to them if you are going to be holding tongs which are likely to be hot or some sample that you need to handle, some sample that is little hot, but these gloves are available, and this adds to the safety equipment in the lab. You should have a set of these gloves handy in your labs, so that you can use them as you would need that.

This is another pair of gloves which I wanted to show you. These are actually, specifically meant for handling and they are more resistant to acids. So, if you are working with acids, then these may be better gloves to have, so that you can handle them in a safe manner, because you don't want anything spilling on your hand, damaging your hand very badly, severely. So, these are simple safety devices to help you in that case okay. These are other form of gloves that are commonly available. This is also a pair of nitrile gloves, which are there. They are just simply different versions what I am currently wearing, different sizes, different thicknesses and so on are available. And they do provide you with fair bit of safety. So, as I mentioned, as I have just shown you, there are a set of gloves which handle, which provide safety for your hands, and then, there are masks which provide safety for your nose, for your breathing process, and there are set



of shields and goggles which provide protection for your eyes, in addition to the uniform that you wear, which then provides you with general protection.

So, we will now talk a little bit about gas bottle safety, and as I mentioned, the point here is that I am just highlighting some very important aspects of safety. And in all cases, there is a much more elaborate treatment of these topics. So, the point that we need to pay attention to, when we are dealing with the gas bottle, is that this is a structure inside which there is pressurized gas, and to access the gas, we do put a regulator on top of the gas bottle and this is the neck, so to speak. So, now, once you put the regulator, this is, the neck is in a delicate situation, meaning if the bottle were to fall, there is a great chance that the neck can break. If the neck breaks, the gas can come out in an uncontrolled manner, and then, it's a very dangerous situation. So, therefore, a very important thing to do when you are dealing with gas bottles is to ensure that the gas bottle is constrained, restrained such that it cannot fall down. So, a simple thing that is done is to put a chain like this, so that it is then attached to the wall; there should be a good securing system on the wall from which there is a chain that comes around, and then, goes around the gas bottle.

So, any time you walk in to a lab, if you see a gas bottle, you should see it restrained in this particular manner. It is also important to remember that is not sufficient that you simply have a chain that is around the gas bottle. You should make sure that the chain is roughly about two-thirds or above the... closer to maybe say three-fourths of the height of the bottle. So, that ensures that the bottle cannot be topple down. So, if you have the chain too low, then you have actually not served the safety purpose, because if it is too low, the bottle can still fall down. Only if it is reasonably high, the bottle will not fall down. So, a very important, a very simple thing that you can do when you are handling gas bottles, but makes a great difference to the safety in the process of handling gas bottle, is to make sure that you have this kind of a chain arrangement, which then, you know, holds the gases bottle securely. It doesn't allow the gas bottle to fall down. This is particularly important the moment you put a regulator on the gas bottle. You must never have a gas bottle in your experimental setup or in your lab, where you have put a regulator, where this neck is not covered, and you still don't have it constrained or restrained or secured.



So, you must have it properly setup this particular manner and that is a very important aspect of safety associated with gas bottles. Incidentally, the same safety would also hold when you are trying to, if you have to transport the gas bottle. Typically, to transport the gas bottle, there are trolleys. Those trolleys will have an identical arrangement in them. So that the gas bottle is loaded on to that trolley, and there will be a system which secures the gas bottle and prevents it from falling down, and that would be very similar to what you are seeing here. That would be some kind of chain that those goes around the gas bottle at an appropriate height and secures the gas bottles, so that it does not fall down.

So, this is a very simple thing that you can do, but greatly enhances the safety of your laboratory and your experiment and it is an absolute must; you should not have a gas bottle which is not secured in your lab.



Introduction to Research
Prof. Prathap Haridoss
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture – 19
How to Make a Technical Presentation

Hello, in this class, we will look at how to make a technical presentation. So, I will begin, first, by telling you that there is no specific single way in which you make a technical presentation. What I am going to do, through this class, is show you what are the several of the best practices, so to speak, in a how to make a technical presentation, and also, several of the mistakes that have commonly seen in presentations. So, these are things that I will highlight to you and so that would help you make your talk better. This is especially true if you are a first time presenter of technical content, there are common mistakes that you make, and I hope that this talk will help you make your first or initial presentations better than what they would be if you had no idea of what was involved okay.

So, this is a talk about talks so to speak and so that's how I am going to present it. So, many of the aspects that I am going to tell you, I am also going to show you that even in this talk we are following it. So, one of the aspects that we need to focus on is time; this is a fifty-minute talk - intended as a fifty-minute single talk on how to make a presentation, and we will see how well we pace ourselves through this talk okay. So, that is the way we will go about it.



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A presentation slide with a light gray background and black vertical bars on the sides. The text 'Outline:' is at the top. Below it is a bulleted list of five items. At the bottom, there is a blue italicized phrase 'What to do and what not to do!' followed by a green checkmark and a red prohibition sign.

Outline:

- Technical presentation Vs journal paper
- Constraints
- Connecting with your audience
- Illustrations
- Summary

What to do and what not to do!

✓ ✗

So, first thing we do is a good practice is to have an outline, and which is what we have here - an outline for our talk. And this outline helps us accomplish several things. Our listeners or viewers or, you know, people who are attending our talk get an idea of what are all the major aspects that we are going to cover in the talk. You can see here there are five major aspects that are listed and so that's what we are going to cover. The audience gets a sense of, you know, what are things that they need to focus on. So, for different people in the audience, there may be a particular aspect of your talk that is more interesting, and so they may pay more attention to that aspect of your talk relative to the rest of it. So, an outline helps them understand that.

So, for example, here we have five items listed here. The first is a technical presentation versus a journal paper; this is interesting for us to know, because these are activities that most people in the technical field do, and there are some variations and differences between them, and so that's something we will look at. Constraints is the next item we will look at; that's sort of the boundary, so to speak, of a technical talk. What are things that sort of define what you should be trying to do, so that's the constraints. The most important thing perhaps then is connecting with your audience, because that's what we do. There is an audience sitting in front of you, they have to feel comfortable with the material you are presenting, and that's the purpose of your presentation, and so you need to connect with them; so, that's something that I will highlight.



An important aspect of any talk is illustrations, and we all routinely put up illustrations. What I am going to show you is that there are varieties of illustrations that we can put up, that we do put up; and if you are aware of it, you can choose which is the illustration - type of illustration - that you wish to put up, and what are kinds of details that you need to look at. And finally, we will do a summary, which is the good thing to do for any talk, because a towards the end of the talk, it's nice to show in just may be a one slide, one slide is pretty much all you should put up, maximum two slides, but ideally one slide in which you sort of summarize the key aspects of your talk, so that people who finish listening to your talk are able to, you know, quickly get an understanding of how various things relate to each other. So, this is what we will do.

And throughout the talk, the idea is to tell you what to do and what not to do; so, that's what I will highlight through this talk. And just to emphasis it further, whatever it is that is worth doing, I will put a tick mark like this on that slide which would tell you that that's a good thing do; good practice to do. And where there is something that you shouldn't be doing, I will put this symbol which basically indicates no, and so that when you review this material, you will have a quick understanding of what is the better practice to do and what is something that you need to avoid okay.

So, with this we will start. So, by the way, one of the good practices to do is to have this outline slide, and to flash it a few times through your talk, so that people can again, you know, recalibrate themselves with respect to where you are in your talk, and so that's what we will do here.

You can see here, we are now going to focus on the first point, which is technical presentation versus journal paper. And based on the software you use, based on the package you use for making your presentations, there are different ways in which you can incorporate or put in your outline slide with the rest of your talk. Right now, I am just going to flash the slide a few times, that helps us to keep, you know, some kind of a frame work through which you are proceeding with your talk.

So, the first point, we will look at is technical presentation versus a journal paper, because both of these are activities that researchers do, and, typically, we are using the same kind of data. You are working in a lab, you are generating data, and sometimes you



make a technical presentation using that data, sometimes you publish it a journal. So, we just see what is the difference between them.

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A presentation slide with a light gray background and two thick black vertical bars on the left and right sides. The title "Journal Paper:" is centered at the top in a black, sans-serif font. Below the title is a bulleted list of six characteristics of a journal paper.

Journal Paper:

- Is a one time activity
- Often has formal structure – minimal flexibility
- Peer reviewed
- Adds to the repository of knowledge
- Not interactive
- No control on audience

So, in the journal paper, some of the aspects that makes a journal paper activity unique relative to a technical presentation are listed here. So, the first is that it's a one-time activity with respect to that data. So, with respect to that body of the data it's a one-time activity; you send it to a journal; you are supposed to sign an agreement saying that you have not published it anywhere else, they look at it okay. And it often has a formal structure; there's very minimal flexibility in it, because the journal will have some guidelines and some constraints saying it cannot be more than so many words or not more than so many pages; there may be restrictions on how many figures you can include and so on; and there is a certain style that they will expect their articles to be presented in - certain aspects of the information should be presented first, some aspect should be presented next and so on. So, there is usually a very rigid structure that you have to follow with respect to every specific journal. So, you should, when you decide to publish you have to contact the journal, get their style and then follow that style.

The other important aspect of a journal paper is that it is peer reviewed. Okay so, peer reviewed means it has been looked at by at least two or three other experts in the field - independent experts - and they agree that what your presenting is something unique, is something new, is something valuable to the field, and only then it ends up being a



journal paper. So, it is peer review. The general idea of a journal paper is that it adds to the repository of knowledge in the field, and therefore, it's something that goes there and joins the body of information that people can look at for several years. In fact, journal papers are around, I mean, and you know, a permanent record of what work has been done. So, you look up papers which are, you know, one hundred years old which are classic papers in that field and they are still available for us to see.

May be the two other major things that you need to understand about a journal paper are these last two points here - that it is not interactive and you have no control on audience. It's not interactive because you write it and you leave it out there, people read it at their own time, you are not present when they read it. So, if they have doubts they cannot just like that ask you, and that's part of the reason why it has such a rigid structure to it, and so that you sort of cover all the aspects so that person reading it can independently read it. So, it's not interactive; so, that something you should aware of. And you have no control on audience; you could have a high school student reading it, you could have a undergraduate student reading it, a postgraduate student reading it, a post doc reading it, a professor reading it or even a senior scientist reading it. So, you have no control, so you have to keep it reasonably complete, you have to sort of decide for whom you are pitching the paper and do it okay.

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Technical Presentation:

- Is an event that can be repeated
- Significant flexibility in structure
- Not peer reviewed
- Interactive
- Specific audience each occasion



Technical presentation is quite a bit different, even though it's pretty much the same work that you have done in the lab, which you are now presenting as part of a technical presentation. First and most important thing about it is that it is an event. So, it happens at a certain instant of time, there are people around you, you are present, and then, you make the presentation. So, it's an event and that is repeated. So, today you may make a presentation in your group, tomorrow you may make a presentation in a conference, a few days later you may visit a university and make a presentation to an allied group there, you may go to a funding agency make a presentation and so on. So, there is a lot of possibilities of repetition and the audience can change. So, your presentation changes based on your audience.

There is significant flexibility in the structure because you decide how you want to present it based on the audience that is present. You may highlight specific, you know, may be there are three slides that you want to highlight for one audience, you may add couple more slides when you present it to another audience, you may completely change the format of presentation when you go to a third audience. So, there's a lot of flexibility. It's not peer reviewed in general, any time you present it, I mean, you are just there, you are discussing with people in front of you, and you are presenting it. There are no three experts who look at it first before you come and present it in front of the audience; so, there is no peer review there.

It's typically very interactive; there are people who will ask you questions during the presentation, they may ask you questions after the presentation and so on. So, you have to mentally prepared for the fact that it is interactive. And as I mentioned, there is a specific audience in each occasion. So, you have to be prepared for the audience and pitch it at that audience level okay, so that they stay with you. So, these are the main aspects; we will look at them again as we go along and keep track of ... Some other slides will help you relate to some of these aspects as we go along.



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Outline:

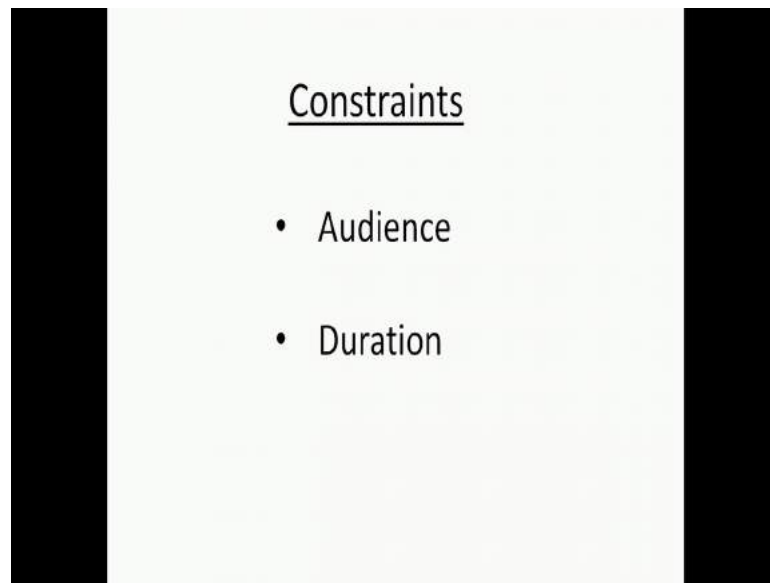
- Technical presentation Vs journal paper
- **Constraints**
- Connecting with your audience
- Illustrations
- Summary

So, we are back to our outline now - just to show you that we have completed the first topic that we wanted to look at which is technical presentation versus a journal article. As I mentioned, our intention is to do this over fifty minutes, we have five topics here: one, two, three, four and five. So, at first glance, one way to pace ourselves is to set about ten minutes for each of these topics, and as you can see, we have just about completed ten minutes, we have looked at that first topic. The summary may not take ten minutes; so, we can allow us ourselves a little extra time on these other three topics. Another way to pitch your self is to also look at the number of slides but that we will look in just a minute, but I think with respect to topics is a better way to pitch ourselves, pace ourselves, and somewhere here, we should be in the mid-way mark okay.

So, now, we will look at constraints, which includes the duration, which we will see in just a moment.

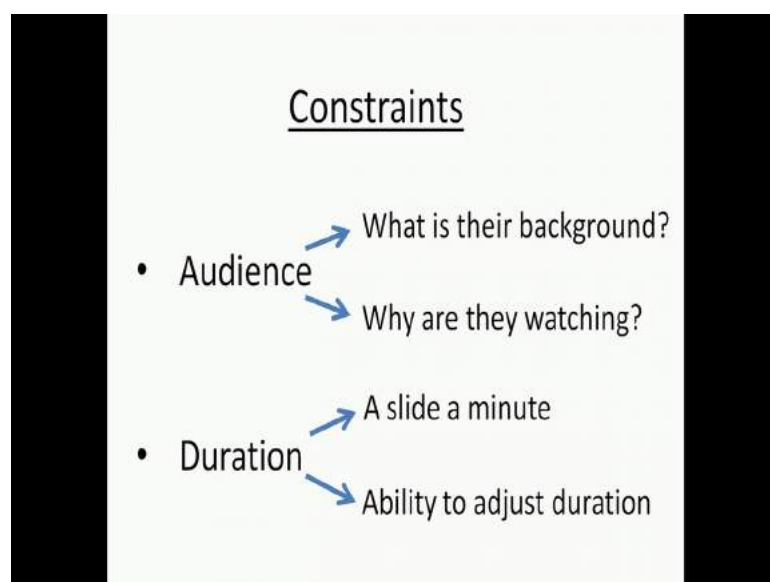


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There are really two constraints for a technical talk: the first is audience and the second is the duration okay. So, these are two things that we will look at and highlight a little bit more.

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So, the most important thing with respect to the audience is - what is their background? Okay so, you are a technical researcher, you are a researcher in a lab, you are in a university, you go to... maybe you go to your high school when you visit your home and your school teachers may ask you to make a presentation to say the tenth-grade students



or tenth standard students in your school. So, you should go ahead and make a presentation there. So, that audience is very different; they are interested in knowing what are new things in science that people look at and also get a sense, and they should be able to relate it to aspects that they are already aware of. So, you cannot just make the same presentation that you make in a conference to your school audience. So, you may have just returned from a conference, you have a presentation ready from the conference, you cannot just directly show that to your school. You have to make enough modifications, you have to ensure that there is enough time to explain technical terms if necessary, enough time to highlight the kinds of things that the audience may be interested in. So, it's very important to understand what their background is, so that you can pitch the talk at the right level okay.

The other thing that you need to understand with respect to the audience is - why are they watching? So, if it is a funding agency that's watching your presentation, then you have to highlight what is unique, that new aspect that you are going to study - which is what you need to highlight. If it is a conference, again you have to highlight after doing some bit of background presentation, you have to highlight what are the new results that you have got which are, may be, different from what other people have obtained or highlight a new aspect of that area of work which other people have not explored; so that's something that you need to highlight. As I mentioned, if it is your high school, then you need to highlight those general concepts that are interesting in that field which students can relate to, and which they can be enthused by, so that they may also consider a career in technical activities okay. So, these are the various important aspects that you need to keep in mind when you look at an audience, and try to see what it is that... how you need to reposition your talk with respect to that audience.

The other important aspect, as I mentioned, which forms part of the constraints which sets, you know, the framework within which we make the presentation is the duration. So, keeping track of duration and paying importance, showing importance, to time is a very valuable characteristic that is expected of any presenter okay. So, it conveys that you value the time of the audience who are listening there, sitting there and listening to your talk; it also conveys that you value your own time. So, it's very important that you, when you are called to give a talk, to discuss with the person who is your host to figure out how much time you have for your talk, and prepare your talk accordingly. So, this is



a part of the customization of your talk for the audience. So, you have to customize it for the audience, so that you pitch it at the right level; you have to customize it for the occasion because you have to pitch it within that time, you have to stay within that time. So, these are two important constraints that you need to keep track of. You cannot, it is not appropriate for you to just show up there and start talking indefinitely. So, you have to really pay attention to it, give importance to this aspect.

So, one general rule you can use - rule of thumb - is a slide a minute; that's a very reasonable starting point. But actually, if you get more and more confident with what you are presenting, and you have a more valuable, I mean, important things to say about the specific aspects of your work, you can actually have less number slides and more number of minutes; then you can actually stay on a slide, allow the audience to absorb what is on the slide, discuss the slide in greater detail. It is not a great idea to keep flashing slides very fast. So, this is one thing that we need to keep in mind. So, slide a minute is the good idea. So, in this case, for example, it's a fifty-minute talk, we have about thirty-five slides. So, that is what we will work towards - fifty-minute talk and thirty-five slides we have. So, we have more than a minute per slide, but I am also looking at it from other perspective, that we have five topics, and we have fifty minutes, so we are keeping track of that 10 minutes by topic kind of constraints.

The other very nice thing that a presenter should learn over the years of the presentation, which you should get, you know, prepare for mentally is that you have to have the ability to adjust the duration of your talk okay. So, this means, first of all, you have to be confident about the content of your talk. The most important aspect of all your work is your content of your presentation, all the other things help you make it a make it, present the material correctly, appropriately, and so that the audience can absorb it, but content is most important. The more confident you are about your work, you will be able to adjust the duration of your talk very easily without much trouble okay.

So, for example, you arrive at some place where originally, they told you that you are supposed to give a forty-minute talk. And then, for some reason, there is some other aspects, may be there are technical glitches in the system before you get started with your talk, and suddenly your down to 25 minutes, and it turns out that your audience is unable to extend the closing time of the talk, the hall is required for something else, so you have only 25 minutes. So, you prepare for 40 minute talk, you now only have 25



minutes; you should be able to comfortably reassemble your talk as you speak, reassess your talk, and represent your talk, so that in 25 minutes you have conveyed the essence of your work.

So, ideally, if you know your work very well, you should be able to present it in 5 minutes, you should be able to present it in 20 minutes, you should be able to present it in one hour. Naturally, all the information that you present in one hour, all that information you will not be able to present in 5 minutes or 20 minutes, but you should know what are the key things that you need to highlight. So, if it's a 5 minute talk, you should be able to pick 2 slides, may be 3 slides which capture the most important things of your talk, present it, and be done. If there's a 20 minute talk, may be take 12 slides and complete your talk in about 20 minutes. If you have a 50 minute or one hour talk, you can, you have much greater of flexibility, you can look at 30, 35 slides. So, but it's important to remember that when you suddenly move from a one hour talk to a 5-minute talk, you shouldn't take the same 50 slides, and keep hitting enter very fast, so that you flash all the slides. It's very important that the people understand what you are trying to say and quantity of information you provide is not as important as the ability of the audience to accept that information.

So, I will give you a simple example. I can take a book, many people - many first-time presenters - make this mistake, they try to show too much and that's not a great idea. And a simple way you can understand that is if I take a book, and I simply flash the pages of the book in front of you, I just, you know, twirl the pages of the book in front of you; I cannot just finish off by saying - look, I showed you the entire book, now you know the whole subject right; it doesn't work that way. You have to have time with each piece of data, you have to have time absorb the data, understand the how the data relates to something that happened before, and something that's you are going to present next and so on. So, you have to give audience time with the information you are presenting, and therefore, you have to keep a certain a reasonable number of slides that the audience can see okay. So, these are the major constraints. So, we are done now with constraints that I wanted to highlight.

We are back to our outline slide. We are doing quite well with respect to time; we spent about 7 minutes or so, on our previous topic on constraints. So, we still have about 32 to



33 minutes left on our talk, so that gives us fair bit of time to look at connecting with our audience and illustrations okay. These are two important aspects of a talk.

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Outline:

- Technical presentation Vs journal paper
- Constraints
- **Connecting with your audience**
- Illustrations
- Summary

So, we will now look at connecting with your audience. In particular, I will try to highlight aspects that you should avoid, so that it helps you connect to the audience better. So, how busy is your slide? It's a very important aspect of how well you can connect with your audience okay.

(Refer Slide Time: 18:36)

Experimental setup

CNT Properties : Mechanical
 Mechanical properties of carbon nanotubes Tensile strength ~ 200 GPa
 Young's modulus ~ 1 – 5 TPa
 Steels: $T_u = 1.2 \text{ GPa}$; $E = 0.2 \text{ TPa}$
 Intrinsic low weight characteristics: $\rho = 2 \text{ gm/cc}$ or less
 The extent to which such reinforcement can successfully improve mechanical properties has not been explored.

CNT Properties: Electrical
 CNTs can be metallic or semiconducting or insulating
 When $2n + m = 3q$, it is metallic
 $n = m$ satisfies this, so all armchair nanotubes are metallic
 Metallic nanotubes can carry 1000 times the current densities that Al or Cu can



So, we will look at that. Take a look at this slide. It's from a work that I have done. I have deliberately made the slide very busy; primarily to highlight that this is not the way to do it. You can see the no sign that is out here. So, what do we have here? We have here the sketch of an experimental set up; it's all well abled and so on; we have a photograph here which is part of this experimental set up, part of this region of this experimental setup. Then we have some photograph of a sample produced, photograph of another sample produced, electron micrographs of these two samples. And then, there is something on mechanical properties of the material synthesized, and we have properties of electrical properties of the material synthesized. This is a very busy graph; a very busy slide. There are way too many things on this slide; although, in principle, you can put so many things on the slide.

What you need to understand is just because you have a slide it does not mean you need to fill it from end to end; open space in the slide is particularly useful, it helps people focus on something that you are trying to show, it helps them understand that concept better. So, just filling it with too much material is very similar to the previous analogy I showed you, where I am just flashing the pages of a book in front of you; it does not work. So, for example, the same slide that I am showing you here could have for easily been four or five slides, there could have been one or two slides on just the experimental setup, may be a slide on the samples produced, slide on characterizing the samples, the electron microscopy of the samples, and then, a couple of slides on - one on the mechanical properties, one on the electrical properties. And these would have to be reorganized, so that they form a certain flow in the information you present. So, these four or five slides of information here; it's not a great idea to show so much in a single slide. So, you should understand and spread it out across a few slides, so that people can focus on one - a specific concept - in each slide, accept it, understand it, and reflect on it, before they see the next slide.



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Judicious use of 'special effects' available in presentation software

Judicious use of special effects okay; something that I want to touch upon because I think first time presenters make this mistake, they put in... much of the software that we have enables us to use a lot of special effects. So, I will just show you one quick slide just to...I mean I have exaggerated it just to show you all sorts of special effects that can be thrown in into a slide. Please take a look at this.

(Refer Slide Time: 21:08)



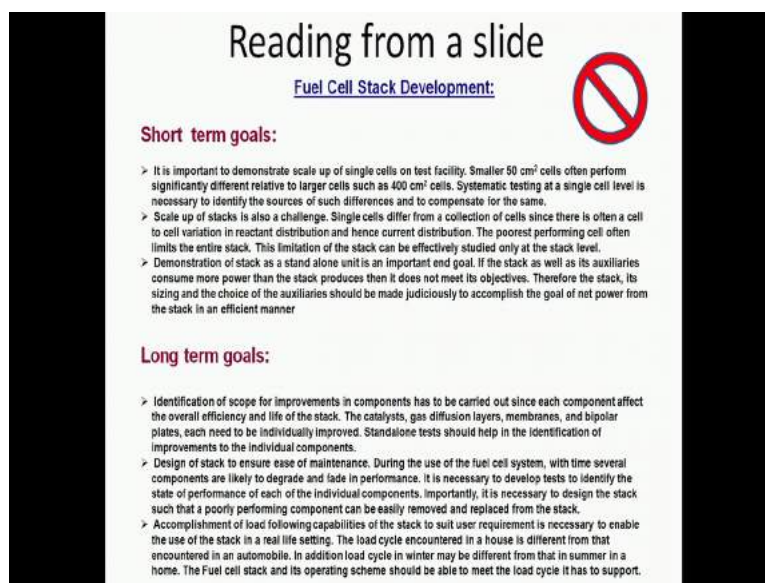
I will just repeat it for you. So, this slide has a lot of special effects. I think first time presenters may be tempted by this idea that there are lot of special effects available in



software which you can throw into your slide and so on. In a technical presentation, it distracts from the presentation. So, in a technical presentation, the focus of the presentation should be the content - the technical content - that you are presenting; special effects tend to distract from it. So, ideally, it's a good idea to avoid putting in too many special effects; some special effects may add value to your slide because they may highlight something interesting that you have got there. So, flashing some information may be a good idea, but you have to use this very judiciously, pick only those specific things that you want to use and use them.

Also, you can see on the slide, I have used few different font and so there is lot of different font that is available in the software. So, there is a tendency to try out various fonts, and various fonts in different slides; that is also not a good idea; please stick to standard, you know, one or two fonts, and then, use them through your slides. So, judicious use of special effects is a good idea; it's a better practice to do and so that's something that I wanted to highlight.

(Refer Slide Time: 22:29)



Reading from a slide okay. Again, this is what I have put up here is a slide that is not the right way to do it. So, you see the no sign there. Again, first time presenters, just to be sure that they are not missing out on a point, have this tendency to write everything that they want say on the slide, and then just read the slide, and that's not a great idea. So, first of all, when you put everything on a slide, the size of the font becomes very small.



So, it's not at all readable for somebody who is even sitting, you know, 10 meters away from the slide. And then, it's very monotonous; if you just look down on the slide or look at the slide and not look at the audience, and just keep reading.

So, for example, if I simply read – it's important to demonstrate scale up of single cells on test facility. Smaller 50 centimeter square cells often perform significantly different relative to larger cells such as 400 centimeter square cells. Systematic testing at single cell level is necessary to identify the sources of such differences and to compensate for the same. Doing this, this would be a one way of presenting it, which we will just look down on this slide and you keep on reading it; that's not a good idea; it's a very poor way of presenting; it more or less ensures that you don't connect with your audience. All these are points that I am indicating with respect to connecting with your audience.

Important aspect of connecting with your audience is to look at your audience, face your audience, not face away from the audience. So, if you have a slide up on the wall, you should not be showing your back to the audience and only reading that slide; that's not a good practice. You should face the audience, you should look at them as you talk, and so, what is there on the slide should simply be pointers on what concept you need to talk about, and then you elaborate - you conversationally elaborate on that point. You should feel comfortable to conversationally elaborate on it. Don't worry about the exact words that you are using; if there's an error you can always correct it; you should conversationally elaborate on the point. You look at the audience. If you see them nodding and they understanding, you can look at their eyes, you get a sense of, you know, how well they are staying with you, if necessary you elaborate a point a little more, if you want you can speed up a little bit. So, these are things that you can do, and so, you need to do that.



(Refer Slide Time: 24:33)

A presentation slide titled "Fuel Cell Stack Development:" in blue text. To the right of the title is a large green checkmark. Below the title, there are two sections: "Short term goals:" and "Long term goals:", both in red text. Each section contains a bulleted list of three items in black text. The slide is framed by two vertical black bars on the left and right sides.

Fuel Cell Stack Development:

Short term goals:

- Scale up of single cells
- Scale up of stack
- Demonstration of stack as a stand alone unit.

Long term goals:

- Scope for improvements in components
- Design for ease of maintenance.
- Load following capabilities of the stack

So, for example, a better way to present this exact same slide where it says, you know, fuel cells stack development - and that is an area I have work on - is to put up something like this. There are only some short-term goals that are presented, and then there is another topic called long-term goals, and there are just three points on each of these under these headings. And so, that's very easy for somebody to look at and get an idea that those are three points that you are going to look at under short-term activities and then long-term activities - three points. And that just gives you a hint on what you want to talk about; so, you go ahead and talk about it. So, this the right way to do it; you can see the tick mark on it and that's a right way to go about it.

Right so, now we are on to our... we are halfway into our talk. Actually, we have sort of about completed 25 minutes for our talk. We are moving into the illustration section of our talk, which is a very important aspect about what you do in a talk, because most of our talk we have slides, there is some text, and there are some illustrations. So, that's the thing that we need to look at and we will do that.



(Refer Slide Time: 25:32)

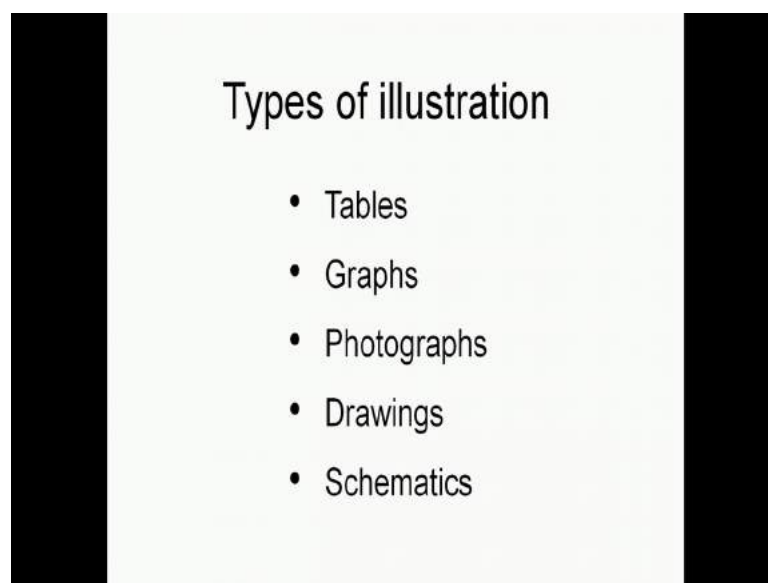
Illustration

- Choosing the right type of illustration
- Paying attention to details

So, with respect to illustrations, there are two major aspects that we need to look at - the first is choosing the right type of illustration okay. So, what we will do in the next several slides is to look at different types of illustrations that you can put up and that itself is something that we need to be aware of. Because, sometimes, a software comes with a certain illustration - type of illustration - comes as an output and we simply use it; we don't apply too much thought on it, and I would like to alert you that you should apply some thought on it - on what kind of an illustration to put up. And then, the other important aspect is for every type of illustration that you put up, you have to pay attention to details. In the illustration, there are specific details that you should pay attention to. Many first-time presenters, many students who present - make presentations - often miss out on important details that need to be looked at. And if you look at those details your presentation automatically looks a lot better okay. So, what do we mean by choosing right type of illustration. So, in the slides that follow both these points I will touch up on and highlight specific aspects in each of those slides.



(Refer Slide Time: 26:37)



So, for example, there are tables; different type of illustrations are all of these things that I am listing here. There are tables; you can have a table which shows different data for different, different samples. There are graphs; you can draw graphs with the data that you have and graphs again there are wide range of data - you can have a line plot, you can have, you know, just a scatter plot, you can draw pie chart, lot of different things you can do with graphs. There are photographs - actual experimental equipment you photograph and you put it up; so, that is something that you do. You can make drawings; drawings of an experimental setup or drawing of a flow that is happening etcetera. You can have schematic; the schematic of some system that your putting together. So, lot of different things you can do, that you have choice of.

Now, the point to remember or the point to keep in mind is that when you are trying to make a presentation, you are trying to convey some information to the audience. You want to do this effectively; you want this when you show something and you are trying to make a point, the illustration should help you make that point; so, that is the purpose of saying that you need to select the right kind of illustration. Whatever illustration you put up should help the audience understand the point that you are trying to say okay, and ideally agree with you, but even if they have a disagreement, they should able to look at the plot and then discuss with you what is that they are disagreeing with okay.



So, for example, if you simply have, you know, you want to show that a wide range of samples have been studied, and specific, you know, values have been obtained from these samples, may be a table is fine; may be a table may be able to convey that information very well. On the other hand, if you want to convey that, you know, in the range of experiments that you have done, a certain parameter has a low value for some set of experimental conditions, then it reaches a maximum, and then it comes down again to a lower value, and the purpose of your presentation is to highlight that there is a maximum for that parameter; in that case, a table is a very poor way to present it. A table does not very nicely convey that there is a maximum; you cannot just a look at a string of numbers, and then, mentally make up in your mind that there is a maximum. So, it helps to then draw the graph; so that you can see immediately that, you know, there is a difference between what can be shown on a table, and what can be shown on a graph. Even though it is your experiment, it's the same set of numbers that you have and so on, but definitely to show certain features the graph is much better than a table is. So, similarly for photographs and drawing; photographs sometimes show you the actual object, so we will see that, but there is a choice. So, you have to make that choice. I will highlight that a little bit more in our next few slides.

(Refer Slide Time: 29:06)

Table



Time (Min)	First Peak position (2θ)	First Peak width	Second Peak position (2θ)	Second Peak width
Fresh	19.967134	3.203901	40.93	1.921001
5	20.044281	3.299972	41.039333	2.151008
10	20.015936	3.612154	41.006262	2.272935
15	20.652753	4.837113	41.2	3.345802

So, for example, look at this table, I have put a NO sign against it, mainly because it has some errors. I want to use this table; deliberately it has some errors, and I am going to use this table to highlight that for you okay. So, for example, if you see here, somethings



are ok; you do have heading, you have a time, first peak, first peak width, second peak, second peak width positions here, and some timing listed here, and then some values put down here. So, few errors that are there immediately are that, you know, not all of them have the headings, not all of them have units. So, any time you put up something on a slide, you should put up the appropriate units. Please remember you are making a technical presentation, so attention to details such as units is important, because that conveys something... that conveys the actual information that you are trying to convey. So, there must be units; so, that is missing here.

And you also see here a common mistake that many students would make in their first-time presentation; look at the number of significant digits out here; it's a very large number here. So, you have, for example, after the decimal point, you have six digits here. So, it is not that automatically six digits is wrong, but you need to look at your experimental process, you need to look at your data analysis to understand if showing so many digits after your decimal point is appropriate or inappropriate. So, for example, if you are using a ruler to make a measurement and in the ruler the least count is 1 millimeter. So, in principle, you can measure a quantity which say a 19.1 millimeter. You may be able to gauge with your eye and say it is 19.15 millimeter, but that is about it. If you are write nineteen point... now, for example, although this is two theta position, if you actually write 19.967134 centimeters, then that's a very absurd piece of information to present using a ruler okay, your standard ruler.

If you say my experimental set up is to measure something with a ruler and you give the final value as 19.96 something 7134 or something like that centimeters that's completely absurd. So, you need to ensure that the data that you show on your table is consistent and meaningful with respect to the experimental process that you have got, so therefore that is an error; that is now a second error that you see in this that table. And even then, over and above that, you have stay consistent. So, for example, look at the quantities on this particular column. So, here you have the number of significant digits is actually different for each of these values that is shown here. So, that's again not a good practice. Ideally, if you are using a similar kind of experimental set up and so on, you should actually put them up to the same number of significant digits. If for some reason, you know, technically you were unable to get it to the same number of significant digits, you should put an explanation below saying why some quantities are now precise to some other



value and some others are not to the same degree of, you know, precision; so, that is something you would like to convey. So, these are all various aspects which it is wrong, the units are wrong, the number of significant digits is wrong, consistency is not there. So, these are all at least few different ways in which this slide has errors okay.

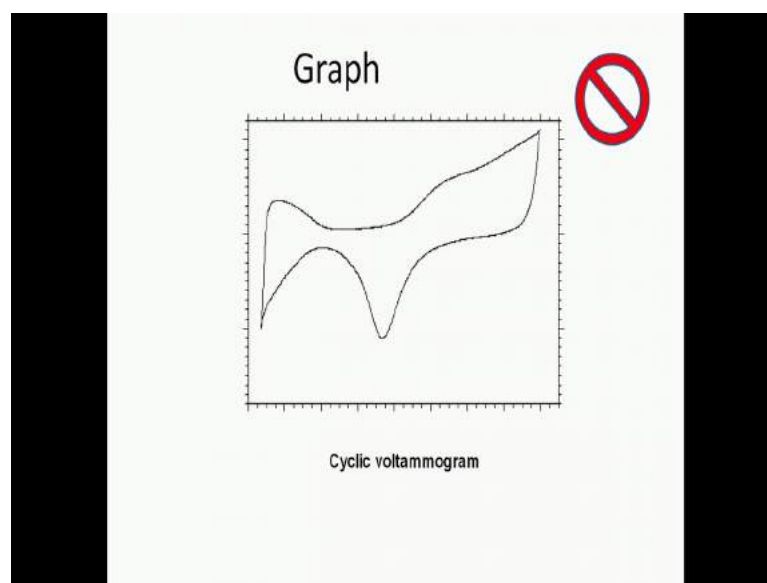
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Table 

Time (Min)	First Peak position (2 θ degrees)	First Peak Width (2 θ degrees)	Second Peak position (2 θ degrees)	Second Peak width (2 θ degrees)
Fresh	19.9	3.2	40.9	1.9
5	20.0	3.2	41.0	2.1
10	20.0	3.6	41.0	2.2
15	20.6	4.8	41.2	3.3

So, the same slide is now being presented in a much more elegant way with lot of things corrected. So, for example, you do have minutes, you have 2 theta degrees, 2 theta degrees, 2 theta degrees, 2 theta degrees. And the number of significant digits is limited, it's consistent. I am not saying this value, this is just an example that I am giving you, there may be experimental conditions where you can show vastly more significant digits than what I am showing here. I am just showing you that this is... there is something consistent about this whole table, which was missing in the previous table. So, if you do select to show some information using a table, you can, there may be specific kinds of data for which a table makes sense but there are things that you have to pay attention to. So, the choice of a table as a form of, you know, illustration is yours, the level of detail and attention to detail that you need to pay are some of things that I have highlighted on these two slides that you have just seen. So, table is one way in which you can illustrate something. Now, we will move to the next one.

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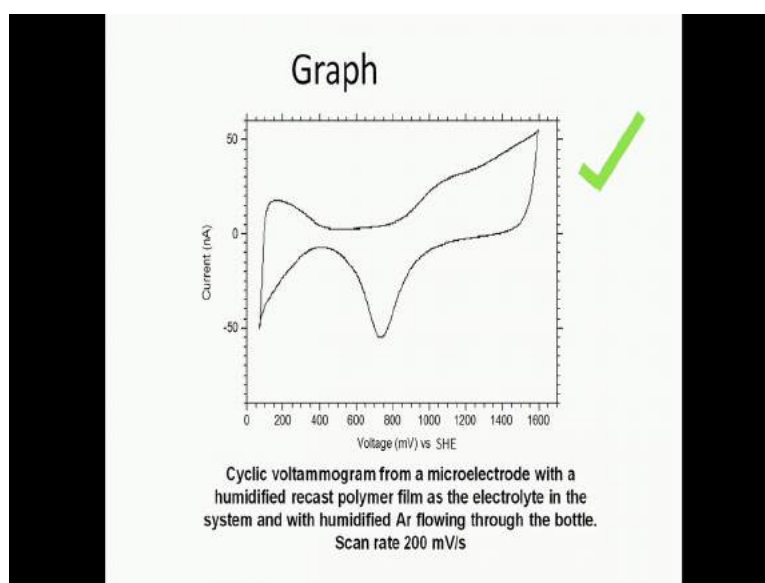
Let's take a graph right. So, again I have put a NO sign on this particular graph; many things are missing on this graph; and that something that I wish to highlight. So, if you look here, whenever you put up a graph okay, so a person looking at a graph should first and foremost know what is the quantity you are plotting on your x-axis and what is the quantity plotting on your y-axis. So, that basic piece of information is already missing on this graph. So, you have got some plot here, there is some shape that shows up here, but there is no idea what's on the y-axis, no idea what's on the x-axis. So, you have put that up. Then, you also have to put up what are the values, what are the values for these x component values that are here, the x-axis values and the y-axis values. So, there is no sense of, you know, how big a value it is or how small a value it is or even what value it is. So, this is significantly wrong with this graph right; so, that something we need to correct.

And this is a common error; actually, students put up a graph without even thinking that they, you know, because they just look at raw data from some experimental setup, and they just plot the graph without realizing that they have to do some additional work with a graph, so that is there. Sometimes, in fact, strangely enough, in some conferences you will see, people who come from a may be some R and D setup or where they are very secretive about what they are allowed to show, and therefore, because the constraint from by the their employers, they may show you a graph where only the shape of the graph is visible, no values are visible. In my view, that's extremely bad practice to do that,



especially in a formal conference of any sort. So, it is better that you don't show the data rather than show it without having any values or any units or any numbers on it. So, that's something that you really have to pay importance to, because you have an audience in front of you, you don't want them to feel that, you know, you are just wasting their time. And when you show a graph with no values on it, no units on it, then you really are wasting their time.

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So, the same graph with many things right about it, so that you get a sense of what's correct. So, you do see now here that there is current, and it says here nano amps, nA current in nano amps that is on your y-axis. And then on your x-axis, you have voltage, in millivolts, and it says against S H E that's a standard hydrogen electrode, and so that's important from an electrochemical perspective, which is what this data is presented in the context of that electrochemical perspective. So, it's not simply voltage, it's voltage versus some reference electrode; so, that detail is also highlighted there for you. And therefore, it gives you some information about your setup - it says cyclic voltammogram from a microelectrode with a humidified polymer film. So, some description of what the setup that's being examined here. And you are also given the scan rate, which is a very important parameter to give in the context of this data.

So, when you present a graph with this level of information, your graph is quite complete okay. So, somebody looking at this graph can really get many of the points that you are



trying to convey. And also, it's a good idea that whenever you present an illustration, I mean from the perspective of say the legend that you put up, in this case there's only one plot that I am showing; if there are multiple plots there should be a legend saying what is which sample. So, using the legend, using the title, using the, you know, caption that you have below it, using the label on the y-axis, using the label on the x-axis, using all these parameters which are now available at your disposal, you should make that slide complete, so that a person looking at it already has all the information up there. You are, of course, there to present and to talk and discuss it and so on, but having all that information helps the person focus on it better. Because you may be talking about something, they may be still stuck on some other aspect of your the graph that you are showing, and having all this data on that graph really helps them have a better sense of what you are trying to convey okay. So, this a much better way of showing a graph.

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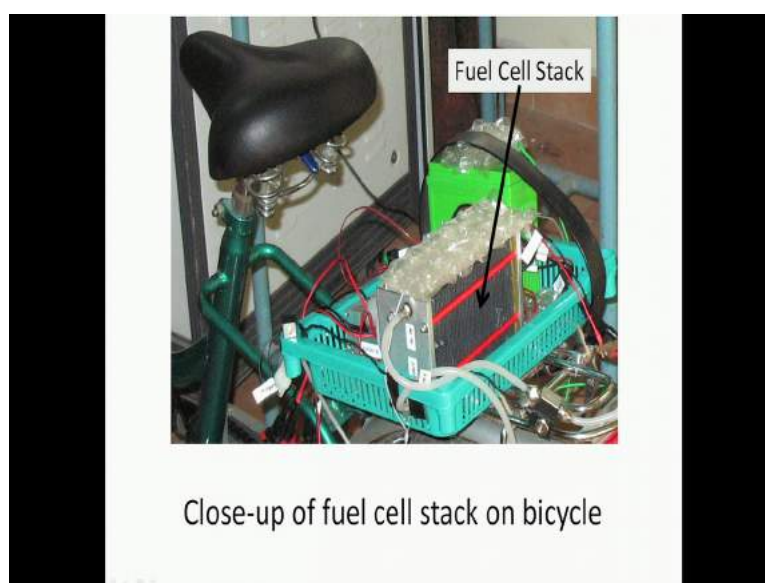
The next example, I am going to show you, is a photograph. A photograph has a great sense of conveying something visually that, you know, this is really what the object is. So, for example here, this is a bicycle that's powered by a fuel cell, an area that I work on. And I am just showing that there's the fuel cell placed out here on top of this, you know, carrier rack that is back there. And the two things that I am trying to highlight here are - that there is a fuel cell there and that there is a hydrogen storage tank which has been mounted here; that you can see. So, that's the layout of these two devices which are necessary for the cycle to work in, and so, that's what is highlighted.



I would say, the main thing that you should keep track of when you put up a photograph, which again, first time presenters forget, is that not only should you be aware of what is visible in the photograph, you should also make an effort to ensure that extraneous things are not seen in the photograph. This is a common mistake many people make who are not familiar with how photography is done or what you should pay attention to. So, you simply go to a lab, and you take a photograph. In the lab, lot of other things may be going on; there may be a beaker in one table wherein one experimental setup which is running, some analysis going on; you may have some other instrument, you know, flashing some number which represents some current it's measuring and all that. And in front of it, if you keep this bicycle, and you take this photograph, how does that beaker relate to the bicycle; how does that current value relate to this bicycle - these are all questions that someone who first glances at the photograph may get distracted by right.

So, you should ensure that you do not have extraneous things in your photograph. So, you should move the object that you are trying take to a location where nothing else is there; may be just the wall or the floor or even put a sheet of paper, so that that's all the object they are looking at. So, that's something that I wanted to highlight. So, in fact, in the next photograph I am going to show here little bit of other things in the lab will be there, primarily, to alert you to the fact such things happen. So, in the next photograph is simply a close up of the fuel cell.

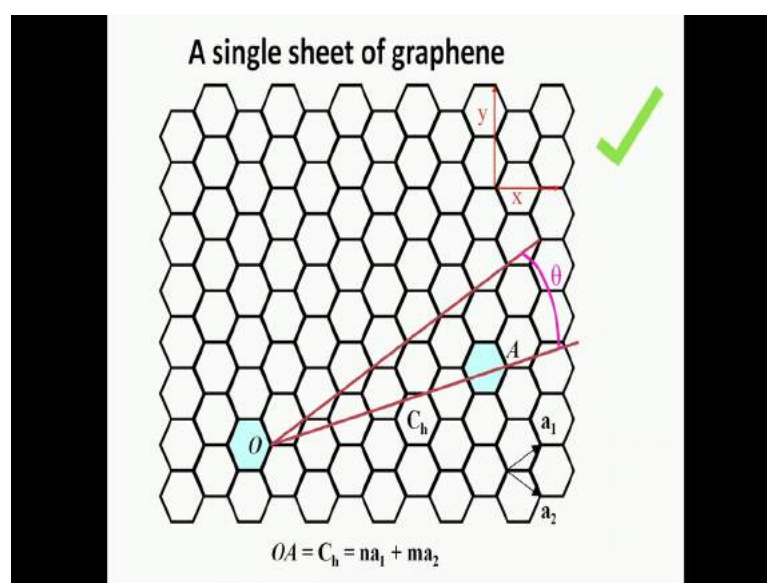
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So, you can see here a close up of that fuel cell. Again, I have marked a fuel cell stack, you do see this seat of the bicycle, so you know how it is relates everything else that you previously saw. But you can see in the background, some other, you know, there is this experimental setup that is there, there are no values flashing, but there are a few other things that are there in the background. Ideally, this is not a way to take this photograph. You should again reposition the bicycle, so that you are able to take the same shot without anything else from the lab appearing in the photograph. So, this is not the ideal way of taking it, even though they are not as distracting as some of the things that I was mentioning okay. So, it helps to have... so that's something that you need to be aware of. And in a photograph, as you can see, it's also good idea to have both an overall photograph and a close up of what you are trying to show. So, that's the overall photograph and this is the close up of what you are trying to show right. So, a few things I have highlighted here.

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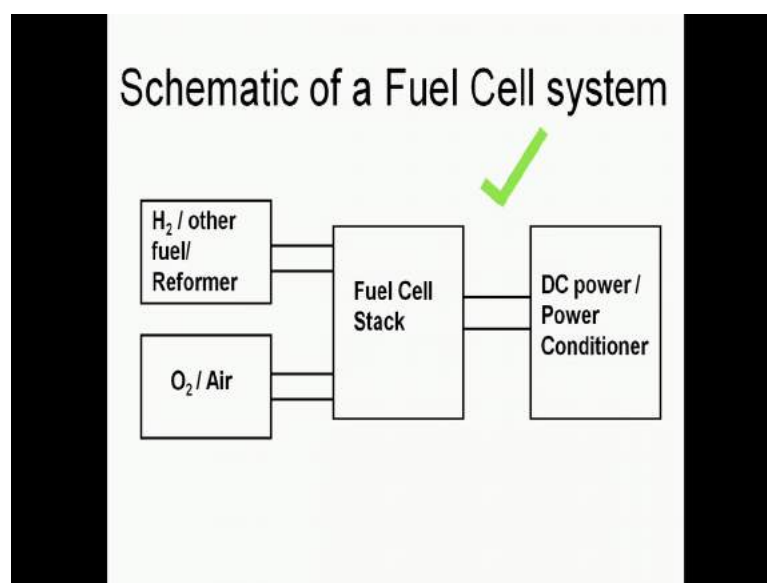


This is for example, a drawing of say a graphene sheet here. And this is a nice way in which it has been done. This is another way in which you can illustrate things. This is an illustration at an atomic level; a drawing of something that's present in an atomic level, in a very visual way which somebody can relate to. So, all the... many things are right about it. I just want to highlight it and that's why I put the tick mark here. You can see the y-axis and the x-axis marked here. You can see the unit vectors a_1 and a_2 marked here. You can also see location marked as O, and if you follow this vector here, you see



another location marked as A, it's a vector which is marked as C_h indicated as C_h designated as C_h . And you can see here OA equals C_h and how it relates to a 1 and a 2; n times a 1 and n times a 2. So, this is something that relates to how people analyze how a graphene sheet is, and how it can be folded and so on. So, this is an illustration, sort of a drawing which shows you at an atomic level how things are laid out and how you can relate to various quantities. This is a good way to make a drawing of this nature.

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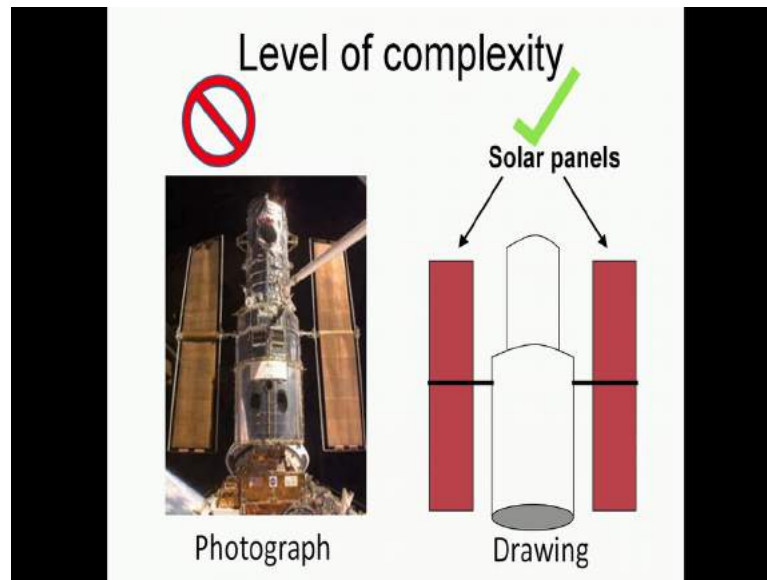
Another nice drawing that you see here is a schematic of a fuel cell system. The previous drawing was at an atomic level, this is at a, you know, much larger scale system level. You can see here the central part is a fuel cell; there is a fuel stream that's coming in; there is an air stream that's coming in; both of these feed into the fuel cell stack and then you get DC power out. So, the flow of, you know, how these things - various major components of a fuel cell - relate to each other are very nicely indicated in this schematic.

And, by the way, as I always mentioned, we need to keep track of our time; we are now down to about 10 minutes of our talk; we have finished about 40 minutes. We are just under 10 minutes now, and we have progressed quite significantly through our talk. We are talking of illustrations which is the major aspect of our talk as we move towards finishing this talk.



So, this is a nice way of doing a schematic. So, I have just shown you a drawing at an atomic level and I also shown you something at a much larger scale. So, these are good ways to present some data.

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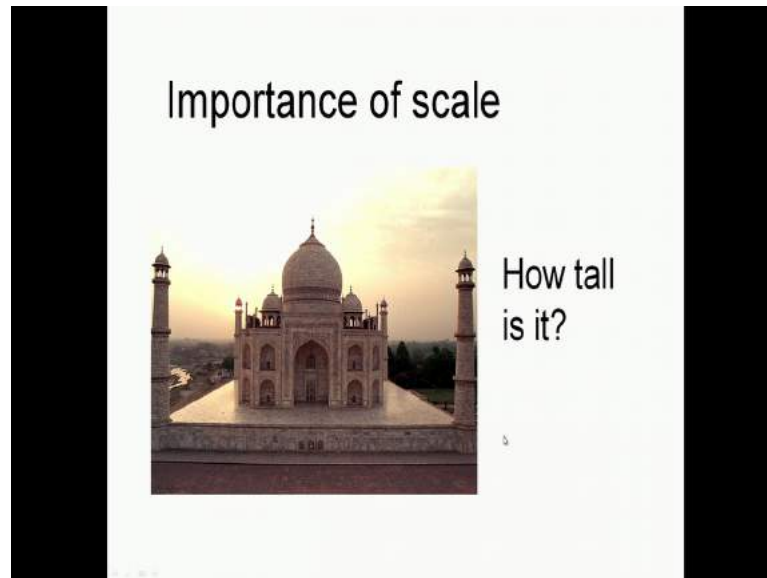
Another detail that you need to pay attention to is the level of complexity that is there in your illustration okay. So, I am showing you two illustrations here; both are the Hubble space telescope; this is a photograph of the Hubble space telescope; this is a drawing. Now, I have put a NO symbol here and a tick symbol here, primarily to indicate one particular point. If, let's say, the purpose of your talk is simply to say that the Hubble space telescope uses solar panels to power itself okay. So, when you show this photograph, it's not that it's completely wrong, but there are certain aspects of this photograph that make it distracting. And that's because you see this space telescope here, you see the robotic arm that connects it to the space shuttle, you see part of the space shuttle here, you see a little bit of earth here, you see the sky there and you also see the two solar panels; so there are many things that are there in this photograph and so sometimes it's not immediately apparent what you are trying to highlight. Even on the telescope you see lot of, you know, wiring, you see ports, and lot of different things are there.

So, a photograph has the nice aspect that it shows you the real object. So, people can really relate to it, because here you are seeing the real object, but it has this issue that it



can be distracting because there are way too many things that are visible on a photograph. On the other hand, if you make a drawing, you can really, you know, simplify all the rest of your telescope and you really highlight your solar panels. So, if the whole purpose is to just show that there are two solar panels on a space telescope, then a drawing may be a much better way of conveying this information.

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And the other major aspect, I would say the really important aspect from a technical perspective, which many students miss out on is the importance of the scale. So, here we have a monument which I have taken. I am going to show you both from the perspective of a general monument that we can easily relate to and also a technical perspective what we mean by importance of scale. So, I have managed to find a photograph where you don't seem to see many other things around the Taj Mahal which would indicate any sense of scale. So, if you have never seen it before, if you have never visited it before, a fair question to ask is, basically how tall is it? How tall is it going from base to top? And let's say, you have no sense of how tall it is, and so when you just look at the image, may be this conveys that these, you know, these features here look like they are windows, this feature here looks like it's a door and it has some four windows around it, and so, this door is, let's say, you know 2 meters high. So, may be this is may be 4 or may be this is 8 or 10 meters tall - is one, you know, perspective that you may get, if you have no sense of scale right, because there is nothing to relate to; so, that's one issue that we face.



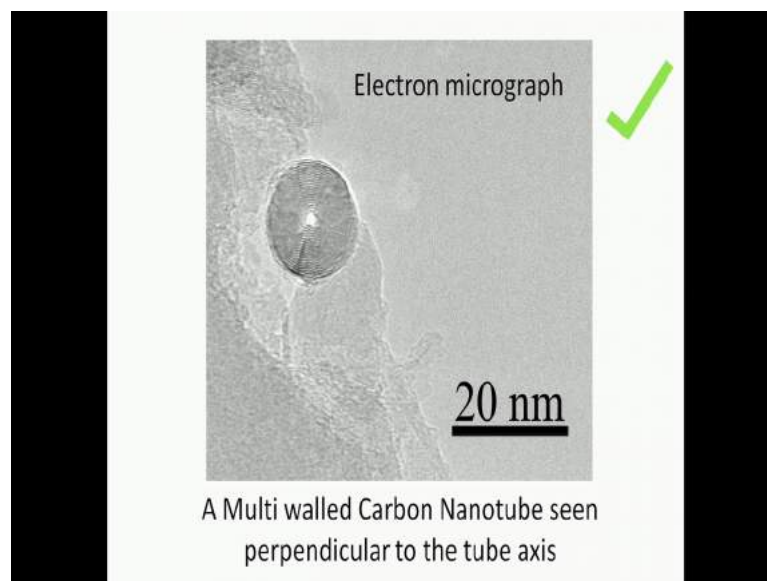
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Let's look at another photograph of the same monument. Now we have a sense of scale, because you see a lot of people here; there are a lot people here. And we know that, you know, they are all about, say, one and half meters tall, one and half to two meters tall; so, that is how tall they are; so, that's one and half to two meters tall people out there. So, if you actually scale with respect to them, you will find that this monument going from, you know, from the base all the way to the top is something like 75 meters tall okay. So, that's quite a tall monument relative to our previous estimate about 10 meters tall. And, really, the major difference between the previous slide and this slide is that we have something that we can relate to in terms of scale. In both cases, you saw the same monument - a world famous monument that most people would be familiar with, but because we have a sense of scale, we are able to identify what we are talking about, and so that highlights the importance of scale. Any time you put up a photograph or an image always try to make sure that at least there is a ruler there or some familiar object that people can look at to see what is the sense of scale.



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From a technical perspective, for example, this is carbon nanotube seen head on. So, you can see the rings of carbon tubes and you see that this is about 20 nanometers across in scale; so, this is, may be, the actual nanotube is may be 17, 18 nanometers across. And then, you can see this central tube, you can see all the concentric tubes; if you want, you can use this scale to find out the interplanar spacing okay. So, this is something that we need to pay attention too. So, these are the major things - aspects - associated with illustrations and the types of illustrations, and the kinds of things that you can pay attention to, how you select between various forms of illustrations, the particular details that we need to pay attention to.

So, how are we doing with respect to time? We are down to our last five minutes; we have completed all major aspects of our talk and so we are doing perfectly fine with respect to time.

So, what do we have remaining? We come back here to our outline slide and we find that we have completed all the four major aspects of your talk; you are down to the summary. So, we are doing perfectly fine.



(Refer Slide Time: 46:53)

A presentation slide with a white background and black text. The title "Summary:" is at the top. Below it is a bulleted list of points. The slide is flanked by two vertical black bars on the left and right sides.

Summary:

- Provide an Outline
- Technical presentation is an event
- Audience and duration are important constraints
- Connect with your audience
 - How busy are your slides?
 - Judicious use of special effects
 - Avoid reading from the slide
- Illustration
 - Choice of illustration
 - Pay attention to details
 - Level of complexity
 - Importance of scale
- Make a Summary

So, let's look at the summary. This is a talk about talks. So, the summary conveys that to you. We need to provide an outline which we did. And we used that outline at several times during this talk to keep track of, to convey, where we were in this talk. We spoke about technical presentation being an event; how it contrasted with the journal paper. We discussed, you know, how audience and duration are important constraints; that you have to tailor your to talk the audience, and therefore, you cannot simply present the same talk everywhere. And you should understand that you can present the same content with different levels of detail, and therefore, you can adjust your talk to differing durations. And I also pointed out you should be agile; you should be agile in the sense that if somebody gave you a 40 minute slot; and suddenly changed it to a 20 minutes' slot, you should show your presence of mind and represent it in 20 minutes. And the audience would be much more impressed when you do that; so, that's important.

We spoke about connecting with your audience. I pointed out that you need to pay attention to how busy your slide is; if you put up a very busy slide, then that throws off the audience. So, the amount of information you show on your slide should be reasonable and that's something that you should look at. You can use special effects to highlight a particular aspect of your talk, but you should be judicious on how much special effects you use, because that can otherwise become very distracting. You should avoid reading from a slide; again, that's very boring and monotonous. This slide for



example, has a lot of text, but that's because it is a summary slide but otherwise we would try to avoid a slide which only has so much of text.

We spoke about illustration - a major aspect of any talk; certainly a major aspect of technical talks. I told you that we have a wide range of choice in terms of what kind of illustration we can show. And therefore, you should be... and each of them has certain positive aspects to it and may be certain short comings; graphs show certain trends which are not visible when you see tables; photographs give you a sense of realism of some event or an object, but they may have too many details which can be distracting; drawings may simplify things and may help you highlight specific aspects.

I also pointed out that in terms of paying attention to detail scale is very important; choice of what illustration, level of complexity, and scale. Without scale, it's very difficult for people to understand what you are trying to present. In a graph, you should pay attention to what are the quantities you are presenting, what are the units, what are the, you know, values that they have. So, all of these things that you have to look at. In a table, you have to look at, you know, number of significant digits and so on; so, many things. So, we have a choice of illustration you need to pay attention to details, a level of complexity, and importance of scale.

And finally, you need to make a summary which is what we have done here; so, that in this single slide all the major aspects that we discussed in the 50 minutes of this talk are all captured here. So, just when you look at this slide, and you look at this discussion, it quickly helps you recollect all the major things that we have discussed okay. Again, how are we doing with respect to time? We are now down to a minute in the half. So, we have done perfectly well with respect to our time.

And so, I would like to finish by just showing you this slide, which is thank you; thank you for paying attention. Mainly again this is a very cultural thing in I think in Indian conditions, people do put this up quite a bit.

Thank you for paying attention and have a great day.



Introduction to Research
Prof. Prathap Haridoss
Department Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture – 20
Technical Writing

Hello. In this class, we are going to look at with some detail at this topic on Technical Writing. Now, technical writing is a very important part of doing formal research and that's the reason why we are going to spend this class looking at technical writing. It's also important to go through this class because technical writing is very different from other forms of writing. And so, through this class, I will try to highlight, what is special about technical writing, what are aspects that you should be aware of, that you should pay attention to as you do technical writing, so that, it's something that comes off well when you attempt to do it.

Okay so, when we say technical writing that's actually a more general phrase that I have put there. More specifically, what we tend to do is to publish - as it is called - publish articles in journals okay. So, there are journals in which we publish articles. So, this is what we refer to as technical writing in the context of research okay. And incidentally, while I go over this class, there is one very good reference book for this, which I have personally found very useful. So, I would encourage you to take a look; it is called *The Craft of Scientific Writing* by Micheal Alley and it is a Springer publication book. So, if you get a chance, please take a look at the book. It's a very nicely written book which very methodically explains to you, what is scientific writing and how you go about it. Actually, it talks about the broader range of scientific documents that one might write.

In this particular class, I would focus more on journal based writing, but this talks of a much broader range of kind of writing that scientists might do including publishing in, you know, more general audience kind of articles. So, if you get a chance to take a look at the book, it's a very well written book, very easy to read, doesn't get it into too many difficult technical terms, you can follow it very smoothly, and benefit from it, and implement it pretty nicely. So, that's something that when you get a chance take a look. In any case, we will go over this topic in a more focused manner with respect to journal



articles. So, that's what we will do.

So right so, one of the things that we will look at, first of all, is we will have to revisit some of the ideas that I have presented right at the beginning. Towards the end of the class, we will look it again, but just to get you an idea of what is involved here. When you say a journal article, there is a certain process involved. So, it is very different from the process involved from various other things; so that's why I thought I should first highlight the process for you and then we will get on to the topic by itself.

So the general process is this. So as a researcher, you run some experiments in the lab or you run some simulation in the lab and you get some results. So, if you have found something new, so that's one of the important underlying concepts here. That it is something that is new; new with respect to what? New with respect to all the body of information that is out there okay. So, these are things that we will have to revisit in a moment, but I will just tell you in any case. So, it is something new that you have discovered in your lab and that you want to communicate it to the general scientific audience out there okay. The process by which you do that is called as journal article.

So, you write an article in which you write, you know, this is the experiment you did; these are results you obtained; this is the significance of the result that you obtained; this is the context in which the result has to be understood, and then, you send it off to a journal. So, there are many journals out there. You have to pick a journal which pertains to, which publishes articles in the same area that you are trying to publish, and then, the journal will receive your article. It will then send it to some number of independent experts in that area. So, they don't just publish it simply because you have sent it, they will send it to some bunch of independent experts, who will review that article, who will look at the article that you have sent, and then make an assessment - whether that article indeed represents something new in that area; is it something significant in that area? Is it something worth publishing? So, they will send their recommendations. So, they may say, you know, that it is very good, you can directly publish it or they may say, no, specific things we need some clarification on, and once you provide that clarification, it can be published. So, they will give some recommendations; you will have to respond to those recommendations. And when that process is complete, if the result is positive -



meaning the reviewers, those people who are the experts, who reviewed your article, if they are happy with the article – in that it represents new scientific work and it is significant work, then they will recommend to the journal saying this is worth publishing. So, then your article gets published; it gets printed. These days with online journals, with many journals having online versions, it will first appear on an online site as an official publication, as an official paper published in that journal okay. So, that is the general process.

So over the years, people have been publishing like this in established journals. So, journals get established in specific areas; so, there could be one in mathematics, there could be one in physics, there could be several in physics, several in mathematics, several in material science, and even in material science, there may be something related to welding, there may be something related to nano materials and so on. So, for various topics, there are journals okay. So, in those journals, historically, articles have been getting published in a very specific area okay. So, that constitutes a body of formal information that is available of experiments that have been conducted by various researchers all over the world okay. You may do research in a lab which you have never ever informed anybody, so that is not something then that is formally available for people to refer to; only when you publish in a journal it is something that other researchers can look at.

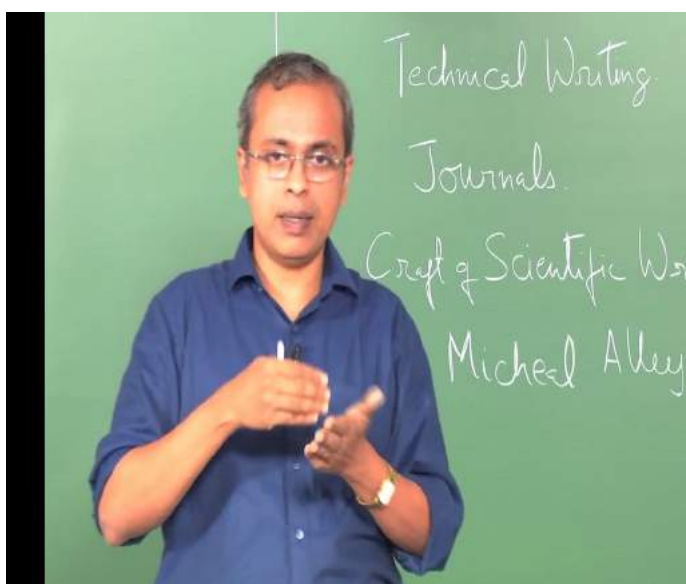
Similarly, when you do research, you look at these same journal articles published by other researchers, so that's the way in which we first gauge that we are doing something that is new. I mentioned right at the beginning, that for you to publish in a journal, one of the defining criteria is that it is new work okay. So, how do you first of all prove that it is new work. One way in which you prove that it is new work is you compare it with the work of other researchers in the same area. And you do this comparison using articles published by them. So, you use prior published journal articles to justify your current article okay. This is the process by which you are now comparing your work with previously published journal articles. And if it compares favorably, if there is something nice in it, something new in it, then it adds to that same body of information and your article also now becomes a journal article okay. So, that is the process. So therefore, journal articles, and therefore, technical writing is a very important aspect of research.



And as researchers, this is one common activity that we participate in.

In fact, in a very general sense, one of the measures of scientific contribution is the journal articles that you publish. It is not the only way to measure how good your work is, but one of the measures is how many journal articles you have published, in which journals you have published because some journals set standards, which are considered very high, and therefore, it's considered very difficult to publish in those journals. And so if you publish in those journals, it is automatically assumed that you have done a very high quality work okay. So, that's the general context in which we talk about publication in journals.

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So now, one of the issues that we... okay so, we are now going to talk about this journal article process, and the journal technical writing process, and some of the details associated with it, but before we do that we will take a step back, and I will just, very briefly, introduce to you things that you already know. And, the reason I will do that, is it is exactly for this reason - that you know these other things that I am going to show you - that technical writing is challenging. And that's the reason I am going to show them to you.



So as a general person, many of the things that you read are different forms of literature. For example, here are a set of story books. So, these are novels, famous novels which you have lightly read and there are many more. These are old classic novels, there are many more recent novels and so on. This constitutes a form of literature. This is available; you can buy it; you can read it; this is also published okay; so, this is published literature. The other thing that you are very familiar with on a daily basis is newspapers. So, you get newspapers everyday, delivered to your home or to your office, and we will tend to look through them, leaf through them, read them, and there is a wide range of different types of articles that appear in newspapers. This is again something that you are very familiar with.

These are magazines okay. So, this is again another form of literature that you are likely familiar with, and you have seen many times. This is just some samples that I have picked up, but there are many other journals and many other magazines that you would have read. So, these are common place literature okay; that's commonly available and this is the kind of written material that we are more familiar with. Growing up, these are the kinds of things that we read okay. So, I have some more examples here. So, these three, for example here, these are journal articles. So, these three that I am going to show you here are journal articles okay; these are journal articles; this is what we are going to look at in greater detail okay. This is a journal article. I will again get to show it to you detail and also explain various parts of it and so on, but this is a journal article. There are three articles here: one, two and three; and you can look up many more such articles on the Internet.

So right, the reason why I showed you these different forms of written material that you can access, is that they represent totally different styles of writing okay; that is the first thing that you should recognize - that you have literature out there, that you are likely reading very often, but much of that is very different. First of all, they are different from each other, and they are certainly very different from a journal article or a technical scientific document, and most of the time the greatest difficulty that students face is recognizing this difference okay.

For example, if you take a story book, so when you look at story books, the intent is to...



they are all written for various different intentions, but then basically one of the underlying ideas that you see is that the author would like to keep you interested in the book, try to make any person who picks up the book, get engrossed in the book to read all the way to the end of the book. So, that's intended to be something that any person picking up will stick to the book and try and read to the end of the book. So, that's the kind of intent with which the book is written. Often, if it's a mystery novel or a suspense novel, etcetera, specific details will be hidden; specific details will be hidden so that all the way to the end as a reader you are confused, you are unable to latch on to that specific technical detail which they have hidden somewhere very nicely in that story, and at the end, the greatest pleasure that you get from the book is how well they have hidden the detail from you, and you know, you feel thrilled that, you know, it was right there in front you, and you never noticed it, and the author springs the surprise on you. So, that's how a typical suspense novel is written.

Newspaper articles, on the other hand, are written on day to day events that occur. So there the focus is the urgency with which that information is being conveyed to you. The key details that are being presented to you, and the you know, the concise way in which they are able to get expert opinions and put it out there for you; so, that's how a newspaper article gets put together.

Magazine articles extend what the newspapers provide to you; they give you much more, you know, they give you some context in which issues are being discussed. Again, they are written by various different people and presented to you based on the nature of that magazine; so, there may be a magazine on interior decoration, then they only talk about interior decoration articles; there may be others on current events and so on.

So these are all different, they are all written for different audiences, different level of depth, different level of reading patience that they expect from you and so on. But one of the things that you notice about these is that they are not necessarily sent to three reviewers to get reviewed, etcetera. They may have some review process, the publisher may have some reviewer process to understand whether it is something that they want to get into, but in a fundamental sense they are not in quite the same way that a technical article is written. Technical article formally requires the multiple experts to review it to



say that it is new work, and that's how it becomes a technical article, but there is more to it. To enable this process to occur smoothly there are various aspects associated with technical writing that we have to become familiar with okay. So that's what we will look at.

Right so we will now look at why we do technical writing? So, why should you do technical writing? The primary reason why you do technical writing is to get credit for your work. Okay so the primary reason for technical writing is to get formal credit for your work okay. As a researcher, as I said, one of the things that people look at is the number of publications you have, as some measure of how much you have been contributing to the scientific community. It's just one measure, but that is a measure and when you publish in a journal, it means that you have done something new, which other people have not done before, which certainly other articles and journals have not mentioned before, and therefore, you are now credited with it.

So in future, people will say that so and so was the first person who did this work okay. So, that is the important reason for this - to get credit for your work and that is the particular reason why you should be coming up with technical writing. Of course, what you do when you credit for your work is also you are informing people about new results. So, that's another important thing that you are doing in the process, so you are conveying information, conveying important new information okay. So, you are conveying important new information.

So what this information gets used for is a different story. So for example, based on what you have discovered some public policy may be changed okay. For example, if people are trying to fund one form of fuel verses another form of fuel, and your work is the first work that shows that one form of fuel is cleaner than the other form of fuel, and you are able to convincingly prove that, then may the government will then fund the first kind of fuel in preference to the second kind of fuel or at least this will be one of the factors that will help them decide. So, there is a lot of public policy that is affected by research that is done. And in all cases, it's not just a person's view that gets taken into account; it is what they have actually published that really makes the difference. So, not only you get credit for your work, your work then gets formally used for other purposes including forming



polices and so on. Therefore, technical writing is important; it is good for you, for your profession. It is good for the scientific community because it helps the scientific community in a systematic manner. Keep track of all the accomplishments of various scientists around the world and to help the scientific community share this information between various scientists, and therefore, it is very important, and then, it also gets used. So this is the reason why you should do technical writing okay.

So now, having understood why you need to do technical writing, we immediately, I think, should address this point - as to why it is difficult? So, why is technical writing difficult? So first of all, I must step back a bit and tell you that most students - most research students - if you ask them, most graduate students, who are writing are in the initial years of their graduate studies or post graduate studies, who are writing their first paper or second paper, most of them will tell you that it is a difficult process. It's not an easy process and there are various reasons for it. The main reason is that most students are not aware of what is required in a technical document, what is required in a journal paper, and therefore, they write things that are perhaps not appropriate for a journal or not in the appropriate format for the journal. And one of the primary reasons for that is exactly what I showed you right at the beginning, is that we are very familiar with things like magazines, newspapers, novels, mystery novels, and so on. Those are the more common forms of literature that we are very familiar with.

And as I mentioned, they all have a different style of writing, and because they have a different style of writing, and they are intended for a different audience, and more importantly they are intended for a different purpose, because it is like that, when you read that, that is in the back of your mind saying, you know, that is how an article is written. I read this book, it felt really good; so, when I write a journal article, for it to be good, I should write along those lines. That's kind of imagination that we end up having, and that's exactly wrong, primarily because that was intended for a totally different purpose, a journal article is intended for a totally different purpose. So, the manner in which we write these two ends up being very different. You may think that may be even though the purpose is different, you can still write the same way; it doesn't work that way. If you write the same way, the journal article becomes a very poor journal article; it will get returned to you saying that this is not the way you write it okay. So therefore, the

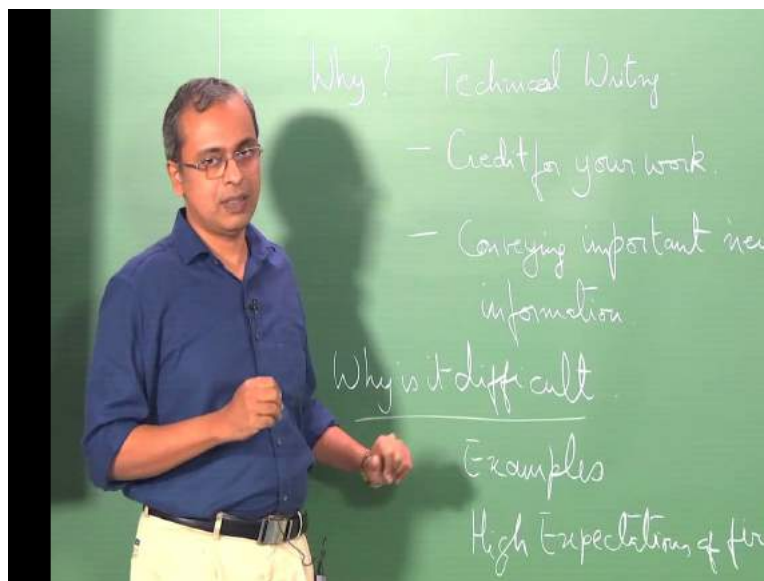


familiarity we have with other forms of writing actually works to our disadvantage in the case of technical writing at least till we become aware of it. Once you become aware of it, it's fine; you can always read all forms of writing, recognize that they are different from a technical journal article, and then, continue writing a technical journal article. So, that's, therefore, the primary thing that we have to become alert to, aware of, and implement appropriately in our activity. So, therefore, it is generally very difficult, and so mainly because of, as I said examples that we are familiar with, examples that we are familiar with - which mislead us – inadvertently mislead us - because they were just written for a different audience. Unconsciously, we just pick up those habits and try and pass it off to journals, to our technical writing, which doesn't work correctly.

And very high expectations of first draft. If you talk to a graduate student this may be the most common complaint that you will listen to from a graduate student or a post graduate student, who has just started writing papers about their work. They will say, you know, I wrote something, I gave it to my advisor and then he wanted some changes; I made those changes, I sent it back, he wanted some more changes. So, this is the process that goes on and the student feels frustrated thinking that, you know, I have finished the work, why are we wasting time, let's just put it out there. And primarily, the issue is that they feel that the very first draft that they wrote is already fine, all the results are there, let's just go put it out there; but with experience you realize that often the first draft - certainly, the first draft of your first two-three papers, after which you will learn, for sure you will learn how to write it better; the first draft of the first two-three papers that you write tends to be very poorly written from the perspective of technical writing. And therefore, it becomes very frustrating to the student because they don't know that they are not doing it correctly and they end up writing the wrong thing all the time. And it gets corrected and they don't completely appreciate what is being corrected in the paper. So, those are the things that I hope that when you get done, you will have a better sense of what is expected in the technical paper, with various sections of the technical paper, and also therefore, have a better sense of why there are changes being suggested in your paper and how to incorporate those changes okay.



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So what we will do is, we will look through this journal paper and look thorough what we can learn from the paper. So one of the points that I will make here is actually a singular, a specific point that I want to alert you to, which if you keep in mind will be the background idea which will affect all the things that I am going to tell you about a technical paper okay. And, that point is simply got to do with the purpose of technical writing. So what is the purpose of technical writing? This is the thing that we have to understand - what is the purpose of technical writing? Once you understand the purpose, all the other details that I have mentioned will make sense okay; so, that's what we are going to do. The purpose is basically, you can say, okay you are trying to convey your work. You are trying to inform some reader; you are informing your reader of some work that you have done okay. So, that is the purpose. And sometimes, in the process of informing the reader, you are also trying to argue something about that concept that you have presented, but in principle you are trying to inform the reader, may be persuade the reader about a particular point of view.

For example, as I mentioned, if you are trying to say one fuel is cleaner than the fuel, you are going to inform the reader of the experiments that you did, and then, use some arguments to say why that the results of experiment clearly show that one fuel is cleaner than the other fuel. So, you are also doing some level of persuasion of the reader; so that



is something that we will look at. So, inform and persuade. But may be the most important thing that you have to understand of technical writing is that you have to do these two things or you have to do all of these efficiently okay. This is the important work: you have to inform the reader of the work that you have done which means you are presenting the data that you have collected; you have to persuade the reader which typically represents a discussion of the data that you have provided. So, those are two different things; that itself is something that I will highlight as a difference. You inform which is to provide the data, persuade which is discuss the data, and then, both of these you have to do in an efficient manner. So, the key here is efficiency; how efficiently are you going to do this process and that is the most important aspect of technical writing. So all the other details that I am going to tell you, I will keep drawing your attention to this word of efficiency okay.

So, with that we will now take a look at the journal paper, and then, we will go over what are all the parts there, and how this idea that I am mentioning starts appearing at different places in the document. Okay so, what we are going to do is like I said, go through some parts of a document, and the reason I also want to do this is to encourage you to keep reading journal papers with all these ideas in mind, and once you understand these ideas, you can evaluate, you can access the journal papers to see if they are all also meeting these criteria. In that process, you will become a much more critical reader of those papers. When you become a critical reader of the papers you become a critical reader of your own paper also.



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So, then, when you write your third paper or your fourth paper, automatically the first draft of your third paper or fourth paper ends up being much better than the first draft of your first paper or second paper okay. So, it's important to understand these things, look at papers with these ideas in mind, and then, slowly improve your ability to write your own papers. So, that is the thing that we are going to look at.

So now, if you look at a journal paper there are various parts to it. Broadly, you can look at it as a beginning for a journal paper which would consist of a title okay, then you will have an abstract, and then, you will have an introduction okay. So, these three parts constitute the sort of the beginning of your paper okay.

And in this also, there are certain things that are normally expected. So for example, in the title, you sort of indicate what is the area in which you are doing research okay. So, for example, it could be on, say, rechargeable batteries okay. So, you have to talk about the area of research and also what is unique about your work. So, that should come somehow get highlighted in the title. So, if you just write area of research, and you simply say rechargeable batteries, that does not immediately convey to a person picking up the paper what you are talking about. So, if you are talking about, say, a specific failure mechanism, let us say, you know, mechanical failure, you have done some test



on, you know, if the battery – that rechargeable battery - is subject to some mechanical failure. How does it survive the mechanical failure? So, you have to talk about the failure mechanism; you have to say, you know, mechanical failure of a particular version of a rechargeable battery; then that will be something that both identifies the area of research and also highlights what's new about it. So, what is new in that area that you have been working on which is what your paper is about okay. So, a title should immediately convey that to the reader, so that is the thing.

Now, the next point here is the abstract. So, normally if you take a journal paper, on top is the title, and then, immediate next section is an abstract. You can pick any journal paper; you can see it; this is just an example here. So, it doesn't matter if you cannot really read the text that you see here. You see a title, and then, you see an abstract, and then, you will see an introduction; the immediate next section will be the introduction. So, those are the three that we are talking of here okay. Now, the important thing about an abstract which is where most students have the greatest difficulty is that in an abstract you have to sort of give away all the important research. So, in an abstract, you have to give it all away. So, that's the important thing of an abstract.

So, you will be very surprised to note that in any journal paper that you take, the abstract that you see here, right below the title, contains all the major findings of your work, will be there okay. So, an abstract gives it all away okay, but there is some catch there; it is aimed at an reader who has not read the rest of your paper. So, you cannot assume certain things about the terminology that they may be aware of etcetera, but still it gives all the results away. This is where most of the students have great difficulty, because almost any other form of literature that you read, does not give away all the important results right at the beginning okay, particularly a novel. If you read, if you have been enjoying good stories, the thrill in the story is that the most important result or the most important point of the story is hidden from you. So, only towards the end of the story you get that important point. Here, in a journal paper, the exact opposite is true; right at the beginning you give the most important result away. You don't try to hide anything; in a journal paper, you do not follow the style of hiding your result, and then all the way to the end of the paper, and then suddenly springing the result towards the conclusion. So, what you have thought as a very good style of writing is exactly the thing that is not acceptable in



journal writing okay. So, you have to give your result away. And this point has got to do with it - completely and entirely to do with this; specific word that I highlighted earlier, that the purpose of technical writing is to convey your work efficiently. So, this is the most efficient way of conveying your work. Right at the top of the document you give away your best result.

Now, students typically become very apprehensive that, you know, if I publish a paper, and then write at the top of the paper I have given my best result away, why will anyone else want to read the rest of your paper? So, this a concern that students have, so they become very uncomfortable with this idea that you should give your best results away. If I give it right away on top, then why will anyone read the rest of the paper. This is where again the purpose of technical writing differs from other forms of writing. The purpose of technical writing is not to make people read the entire paper; that is not the purpose of technical writing. The purpose of technical writing is to give information of your work efficiently. Only if a person is interested in more details they will read the rest of your paper, and that is something they accept, we also accept.

So, when I pick up a paper published by somebody else, that's exactly what I am looking forward to in that paper. If somebody else picks up that paper published by me, that is exactly what they are looking forward to in that paper. They want to see the important result at the beginning. If they are also working in that area, if they want to do an experiment which is along the lines of what I have done or want to improve on an experiment that I have done, then they may read more details of it in the paper, but the purpose is not for me to somehow force them to read the rest of the paper. So, up front, when you accept this idea that, that is the way in which technical document is written, it becomes much easier for you; you breathe easy knowing fully well that your best result is out there. If somebody just wants to read the abstract, and move on, and do something else, that's fine. We have no problems with that; we should be happy with that okay. So, that's what an abstract is. So, you give away all your important results here; you give it all away in your abstract.

An introduction talks about what is your work, you also talk about why is it important, you may say something about what is required to understand your work etcetera okay.



So, what is your work. So, you will give some background, you know, I am working on, say as I mentioned, rechargeable batteries. And, why is it important? I mean there are lot of people who are interested in having rechargeable power sources for portable applications and so on. And you also want to indicate what is required to understand your work; meaning, if there is some background information that people need to be aware of. Let's say you are looking at a particular kind of model for rechargeable batteries, and you are using that model to assess your data, then you mention that saying that, saying that, in this paper we are using this particular model. So, that alerts the reader that if they want to understand your paper, they should be familiar with that model. They may need to go on look up some other paper or may need to look up some other book to make sure that they know what that model is. So, that they can understand your work.

And also, when you talk about, why is it important, this is another specific place where you draw the attention of the reader to other published literature papers, other technical papers in that area, saying that you know, so and so researcher x has done the following work, researcher y has done the following work, and researcher z has done the following work, and then your work. In between all of the important work that other people have done, a particular aspect has been neglected; somehow, they have not looked at a particular aspect for which you have now created some new nice experiment and that is why your work is in that area. So, that is where you compare your work with other journal articles okay.

As I mentioned, in journals that is the process. To publish, you have to first of all convince yourselves and convince the reviewers that you have done something new. So, this is a important. Why is it important is the location where you try to highlight that – that how is it that your work is new? You compare it with what is already in the literature, you put it down there in your paper itself, and then, in that process you convey that you know you have done something new.

In the end of the paper, at the back of the paper, you will have a list of references. So, typically, if you take the last page of a journal paper, journal article, there will be references. So, those are all articles that you are referring to in your introduction, in your



experimental details, etcetera. You are referring to those articles, so that you can convey what this idea; so, that's how you link up with other work that is done. So, this is the beginning of the document; this is the title; the abstract that gives everything away; and then, the introduction which mentions – what's your work? Why is it important? And then, what is required to understand your work.

And then, you get to the middle of the document. The middle of the document, basically, normally much of the work that we do, if it is experimental work, it will consist of..It will have your experimental results. It will have experimental details, it will have results, and it will have discussion.

You may have a modelling kind of work, so in which case you will talk something about the theory behind that model. How you have gone about setting up the model? How have we implemented the model? And so on, I have put down something here which relates more to experimental work, but you can correspondingly think of some parameters, which would be relevant to modelling. So, normally, many of the experimental papers will have, you know, experimental details. So, that they will simply talk of - what is the experimental set up they had, what are the different standard techniques that they have used, if those techniques had particular parameters that had to be set at some particular values - they will mention all that. So, all of the details of how each experiment was conducted will be mentioned here. Again, the intention is to make it clear enough that somebody else who wants to learn your experiment should be able to look at your experimental details and run a similar experiment. So, you hide nothing here, you make every detail of your experiment clear okay; unless you are trying to patent something, in which case, you indicate that it is being patented and there are some details which you have withheld, but that has to be very clear. Normally, the experimental details are written clearly enough that somebody else can reproduce it.

Then you have results and then you have discussion. These are two separate points; lot of students actually don't understand this difference. A result is something that any technician operating an instrument can convey to you okay. For example, if you are looking at say, the hardness values of ten samples and with, say, decreasing grain size. So, if you find that, you know, the hardness is increasing as you decrease the grain size



then that's simply a result. Okay so, that's simply a factual statement of data. So, a result is simply a factual statement of data. So, you see a trend that is going up, you simply say that I see a trend that's going up; you see a trend that goes up and then comes down, you can mention that I see a maximum in that trend. So, these are all things that any person looking at the graph can mention, can immediately recognize. So, a result is typically a statement of fact okay, there is no judgment there. It's simply a statement of fact. So, strictly speaking, if you have written your result section correctly, there can be no controversy on the result. Meaning, if I look at your results and somebody else looks at the result, they cannot have two different conclusions because you have simply stated a fact and the facts are as straight forward as that as something is a straight line, something is a curve, something shows a maximum, and something shows a minimum. So, that's a result. So, that's is one thing.

A discussion is where you try to understand the implication of your result okay; so here you are making some argument about that result. When you say that you are seeing a maximum, you are making an argument as to why is it that you are seeing a maximum. You may say that there are two parameters and conflict, one parameter is causing something to go up, another parameter is causing the same thing to come down, and as a result when the second parameter takes precedence, it starts causing the curve to come down. And so, it will be something that you are analyzing, that you are using all your scientific background to analyze and come up with the answer. So that is a discussion. And, in general, the idea is that result is something that nobody can argue with you about, discussion is something where different people can have different opinions. So, a discussion is to some degree an opinion. It is not some arbitrary opinion; it is based on some scientific background and some scientific analysis. So, you are going to apply thought in coming up with your discussion, but in general, it is something where you have much more freedom in what you are doing okay.

So, normally, this is the middle of your document and you can actually adopt different strategies for this document. So, for example, one standard strategy that many people employ for a document is simply chronological, which simply means that I did first one particular experiment, then I did another experiment, then I did the third experiment, that is how you present your result – that's chronological. You can also say that, you know,



for example, you are discussing say something about a nuclear reactor. So you may want to start from in terms of temperature, the outermost shell of the reactor is at room temperature, where let's say something outside the nuclear reactor, all the way outside, is at room temperature. Inside you may end up seeing a temperature of several thousand degrees C or tens of thousands of degrees C, even million degrees C (may be there right inside)). So, you can follow temperature, you can go layer by layer, and you can follow temperature. You can follow a particular variable, and you can say, you can do a spatial approach, you can say, you know, outer most; it is going to be something all the parameters, materials, etcetera; outer most are going to be something. You go inside, inside, inside, the materials change, the conditions change, and the implications change. So, this is something that's spatial.

You can also look at something else like the variation of a particular variable. Let's say, you have some fluid flowing through some particular process. You can just simply look at, say, the viscosity of the fluid. May be the viscosity of the fluid changes based on the conditions that is experiencing at different locations in its path; so, you can follow that. So, you will have some schematic in which you follow that variable, and you put up the key values of the variable at various places, and you imply indicate what they are. So, my point is the middle of the document has a purpose that is the primary, you know, the greatest detail of your work is presented in the middle of the document, which is summarized in your abstract, but it's presented in the middle of your document. And you have different options available to you, how you can go about to the middle of the document. I mentioned about the spatial way, you can follow chronological way, you can follow the flow of an variable. Lot of different ways in which you can see it happen, when you pick up a journal paper you can also understand that is how they are doing it. So, that is the middle of the document okay.

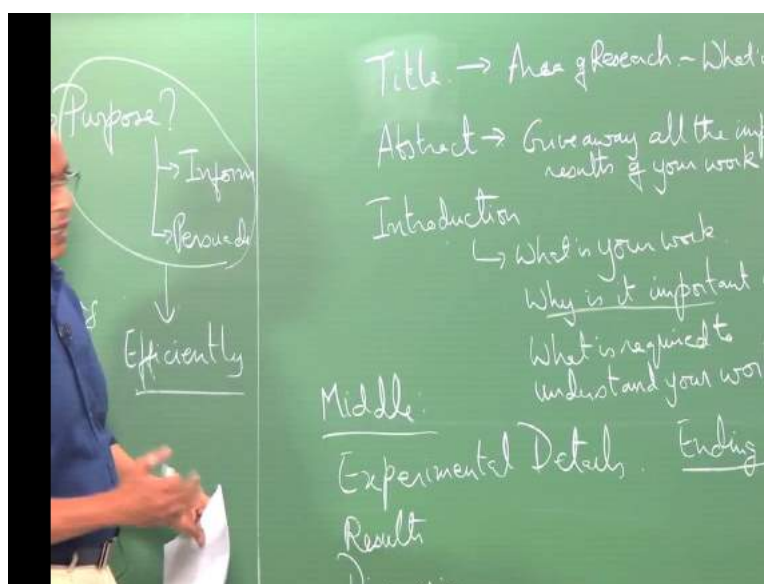
And then, finally, you have the ending of a document okay the ending of the document, which is typically the conclusions. So, conclusion of your document, where, again, you conclude the major results of your work. So, now, in some ways, the ending and the abstract have similarities; both of them show the results of your work, they summarize the results of your work, but there is one difference. An abstract is aimed at an audience that has never seen your work before; so, therefore, some of the terms that you use.. you



have to be careful with what terms you use here. You have to be little bit more restrained on what terms you use here. You cannot use abbreviations which you have actually mentioned later in your work. It helps to keep it in a manner that any new person reading it will understand what's going on here. So, that is the thing that you have to remember that the reader has not read your work.

In an ending, it's different; you can assume that the reader has actually read through your paper you have greater of flexibility in presenting your work towards the ending, still you are going in a concised way, mention all the important results. And you can also give a future perspective. You can say that you know, for example, I mentioned that those two comparisons between two fuels and that one fuel is better; you can say that one of the additional aspects that needs to be looked at is the operating condition of the fuel. To see if the fuel that is better can be made even better based on the operating condition that you are working on. So, some future perspective which you don't necessarily throw in the abstract okay. So that's how the abstract differs from the ending. So, these are the major things associated with the technical document, and in all these cases, as I mentioned, the idea is to convey your work efficiently right. So, the idea is to convey it efficiently, so that's how in each part you will find that theme being repeated. We don't want to waste words and so on.

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In the entire document, there are couple of things that show up which affect, you know, this idea of efficiency throughout your document. And those two are: language and illustration okay. So, language and illustration are two parameters that appear in various parts of your document; language and illustration are two parameters that appear in various parts of your document and they are very critical in this idea of efficiently conveying an idea. So, for example, with respect to language, again, in common literature that you read in, say, novels and other forms of literature, may be poetry you read, there is less emphasis on trying to write simple straight clear sentences. People would like to write some flowery language, etcetera, and so when you read that kind of literature, it's the wrong example to use for a technical paper.

In technical paper, you want to keep it very straight, straight forward, clear cut description of what work you are doing. Some of that common mistakes that people make are first of all, writing very long sentences. A simple gauge of whether a sentence is long or not is that when you read the sentence, you should not have to, when you reach the end of the sentence, you should not have forgotten the start of the sentence; it should not be that you read all the way to the end, and then, again, suddenly you are not sure how the sentence started, so you go back and read the sentence again.

Basically, if you write it such that the reader can read it efficiently, in a single reading they can go from sentence to sentence to sentence..So, they can keep doing that.

And other mistakes that people make are using phrases such as it is obvious. So, those are considered to be improper and arrogant ways of saying conveying some research. So, ideally, those are phrases that you should avoid. So, language is something that you have to pay attention to, which with experience you will gain as read papers on what kind of languages is acceptable, but basically you have to keep it simple. Keep it simple, keep it clean, so that a person reads a sentence, understands exactly what you are trying to say in the sentence, and moves on to their next sentence, and they are not forced to reread it again.

And then, there is illustration, illustration you have wide range of illustration available to you. You can have graphs, you can have schematics, you can have photographs, you can



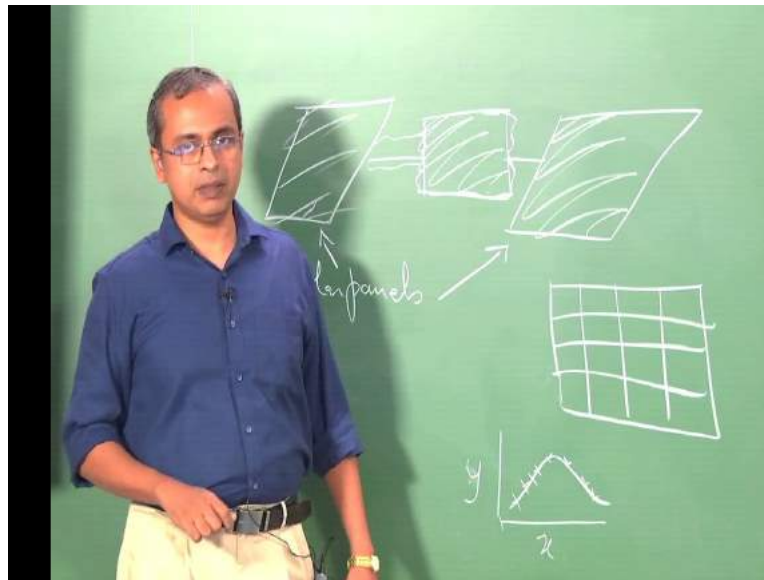
have charts of different kinds, you can have micro graphs of different kinds, etcetera. So, you have to judiciously select - what is that form of illustration that best conveys the idea that you are trying to convey okay. Often, putting an illustration together is the place where you actually put in your creativity to come up with a very interesting and nice way of efficiently conveying the information. You have a large body of information of experiments conducted in various conditions, but if you make a very nice simple table which summarizes all of those experiments, so that when a reader just looks at that table, at a quick glance they get an idea of all the range of parameters that you have done, and what are the important points that you are trying to highlight, then that makes a big difference in the reader's ability to read the rest of your paper. You can keep referring to that, you know, table often; say in table, this table such and such, you have got a particular item highlighted and the description that follows in that paragraph is relative to that item. So, as they read the paper, they can use this table as a reference, and when they get done, this table will give them a very good way of summarizing that result, and also to agreeing with you. You know when you are trying to make some argument based on a wide range of results, this table helps them focus on your argument from your point of view okay. So, therefore, making that illustration is very important and you would also understand the level of complexity of the illustration.

So, for example, so sometimes you have to make a judgment. So, you would... let's say you are trying simply say that there is a solar panel. So, you have some satellite and associated with that there is a solar panel. If the only purpose of this schematic is to convey that a particular satellite has two solar panels, and they are deployed in this manner, then a schematic of this nature is sufficient, if that's all you are trying to do. On the other hand, if you actually put the photograph of the satellite, you may find a lot of wiring, etcetera; other details - you will have the reflective surface on top of this satellite, this will have some shade associated with it, some other shade associated with it, lot of other information will be there. There will be plenty information here which will then be very distracting okay. So, this may not even be uniform, it may be of some odd shape and so on. So, lot of other detail is there which when you see can be distracting. So, you have to decide what is the kind of illustration that very efficiently conveys that one information that you are trying convey to the reader.



So, that is the reason why you have to select whether a photograph makes sense or a schematic makes sense, whether a table makes sense or a graph makes sense. These are all the ways in which you have to... because a table will give a whole range of numbers right. So, you have lot of different numbers here. So, if you have lot of different numbers in a table, sometimes that does not immediately convey the idea that you are trying to say. On the other hand, same table when if you have plotted it as some kind of a graph, you know, and it showed you this; this graph is in a much better position to convey the information that there is a maximum, than this table is in a position to. If you just have various x , y values here, a wide range of x , y values here, you know, you have a 4 by 4, 16 coordinates you have here. If I have put 16 coordinates down there, of x and y values, that does not immediately convey that there's a maximum; whereas, the same 16 points, if I plot here, if I have all those points plotted here, and I show a maximum, that's a very nice way of showing that there is a maximum. So, there is a choice, you have a choice and you have to intelligently use this choice okay.

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So I think to summarize, we have now spoken today about technical writing. I have compared technical writing with other forms of literature and writing that you may be familiar with. Tried to highlight what differences are there between these different forms. We have also walked through all the various parts of a technical document and you have



seen that there is a beginning, there is a middle, and there is a ending, and then there are certain things that are excepted out of each of these points. And your familiarity with what is excepted is what makes your writing better. If you become familiar, then when you write a technical document it becomes easier to make the document closer to what is expected okay.

So, I hope that this information will now help you become a more critical reader of technical documents and will also help to become a better writer of technical documents and so that you become happier that your drafts are much better and much closer to being the ones that are accepted okay.

Thank you.



Introduction to Research
Dr. C. Balaji
Department of Mechanical Engineering
Indian Institute of Technology, Madras

Lecture- 21
Creativity in Research - Part 1

Okay so, good morning. Welcome to this MOOC course on Introduction to Research. So, my name is Balaji. I am a professor in the Department of Mechanical Engineering. I specialize in optimization, and e-transfer, and numerical weather prediction. So, these two hours we will talk about Creativity in Research. Creativity is often misunderstood or not understood properly; particularly, in engineering, generally people have a feeling that creativity is very less. So, the moment you talk about creativity, you talk about arts, you talk about movies, you talk about music and so on, but actually there's a lot of creativity in engineering also. And in research, we will have to see how we can use these principles of creativity which have been developed for the past few centuries, so that we can make our research more exciting and meaningful alright.

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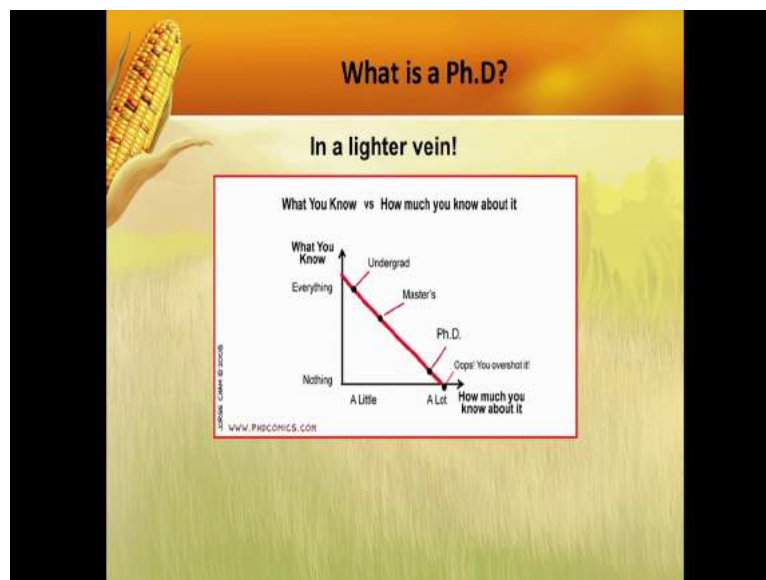


So, the outline of the two lectures will be as follows: first, we will see what are the attributes of becoming a fully professional researcher. Then, we look at Sternberg's theory of creativity. He is a professor in Yale university in US. Then, we look at the 10,000-hour rule by Herbert



Simon, a Nobel Laureate. Then, what is the role of stress in research. All of you are doing research; so, obviously, stress is an integral part of research. So, we need to be stressed, but we should not be too much stressed. So, Peter Medawar, Nobel Laureate in medicine, in 1960 he got the Nobel prize. So, his ideas on research problems. Then, what are the lessons we learned from lives of Nobel Laureates. If you look at the lives of Nobel Laureates - can we draw some inferences? Can we learn some lessons, so that we can become better researchers. Then the traits of a creative engineer or researcher. And then, summary and we will see what are the ways to improve creativity. At the end of the day, there is no point in saying all these people were great, all these people were great okay; they are great; from their lives, can we learn some lessons? Can we also improve? okay

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What is a Ph.D.? In a lighter vein... So, the y-axis, the ordinate, is what you know and the abscissa or x-axis how much you know about it okay. So, if you are an undergraduate student, you almost you get to know something about everything, but you get to know little about everything okay. And when you come to Masters level, you get to know something about... you get to know reasonably enough about many things okay, and slowly, your confidence goes down. You are having a doubt whether you know anything deep or not - that is the situation of Masters. When it comes to Ph.D., you know you know a lot; you know a lot about a very narrow area of research. When you become a fully professional researcher, then you know completely about nothing. So, jocularly, often refer to it as the exponential law or the inverse law. If you think you know everything you will get a B. Tech; if you begin to

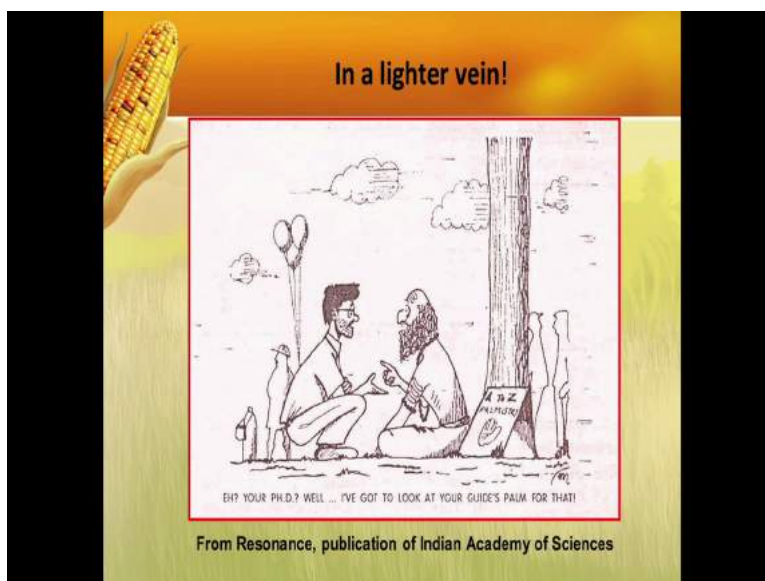


doubt that you know anything at all, you will get a Masters; but if you are convinced that you don't know anything, but you are also convinced that others also don't know anything, then you get a Ph.D okay. The last step will be if you are convinced that you know everything, if you are convinced that you don't know anything, if you are convinced that others also don't know anything, but you are also convinced that in your whole life time nobody can actually find out that you don't know anything, then you become a Prof. I know I am on air, doesn't matter alright.

So, this is the story. We try to know completely about such a small this thing, that it's almost like nothing. You take a particular problem – solar energy and like something parabolic solar collector - this thing you want to concentrate, concentrate and concentrate okay. So, sometimes, you have a doubt - when you proceed so much, when you getting so much of depth, will you not forget other things and all that, but the whole approach is you get so much depth that your clarity about a particular thing, you keep doing a tapas on that for many years, you are getting so much depth into that, but through that depth, the darshan or vision comes, then your ability to expand to other things radically improves, which you will not get if you just stop with... if you are not doing research. So, through the depth, your ability to perceive, understand other things will improve. So, we gain width through the depth; this is about it. That's why, once you have your Ph.D., and once you have... once you have gone through this process, your ability to appreciate, even you can sit in a presentation on electrical engineering, suppose all of you are in mechanical engineering, you can sit in presentation on electrical, metallurgy materials, whatever, then, quickly, you will be able to... you will be able to grasp what somebody is trying to say okay.



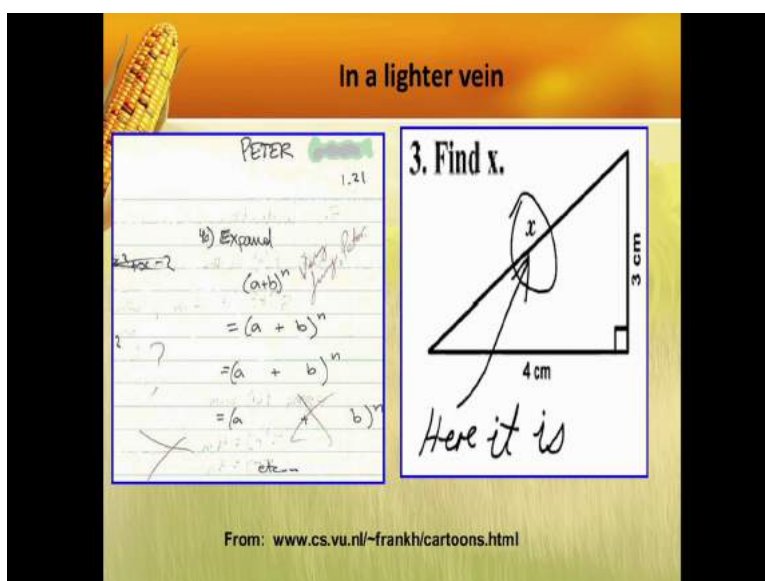
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In a lighter vein okay, somebody has taken lot of years, his Ph.D. is not over after 7 or 8 years; he is growing a beard okay. He is going to an astrologer and palmist; he is saying to please look at my hand, please tell me, when I will get my Ph.D.? That fellow says there is no point in seeing your hand; please bring your guide; I want to see his hand. So, these are all in lighter vein.

Now, this is also, this is also super creativity. Teacher asks Peter to expand a plus b to the power of n; he keeps on expanding like this.

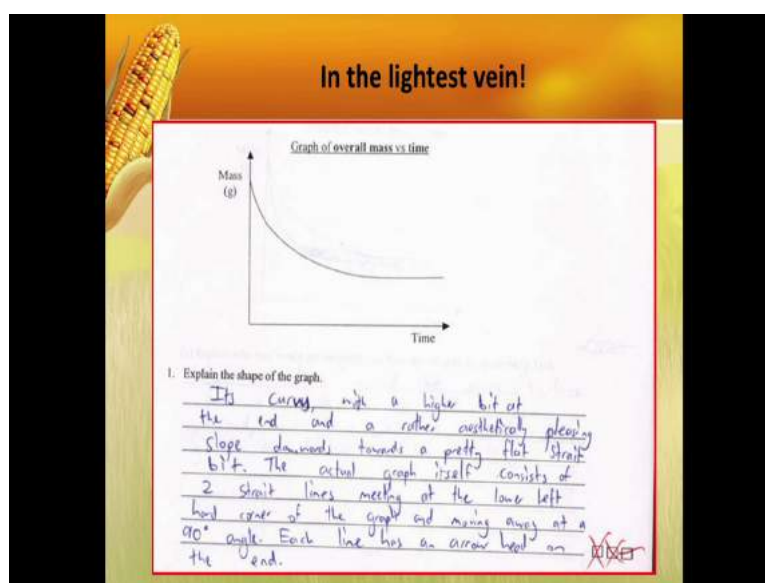
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Are you getting the point? a plus b, a plus b, he is expanding that okay. This is also an instance in a trigonometry problem, where this is 3 centimeters, this is 4 centimeters. Find x. x is root of 9 plus 16, 25; he says here it is. So, you would have wondered. So, you can very well imagine what grade Peter would have got in the exam; this is also an instant of creativity; these are also instances of creativity, but we are not looking at this; we are looking at serious creativity okay.

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So, this is familiar to you. So, this a variation of mass with time. It is exponentially decaying okay. This boy does not know anything about it okay. Well many times people do that know; then he starts explaining – it curves with a higher bit at the end and a rather aesthetically pleasing slope towards a pretty flat straight bit. The actual graph itself consists of two straight lines meeting at the lower left and all that. That means he does not know anything about this curve. This is basically a, y equal to a into e to the power of minus b x kind of curve okay, alright. So, it's an exponentially decaying curve, which have cooling of a first order system okay.

If there is a hot object placed in cold surroundings, we will look at heat transfer. Initially, the temperature difference between the body and the surroundings will be very high; therefore, it will cool very fast; then, the driving temperature difference comes down; therefore, slowly the cooling will reduce; asymptotically it will reach the...; asymptotically it will reach the ambient temperature. Learning is also like that. Learning will follow an inverse this thing



okay. It will be opposite of this. Initially, there will be explosive learning which is applicable to Ph.D. also; initially, you learn a lot of things. After 6 years, 7 years, 8 years, the learning will become very flat. Then it is a time to go okay, because even if you spend many years it will get saturated okay. So, that will be the inverse; you can very well imagine how that is; that is also e to the power of something curve okay.

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Now, we are not talking about those instances of creativity where somebody writes something here, somebody for fun does something, but we are looking at some serious creativity.

How do we become a fully professional researcher? See, if you are saying that... if you are saying that you are professional researcher - what is expected of you? We have something to say that our peers want to listen to; that is very, very important. We cannot keep on claiming that I am a great researcher; I am a great researcher. In an international conference, in an international meeting, that is ASME or IEEE, are you invited to give a key note lecture? Are hundred people ready to sit down and listen to you; will they give that half an hour or one hour to you? We should come to that level. There is no point in saying that I am great; I am great; it will not work. And that peer group must be truly international. It should not be peer group only in Chennai 36 or Chennai 20 or in Coimbatore or Nagapattinam or Visakhapatnam or whatever; that peer group must be truly international.

If there are 10 big experts, if there are 10 big experts in robotics or artificial intelligence, are you one of the 10, are you one of the 20 - that is what we want to aspire; that is what we want



to get there. So, to do this, command of our subject is required for this. So, Rome was not built in a day; you cannot do, you cannot achieve this over night. So, soaking is required; it takes many years to get to this position okay.

For example, Nobel price is not given to upstarts. After you finish your Ph.D., you found something very interesting. After 5 years, they don't give Nobel prize. The first thing is you have to have a long life to win the Nobel price. Why? Whatever you figured out at the age of 25, 30 or 35, they will wait for 20 or 30 years. So that there is an opportunity for others to find out and prove or disprove whatever you said, whatever you said some 20-30 years back; therefore, whatever concept, whatever theory, whatever experimental results you have generated have they withstood the test of time okay, because time is a violent torrent okay. So, there are many things which get swept off by time. So, are your results having enough shelf life, that even after 20 years or 30 years, people are ready to accept it, and I mean, no other paper has come or no other paper has come which tries to say that whatever you have generated is erroneous, and they are getting results much different from what you got and so on. May be the accuracy may not be there, and after some years you may have better computers, you may have better experimental methodology, so that you can get some percentage improvement, but if somebody gets radically different results, and they contest your findings, then we will have to... then what happens is it opens up, that means the issue is not settled; when the issue is not settled yet, then it's still an open problem for further investigation okay. So, in order to do that you need to have a mastery of appropriate techniques in the field. You should have a mastery of all the techniques in your field.

For example, if it is thermal sciences, you should have a mastery of analytical techniques; you should have a mastery of numerical techniques; you should have a mastery of experimental techniques, because in your long life, you cannot stay as just an analytical person, because the problems which are available, which lend themselves to analytical investigation are slowly going down. In the fifties and sixties, you can do momentum integral method, this integral method, approximation perturbation method and all that, similarity transformation, η is equal to y by two root αt all that kind of things - the kind of problems which can be solved using that approach they have come down. Therefore, analytical methods have some place, but still you cannot, I mean, just as they say you should not put all your eggs in one basket, then if one egg is spoilt, all the eggs will get spoilt; similarly, you cannot invest your whole career on analytical method because the area -



analytical methods - may themselves vaporize or evaporate. So, your skill set must be in... your skill set must be all pervading, skill set must be diverse enough okay, so that the risk of obsolescence is not there okay.

So, if you proceed from analytical techniques, the next will be a..next will be numerical techniques. You should have mastery of numerical techniques, because we don't have the luxury, the time and the money, to have a research career, where all our investigation will be based only on experiments okay. At the same time, you cannot assume that your life - whole life - you will be working on some software like Ansys Fluent. After 5 years, 10 years some other different software will come or those problems themselves will not be very important; flow over a flat plate, flow over a flat plate; nobody worries about flow over a flat plate now okay. Therefore, you should have mastery of experimental techniques also. So, if you need to have this mastery, if you need to have this mastery of all these techniques it takes time okay. So, you have to spend time.

Then, also, once you are a fully... after you are a Ph.D., in your initial years after your Ph.D. and all that, you should settle down to what is called a line of investigation. What is a line of investigation I want to pursue in the next 5 to 10 years? 30 years may be a difficult time, because it may be a difficult proposition, because lot of things will change, but in the short term - short to medium term, the next 5 to 10 years - where is it that I am finding lot of gaps? Where is it I feel, there is that there is lot of excitement? With my skills in analytical, numerical, and experimental, my interest and my background, where is it I can make a contribution? This is a very important decision which you have to take alright.



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You should be able to communicate your ideas through papers and presentations very effectively; communication is very, very important. We should be able to give good poster presentations, when you are research scholar or a budding researcher or you should be able to give good oral presentation in conferences, and then, you should slowly develop the knack of writing very well, the art of communicating your results, your findings, results. It's not enough if you just got some great results, you should also be able to communicate okay. Same thing, you would have, you figure out that you have invented something which is highly creative; you may say it is creative creative, but other people have to accept it, isn't it? Other people have to accept; after all, we live in a society and so on.

The Apple iPhone became such a massive hit. Apple iPhone is an icon of creativity because many people accept it, whatever they said, because they did something, they did many things, which are unthinkable - like a phone without any keypad, and then there is no stylus, the finger will be used as a stylus. So, when the market, at the time, was flooded with phones, which had big big keyboards, and the screen was very small okay, and there was always a stylus at the back. So, this ability to communicate your ideas is very, very important. So, this, for this maybe you will have to do some reading; you will have to do a lot of reading which is outside your subject. You may do fiction, you may read fiction, you may read nonfiction, whatever; you have to do a lot of reading to find out how different people communicate their ideas in different fields - how a newspaper correspondent communicates his ideas; how a science writer communicates his ideas; how a person like Chetan Bhagat communicates his



ideas. So, you should get an idea, and after reading all this, you will get an idea how different people communicate in different ways, and pick and choose the best among all these, and then put your own spin, and come out with your own formula okay for writing and presenting.

And if you are in academic setting, you should be able to guide Ph.D. students sooner than later okay. So, your life doesn't stop with your Ph.D. If you are working in any... now even in CSIR labs and all that, you will have to guide students. So, you should be a fountain head of ideas; that means, you should be able to generate ideas throughout. So, the successful researcher is not somebody who is just very brilliant in Maths or very brilliant in Physics or whatever; the successful researcher is one who has the ability to keep on generating new ideas throughout his life time okay. So, it should come, for that for that soaking is required. And then, you should know what is going on in other fields, and then, you should have passion, you should have hard work, all these things will follow now okay. We look at all these in a little while.

If you are in the industry, you must be able to write and defend your proposals okay. You should be able to defend your proposals to your boss. So, you are a head of a R&D division, you have your finance and other marketing - all these people you will have to convince that please make this, I want to make a new engine. Then, they say, I can get the engine from some other company; why should I make a new engine? Then, you say that my USP is this; you say that I want a light weight, this thing, whatever be your USP. So, you should be able to defend your ideas.

Then, finally, the peer group must be truly international. Even in the case of industry he is the top, he is one of the top ten guys in emission control, he is one of the top ten guys in steering, he is one of the top ten guys in ABS, he is one of the top ten guys in networks; I mean some... So, we should try to get there. Only then, you can say that you are a fully professional researcher.



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What are the attributes of a good researcher? First you should have sound grasp of fundamentals, very, very important. You should have sound grasp of fundamentals. That is why you cannot do research immediately after clearing your plus two; you have to do your B. Tech; you have to do your M. Tech; then you do some courses. And so, you should have a lot of grounding.

Then you should have mastery of techniques in the field. For example, all of you are doing heat transfer; there are certain techniques which are in heat transfer - the integral method, similarity transformation method okay, how to solve partial differential equations, finite difference method okay, measurement of temperature, measurement of pressure, measurement of flowrate - also, these are all, I mean, these are all specific to heat transfer. Like that, if you are doing in electrical networks or communication engineering, there will be a set of things which you will have to master. So, you have to list out what are the things which are to be mastered, so that you can become a fully professional researcher in your field.

Then tenacity and perseverance. The experiment will not work. The program will not converge. The paper may get rejected. It may take a long time; the paper may be sitting with your guide for a long time. There may lots of frustrations, but in all these times, all these times you should show your resilience and see how optimally you will make use of your time; stay positive, keep learning, keep improving.



Then scholarship. You should you should have scholarship; that means, you should have a deep knowledge; you should have a deep knowledge about your subject; you should also have a good knowledge of what are the problems other people are working in; what are the kinds of problem in this field, where other researchers are working; you should know what are the problems your friends are working on - that is the scholarship. I know everything only about my problem; I don't care, I don't have interest in other things; this is not a good attitude. So, you should know, for example, the top ten, the top fifteen, the top ten or top twenty researchers in heat transfer or thermal sciences; what are the kinds of problem they working on now. What are the kinds of problem which are going on in your laboratory? Not in your group; your group, any way, you will know; you will go for coffee with them and you will know. What are the kinds of problem being investigated by other groups okay, other groups in other research institutes in our country, in our continent, in the world okay.

What are the kinds of papers which have been published in the latest journals – in IEEE or ASME or in Elsevier. What are the kinds of work? So, what are the kinds of work? What are the South Americans working on? What are the Americans working on? What are the North Americans working on? What are the Japanese working on? What are the Chinese working on? So, for that, you have to get scholarship; you have to read. Just I will work only on my problem; it will give you... I mean, it is like frog in a well okay; you should have a world view of the kind of... because lifelong you cannot be working in this problem. 3 to 5 years down the line, your problem will be well settled problem. Is it, okay, you have done enough, you got with your guide, you got the... you should not keep... you should not think that with that experimental setup, with that idea I will proceed for 10 or 20 years, then slowly you will get more and more rejection; then, there will not not be much fun and excitement in that. So, sooner than later you have to come out of your comfort zone and leave your Ph.D. problem and do something, but whatever techniques you have learned will stand you in good state in taking you forward.

Communication skills I have already told you okay; you can also look at YouTube lectures; you can look at NPTEL; you can look at open course, which are on MIT; you can look at other YouTube lectures. How different people communicate, present their idea; not necessarily in engineering; how somebody present his ideas in law, justice, psychology, physiology, whatever. So, you get a perspective okay.



NPTEL (Refer Slide Time: 21:08)



Creativity is often neglected. So, now, you are going to talk about creativity. What is creativity? A creative person does things that have never been done before okay. The first thing is it should not have been done before. There should not be a... there should not have been an iPhone before that Apple iPhone, then only it is highly creative. Important instances of creativity; discoveries of new knowledge in science and medicine - discovery of penicillin okay or acceleration due to gravity, universal law of gravitation okay, f equal to $\frac{G m_1 m_2}{r^2}$ okay, it is directly proportional to the mass of the two objects, it is inversely proportional to the square of the distance between them and the constant is G okay.

So, invention of new technology - the fluidized bed combustion, the fluidized bed boiler - where you can burn anything; if you have a bed which has got a high specific heat made of sand, then you put whatever you want, you put biogas, this thing and all that, it will burn; this is a new technology. Composing beautiful music - whether it is by Mozart or A. R. Rehman, that is also creativity. Or take Indian penal code, take a particular section, some supreme court lawyer argues it in a way which nobody else has argued; that is also creativity. So, creativity can be in law, philosophy, history or interpreting history in a different way. All these are instances of creativity. We should not think that creativity means new mobile phone, new computer okay, internet and all these, they are all pervading and many instances of creativity which are constantly going on okay.



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Sternberg's theory of creativity

- Numerous hypothesis and theories of creativity are available
- We will see the one by Prof. Sternberg, Yale University
- He gives a list of "must have" for being creative

This Prof. Sternberg of Yale University, numerous hypothesis and theories of creativity are available. If you put creativity, you go to Google, it will give, it will return you millions of sites, articles, and all that. So, I picked up this because I find it very it gels with the theme of this lecture. So, this is Prof. Sternberg. He gives the list of what all qualities one should have or must have qualities in order that somebody is creative.

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Intelligence

- **Analytic intelligence:** Problem solving. Getting the correct solution, when we know that there is only one solution (academic)
- **Synthetic intelligence:** Ability to combine existing information in a new way. Giving off-beat solutions
- **Practical intelligence:**
 - (a) Ability to adapt to everyday needs with existing knowledge and skills
 - (b) Ability to sell one's ideas to the funding agencies, managers, editors, and reviewers.

Sternberg, R. J. (1985): *Beyond IQ: A triarchic theory of human intelligence*, Cambridge



He classifies intelligence into three types, analytical intelligence, synthetic intelligence, and practical intelligence. This is from his book, 1985, *Beyond IQ: A triarchic theory of human intelligence*, published by Cambridge University Press; that is indicated at the foot note.

Analytical intelligence is what you learn in college; Mechanical Engineering divided into 4 years. Each year divided into 2 semesters; every semester you have 6 theory courses and 2 labs. Science of material, dynamics of machinery, heat transfer, fluid mechanics, turbo machinery - if you learn all this, if you get some... if you pass in all these courses, then you have learnt mechanical engineering; that is the assumption okay. And then how, how do you test it? There is a question paper; everybody is given the same question paper; everybody is given some time; nobody should look at each other's this thing; you should not look at books and this thing; and then you solve some problems. And the teacher knows the has the answer to the problem, and there is only one correct answer to that. So, it's a highly simulated environment. In this thing, you then put first class, distinction, some gold medal, silver medal, whatever. And then, you say, that he is a mechanical engineer.

So, this analytical intelligence is, basically, is we test your concepts, your fundamentals on problems for which solutions are already available; you are testing whether you can get the same solution. We now rarely we give problems in which... rarely we give problems which are open ended, where there can be more than one solution; then it becomes difficult, then somebody will say he got more marks, I got less, there are lots of problems okay; therefore, often this analytical intelligence, analytic intelligence, is theory class, lab class, exam, grades, and degree certificate okay.

Synthetic intelligence is the ability to combine whatever you have learned in a new way. The only place where you get, where you have a idea of the synthetic intelligence in your course, is basically in your B. Tech and M. Tech is your project work, where you try to apply what all you learnt, to solve something. Research is all about synthetic intelligence right; whatever you have learnt, you are making use. In a course-based program like B. Tech or M. Tech, so the only way where you learn is doing project or in a design project and so on. And also, in a synthetic intelligence, you can give off beat solutions and different people can get different solutions; that is often the case in life. In life, generally different people will take... Now, after this lecture is over, from 11:50 to let's say 12 'o' clock to 1 'o' clock, you can... somebody can go straight for lunch, somebody can go back to lab, somebody can do... life is



a set of possibilities; everybody can choose his own okay; then what the cumulative effect of all these choices is what you become.

So, the synthetic intelligence, let's say, gives you the freedom that the solution finally is not fixed. There are different solutions; that's what normally happens in design. For example, in heat exchanger we always talk about two kinds of problems. There is a heat exchanger which is already available, you have what is called the analysis or rating problem, where you want to find out, you want analyze and find out how much of heat can be transferred with this heat exchanger okay; then everybody will get the same answer. But now I have 100 kilowatts of heat to be transferred, I decide the type of heat exchanger, but how many numbers of tubes, what is the length of the tubes, what is the diameter of the tube - that is a design problem, different people can get different solutions, is it okay.

Practical intelligence is the most difficult. Ability to adapt to everyday needs with existing knowledge and skills, and ability to sell your ideas to funding agencies, managers, editors, and reviewers, practical intelligence is never taught, it's also extremely difficult to teach; you have to acquire it over the years. So, practical intelligence is as important as analytical and synthetic intelligence. But if you see the triarchic theory of intelligence, out of the three, only one is excessively focused in all your education. So, the other two, you will have to pick it as you proceed. So, your success, eventually, will depend a lot on the other two also okay, and these are to be self learned.

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Knowledge. What about knowledge? Knowledge gives us the ability to recognize what is genuinely new. Why cannot you start research immediately after plus two? I have lot of brilliant ideas. Let me start research. The basic problem with that approach is, first, we should know what all have already been done in a particular field; otherwise, you will say the best way to transport is to have a circularly shaped object, with hub in between, and spokes, and then, you will come out with the design of a wheel which is of 500 or 500 years old or something; you should not reinvent the wheel okay.

So, only if you go to journals, only if you go to reference books, only if you go to text books, for example, in heat transfer you will know what has been done in the last 100 years; only after you know what has been done, there is possibility for you to do something new. How will you know, how will you get to know what has already been done? For this, knowledge is required. You have to go through the basic courses. Then, you have go through the advanced courses. Then, you will do the literature survey. Then you will get to know what is already done. So, some humility is required. You may have brilliant ideas; that brilliant ideas hundred people would have got, and then twenty people would have tried, and they might have even published, and it would have also been a failure. So, knowledge is very important because it gives you the ability to recognize what is new, you should know what is new okay; it also gives you the skills to design experiments, design new products okay.

How do you design your experiment? How do you measure a temperature of 500 degrees centigrade? You may have a lot of ideas, you remember how to measure temperature, when i went to the clinic, doctor put a mercury in glass thermometer in my mouth. So, you are brilliant; you got some brilliant ideas. Have you used this mercury in glass thermometer? I will use this mercury in glass thermometer to measure the temperature of liquid which is 500 degrees. Your brilliance will not let you know how to measure temperature. You should have knowledge; you should have platinum resistance thermometer or if it is a surface you will have infrared, therefore, you must know what other people have done. You should have knowledge; creativity is one, but knowledge is very important okay.

How do you design the experiment? How do you measure a flowrate? How do you measure pressure drop? If you will use special draft, you cannot use U tube nanometer. H rho g. What's the maximum h rho g? How much or you will climb a ladder to find out what is that h? Are you getting the point? So, knowledge is very important.



Then analyze experimental results. You are getting huge amount of data. Data logger gives you so much of data, how do you analyze these results? You should know there is something called heat transfer coefficient, there is something called nusult number, there is something called coefficient of performance, there is something called efficiency; so, that comes from knowledge. And then, you require knowledge to do scientific computing. I want to solve the equation $d^2 y/dx^2 + 5 dy/dx + 5 y = 0$. First, you should know that it is a partial differential equation, it is linear, it is elliptic, it reverse boundary conditions on four sides. Three kinds of boundary conditions are available. Then, there are various ways of solving: you can use a Fourier series or you can use finite difference or you can use finite volume or you can use finite element. You should know the formulation. All these dosen't come from your brilliance; with brilliance, only to certain extent you can go; it has to be supplemented by all this okay.



Introduction to Research
Dr. C. Balaji
Department of Mechanical Engineering
Indian Institute of Technology, Madras

Lecture - 22
Creativity in Research - Part 2

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Thinking styles - creative people often question conventional wisdom, assumptions, and rules okay.



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Because of this, creative people get into conflicts with the society around them. I am not suggesting it you that you should start fighting with your friends and all that, but generally these are the characteristics of creative people.

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So, to be creative one needs to be persistent; you should be at it; you should not give up okay; that is why you have to be tenacious. You have to be uncompromising. You have to be stubborn. You have to believe that it will work and keep on working at it. Arrogant - I put question mark. There is a thin dividing line between arrogance and self-confidence. So, you



can be arrogant in homeopathic dose, but not in allopathic dose okay.

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Hard work okay. Hard work is very, very important. There's something called the 10,000 hour rule, which was first proposed by Herbert Simon, who is a Nobel Laureate, who says, who said that it takes 10,000 hours of extensive training to excel in anything. Bharatanatyam, cricket, tennis whatever, you have to put in 10,000 hours of quality time before you can make a mark okay. So, the reassuring thing in this is, contrary to popular perception it is not innate talent or genius that alone matters. If you are having above average intelligence, then if you put in this 10,000 hours anybody can become an expert. All these great people you are thinking right - Mozart, A.R. Rahman, Viswanathan Anand, Bill Gates - all these people put in 10,000 hours in their respective fields before they became famous. Rahman started learning key board at the age of 3 right. So, it actually debunks this so-called Genius Theory. Lot of it is only perspiration; just hard work okay. So, the most reassuring, I come again is, hard work alone matters, which means any one can do it. Basically, do you have the tenacity? Do you have the tenacity? Okay are you ready to run this marathon, steadily without giving up okay? If you put in these 10,000 hours, you can be a master in any field okay.



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The 10000 hour rule

- What exactly is this rule?
- It takes approximately 10,000 hours of deliberate practice/study to master a skill/area of research

Hours of work/practice per day	Period required, years
3	10
8 (full time employment)	5
10-12	3-4

Ph.D.

What exactly is this rule okay? It's so simple. It takes approximately 10,000 hours of deliberate practice/study to master a skill or area of research okay. So, if you practice tennis for three hours a day, it will take about ten years for you to become a champion in school or district or whatever it may be. The same is the case with karate, Bharatanatyam, whatever, okay violin or any instrument or whatever.

Full time employment. We spend 8 hours, so that it takes about approximately 5 years. That's why they say to become Associate Professor you have to Assistant Professor for 5 years. 5 years is also the time people get their Master's degree - M.Sc., MA. So, in 5 years you would have put in some 10,000 hours in that field. Now, many of you research students, put this 10,000-hour rule. If you spend 10 to 12 hours a day, 6 days a week, leave the Sunday for your personal things you want to do - sport or you want to watch movie whatever or you want to just sleep off in the afternoon - if you spend about 60 hours a week, it takes 3 to 4 hours, 3 to 4 years before we get that vision - that dharshan - and you say, that yes, I am completely confident about what I am doing. I need my Ph.D. now okay. And the guide also says, ok, you are ready, now please go out and conquer the world okay. You just get past your guide in that field and then you talk more confidently. So, I say Ph.D. is also about this angle. Initially, you will say good morning Sir, then till compri it is like this; then, slowly it will become like this; then, first paper it is like this; when you are finishing, it is like this okay. So, that angle, the theta, Ph.D. is all about the theta okay.



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The 10000 hour rule

- Man behind this: Prof. Anders Ericsson, Florida State Univ.
Chase and Ericsson, 1982, Psychology of Learning and Motivation (vol.16), Academic Press, New York
- Popularized by Malcolm Gladwell, Outliers, 2008, Allen Lane

Mozart
Beatles
Bill Gates
All Chess GMs

Okay Professor Anders Ericsson of Florida State University has also done lot of work on this, subsequent to this Hebert Simon's proposal. There's a good chapter on Psychology of Learning and Motivation in Academic Press Volume Sixteen; you can take a look at this. And a very popular book on this is by Malcolm Gladwell; his book, the title of the book is Outliers; he published it in the year 2008, and he looks, and he looks at the success of Mozart, Beatles, Bill Gates, all Chess Grand Masters; he has conclusively established that all these people have spent 10,000 hours before they became famous.

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Hard Work is the Key

- Science progresses through hard work, failures and frustrations!
- The human element of these "travails"- Not seen in publications!
- Society attaches way too much importance to success.
- Its importance to hard work is conditional upon some "final" success!
- **This is sometimes the tragedy!!**



So, therefore, hard work is the key. First you have to acquire knowledge then hard work is the key okay.

Actually, if you look at the progress of science, science has progressed through hard work, failures, and frustrations. First time when we are writing a computer program, if it works, I can give it in writing that you will get absurd results. First, first the program, you will not be able to compile; then, you will be able to compile, you will be able to run; when you are running, it gives garbage; then one stage will come, I have done everything, this stupid fellow he is not giving the correct answer; that peak of frustration, at that time you just take a break, and visit a temple or just take three four days break do something else; then, when you are taking bath or you may be reading a newspaper or you are watching a movie, ah! in line 542 I didn't put the square root instead of the... you will realize, you will come back, and then, it will start working okay. So nobody gets it right out.

If you look at a guide, some particular X professor, 450 papers, what a great person he is. All people who have achieved this greatness also, those people also have rejections. You have to talk to them, but the thing is even after this rejection and all that, they are not giving up; you should continue. See when you write, I have said this in my Joy of Research, Joy of Research book, when you write ten papers, one or two papers will get rejected, because it is statistical. What is the probability of acceptance? For example, I am a reviewer; if I get to review ten papers, I reject 2 or 3 statistically. So, my probability of rejection is 0.3 okay. Some other process may be 0.4. So if your paper is reviewed by three professors, find out the joint probability of acceptance. For the first reviewer, for him to accept it is 0.7 and, second reviewer is 0.7, say third is 0.7. So, your joint probability is 0.7 into 0.7 into 0.7 which is a low quantity. And then, when we write more and more papers, from our group some paper will get rejected; you should not worry. So long as it is original, you have done the experiment, you have got correct results, then the contribution is something very subjective; different people will... it is just that the set of two or three people viewed your paper differently. So, you try in some other vehicle.

So, if you are confident that it is original, it is not copied, there is no plagiarism, you have put in your hard work, then finding the right vehicle for your this thing is so easy now, because you have got so many journals okay. So just like a mother giving birth to a child, the most important thing is you have done your part and that baby has come out - the paper has come out - then we will find out something, we will find out some mechanism. So, that joy, that



thrill is already there; you have already done it; you feel, yes, it's good; you feel, yes, it's good; my hard work everything is there; we will find, we will find a journal to accept it okay.

So, the human element of these frustrations, how many times the guide corrected, how long it has taken, how many test tubes are broken, how many this thing volt meters went off okay, how many times program cupped all these things are not seen, because there is no time for the reviewer or there is no time for reviewer or the editor or the reader to look at all this. Why? The society attaches too much importance only to success okay. So, if you want to know how science has progressed, you have to look at the biographies of scientists. Benjamin Franklin, you will get an idea of what... or Thomas Alva Edison, you have to read; then you will understand the trails, the tribulations, and then the failures, which they went through before they could turn into success.

Therefore we often correlate hard work with success. If it is not successful, there is no hard work, but this is a dangerous correlation. If somebody has not scored marks, we automatically conclude that he has not studied well; there may be many other reasons okay. So, this is sometimes the tragedy, but we have to live with this, because it is a part of us, we are part of this system, we have to live in the society okay.

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Let's now discuss about motivation. Motivation, sometimes, can be a big problem, particularly for research students, because we go through phases where sometimes the motivation is very high, sometimes the motivation is moderate, sometimes the motivation is



very low okay. Whenever the motivation is very low only during those periods you require something, so that it gets back to normalcy, and then, because you have to be motivated constantly, and you have to get... you have to be self-motivated, because motivation propels hard work okay. So, how to stay motivated is very, very important; motivation propels hard work. There are two kinds of motivation: first is extrinsic okay. So, you want a reward, you want money you want praise, you want promotion, you want prizes, fame - all these extrinsic motivators. If you get that then you will work more, you will work more. That is ok. These are all called – extrinsic; extrinsic motivators okay.

The most difficult, and the most, and the more important of the two is basically the intrinsic motivation, where you set your own goals, achieve your goals, and then don't stop there, and then don't stop there; set new goals, achieve; set new goals and achieve; and constantly, you are propelling forward; this is intrinsic motivation. In highly creative people the intrinsic motivation is very high; therefore, you constantly... you enjoy what you do; the sheer joy of solving an unknown riddle; when you are in ninth standard you want to solve a unknown riddle, there is no prize, there is no motivation, and you are struggling with that for two hours; you are struggling with it, it doesn't come; then, finally, when you get it; finally, when you get it okay, then you, say, put this is theta, that is theta by 2, $\sin \theta \cos \theta$ two $\sin \theta$, $\cos \theta \sec \theta$ divided by multiply by $\sin \theta$ divide by $\sin \theta$...; finally, L.H.S is equal to R.H.S proved okay; then you get the Eureka feeling okay; that Eureka feeling is not related to money, is not related to degree, it is not related to prize, that is your own feeling; at that time your mother might have kept Bournvita, all that you forgot, all that, even food was not important, sleep was not important. You, in fact, you were not conscious about your body also; you were just in, you were just freaking out, it is just your mind which was working at that time; that means, you are at the peak of your intellectual abilities okay. So, the challenge, the challenge and the skill got balanced, and a sense of timelessness set in your activity; you really don't.... the time become infinite, at that point in time you are not looking, you are not looking at even the watch okay.

So, if you are intrinsically motivated, in many of your activities, you can reach that peak levels of achievement. In very highly creative people b is not so important as a, but initially you will start with, initially you will start with extrinsic motivator. The same thing with prayer; when you go to a temple, what do you ask? Initially, I want this, I want that, I want that; whether God will give or not give is not our this thing okay; that is another point. If you



really believe so deeply, then at some stage you will understand what is it you will not give which I deserve if you don't ask, even if you don't ask, if it has to come to me it will come to me. So, you will... so, your own ideas about prayer may change right. So, I will just come, I want to see you. Oh! today is advance heat transfer; so many coconuts I will break. So, it is not an insurance policy, you know. Why he should solve your advance heat transfer? You study your conventional and conduction properly.

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Environment

- A **conductive environment** is a genuine plus
- May not be "absolutely" necessary.
- Srinivasa Ramanujam (Indian Mathematician and number theorist) carried outstanding work under adverse conditions, most of his life!
- **Sometimes, adversity brings out the best in a person!**

Srinivasa Ramanujam
(1887-1920)
FRS - 1918

Environment. A conducive environment is a genuine plus okay, but it is not absolutely necessary. Srinivasa Ramanujam FRS, you know, one of the India's great mathematicians. So, he was born in Kumbakonam and most of his life he went through lot of struggle. He did not have access to journals and this right, and then, he was not in a big city and all that, but still under this adverse conditions the best science came out of him, you know, the best mathematics came out of him okay. So sometimes adversity brings out the best in a person okay. You should not say I will put these kind of conditions; I want a beta flop machine; I want so many these things; I want such a big RAM; only with all these things I will work okay. Even if you don't have this, what are the other alternate ways, what are the other ways of seeking solutions to these problems? Then it will make you more and more creative okay.



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Role of stress in Research

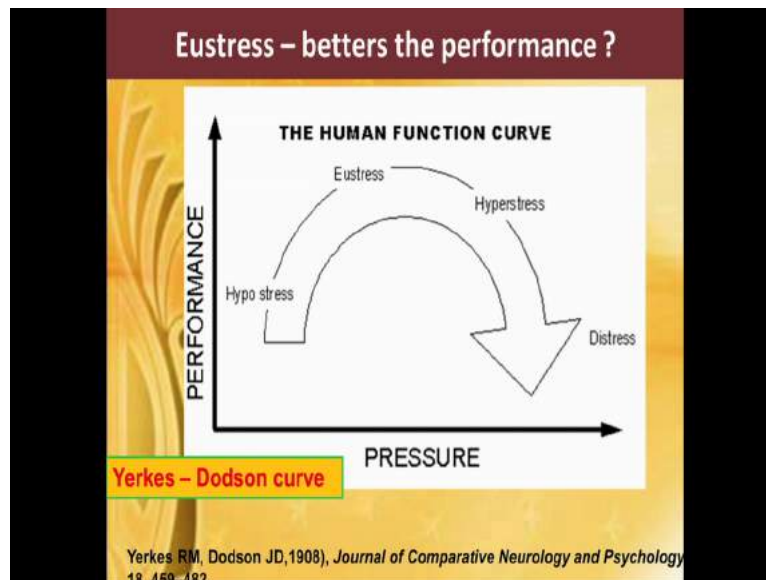
- Discontent with the present is the first step towards progress
- Hence, progress is all about “Optimal disenchantment” and channelizing this disenchantment
- Achievement is all about “Optimal nervousness”.
- “Eustress” is required for bringing out the best in us.
- Stress by itself is not stressful until it becomes distress



Role of stress in research. The first thing is you should be somewhat unhappy about your current state. If you are too unhappy, then you will sit in the hostel room and you will not come out, that is the extreme case; but you should have some this thing, oh! it is not good, it is not good okay; there should be some optimal unhappiness with your current state which will propel you to improve and move forward okay. So, hence, progress is all about optimal disenchantment okay, optimal unhappiness, and channelizing this disenchantment. So, achievement is all about, achievement is all about optimal nervousness; I mean you should be optimally nervous, you should be optimally stressed, and then, with that, this propels you to work harder okay. Eustress is required for bringing out the best in us; u plus stress is good. So, there is also something called good stress; for some of you this may be news; you also have, you have good stress and bad stress. So, stress by itself is not stressful until it becomes distress. I have confused you enough okay? Stress it is not stressful until the stress becomes a distress. So, there's some... So, a little amount of pressure, a little amount of stress will make us work better okay.



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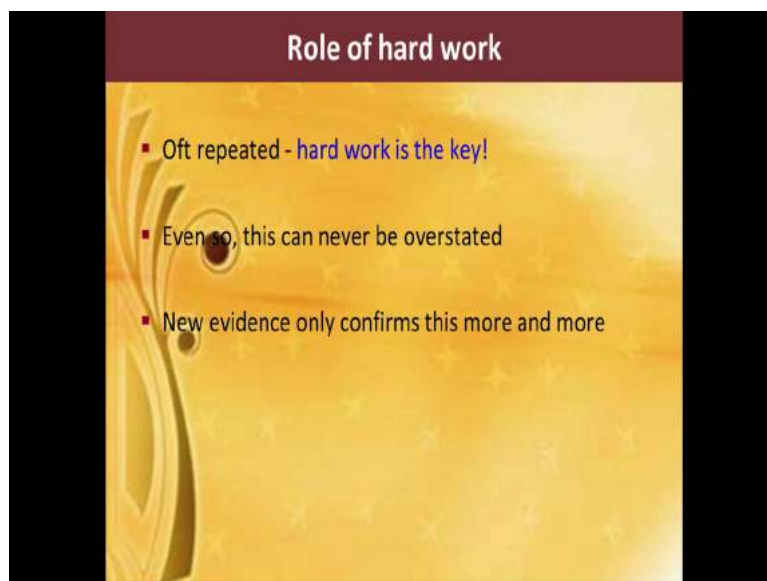


So, this is a human function curve. This is called Yerkes-Dodson curve. This is Journal of Comparative Neurology and Psychology, published in 1908, where they did studies on rats, and then they give electric shocks to rats, and ask the rats to do some activity. So, rats were asked to do some activity or exercise or whatever. So, that was the target. I mean, there is some sort of a activity and this was benchmark, and now they figured out that if they are not stressed at all, if they are in hypo stress, then the performance is very low; the y-axis is performance, the x-axis is pressure; when they are in eustress, when they are reasonably stressed, they did very well okay. And then, once they are over stressed - too much of electric shock - again the performance goes down; then finally, distress, and finally, it will even lead to death and so on.

So, this has been mapped on to a human performance curve and they say human performance curve also resembles this. So, if you are extremely stressed, at the same time you are not stressed - under both these cases the performance will be very less. You should be reasonably stressed; you should be in the eustress to bring out the best in you. What is the eustress for you is something which will you have to introspect and figure out. This depends on different people, how many activities you can take at the same time okay, how much time you want to sleep, how much time you want for leisure, how much time you will devote for research, how much time you will devote for family, whatever. So, you will have to figure out your eustress point okay.



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Role of hard work -we have already seen. Oft repeated hard work is the key. Even so, this can never be overstated. Even if you say hundred times, we cannot say that it is over stated, because new evidence only confirms this more and more okay.

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Now, how does creativity occur? The conception of a new idea often occurs in a... often occurs in intuitive flash of insights; suddenly, some insight comes that I have found the answer to this problem. Sometime it comes, it can come when you are not in the lab; it can come when you are not in the experimental setup; it can come in somewhere when you are



jogging or it can come when you are watching a movie or even in your bus or something, where more or less the complete idea is revealed to you, but it doesn't mean that mathematically you can prove it or when you do an experiment you can get this, but sometimes that intuition tells you where more or less the idea is completely... in which more or less the complete idea is revealed – it's not related - the complete idea is revealed okay.

Scriptures we called it as Dharshana or we call it as Dharshana. Dharshan which is, say, in a temple; dharshan means you have that vision; the vision of that heat transferring something fluid flow or something, you have a complete idea of what that. After that, you have to note down and jot down, and then, you will have to work hard to do this thing.

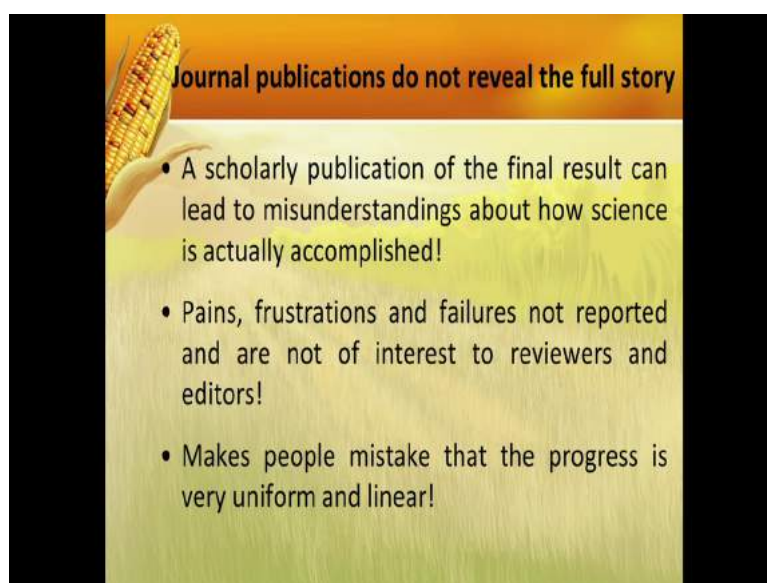
See, Max Planck got this in 1901 okay. b_{λ} , the black body distribution - the spectral black body radiation into the $c_1 \lambda^{-5}$ divided by $e^{-c_2/\lambda T}$. Some people already figured out $c_1 \lambda^{-5}$ by $e^{-c_2/\lambda T}$. Max Planck figured out that $e^{-c_2/\lambda T}$ if I put minus 1, then exactly that theory was matching with experiment. So, he got that dharshana - that is the correct black body distribution. He published that paper; it was accepted, but that bothered him - why that minus 1 is coming? From where that minus 1 came? Then, he figured out if he has to go back to minus, how to get that minus 1? He has find out that e is equal to $n h \nu$, e is equal to $h \nu$, e is equal to $n h \nu$ or s equal to $n h \nu$; energy transfer can take place in finite multiples of $h \nu$ only, it should be $n h \nu$ where n can be an integer. Therefore, h is a fundamental constant of nature, which cannot asymptotically approach zero; it is 6.627×10^{-34} joules second. So, this announced the birth of quantum mechanics, but first he got the idea - it was just a curve fitting idea; he just got this dharshana, oh! I put minus 1. So, that minus 1 got him the Nobel prize, but it took 17 years for him. 1901 was this thing, and then, he worked, and then he proved, and then 1918 Quantum Statistics was his Nobel prize okay. So, then Einstein, other people have worked together, and then, you know now, this can also be used for splitting the atom; e is equal to $m c^2$ and all that right.

So, when you are doing your research, when the dharshana comes, take a note book and write down this idea. Suddenly in your heat pipe, you get a brilliant, you will get a brain wave; then you write it down, then you have to logically follow it up it; it may cup also; doesn't matter okay. Then, what is the lesson you have to learn? If some brilliant idea is coming, and you work on it, and it doesn't work, then slowly what do you learn is, far many times brilliant



idea will come, but which brilliant, which of these brilliant ideas I have to pursue is something you will have to figure out. So, you say the next two three weeks I will work, if it is not coming out, I will discard. So, you will have a mechanism by which you will judge, this is the path I will take, I will spend one month; if it's not coming, I will go to some other path. You should have your own back up, multiple backups, second line of defense, third line of defense, you should internally evolve all this alright.

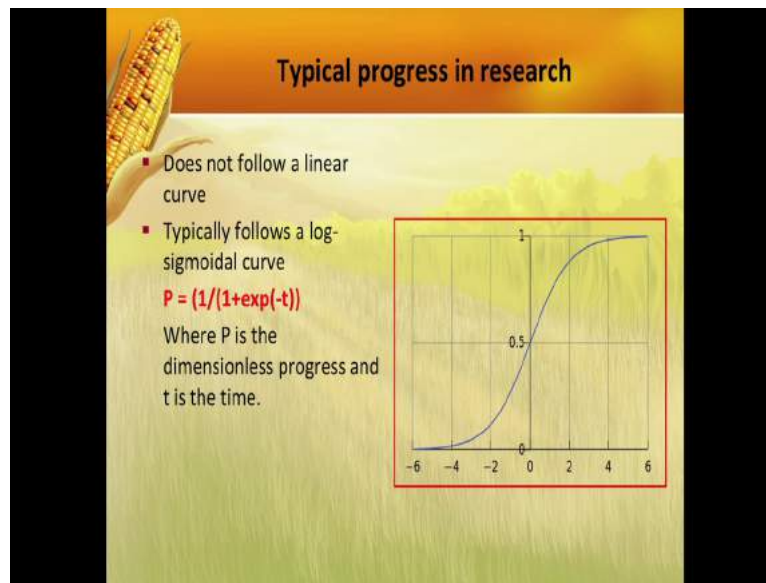
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Journal publications do not reveal the full story. A scholarly publication of the final result can lead to misunderstandings about how science is actually accomplished: abstract, introduction, literature survey, experimental methodology, results and discussion, conclusion, references, in between nomenclature; how nicely he got it, but how much suffering the authors went through to get that paper, that we don't know, because that is a way, because everywhere that is people don't have time, reviewers don't have time, people don't have... I mean it costs a lot of money to write your own story and all that right. The pain, frustrations, and failures, are not reported okay. So, it makes people feel that progress is very uniform and linear; it is not the case.



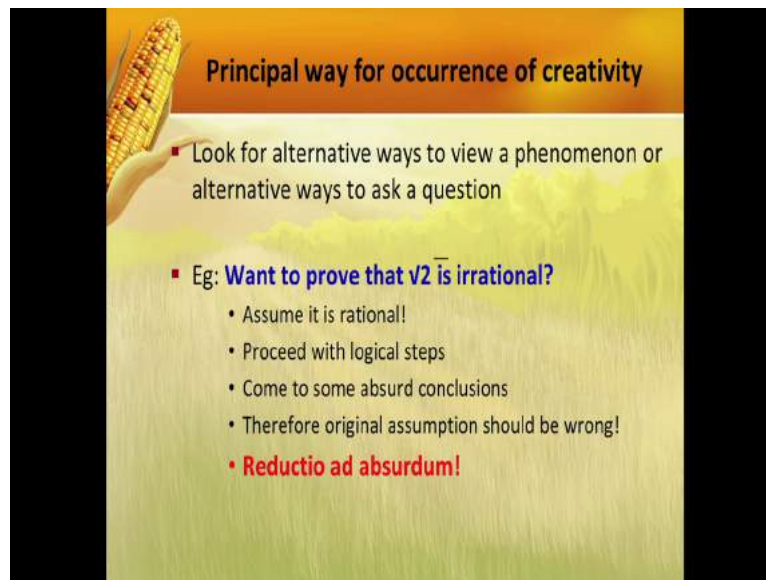
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Okay let us look at this. Typical progress in research follows what is called a log sigmoidal curve okay. In the initial years, your progress will be very less, you should have patience, then suddenly the graph takes a turn; isn't it? The graph takes a turn okay where you get a rapid progress, then afterwards if you keep on spending more time, again, the progress is very less; therefore, initially, one, two, one, two, three years whatever, you are learning the science, you are learning your field, you are going through your qualifiers and all that. And you are trying to write a program or you are trying to develop a setup, it is not working; some joy is not there or some workshops something is not there or something is not moving or somebody is not signing; I mean so, many things are, but after sometime suddenly the flood gates will get open; one day to your surprise, you will see the experiment is actually working, and it is giving good results. The program is actually working; then you should capitalize on that, and try to extract maximum, and after that, again milking it dry is not a good idea. I will write the fourth paper, fifth paper, sixth paper, then the reviewer will say I don't want any more paper on this. So, this is the right... So, initially if it's not working, don't worry, give it some time.



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What is the principle way for occurrence of creativity? I mean, we should also look for alternate ways, alternate ways in which you view a phenomenon okay or how do you rephrase a question okay. For example, you know that you want to prove root 2 is irrational. How did you do it in mathematics? If you want to prove that root 2 is irrational, we assume that root 2 is rational, that is root 2 can be represented as p by q okay, where p and q are integers; then you start working various operation; finally, it will lead, it will lead to some observed conclusions, which you believe is not correct; therefore, if this is not correct, you go back and find out where you make the mistake; you figure out all the steps are correct. So, what is wrong? To assume that root 2 is equal to p by q and is rational was wrong; therefore, root 2 is irrational. This is a way of proving; this called reductio ad absurdum, reducing to absurdity. So, used by ancient Greeks; this is an exercise in logic okay. So, in your this thing also, you have a multi objective optimization problem how are, what is the, what are the different ways in which you propose, in which you can impose the problem or you want you want a figure of metric for the performance of your device; it could be your solar collector, it could be a heat pipe or it could be a jet impinging jet whatever, can we think of some other this.

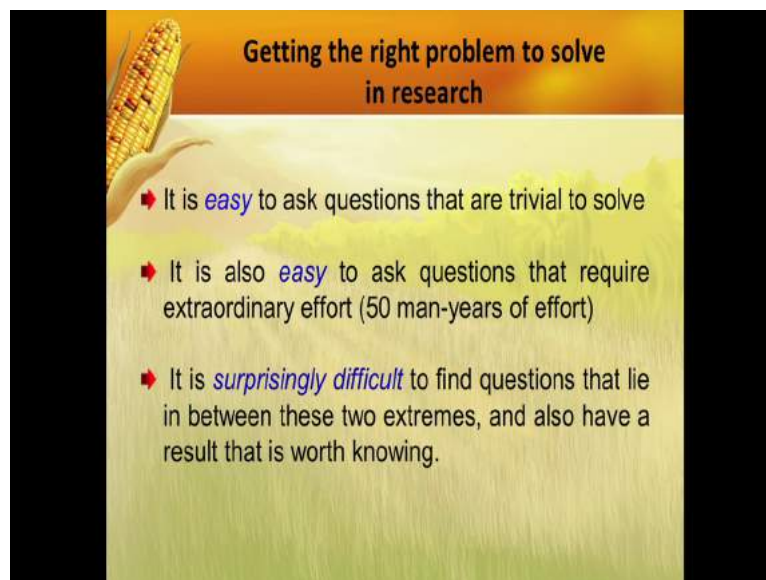
Traditionally, so many things are used. Can you deviate and come out with a new performance metric? Think about it okay. And then, literature, everybody knows Reynold's number, you keep plotting Reynold's number. As Reynold's number increases, Russelt number increases; what else it will do? What a brilliant conclusion. So, we should not restate



the obvious; it is already known; it is already known. Okay we should say something, so that's what we say know, dog bites man; what is news? Man bites dog; that is news; how your results and discussion and conclusion will be man bites dog.

Contrary to popular belief as I reduce a tube diameter, though the pressure drop increases, the heat transfer is increasing significantly. Therefore, heat transferred to pressure drop is increasing; therefore, it is more advantageous to use a smaller tube diameter, and yes, then people will say, what he is saying, what he is saying, what he is saying, surprise; unsettle all the people who read your work; do not allow people to read your things in peace; that should be your goal okay - *reductio ad absurdum* is clear right.

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Getting the right problem to solve in research. So, the happiness comes out of creativity, because it gives you a sense of achievement or accomplishment that you are able to do something that it is your contribution. You are contributing something or you are able to find out something which was not there; it comes out of that; you don't even want that; what is it? where do you want to get happiness from otherwise? You tell me.

Student: The joy is intrinsic, you are doing.

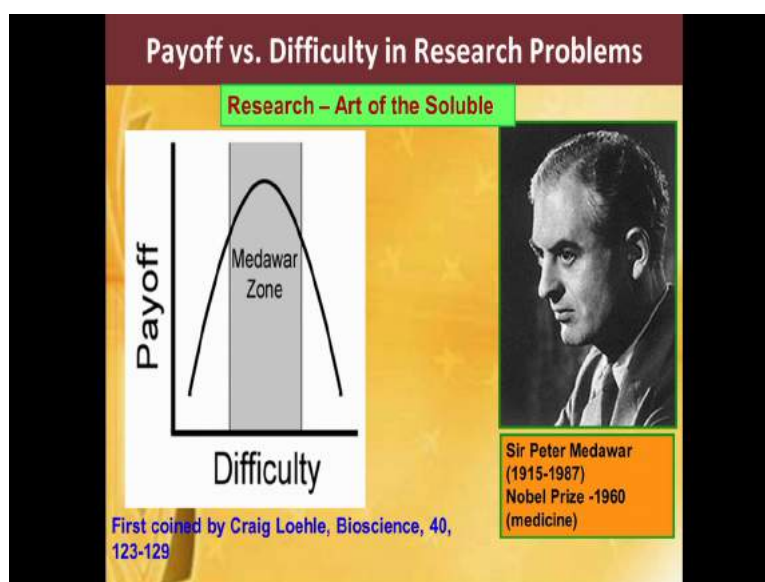
Yes, yes, the joy intrinsic means because you have improved, you have improved over what you are thinking you can achieve; you have improved compared to your present level. So far, you are not able to solve this unknown rider. So, far you are not able to... you are improving,



you are improving, you are improving; through this thing, other things may also improve; that we don't know now, but you get some sheer joy out of finding the unknown; darkness to light Asathoma sadgamaya , thamasoma jyothirgamaya, I mean it's light, from darkness it is light. When you are finding something new, it's like thousand watt bulb; isn't it?

Getting the right problem to solve in research okay. It is easy to ask questions that are trivial to solve. Find the inverse of a three by three matrix; everybody will solve; that is not a Ph.D. problem. It is also easy to ask questions that are extremely difficult to solve; Ph.D. problem using thermodynamic arguments, using thermodynamic arguments prove that God does not exist - can that be given as a Ph.D. problem in thermodynamics or radiative heat transfer? Or I will develop a new theory for the evolution; high risk problem okay. It's surprisingly difficult to find the questions that lie in between these two extremes. You ask the right question, you solve, you work on this question for three to five years, you get a good answer, you get a couple of publications, you get your Ph.D.; that is the right problem for research; are you getting the point? And it surprisingly... and if you are an academic, a full-time scientist or a professor later on, how to constantly generate these questions which are neither easy to solve nor difficult to solve, but are worth solving. So, it's a challenge okay.

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So, this Sir Peter Medawar; he got the Nobel prize in 1960. So, he talks about the return in investment on working on a research problem; that is payoff y-axis, and the x-axis, the difficulty level. If the difficulty level is very low, the return is very low; if the difficulty level



is very high, the return is also very low, because you may not be able to solve it, but if it has the right difficulty level, then you have the return is very high; that is indicated in the grey area that is called the Medawar zone. So, what is your Medawar zone in your field of research you will have to figure out.



Introduction to Research
Dr. C. Balaji
Department of Mechanical Engineering
Indian Institute of Technology, Madras, Chennai

Lecture - 23
Creativity in Research - Part 3

Look at this velcro.

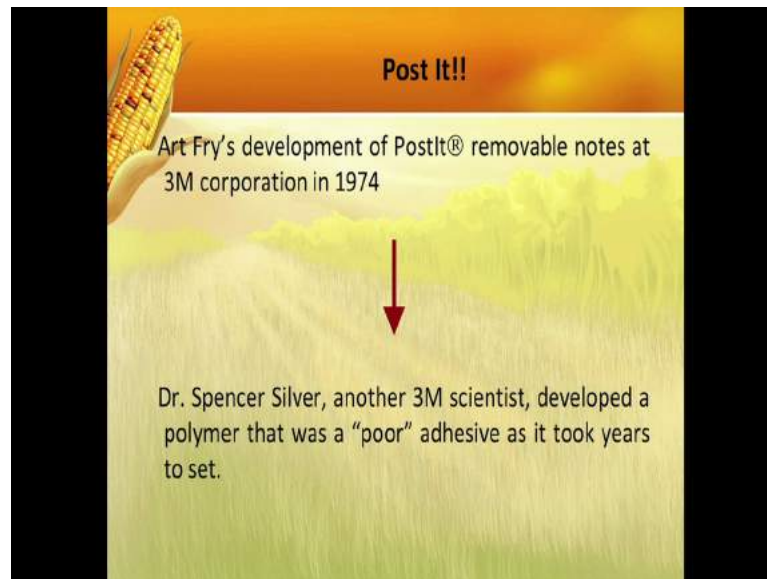
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All of you know Velcro®, which is used for BP meter, some times it is there in watches and all that. So, George de Mestral's observation of how cockleburs - a kind of plant - it got attached to his jeans pant; it was a constant nuisance. He was always trying to remove it. Then he said - can I invent a new product which can make use of this? And, can I convert the nuisance into a useful product? That's how the Velcro® came; this is a great... this is a great instance of creativity.



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Next, post it okay. You know the PostIt®, right? The bookmark; the removable bookmark which comes in various colors, and shapes, and sizes okay. So, Art Fry developed this PostIt® in 3M Corporation, in the US, in 1974. Doctor Spencer Silver - another 3M scientist - developed a polymer that was a poor adhesive; it is considered as failure because all polymers were considered to be like... - it was successful only with like Fevicol, they said two elephants will attach and that is the advertisement - but this was failure because it is not permanently attaching. Art Fry said, if something is there which attaches, but does not attach permanently, I can use it as a bookmark; I can put it on files, and I can write some notes - please clear this file and all; then, it can be removed whenever the job is over. It has become such a big, successful office product - PostIt®.

By ignoring conventional wisdom that any adhesive has to permanently set, Art Fry deviated from conventional wisdom and got a successful office product okay. This is also an instance of creativity.



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There are several other instances, for example, the Japanese bullet trains; the bullet trains travel at a very high velocities - 250 kilometers per hour and all that; whenever they are entering a city, whenever they are entering a station, that tunneling effect is there, they used to produce a lot of noise, and all the people close to station, they complained that the bullet trains are generating excessive noise, and in Japanese, every 10 minutes there is a bullet train okay, and then, JR - Japanese Railway - started working on this. They figured out how the kingfisher catches its prey okay, and then, without any noise, if it makes a noise, the fish will escape. So, it dives and this thing such that it makes very little noise. So, they figured out the aero dynamic body, the precision all, this; so, it is bio-inspired, bio-inspired engineering and the Japanese Railway figured out a way by which now the nose cone of the new generation Shinkansen or the bullet train, it is shaped like the beak of a kingfisher. Once they did it, the noise considerably reduced; no complaints from the residents. So, you can, so, you can draw inspiration from nature also alright okay.



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Flow and creativity in Research

- Flow – a concept in psychology
- First proposed by Mihály Csikszentmihályi, Professor of Psychology, University of Chicago
- Flow - fully focused motivation
- Single-minded focus - represents the ultimate in harnessing the emotions in the service of performing and learning.

Flow and creativity. Flow is a concept in psychology; this was first proposed by Professor Mihaly Csikszentmihalyi; Professor Mihaly Csikszentmihalyi - very difficult name to pronounce. He is a professor of Psychology, University of Chicago. So, it is like your flow is, basically, like the unknown rider, and you forgot your Bournvita; that is basically then you are in flow okay. So let's leave all this.

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Components of flow

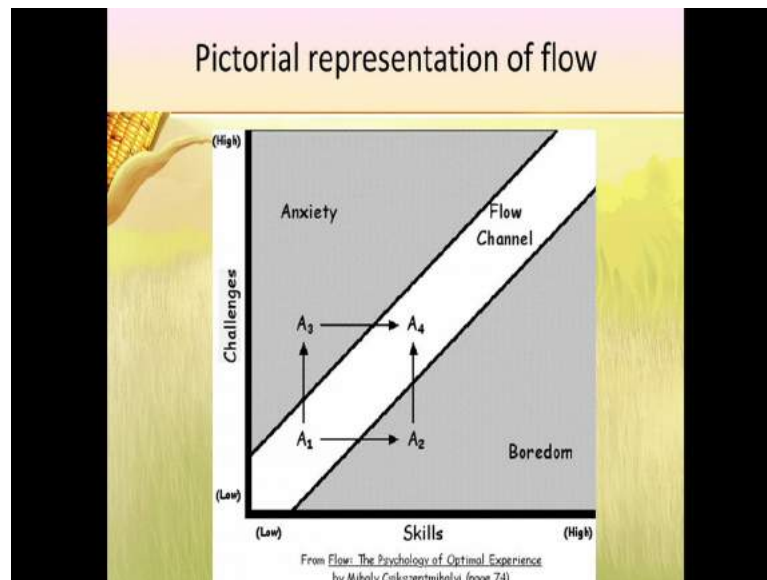


- Clear goals
- High degree of concentration on a limited field of attention
- A loss of the feeling of self-consciousness, lack of awareness of bodily needs - **Timelessness**
- Balance between ability level and challenge
- A sense of personal control over the situation or activity.
- Activity is **intrinsically rewarding** - no worries about prize or reward



So, let us leave all this.

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So, this is a flow curve. If you see, y-axis is challenge and x-axis is skill; y-axis is basically the difficulty level in the problem you are pursuing; pursuing x-axis is your skill. If the skill is very high, if the skill is very high and the challenge is very low, you will feel bored; that is in the grey color. If the skill is very low and your challenge is very high, if your challenge is very, very high, then it can happen to you, it can happen to you in your Ph.D. problem also, then you will be highly anxious, but if your skill and challenge are in balance, you are in that white path which is called the flow channel, and there is an exact matching of challenge and skill, the best thing you will come out, and you will enjoy what you are doing; this is the basically the Flow Theory okay.

Let us consider that a ten-year-old, a ten-year-old boy, wants to learn tennis. He is at the level A one. His skill is very low, and his challenge is very low, because first few days the coach will ask him to play against the wall. He will tell him what is the racket, what is a ball, what is the height of the net, what is the height of the net at which you play and all that, and then he will start returning this serve and all, that he will learn. Slowly, his skill will improve, and he will go from A one to A two. After two months also if the coach will say's that you will play against the wall, then his skill is very high, his



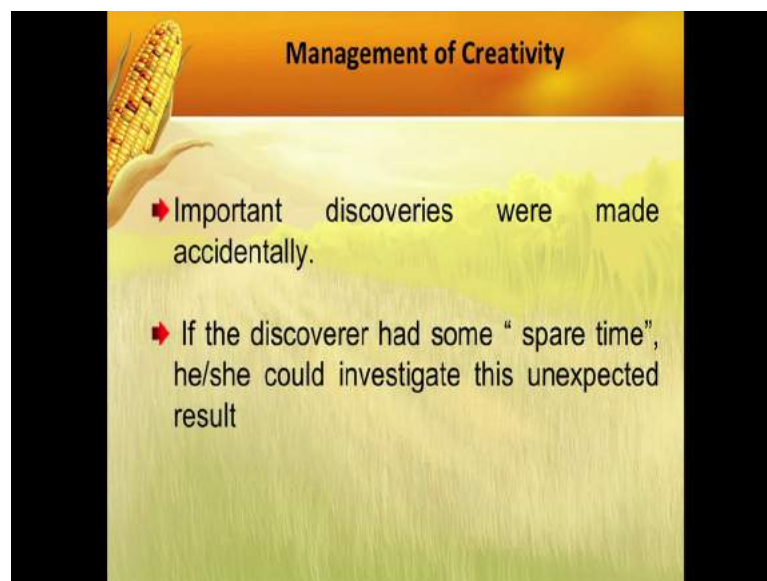
challenge is very low, he will get bored, and he may drop off from tennis. Other possibility is he will tell his coach, no, no, I want my challenge to increase, then he will ask him to play against the school champion then suddenly or he may ask him to play against another student, then suddenly his challenge is increased to A four, and he finds that there is the sheer unpredictability of another human being playing as against the wall, and the variety of new things which he is learning because he is playing against somebody else makes the whole process of playing tennis more exciting and more enjoyable. Then, at A four he is very happy, and then, you cannot stay happy at that A four for a long time, again he will learn, he will go to A five or A six. Then, he will get bored; he will start playing inter-school, intra-school, inter-district, national and this thing and go on.

Let's go to the other end. The first day itself the teacher asked him to play against the school champion. So, from A one suddenly it is increased to A three. A three, the challenge is very high, because the other fellow is smashing left and right, he doesn't even know, this fellow gets scared, he will tell – mummy, from tomorrow I am not going to tennis. Like, some coaches will just put the child into the swimming pool, into the water; that is a last day the child will go to swimming pool. Next day onwards... Sometimes, it happens, sometimes, you know, they just throw out the child into the swimming pool okay. So, the challenge is very high in A three. So, he may either drop from tennis or he will ask his coach - how that fellow is playing like that, and the coach will say, if you have to play like that fellow, you have to practice. So, you go from A three to A four. So, that route also A three to A four he will go. Then A three to A four

Here the funda is you cannot stay in A four for a long time; you will either learn or somebody will... suddenly your challenge will increase. In an office, the challenge will increase because your boss will give you something, which will make it very difficult for you to work or you keep on learning and it becomes something is boring for you. Only when, so, therefore, life itself will have situations where you are either at the border of the boredom or anxiety, in between boredom and anxiety there is a narrow window, you should navigate life in such a way that your challenge and skill are matched, so that you are happy. Only if you are in that, out of the grey region, you will be happy. Is it okay? This is a theory by Csikszentmihalyi okay.



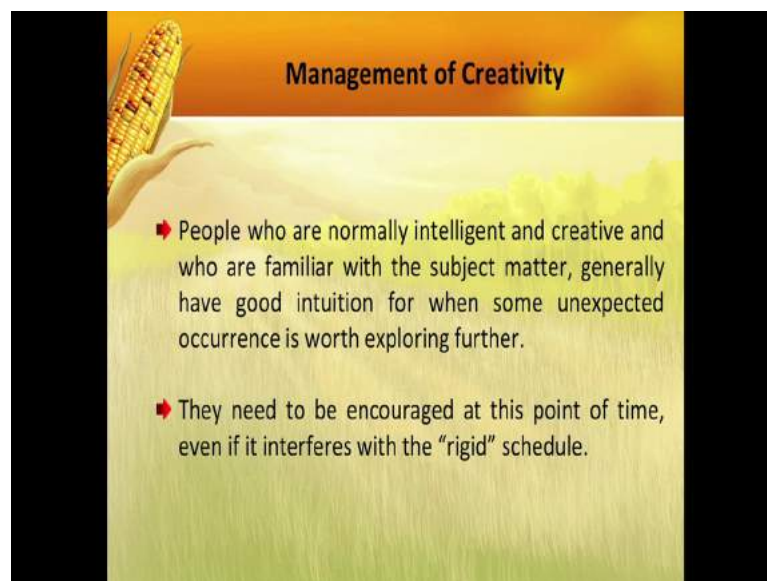
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Management of creativity. Suppose, you have too much creativity - how to manage? Now we have to answer a question, no? Okay so, important discoveries were made accidentally. If the discoverer was given more time, and then, he or she should, could investigate this unexpected result. Therefore, your boss or guide or whatever, if you are pursuing something, as advisers we should give okay, even though it may not be central to your work, we will say, ok, another one or two months we will go, it doesn't matter; you try this line; suddenly, if something exciting comes, then we can, we can pursue further in that line; is that ok? That's what I am trying to say.



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People who are normally intelligent, and creative, and who are familiar with subject matter, we have generally a good idea on when some idea is worth pursuing. You may get lot of brain waves, but if you are intelligent and smart, then you will know that, ok, some many ideas come, this idea is not very good. So, you will filter out yourself which one is worth pursuing and which one is not okay. And then, finally, you have come to conclusion that this is worth pursuing; then your boss or who ever it is, must encourage you and give you sometime to pursue this.



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A presentation slide with a yellow and orange background. On the left, there is a vertical illustration of a corn cob. The title 'Nobel prize winners and famous scientists' is at the top. Below it, there are three bullet points. In the bottom right corner, there is a small portrait of Albert Einstein.

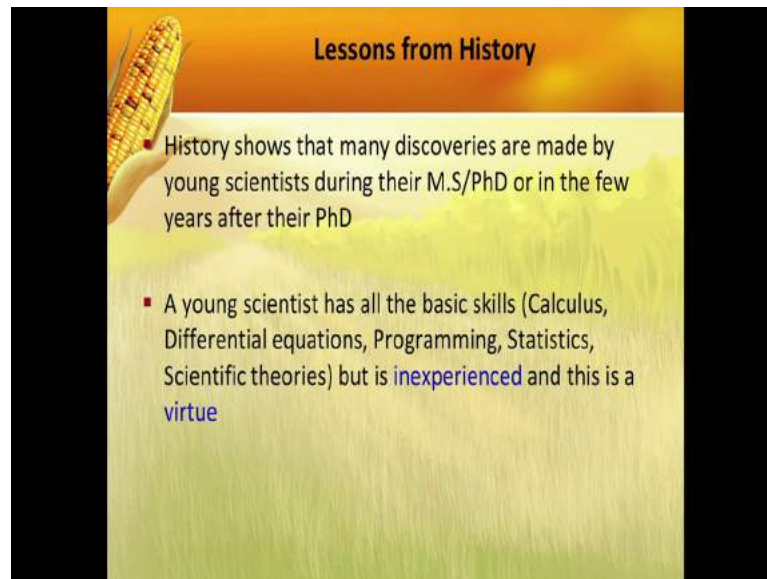
Nobel prize winners and famous scientists

- **Competent Scientists:** Those who were in the right place at the right time.
- (b) **True genius:** Ability to repeatedly develop significant innovative ideas (e.g. Einstein)
- How do we then encourage discoveries to occur more frequently?

Let us look at the lives of Nobel prize winners and famous scientists. Competent scientists. There are two types of scientists: competent scientists - those who were in the right place at the right time, and true genius. I am not saying that those who were in the right place, in the right time, are not great people or whatever, they are also very competent, but a true genius is one who repeatedly generates lot of new ideas; Albert Einstein was one. If you look at these lives of these scientists, let us see, let us try to figure out, how can we encourage discoveries to occur more frequently.



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History shows that lot of discoveries are made by young scientists who during the MS or Ph.D. or in first the few years of the Ph.D. they found something remarkable okay. So, a young scientist has all the skill calculus, programming okay, measurement techniques, whatever, differential equations, statistics, and scientific theories and so on, and but he doesn't have a baggage - he doesn't have a reputation, that he is not scared of failure, but if a professor or somebody who has worked in a field, who is already established, he is worried - what will other people think when you write this thing. So, generally, you tend to get close, then you will say, no, no, I have seen, I have seen, I know from my experience this will not work, this will not work, this will not work, this will not work okay. So, being inexperienced, a young scientist is fearless; a young scientist is fearless, then we should encourage him or her at that point in time. When someone has worked, or lived in an environment for more than ten years he or she tends to be less curious because of familiarity.



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Lessons from History

- When someone has worked or lived in an environment for more than 10 years, he/she tends to be *less curious because of familiarity*
- *One solution:* Change fields once in ten years(?)

Slide 39/75

So, generally okay, so, one solution is change fields once in ten years. Many people have done that. So, I just talked about timelessness, and flow, and all that; therefore, I am putting this slide number also. So that one way of how to find out in a class that the people are in flow is - how many times students looked at the watch okay.

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Prof. S. Chandrasekhar (1910-1995)
Indian Astrophysicist

- **1929 – 1939:** Stellar structure, theory of white dwarfs
- **1939 -1943:** Stellar dynamics
- **1943 -1950:** Radiative transfer, quantum theory of the negative ion of hydrogen
- **1950-1961:** Hydrodynamic and hydromagnetic stability
- **1960-1970:** Equilibrium and stability of ellipsoidal figures, general relativity.
- **1971-1983:** Mathematical theory of black holes
- **1983-1995:** Theory of colliding gravitational waves

FRS – 1944
Nobel Prize- 1983



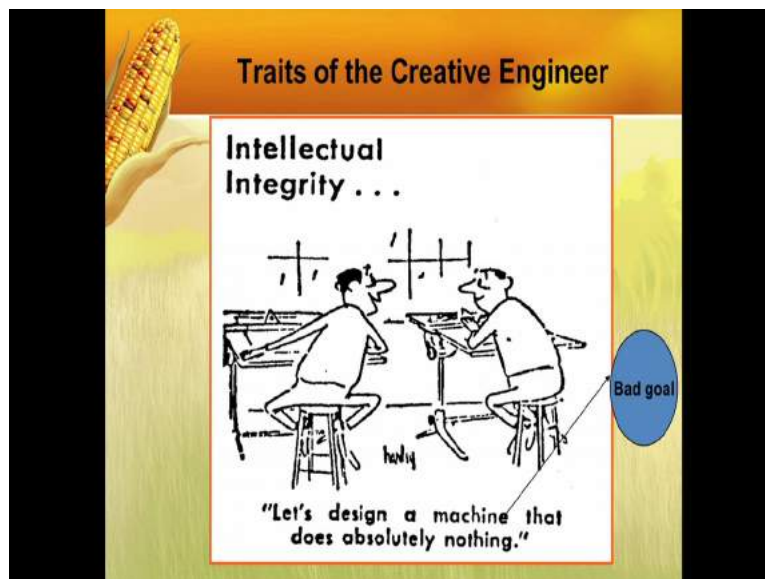
Professor Subramaniam Chandrasekhar, Nobel Laureate, 1983. He got the Nobel prize. We can see that every ten years he has changed the field; 1929 to 1939 - Theory of white dwarfs, theory structure of stars; 1939 to 1943 - Stellar dynamics; 1960 to 1970 - Stability of ellipsoidal figures; 1971 to 1983 - Mathematical theory of black holes; and 1983 to 1995 - Theory of colliding gravitational waves. Every ten years he has worked in a new field, wrote a book, conquered the field, and then, moved on okay. Don't get stuck to what you learned in Ph.D. throughout your life. Then, the contribution will become marginal then, you should improve okay.

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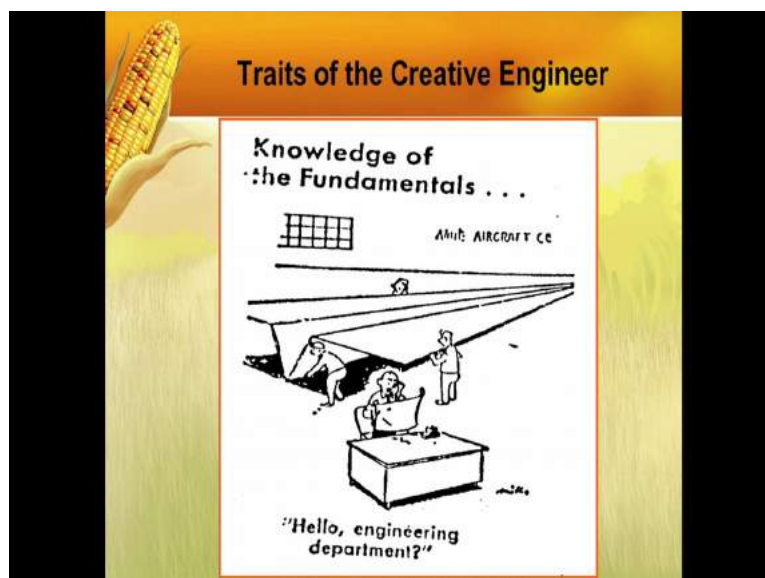
Why are designs, concepts, and all these things not creative? Why is not there much creativity? Now, that we have seen all the theories and all that – why? Now, let's try to the answer the questions - why are designs or concepts not creative? We resist new ideas; no, no, this will not work.

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And, sometimes, integrity is not there - this fellow says, let us design a machine that does absolutely nothing. That is a very bad goal okay.

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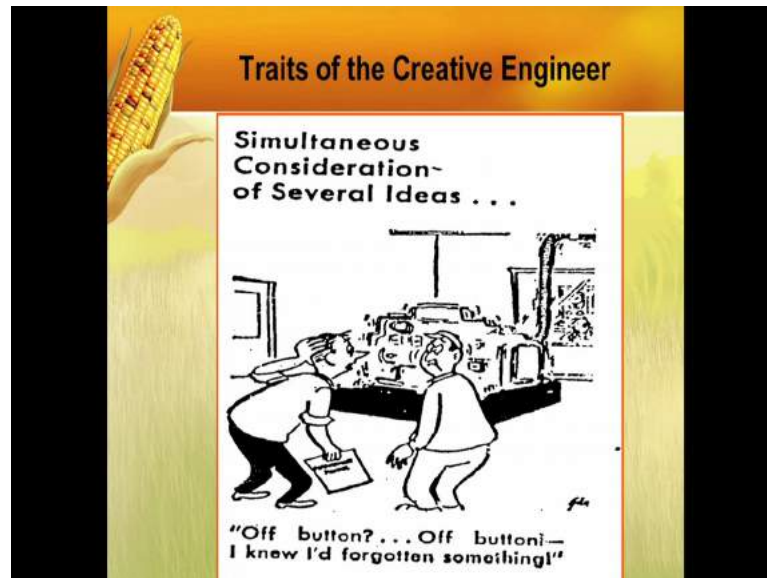


Knowledge of the fundamentals. So, he is in the aircraft wing division. This is like the rocket which is thrown at teachers. So, then, hello engineering department, we should



know that this is not the way to make a wing okay for an airplane.

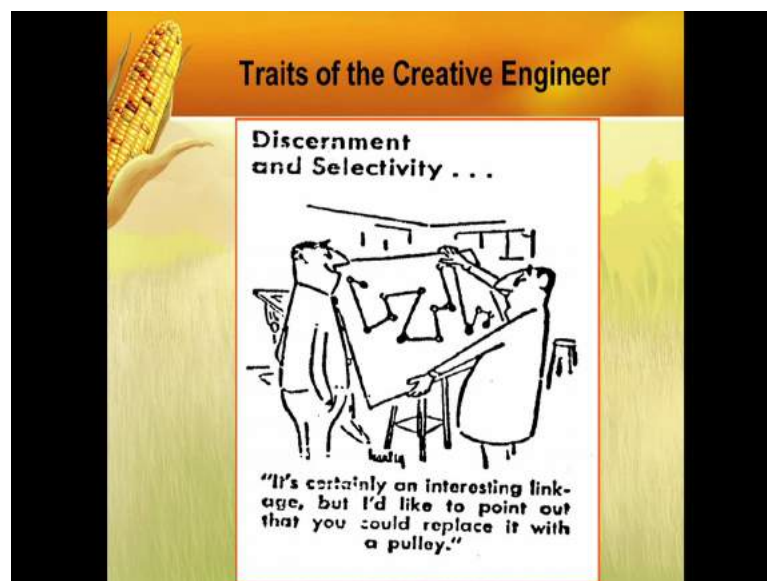
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Simultaneous consideration of several ideas. They built a such a wonderful machine, but they forgot the On Off switch. They do not know how to switch off this okay. In your, under graduate lab, the engines lab and all that, four people will do the experiment; only one fellow will do; all other three fellows will be chatting okay; only that fellow will know that in the Kirloskar engine where is On Off switch. So, in my under graduate days, in the Guindy Engineering College, the professor, in the viva, he will say, where is the On Off switch for the Kirloskar engine; he will find out who was chatting away and who did it? So, it will be there some where behind na, yes okay. So, this On Off switch is very important.

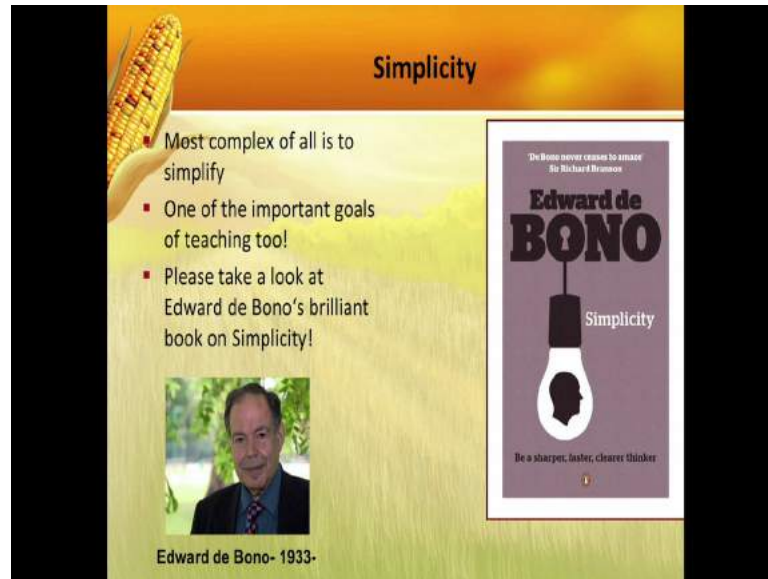


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Then, discernment and selectivity. We have something called Contraption in our Shastra, in our IIT Tech Fest. That is to make a simple thing more complicated. So, if a pulley can do a job, there is no need to put a complicated arrangement like this. Therefore, simplicity is very important. You should try to make things as simple as possible. Edward de Bono okay, who is a protagonist of this lateral thinking, he is written a 500-page book on simplicity, how society has become very, very complex, how do decomplexify; it is a book on simplicity, Edward de Bono okay.

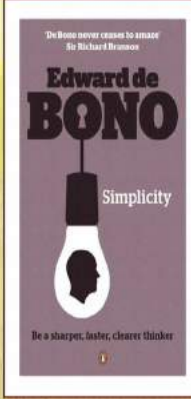
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Simplicity

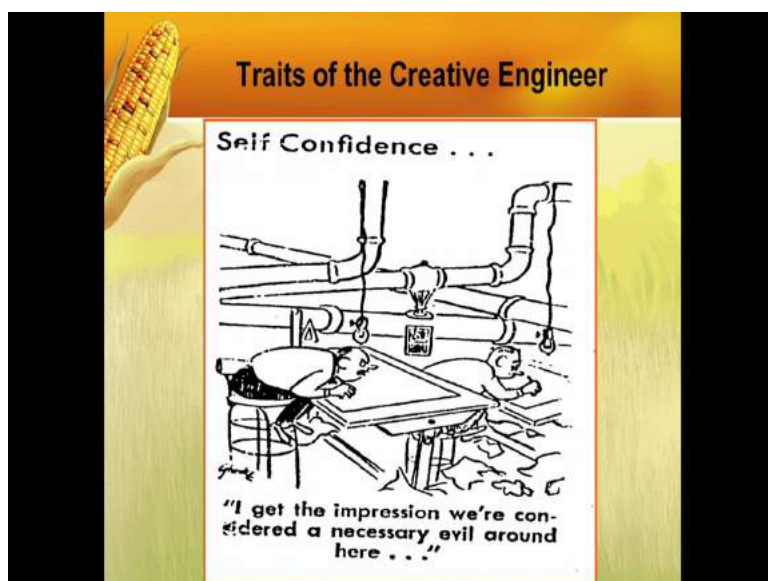
- Most complex of all is to simplify
- One of the important goals of teaching too!
- Please take a look at Edward de Bono's brilliant book on Simplicity!


Edward de Bono- 1933-



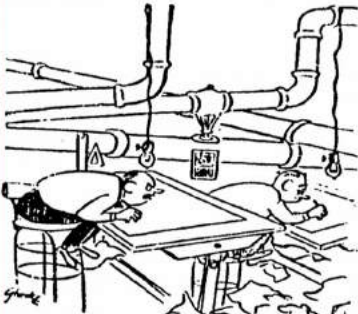
So, he says the most complex of all goals is to simplify; that is also one of the important things in teaching. We should make it very, very simple, that to such an extent, that somebody cannot make it any simpler. Then his brilliant book on simplicity you can buy; it is available; you can buy it on Flipkart or Amazon.

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Traits of the Creative Engineer

Self Confidence . . .

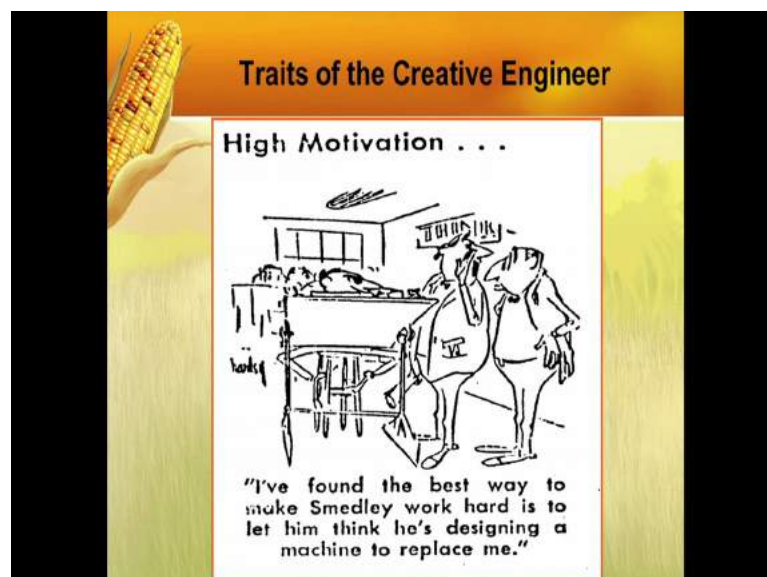


"I get the impression we're considered a necessary evil around here . . ."



Self confidence. You should not start out, I get the impression that we are considered a necessary evil. I get an impression that I am unfit for research. You just leave. Then, you should have a chat with somebody who is very enthusiastic and this thing right. Do remember that 10,000-hour rule; if you spend 10,000 hours, you have it already. How many people have already done their Masters? How many people have cleared the comprehensive viva. You come up to this stage, then you will do the other things also. That confidence you should have. All of us go through that this thing, but that when you go through this, you should not think that is permanent okay.

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High motivation. I found that the best way to make Smedley work, work hard, is to let him know he is designing a machine to replace me. That's the boss telling his subordinates, you design a machine which will replace me; then, it gives so much kick and energy to the subordinate okay.

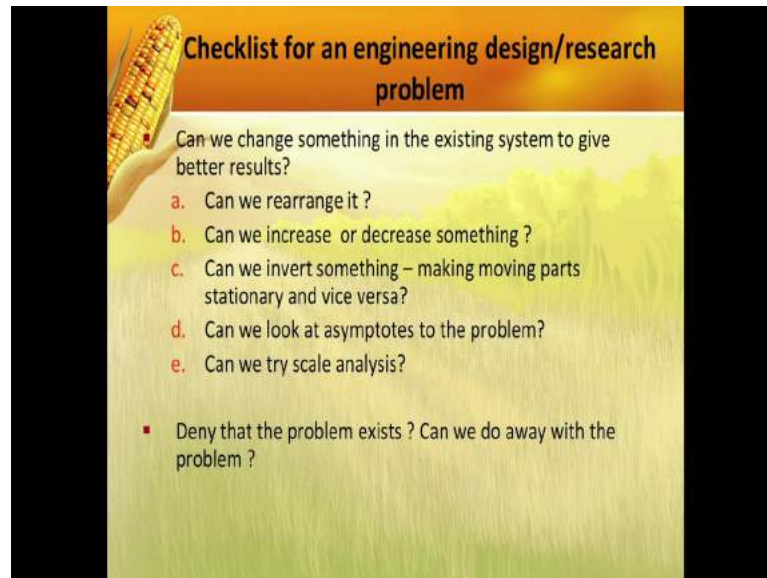
Constructive Nonconformity. You should not, just because you keep four pencils or you have a haircut like this, you are different. There is no point in just looking different; you should do things which are genuinely different okay.

Ability to think in images. We should be... your Autocad, Three-D, these things... that is



why drawing is the language of engineers. So, we should be good in that.

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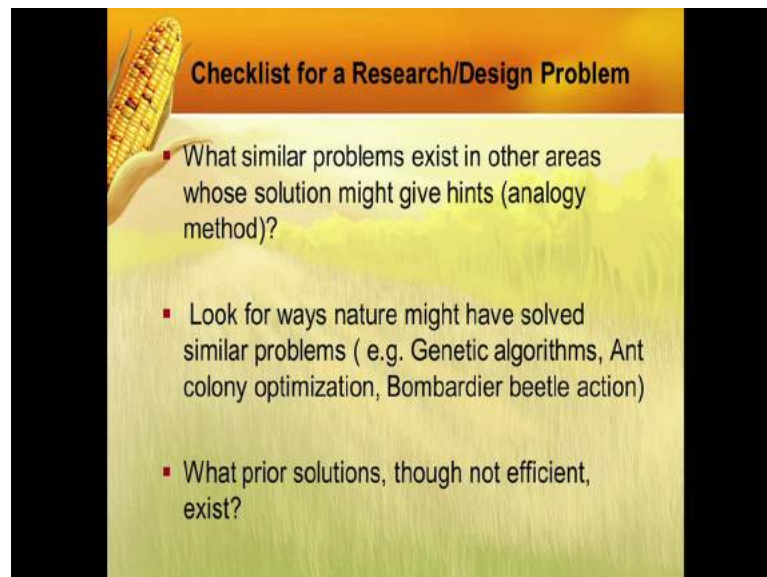
What is a checklist for an engineering design problem? Can we change something in the existing system to give better results? Start asking these questions in your research problem. Can we rearrange something? Can we increase or decrease something? Can we make moving parts stationary and vice versa? Can we look at asymptotes to the problem? If you are working, can I look at asymptotes? Can you look at Reynold's number tending to zero? Can I look at Reynold's number Ten into infinity? Can you look at $g r$ by $r e$ square tending to zero; $g r$ by $r e$ square is equal to Ten into infinity; epsilon tending to zero; thermal conductivity tending to zero; thermal conductivity tending to infinity, and then work out some maths, and find out some asymptotes, which can be used to benchmark your solution. Can we just work with paper and pencil instead of always going to the computer? You should ask these questions okay.

Can we do scale analysis? Can we look at approximal scale? So, how much time it takes for cooling? How much time it takes? Can we look at approximate scale for what is the maximum efficiency of your collection? Actual collection efficiency divided by the maximum collection. Can we call it as a new parameter, new paradigm or something, whatever in your field, and the deny that the problem exists? Ok, it is too much; take



some ten days off; don't work on it; then come back; you may get a solution.

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What similar problems exist in other fields? Analogy method. We use that analogy method. Electrical resistance network we used in conduction. Can we look at analogy? Can we look at how nature might have solved this? Nature, see, this called biomimetics, biomimicry, the genetic algorithms, ant colony optimization, bombardier beetle action, all these are inspired from nature. What prior solutions exist in other fields?



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Ant Colony Optimization

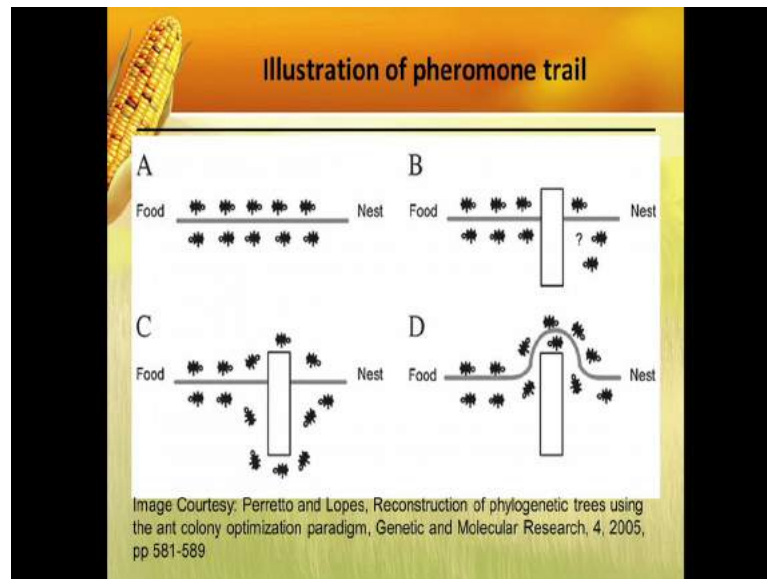
- A probabilistic technique to solve computational problems.
- Inspired by the foraging behavior of ants.
- Well suited for combinatorial problems and multi-parameter optimization.
- Uses the concept of pheromone trail to identify the path that results in the best value for the objective function.

The slide features a yellow background with a green field at the bottom. On the left, there is a corn cob. On the right, there is a small inset image of an ant on a branch.

Ant colony optimization. It is a probabilistic technique to solve computational problems. It is inspired by foraging behavior of ants; foraging behavior means how they behave when they look for sugar or food. So, well suited for combinatorial problems okay. So, when the ants go in search of food, they find out the previous ants in which path they have gone. So, after the ant has taken the sugar, it will leave that... it will leave a chemical, what is called pheromone, which is like formaldehyde. So, out of the multiple paths, the paths of the other ants, the next ant which will follow, will look at the paths where the pheromone concentration is very high. So, like wise. So, automatically, it ignores paths where the concentration is low; therefore, this can be used for a minimization function. So, this, in operation research, they will use this to solve the travelling salesman problem and so on okay.



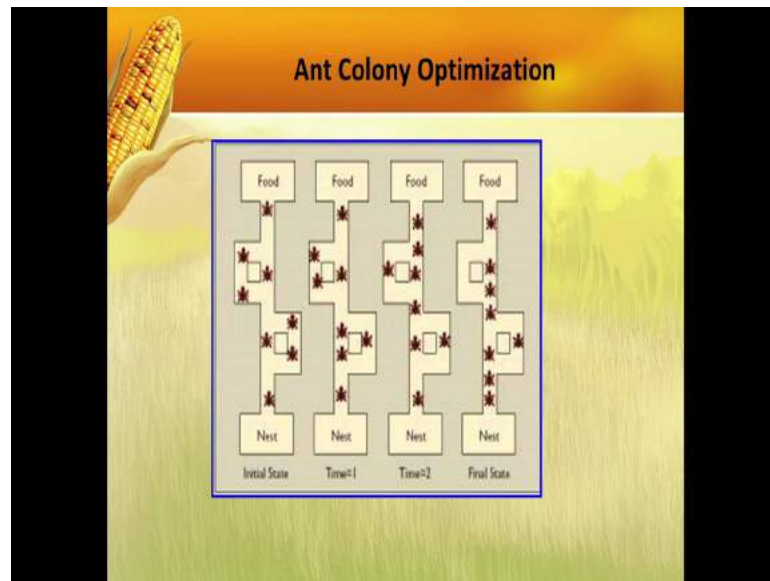
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In optimization engineering, we are using this in regular optimization problems also. So, this can be used to find out the best value of objective function. So, you try this experiment. If ants are there... if there is some sugar, ants are coming, you place an obstruction, then after sometime the ants will figure out the least path - the shortest path - automatically. So, we can see, you can keep obstruction, then first it will get confused, in C it is getting confused, but in D it is again choosing the shortest path. So, you can develop an algorithm based on this okay. So, that pheromone concentration; assume an initial pheromone concentration; the pheromone concentration in time is e to the power of minus pheromone concentration at this thing, you need to divide by pheromone concentration at time t equal to t_{naught} , and then you can develop an algorithm. One of my students, some years ago, did this. So, as a B.Tech project. So, this is the... basically you see after the obstruction it becomes clear, first it becomes unsettled, then it becomes clear.



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The diagram illustrates the Genetic Algorithm. It shows a grid with four columns representing different stages: Initial State, Time=1, Time=2, and Final State. Each column has a 'Food' source at the top and a 'Nest' at the bottom. Ants, represented by small black dots, are shown moving from the nest to the food source. The paths of the ants are marked with red lines, showing how the shortest path is reinforced by the most ants. The background features a corn cob on the left and a sunset sky.

Genetic Algorithm

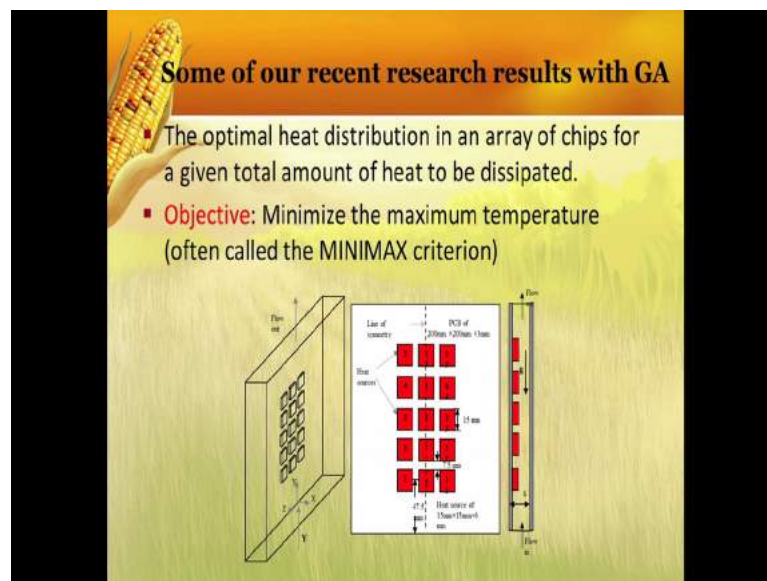
- A search technique used to find exact or approximate solutions to optimization problems.
- An evolutionary approach to optimization.
- Based on the processes of selection, crossover, recombination and mutation to arrive at an optimal solution.
- Very useful in obtaining global optimum in complicated situations.

Genetic algorithm. Genetic algorithm basically mimics the process of evolution. Evolution can be treated as an optimizing process. Best parents mate, and get, and produce the best children. From the best children, new sets of parents are created, like that... If you have multiple solutions, you find out which solution gives satisfies objective function, maximizes your objective function. In the case of a maximization



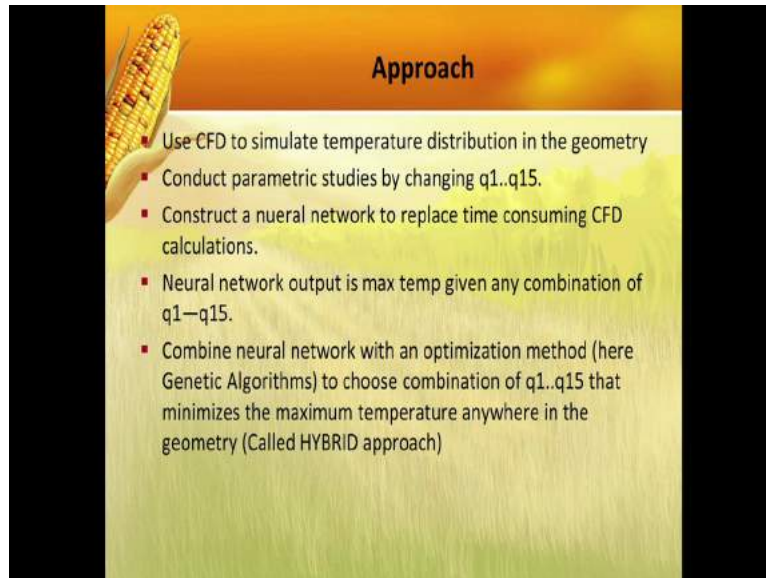
problem, convert your variables into zeros and ones, mix and match chromosomes in similar to genetics, and then, produce new children, check for that fitness – that's objective function mix and match. Randomly, alter some bits - that is mutation - and then proceed. So, you are getting inspired by nature. So, genetical algorithm is a very powerful tool now used in optimization okay.

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So, some of our research, for example, there are 15 chips which are generating heat. If the total is 15 watts, how will I distribute these 15 watts among these chips, such that the maximum temperature is minimized? This is called a minimax problem. So, this is a fundamental problem in electronic cooling. So, the base line could be, you put one watt to all the 15 and find out what is the maximum temperature. Then, you do your genetic algorithm, what have you used up whatever optimization, and find out what is the best, and compare to this case where you got uniform one watt for everything - how much is the temperature decreased okay? That is the benchmark okay.

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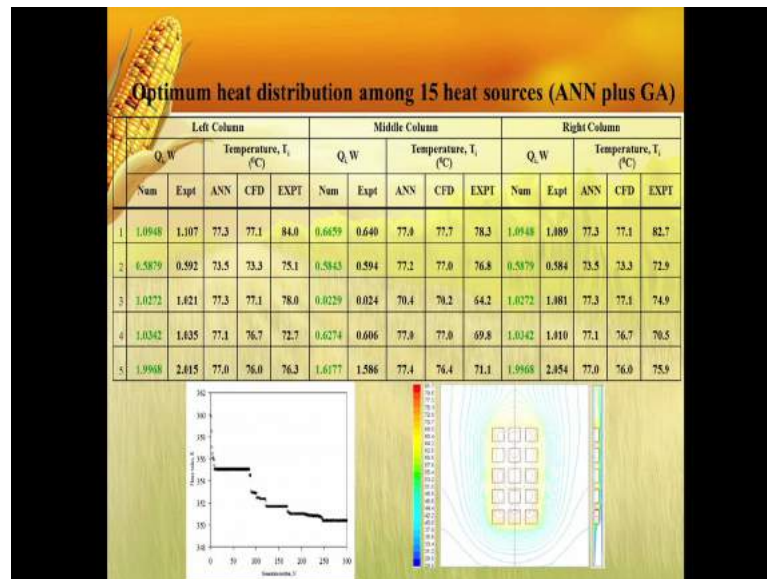
Approach

- Use CFD to simulate temperature distribution in the geometry
- Conduct parametric studies by changing q_1 .. q_{15} .
- Construct a neural network to replace time consuming CFD calculations.
- Neural network output is max temp given any combination of q_1 — q_{15} .
- Combine neural network with an optimization method (here Genetic Algorithms) to choose combination of q_1 .. q_{15} that minimizes the maximum temperature anywhere in the geometry (Called HYBRID approach)

So, approach is, use CFD to solve this, and then, give different combinations of q_1 to q_{15} , generate a neural network, because the CFD takes a lot of time, it is computationally more involving, more involved. So, generate a neural network, which is called a surrogate model or a fast forward model. Then, whenever the optimization genetic algorithm wants solutions, want what solution means, if I give q_1 to q_{15} , what is t_1 to t_{15} ? That the neural network will give. Use this neural network to drive your optimization engine or genetic algorithm. Try various combinations of q_1 , q_2 , q_3 , to q_{15} ; find out which combination of q_1 , q_2 , q_{15} , gives minimum of the t_1 to t_{15} , subject to the condition that q_1 plus q_2 to sigma q is equal to 15 watts, 10 watts, 20 watts, whatever has been assigned by you. Is it okay. It is the classic optimization problem; use genetic algorithm.

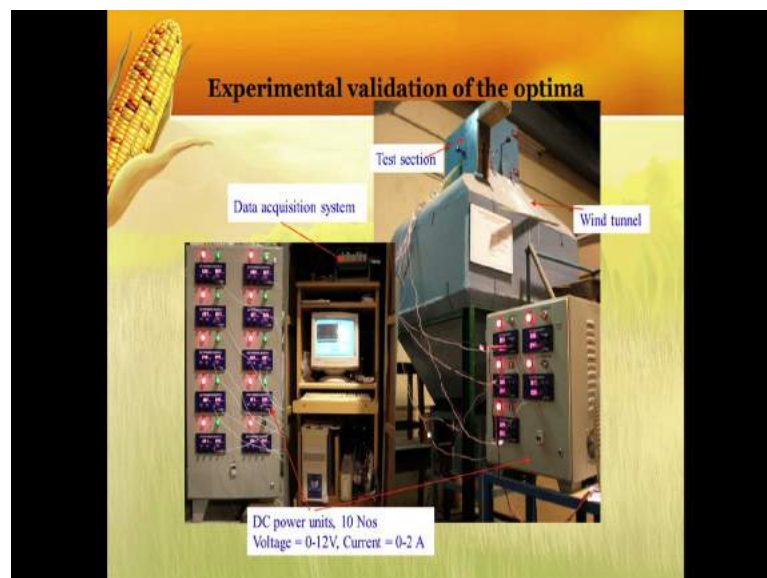


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So, you can... So, you can see that.

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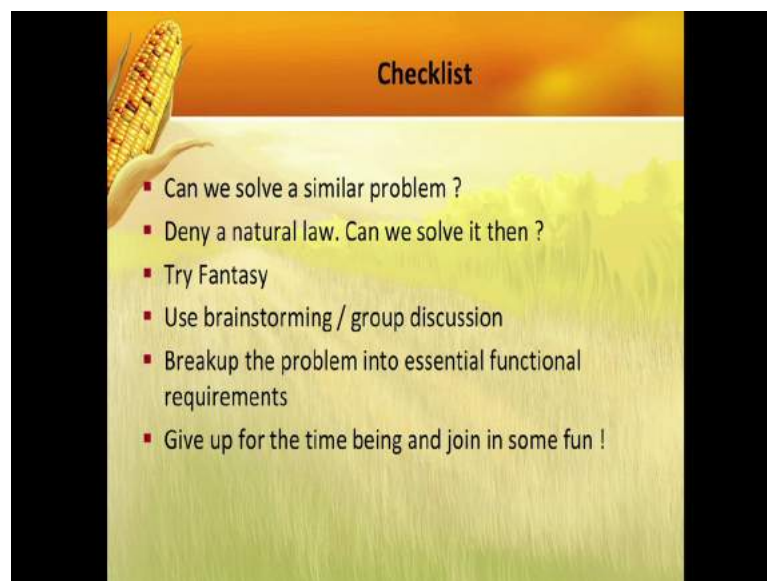


So you can see that finally, the optimum. So, you can generate the solution from ANN - artificial neural network; you can generate from CFD, and you can, finally, confirm, we confirmed with the experiments, we get within an error of 7 degrees; you get within an



error of 7 degrees. You can see that the q , the q is not one on everything. So, for some heater it is 0.5 watts, for some heater it is 1.091, for some heater it is 2 watts and so on. So, that distribution gives you, that gives you, that gives you minimax. Therefore, putting one in all the fifteen which is a trivial thing, which somebody will say no, no, I will know by this thing, I will know by dharshana, I will know all that - it is not possible. You have to go through this; you have to go through this optimization procedure to achieve this. Finally, after we obtain the optimum configuration, you have seen this wind tunnel in our laboratory, so we put that all this - there are 15 DC power supplies - we energize q_1 to q_{15} , gave what was experimentally, what was determined by the optimum or by the optimization program, and we figured out, that finally, the experiment gave less than 10 percent error. So, the optimum is confirmed by or validated experimentally.

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Check list. Can we solve a similar problem? Checklist for a research problem okay or a creative design problem. Deny a natural law. Can you ignore g and proceed and see what happens? Try fantasy. Use brain storming - discuss with your friends in the coffee table. Breakup the problem into essential sub problems and start attacking only the sub problems or just give up for a week or ten days and then get back okay.



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What are the key road blocks? Basically, fixity or mental block. I cannot solve this it is, so difficult. You have to get rid of that mental block. Analogy is a very good aid. So, for example, ant colony and genetic algorithm analogy kingfisher analogy to nature, electric analogy resistant network. So, keep your ears and eyes wide open. If you look at lot of journals - what are the electrical people doing? ANN and all this computer science people are doing. So, you have to look at some computer science journal. Artificial intelligence, some ideas from there you quickly put it to heat transfer and your first paper, your paper will become highly cited. So, you should know what other people are doing also. Instead of just seeing, you are looking always at what heat transfer people are doing, what solar people are doing; instead of that, you should also look at... get some best ideas from other fields and apply it to our field okay.



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Summary. The inventive/creative phase of the design research process. The inventive/creative phase of the design or research process is the key where we generate ideas and concepts to solve the problem. Once you generate the concept or idea to solve the problem, then the next stage is you will either write a program or you will develop an experimental setup and all these. So, the generation of ideas and concepts is where the creativity is involved. After that, after it is generated and all that, you build your experimental setup, then you get bogged down with details; then you look for a mechanic, you look for a technician, you look for those things, you look for a dark room, and then, you will do ventilated, this thing, night I don't want temperature to change, but the quality of whatever you get, whatever you get the usefulness or not, the creativity or not, the innovativeness or not, depends on the quality of your ideas. This experiment will give this.

According to you it is dharma, it will give the result, but whether it is useful or not depends on what is the quality of the, quality of the... what are the ideas behind the setup itself? You get what I am saying? Whether that itself is worth or not, that's where the idea generation is important okay. So, the quality of ideas is very, very important. That's why we need to be more creative. So, therefore, creativity, what is creativity ultimately? All the elements are already in your mind; this is the combination of all these things and



something comes out. As they say, a good teacher is somebody, a teacher is one who figures out what is already there in the students mind and allows that to come out. We cannot implant new things into this. We can only help whatever is there to come out okay.

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The creative process is you soak the mind with information. You try various combinations, and then, throwing up of the solution, and then, you say: this will be my experimental design; this will be the governing equation; this will be the problem I will solve. So, the creativity depends on the communication between conscious, again there is a typo, it's not unconscious, subconscious; there is a constant dialogue; this dialogue may take place when you are sleeping, in dream also it may come, in the dream also sometimes it may come, sometimes the dream also the computer go to line 640 I didn't divid it by 2; morning when we are there, at nine 'o' clock only the computer center will open; eight 'o' clock, eight thirty you will go, you will change, but you put that 2 also it may not work, that's a different matter, but atleast you know that problem that I solved; line 620 problem, again you sleep, line 1424 that square root is not there okay, and then what do you it's not working, write statement, write statement, write. So, you keep on every two lines, you put write statement okay. You will have to go through this journey okay. Three D visualizatton is very important. Talking to others, developing your



scholarship, and then, large library of ideas, you should look at biographies, you should get inspired; you should have large library of ideas, where you are able to generate many ideas and pursue them okay.

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How to improve creativity? Start playing games or puzzles; have some hobbies which will release your imagination; write some poetry or maintain a daily journal, just maintain what all you have done everyday; read short stories; read poetry and biographies, it provokes your imagination; write, not write not journal paper, write about your experiences or something you dont have to show to others also, show it to only your close friends; you are going on a three day trip to some place, write it down on word and share it with somebody else. And then, experiencing the process. So, this I will skip; you want this? We will skip this.



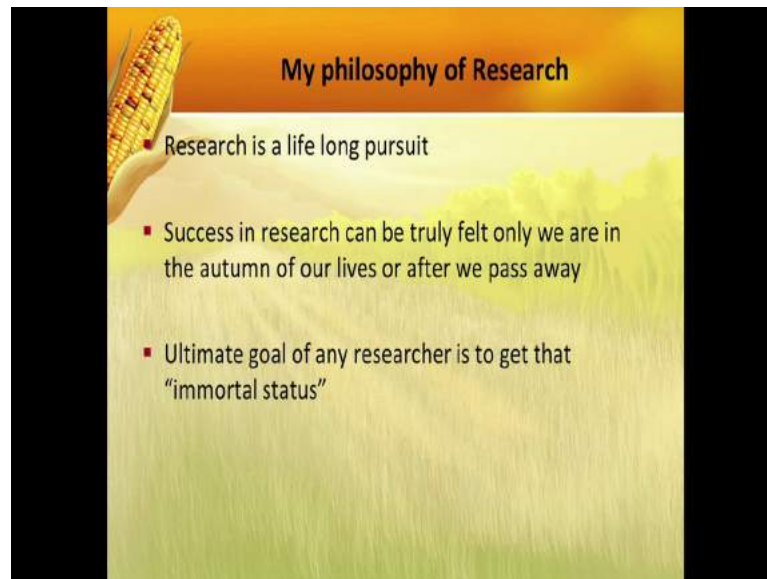
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Okay, finally, research is a serious activity. It must be treated with great respect. It's a life long activity. It is a creative pursuit. So, it is a penance or tapas okay you have to keep doing it. It requires intense focus, energy; it doesn't mean that you cannot relax, you can relax, but generally just like a compass will always point towards true north, whenever there is no activity automatically the mind should come towards research. You allot some time, you want to do a sport, you want to jog, you want to walk, you want to play tennis, you want to watch a movie, all that is fine, but the compass left undisturbed it will come to true north. Mind must pursue that, within five years I want to do, that ten years I want to do that okay.



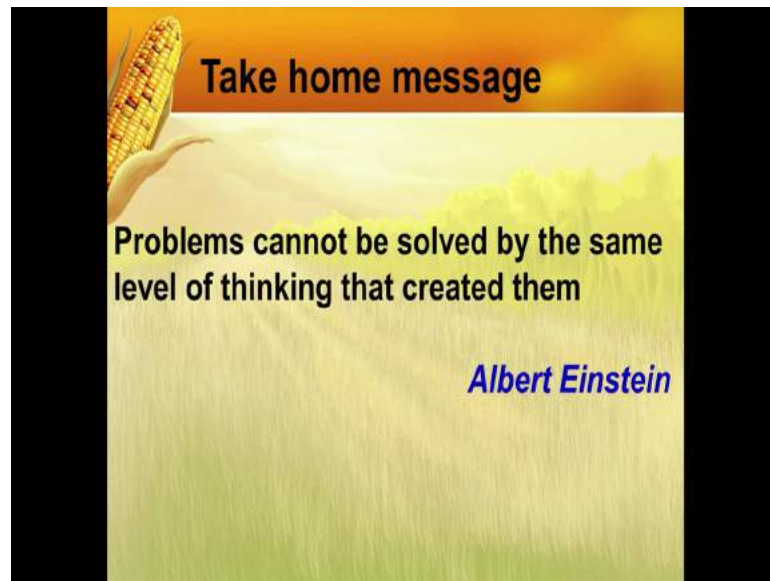
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Success in research is very difficult. Other people should tell whether your successful or not; we should not keep on talking. Time will decide whether you are successful or not. We have to freak out, we have to enjoy what we are doing; that's very important. So, the ultimate goal of any research is to get that immortal status; everybody wants to know - will I become another Prandtl? It's very doubtful; you may not become, but at least the goal is there. Will everybody talk about me, just like they are talking about Prandtl? 1905 Prandtl's Boundary Layer Theory; 2016 also any professor going to the class Prandtl's Boundary Layer Theory. We don't know what you do; we may not even come to one percent or half a percent of that, but inside you have - can I become the next Prandtl? Okay. It's good to have that goal and proceed okay. You want to attain that immortal status okay; you call it as anantha.



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Take home message. Problems cannot be solved by the same level of thinking that created them okay. So, you require a higher level of thinking to solve problems.

Thank you very much.

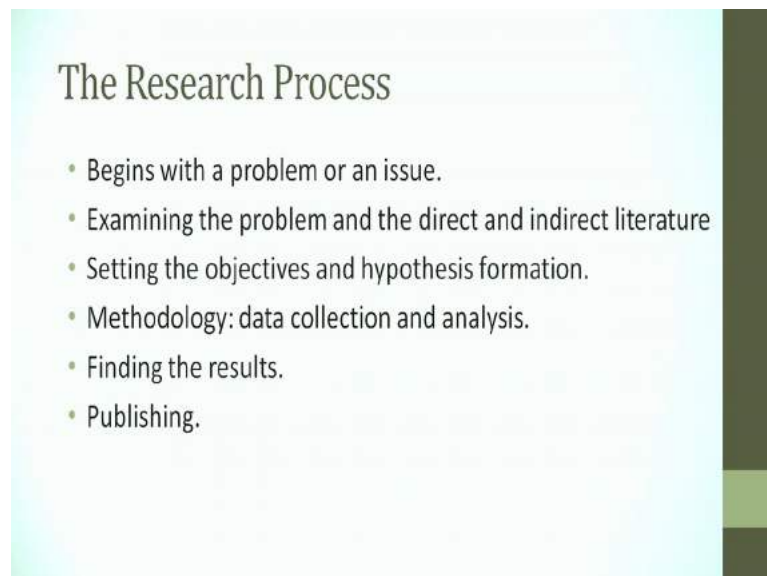


Introduction to Research
Prof. Dr. Sreekumar Nellickappilly
Department of Humanities and Social Sciences
Indian Institute of Technology, Madras

Lecture – 24
Ethical Practices in Research

Hi, welcome viewers. This is a lecture on Ethical Practices in Research. I will try to reflect upon some of the important ethical issues involved in conducting research. So, we will begin with a very brief explanation of what happens in the research process, just to see how ethics becomes important there. Then, afterwards, we will try to see the historical evolution of research ethics or rather ethical issues in research, how this has become an important issue in today's world. Then, finally, we will see the some of the issues as such. That is the way which this lecture is planned.

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So, first, let us see the research process. This is not a very scientifically classified account of what happens when we do research; it is just a commonsensical account of it - the research process. We begin with a problem or an issue, normally. This need not be the case always, but usually this is a case that a researcher starts with a problem. And quite often we can see that here itself ethics becomes important. Many of these issues which a researcher tries to understand would be resolving an issue in the society, which would



drastically improve, sometimes, the lives of people in the society. So, in that way there itself, we find a role of ethics. Then we will start with examining the problem, and the direct and indirect literature related to the problem, which we normally describe as literature review or you conduct a review of the already existing research in this domain to understand your problem from a right prospective, and also to situate in the correct angle.

Then, we will go to the setting the objectives and hypothesis formation, where exactly some of the problem and its outcome are narrated. Finally, again, you also discuss about the methodology, where how data is collected in various disciplines, this process varies from discipline to discipline. And then, also how an analysis is done and then finding the result; what kind of results, what kind of findings are you planning to come out of with, and then, publishing them in a research journal; and this is very crucial part of it - publishing.

So, it begins with identification of the problem and sort of ends with publishing; one stage of research is over. The other stage is now when you apply this research in actual situation - may be in the industry or in somewhere else. I am not discussing those aspects here, but we could see that all the steps - the six steps described here or narrated here - we can see that ethics plays a very important role, starting from identifying the problem till publishing. For instance, in publishing, there are several issues like authorship - issues related to authorship, whom should be given credit when a paper is being published in a scientific journal or a technical journal, again plagiarism, issues related to publishing; all these are very important ethical issues. So, we will very briefly try to analyze some of these problems today.



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Ethics and Research

- The term *ethics*: from the Greek word *ethos* or "character."
- Good character is associated with his/her moral integrity.
- This is important in ensuring a reliable process with trustworthy and valid findings.
- Public trust is crucial for research and hence researchers must be responsible for their work.
- Many professional associations for different disciplines have their own general ethical guidelines relevant to their disciplines to ensure integrity and trust.

So, ethics and research - the term... Before we really get into the lecture, get into the problems, let us see - what is ethics? So, ethics - the term ethics - is derived from the Greek word *ethos* which means character. So, here itself we could see that good character is associated with some sort of a moral integrity of a person, and when you try to apply this into the research domain, we could see that the moral integrity of the researcher is extremely important. So, in that sense, we can connect these two terms. Again, this is important in ensuring a reliable process with trustworthy and valid findings - these two things are very important. The valid findings of a research which sort of concludes such activity at certain stages is very important, and this is because research depends a lot on public trust; it is crucial for research; and hence, researchers must be responsible for their work. They have to convince the public, they are indebted to the public. See, many professional associations in engineering or in medicine or various other domains, they have come up with their own ethical course and their own general ethical guidelines which are relevant to their disciplines to ensure integrity and trust - this public trust.



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Ethical Dimensions of Research

- Guiding moral principles in research.
- Selection of a research problem.
- Ethical issues affecting the conduct of research.
- How to treat the participants in research: human and animal.
- The need for being truthful and transparent.
- Preparedness for risk management.
- Willingness to share the benefits: with fellow-researchers, other stake-holders and the society.
- Act with responsibility.

Then we try to understand what ethics has to do with research, the ethical dimensions of research. We can see that on most occasions that there are guiding principles - general guiding principles - in research and ethics have ((a very)) role to play in the context. As I already mentioned, right from the beginning, the selection of a research problem itself ethics plays a very important role. And again, there are so many important ethical issues that affect the conduct of research, the very conduct of research and also researchers, how researchers are related to each other. Particularly, in today's world, where there is a lot of collaborative work taking place people have to engage in very important professional relationships, there is a lot of, you know, questions of sharing certain resources, laboratories, and finally, also publishing the kind of findings they have come out of a research work. In all these context, the entire conduct of business need to be regulated by very strong ethical guidelines.

And how to treat the participants in research - these questions are important. There are a set of questions which deal with how to treat the participants in research. Particularly when human beings are involved and also when animals are involved in many domains we know in science, in many domains human beings and animals are involved in research; particularly animals. And when it comes to medicine and biotechnology, human beings are also involved in research; and even in social sciences human beings -the participation of human beings - are unavoidable in today's world. So, in that case there



are very important issues - ethical issues - that have to be taken into account. And again, the need to be truthful and transparent; one has to be truthful - these are basic, very elementary, ethical values.

And again, another very important domain is risk management, because every research involves, particularly if it is scientific research, many research activities involve the question of risk and we cannot avoid that. Our aim should not be avoiding all the risk that are involved in, but rather how to properly take care of, how to properly manage risk. Be aware of the fact that the risk involved, there are risk involved in the research process, and we should also be prepared to tackle them, to manage them, to properly intervene and interfere at the right time, so that we can minimize risk. So, these are some of the important issues.

Then again, willingness to share the benefits; that is very important. Every research will have social a benefit, a personal benefit. And how far researchers are willing to share this. This might come in terms of publications, sometimes in terms of patents or creation of intellectual property and many aspects of findings of research. So, whether researchers are ready to share this with others - those who are stake holders, the fellow researchers, other stake holders, participants in research, subjects in research, even people who help them in conducting those research, this has to be shared. But it need not be shared in the same way. For example, with participants in research or rather with collaborators in research, there is a kind of intellectual sharing. When you publish an article, in a scientific journal, you need to consider many serious authorship issues here. But when it comes to subjects of research, the sharing might be having certain very important financial implications and other implication. So, all these things have to be taken into account and properly taken care of when you do research.

And what is more important is - very simple - act with responsibility. And this term – responsibility, has a very wide meaning in the context of ethical research. And in one sense we can say that ethics is all about responsibility. So, one has to be responsible and show this responsibility in various ways when research is conducted.



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The Case of Prof. Hwang Woo-suk

- In an article in *Science* in 2004 the South Korean scientist Hwang Woo-suk claimed that he cloned a human embryo and extracted stem cells from it.
- A remarkable accomplishment if it were true.
- Many implications to the science of medicine and treatment of various ailments.

A portrait photograph of Professor Hwang Woo-suk, a middle-aged man with dark hair, wearing a dark suit, white shirt, and a patterned tie. He is looking slightly to the right of the camera with a neutral expression.

See, I will just, very briefly, explain the case of Professor Hwang Woo-suk, who is being celebrated as the most important Korean scientist, who shot into fame due to certain inventions he made or rather he claimed that he had made. In an article in science, in 2004, the South Korean scientist Hwang Woo-suk claimed that he cloned a human embryo and extracted stem cells from it. It was regarded as remarkable accomplishment, if it were true at that time. And many implications to the science of medicine and treatment of various ailments, this discovery or that or rather this invention would have helped a lot of people all over the world who are suffering from various diseases and ailments.



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The Consequences

- An individual's skin cells could be used to make a cloned embryo and from that extract kidney, liver or heart stem cells.
- These cells could then be used to replace diseased tissue - with no fear of organ rejection - and cure diseases such as Alzheimer's disease and diabetes.
- Prof. Woo-suk shot into fame.
- Became his country's most prominent scientist and was called its 'national treasure'.

The consequences are very briefly: an individual's skin cells could be used to make a cloned embryo and from that extract kidney, liver or heart stem cells. So, this definitely you can see, quite obviously, will have a lot of implications, positive implications, in the domain of medicine. And these cells could then be used to replace diseased tissue with no fear of organ rejection, because it is extracted from one's own body and cure diseases such as Alzheimer's disease and diabetes. Prof. Woo-suk shot into fame; no, no surprise, and he became his country's - that is South Korea's - most prominent scientist and was called its national treasure at that point of time. He has been celebrated.

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The End of the Story!

- An academic panel investigating his work found that he faked his claim.
- His claim to have efficiently developed 11 patient-specific stem cell lines was false.
- He admitted that female researchers in his own lab had supplied eggs for his research.
- The 2004 paper was written on fabricated data to show that the stem cells match the DNA of the provider although they didn't.
- In March 2006, he was fired from his professorship at Seoul National University and was charged with fraud and embezzlement.



But the end of the story is really sad. An academic panel investigating his work found that he faked his claim. I am just reading it. His claim to have efficiently developed eleven patient-specific stem cell lines was false. He admitted that female researchers in his own lab had supplied eggs for his research, which is actually a very important unethical practice, he should not have done that. Then again, the 2004 paper was written on fabricated data to show that the stem cells match the DNA of the provider although they did not. And again, what is the consequence? What had happened finally? In March 2006, he was fired from his professorship at Seoul National University and was charged with fraud and embezzlement. So, this is the sad end of the story.

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Ethical Issues	Associated Issues
• Unethical and coercive behaviour. • Fabricating data. • Dishonesty. • Lack of trust. • Lack of integrity. • Cheating.	• Plagiarizing. • Making unfounded charges about another researcher. • Breach of confidentiality. • Discrimination: gender, caste, race, national etc. • Negligence. • Truthful and responsible data management. • Research involving human and animal participants. • Social research.

So, let see some of the ethical issues particularly pertaining to this case which we have discussed, and also some of the associated issues which we can consider in this context. One is the unethical and coercive behavior of the scientist and also his team members. Then, fabrication of data is one of the most prominent ethical violations which this particular case is involved in. Dishonesty, no doubt in that; lack of trust because he has violated all basic principles of trust, and the public trust is completely lost in this particular case; a lack of integrity; cheating. So, all these are some of the specific issues pertaining to this particular case, which we have discussed.

But there are certain associated issues which we can find, which would be relevant, when we discuss research ethics, something which has to do with plagiarism or plagiarizing. A



plagiarism is a situation where you take someone else's work and present it as your work either through writing or presenting it as in publishing in journals. When you publish a research article, you publish it as your original work, but you are lifting it or you are taking it from someone else's work without properly acknowledging.

Again, making unfounded charges about another researcher. This often happens in research laboratories, and also collaborative research, and also in certain other context research takes place.

Breach of confidentiality. In many domains of research this happens, because particularly, for instance, in certain domains like medicine, a medical research or even in social sciences this can happen, because confidentiality of participants or subjects are very important; privacy of participants have to be maintained. And, they have trusted the researcher, and revealed a lot of information about them, and researcher now has a lot of information about the subject, and this is happening because the subjects trust the researchers. So, in hope that or rather they believe that researcher will maintain confidentiality, but breach of confidentiality in that sense is a very severe unethical practice.

Discrimination. This again is a very common thing that happens in many context. Discrimination on the basis of gender, caste particularly in the Indian context, race, nationality, and the language you speak, and several other factors which decide our personality or our character.

Negligence. This is what I have already mentioned. A researcher should show responsibility. Responsibility is one of the important things and researcher should conduct his research with responsibility, so that avoid all kinds of possibility of negligence, because sometimes negligence can lead to harm. Truthful and responsible data management.

Research involving human and animal participants; so, this is another very important domain, very sensitive domain; when animals and human beings are involved one has to be extremely careful. First of all, try to avoid all situations of harm wherever it is possible to avoid, and where it is not possible to avoid harm, as I mentioned earlier, one has to be prepared for managing it properly. All care has to be taken, because life is



precious, and it is important, and researcher should respect life, should show at most respect to human and animal life.

And then, in the context of social research there are several other issues, which we may not be discussing, in detail, in this lecture.

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A presentation slide with a light blue background and a dark green vertical bar on the right side. The title "Research ethics" is at the top in a dark green serif font. Below it is a bulleted list of five points in a dark green sans-serif font.

- Deals with **norms for conduct** for researchers.
- How to distinguish between acceptable and unacceptable behavior?
- Critically examining the ethical questions researchers face and how they ought to respond.
- Trying to understand the moral, cultural, social, legal and political implications of the research.
- Basic responsibilities and commitments of researchers.

Now, again, when you try to specially understand the domain of research ethics, we can see that it deals with norms of conduct, norms of conduct for researchers. How to distinguish between acceptable and unacceptable behavior? For example, plagiarism is unacceptable; there is a reason for that; or discrimination is unacceptable; there are reasons for why they are unacceptable, because they are all impediments in way of truth, finding truth in research. Again, critically examining the ethical questions researchers face and how they ought to respond.

Trying to understand the moral, cultural, social, legal, and political implications of research. Here is again we could see responsibility coming in; one has be careful about the cultural factors, because when a research is conducted in a one particular culture, the implications of that research, the way in which it is understood, all will be different when it is the same research is conducted in some certain other culture. And this is very important in today's world, because we are living in a globalized world, and a research which takes place in one of the distant corners of this globe will have implications to all human beings in this world. Say, for example, a research in the pharmaceutical industry,



the medicine which is going to be invented or designed will have implications on the health of all human beings in the globe, so, in that sense, but certain experiments cannot be conducted in certain areas, and one has to be sensitive towards the cultural aspects then social and legal; certain countries... Law will be different in different countries. So, certain experiments or research cannot be conducted in certain countries, but while in certain other countries the law might be different.

Then, political implications of research. So, all these things a researcher might be aware of and should be sensitive to.

Basic responsibilities and commitments to researchers. This involves one's commitment to other researchers, one's students, one's professors, one's fellow researchers, to the society, to the scholarly community, etcetera.

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Why Unethical Behaviour?

- Research is frustrating; hence the look out for an easy way out.
- Self-interest, ignorance and relativism.
- Pressures to publish or obtain grants or contracts.
- Career ambitions guided by self-interest.
- The pursuit of profit or fame.
- Poor supervision of students and trainees, and poor oversight of researchers.
- Insensitive to potential harm.
- Irresponsible behaviour and attitude.

And, when we examine why unethical behavior happens, we could see that there are several reasons for this. Particularly, in today's world, there is the chances of, you know, research world encountering unethical practices are more compared to some 50 years ago. The reason is that research is frustrating; hence, researchers might look out for an easy way out. So, there are many such attempts might be severely unethical.

Self-interest, ignorance, and relativism. People do not know that certain practices are unacceptable and unethical. People may not be knowing; they do not have the awareness.



Again, relativism. They think that it is ok and which may not be ok in certain other places or certain other individuals. So, when they collaborate, this might create a problem. So, the ethical practices, when particularly in collaborative research, it is important that all the collaborators agree upon certain practices, before they really get into the process.

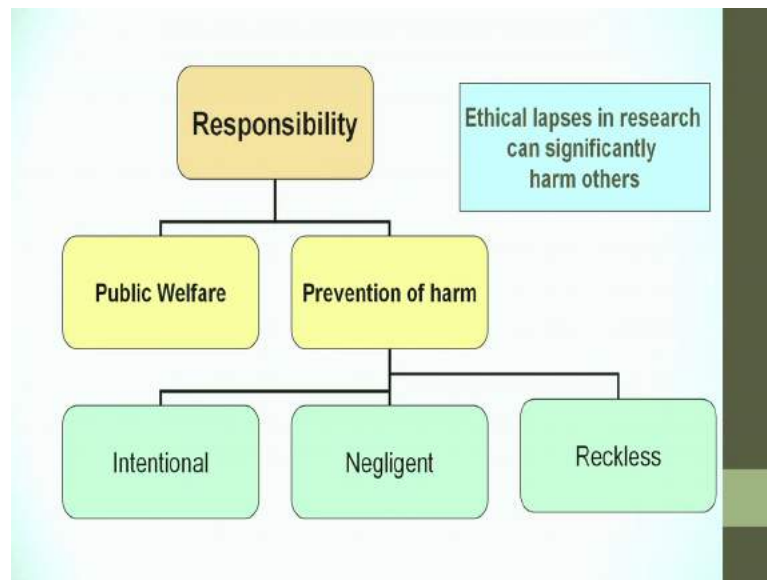
Pressures to publish or obtain grants or contracts; this is another thing. Today's world there is a lot pressure to publish; publish or perish has become the mantra in many places. Many research organizations and universities appoint people with the expectation that they come up with wonderful results in research. They are ready to fund the researchers with money, but they also expect results, and if this does not happen that will affect the job security of the researchers. So, naturally, there is a lot of pressure to publish and also to obtain grants and contracts. So, in this context, out of frustrations, researchers might give up for self-interest and look for easy ways out, which might result in unethical behavior. So, one has to be very careful in dealing with that.

Again, career ambitions guided by self-interest. The pursuit of profit or fame. Then, poor supervision of students and trainees and poor oversight of researchers. This is again, very important, because as professors or as senior scholars, researchers might have certain responsibilities towards the scholarly community, and one of such responsibilities is towards one's students. And it is very important that people should guide their students properly, their trainees also properly.

And then, again, insensitivity to potential harm. People have to be extremely careful about this, because certain actions or certain consequences of research might end up in harming human beings or otherwise even nature, animals. So, one has to be very careful try to avoid it.



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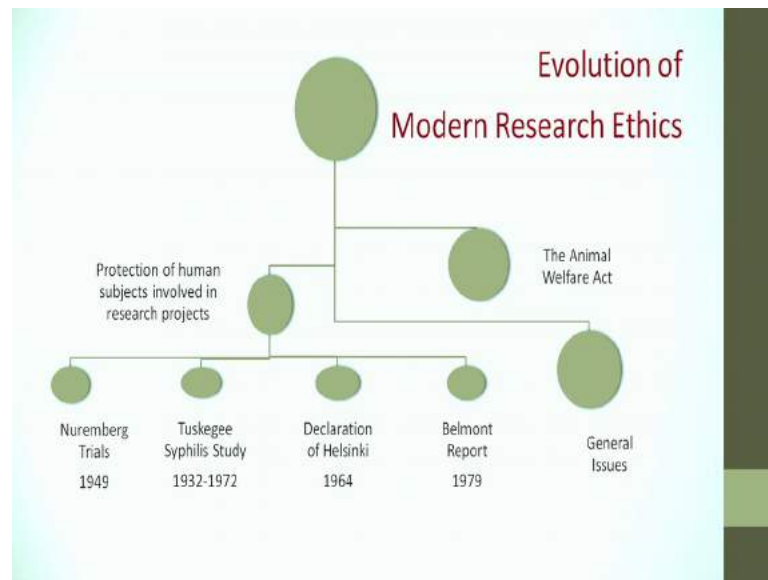


Irresponsible behavior and attitude. So, as I mentioned, responsibilities so key term in research, and because ethical lapses in research can significantly harm other people. So, when we talk about responsibility, every researcher has a responsibility to negatively to prevent harm; I would start with that, I say negatively, because researchers, research work or the findings of research or the products and consequences of research, might harm people or even the process itself might be harmful. So, it is very important that researchers are aware of it, and there are different types of harm: intentional, negligent, and reckless, you should avoid all the three types of harm and try to prevent it.

The second one is the positive aspect of responsibility where researchers should understand that their entire work and career is based upon public trust. So, they have a responsibility towards public; they are indebted to the public and public welfare is very important. They should know that public supports or rather the trust is built upon the expectation that research work might end up or result in some sort of development or progress in the society, it might benefit the society. So, with that expectation researchers are supported by the society. So, researchers should keep this in mind that they have a positive ethical, moral responsibility towards public; that their work should benefit the public in a significant manner.



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And now let us very briefly try to understand, the evolution of modern research ethics, because research itself is a comparatively modern phenomenon. Earlier also people used to conduct research, but the way they use to conduct it was very different. Now around eighteenth century onwards the way in which research as we understand today came into existence. Prior that also research was there, but they used to perform it in a very different environment with different priorities. Most of the researchers of earlier days were very spirited people, and they really enjoyed what they were doing, and they were not constrained by several factors which a modern day, contemporary day researcher, are constrained by.

So, if you try to understand the evolution of modern research ethics, we could see that the protection of human subjects involved in research projects was one of the primary concerns of research ethics in the modern day. Primarily after the Second World War this has become a major issue; and all over the world this is being discussed. Immediately after the Second World War, with the victory of the allied forces, lot of atrocities and cruelties, the Nazis - the Nazi Germans - have done to the Jews were exposed. Jews and the gypsies, and old people, the sick people - all of them suffered immensely under the rule of the Nazis. So, in this context, lot of discussions happened all over the world.

So, there are certain milestones we could find in the development of modern day research. I will very briefly explain them.



The first one is the Nuremberg trials which took place somewhere around 1949, which has ultimately some with certain codes. Then, the Tuskegee Syphilis Study which lasted for nearly about forty years in the US, and which also exposed some grossly unethical practices that took place in the state of the Alabama. Then, the declaration of Helsinki which has come up with some of the very important guidelines of how research needs to be performed. Following that in 1979 the Belmont report. So, these are some of the historical milestones; and then, later on, we could see that all these important events have taken place in the domain of or rather they are more directly related to medicine or health where human beings are particularly involved in.

As I mentioned, the explicit concern was protection of human subjects involved in research projects. And then, again, we could see that in this context the animal welfare act in the US, which has come up with some of the or rather expressed some of the concerns when we conduct research on animals or when animals are involved as subjects. Proper care has to be taken and the researchers have responsibility to value the life of animals as well; we cannot just ignore them. So, those things are articulated in the animal welfare act, and many countries in the world today have laws or legal suggestion, how animals have to be respected. And particularly, in research institutions and organizations we have ethics bodies today or ethics committees or institutional review boards; all of them will take care of or will look into that kind of research that is happening in within the institute, within the organization, and try to suggest ethical guidelines to be followed when researchers conduct their research. And then, of course, some general issues, some of them which we have already seen.



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Nuremberg Trials

- Examined the medical experiments conducted on concentration camp prisoners by the NAZI physicians.
- The effects of many toxic chemicals and other substances were tested on Jews.
- Euthanasia performed on the sick and disabled civilians.
- Underlined the need for postulating certain universal codes of conduct that would guide any future medical research in the name of science.



- Creation of the **Nuremberg Code** in 1949.
- Emphasis on voluntary consent.

And when we talk about, very briefly about, Nuremberg, when we talk about Nuremberg trials, which I have already indicated, these trials have examined the medical experiments conducted on concentration camp prisoners by the Nazi physicians. The effects of many toxic chemicals and other substances were tested on the bodies of Jews and gypsies. And euthanasia was performed on the sick and disabled civilians without their consent and permission. This, underlined the need for postulating, these trials actually underlined need for postulating certain universal codes of conduct that would guide any further medical research in the name of science. So, scientific research cannot be blindly carried out by scientist, just because the consequences are going to be good for humanity, just because the end is good, you cannot conduct it in any manner you want. The means also are very important; the ends and means are intimately interconnected; one has be careful about that. So, the creation of Nuremberg code took place in 1949, where there is an explicit emphasis on voluntary consent. People have to be consented, people have to be taken... the subject's permission or consent has become so central for research involving human subjects thereafter.



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The Tuskegee syphilis study (1932-70)



- Conducted by the U.S. Public Health Service (USPHS) on the blacks.
- The medical study monitored to discover the effects of untreated syphilis.
- No consent was taken.
- The intention of the study was not revealed to subjects.
- No treatment was given even though it was available.
- Race-based study.

The Tuskegee syphilis study, another unfortunate incident that had happened in the United States; one will be surprised to know this. It has happened for forty years from 1932 to 1972, when it was exposed by the media to the public. And it was conducted by the US public health service on the blacks. So, only blacks were selected as subject for this study, which also highlights how racially motivated this scientific study was. And the medical study monitored to discover the effects of untreated syphilis. So, people with syphilis and without syphilis, they were all monitored by the physicians. And the people who had syphilis were not treated even though medicines became available slightly later, they were never given, they were never administered this medicine because a purpose of the study was to see the effects of untreated syphilis.

And no consent was taken, because many of these participants did not know for what this study was conducted. And they were never been told. They were only told that they were treated for bad blood. No other information was given and no consent was obtained. The intention of the study was not revealed to the subject at all. And no treatment was given even though it was available and in that process many people lost their life; many people were injured, and finally, there was a public outcry against this when it appeared in the newspapers, and the study was stopped. The research study was stopped by the US public health services. So, it was obvious that it was a race based study. So, sometime back I mentioned about discrimination. Here there is a gross violation of the principle; it was based on discrimination; no whites were studied as subjects in this research work.



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Declaration of Helsinki (1964)

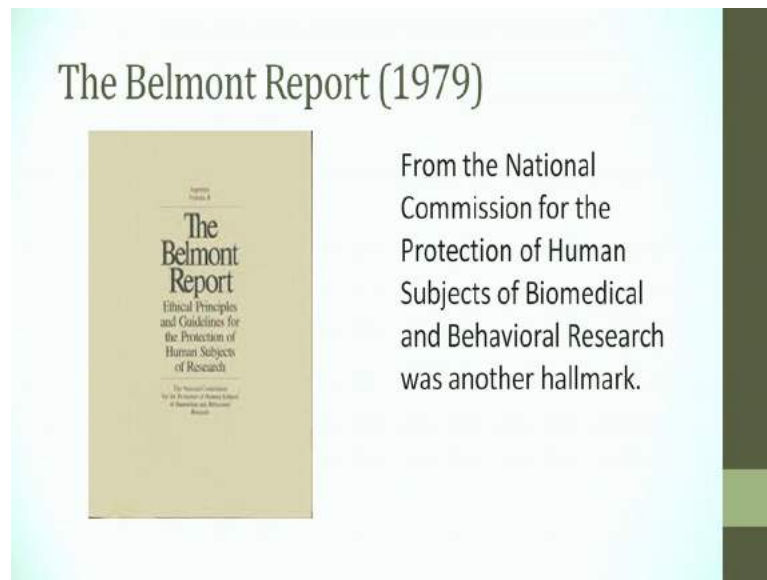
Mandated that all biomedical research projects involving human subjects carefully assess the risks of participation against the benefits, respect the subject's privacy, and minimize the costs of participation to the subject.

A photograph showing three individuals (two men and one woman) standing behind a table, signing a document. The man on the left is wearing a dark suit and glasses, the woman in the middle is wearing a colorful patterned dress, and the man on the right is wearing a blue suit. They are all looking down at the document on the table.

Again, the declaration of Helsinki in 1964 mandated that all biomedical research projects, involving human subjects, carefully assess the risk of participation against the benefit. So, there should be a risk-benefit analysis done, and if the benefits are more the study can be conducted. Respect the subject's privacy. So, this is what I mentioned about - confidentiality. Every individual human being who participates in this subject, in the study, in the research study as subject, has to be respected as a human person; his or her privacy should be maintained. So, all information about him or her should be maintained confidentially, and minimize the cost of participation to the subject. So, risk management - proper risk management has to be taken into account.

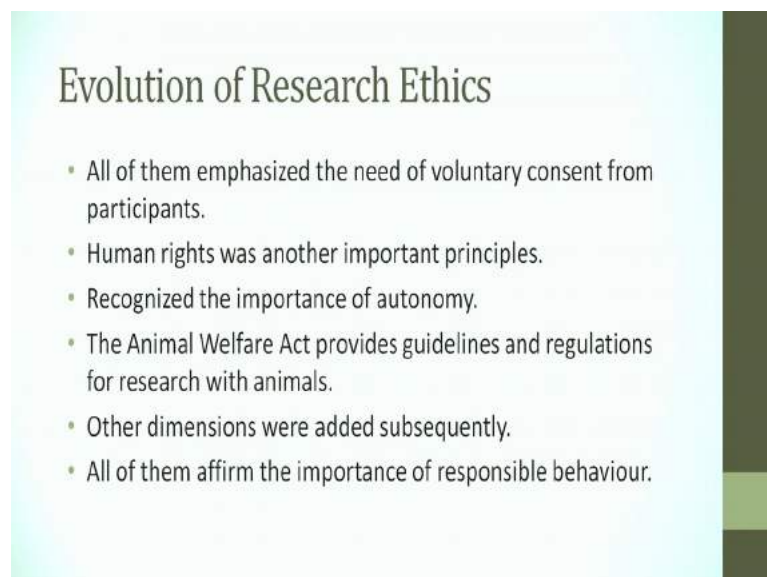


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Then Belmont report in 1979. From the National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research was another hallmark in this process of evolution of research.

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Again, all these things which we have seen right from Nuremberg trail to Belmont, all of them emphasized the need for voluntary consent from participants. So, people have to be respected, their human rights have to be respected, their permission has to be taken, their consent has to be taken, their information have to be kept confidential - all these things



are highlighted. And recognize the importance of autonomy of individual; each individual is recognized as autonomous entity that we have to respect him and her by virtue of the very fact that he or she is an individual human being.

So, the Animal Welfare Act provides guidelines and regulations for research with animals. Other dimensions were added subsequently. There are many other issues, which got, which actually made the domain of research ethics stronger and stronger. All of them affirm the importance of responsible behavior, and different aspects of responsible behavior from the side of researchers involved.

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A presentation slide with a light blue background and a dark green vertical bar on the right. The title "Human Participants as Subjects in Research" is at the top in a dark green serif font. Below it is a bulleted list of four ethical principles, each with a red square bullet point. The text is in a dark green sans-serif font.

Human Participants as Subjects in Research

- **Consent** to participate, withdraw from, or refuse to take part in research projects
- **Confidentiality**: personal information or identifiable data should not be disclosed without participants' consent;
- **Security**: data and samples collected should be kept secure and anonymised where appropriate; and
- **Safety**: participants should not be exposed to unnecessary or disproportionate levels of risk.

And when you try to address some of the specific concerns like human participants as subjects in research, as I mentioned, some important principles are: consent to participate in the study; for that you know all information about the study, whatever is available should be given to the participants; withdraw from the study at any stage of the study and or refuse to take part in research projects. So, these things have to be respected. Confidentiality, I repeat, personal information or identifiable data should not be disclosed without participant's consent. For example, there are several things when a study is conducted this has to be published in a journal. And several information about the subjects need to be published, but certain information about the subject should be withheld. For example, the name, the place from where the address, and all such details which might help others to identify the identity of the person should not be published by



the researchers. This is very important to maintain confidentiality. Security - data and samples collected should be kept secure and anonymized where appropriate. And safety - participants should not be exposed to unnecessary or disproportionate levels of risk.

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A presentation slide with a light blue background and a dark green vertical bar on the right. The title "Issues in Using Animals" is at the top in a dark green serif font. Below it is a bulleted list of seven points in a dark green sans-serif font.

Issues in Using Animals

- Using animals as models for humans to understand disease processes and to develop effective preventative and therapeutic measures such as vaccines or medicines.
- Using animals as models in toxicity testing.
- All these involve harming them.
- Do we have the right to do that?
- Under what circumstances?
- Are we not using them for food?
- Can we use all species of animals? Criteria?

So, but when you come to animals, research where animals are used as subjects, there again, we have to understand that animal life is also precious. Using animals as models - there are different ways researchers use animals in today's context. People use animals as models for humans to understand disease process, particularly in the domain of biomedical research, in pharmaceutical research, animals are used extensively. And to develop effective, preventative, and therapeutic measures such as vaccines or medicines. Using animals as models in toxicity testing; certain toxic substances, if they are exposed to such substances what are the impact and how they can be cured; all these aspects have to be studied, for that animals are being used. All those involve harming animals because eventually they will get harmed in this process. Do we have the right to do that? That is a very important question, because in the case of human beings at least the consent is being taken. We take the permission of the human being, an adult human being, but in the case of animals this is not the case; we do not take the permission of animals to conduct study upon them. Can we do that? Under what circumstances can we do that? Again, that is another thing. We cannot just abruptly do it; only when it becomes necessary we can do that. Again, are we not using them for food? So, one can definitely come up with such an argument; we are anyway eating them, then what is wrong in using



them for experimental purposes? Even if they are killed or harmed what is wrong in it, because we are using them as food, and to use them as food, we are killing them, but is there any substance in such an argument? Is using animal for food or using an animal for research experimentation are there - are they at the same level? Can we compare them with one another? See these are the some of the issues.

Can we use all species of animals? What is the criteria? See, for instance, for example, some of the animal which are very commonly used are mice, then rats are used very commonly by pharmaceutical research; monkeys are being used, because they are very close to the human species, but can we use chimpanzees which are very close to us. If we cannot use chimpanzees, because some chimpanzees exhibit certain skills, which are evidence which suggest that their brain is quite advanced compared to other animals. So, can we use them? So, these are some of the issues which we have tackle when we discuss ethical issues in research in model animals.

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A presentation slide with a light blue background and a dark green vertical bar on the right side. The title "Researchers' Obligations" is at the top in a dark green serif font. Below it is a bulleted list of nine items in a dark green sans-serif font.

- Honesty.
- Integrity.
- Minimal possible risk to participants and to themselves.
- Cultural sensitivity.
- Transparency.
- Readiness to share with participants the benefits of research.
- Social benefits are important than individual benefits.
- Proper risk management: not complete avoidance of risk, but its proper recognition and preparation for them.
- Be risk aware, not risk averse.

But whatever it is, there are certain obligations which researchers have, which we might underline, before we conclude. One is honesty; honesty of researchers to himself, to the work which he does, to the society, to fellow researchers. Then integrity. Minimal possible risk to participants and to themselves - this I am repeatedly underline this. Cultural sensitivity, I already mentioned this. Transparency for what this research work is being conducted, it has to be published. See, for instance, if a researcher is working in a



university or a research organization, it is very important that before the research work begins, the permission of the authorities - the organization authorities - have to be taken; in the sense that the organization authority would have appointed an institute review board or an ethics committee, their permission is mandatory. And there the researcher has to be transparent, they should reveal whatever information they have about the research process, wherever they anticipate a risk or a harm that is involved that has to be reported.

Readiness to share with participants the benefits of research. This I have already indicated in the beginning. This benefits of the research can be of different types; on some occasions, credit of the research work needs to be shared with fellow researchers and collaborators. On certain other occasions, we need not do that, because, again, when you publish an article you can keep your colleagues as co-authors. The whole issue of who is the first author, second author, and third author - that is an issue. At the same time, whether a particular researcher who collaborated with your work, can his name be included as an author - that itself involves an ethical issue. Just because somebody helped you at a couple of occasion does not mean that you should use or we can use that person's name, keep that person's name; that person might get a credit, a publication, but that is not the objective of the research. The objective of the research definitely is not for people to get name and fame; it is the social benefit that matters.

So, you have to recognize contributions of people based on the merit. If there is substantial contribution in the development of the ideas involved in the paper, which is published, you should definitely consider giving that person authorship, but if there is no substantial contribution, but only help here and there, then you can just acknowledge that person. So, these are the things which researchers have to keep in mind when they take decisions about it.

Social benefits are important than individual benefit; this is one very important thing we have to keep in mind. We all have responsibilities towards the society, and social benefits are very important, because, as I mentioned and I repeat now, research is based on public trust.

Again, proper risk management which has already been discussed extensively in this lecture. Be aware of the risk, but not one has to be risk aware and not risk averse.



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And other important issues, if you reflect upon some other important issues in research - issues related to society, where social benefit, development and progress of the society is positively some of the outcomes, because a particular research will definitely have some very positive impacts on society. Say, for instance, many research endeavors may not have a very explicit and direct positive impact upon the society; they might definitely have some impact, but certain other forms research will have a very direct positive impact. So, in such cases, what should be the responsibility of the researchers, of the funding agencies, and several other considerations have to be taken into account.

To other researchers, be responsible mentors, I already mentioned this.

Collaborative research where publication and authorship issues are involved.

And respecting intellectual property rights; that is very important, because as a researcher, you should be aware of the fact that you are creating something which is going to contribute to the improvement of the scholarly community. And this can be done only if you respect intellectual property rights of other researchers.

Do not discriminate, be responsible, and take responsibility; one has to be this. Probably I will conclude with this word that what is one of the most important or rather the most important idea in ethics or in research ethics is - be responsible for what you do and take responsibility for what is happening. These are the two terms or rather one single term -



responsibility, which is very important. I will conclude with these remarks.

Thank you.

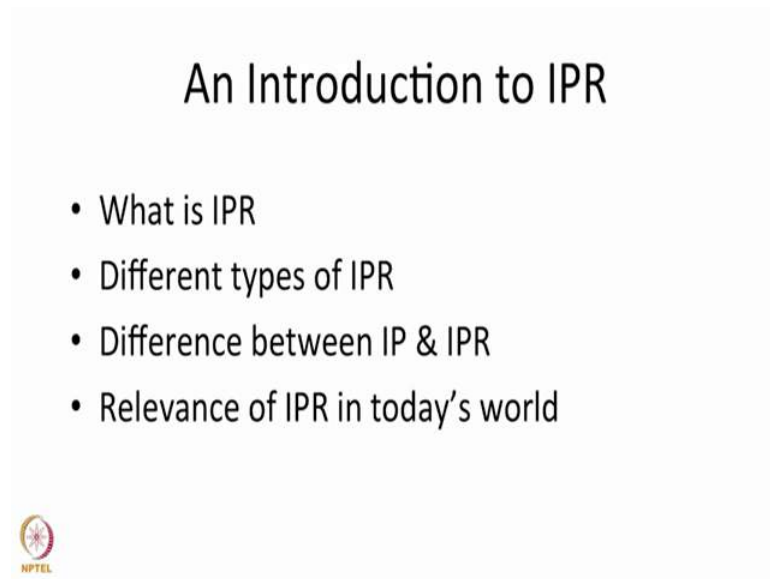


Introduction to Research
Prof. Feroz Ali
Department of Management Studies
Indian Institution of Technology, Madras

Lecture – 01
Intellectual Property Rights

An Introduction to Intellectual Property Rights.

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In this lecture, we will be looking at some of the fundamental principles that govern intellectual property rights. We will be looking at what is intellectual property and looking at the individual things that constitute intellectual property rights. We will also be looking at the difference between intellectual property - that is IP, and how that stands different from IPR, which is intellectual property rights. For this, we will have to first look at what rights are. What do we mean when we say rights? And we will have to also look at the concept of property. Then, we will have to explain what we understand by the term intellectual. Now, when we do these things, we will find that it is possible for us to identify certain fundamental things as characteristic of an intellectual property right. After that exercises is done, we may try to give a definition to intellectual property rights. And I must tell you that there is no universal definition on intellectual property rights as of now.

Then, we will look at the types of intellectual property rights; the list, there is a long list;



it is a growing list, in fact. We would try to bring out the need for intellectual property rights in today's world. And we will conclude with the relevance of intellectual property rights in the modern world.

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What is intellectual property (IP)?

- IP may include original ideas; research results; critical business information
- IP emanates from human creative labour
- IP is different from 'real property'
- Real property is tangible, has defined boundaries



What is an intellectual property? Intellectual property relates to original ideas, results that emanate from research, and generally, it covers some kind of critical business information. Intellectual property refers to that set of property that emanates from human creative labour. Now, when we mention intellectual property, we are specifically trying to define intellectual property as that is distinct from real property; for example, land or a laptop or a pen - these are instances of property that exist in the real world; you can touch and feel them; they are tangible; you can feel them; there are borders to it; there is no perceivable dispute with regards to where the contours of these properties are. A land is defined by its boundaries; a pen is an object that exists in time and space. So, we do not have problems in ascertaining the boundaries of this property; whereas, when we come to intellectual property, we do have certain issues as to understanding the outer limits or the boundaries or the private space of intellectual property.



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What is intellectual property (IP)?

- There are different kinds of IPR, for e.g., inventions are protected by **patents**
- Literary works are protected by **copyright**
- IP is different from IPR



Now, intellectual property specifically refers to things that emanate from human creative labour. Intellectual property manifests itself in various forms; for instance, if you look at, if you look at, invention, an invention is an intellectual property which can be protected by a patent, which is an intellectual property right; a literary work, a book or the work that is written or expressed in words could be an intellectual property, which can be protected by an intellectual property right - that is copyright. So, there is a distinction between intellectual property or the rights of property that manifest in certain creations made by human beings and the right that protects these manifestations.

Now, for us to understand the concept of intellectual property better, we need to understand what these words stand for. Now intellectual property and intellectual property rights, that is IP and IPR, are in most places used interchangeably, and there is nothing wrong with that, because many a times when we talk about intellectual property we also want to cover or encompass the corresponding rights, but for the purpose this lecture, we will try to explain the independent ingredients that constitute intellectual property rights.



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Meaning of 'right' in IPR

- What is a 'right':

"A **legal entitlement** which is **recognized** and can be **protected** and the **violation** of which is deemed as '**unlawful**' and leaves the right-holder with a **remedy**."



Now, let's start with the right part of it. What do we mean when we say we have a right? A right refers to a legal entitlement, something which you are entitled to get legally. It is something that can be justified, that can be a recognized, and that can be protected, the violation of which is deemed as unlawful, and the violation of which leaves the person whose right is violated with a remedy.

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Meaning of 'right' in IPR

- Rights can be used in 2 ways:
 - as a liberty (gives the right-holder freedom to do certain acts)
 - as a licence (right to do something because someone has given you the consent)



Rights can be used in two broad senses: it could be used in the sense of a liberty, your ability or freedom to do something; it could also be used in the sense of a license, you are



right to do something, because somebody has given a consent or somebody has been allowed to do certain things.

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Meaning of 'right' in IPR

- Certain rights manifest are inherent in a human being (Eg: human rights, right to vote, right to privacy), while certain rights manifest themselves outside human beings (Eg: rights in property)
- A right must be recognized by law.
- Rights can be **general** i.e. claimed by every citizen (e.g. right to protection by the State) or **exclusive** i.e. the right-holder can stop others from doing certain acts without his consent (e.g. right to property)



Rights may express itself inherently in a human being like what we say about human rights: your right to vote, your right to privacy - these are things which are inherent in a human being. There are also rights that manifest in things beyond the human being were in, for example, property. So, we have a set of rights that manifest in a human being because of his character of being a human being - those who are what we called inherent rights; and we also have rights that express on things, because human beings can possess own transferred things; in that sense rights are created by the law. In fact, you need a legal recognition for something to be regarded as a right. Rights can be of a general nature which are shared by people like the right of protection a citizen expects from the government, is a general right, legal right which every citizen can claim and every citizen will get; whereas there are certain rights that could be operated or that could be exercised in an exclusive manner. Now, rights offer you... when rights offer you exclusivity, it means that you have a right to stop people from doing certain acts; for instance, if you own a property, then you have a right to exclude people from getting into the property or enjoying that property.

If you own a book, you have the right to exclude people from reading that book or from looking into that book or from using that book in anyway. So, exclusive rights are rights



which confer the ability on a person to stop others from doing things without his consent.

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Meaning of 'right' in IPR

- IPR refers to rights which emanate from IP which are capable of being protected (Eg: copyright, patent).
- Violation of IPR refers to 'infringement'.
- Patent law protects the right to use/make/sell/import the invention.



In that sense, when we talk about rights, when in connection with intellectual property, we are talking about rights that emanate from the intellectual property, which are capable of being protected. For instance, if there is an artistic work, in the form of a book, then the creator of the book or the person who owns the book or who created the book could stop other people from using that book, from making copies of that book, from disseminating information from the book by different ways, because he has an exclusive right over it. In this case, we call that right a copyright. If the exclusive rights vests in an invention, vests in an invention, we would call that right a patent right, and the patent right gives a person the liberty to make or manufacture, to sell, to offer for sale, to import, and to use the right in that invention, and any person, who does these things without the consent of the right owner, we would say that person has violated the right of the right owner. A violation of a right of a right owner is what is called in intellectual property law as infringement. Infringement technically means trespass; trespass is getting into the property of someone else. So, when you would have read, you would have seen these notices in front of some private property- trespassers will be prosecuted; it simply means that if somebody intrudes into the property that is an violation of that person's right. You could have civil remedies to trespass, you could also have criminal remedies to trespass.



When trespass happens on an intellectual property, then we call that by the word infringement. Infringement is nothing but trespass into the intellectual property owned by a person. Now this could relate to a set of rights that come out of the intellectual property. As I said, in the case of an invention, patent law protects the right to make, the right to use, the right to sell, the right to offer for sale, and the right to import the invention. So, if there is an intrusion into any of these rights vis-à-vis the invention, then we would say that there is an infringement of the intellectual property rights. Now, that's a short summary of what rights are.

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Meaning of 'property' in IPR

- Property is a form of **regulation**
- Property can be of 2 kinds: public and private
- Public property is property which is held in common (also known as 'the commons'), whereas, private property is owned by an individual.



Now, let us look at what we mean by the word property. Now, we just had a short understanding of what we mean by right. Property is a form of regulation, and when it comes to intellectual property we are talking about a form of regulation of creations that come out of the mind. Property can be either public property or private property - these are two broad classifications of property.

For instance, public property is something which is held in common; what you called the commons, and private property is something which a person holds for himself individually.



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Meaning of 'property' in IPR

- Real property (land) has **physical boundaries** & can be distinguished from the property of another
- The owner of land & its boundaries can be ascertained from the property deed.



Now, to understand the concept of property, we need to take the analogy of land or landed property. Any landed property exists in time and space. You could go to a particular location and analyze or look at the land, and see its boundaries, and you will get a fair understanding of the limits of that land - where is the land, what are the boundaries of the land, where are the boundaries of the neighbor's land, and where it is, where is the path way that leads to the land - all these things can be ascertained in the physical world. So, property is something that we understand as - especially private property - we understand it as something that can be differentiated from other's property and something which is capable of being distinguished from other's property. And this hall mark of property - that it can be distinguish from other's property - comes from the limits of the property.

In the case of the land, we call them the boundaries of the land. The boundaries can be ascertained physically by going to the location and measuring it or looking at it; if the boundaries are protected and it gives a much clearer view of what is the extent of the land. Land can be measured; it can be ascertained in numbers; we can compute it and we can say what the extent of the land is. Land is also - the title in a land - also manifests itself in a piece of a document what is or what we call the deed or the sale deed by which a land is conveyed from one person to another. In a sale deed, typically, you would find towards the end of the sale deed, a portion of the sale deed - what we called the schedule - the schedule normally has the description of the land, the extent of it, and also the



boundaries of it. As I said, in the case of a land, you can doubly be sure: you can be sure by looking at the document, where the boundaries of the land are; you could also go and look into the property deed or the title deed and see what are the boundaries of the land. So, two ways to check it: one you could look physically and check the boundaries or you could look at the title deed and look for the boundaries of the property.

Intellectual property does not exist in time and space like real property. As I said, because intellectual property deals with creations of the mind like an invention or an idea or an expression of an idea, we need to have some form by which we can ascertain the limits of property. When a person comes up with an invention, the invention could be improvement in an internal combustion engine - that could be an invention; the improvement rests inside the engine; it is not possible for a person who sees the engine from outside to ascertain where the invention is or which part of the improvement actually makes the engine better, unless it is described in detail, in writing, it will be very hard for a person to understand the improvement in the engine, unless he analyzes it or he breaks it down or he reverse engineers it, it will be very hard for that person to understand where the contribution is with regard to that particular invention. So, though there is an improvement that the inventor has made with regard to the internal combustion engine, it will be very difficult for people to ascertain where that improvement is unless the inventor himself tells the world as to what was the contribution that he made.

In patent law, every invention needs to be described in writing. The descriptive part of the invention is contained in a document what we call the patents specification. And the patents specification, much like the land deed, would convey the details about the property and would end with something what we call the claims. The claims will demarcate the boundaries of the intellectual property when it comes to a patent right. So, you could look at a patent specification and the last portion or the concluding portion of a patents specification is what we call a claim or many claims; it could be a multiple claims or it could be a single claim. And these claims when you read will give you an understanding of what are the boundaries of the property, what are the limits of the property. By property we understand as something that can be processed and something that can be transferred. We also understand property as something that has boundaries, which makes it possible for you to distinguish your property from another person's



property.

Take a pen, for instance; the pen has a boundary in time and space, and the fact that pens are possessed by people, we don't have title disputes with regard to ownership of pens. Assume a scenario where you and your friend have a small tussle with an ownership of a pen. How are we going to settle this ownership dispute? Now, if your friend is smart enough, he could just pull up his pen, just open it up and say, you know, I had written my name somewhere discretely. He could show that and he could get away with it. He could say that is one way to prove ownership, you had given, put in a secret mark. The other way to prove ownership is mere possession; you could say that I have been possessing this for a long time, this is my pen. So, it's hard to make a claim. Now, if your friend wants to really make a claim, he would come up with something called a bill showing that this pen is actually mine, because I purchased this for consideration. He could produce a bill. Now, that may put you in a back step, on the back foot; you may wonder whether he has a better title than you.

Now, you could further circumvent that by saying that no, this bill is fabricated or the bill does not correspond to this pen. Now, all these things are disputes with regard to title on the ownership of a pen. Now, eventually, let us assume that you convince your friend on the ownership of your pen. You could have complete possession, and once you clear it off, then you could have complete possession on ownership of the pen.

Now disputes with regard to property are normally settled in this way. You could either show possession for a long period, uninterrupted period of time, or you could bring evidence of title - a bill or a receipt or something of a similar nature - to show that or the fact that you received it as a gift from someone else. You could show evidences by which you can claim title to a particular piece of property. Intellectual property because it is not readily discernible, because the rights don't manifest itself in time and space, on a tangible form. We find it difficult to settle disputes on intellectual property rights.

When it comes to patent, that is why we have a requirement of every invention to be expressed in writing, disclosed in great detail, differentiating itself with every other invention that went before it, which is close to it or proximate to it, and then, explaining the contribution made by the inventor with regard to what has gone before; what has gone before is generally referred to by a term called the prior art, and then,



demonstrating what is the inventive contribution that he made. All this is done in the patent specification, and after the inventor distinguishes his invention from what has gone before, he claims as something to be his own property; that is what it is contained, as I said, in the concluding part of the patent specification, what we call the claims.

The claims are actually for what the protection is granted. The protection in a patent is confined to what is mention in the claims and not what is mentioned in the descriptive part of the patents specification. Now so, the proof of the existence of intellectual property, if it is a patent, if it is an invention, then we would look at the patents specification - the document that encompass it. Now, you may be wondering - so, is it easy to get an intellectual property right? I just need to put everything in writing; that's not the case. If you put everything in writing, there is a scrutiny that is done by the patent office; in India, it is done by the Indian Patent office and only when you pass the tests that the Indian patent office will subject your written description to, will you be granted a patent, which will give you a title and a protection for a limited period of time, provided you keep it alive by paying the required fees.

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Meaning of 'intellectual' in IPR

- 'Intellectual' refers to 'products that come out of our intellect'
- Intellect is our ability to think and understand ideas
- An idea is a product of careful thinking



Now, we just saw what we mean by the term rights and what we mean by the term property. Now, let's take the intellectual part of intellectual property rights. When we mean intellectual, we refer to things that are coming out of our intellect; for instance, products that come out of our intellect. By being intellectual we refer to our ability to



think and understand things; sometimes these things are complicated. So, it is the ability to think and understand ideas, sometimes which some of those ideas could be complicated ideas. So, an idea is something that comes out of a mental effort. We could also say that an idea is a product of a careful thinking.

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Meaning of 'intellectual' in IPR

- Inventions are products of creative human labour.
- Intellectual effort cannot be perceived by others.
- Intellectual effort is special to the inventor and cannot be attributed to machines.



Now, all these things, creativeness that is inherent in human beings, creativity that comes out of an intellectual effort, and idea that comes from constant thinking or careful thinking - these things is what we refer to as intellectual, in the context of intellectual property rights.

Intellectual property rights are descriptive of the character of the things that it protects; for instance, when we say patents protect inventions, the intellectual property rights - that is patents - they protect the intellectual property - that is invention. We refer to the invention as something that comes out of creative human labour. An invention, it could be a serendipitous invention, but never the less we attribute it as a product of the intellect. Now, if we need to understand intellectual property, how it can manifest, we could have the example of a carpenter creating a chair. Now, if you get a... just imagine a carpenter who is creating a chair out of wood. He saws the wood, cuts them into pieces, then he aligns them, he could do the... and he could construct an entire chair without even having any drawing with him, without even putting pen on paper or even without even making others feel that there is some kind of an intellectual effort that is coming out



of him. What a person gets to see when a carpenter is actually making a chair is the fact that he is involved in human labour; what we call physical labour.

The mental labour is not discernible, though it is there, but at the end of the product, when the product gets done, you will be able to see a beautifully crafted chair, where you may not be able to discern the mental effort that went in, but while you were watching him, you were able to ascertain the physical effort that he has put. Intellectual covers that mental element; it would also cover skills that come out of that mental element; it sometimes hard to segregate the physical and the mental element, because when there is a particular move or a skill that comes out of the carpenter's hand, it is a combination of certain physical skills and mental skills. So, it's very hard for us to cut and say the physical part is different from the mental part, but for us to understand this concept better, though we saw the carpenter exercising physical effort and physical labour, we were not able to see the process that went in conceiving the chair and in putting the chair together, which went in his mind.

Similarly, when we see an author writing a book, what we see him do is putting the pen on paper, and you see him laboriously doing that days and days together, till his book is done. This sight that we see could be the same of all the authors, regardless of who it is; it could be an author of a text book, it could be an author of a letter being sent to someone else, it could be an author of a master piece, but the mental effort differs from all these is different and it differs, and the quality that comes out will also differ. So, by intellectual we mean or we are looking at the mental effort or the creative effort which could be a combination of physical and mental skills; we are looking at that part which is special to human beings, if we need to put it that way. So, whenever we talk about the property that comes out of intellectual effort, it is this trait that we are talking about. For instance, we do not say an artist or a tailor uses colorful threads to create an embroidered piece of cloth, the same could be done today by a modern sewing machine; we don't say the work of the sewing machine as something intellectual property, because the intellectual property went in much before, in the creation of the sewing machine which was capable of doing this intricate designs on cloth using colorful thread; whereas, the tailor who did it or the person who actually embroidered a piece of design on a cloth, we would say that, that involved an intellectual effort. So, we distinguish intellectual as something that comes from human beings and not something that comes from machines;



we also distinguish intellectual as something that is special to human beings and not to other beings.

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Common Characteristics of IP

- Protectable by law i.e. enforceable right
- Intangible
- Capable of being described and registered (copyright does not require compulsory registration).



Now, that we have understanding of these three terms - what rights are, what property is, and what we mean by the term intellectual, we should be in a position to move forward and give a definition for intellectual property rights.

Intellectual property rights generally protects applications of idea and information that are of commercial value. Now, this branch - intellectual property rights - generally protects applications of ideas; they don't protect ideas per se - this is something which we need to understand; ideas in themselves, by themselves cannot be protected, but applications and expressions of ideas can be protected. Now, the best instance would be or a one illustration would be the genre in literary works; crime fiction, you would have seen that almost every novel in crime could either be a who done it or how somebody did something. So, this is the genre, there is a plot, there is an event that happens, and there are a set of characters all looking suspicious, and there is a person who would come to solve that thing, and eventually, there could be some twist in the tale; eventually, the crime is solved or at least the person who reads the novel is given an impression that he was a part of a process of solving something.

Now, this genre has been in existence for a long time. People of different calibers have written crime novels and they all follow the same plot - there is an event which has to be



solved, and somebody solves it, there is a suspicion on the most possible characters, and somebody else turns out to be the person who actually did the crime. So, that is the who done it part, the how he did it part is the - crime is there, and then somebody explains or tells the readers how this crime was done. So, these are the... they could be other ways of it, but it is this genre, this group of fiction, is predictable by its nature, because the setting is always a crime, and there is an explanation of why the crime happened or how the crime happened or who did the crime, but look at the number of the books that have come out of these genre and look at the number of authors; some authors like Agatha Christie, she has written most of her books on this genre and still each book is different; it could be the same idea, it could be the same main character who solves these plots; nevertheless, each idea is expressed in a different form, and because it is expressed in a different form, each idea as it is expressed in a different form can have a separate right.

So, copyright subsists in all the books of Agatha Christie and they cannot be an objection that they are all crime novels or they are all works of fiction. So, they are all the who done it model. In some cases, it's a same protagonist to solve these problems. So, that cannot be a reason to say that the idea is the same; as long as the expression is different, you can have different rights subsisting in those ideas. Now, having said that, now we can venture to look at the nature of intellectual property right.

Now, we have analyzed these terms, and these terms have actually thrown up certain traits which we can identify as traits that are common for intellectual property rights. If not common, traits which actually characterize this group of rights. The first thing is that they are protectable by law – that's the first thing; because they could be very valuable information, an idea, which the law does not protect, but when we talk about intellectual property rights, we are talking about a specific right - be it a patent, be it copyright, trade mark, design - there are a list of rights which are granted protection. So, the first thing is that these are enforceable. So, when we say right we mean enforceable; so, that's the first thing. For something to qualify as an intellectual property right it should be enforceable; it should be in the nature of a right, there should be an entitlement - somebody can claim something used on it, and if there is a violation that right, there should be a remedy. So, patents offer protection if there is infringement or violation of a patent there is a remedy; you can stop the person from asking him not to do the thing or you can claim what we call monetary compensation or damages. So, the first thing is they are enforceable.



The second trait is that they are different from normal property. We can use the word intangible; these rights are intangible; you can't touch and feel them, but they manifest in the end product in some way, and because they are intangible, they need to be described by some means. So, the description part has led to various forms of recognizing the descriptive part. In a patent it is described in writing by something called a patent specification, and this patent specification undergoes a process what we call patent prosecution within the patent office, and it emanates as a patent right. This entire process in a very simple language we can call it registration. So, rights they are enforceable and they are capable of being registered. The same is for design; designs can be described and they can be registered. Trademarks; trademarks can be described and they can be registered. Copyrights; copyrights can be described and they can be registered, but because of an international arrangement call the Burn Convention it is not necessary to get a registration of a copyright to enforce it.

Mere publication; you would have seen in many books, there is a copyright notice in the initial pages where the copyright is held by the author or by the publisher. You might find in various websites, in the footer, there is a copyright notice. The fact that you have put a copyright notice, and most likely with an entity's name and the year in which it was published, and the fact that you have published it, is good enough for you to say that you are the owner of the right. So, copyright is the exception where you need not need to register it for enforcing it, but registration, there is a possibility, there is a way in which you can you can register it. So, that is the second part. The first thing about intellectual property rights is that they are enforceable; the second thing, they are capable of being described and concomitantly being registered. So, that is the second part - they can be registered, which brings a whole lot of things. There is an office, there is an authority which analyzes the right and gives and scrutinizes it, and then verifies it with other things, and then grants you a title. So, the process of registration in patent law is called patent prosecution. It happens at the patent office. Only when the written part, that is patent specification, goes through the process of prosecution, does a person get a grant; when we mean by a grant? We mean a title of patent so, enforcement, registration.

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Common Characteristics of IP

- Can be easily replicated
- Requires intellectual effort for its creation.
- Has commercial value
- Economists view IP as 'public goods':
 - non-rivalrous
 - non-excludable



The third thing about an intellectual property right is the fact that you can easily replicate it. Now, I ask this question - how much does it cost Microsoft to come up with the first copy of windows, say windows 7? It will be few million dollars, because you look at their R&D, their investment, their staff salaries, a whole lot of things would have gone into creating the first copy of windows 7. How much does it cost anyone to make another copy of windows 7? Next to nothing. You can just copy it, whatever is there in a CD on to a hard drive or to a pen drive, if it has the capacity to do it. So, the cost of making the second copy is next to nothing.

So, this defines yet another trait of intellectual property right - it is difficult to get the first copy out, but once the first copy is out, it is easily replicable. Now, this is also true for books. Books may take, an author may take years to write the book, it does not take much to scan the book and put the copy of the entire pirated book somewhere in the cloud. Similarly, a movie may take months or years of effort to complete, but we hear constantly about pirated versions coming out within just few days of screening of the movie in the theaters. All these things tell us, that though it is hard to create a intellectual property, it is easy to replicate. Now this is another trait, that it can be reproduced easily.

Yet another trait which is not common, but nevertheless it's a trait, is the fact that it requires effort to create intellectual property. As we said, there is intellectual effort involved in it, but in today's world, in a modern world, sometimes this effort comes from research and development, sometimes. And sometimes, it requires many people coming together to create intellectual property rights, especially when we are talking about



inventions, and today, gone of the days where an inventor could be in his garage and come up with something new. Today, the science and technology has progressed to such a level that most of the inventions - and this is also true about academic writing in science and technology - most of the time they are co-authored; we would very rarely find papers written by a single person. So, this is true for patenting too. Many of the patents are group efforts, done sometimes by teams that are spread across in different countries, they put their efforts together and they come up with an invention. So, they are group efforts. So, the fourth point is the fact that it requires an intellectual effort to create these rights. So, they are creations of the mind, but it requires an effort to create these rights.

Another trait or something inherent in the nature of intellectual property right is that these rights once they are created, they are valuable; there is commercial value or the fact that these rights can be used in trade and in business makes them valuable.

So, they are capable of being enforced; they can be registered; they can be duplicated, reproduced in many numbers; it involves a effort to create them - a creative effort, sometimes by many people put together; and it has commercial value. Now these are five points which we have tried to cull out which define the nature of the intellectual property.

Economics have also used two more terms to understand intellectual property. They say intellectual property by its nature is non-rivalrous and it's non-excludable. Non-rivalrous means the use of intellectual property by one person, does not cause rivalry or does not take away the enjoyment of the same intellectual property by another person. Now, to understand non-rivalrous nature of intellectual property, let us imagine empty room; an empty which is in the size of a normal class room; you just imagine an empty room; and imagine that you own this empty room; you have complete power over this empty room. Now, if there is a entrepreneurial spirit in you, and you want to make some money with this room, there are multiple ways in which you can use this room to make money. You could rent this room out for a couple of people to sleep, you can have some bunker beds, and you can make some money out of it. And if you want to stretch it a bit more further, you can convert the room into a classroom, by which you can make more people sit than to sleep, and you can make some more money or you can change your enterprise into a different one.



Now, if you get more enterprising, and if you want to get in more people into it, you can remove all furniture and make it into a place where people come and generally socialize with each other for whatever reason - they could eat, they could talk to each other, and if you really want to stretch this forward, you could have some kind of a party where people don't mind being at close proximity with each other. Now, these are things which you could do to use or maximize the use of your real property - which is your room. Now, at best and depending on the tolerance level of people, you could cram in close to hundred people into that room; yes, it gets tough, the air gets hot, ventilation becomes a problem, but still if people's tolerance limit is high, you may be able to push in hundred people into a room of a average size.

Now, if you try to put 500 people into that room, they could be police at your door step, because that's simply not possible. So, that tells us real property has it is limits on enjoyment. These limits are not there when it comes to intellectual property rights. Imagine a bestselling book, how many people can read that book at one time, yes, the person who is reading it, another person who sitting next to the person is also reading it, yes; they could be a group of people around that person; say, it's a new book which has been released for the first time, somebody stood in the queue for twelve hours and got the book; assume it is a very, very popular book; you could have a group of friends would do not mind being pushed around looking into the page and reading it. If you are more creative, you could scan the page, and put it on a computer screen or project it on a computer screen, and a whole lot of people can read it. If you can even stretch it further, you could scan the page, put it on cloud, and anybody who has access to that document could see it, and that could run into millions or as many devices you have you could see that. So, this is a trait of intellectual property which distinguishes it from real property. Real property has limits in enjoyment; there are no limits in enjoying an intellectual property right. This is what we mean by non-rivalrous - use by one does not take away the enjoyment by another. So, all the people who crowded around a person who bought a copy of a bestselling book equally enjoy the intellectual property right. They were able to use it without affecting the right or without affecting the enjoyment quotient of the other person to use it at the same time, which would had been different if people were crammed into a room; there are limits to which people can use a room, whereas there are no limits to how an intellectual property can be used. So, this is what is referred as the non-rivalrous nature of intellectual property.



Intellectual property also becomes non-excludable; this is another trait which it has - it is non-excludable. The fact that the intellectual property can be used by some, you cannot stop others from using it. For instance, somebody scans the book and chooses to put it on a live telecast, on youtube. So, what he is doing is, he is just holding a page, and giving sufficient time for people to read the page, then he is moving to the next page and he is just holding it in front of a camera, and he is live streaming this on youtube. So, the book is available for live viewing for any number of people. Now, the fact that he has already put it on youtube, there is no way you can exclude a person who has got access to youtube from enjoying it. So, there is once it is put in a way in which others can see it, there is no way you can stop others from or exclude others. So, this is again a trait which intellectual property enjoys. Once a medicine, say it's a lifesaving medicine, is out in the market, it possible for the competitors of the manufacturer – who ever manufactured this medicine - to take the medicine, analyze it, and even without going back to the manufacturer, to create a copy of it. So, once it is out, there is no way you can exclude people from taking effect of it or in understanding how that particular thing was done.

These two traits - non-rivalrous and non-excludable nature - is something that is shared by what economists call public goods. Public goods, by definition, are non-excludable and non-rivalrous. For instance, national security; you cannot exclude people from national security; everybody gets it and you cannot specifically say that national security is only for few people, because if a country's boundaries are bordered and everybody inside gets the benefits of it.



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Definition of IPR

- Rights which protect application/ expression of ideas & information which are of commercial value
- Rights which protect products of human creative labour (E.g.: patents, copyright)
- Inclusive definition given by WIPO
- Patents protect technological inventions



Now, definition of intellectual property right. There is no agreed definition of intellectual property right. We do not have an agreed definition, in fact, as I mentioned, if you look at a dictionary meaning intellectual property rights can be defined as rights that protect applications of ideas and information that are of commercial value - is one definition. The rights that protect the products of creative labour – it's another definition. You will find multiple definitions around this and some definitions would actually give you a list of things - it is more like a glossary/grocery list - a list of things that say that these are all intellectual property rights.

For instance, the WIPO, which stands for World Intellectual Property Organization, which is an international organization under the United Nations, which deals with intellectual property rights. The WIPO has a definition on; it's an expansive definition or which is more like a glossary/grocery list. It includes many things under the ambit of intellectual property rights. It talks about the things on which the rights are manifested and it also talks about the actual rights.

Now, you will find, in the WIPO's definition, that they talk about patents, they talk about copyrights, they talk about trademarks, designs, and similar rights. The problem with the WIPO's definition is it does not take care of the new and emerging intellectual property rights. There are some intellectual property rights that are emerging as we speak, it does not take that into factor. Secondly, there is no yard stick by which you can understand



intellectual property rights; it's simply not there. So, we could come up with the definition of intellectual property right either as rights which belong to a particular nature, which protects creative and intellectual products that come out of human labor or you could have a list of things on which an intellectual property right may manifest, and so, the WIPO's definition is a definition which is an inclusive list, and they could be new things that come into the list, but the WIPO definition does not offer, apart from referring to it as products of creative labor, it does not give us anything more for us to have a uniform definition for defining intellectual property rights.

Types of intellectual property rights. Intellectual property rights, there are many different varieties of intellectual property rights, and they are distinguished by the nature of the product on which they manifest or the end products that comes out of human creative effort. When we peg the definition of intellectual property right human creative labor, that itself creates some problems, because we have today something called geographical indications where no human creative effort is technically involved; but nevertheless, when we talk about, when we put the emphasis on human creative labor, we are referring to the origin of intellectual property, because intellectual property originated through copyrights and patents, that was the two initial rights that emerged.

So, there are different types and each type refers to a different category of products. Patents are used for protecting inventions and when we talk about inventions, we are referring to technological inventions.



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Definition of IPR

- Copyrights protect creative works (e.g. literary & artistic works)
- Designs protect visually appealing works & not their functional aspect
- Trademarks protect works/symbols which act as source indicator of goods/services
- Geographical indicators (GI) protect source of goods
- Trade secrets protect secret information related to trade.



Copyrights - Copyrights are used to protect literary and creative works, literary artistic works, sound tracks, videos, cinematography, computer programs, and a whole lot of things.

Designs - Designs are used to protect what can be distinguished by the eye. For instance, the law of designs is different from the law of patents, because designs are not used to protect something that is functional; designs are used to protect something that appeals to the eye, something that is visually appealing to the eye, but does not have a function behind it.

Trademarks. Trademarks are used to protect words, symbols that can be attributed to trade. Geographical indications can be used to indicate the origin of certain goods. Trade secrets can be used to protect information relating to trade, which the owner of the information wants to keep as a secret, and there are other intellectual property rights, which we will discuss in some detail as we go by.



Introduction to Research
Prof. Feroz Ali
IPR Chair, Department of Management Studies
Indian Institute of Technology, Madras

Part - 02
Defining IPR

Speaker 1: Defining IPR, we had seen that there are certain elements that together constitute the concept of Intellectual Property Rights.

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Defining IPR

- Subject matter
 - Invention – Protected by Patent
 - Expression – Copyright
 - Aesthetic Design – Registered Design
- Registration (Government)
- Exclusive Set of Rights
- Duration



We saw the fact that the subject matter can be replicated, it can be repeated. The fact that the right krance, the right holder, the option of enforcing it against others. And we also saw the other characteristics or traits of intellectual property right. Now, one of things in understanding or in defining intellectual property right is to first understand the subject matter on which the intellectual property right will manifest itself on. Because, as I said earlier, intellectual property rights are a group of rights, which manifest on different expressions of ideas - that is one definition of it. It also manifests itself on... the new



rights are not actually on ideas; like a geographical indication is not actually on some idea; it's the fact that there are certain products that come of a particular geography, which are valuable and it's an attribution to the origin from that place. That place may have certain special weather or it could have some climatic conditions which makes the product or contributes - the geography contributes - to the product, so you identify the product with a particular place. Now, there is not much idea involved.

Speaker 2: Who will be the beneficiary of this geographical indication?

The beneficiary of a geographical indication are the people, are the community which uses its right; the producers from that place, they are the the people who can use that right. For instance, Darjeeling tea - the people who are having plantations in Darjeeling, and who are actually in the manufacturing and production of the tea, they can use that label to say that this tea is from Darjeeling, because the Darjeeling tea, it has been found out that has certain properties which is not there for tea that is grown in Coorg or in some part of Srilanka; it's not there. The people who are able to manufacture from that particular region can claim a GI – A Geographical Indication.

Now, coming back to our definition. So, we know an intellectual property by the subject matter on which it manifests itself. If an idea manifests itself in the form of an invention, then that is protected by a patent. If the idea manifests itself in the form of an expression - a literary work or an artistic work or a cinematographic work - then that is protected by a copyright. If the idea is an aesthetic design - a design that pleases the eye with no functional component to it, it's just that it pleases the eye – then, we say that subject matter can be protected by a registered design. Now, to understand a registered design, I will give you this example - a shoe that is designed by a company like Nike or Adidas will have certain design elements which also contribute to the functionality. It makes it cut the air faster, it gives it grip, it gives certain support in certain parts of the foot. So, those design elements have also a functional component. So we do not say such design should be the subject matter of a registered design, because they could be an aesthetic part to it, but if there is a functional part right holders will not go for a registered design. Whereas, if a shoe is designed to look like a bunny or a rabbit, you know many children's shoes are designed like a rabbit or like a character in a cartoon, in such cases, we do not say that design has a functional element. Try to understand; the scope of a registered design is for things that are aesthetically pleasing to the eye and the test is -



what the eye can see. So, when functional elements are tied to the design, right holders normally do not go and get a registered design for it. They have multiple rights, first it is a trade mark; and then the thing can also be copyrighted the way it looks, because almost all Disney characters were at some point subject matters of copyright. So you cannot create that image in any other form. And that violates a copyright, and copyright, as you know, is the life of the authors plus 60 years in India; so, it is quite a substantial right.

In defining intellectual property we first look at the subject matter. Then, we look at the form by which it is protected. Now, we said that intellectual property rights are intangible rights, and it is sometimes difficult for us to ascertain the contours of this right, the boundaries of this right.

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Registration

- Form by which IP is protected
- What the right holder has claimed
- Process by which IPR is recognised
- Backed by law
- Done by the Government



Registration, is a way in which we can understand what the right holder has claimed. Registration could be registration of a patent specification by which a patent is granted. Registration could be registration of a literary work; literary works can be registered though it's not mandatory for enforcement purposes. Software code gets registered by



way of a copyright. Trademarks are registered. So, registration is the process by which these right become court and court official; it's recognized; people can verify it; and it is also, because it is backed by a law - the Trademarks Act of 1999, is what gives the trademark holder a right to enforce it. The Patents Act 1970 gives the patent holder a right to enforce a granted patent against others. So, registration is done by the government or by the state.

So registration confers sanctity over the intellectual property right. So subject matter can vary, and depending on the subject matter, you can have different rights; kind, type of registration, the details of registration also varies because if you are filing a design – a registered design - or a registered trademark, you are just filing forms and figures, you know, you are just filing some papers with some marks in it, and the registry does a check, and the registry... if the check is cleared, then you get your right. It's more like matching what has gone before, - are they similar terms, then the registry may raise an objection; if there are no similar terms you invented or you coined a word for the first time, nobody else has done it before, most slightly you will get the mark. It's a straight forward process. Design, again, the design looks unique, it is original, and you are the first person to file that design, it gets a registration.



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Patents Specification

- Has to explain the entire invention in detail including:
 - Working of the invention
 - Features
 - Advantages
 - Variations
 - What went before the invention
 - Illustration
 - Claims



But patents follow a different path. The patents specification will have to explain, in detail, the entire invention. And it is not just the invention, the working of the invention, the features of the inventions, the advantages, the various variations in the invention. And the right holder will also have to explain what went before the invention. So it is a narrative, he explains his invention, and he also before that, he explains the background of the invention - what were the technologies before him and how it was not possible for a person skilled in the art, which is his sphere, to come up with this invention, but he with his inventive effort was able come up with it.

So, it's a narrative which starts with the background art, then it goes towards describing the invention, the purposes of the invention, what are the problems the invention solves, various other things, and it also has embodiments, illustrations - how the invention works or what are the parts of an invention, if the invention has a mechanical invention. And finally, it ends with the claims.



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Scrutiny by Patent Office

- Done in great detail
- First Examination Report
- Right holder is asked to justify his invention.
- Two types of Objections - Technical or Substantial
- E.g. of substantial objection:
 - Not patentable
 - Preceded in the art
 - Obvious for a person skilled in the art to do



So, this entire process of explaining the invention when it is captured in what we called a Patent specification. The Patent Office does the job of not just registering it. It also scrutinizes the details in great depth.

Now, the scrutiny is done by giving a report to the right holder; in India, we call it the first examination report. The first examination report is given where the right holder is asked how he can justify his invention in the light of, if there are any objections, but most likely there are, if there are any objections either it could be technical objections or it could be some substantial objection. If the patent covers a subject matter which is not patentable under the Indian law, there will be an objection. If the patent covers something which is already preceded it in the art, it is preceded, it's already come or the patent covers something which is obvious for everybody to do. In all these cases, the office is going to raise these substantial objections, and these substantial objections, and these substantial objections takes time to get over them.



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Patent Prosecution

- Process of raising an objection over a patent
- Different from other IP prosecution
- There is a lot of analysis is involved



And this process of the office raising objections over a patent application is what you called patent prosecution. And Patent prosecution, is a very detailed and sometimes a complicated process. Patent prosecution actually is different from a trademark prosecution or a design right prosecution in the sense that there is quite a lot of details involved, there is a quite a lot of analysis of what the person says which goes into it before the right is granted. So, registration is one of the key elements by which we can define intellectual property rights.



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Copyright

- Right to print, publish, perform or record the subject matter
- Expression of idea
- Forms of expression be captured in a medium
- Right to make copies
- Copyright regime requires the work to be recorded or copied to some medium



And registration itself varies, just how the subject matter varies when it comes to different types of intellectual property rights, the registration process also varies when it comes to different types of intellectual property rights. Then, these rights offer a set of rights; intellectual property rights be it patents, trademarks, copyright, designs - they offer a set of rights, it is actually a bundle of rights to the right holder. And these bundles of rights also varies. In the case of a literary work, the right may involve the right to print, the right to publish, the right to perform, film or record the subject matter. Because, here is an expression of an idea, and all the forms of expression - most likely which can be captured in a medium - this is critical when you understand copyright, because copyright is the right to make copies.

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Set of exclusive rights

- Making copies implies a medium being there
- E.g. print is recorded on paper, films on tape
- IP rights first came up in the industrial revolution
- IP rights are commercially valuable
- Tied to an idea that can be shared, expressed or have an application out of it
- Industrial revolution was tied to ideas and dissemination of ideas



And when you are talking about the right to make copies, where are you making those copies? There is always a medium right. So, before copyright came - I mean this is just before industrial revolution - the world followed a means of transmitting knowledge what we commonly call the oral formulaic tradition.

Now, if your children are studying nursery rhymes, which you studied, and which probably your parents studied in school, that's because of the oral formulaic tradition. They are the same ones which get repeated over the years, and this used to be... just before I mean the industrial revolution, which is seen as a point in history where these rights come into being. They were, I would say, remote instances of rights even existing before that, some kings granting these rights, but the rights attained international recognition with the advent of the industrial revolution. There are reasons for that, historical reasons, we will get into it soon.

So, before copyright actually came into existence or before copyright became a recognizable right, there was the oral formulaic tradition. The oral-formulaic tradition was a tradition by which people just memorized things and passed it on to the next generation. Now, if you do that, technically, are you violating a copyright? Because now



the medium is your mind. You remember things in your mind, and you pass it on to the next generation, and the next generation or the person who hears from you he or she remembers in her mind, memorizes it, and passes it on.

So the oral formulaic tradition, unless the substance is captured in some medium, can remain outside the purview of the copyright regime. Because the copyright regime requires things to be recorded or expressed on some medium. So, print, the medium is paper. If it is a film, it is recorded on tapes or now it's digital. You will understand that whenever we are talking about making copies, it implies a medium being there. We don't not say that if you heard somebody reciting a poem to you, and if you memorize it, you say that I made a copy of it; you do not use that even in common parlance. We have different terms for that, we would just say that I memorized it or I learnt it by heart. So, this is a critical point that you need to understand and it has some history behind it. In the sense that copyright, when it evolved as a right, if you look behind it or before it, you will find that, Europe especially, had an oral-formulaic tradition. And there are enough number of studies, which some of which is say that one of the biggest beneficiaries of the oral-formulaic tradition was Shakespeare, you know, just before Shakespeare came in he actually inherited from a oral-formulaic tradition. And there are studies which say that some of the things that he wrote were already there in the oral-formulaic tradition.

So, that was a point in history where the right became recognized, and because of the industrial revolution, it spread far and wide. And the industrial revolution also contributed to it, because all the things that we were talking about intellectual property rights are first emanated in the industrial revolution. We said these rights are duplicable; you can make multiple, you can reproduce them; if you have the first copy, you can make multiple copies of it. Industrial revolution actually mechanized manufacturing; it made producing many copies of products easier.

Industrial revolution gave value to things because they were manufactured in large numbers; an entire industry of marketing came with the industrial revolution. Before that people were not marketing the way or the scale in which they were doing. This, again, is similar to what we discussed about intellectual property right. We say that intellectual property rights are valuable rights, they are commercially valuable. And intellectual property right is tied to an idea which can be shared to others or which can be expressed or which can or you can have an application out of it. If you just see the advent of



industrial revolution, there are certain things that contributed to the growth of industrial revolution itself. One of the things includes the advancement in printing technology. Now, this led to the quick spread of ideas, because printing technology advanced, and you could make copies of what was disseminated; big ideas, great works could now be copied and it could be sent to other places. And because of the invention of steel and our ability to cross the seas using ships which could withstand long voyages, we also found that people were meeting each other or crossing borders at a much greater pace than they ever did in history and at a much bigger number. And in Europe, this actually lead to the need to translate works. So, when you found new a German philosopher coming out with his work or a German artist coming up with his work you found quickly people translating them and that ideas spreading in England.

So, when people started moving that also contributed to them understanding each other and the entire field of translation or interpreting other languages came up, which also lead to the quick spread of ideas. So the reason why we take the industrial revolution as the point where intellectual property rights actually took off, is it was tied to idea, and it was tied to dissemination of ideas.



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Kind of IPRs

- Copyright gives the creator the right to make more copies in different forms
- Trademark – using a symbol or word legally registered or established by use
- Patent – right to make, sell, use, offer for sale or import an invention



Now, we were mentioning that the exclusive set of rights that an intellectual property right encompasses would also vary depending on the kind of intellectual property right. And we were saying that is if it is a copyright, then the copyright gives the creator a fixed number of years to print, to publish, to perform, to film or to record, literary, artistic, musical or other cinematographic works. So, the set of exclusive right in copyright pertain to making more copies in different forms. The set of rights - exclusive rights - in trademark pertain to using a symbol or a word that is legally registered or established by use as representing a company or it's products; could be products or even services. In trademark, the right is for you to ascribe a symbol or a set of words - the Nike mark or the Mercedes Benz logo or any of these things - to a manufacturer or to a company which owns it. So, this attribution gives you a set of rights that are different.

Now, you could use this right for all your products; you could use this right for if you enter new industries in which you were not there before; you could stop people from using it, even if you do not have products, because your rights have goodwill. Now, there was a case where someone used the Benz mark - the Benz Tristar logo - for selling



undergarments in India. The court came Benz had no idea, as far as I know, of entering into the garment business, but still they stopped it, because that was the right holder has the right to use the mark in the way he or she wants. It need not be if the business is that in which you are, where you can stop other from using it, you could also preempt people from using it if the mark is well known, if it's a reputed mark. The exclusive set of rights, when it comes patent law on a patent, pertain to the right to make, the right to sell, the right to use, the right to offer for sale and the right to import an invention.

Now, look at that. In patent law, we are talking about rights relating to manufacture, marketing, sale, and in some cases importation. So the nature the subject matter - we are trying to define intellectual property right and we are trying to understand that the subject matter can be different for different rights; the registration process is different for different rights; the exclusive set of rights that patents and trademarks and copyright encompasses - they themselves can be different, because of the subject matter being different.

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Duration

- Can be for a fixed term or renewed
- In trademarks a right that has not been renewed or used it cannot be enforced
- Right as long as right holder wants to keep it alive
 - Pay official fee
 - Needs to take action against people who are using it
- Trademark is an unlimited life IP





And the duration can also be different. Trademarks are for a fixed term, but they can be renewed as long as the right holder pleases. Some of the trademarks that are in operation like the Coca Cola mark are very old - close to a 100 years or more. Some of the trademarks, trademarks can be situations where if the right is not renewed or if the right is not used, then that marks cannot be enforced. But if you look at a business that has survived for 100 years, then you would actually see that the marks have been kept alive, they have paid the charges, official fees for keeping the marks alive, and you will find that it can be. The trademarks, in business parlance, we say these are kind of unlimited life intellectual property. There is no limit to this category of intellectual property, because their life is as long as the right holder wants to keep them alive; he just needs to pay money to the government, as a official fee, and he also needs to take action against people who may use this right. So there are two things the right holder needs to do: 1. pay the official charges, 2. he needs to be vigilant and needs to take action against people who are using the right without his consent. Otherwise, it could be assumed that he has given up his right.

Speaker 2: In case, if you see Coca Cola, it has large number of design rights; right from 1900 to 2000 (Refer Time: 21:57). It has changed its design for a quite a period of times; I mean every 10 years, every 5 years it changes its design. Say for example, in 1910, if there is one design for Coca Cola, then that is not been renewed. The trade mark for the design has not been renewed. Can they be used by some other party?

Speaker 1: Okay See Coca Cola has designs on its bottle. Coca Cola has copyright protection in the way in which its products are displayed; it is an artistic work. Coca Cola has trademark on the word. The trademark without doubt is an unlimited life IP - Intellectual Property - it is unlimited; there is no limit to it. The other rights come and go. If they had assumed that they had registered a design, and there was a limited life for it, it would come and go, but the mark can always and only be used by the right holder. So, it is combination, there are multiple rights, you will see that there will be multiple rights, but the way in which they take it forward is to get an unlimited life IP.

Trade secrets - they are, again, a category of intellectual property right which fall within the unlimited life IP; they are not limited by, if you can keep the trade secret confidential, then you can enforce it as long as they are kept confidential.



Speaker 2: (Refer Time: 23:24) This is also valid internationally right not only to specific countries. Also In Germany, United States

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Two types of IP

- Limited life IP - can go to public domain
- Unlimited life IP - subject to renewal
- Duration of patent 20 years from the date of Application
 - By virtue of TRIPS Agreement
 - A Patent is not technology specific
 - Originally the patent term was 14 years
 - It was an exclusive privilege



Speaker 1: Yes. Yes. Yes Yeah The duration is true for all the countries. So, you could classify the entire lot of IP into two categories from a business perspective. You have the limited life IP, where the duration is limited, after which it falls into the public domain and people can use it. And you have the unlimited life IP, where subject to renewal; it can be kept alive forever. So this is true across the globe.

Now, let us look at duration. Just a few examples - Today, across the globe, duration of a patent is 20 years from the date of application. Now, we have this uniform system, thanks to the TRIPS agreement; the trade agreement on the trade related aspects of intellectual property rights, which is TRIPS, which is an agreement under the World Trade Organization WTO. WTO is an organization and it is also a list of agreements. So, the TRIPS agreement, is the agreement which covers intellectual property rights, and every WTO member is bound to implement that agreement, because it was an



international arrangement. And most countries, who are all members of the WTO, have across the board 20-year term for patent. This was not so before; the duration of a patent was not 20 years. In India, it used to be 14 years, and for pharmaceuticals in India, the term used to be upto 7 years; it was a flexible term; it used to be different; for pharmaceutical, food, and agricultural products the term use to be even lesser.

Now, you may ask why 20 years? I mean, where did we get the 20 years from? I mean there should be some logic behind 20 years. Because we know that 20 years is not only true or not only applicable for all countries, it is applicable it is technology agnostic. Every technology is now protectable by virtue of the TRIPS agreement; every technology, even if the technology has a life span of 3 years, it still gets a 20-year term. So, patents or the patent term as we understand it today, internationally, is not technology specific. Whether it is pharmaceuticals, aeronautics, biotechnology, software - for countries which grant patents on software – it's a 20-year period. And you and I know that a 20-year period does not make sense for all technology. Some technologies are so quick, they are, and you know, 20 years may be many generations, for all you know, it could be many generations.

So, how do we understand this 20 year period. 20-year period came in because of some international lobbying because WTO, before it came into being, there were 8 years of negotiations, what we call the Uruguay rounds, you know, countries participated in it, and it was a long drawn process after which the World Trade Organization was formed. And in that time, there were stake holders putting up their interest and pushing things to it and some how we had this agreement on 20 years. Now, mind you, the predominant time period before this 20 year across the globe used to be 14. And there is a small explanation - historical explanation - as to how it came to be 14 years. In England, it is said that it took 7 years to train an apprentice; that was time of an apprentices under a master or a person with whom he learnt was 7 years. Initially, the British kings when they started granting patents, and they did not actually grant patents, in the initial years they granted exclusive privileges.

Now exclusive privileges were granted by the King or the Queen to enable craftsman, very talented craftsman, to come from Continental Europe and to set up their businesses here. Now, imagine, if these craftsman - some of them made soap, some of them made glassware, some of them on perfumes, playing cards, n number of things. If these



craftsman, were asked to come without the protection of an exclusive privilege, then they come in here and immediately their trade gets copied, isn't? This is the historical part of patent law. These exclusive privileges actually came in a way in which some protection was granted to people with special skills.

Now, at that point, we were not even talking about inventions, we were not even talking about technologies, some people had these exclusive privileges to import playing cards, and it was a royal privilege, the King or the Queen could give a royal privilege for any thing. Sometimes the royal privileges could be given for exploration of minerals - privileges were given. In fact, this country was ruled by the British for a long time, because one company - the East India Company - came with a patent charter, they came with a charter and that was an exclusive privilege that the Queen gave to that company to explore business opportunities here, and they came, and we know rest is history; you know they colonized, they came here and they colonized. But this, the origin of their charter was an exclusive privilege given by the ruler.

So, exclusive privileges were there at all times. In the early days, we find we do not find patent grants, we find exclusive privileges. So, when a technician or a craftsman was given an exclusive privilege to manufacture soaps, for instance, the King or the Queen would ask the craftsman to train two British nationals, because if you train to British nationals eventually they will learn the trade, they will set up shop, and this was a way in which technology transfer or rather let us call it skill transfer happened in those days. So the privilege would be given in return of training two apprentices. So, apprentices normally take 7 years, 7 plus 7 14, that is how we came with the first term. This one explanation given as to why we had a 14-year patent term before the TRIPS agreement; so this is one explanation.

So, patents have a 20 year term which is universally applicable across the board, all the WTO member countries have to honor that commitment, and grant patents for 20 years, regardless of the field of technology. It is 20 years from the date of application. Copyright in India has a term which is computed as life of the author plus 60 years.



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- Copyright – Life of author + 60 years
- If institution – from date of publication + 60 years
- Trademarks – granted for 10 years can be renewed thereafter



If the author writes a book very early in his life, and if he gets to live long, then the books get a longer right. In fact, I think it was in 2009, all the works of Mohandas Karamchand Gandhi, Gandhiji, they came into the public domain, because his life plus 60 years I think it expired in 2009; I can check and tell you the date. So, the duration of a copyright is life of the author plus 60 years. If it's not an author, if the author or the creator is in institution, then the institution from the date of the publication or from the date it is for another term, I think it is 60 years.

Trademarks, as I said, trademarks the duration is it is granted for 10 years; it can be renewed every 5 years.

This makes intellectual property rights; it puts intellectual property rights into two broad categories: one as I said is the limited life IP - copyrights and patents. The other will be the unlimited life IP - trade secrets, trademarks. Now just a quick run through on what these rights are, what they manifest. Patents grant exclusive right; they are conferred by the government.



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Patent

- Exclusive Right
- Conferred by government
- Ability to stop others
- Make, sell, use, offer for sale, import
- Duration – 20 years from date of application



As I said, exclusivity means it gives the ability to stop others, and the right pertains to making, selling, using, offering for sale, and importing. And the right exists for a time period, as I said is 20 years from the date of application, after which it falls into the public domain. So, whatever was covered by the subject matter of a patent will then be free for everybody to use it.



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Copyright

- Exclusive Right
- Conferred by the Government/upon creation
- Print, Publish, Perform, film, record etc.
- Duration: Life of Author + 60 years
- No separate registration needed for enforcement



Copyright, again it's an exclusive right. It is conferred by the government in cases where you seek a registration or upon creation - I have mentioned this before, if you are publishing something, all you need to do is put the copy right notice, which is the C within a circle, your name, and the year on which it was published. You would see a copyright notice on most works. That itself gives you a right and that is done because of an arrangement where you can just publish it and claim to be the creator. So, copyright comes into effect either upon registration by the government or upon creation, creation with the notice.

Speaker 2: I don't even have to register.

Speaker 1: No, no need; for enforcement it's not required. In fact, all the books that are there in our library just the copyright note is the sufficient; there is no need to separately register.

Speaker 2: Notice in the sense C copyright.



Speaker 1: Yes. Copyright almost most websites today have it, if you go to the bottom, privately held websites, company corporate websites they will say copyright from this year to year in the name of the company. So which means every thing that was put on the website is copyrighted and reproduction should be done only technically with permission. But the point with the website is I need not copy or paste anything. I can give a hyper link directly to that website. So, in that sense it is irrelevant; copyright is irrelevant on the internet, because I could give a an hyper link from my website and take to that person's page.

Speaker 2: But still in that page that the other persons logo or whatever.

Speaker 1: Logo is protected by trademark, but the point is the copyright notice is put so that people don't enmasse copy things, and put it, and pass it off as somebody else's website; that's objective of it. But if I need to refer to something, say Mercedes Benz's website or Tata motors website, something, I can just hyperlink and people can read from that page.

Speaker 2: Films like let us say Tamil, suppose they have taken a film or in Hindi let's say Don is the name of the film the taken. Now, what is the duration that they cannot take Don in that same name.

Speaker 1: That does not come under copyright. It's a separate registration, like registering a company's name does not come under the copyright law. The Companies Act has a norm for it. Registering a domain name does not come under copyright law. There is a domain registration service, where you can go like privately godaddy manages it, you can get it, as long as you keep renewing the name, you can have it. So, those things are it's not of subject matter of copyright. There may be some industry arrangement, there may be some other statutes like the Companies Act by which you can register company names.

Now, these rights pertain to multiple things - you can print, as I said, publish, perform, record, and depending on what the subject matter is and the duration of the life is life of the author and plus 60 years. So, from the date of creation till the author dies and 60 years from there on.

Speaker 2: In case of an institution?



Speaker 1: Institution, the date of creation plus 60 years. In some places, in some countries it's 50, in some countries it's more than 60, in America it's more than that, so universally you will see that there is some leeway as to what could be. The copyright term can vary if you create a copyrighted product and have copyrights in multiple jurisdictions. It can vary because the countries have, we don't have something like your TRIPS mandate that the copyright term has to be same across all the countries; we don't have that.

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Trade Mark

- Exclusive Right
- Conferred by Government/Established by use
- Symbol, word, mark etc.
- Marks help to identify the company
- Duration: Unlimited; subject to renewal



For trademarks, trademarks are exclusive rights, they are conferred by the government, possible for you to register them or it is established by use. Now, establishing by use is if there is a mark that is used in Europe like Ikea, Ikea they are slowly coming here, but Ikea has been used in Europe and assume that Ikea did not sell anything in India. If someone, an Indian manufacture, start using Ikea, Ikea could still come and stop that person, because their right to the mark is established by use - international use. And they have a reputation, and there is another branch of law, as I mentioned earlier, a law of passing off can come to Ikea's rescue to stop people from using it, though they may not



have business here and they may not have registered it in India. So it's possible. So, that is why we say the trademarks, when we say trademarks in that sense we also include the right to use, which comes by way of, you know, you can protect your right by way of the law of passing off.

Speaker 2: With law will be applicable if I don't register the trademark in India and still using for a prolonged time it may be like 20 years or 30 years can I use it now.

Speaker 1: Yes. The law of passing off will still come to your help, because you have been using it in trade, though you have not registered it, and you have some kind of reputation that has come out of it. So, we know some cases where people involved in old businesses, they do not register their mark for whatever reason either the business was too small then or whatever. Then, the next generation takes the business and they are able to broaden it and take it to all places. At that point they go in for a registration, because the next generation - the new generation - they are aware of these rights and they go in for a registration. But still, as they enjoy the mark, they would have been able to stop any person who would have intervened even without registration. That is the fact that you have been using something, and people know you by that, and you can stop people from using it, so even without registration. The right expresses itself in the form of a symbols, words, and marks. So the trademark is something, it is a mark by which your products and services are identified by the world. When we see the Tata mark on a packet of common salt we know who created it.

When we see the same mark on an automobile we know who created it. We see the same mark on IT company – TCS - you know the Tata mark, we know that it belongs to a conglomerate. So, marks help us to identify the producer or to know more about the origin of goods and services. In a world where marks are not respected, they will be rampant piracy and counterfeit happening all over the place, because people would now pass off cars as Tata cars or they will pass off products as Tata's products. So there will be issues with regard to reliability of those products, and also the company's reputation can get affected, because if you see businesses are run on reputation, and a reputation is built over years; sometimes it takes many years for somebody to build a reputation. And the marks become some kind of an ambassadors for a business.

So, when they see the mark, there is so much of reputation that is attributed to the mark,



because this mark is how the company identifies itself. When others use the mark, and they are not able to give the quality, even if they give the quality still the mark holder can come and stop others from using it.

Now branding, it also has certain issues. Sometimes, a mark that is well known and well established may go for a re-branding; they may decide to change the way they look and that has whole lot of trademark issues, because the existing mark is still held by the company. Because, they don't want others, when a mark like that is discontinued, it will not be open for others to use that mark, unlike a patent which expires and comes into the public domain, because still this company would want others not to use the mark. They will still keep it alive. For instance, I don't know whether you noticed, Airtel used to be known by a different logo, few years back, Airtel used to be written. Now they have a symbol. Yeah now they have a symbol. Now, they made this transition, obviously, it was an a transition done for business reasons, but just because Airtel made the transition, Airtel will not allow people to use it's old mark. Yeah and Airtel can stop others from using its old mark. This is not the case with the patent, if the patent is expired or revoked for whatever reason, then everybody can use that technology.

Speaker 2: Airtel paid for the renewal of the old mark?

Speaker 1: Yes, if it is keeping it alive, and I would guess that they are keeping it alive, I would guess that because if they don't keep it alive, then there could be a reason for others to use it for what ever. But it's difficult, because Airtel, not just the mark, the word itself is a subject matter of trademark. So anybody who uses Airtel in any font can still be caught by their trademark. But but the symbol you know it underwent a transformation.

Speaker 2: Can the medical therapies or processes or procedures can be protected by intellectual property right?

Speaker 1: There is Medical methods of treatment are exempted from protection under the Indian Act. Section 3 does not allow methods of treatment to be protected. And there are some sound reasons for that, when we come to the issue of patents in detail we can discuss that.



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Design

- Exclusive Right
- Conferred by Government
- Shape, configuration, pattern etc. (aesthetic)
- No functional element
- Not protected by Copyright
- Duration: 15 years



Design. Again, it is an exclusive right; it is conferred by government which means it involves registration. This right pertains to shape, configuration, pattern, etcetera, which are aesthetic in nature, which has pleasing to the eye. Which does not involve functional element, because if there is a functional element, then it goes to the domain of another right - patents. These rights are not rights which can be protected by copyright, because if something can be protected by copyright, then that's a subject matter of copyright; nobody will go for a lesser term, look at the duration of this.

So, if you have a design, and if by some means you can protect it by a copyright, then you can stop people from using that design through your copyright, because it's an artistic work; you can say that there is infringement of your artistic work, then the protection is your life plus 60 years.

Speaker 2: Then why is that there is separate entity for design?

Speaker 1: This is for another purpose industrial designs; we are not talking about artistic works, industrial like interlocking tiles ok or TMT steel bars, there were some novel



designs over them. Engineering again aesthetic. Shape of a bottle for instance okay. You may argue that the shape of the bottle has a functional element, but we are looking at all bottles have function in that sense, but we are looking at some kind of a design which makes a bottle look unique. This is largely mass produced goods which come under this category. And and, again, this is the weakest of intellectual property rights, because there are restrictions on enforcing it, you cannot stop people from doing things, and your damages are also limited; we will get to it.

Speaker 2: Dresses.

Speaker 1: Yeah; mass produced dresses, anything that is aesthetically appealing, anything can be done. Jewellery.

Speaker 2: Somebody says blouse is my design nobody told ... (Refer Time: 44:03)

Speaker 1: Yes, jewelery. Yeah mass produced; anything that is mass produced in a particular thing can come under this.



Introduction to Research
Prof. Kannan
Department of Chemical Engineering
Indian Institute of Technology, Madras

Lecture – 01
Design of Experiments

Hello, welcome to the introductory lectures on Design of Experiments.

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First, we look at the relevant books that are useful for this topic. The first book is by Montgomery and Runger, *Applied Statistics and Probability for Engineers*; Fifth Edition, New Delhi, John Wiley India, 2011. This is a basic book which starts from Statistics, and then goes on to Probability, and also later on into Design of Experiments and Regression. The next book is by Montgomery, which deals with the Design and Analysis of Experiments, the various possible designs, and their method of implementation, and analysis.

The third book is a slightly more advanced one; it requires a background in linear algebra, and with sufficient background in this particular linear algebra you can work through most of the contents in the text book, and discover the fascinating aspects of design of experiments. Then, you also have the book written by Ogunnaike; this is a new book, and some of the complex topics are dealt in a nice manner; there are some interesting examples also in this book. For those of you who want to get started, the first



book is recommended, and once you develop interest in the subject, and you want to really implement the design of experiments concept in your experimental work, then you can start looking at the second book. And those of you who want to give a further theoretical background to your results and explore into more advanced design of experiments options can choose the book by Myers et al.

So, why do we do experiments? Experiments are necessary in order to prove the theory, and also, when sometimes theory is not developed or it's very difficult to develop the theory for a particular process or application, then experiments are necessary. Experiments are necessary also as proof of concept. So we all like to do experiments and we enjoy doing experiments, but sometimes we get frustrated with the results.

But design of experiments help us to overcome these frustrations and present the results in a scientifically acceptable manner. So once you have done the experiments, you have a certain amount of data with you, and you will be studying about representation of the experimental data. You will be looking at Histograms, Box Charts, Scatter Plots, Normal Probability Plots and so on. So, when you do experiments most of the time you do repeats just to convince yourself that the results are pretty much the same or more or less the same during each repeat, and then, you present the data in an average form. You take the average of all the responses from your experiments or output from your experiments. Next step is to quantify the scatter in the experimental results.

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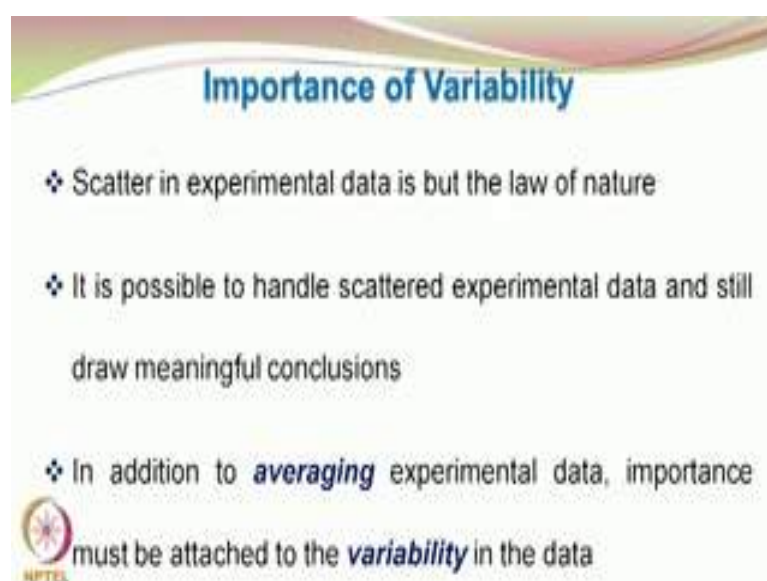




It is very unlikely that when you repeat the experiment you will get the identical response or the identical output, there is going to be some amount of deviation between the responses. So, it is very important for you to quantify the scatter in the experimental data in a suitable way. Once you have done these, it's very important to see which variables in the process are actually influencing your experiment. If a particular variable is not having that much of sensitivity, probably you can fix that variable at a certain level, and then, look at the other variables for their effects.

If a variable is not having an important or a significant effect on the outcome of the process, then it may be kept fixed, and you also reduce the number of experiments you are planning to do further. So once you have done a basic set of experiments called as Screening Experiments, you want to know where exactly you should do the next set of experiments so that you get the maximum yield or minimum power.

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So, depending upon your objective, you would like to explore the experimental design space, and see where the next set of experiments are going to quickly lead you to the optimum set of conditions. As I said earlier, the variability in the experimental data has to be accounted for. The scatter in the experimental data is inevitable, because there are lot of random effects which are influencing your experiment and you cannot control all of them. So you don't have to get frustrated when you are unable to identically get the response when you repeat the experiments, and if there is scatter in your data, sometimes



you may feel – ok, my data is not good and I am doing something wrong in the experimentation, so you may want to stop.

But it's very important to note that you have to live with this scatter experimental data and still draw meaningful conclusions. So in addition to averaging the experimental data importance must be also attached to the variability in the data. The design of experiments you will see how to account for the random variability, and also the systematic variability caused by changing the variables, and compare the differences between the two.

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Variability in Experimental Outcome

$$y_i = \eta_i + \epsilon_i$$

- ❖ Linear combination of error and true but unknown process response is assumed.
- ❖ The random component (ϵ_i) is responsible for the observed variations when repeating the same experiment.

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Let us take a simple case where you have the response for the i th run y_i that is equal to $\eta_i + \epsilon_i$. For a moment imagine that ϵ_i is 0, what is ϵ_i ? ϵ_i is the random error component. If it is 0, then when you repeat the experiment, and assuming that all the components in your experimental setup are working perfectly, you will get the same response. But, because of this random error component you are having different responses when you repeat the experiments. Thus, ϵ_i may be either positive or negative. So, your response may be varying on the positive side and also on the negative side.

What I am trying to say is, suppose in the first case you are getting 100, and the next time you will repeat the experiment you may get a 101, in the third time you may get 97. So, the variability can be on either side that's because ϵ_i may be either positive or negative. η_i is the true value, it is the absolute value, but unfortunately, we do not



know that, and so we need to estimate what the true value may be by doing experiments, and also living with the associated variability.

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A presentation slide titled "Random Variable" in blue text. It contains three bullet points, each preceded by a blue diamond symbol. The text of the bullet points is: "Denoted by X and once its value is known after the experiment, it is termed as x.", "Is the genesis for probability distribution functions", and "Probability distributions are used to predict the behavior of a group of randomly behaving entities". At the bottom left of the slide is the NPTEL logo.

So before we jump into the design of experiments, it is really worthwhile to invest some time in understanding the basic concepts. Only then we will be able to appreciate the design of experiments concepts and understand what the results are actually telling us. Without doing this background material or without knowing this back-ground material you may not be able to understand what is meant by P value or confidence intervals or level of significance, mean square error, and things like that. So, let us spend a bit of time in looking at the basics.

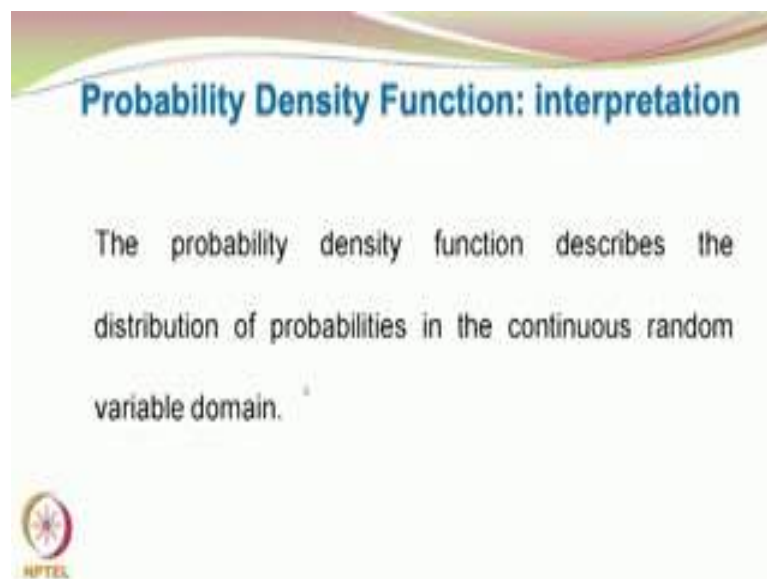
First concept is the random variable. It is a kind of an abstract entity; the random variable is denoted by capital X and once its value is known after the experiment it is labeled as small x. So this is the nomenclature and this is the genesis for the probability distribution functions. You might have come across probability distribution functions earlier, so this is very, very important, and this gives the distribution of the random variable. And Probability distributions are used to predict the behavior of a group of randomly behaving entities. Let us take a class, and if we want to predict the performance in the class of a single individual, then it is very difficult, and if you assume that all the students in the class are independent of one another, each student is going to be difficult to predict beforehand on how his performance is going to be.



But an experienced teacher will know that when you have a collection of students, then their behavior is easier to predict, their collective behavior is easier to predict than the individual behavior; because, most of the collection of individuals form on or fall under one distribution or the other. So, the instructor may be able to say, okay in this class may be about 10 percent of the students will do very well in the exams, may be 5 percent of the class will do pretty badly.

The average level of the performance in the class is likely to be 60 percent, and more often than not you will find that his predictions are valid because he has handled large number of data sets. So, the random variable is the originator for these probability distributions. So we will be looking at continuously varying random variables, there may also be examples of discrete random variables, but in our design of experiments we deal with usually continuously varying quantities.

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So the probability density function describes the distribution of probabilities in the continuous random variable domain.



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Definition

The probability distribution function $f(x)$ for continuous random variables is defined as

$$P(X \leq z) = \int_{-\infty}^z f(x) dx = F(z)$$

$F(z)$ is the cumulative distribution function.

The NPTEL logo is located at the bottom left of the slide.

So what is the definition for the probability distribution function? It is defined as probability of the random variable taking the value less than or equal to small z . Small z is any particular value; it can be 0.5, 1, 3 or even infinity, is equal to minus infinity to z , integral of f of x dx , and that is denoted by f of z , capital F of z . So, what you are doing here is the upper limit you plug in the value of z and then you integrate the probability distribution function. So when you do that, you get a particular value if you know the form of the function f of x ; either you can do this integration analytically or if the form of the probability distribution functions is quite complex, then you may want to do it numerically.

Any way, you will get a value and that value is representing the cumulative probabilities from minus infinity to z and that is given as the cumulative distribution function. What will happen when z goes to plus infinity? By definition the total area under the curve is representing the sum of the probabilities which we know is equal to 1. So, the cumulative distribution function is referring to the sum of the probabilities. If it is plus infinity, it is the overall sum of the probabilities or if it is value less than plus infinity, it will be the sum of the probabilities up to that value z . We are not doing direct addition, but we are doing integration.



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Features of Probability Density Functions

What is the probability of x lying between values ' a ' and ' b '?

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

$$= \int_{-\infty}^b f(x) dx - \int_{-\infty}^a f(x) dx = F(b) - F(a)$$


So, people who are familiar with integral calculus can very easily figure out this particular slide. Probability of a less than or equal to X less than or equal to b ; that means, the probability that the random variable is lying between a and b . And that is given by integral of a to b f of x dx which may be shown to be the cumulative distribution function at b minus cumulative distribution function at a .

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Mean and Variance of Probability Density Function

The mean for the continuous distribution is given by

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x) dx$$

The variance of the continuous probability distribution is

$$\sigma^2 = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$


Now, whenever you have a probability distribution function you can imagine that there is a distribution of the probabilities. So, when you have a distribution you may want to



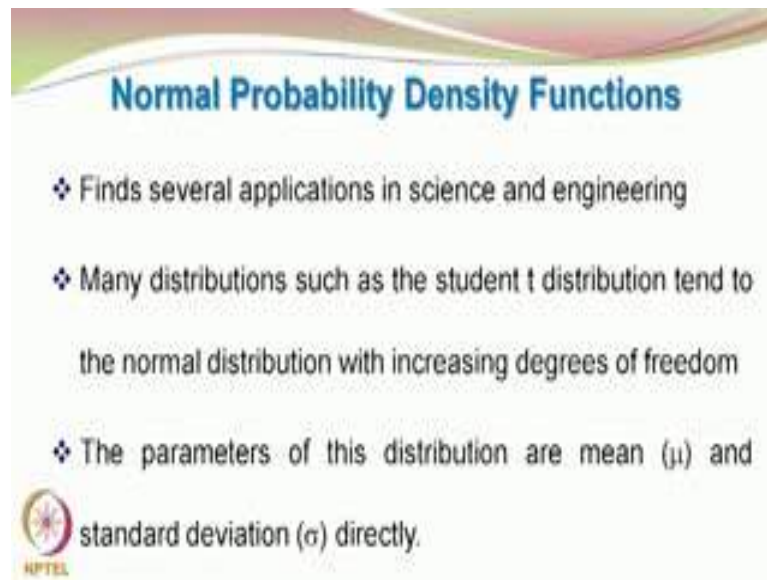
know where the center of the distribution is; you may also want to know the extent of spread of the distribution. So you would like to know the mean and you would like to know the variance. Mean is the representative of the average value of the distributions. Sometimes, depending upon the shape of the distribution, the mean may be at the geometric center of the distribution. Sometimes, in the case of skewed distributions, the mean may not be in the center. What I am trying to say by center is suppose the random variable x is going from minus infinity to plus infinity, in some symmetric distributions the mean may be at 0, but in some other cases it may be at minus 5 or plus 10 and so on depending upon the nature of the curve.

So, before we get too ahead of ourselves let us have some definitions out of the way. What is μ ? μ is the mean of the continuous probability distribution function. The mean for the continuous distribution is given by μ is equal to expected value of X . So, what is the expected value of the random variable X and that is nothing but the μ or the mean of the distribution. That is given by $\int_{-\infty}^{+\infty} x f(x) dx$; this need not be equal to 1 because we are multiplying the probability distribution function by x and so it will not be always equal to 1, only $\int_{-\infty}^{+\infty} f(x) dx$ will be equal to 1, but here it can be any value. It can even be a negative value, because if the distribution is predominantly on the negative side of the x -axis, then you will have a negative mean.

To quantify the spread of the probability distribution function, we have σ^2 representing the variance; so σ^2 is the variance and that is defined as expected value of the X minus μ whole squared okay. So we are finding the deviation of the random variable from the mean, and then we are squaring it, and then we get σ^2 . So, we have $\int_{-\infty}^{+\infty} (x - \mu)^2 f(x) dx$.



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So, now, we will be looking at one of the most important and popular Probability Density Function. This is the normal distribution; you might have come across this normal distribution several times in the past. It finds applications in science and engineering and many of the other distributions also tend to the normal distribution under certain conditions. For example, the T distribution tends to the normal distribution when the degrees of freedom tends to infinity. What is meant by degrees of freedom we will see a bit later. And as I said earlier, whenever we look at a probability distribution function we are interested in knowing its average and also the spread.

In the case of the normal distribution, the parameters of the distribution themselves are mean and standard deviation. The standard deviation is nothing but the square root of the variance. There may be other distributions where the parameters may be a and b , for example, and you have to manipulate these parameters mathematically to get the mean and standard deviation. But the advantage in normal distribution is that the μ and the standard deviation σ are themselves the parameters of the distribution.



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DEFINITION

A random variable X with the following p.d.f. is a normal random variable with parameters μ ($-\infty < \mu < \infty$) and σ ($0 < \sigma < \infty$)

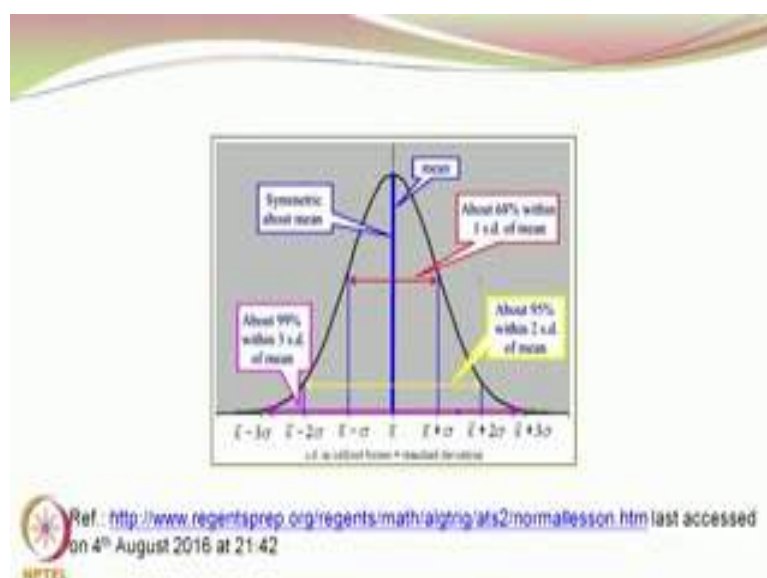
$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \text{ with } -\infty < x < +\infty$$

The notation $N(\mu, \sigma^2)$ is used to represent the distribution.



So what do you mean by parameters? I will show in the next slide. So, this is the form of the probability distribution function and that is given by 1 by $\sqrt{2\pi}\sigma^2$ exponential minus x minus μ whole squared by $2\sigma^2$. So this is a mathematical form for f of x and the parameters in this distribution are μ and σ . So, you can get different types of the distribution by changing the μ and σ . That is why these are called as parameters.

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So how does a normal distribution look like? This is the shape of the normal distribution. As you can see, this is nicely symmetrical and it is centered at $\bar{x}$, $\bar{x}$ is also the mean μ . Why it is called $\bar{x}$? Again, we have to wait a bit; that represents the sample mean, but normally we put μ here. So, you can see that the area of the curve would be 50 percent below the mean and 50 percent above the mean. So, that the total area under the curve is 100 percent, and if you look at it from a probability point of view, it would be 1. So, you are going from minus infinity to plus infinity, but the tail becomes very thin when you go further and further away from the mean value.

So you can say that 68 percent of the area is within one standard deviation from the mean. So suppose this is the mean value, so you have one standard deviation on either side, so $\bar{x} + \sigma$ and $\bar{x} - \sigma$, where $\bar{x}$ is the mean of the distribution. We can call it as μ also, but here it is shown as $\bar{x}$. And so 68 percent of the area is tucked between these two limits, and if you go to $\bar{x} - 2\sigma$, that means a two standard deviations in the mean value are below the mean value, and then, you go two standard deviations ahead of the mean value, then you pack something like about 95 percent of the total area okay. So when you go three standard deviations from the mean on either side, then you are packed something like 99 percent of the area.

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DEFINITION

A random variable X with the following p.d.f. is a normal random variable with parameters μ ($-\infty < \mu < \infty$) and σ ($0 < \sigma < \infty$)

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \text{ with } -\infty < x < +\infty$$

The notation $N(\mu, \sigma^2)$ is used to represent the distribution.





The notation for the normal distribution is capital N mu comma sigma squared, where mu is the mean and sigma squared is the variance. This is used to represent the distribution.

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A presentation slide with a decorative wavy header in shades of green and blue. The title 'Standardizing a Normal Random Variable' is in blue. The text explains that a normal random variable (X) with mean (μ) and standard deviation (σ) can be converted to a standard normal random variable using the transformation formula. The formula is displayed as
$$Z = \frac{X - \mu}{\sigma}$$
. The NPTEL logo is in the bottom left corner.

Standardizing a Normal Random Variable

A normal random variable (X) with mean (μ) and standard deviation (σ) may be converted into a standard normal random variable by the transformation

$$Z = \frac{X - \mu}{\sigma}$$

Now, depending upon the value of mu and sigma squared you can have infinite number of distributions. Sometimes, the mean value may be negative, the distribution may be centered at a negative value, but the variance which is sigma squared is always obviously positive and so is the standard deviation - square root of sigma squared which we call as sigma or the standard deviation is also positive. The mean may be negative, but the standard deviation is always positive. So we want to create a single standard normal distribution irrespective of what the mean mu and the standard deviation sigma are. So suppose you have any arbitrary normal distribution with any arbitrary value of mu and arbitrary value of a sigma, then you can standardize it in a very simple way.

So, when you redefine the random variable x by subtracting it with the mean mu and dividing this difference by the standard deviation sigma, we get a standard normal random variable called as a capital Z, and then, we have x minus mu by sigma is equal to e Z. When you make this transformation, the normal distribution gets magically transformed, and it is centered at a 0, and has a unit standard deviation or the variances equal to 1. And once you transform any normal distribution in to the standard form, and you want to look at the probabilities, all you need to do is to look at a single table



containing the probability distributions of the normal distribution with mean 0 and standard deviation 1.

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Standardizing a Normal Random Variable

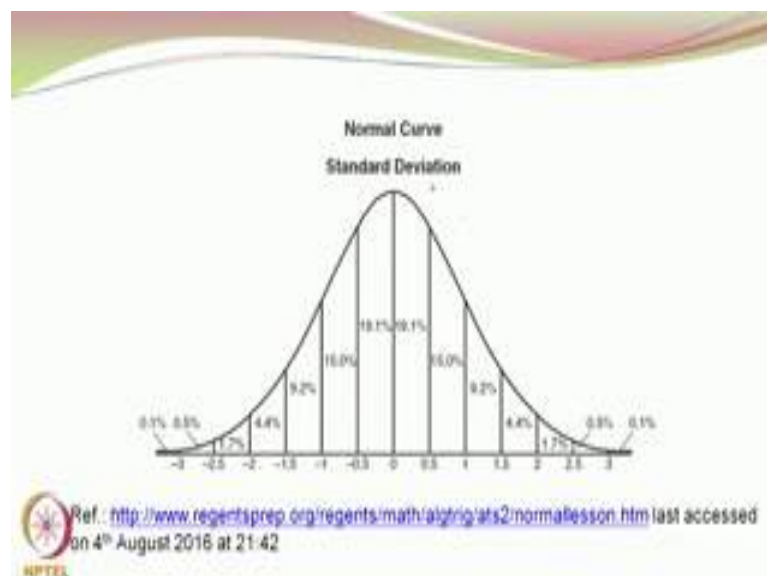
Hence this transformed random variable X has zero mean and variance 1 and hence is a standard normal random variable.

The cumulative distribution of a standard normal random variable is denoted as $\Phi(z) = P(Z \leq z)$



So as I said earlier, once you standardize a normal random variable, it has 0 mean and variance 1; so this is called as a Standard Normal Random Variable. The cumulative distribution of a standard normal random variable is defined as $\Phi(z)$ is equal to the probability of capital Z less than or equal to z ; remember that capital Z represents the random variable z and z is any numerical value.

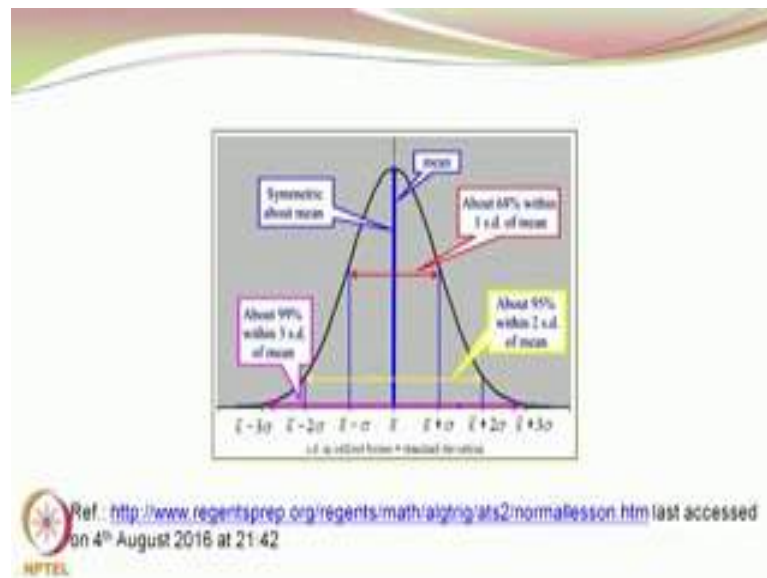
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So, when you look at the standard normal curve, you can see that it is centered at mean 0, this minus 0.5 lies 0.5 times the standard deviation value from the mean. And obviously, the standard deviation value for the standard normal will be equal to 1. So, this represents half a standard deviation below the mean and this 0.5 represents half a standard deviation above the mean; the mean being 0. And you can see that the area under the curve corresponding to random variables lying one standard deviation among either side of the mean comes to 15 plus 19.1 is 34.1; and so, again, you have 34.1 here, that is 68.2 percent. So, 68.2 percent of the area under the curve is covered within one standard deviation on either side of the mean and that's what we saw in the previous slide.

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So about 68 percent within one standard deviation of the mean, and when you say one standard deviation of the mean, we mean on the either side of the mean.



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Why do we need all this? It would be ideal if we are working with the normal distribution because it is nice and symmetric, and you can also show that the mean is equal to median is equal to mode, and then the properties of the normal distribution are well understood, and very elegant to work with. So it would be nice if the data we are dealing with is described by the normal distribution, but it need not be the case; when we are dealing with the large data set, we really do not know how the data is distributed, and what kind of shape it has, and what would be the probability distribution it is following; so, these things are not known to us. So this essentially defines the population.

Why should we deal with population? Because, we want to understand the behavior of large number of entities which come under the collective term of population; for example, you can have the student population or you can have the population of waters or population of people who are preferring the particular soft drink and so on. We need to estimate the characteristic features of the population, so that we have a good idea about the population's preferences or habits or abilities and performances. So, these are required from a decision-making, quality control, and marketing points of view.

So once you understand the population we can set our goals, objectives, process, and settings etcetera. Obviously, if you want to understand the population you cannot go to each and every entity of the population and collect data from that entity, it is practically



not possible. So, we need to have samples taken from the population. And in order to get a liable estimate on the population the sample should be done carefully.

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SAMPLE

Since questioning or testing each entity in the population is not practical, we need to **sample** the population and get to know from the sample attributes **estimates** of the population mean, variance, nature of distribution etc.

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So, once you get the sample, you can estimate the population mean, variance, what kind of distribution it may be having and so on.

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Population and Random Samples

The sample should comprise of independent observations which are coming from the same population.

In other words, they should represent the same probability distribution.

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So how should we sample? Whenever we sample, the observations which are considering the sample should be independent of each other and each entity which we



sample should have equal probabilities of getting picked. Further, they should also represent the same probability distribution. They should all come under same distribution. So now, you have a population and random samples, the sampled element must be independent of each other. Each element in the sample should have equal chances of being.

Let us say that you have a collection of 1000 balls, and you want to know an idea about the distribution of the different colored balls in this box, and if all the blue balls are packed in the top most layer, and you pick up all the blue balls, you may conclude that all the balls in this box are blue in color. You may be taking a sample of 10 balls, and if all of them happened to be blue, then that is the conclusion you will get. That means the red balls and the white balls were not given a chance of being picked, but if you randomly distribute all the balls, and then you start picking from the box, if you take a sample of 10 balls, then you will get a reasonable distribution of the balls as they are in the entire box.

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A presentation slide with a white background and a blue header. The title 'Population and Random Samples' is in blue. Below it are three bullet points, each preceded by a blue diamond symbol. The first bullet point says 'Sampled elements must be independent of each other'. The second bullet point says 'Each element in the sample should have equal chances of being picked'. The third bullet point says 'More number of sample elements, more confident and precise we feel about the responses'. At the bottom left of the slide is the NPTEL logo.

Population and Random Samples

- ❖ Sampled elements must be independent of each other
- ❖ Each element in the sample should have equal chances of being picked
- ❖ More number of sample elements, more confident and precise we feel about the responses

And again, the third criterion also makes a lot of sense. If you have more number of sample elements, more confident and precise we feel about the responses. The more number of balls you pick from the box, better idea you will have about the distribution of the balls in the box.



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A presentation slide with a white background and a decorative wavy header in shades of green and blue. The title "Population and Random Samples" is written in blue. The main text explains that properties of random samples like sample mean and variance can be treated as random variables and are called statistics. A small NPTEL logo is in the bottom left corner.

Population and Random Samples

The properties of random samples (i.e. sample mean, sample variance etc.) may be treated as random variables themselves. These are functions of random variables and termed as **statistics**.

The properties of the random samples we are interested in are - the sample mean, sample variance and they are also treated as random variables. If the entities of a sample are random variables, then any mathematical combination of them will also be a random variable; and these functions of random variables are called as Statistics.



Introduction to Research
Prof. Kannan
Department of Chemical Engineering
Indian Institute of Technology, Madras

Lecture – 03
Design of Experiments

Now that we have taken the sample, we can find the sample mean and sample variance.

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Sample Mean ($\bar{X}$) and Variance (S^2)

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n} \qquad S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$$

- ❖ Here n is the **sample size**.
- ❖ In definition of S^2 , not all $(X_i - \bar{X})$ terms are independent as the constraint $\sum_{i=1}^n (X_i - \bar{X}) = 0$ has to be satisfied.



Sample mean is denoted by $\bar{x}$ and sample variance is denoted by s^2 . So, $\bar{x}$ is defined as $\sum_{i=1}^n x_i$ by n , x_i is the i th random variable. Similarly, a sample variance s^2 is defined as the square of the deviation of each random variable from the sample mean. So, each of the deviations is squared and then summed. We get $\sum_{i=1}^n (x_i - \bar{x})^2$ by $n - 1$. Here, n is the sample size, and it also denotes the number of degrees of freedom. The degrees of freedom is a very interesting concept and refers to the number of independent entities in the collection, you are looking at the collection.

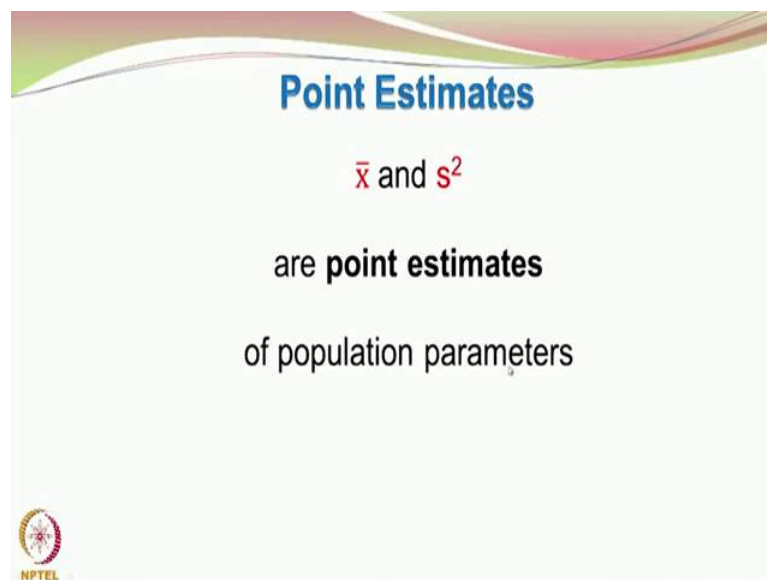
The collection we are looking at is $x_i - \bar{x}$, we have n such terms, we have n random variables, but not all the $x_i - \bar{x}$ terms are independent because we know that the sum of the deviations from the mean is equal to zero. So, when you have $n - 1$ deviation, the n th deviation should be such that the sum is having a value of zero. So,



there are only $n - 1$ independent entities. If you have n independent entities of the deviations, the sum may not be equal to zero and the constraint is violated.

Another thing to remember here is $\bar{x}$ and s^2 are called as the estimators of the population mean and variance respectively. So, these are estimators. So, the formulae for these estimators are defined as shown in this slide. So, now, once you have actually taken a sample, and found out the values, and then calculated the mean and variance based on the sample values, and then we have what are called as sample estimates of the population mean μ and variance σ^2 . So, point estimates are denoted by small $\bar{x}$ and small s^2 and these are point estimates of the population parameters.

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So, the definition is sample mean involving the random variables before their values are known is given here. And, similarly, for the sample variance it is given as shown here. Sum of the square of the deviations divided by $n - 1$, where n is the sample size, and these are measures of the population mean and the population variance. So, we can call them as the estimators of the population parameters. Once the sample has been taken and the values determined, we have the sample estimates and these are also point estimates. Why we call them as point estimates is because these values are single values; for example, you have one sample mean based on the sample you have collected and you have one sample variance. So, you are giving a specific value for $\bar{x}$ and s^2 .



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Sampling Distributions

- ❖ The statistics $\bar{X}$ and S^2 also have a probability distribution associated with each of them
- ❖ They are referred to as the **sampling distributions** of the sample mean and sample variance respectively

The NPTEL logo is located in the bottom left corner of the slide.

Now, we know that the random variable x is having its own probability distribution; that means, it can take a spectrum of values and there is a probability distribution associated with this spectrum of values it can take. When x can take multiple values a mathematical manipulation of x can also take a range of values. If x can have probability distribution $\bar{x}$ and s^2 can also have probability distributions associated with them okay. And they are mathematical transformations of x into $\bar{x}$ and the s^2 , and so, correspondingly you also have a probability distribution associated with the new random variables $\bar{x}$ and s^2 . So, these are referred to as the sampling distributions of the sample mean and sample variance respectively.

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Properties of the Sampling Distribution

We will look at a general case involving n **independent** random variables. However, we shall assume that all of them have come from populations that have the same mean μ and variance σ^2 .



So, now, let us look at the properties of the sampling distribution. We will look at the general case involving n independent random variables and we will assume that all of these have come from populations that have the same mean and variance sigma square. So, now, let us look at the sampling distribution of the mean.

(Refer Slide Time: 05:30)

Sampling Distribution of the Mean

Since $E(X) = \mu$, and $E(X_1) = E(X_2) = \dots = E(X_n) = \mu$

(n times μ for n random variables). This simply becomes

$$E(\bar{X}) = \frac{E(X_1) + E(X_2) + \dots + E(X_n)}{n}$$

$$E(\bar{X}) = \frac{n\mu}{n} = \mu$$



First of all, what is meant by sampling distribution of the mean? From a population, you can take any number of samples okay. And There is no guarantee that the mean you get from the first sample should be identical to the mean you take from the second sample okay. So, in such a situation you have a distribution of the sample means just as you had a distribution of the random variable x . Now, we know by definition, the expected value



of the random variable x is equal to μ , and since all of these have come from population of the same parameter μ and σ^2 , expected value of x_1 will be equal to expected value of x_2 so on to expected value of x_n , and they will all be equal to μ .

And if you look at the expected value of $\bar{x}$, this is also a random variable, it can be shown as given in the slide that E of $\bar{x}$ is also equal to μ . The random variable x is coming from a probability distribution which is having a mean μ . $\bar{x}$ is also coming from a probability distribution which is having the same mean μ as the parent population okay. You are taking samples from a population and that sampling distribution is also having a mean μ . What sampling distribution are we talking to here? The sampling distribution of the means, so the mean of the means is μ ; slightly confusing, but if you think about it, it is pretty simple after all.

Now, let us look at the variance of $\bar{x}$. The variance of the sampling distribution of the means, so the sampling distribution of the means is also having a spread; so when you have a spread, then you have a variance associated with the spread. So, as given in the slide, you have many random samples that may be drawn from a population, and each of them may have a different mean. So, there will be a distribution of the sample means and the variance of this distribution is V of $\bar{x}$.

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Properties of the Sampling Distribution

If the variance of the random variable X is $V(X)$ and is equal to σ^2 ,

$$V(\bar{X}) = \sigma_{\bar{X}}^2 = V\left(\frac{X_1}{n}\right) + V\left(\frac{X_2}{n}\right) + \dots + V\left(\frac{X_n}{n}\right)$$

$$V(\bar{X}) = \sigma_{\bar{X}}^2 = \left(\frac{\sigma^2}{n^2}\right) + \left(\frac{\sigma^2}{n^2}\right) + \dots + \left(\frac{\sigma^2}{n^2}\right) = \left(\frac{n\sigma^2}{n^2}\right) = \frac{\sigma^2}{n}$$




So, as shown in the slide, the variance of $\bar{x}$ is equal to σ^2/n ; the mean of $\bar{x}$ was the same as the population mean μ . On similar lines, you might have expected the variance of $\bar{x}$ to be also equal to σ^2 , but it is not so. As given here, it can be seen that the variance of $\bar{x}$ is scaled down by the sample size from σ^2 the population variance to σ^2/n the sampling distribution variance. How this was obtained? We can see in this particular slide. When you take the variance of x_1 it is equal to σ^2 , but when you have variance of x_1/n , then it becomes σ^2/n^2 . You remember that we defined $\bar{x}$ as $x_1 + x_2 + \dots + x_n$ divided by n . So, when you apply variance of $\bar{x}$, it will be variance of x_1/n plus variance of x_2/n plus so on to variance of x_n/n . So, earlier it was the expected value, now we are applying the variance, and we get $n \sigma^2/n^2$, which is σ^2/n . So this is very interesting.

What it is telling us is the sampling distribution of the mean has a smaller spread than the population distribution of x . So, the larger the value of the sample size the smaller is the variability, which is also making lot of sense. If you collect sample of large size, then several such samples of large sizes may not have much differences between them in terms of the mean - the sample mean - but if you take a very small sample size, and you take a ten such samples, there is a strong chance that all these ten samples have very different means. So, the mean values get spread, but when the sample size is large, then the means are pretty much close to one another and their variability is less. So, variance of $\bar{x}$, the sampling distribution of the means, is reduced if you increase the sample size, and the variance of $\bar{x}$, as I said earlier, is σ^2/n where σ^2 is the population variance and n is the sample size.



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A presentation slide with a decorative header. The title "Properties of the Sampling Distribution" is in blue. The content consists of two bullet points, each preceded by a diamond symbol. The first bullet point discusses the normality of the sampling distribution of means given a normal population. The second bullet point states the assumption of independent random variables. A small NPTEL logo is in the bottom left corner of the slide.

Properties of the Sampling Distribution

- ❖ If the population's distribution is normal with mean (μ) and variance (σ^2), then the sampling distribution of means is also normal with mean (μ) and variance (σ^2/n).
- ❖ It is assumed here that the random variables constituting the sample are independent.

So, to reiterate this, if the population distribution is normal with mean μ and variance σ^2 , then the sampling distribution of means is also normal with mean μ and variance σ^2/n . Here we have introduced another rider to this, if the parent population is normal, then the sampling distribution can also be shown to be normally distributed and the mean and variance are μ and σ^2/n respectively. So, an important assumption is made here that the random variables constituting the sample are independent of each other.

So far we have been talking about point estimators and point estimates, but we can also have interval estimates. I will give a very simple example for this. Let us say that you are going to a remote place where the train runs through the village only once in the day. Obviously, after finishing the work in the village you want to get back to your place as early as possible, and you don't want to miss that train, and be delayed for another day. So, you might ask the people at what time the train arrives to the station and you may get either a point estimate or you may get interval estimates. Point estimate may be the average time of arrival of the train to the station is let us say 2:10 okay; some people may say 2:10 pm, some people might 2:30 pm, some people may say 2:15 pm and so on.

On the other hand, there may be some people who may say that the train is going to come between 2 pm and 2:30 pm. So, a single valued estimate of a train's average time of arrival is a point estimate, whereas the interval specified on the average time of the



train arrival is an interval estimate. So, from the same sample we can also construct an interval estimate.

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Confidence Intervals

Here $z_{\alpha/2}$ is the upper 100 $\alpha/2$ percentage point of the standard normal distribution.

$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

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So, if $\bar{x}$ is a sample mean of a random sample of size n obtained from a normal population with known variance σ^2 , then the hundred into 1 minus α percent confidence interval on μ is given by $\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ okay. Sounds a bit abstract, but it is quite straight forward after all. We will go through it one by one.

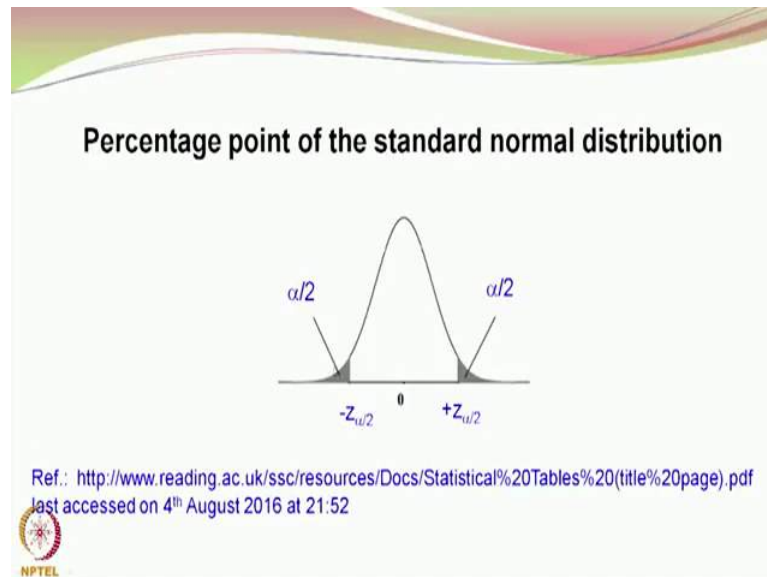
So, here we are assuming that σ is known to us and then we can construct a confidence interval around the parameter μ , the population mean okay. Please note that we are always defining the confidence interval on the population parameter using the sample mean $\bar{x}$. Different samples will give different values of $\bar{x}$, and obviously, the intervals also will be different. And the unknown terms here are α and $z_{\alpha/2}$ other terms are pretty straight forward σ is the standard deviation of the population and as the sample size in $\bar{x}$ is the sample mean.

Okay now let us see what α and $z_{\alpha/2}$ are. We can define $z_{\alpha/2}$ as the upper 100 $\alpha/2$ percentage point of the standard normal distribution. We saw that capital Z is a random variable describing the standard normal variable. We defined capital Z as, if you recollect, Z is equal to $x - \mu$ by σ and that led any normal



distribution to be reduced to a normal distribution with mean zero and variance or standard deviation unity. So, this is the standard normal distribution we are referring to, and then, what exactly is meant by z alpha by 2? Please refer to the following diagram.

(Refer Slide Time: 14:44)



You have the standard normal distribution with mean zero and variance 1. Let us choose points z alpha by 2 and minus z alpha by 2 in this normal distribution. Please note that below the values of zero, the x-axis value are negative and above the value of zero it is positive. And we also note that the normal distribution is symmetric in nature. So, let us define points z alpha by 2 and minus z alpha 2. Due to the symmetric nature of the normal distribution z alpha by 2 and minus z alpha by 2 are equidistant from the origin, and similarly, by the same arguments the area under the curve will also be the same in both the cases. We are talking about the tail end of the normal distribution curve. So, you have this area to be alpha by 2 and this area is also alpha by 2. So, we identify points z and minus z such that the areas beyond those points are identically equal to alpha by 2.

So, this is alpha and this alpha is called as the level of significance. Usually we take alpha to be 0.05; that means, the area under the curve is 0.025 here and this is also 0.025 here. So, we have to look at the standard normal probability distribution table and see what is the value of z such that the area in the tail region beyond the value of z is alpha by 2. And so, if this for example, comes to a certain number minus of that number would be this number here. So, that takes care of this as I said alpha by 2, and this is also plus z



alpha by 2, this is minus z alpha by 2. So, then knowing the value of $\bar{x}$ and assuming that the standard deviation is known of the population, we can construct the confidence interval.

What exactly is meant by the confidence interval? If you take hundred random samples, and the samples have different sample averages okay. So each will have a different value of $\bar{x}$ or most of them will have different values of $\bar{x}$ and $z \alpha / 2 \sigma / \sqrt{n}$ is a constant number. So, when $\bar{x}$ changes from sample to sample, the intervals also will change from sample to sample. So, each interval may have different lower limit and upper limit. If you take 100 such samples, then if you are talking about a 95 percent confidence interval, then we mean that 95 percent of these 100 samples or 95 samples is expected to bracket the population mean μ . So, this is the meaning of the term confidence interval.

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Confidence Intervals

If $\bar{x}$ is the sample mean of a random sample of size n obtained from a normal population with known variance σ^2 then the 100(1- α)% confidence interval (CI) on μ is given by

$$\bar{x} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

So, what we are doing here is when you put alpha as 0.05, which is a common value as I said earlier, if you put the level of significance as 0.05, you get 1 minus 0.05, which is 0.95, you get then 95 percent confidence interval. And again, you can have a broad confident interval, and then, you can also have a narrow confidence interval. It is like a person saying that the train will be coming between 1 and 5. Obviously, this confidence interval is more or less definite to bracket the actual arrival of the train, but this is very



vague. You may be having a tight schedule and we cannot be waiting in the station right from 1 'o' clock.

On other hand, we can also have a very precise confidence interval which says that the train is going to come between 2 and 2:15. That sounds like a very good and precise confidence interval because it is not very big, but on other hand we are not very confident about this confidence interval because the train for all reasons might have come at 1:55 and left okay. So, again we are not very confident about highly precise confidence interval, but very broad confidence interval is less precise, but there is more confidence attached to it. So, these are some of the implications of the confidence interval definition and you will come across confidence intervals quite frequently in design of experiments. So, it's very important to know what really the confidence interval is all about.

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Sampling Distributions and the Central Limit Theorem

- ❖ Even if the original probability distribution of the population is not Gaussian, the sample mean distribution tends towards the normality provided the sample size is high (say > 30).
- ❖ The sample distribution is nearly normal with mean μ and variance σ^2/n

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Now, we were discussing earlier that the normal distribution is preferable, it is unimodal and symmetric, and its properties are well known. So, it's like having good thing to use. If the parent population is normal, and you take samples of any size from the normal distribution, the population distribution is normal, and you take samples of any size. It can be even a small size of 3 or 4, and then the resulting sampling distribution of the means is also going to be normal. On the other hand, if you do not know about the population probability distribution it may be normal or it may not be normal.



Let us assume that it is not normal, then you take samples from this population. We take different samples from this population, and each sample is let us saying of size 35. You are going to get a sampling distribution of the means. This sampling distribution is tending towards normality because you have taken a large sample size. Just because you have taken a large sample size you get an added benefit that the resulting sampling distribution of the means is tending towards normality. This is the a very useful thing to have, and even though we do not know much about the population distribution, we are getting a nice nearly normal distribution for the sampling distribution of the means. This is termed as the Central Limit Theorem.

So, even if the original probability distribution of the population is not normal or Gaussian - Gaussian is another term for the normal distribution - the sample mean distribution tends towards the normality provided the sample size is high, say greater than 30. A sample distribution is nearly normal with mean μ and variance σ^2 squared by n .

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The T- Distribution

- ❖ Used when the **sample size is small** and variance (σ^2) is unknown (as also the other parameter μ)
- ❖ The assumption made is that the population from where the sample is drawn is normal
- ❖ Small deviations from normality assumption is not serious.

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You might have also come across in several papers or books, the t-distribution; we will not be getting into all the details about the t-distribution. This t-distribution is used when the sample size is small and the variance is unknown. So far when discussing about the confidence intervals, for example, we have assumed that the sample, not sample, the population variance σ^2 is known. Now, when you have a small sample size,



and the variance sigma square is not known, which is usually the case, for several reasons you might be forced to take small samples, and also, you may not know the real population variance value. So, now, the assumption made is that the population from where the sample is drawn is normal. So, this is not very serious assumption. So, many populations tend towards normality, and so, making the assumption that the population from which the sample is drawn is normal it is not a very serious one.

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A slide titled "T-Distribution" with a decorative wavy header in shades of green and yellow. The slide contains the formula for the t-statistic, a description of the distribution, its applications, and the NPTEL logo.

T-Distribution

$$T = \frac{\bar{X} - \mu}{S / \sqrt{n}}$$

This describes a t-distribution with $n-1$ degrees of freedom.

Used in hypothesis testing, linear regression and design of experiments

The NPTEL logo, featuring a stylized circular emblem with a flower-like pattern inside, and the text "NPTEL" in red capital letters below it.

Now, we define the t random variable as $\bar{x}$ minus μ by s by root n . This describes a t-distribution with n minus 1 degrees of freedom and this t-distribution is often used in hypothesis testing, linear regression, and design of experiments. So, you can see the degrees of freedom again coming into the picture; s is the sample standard deviation, it is not the population standard deviation. Population standard deviation is given by σ . The sample standard deviation is s . So, instead of using σ , we are actually using s - the sample standard deviation. And again, we saw from the definition of the sample standard deviation, that only n minus 1 entities are independent, and so, we have only n minus 1 degree of freedom.



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Chi-Square Distribution

We may require to report confidence intervals on the **population variance**. It may be done subject to the following assumption

Population is normally distributed



A presentation slide titled "Chi-Square Distribution". The text discusses reporting confidence intervals on the population variance and mentions a key assumption: "Population is normally distributed". The slide includes the NPTEL logo in the bottom left corner.

Now, we also come frequently across the chi-square distribution. Just as we report confidence interval on the mean, we saw it just a few minutes back, we may have to report confidence intervals on the population variance and in this connection the chi-square distribution is very useful, and again, it is based on the assumption that the population is normally distributed.

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Chi-Square Distribution

Let $X_1, X_2, \dots, X_n$ be a random sample from a normal distribution of mean μ and variance σ^2 . S^2 is the variance



A presentation slide titled "Chi-Square Distribution". The text defines a random sample $X_1, X_2, \dots, X_n$ from a normal distribution with mean μ and variance σ^2 , and states that S^2 is the variance. The slide includes the NPTEL logo in the bottom left corner.

So let $x_1, x_2, \dots, x_n$ be a random sample from a normal distribution of mean μ and variance σ^2 ; s^2 is the variance of this sample. So, let us see now



how to define the chi-square distribution. Let $x_1, x_2, \dots, x_n$ be a random sample from a normal distribution of mean μ and variance σ^2 ; s^2 is the variance. So, we define the random variable capital chi-square as $n - 1$ s^2 by σ^2 and we call it as a chi-square distribution with $n - 1$ degrees of freedom.

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Percentage Points of the χ^2 -Distribution

We may find define the probability according to

$$P(X^2 > \chi^2_{\alpha, k}) = \alpha$$

The area of the p.d.f. beyond $\chi^2_{\alpha, k}$ is α .

Formally, $\chi^2_{\alpha, k}$ is an upper 100 $\alpha\%$ point of the χ^2 distribution

with k degrees of freedom.

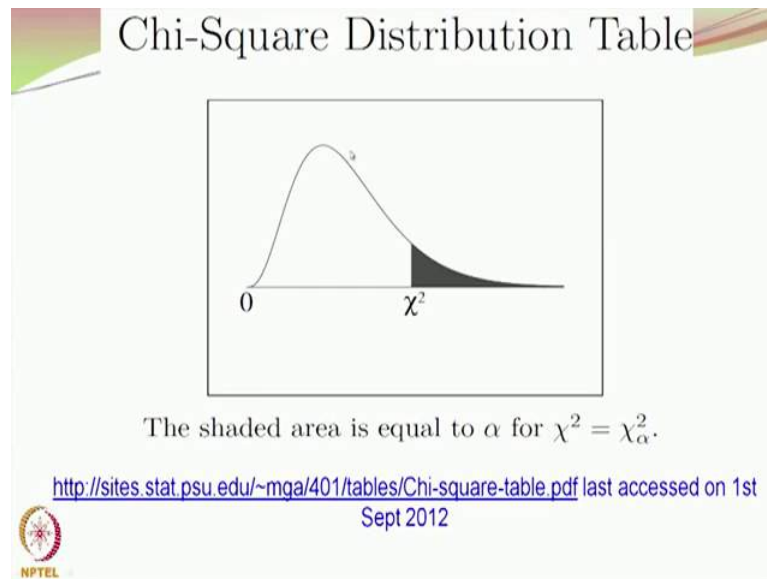
We may find the probability according to the following equation, probability of capital chi-square is greater than chi-squared alpha k that is equal to alpha. In other words, we are finding out the value of the chi-squared random variable such that the area of the curve beyond that particular value is equal to alpha in the chi-squared probability distribution function.

The area of the probability distribution function beyond chi-squared alpha k is given by alpha. We may define formally, the chi-squared alpha k as an upper 100 alpha percent point of the chi square distribution with k degrees of freedom. So, this k degrees of freedom is represented by the subscript k by here. So, this is similar to what we saw earlier in the confidence interval where we defined $z_{\alpha/2}$ with respect to the normal probability curve, and then we said we are identifying two points $z_{\alpha/2}$ and $-z_{\alpha/2}$, such that the area of the normal distribution curve beyond the $z_{\alpha/2}$ and below $-z_{\alpha/2}$ is equal to alpha. Now, in this chi-square distribution called distribution function we are looking at only one end of the curve - the tail end of



the curve. I will demonstrate this in a minute.

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So, let us look at the chi-square distribution. Here it is starting from zero, it is going to have only positive values, you can't have negative values, and then, you are identifying the chi squared value for the appropriate degrees of freedom, such that the area in the curve beyond this chi-squared value is alpha okay - so that is the definition for the probability. Probability of capital chi-square greater than chi-squared alpha k is equal to alpha for the specified k degrees of freedom.

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Scope of Hypothesis Testing

Hypothesis testing concerns with the parameters of the probability distribution of the population **and not with the sample.**

However **it relies on data contained in the sample taken** from the population of interest



Now, let us come to another important application namely the hypothesis testing okay. This is again going to play an important in design of experiments. What do you mean by hypothesis testing? You come up with the supposition or you come up with the claim or a statement and we are going to investigate whether this particular claim or supposition is valid or it has a strong refutation. So, this hypothesis testing concerns with parameters of the probability distribution of the population and not with the sample. You are making hypothesis pertaining to the parameters of the population; you are not making hypothesis with respect to the sample, but with the population; but in order to make suppositions or hypothesis regarding the population we use the information contained in the samples.

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A presentation slide with a white background and a decorative header with green and yellow wavy lines. The title "Formulation of Hypothesis" is in blue. The content includes two bullet points with diamond symbols. The first bullet point lists "a. Null" and "b. Alternate" hypotheses. The second bullet point states that rejecting the null hypothesis implies automatic acceptance of the alternate hypothesis. The NPTEL logo is in the bottom left corner.

Formulation of Hypothesis

- ❖ There are two hypotheses: a. Null and b. Alternate Hypothesis
- ❖ In defining two Hypotheses, we imply that the rejection of the null means automatic acceptance of it's alternate.

There are two hypotheses: one is the null hypothesis and the other is the alternate hypothesis. The nullification of the original hypothesis is the alternate hypothesis. In defining the two hypotheses, we imply that the rejection of the null means automatic acceptance of its alternative. In other words, if you are not accepting the null hypothesis you accept the alternate hypothesis.



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Formulation of Hypothesis

We use the sample data and identify a **test statistic** (which is a function of the sample measurements) using which we try to establish the null hypothesis or its alternate and subsequently make a decision



So, when you formulate the hypothesis we identify a test statistic. The two test statistics we have seen so far are the sample mean definition and the sample variance. So, we identify a test statistic using which we try to establish the null hypothesis or its alternate and then subsequently make a decision.

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Errors in Decision Making

Statistical Decision	True State of Null Hypothesis	
	H_0 is true	H_0 is false
Do not reject H_0	Correct decision	Type II error
Reject H_0	Type I error	Correct decision



Now, let's start of, with an example. You are having particular running plant and a person is newly recruited from a reputed institution. He comes over there and says that this process is not good enough; I have an idea which will improve the process. The



management is a bit skeptical not because it wants to discourage the youngster, but already you are having a well running process and it is making profits. So, why do you want to tinker with the process already existing or running and successfully, and commit money, man power, time, resources, etcetera to try out the new process.

The management is also skeptical that the new process may not be any significant or considerable improvement over an already existing one. So, the null hypothesis in this case would be the processes proposed is actually not producing any improvement. The alternate hypothesis, obviously, would then be the refutation of the null hypothesis - the suggested process is in fact bringing a significant improvement. So, the null hypothesis is the suggested process is not good or not producing any considerable improvement and the alternate hypothesis would be the new process is in fact better than the old process or the existing process. Another, more easier example is, suppose the court is investigating a particular crime and the prosecution is saying a particular person is guilty, the null hypothesis is the person is not guilty, the alternate hypothesis is the person is in fact guilty of committing a crime.

We will continue shortly.

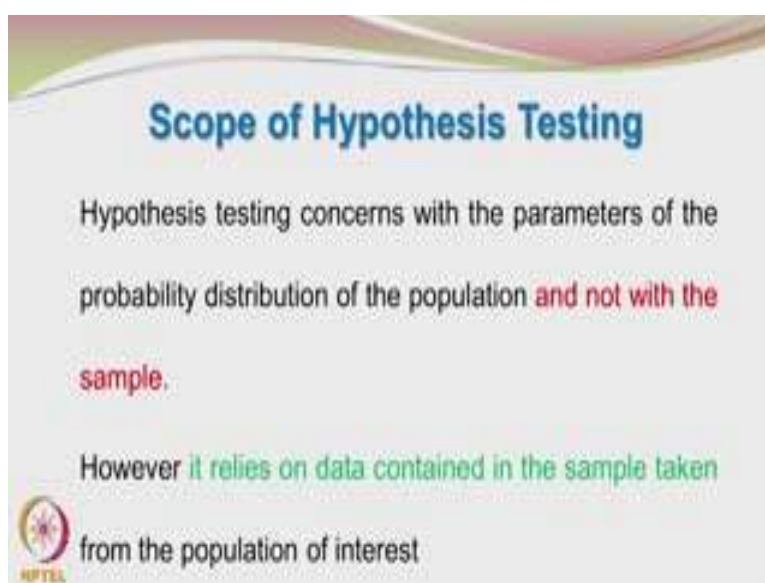


Introduction to Research
Prof. Kannan
Department of Chemical Engineering
Indian Institute of Technology, Madras

Lecture – 03
Design of Experiments

Now, we will be coming to Hypothesis Testing, which is another application involving these statistical concepts.

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This is very interesting and plays an important role, even if a bit understated in design of experiments. Nobody really looks at design of experiments as a hypothesis testing, but this is a very important part of design of experiments. Here what you do is you specify a hypothesis or you postulate an hypothesis, and use the data contained in the sample to see whether your hypothesis is adequately supported. We are using the hypothesis testing to make decisions on the population; we are not making decisions on the sample.



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Formulation of Hypothesis

- ❖ There are two hypotheses: a. Null and b. Alternate Hypothesis
- ❖ In defining two Hypotheses, we imply that the rejection of the null means automatic acceptance of its alternate.

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There are two hypotheses: one is the null hypothesis and the other is the alternate hypothesis. As the name implies the alternate hypothesis is contradicting the null hypothesis. If the null hypothesis says that something is working, the alternate hypothesis will say that is not working. If the null hypothesis says the person is innocent the alternate hypothesis says that the person is not innocent. So, when defining the two hypotheses we imply that the rejection of the null hypothesis means automatic acceptance of its alternate.

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Formulation of Hypothesis

We use the sample data and identify a **test statistic** (which is a function of the sample measurements) using which we try to establish the null hypothesis or its alternate and subsequently make a decision

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Okay so, we use the sample data and identify a test statistic, which is a function of the sample measurements, using which we try to establish the null hypothesis or its alternate and subsequently make a decision okay. So, what we do is, we take the sample, and then extract a sample statistic from it; it may be the variance of the sample or it may be the mean of the sample - these are the two more common ones. And using this estimate, we try to infer about the population parameters. If we are using the sample mean, then we are trying to infer about the population mean; if you are using the sample variance, we are trying to infer about the population variance. So, the decision making is always associated with the errors; nobody can really say that all their decisions have been completely correct. So, we have to see what are the possible errors in decision making.

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Errors in Decision Making		
Statistical Decision	True State of Null Hypothesis	
	H_0 is true	H_0 is false
Do not reject H_0	Correct decision	Type II error
Reject H_0	Type I error	Correct decision

So, let us look at this table. Here the columns are headed as a H_0 is true and H_0 is false. The null hypothesis is true; null hypothesis is false. And the statistical decision may be either do not reject H_0 ; we are a bit careful, we don't accept H_0 , we instead say that do not reject H_0 . And the next one is a bit more unambiguous - reject H_0 . Okay so, if H_0 is true and that the decision made is do not reject H_0 , obviously, you have made a correct decision. But when H_0 is false actually, and you have made a decision not to reject H_0 , a less serious type II error has been made okay. When you reject H_0 , when H_0 is actually true, then you have made a more serious type I error. And again, when H_0 is false and you have rejected H_0 , you have definitely made a correct decision.



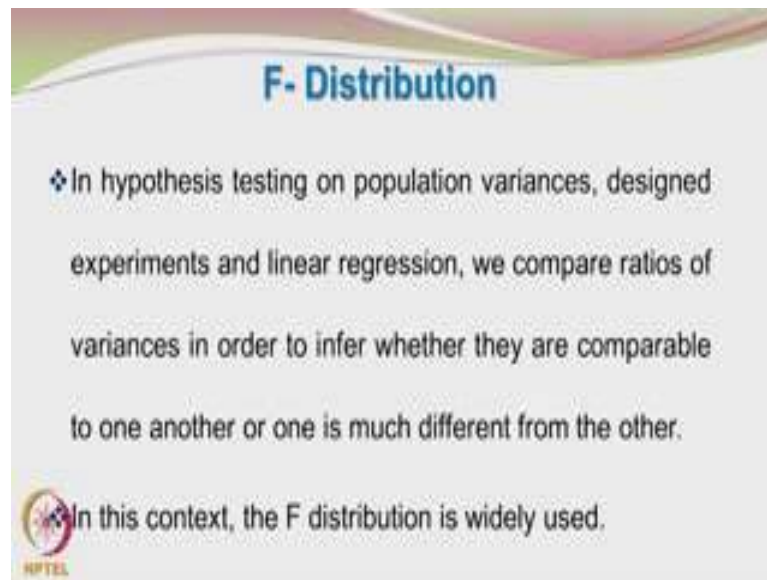
Why is a type I error more severe? I can give a very simple example or may be a couple of examples. If the null hypothesis says that person accused of a crime is innocent, and the court rejects that hypothesis H_0 naught, and instead says the person is actually guilty, and convicts him, then a wrong decision has been made, and an innocent person has been punished. So, a type I error is said to be made. On the other hand, if the person is really guilty, but if the court exonerates him, then the guilty person is getting away scot free, and type II error is supposed to be made. Perhaps you may recall that even though many guilty people may escape punishment not a single innocent person should be wrongly punished.

Another example which I can think of, from an industrial point of view, is a company is having a well established process, and a newcomer to the an industry, perhaps a new recruit or an old hand from another industry, comes and says this process is not really that good, I can improve upon the process. The management is skeptical and says, look we have been using this process for the last 20 years so on, and it has been working fine without any problem, and we are making profits. So, when something is working why tinker with it. So, the null hypothesis would be the proposed process is not a really great improvement on the existing process and the alternate hypothesis would be definitely the proposed process is a improvement over the existing process.

So, if you make a type II error, then even though the original or the existing process is not that good, you are not losing that much; anyway, you are back to where you are previously, and so you are still making profits, and the plant is running. So, a less serious error has been made. On the other hand, if you say that the existing process is bad, it's not good and the new process is better, and that decision is wrong, that means, you are throwing away an actually good process, and then investing lot of time, money, resources, and man power in bringing up a new process, which does not really give any great benefit over the existing process. So, then we say that a type I error has been made. So, this kind of decision making you will employee in design of experiments, for example, you may have several variables influencing your process, upfront you start saying that none of the factors are really significant in the experiment. And then, based on the information provided by the experimental data, you do this hypothesis testing, and then you can conclude if some of the parameters alone are significant and the rest of them are not significant.



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Now, we come to another important distribution called as the F distribution. The chi square distribution is not really immediately used in design of experiments, but the F distribution finds direct application, and the F distribution is similar to the chi square distribution, and it is based on that. So, when you do hypothesis testing on population variances designed experiments and linear regression, we compare ratios of variances in order to infer whether they are comparable to one another or one is much different from the other. So, when you want look at the variability of the data, the variability can be caused by variation in the factor levels or the variability may be simply caused by random effects on which you have no control of. So, when you claim that your experimental factors which you are controlling are impacting the response of the experiment significantly, then you have to prove that this variability caused by these so-called important factors is much higher than the variability caused by noise or random error.

If the variability shown by changing the factors is comparable to the variability due to noise or random errors, then you cannot say that this factor is really making an impact. When you change the factor may be the random effects are causing the experimental response to deviate considerably. So, you have to compare the variability due to the change in factor with the variability because of the prevailing a random error sources, and for this comparison of variability, the F distribution is very useful. So, let us a look at the basics of the F distribution.



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F-Distribution

The assumptions made when comparing the variances are

- a. The two populations from which the variances were taken for comparison are both normally distributed
- b. Both the population means (μ_1, μ_2) and standard deviations (σ_1, σ_2) are unknown.



So, again, we start off with the set of assumptions: the two populations from which the variances were taken for comparison are both normally distributed. We are comparing two variances and we assume that both of them are coming from populations, which are normally distributed. And we also do not know the population parameters namely the means and variances.

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Definition

The random variable F is defined as the ratio of two independent chi-square random variables (CD_1 and CD_2) each scaled with its own degree of freedom.

$$F = \frac{CD_1/m_1}{CD_2/m_2}$$


So, how do you define the F random variable? The F random variable is defined in terms of the chi square distributions. It is simply defined as the ratio of the independent chi



square random variables. Let's call them $C D 1$ and $C D 2$. Each of which is scaled by its own degree of freedom okay. So, we take the first chi square distribution, and then we divide it by $m - 1$; $m - 1$ representing the degrees of freedom for the first chi square distribution. Then, we take the second chi square distribution and then divide it by $m - 2$, and $m - 2$ representing the degrees of freedom for the second chi square distribution.

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Ratio of Variances

The sample variances from these two populations are S_1^2 and S_2^2 . The two sample sizes are m and n .

The associated degrees of freedom are $m-1$ and $n-1$.

Then

$$F = \frac{(m-1)S_1^2 / [\sigma_1^2(m-1)]}{(n-1)S_2^2 / [\sigma_2^2(n-1)]} = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2}$$

So, when you expand the chi square distribution, you see that $C D 1$ is given by $m - 1$ S_1^2 squared by σ_1^2 squared that is the definition for the chi square distribution one and then when you divide that by $m - 1$, the $m - 1$, the numerator and the $m - 1$ the denominator gets canceled, and then you get S_1^2 squared by σ_1^2 squared. Similarly, when you do the same thing for the second chi square distribution, you will get S_2^2 squared by σ_2^2 squared. So the F distribution in fact becomes S_1^2 squared by σ_1^2 squared divided by S_2^2 squared σ_2^2 squared. S_1^2 square and S_2^2 squares are the sample variances of the first and second samples respectively, and σ_1^2 and σ_2^2 are the standard deviations of the first and second populations respectively.



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Percentage Point

The percentage point of the F distribution is defined such that

$$P(F > f_{\alpha, m_1, m_2}) = \int_{f_{\alpha, m_1, m_2}}^{\infty} g(x) dx = \alpha$$

The percentage points in the lower tail may be found from

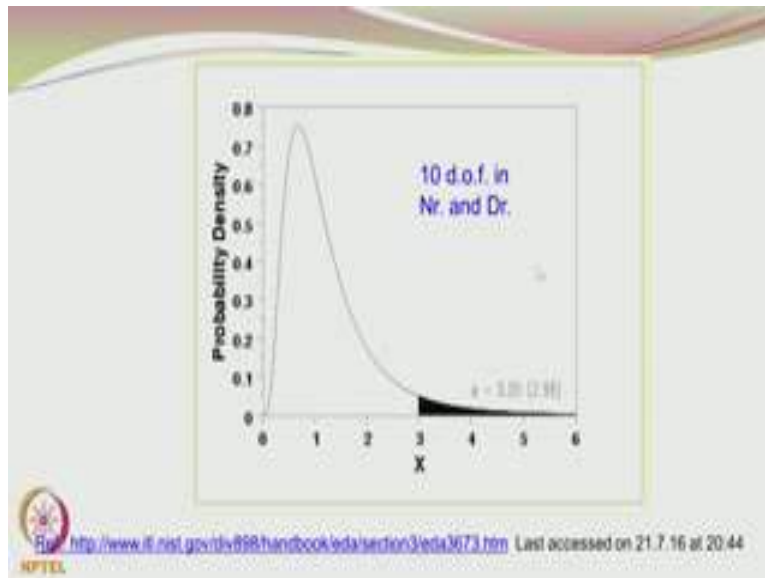
that of the upper tails as $f_{1-\alpha, m_1, m_2} = \frac{1}{f_{\alpha, m_2, m_1}}$

So, we get back to our percentage point. We saw the percentage point cropping up in a normal distribution and in chi square distribution; it's not a surprise that it is now coming in the F distribution. The F distribution is slightly different from the other two distributions we saw previously, because it has two degrees of freedom. One called as the numerator degrees of freedom, so m minus 1 is the numerator degrees of freedom, and n minus one is the denominator degrees of freedom. It is convenient to represent the numerator degrees of freedom first followed by the denominator degrees of freedom. So, then, we have probability of F greater than α m_1 m_2 , and f is the random variable, f random variable and f_{α, m_1, m_2} is a particular value. So, we have to identify f_{α, m_1, m_2} such that we get a probability of α .

So, going back to the mathematics g of x represents the mathematical form of the F distribution, we will not get into it right now. And so, we have the mathematical form here; we integrate it between the limits f_{α, m_1, m_2} to infinity, and we have to identify what is the value of f_{α, m_1, m_2} such that the area under the curve is equal to α - the required probability. So, there is another interesting mathematical manipulation here which I will leave you to go through.



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Now, let us look at the F distribution. For this distribution, we have to specify the numerator and denominator degrees of freedom; in this particular case, it is actually 10. Alpha we set it 2.05; that is usually very popular; for example, you know the 95 percent confidence intervals are based upon a value of alpha of 0.05. So, the f_{α, m_1, m_2} is identified here to be 2.98. So, when you look at the area under the curve beyond the F the value of 2.98, you will find area under the curve to be 0.05.

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Experiments involving ONE factor

- ❖ Hence, **compare** variance due to change in treatments (called as mean square treatments), with variance due to repeats (mean square error).
- ❖ In other words, compare variation between treatments to variation within treatments.



So, now let us get into simple experiments. These are very scarce, but this is a very good basis for illustrating some key concepts. So, what we do here is we are looking at experiments involving only one factor. So, what we look at here the comparison of variance due to change in treatments with the variance due to repeats. So, when you do repeats, you find variability in the data. If all the factors are kept at the same values, then the variability can be due to only random factors. We assume that all the equipment, instruments are all working fine, there are no systematic errors. So, the error is only from the random components and to identify the error from the random components, you have to do the repeats of the experiments. So, the more number of repeats you do, the more or better idea you have about the experimental error.

You may be asking - what is meant by this treatment whereas this treatment is it a medical treatment or what kind of treatment it is? It is just a classical term, which has survived over the years. So, we just call it as the levels of the factors. So, what we have to do is remember that we are considering only one factor and they may be having different levels or different treatments. For example, it can be fertilizer A can be one treatment to the land, and fertilizer B can be the second treatment to the land, and so on. So, essentially, as I said earlier we are comparing the variation between treatments to variation within the treatments. Good !

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Analysis of Variance Table (ANOVA)

Source of variation	Sum of Squares	Degrees of Freedom	Mean Square	F ₀
Treatments	$SS_{\text{Treatments}}$	$a-1$	$MS_{\text{Treatments}}$	$\frac{MS_{\text{Treatments}}}{MS_{\text{Error}}}$
Error	SS_{Error}	$a(n-1)$	MS_{Error}	
Total	SS_T	$an-1$		

a = number of treatments
 n = number of repeats per treatment



Now, we will be coming to a very popular and very important table called as the ANOVA table. ANOVA stands for analysis of variance. So, you can see that finally, everything is boiling down to the analysis of the variability. So, this table has several columns and the first column is the source of variation. So, it can be treatments, and then, it can be the error. As I said earlier, variability from treatments and the variability from error, and this column contains a sum of squares of treatments and this is sum of squares of error; I am not getting into all the details. So, the variability due to treatments is represented in terms of the sum of squares of the treatments.

Normally, we don't take the absolute variability or the actual variability because some variability with the reference to the defined average value can be positive and some variability can be negative. So, instead of taking the variability as it is, we take the square of them. Similarly, we do the sum of squares of the error. So, I am not getting into how these sums of squares were actually evaluated, but I am just indicating that these are measures of variability in a squared form. A is the number of levels for the particular factor or the so-called number of treatments.

So, what you have to remember is the degrees of freedom you have to keep in mind, so that rather than comparing the sum of squares of treatment with the sum of squares of error, because the sample size for the treatments and the sample size for the error may be considerably different. So, to put them on a same basis, you scale each of the sum of squares by the appropriate degrees of freedom. So, you divide by $a - 1$, where a is number of treatments, and for error - to get the scaled error squared, we divide sum of square of error by a into n minus 1; a into n minus 1 is the degrees of freedom for the error component. So, when you divide the sum of squares by the respective degrees of freedom we get the mean square treatment and the mean square error.

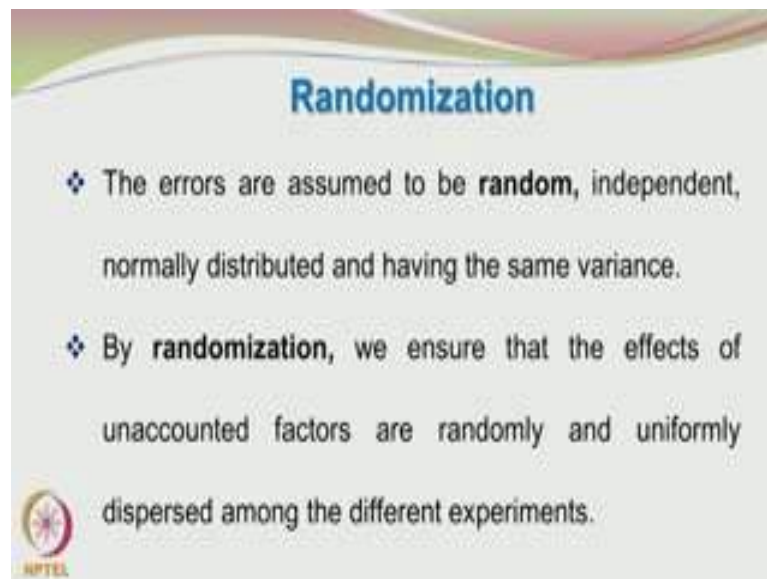
And what we are doing here is comparing the two mean squares, mean square treatments by mean square error. Let us see if I can correct this small typo here. There should be capital S , okay that takes care of it and so here we go.

If you look at the mean square treatment it is similar to the variance. We know the variance is the sum of square of the deviations. Here also when you calculate the sum of squares of the treatments, you define a suitable average, and then, you find the deviation of the response with the respected average, and then square them, and each deviation is



squared, and then all the deviations are added to get the sum of squares of the treatments. And when you divide with the degrees of freedom we get the mean square treatment. So, this is a kind of variance okay. And then, similarly, you also have the error term, where you have the mean square error, where the sums of squares of errors are divided by the degrees of freedom. So, you compare mean square treatments by mean square error to get the F value and based on the probability values associated with this F value, you can make the adequate conclusions.

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There is another important thing you have to remember in mind or keep in mind when you do experiments is the randomization of your experimental runs. Normal tendency, in any experimenter - a new experimenter, let us put it like that is to systematically vary the factor settings. He may or she may go from a low factor setting to a high factor setting for one variable, and then take the next variable, and then again do in a very systematic manner that shows the organized mind of the experimenter, but in reality, it is better to do the experiments in a randomized fashion.

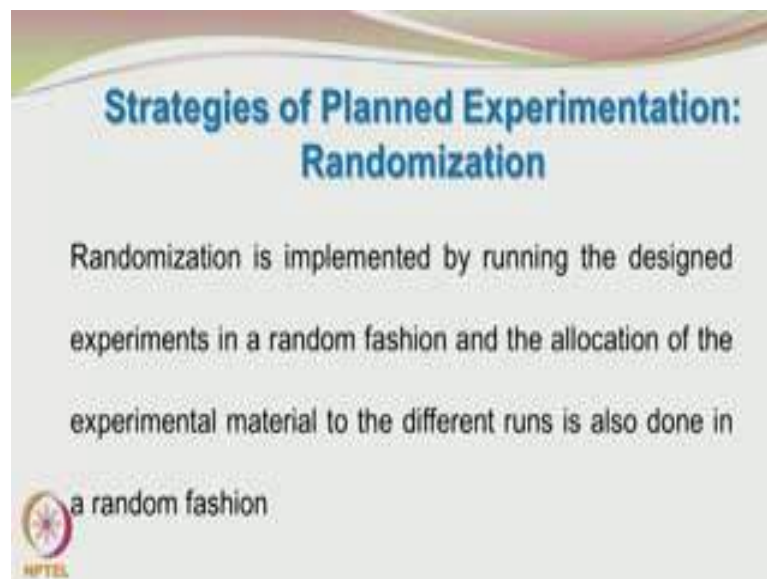
What I mean by randomization is to mix up the order of the runs - why should we do it? Suppose we are looking at a photocopier, and we are looking at three different toners used in the photocopier, let us say that you put for each toner you do three trials, and look at the picture quality, and then decide which toner is better, and the which toner is not so good, and so on. So, in this very illustrative example, what you can do is do toner



A for the first 10 minutes - 3 copies with the first toner, and then 3 copies with the second toner, and 3 copies with the third toner. When you do that, the photocopier may be getting heated up. So, by the time you come to the third toner, the external heating of the photocopier may lead to poor quality copies for the third toner okay. So, this will lead probably to the wrong conclusion that toner C is not good. What you should have actually done is the heating of the photocopier is inevitable and probably not controllable, and you have to finish all these evaluations within 30 minutes.

So, what you can do is you can mix up the order of the runs, you can put toner A or toner 1 first, then followed by toner 2, then by toner 3, and then 2, 3, 1, 1, 3, 2 like that, if you do it in randomized fashion, then the higher temperature effect is also reflected or indirectly influencing the performances of all the three toners than a specific toner. So, by randomization any unaccepted effects is evenly spread or distributed cross all the factor settings. So, randomization is a good idea and I request that when you are doing experiments, you please randomize your order of the runs.

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So, randomization is implemented by running the designed experiments in a random fashion and the allocation of the experimental material to the different runs is also done in a random fashion.



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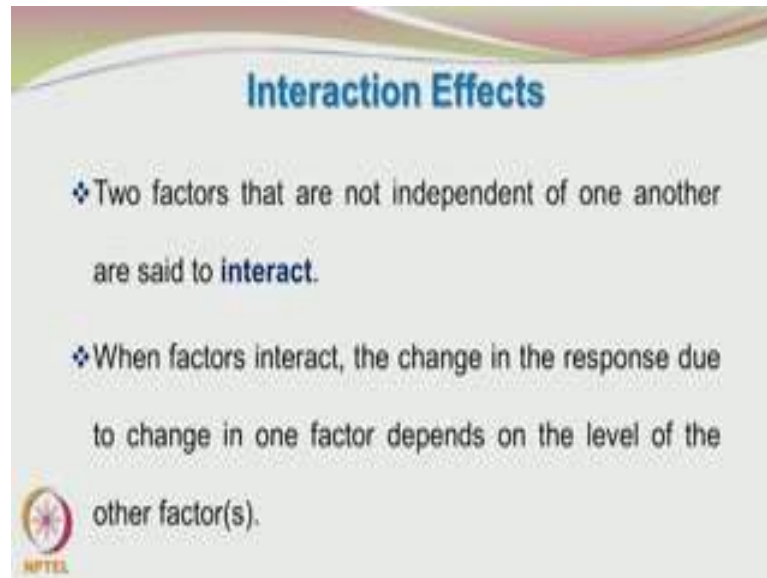
Okay now, we are coming to the design of experiments slowly and one of the important experimental designs is the factorial design; you might have heard about the 2^n factorial design, where 2 is the number of levels and n is the number of factors. So, for example, if you are having three factors a , b , and c – temperature, pressure, and concentration, each factor you investigate at two levels - a higher level and lower level; that is why we have the number 2 okay. So, we end up actually doing 8 experiments; 2^3 which is equal to 8; 8 experiments are being done; and each factor is investigated at only two levels a higher level and the lower level.

So, there are many advantages of factorial design. You may think okay why should I do only at two levels, it looks better if I do it at multiple levels; you are right, but even with the two levels of many factors, you can economize your experimental program and still get valuable information. And even 2^n becomes quite a large number when the number of factors become too large. So, the advantages of factorial design are: you can scientifically interpret your results; it enables optimization approaches like the responder phase methodology. And one another beauty of this factorial design is both qualitative and quantitative factors may be analyzed together. For example, if you are having three different types of catalysts catalyst 1, catalyst 2, catalyst 3, you don't really consider them as any continuous entities, because you are not representing them in terms of the b e t surface area, and poor volumes, and things like that.



So, these are discrete entities. On other hand, you may also want to investigate the effect of temperature or pressure which can vary over a continuous range. So, the design of experiments involving the factorial design, you can have both the mix of discrete and continuous factors or continuously varying factors. And now, it has been realized that for industrial competitiveness factorial design is compulsory.

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And again, you can do limited number of experiments to recover maximum information. The normal way of doing the experiments is to vary one factor at a time; this is not a very good strategy, because it leads to more number of runs and also it has not pick up the variability cost by interaction between the factors okay. One factor can influence the other factor; the value taken by the first factor can determine the response when you are changing the second factor. When you are doing experiments involving, let us say, two factors A and B, the interaction between A and B may also influence the outcome of the response. When you are doing experiments from one level of B and going to the other level of B, the response may depend upon the value of the first factor. Okay when you are changing B, if the value of A determines the response considerably, then the two factors are said to interact.



Introduction to Research
Prof. Kannan
Department of Chemical Engineering
Indian Institute of Technology, Madras

Lecture – 04
Introduction to Design of Experiments

Welcome back to this continuation on Introduction to Design of Experiments. We will start with the simple example definitely a fictitious one.

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A screenshot of a presentation slide. At the top, there is a toolbar with various icons for presentation control. Below the toolbar is a table with 4 columns and 9 rows. The columns are labeled '4', 'C1-T', 'C2-T', and 'C3'. The rows are numbered 1 to 8. The data in the table is as follows:

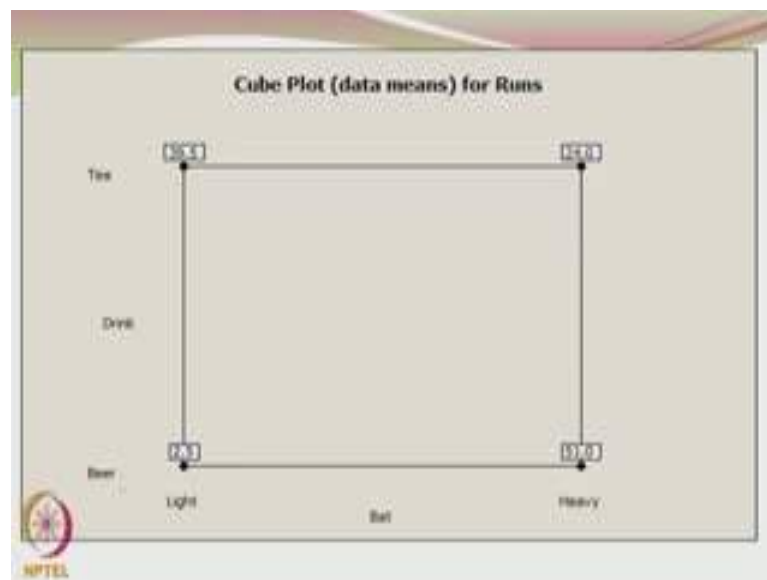
4	C1-T	C2-T	C3
	Bat	Drink	Runs
1	Light	Beer	5
2	Light	Beer	0
3	Light	Tea	40
4	Light	Tea	39
5	Heavy	Beer	50
6	Heavy	Beer	52
7	Heavy	Tea	25
8	Heavy	Tea	23

So, I am presenting data in a table which contains the runs scored by a batsman in the game of cricket, I hope all of you know this game and know its rules. There are two innings and the batsman gets two chances in a match, first innings and second innings. So, the scores are listed in the last column. And in this experiment, what we are trying to do is find out whether the type of drink a person has or the batsman has before coming out to play influences his performance. Another factor we are considering is the weight of the bat he uses. Normally, we will think that the type of drink he has had or the weight of the bat he is using will be independent, but it may so happen that they may affect one another or there may be an interaction between the two factors that is what we are going to demonstrate in this example.



This example has been analyzed using mini tab software; this is a statistical analysis software. It also includes the design of experiments. You can download a trial version and check this for yourself. So, we have the type of bat or the weight of the bat in the first or the second column, you can have light bat and heavy bat, and then the batsman drinks beer, I don't know whether it is allowed in India by I think in foreign countries these things are pretty common. So, the batsman might have had beer before coming out to play, just to put him in good mood or he might have had tea, if he is of the serious type. So, these are the runs scored and you can see that there is a considerable variance in the run scored by the batsman.

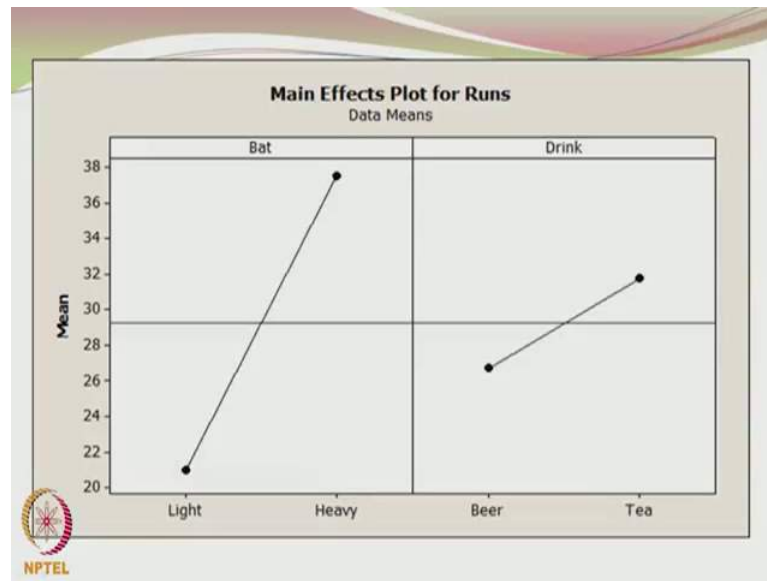
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And you can see that the type of the drinkers represented on the vertical scale, and the type of the bat or the weight of the bat is shown on the horizontal scale. So, this is the average performance of the batsman, when he is having a light bat and beer and this is the average performance of the batsman, when he is having a light bat and tea. You can see 24 is the average performance of the batsman when he is having tea and heavy bat; and then beer and heavy bat the average performance is the highest at 51.

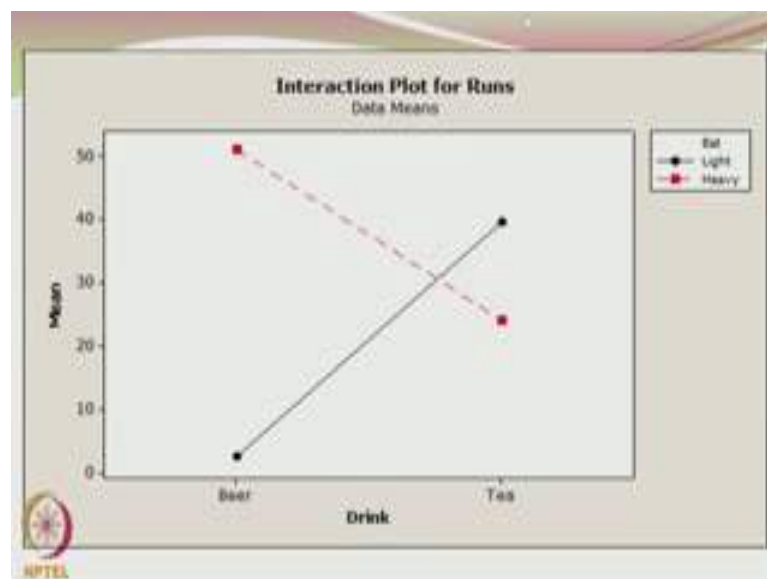


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So, let's analyze this further. You can see that this is the main effects plot; this is the effect of the bat alone and the effect of the drink alone. So, it might appear when a batsman is going from a light bat to heavy bat, the average performance improves, and when he is going from beer to tea the average performance improves, but this just not tell us the complete picture.

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Okay so, we have to see the interaction. Now this is very interesting. When a batsman is going from beer to tea the performance is increasing, when he is using a light bat; but when he is using a heavy bat, the combination of beer and heavy bat leads to better performance than the combination of heavy bat and tea okay. So obviously, under the influence of the alcohol, the batsman is able to swing more wildly and connects more often than not and then scores more runs who knows, but this is an example where there is a strong interaction between the type of the drink and the weight of the bat.

If there had been no interaction between the two variables then the two lines would have been parallel. So, when you do your design of experiments, don't look at only the single effect or the single factor plots; look at the interaction plots. And if the interaction plots are parallel then the variables are not interacting with one another, but if they intersect or they approach one another or they diverge from each other, then it shows that there is an interaction. So, it's a very interesting and important aspect to consider in the design of experiments.

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A presentation slide with a light blue background and a decorative wavy header. The title 'Interaction Effects' is in bold blue text. The main text explains that if a change in the first factor causes a certain change in output response at one level of the second factor, an identical change at the second level will produce a markedly different output response. The NPTEL logo is in the bottom left corner.

Interaction Effects

If the change in level of the first factor causes a certain change in output response at one level of the second factor, an identical change in the first factor level at the second level of the second factor will produce a **markedly different** output response.

So, let us define the interaction effects. If the change in level of the first factor causes a certain change in output response at one level of the second factor, an identical change in the first factor level at the second level of second factor will produce a markedly



different output response. So, you may want to read this couple of times to understand it fully. All it says is when you go from one level to another level for a particular factor, the response depends upon what level you have fixed for the second factor. If the level you have fixed for the second factor makes a difference then the two variables are set to interact.

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ANOVA				
Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares	F_0
A Treatments	SS_A	$a-1$	$SS_A/(a-1)$	MS_A/MS_E
B Treatments	SS_B	$b-1$	$SS_B/(b-1)$	MS_B/MS_E
Interaction	SS_{AB}	$(a-1)(b-1)$	$SS_{AB}/(a-1)(b-1)$	MS_{AB}/MS_E
Error	SS_E	$ab(n-1)$	$SS_E/ab(n-1)$	
Total	SS_T	$abn-1$		

By now we know what is meant by ANOVA table. So, we have listed the source of variation; A treatments, B treatments and then the interaction between the two treatments. The difference between the earlier ANOVA table and the current one is that now we are dealing with two variables, and we also have to consider the interactions between the two variables. So, we find the sum of squares of the deviations for the A treatment, for the B treatment and also for the interaction. And we scale each of the sum of squares by the appropriate degrees of freedom; A is the number of levels for factor-a, B is the number of levels for factor-b. So, we subtract 1 from a and 1 from b and then we divide the sum of squares of a by a minus 1 sum of squares of b by b minus 1 to get the mean squares that is what is presented here.

Similarly, we also get to know that the degrees of freedom for the interaction between a and b is a minus 1 times b minus 1 and so you have the mean squares as sum of squares



of ab by a minus 1 into b minus 1. So, these are representing the variations caused by the actual experiments. Whereas, we also have error which is based on the variation shown when you repeat these experiments, either you repeat all the experiments or you repeat the experiments at the center point. What is really meant by center point, right now do not worry, we will come to it a bit later. We can look at the individual mean squares for example, compare the treatment a , with the variability shown by the error and get the appropriate f value. Obviously, if a treatment is very important then it will completely dominate, what will completely dominate? The mean squares of a will completely dominate over that of the error and so you will have a very high F value.

Similarly for b , if the interaction effect is not important then what might happen is the mean square for ab would be comparable to that of the error, and you will get a value in the order of magnitude of 1 and that would indicate from the F probability chart that this variable is not important okay. There is something called as the p value, but I would leave that for you to understand yourself. It is quite straight forward; it has something to do with the type one error and okay no need to keep it in suspense; p value refers to the probability of committing the type one error. And obviously, if the F value is very, very high the p value would be very small, because you are going on to the tail end of the f distribution and the probability would be very small and so the probability of committing a type one error would be very small. So, I hope you know what is meant by type one error, if you don't remember please go back to the previous lectures and then understand.

Now that we have done some experiments and you analyzed them, carried out the experiments that repeats, put them in statistical software like mini tab, got the response plots, and you also know which variables are significant which variables are not significant and so on. There is another interesting thing you can do with your experimental data. Many researchers and students get pride if they develop their own mathematical model. So, we are not going to discuss about very fundamental first principles model, we are going to discuss about very simple models, these are called empirical polynomial regression models. And even though, they may not have much of a theoretical background they provide useful formulae, if you want to call it like that which can be used to predict the response at different factor settings.



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Multiple Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

β_0 : Intercept

β_1 : partial regression coefficient 1

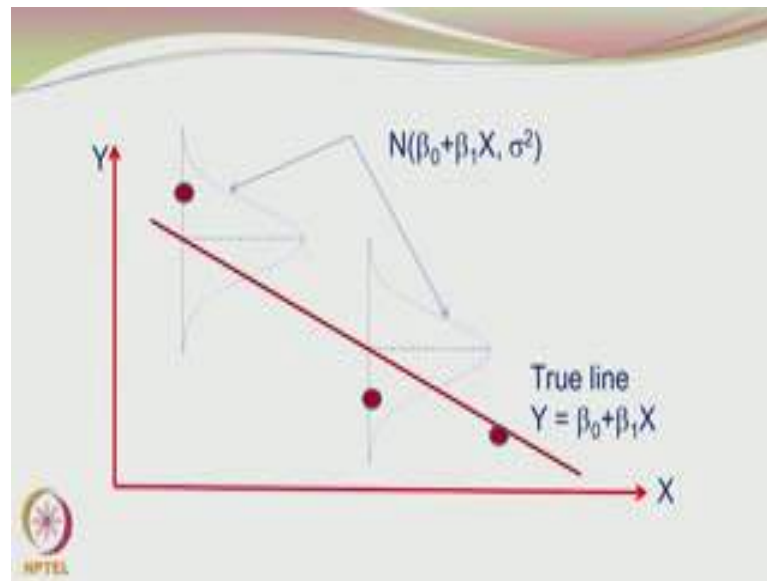
β_2 : partial regression coefficient 2

The slide has a light blue header with the title 'Multiple Regression Model'. Below the title is the regression equation. Underneath the equation are three definitions for the coefficients. The NPTEL logo is in the bottom left corner of the slide.

So, now, we are getting into model building, which is based on experimental data. So, you have the process response, y is equal to β_0 plus $\beta_1 X_1$ plus $\beta_2 X_2$ plus ϵ , here X_1 and X_2 are the variables or the factors. These are values taken by the variable 1 and the value were taken by variable 2. So, X_1 represents variable 1, and X_2 represents variable 2 and β_1 and β_2 are called as the partial regression coefficients associated with these variables 1 and 2. β_0 is not associated with the any variable it has standing on its own it is the so called intercept, for example, if you write y is equal to mX plus b then the b is the intercept. So, β_0 is referred to commonly as the intercept, whereas the β_1 and β_2 are referred to here as the partial regression coefficients.



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This is a very interesting diagram. You can see that the experimental data points are sort of scattered around a mean straight line. This is a true line okay, but the data points do not lie on the straight line that would have been an ideal occurrence or a intentional occurrence, but under usual circumstances, we can always find scatter of the experimental data around the straight line. So, beta naught and beta 1 are the true values of the parameters which unfortunately we do not know; and since we do not know beta naught and beta 1, we also do not know the value of the response y.

Here, for the purpose of illustration, I am just taking a case where there is only one factor or variable influencing the response. So, we have these experimental responses as being distributed around a mean value given by beta naught plus beta 1 X. So, if you take the level for X as X 1 at this point then the actual true, but unknown value would be given by beta naught plus beta 1 X a, calling this point or calling this variable setting as X a. So, this is the true response, but the experimental data because of the randomly varying error component would be scattered and so this is where the point lies.

We assumed that the errors are varying randomly with mean 0, and constant variance, so when you super impose the error effect on the response, the response is also normally distributed, but the mean value given by the true line equation and the variance given by



sigma squared. Sigma squared is the error variance, which is assumed to be constant. Similarly, at the second location or second level for X , let us call it as b you find of response line below the true line. And by random fluctuations, the third value lies pretty close to the true response. So, in reality, we may never get to know, what is the true value of the parameter β_1 and the true value of the parameter β_0 . So, through a linear regression analysis and your experimental data, we can only estimate the parameters β_0 and β_1 .

So, each time you do an experiment, you will get different responses and each time you fit the data to the model, you may get different values of the parameters. So, this is because the random error component is influencing a response in several unknown ways which you cannot predict a priori and so that's the reason, why your model parameters are coming slightly differently each time you do the experiment. So, you do not have to worry, if you do not get the same model parameters if you do the experiments several times; it is expected that this is going to happen.

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Multiple Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$$

This is a multiple regression model with k regressor variables $(X_1, X_2, \dots, X_k)$.

The parameters $\beta_0, \beta_1, \beta_2, \dots, \beta_k$ are called as partial regression coefficients.

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We can have any number of variables okay; you do not get restrict yourself to one variable or two variables at the most. You can analyze several variables together and if you use the linear algebra concepts, it becomes very straightforward. So, when you have



multiple variables, you have the multiple regression model but k regressor variables $X_1, X_2, \dots, X_k$. And again to retreat $\beta_0, \beta_1, \beta_2, \dots, \beta_k$ are called as partial regression coefficients. So, how many partial regression coefficients you have? It is $k + 1$; we have k regressor variables and we have hence k partial regression coefficients associated with these k regressor variables. In addition to these, we also have the intercept which is β_0 , so we have $k + 1$ unknown parameters we want to estimate; in some places we may refer to this $k + 1$ as p .

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Matrix Approach to Multiple Regression

Let there be k regressor variables with n observations

$$(X_{1i}, X_{2i}, X_{3i}, \dots, X_{ki}, Y_i), i=1, 2, \dots, n$$

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + \epsilon_i \quad i=1, 2, \dots, n$$

Note: $n > k$

This may be represented in a matrix notation as $Y = X\beta + \epsilon$

Now, you have as I said earlier k regressor variables, but you are not going to do a single experiment involving different settings of these k regressor variables. You might want to fix some variables and vary some variables; next time you may fix those variables and then vary some other set of variables so on. So, let us assume that you do this in a very systematic manner, and you have n such experiments or n such runs. So, you have now two subscripts i and j ; one subscript referring to the run order okay; i is the run order, you can go from 1 to n and the j subscript referring to the index of the regressor variable okay. So, if I say X_{i2} then I am referring to the i th run and the second regressor variable, I think now that should be clear to you. So, the i th run's response Y_i is given in this way, $\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik} + \epsilon_i$, i running from 1 to n .



Obviously, n should be greater than k , if n is less than k you don't have enough data to analyze problems satisfactorily. If you have n is equal to k even that is not sufficient because the total number of unknown parameters you want to estimate is in fact k plus 1, the partial regression coefficients for each of the regressor variable and then you also have the intercept. So, you have k plus 1 parameters to estimate and you need at least k plus 1 runs. So, if you have k plus 1 runs, and if you have k plus 1 parameters to estimate then you have an exact problem in the sense that your model will fit the experimental data perfectly because you are essentially solving an equation involving k plus 1 variables and k plus 1 unknowns.

But this is not really good because it shows that you have scaled up your regression model, you have introduced many terms in the regression model, and it is fitting your data properly. But this is not really good because you are taking up too many of the parameters, and the regression model may become also very unwieldy and complex to use.

The beauty of a regression model is appreciated when it is very simple and even with the few parameters you are able to explain the process reasonably well. Anyway, we know that this is an empirical model and it is only applicable in the very narrow region occupied by the ranges of these variables; you cannot extrapolate this model all the time beyond the ranges of these variables. So, it's a very simple model. So, better leave it as a simple model involving two or three parameters, and then see how this explains your experimental trend.

Now, if you are familiar with linear algebra, you will easily appreciate what I am going to say it. Now, but even if you don't know linear algebra or you have forgotten, it is very good time and opportunity to go through this concepts quickly and then understand what I am going to say next. In fact, linear algebra you don't have to do research or going to very great depth, all you need to know is how to add matrices, subtract matrices, multiply matrices, and represent matrices first and then also some brief ideas about how to find Eigen values and Eigen vectors.



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Matrix Form of the Regression Equations

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad X = \begin{bmatrix} 1 & X_{11} & X_{12} & X_{13} & \dots & X_{1k} \\ 1 & X_{21} & X_{22} & X_{23} & \dots & X_{2k} \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & X_{n1} & X_{n2} & X_{n3} & \dots & X_{nk} \end{bmatrix} \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} \quad \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

So, when you represent this in the matrix notation, you get nice compact forms. You have the Y column vector, which is comprising of the n experimental observations running from Y 1, Y 2 so on to Y n. Then you also have the X matrix, it's a very important matrix which is having a column of one's this corresponds to the intercept. I will come to this in a moment then you have X 11, X 11 represents the first run and the setting of the first factor or the first variable X 1. X 12 again refer's to the first run, but it refers to the setting of the second factor or the second experimental variable X 2 and so on. Since we have k factors, this one runs all the way from X 11 to X 1 k; similarly you have the other entries in this matrix.

For example, X n 2 will refer to the nth run and the setting of the second experimental factor or the experimental variable X 2. Now, let us look at the beta column vector. By now you should know that this comprises of beta naught which is the intercept and then beta 1 the partial regression coefficient to X 1 so on to beta k, which is a partial regression coefficient to X k. And this is the error term - the mysterious error term which is running from epsilon 1 epsilon 2 so on to epsilon n right.



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Matrix Approach to Multiple Regression

$$Y = X\beta + \varepsilon$$

Using these coefficients in the following model will help us to predict the response for various values of X_i ($i=1,2,\dots,k$)

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \dots + \hat{\beta}_k X_k$$

So, you have y is equal to $X\beta$ plus ε . Now, we have to find the β column vector. We have to estimate the values of β_0 , β_1 , so on to β_k . It looks like a very tough problem, because you are familiar with solving two equations and two unknowns and things like that even three equations in three unknowns becomes quite difficult to solve. But here it looks like, we are going to solve at least 10 variables, it may even be 20, I have seen very elaborate models involving a large number of variables to estimate. So, it looks like a very complex problem, but if you are familiar with linear algebra you will be able to easily appreciate how quickly we are able to solve even large systems of equations right.

So, we do not know the true response because we do not know the true value of β . So, what we instead do is try to estimate the response or try to estimate the parameters of this model, so that we can predict the response. So, we define our regression model as $\hat{Y}$ is equal to $\hat{\beta}_0$ plus $\hat{\beta}_1 X_1$ plus $\hat{\beta}_2 X_2$ plus so on to $\hat{\beta}_k X_k$. So, this is very interesting. Now, we have put this hat on each of the parameters and also the response; this hat indicates that it's a predicted response and not the experimental response. Again $\hat{\beta}_0$ and $\hat{\beta}_1$, etcetera refers to estimated parameters and not the true parameters. So, please be clear about the notation.



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Least Squares Estimators of β

This leads to the solving of the following system of equations

$$X'X\hat{\beta} = X'Y$$

These are the **least squares model equations** in matrix form. Their solution is given by

$$\hat{\beta} = (X'X)^{-1}X'Y$$

So, what do we actually do? We first construct the X prime X matrix, where X prime is the transpose of the X matrix. We saw the X matrix to be defined as shown in the slide, and the transpose of the X matrix is represented by X prime, and the X prime or the transpose of X simply involves interchanging the rows and columns of the X matrix, so to get X prime, interchange the rows and columns of the X matrix.

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Matrix Form of the Regression Equations

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad X = \begin{bmatrix} 1 & X_{11} & X_{12} & X_{13} & \dots & X_{1k} \\ 1 & X_{21} & X_{22} & X_{23} & \dots & X_{2k} \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & X_{n1} & X_{n2} & X_{n3} & \dots & X_{nk} \end{bmatrix} \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} \quad \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$



So, here if I put X' , you will have $1 \times 11 \times 12$ so on to $1 \times k$ as the first column okay. So, I think you know transposes pretty well. So, you multiply X' on X first.

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Least Squares Estimators of β

This leads to the solving of the following system of equations

$$X'X\hat{\beta} = X'Y$$

These are the **least squares model equations** in matrix form. Their solution is given by

$$\hat{\beta} = (X'X)^{-1}X'Y$$

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And then this is pre-multiplying the column vector of the unknown parameters which required to be estimated. Then on the right hand side we have $X'Y$, where X' is again the transpose of the X matrix and Y is the matrix of responses. So, when you do that you are estimating $\hat{\beta}$ to be $X'X^{-1}X'Y$ okay. So, very quickly you should be able to get the estimated parameters. Doing the inversion and multiplying the matrices is a very simple task in many software; MATLAB you might be familiar or Scilab you may be using, where all these matrix manipulations are done very quickly and in a very simple manner.



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Resolution of Error Sum of Squares

$$SS_E = (Y'Y - \hat{\beta}'X'Y)$$

$$SS_E = \left(Y'Y - \frac{(\sum_{i=1}^n Y_i)^2}{n} \right) - \left(\hat{\beta}'X'Y - \frac{(\sum_{i=1}^n Y_i)^2}{n} \right)$$

$$SS_{\text{error}} = SS_{\text{total}} - SS_{\text{regression}}$$

By definition of the total sum of squares, the second term
becomes the regression sum of squares

Okay now, we are looking at something, which is apparently very complicated; it's in fact quite straight forward, you do not want have to worry too much about it. We have the sum of squares of the error. The error component is very essential to see if your regression parameters are significant or not whether the model parameters you have estimated, how many of them are indeed having an impact on the process response and how many of them are dud parameters which are of little value. So, to do that we have to find the sum of squares of the errors; you might have already come across the sum of squares of the errors in earlier analysis of variance tables.

So, how do we find that we subtract beta hat prime X prime Y from Y prime Y. This can also be written as shown here. And the first term in the bracket refers to total sum of squares, and the second term in the brackets is the regression sum of squares. Ideally you would like to have the total sum of squares to be equal to the regression sum of squares, but it never happens the total sum of squares often exceeds our regression sum of squares and more the difference between the two; more is the contribution from the random error component. So, we have the total sum of squares minus the regression sum of squares giving the error sum of squares.



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Analysis of Variance (ANOVA)				
Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Regression	SS_R	k	MS_R	$\frac{MS_R}{MS_E}$
Error or Residual	SS_E	$n-k-1$	MS_E	
Total	SS_T	$n-1$		

So, again we come to the very familiar by now analysis of variance table, where we have the source of variation in the first column, sum of squares in the second column, degrees of freedom in the third column, mean square in the fourth, and the F value in the fifth. So, we compare the regression sum of squares with the error sum of squares and not immediately, we first have to find the mean squares. So, we scaled the regression sum of squares with the degrees of freedom with the regression sum of squares and so we have SS_R by k , which will give you the mean square regression. Similarly, sum of squares of error divided by n minus k minus 1, which is the degrees of freedom associated with the error component and that would give you the mean square error. The ratio of MS_R to MS_E will give you the F value.

I am not really getting into how you calculate the degrees of freedom for each of these contributions, regression and error. It is very interesting I hope you will be sufficiently interested and motivated to find out how, for example, the regression sum of squares has k degrees of freedom, while the error sum of squares has n minus k minus 1 degrees of freedom, the total number of degrees of freedom is simply the addition of these two and it becomes n minus 1; and the addition of sum of squares of regression and sum of squares of error is the total sum of squares.



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A presentation slide with a light blue and green wavy header. The title 'Coefficient of Determination (R²):' is in blue. Below it, the formula $R^2 = \frac{SS_{\text{Regression}}}{SS_{\text{Total}}}$ is displayed. The text 'This represents the proportion of the total variability accounted or explained by the linear regression model' is centered below the formula. The NPTEL logo is in the bottom left corner.

Coefficient of Determination (R^2):

$$R^2 = \frac{SS_{\text{Regression}}}{SS_{\text{Total}}}$$

This represents the proportion of the total variability accounted or explained by the linear regression model

Many researchers in their publications proudly present their experimental results and save they got an R squared value of 0.99 or 0.991, sometimes they may even say 1, what is meant by the coefficient of determination? R squared is the short form or the nomenclature for the coefficient of determination. So, this represents the proportion of the total variability accounted or explained by the linear regression model. Experiments means variations or variability and you are developing the mathematical model. If all the variability or variations are completely described by the proposed model, it looks like you have solved the problem or cracked the problem, and you are getting an R squared value of 1. You will get an R squared value of 1, the number of variables you want to estimate is equal to the number of experimental runs then you are solving the linear algebraic problem and not an actual statistical regression analysis problem. So, you will always get R squared is equal to 1.

Another way you will get R squared close to one would be keep on adding more and more useless terms to your regression model, you can put $\beta_{11} X_1^2$, $\beta_{22} X_2^2$ and so on. It may look as if you are introducing some non-linearity or curvature into your model, but as long as the coefficients are still remaining as β_{11} and β_{22} , it remains as a regression model. So, you add under another or additional columns corresponding to X_1^2 and X_2^2 into your design matrix X and



then manipulate it in the same way to get the regression parameters. So, we are still in the subject of R squared, so don't aim for unrealistically high value of R squared, just because you want to make it as perfect as possible. In fact, a regression model with the R squared value of 0.84 may be probably more robust, and probably more informative and reliable than a regression model, which is having 15 parameters and then R squared value of 1. Such a model may be in fact, unstable and very sensitive. So, how to quantify which is an acceptable R squared?

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A presentation slide with a light blue background and a decorative wavy border at the top. The title 'Adjusted R²:' is in blue. Below it, text explains that mean squares of error and model/total sum of squares are used. The formula for Adjusted R squared is shown, and a note states p=k+1.

Adjusted R²:

Here Mean Squares of the error and model/total sum of squares are used.

$$R^2_{adj} = 1 - \frac{SS_{Error} / n - p}{SS_{Total} / n - 1}$$

Note: $p = k + 1$

We do something called as adjusted R squared which penalizes the R squared, if you are introducing too many parameters. So, now, we have adjusted R squared defined to be 1 minus sum of square of error by n minus p, by total sum of squares by n minus 1. So, we are getting 1 minus mean square error and divided by mean square total. So, essentially when the number of parameters increases, as I told you earlier p is equal to k plus 1. So, when p increases, this term n minus p will decrease, so the numerator term will start to increase. When the numerator term starts to increase then the R squared value or the adjusted R squared value will start to decrease. So, this tells us that even though R squared value is increasing; the adjusted R squared value may decrease because of this n minus p effect okay. Only when you are adjusted R squared also increases with



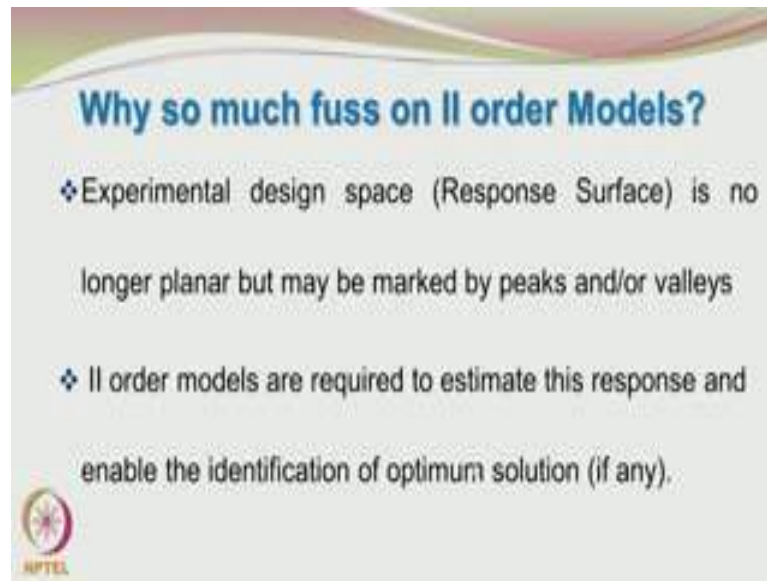
increasing number of parameters then those parameters are required to improve the models predicting capability.

What I am trying to say here is if the mean square error term it has two components, the sum of squares of error and n minus p . When you add the additional parameter, if the sum of the squares of error comes down considerably then that would have a greater reduction than the increase caused by decrease in n minus p . So, I think if you look at it carefully, you will be able to appreciate if SS error decreases at a faster rate than the decrease in n minus p , then there would be a reduction in the numerator overall numerator here, and you are adjusted R squared will in fact increase. So, any way don't only look at R squared, but also please look at adjusted R squared to see whether the R squared adjusted is increasing or decreasing with the addition of more parameters. If adjusted R squared is decreasing with the addition of more parameters then you better not add those parameters, and leave with the simpler model.

Okay now, when you look at experimental design papers of the literature they keep harping on the second order models, why so much first on the second order models, when you look at the experimental design space it is no longer planar, but it may be marked by peaks and and/or valleys. Peak is mountain kind of thing and a valley is a depression kind of geography in the experimental response space.



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So, in order to capture these peaks and valleys, second order models are required and the moment you talk about peaks and valleys you are also looking at the best possible optimum condition or the best possible minimum condition. Sometimes the optimum can be maximum, and in some other cases optimum may be the lowest value okay. So, we have to make sure that we find the correct optimum. There can be multiple peaks and multiple valleys, and we should not be satisfied with the sub optimal solution.



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Why so much fuss on II order Models?

II order models are of the form

$$y = \beta_0 + \sum_{i=1}^k \beta_i X_i + \sum_{i < j} \sum_{j=2}^k \beta_{ij} X_i X_j + \sum_{i=1}^k \beta_{ii} X_i^2 + \varepsilon$$

This equations requires estimation of

$$1 + (k) + {}^kC_2 + k = 1 + 2k + k(k-1)/2 \text{ parameters.}$$

Okay so, now, how do we represent the second order models? So, this is the form of the second order model. It is not very complicated. Even though it looks to be like that y is equal to β_0 plus $\sum_{i=1}^k \beta_i X_i$ and then you have $\sum_{i < j} \sum_{j=2}^k \beta_{ij} X_i X_j$ plus the $\sum_{i=1}^k \beta_{ii} X_i^2$ plus ε okay. This not as complicated as it looks the first one is the combination of the symbol factors; the second one is the combination of all possible interactions involved between the factors, and then you also have the quadratic term effect X_1 squared and X_2 squared. For example, if you are talking about two factors or two variables, you will have β_0 plus $\beta_1 X_1$ plus $\beta_2 X_2$ plus $\beta_{12} X_1 X_2$, and then you will have $\beta_{11} X_1^2$ plus $\beta_{22} X_2^2$ plus the error component. For a general case, you can find out how many parameters you want to estimate, based on this model structure okay. I think we will stop now and continue in the next class.

Thanks for your attention.

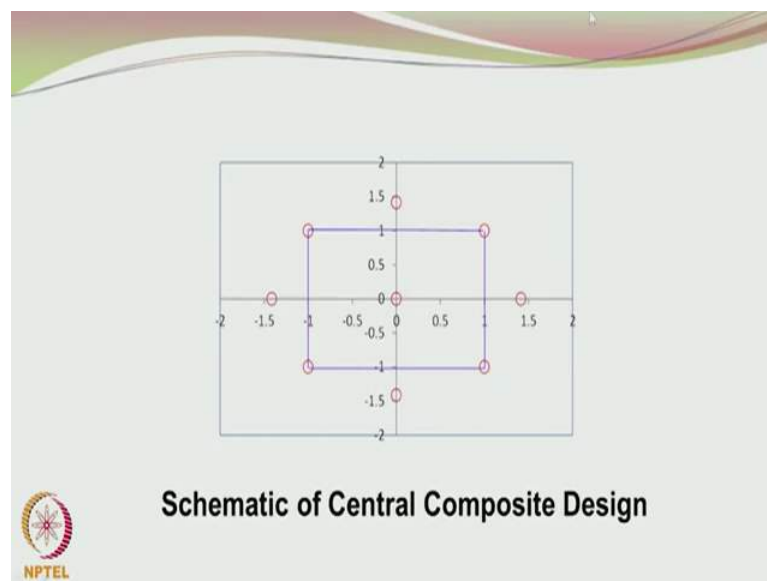


Introduction to Research
Prof. Kannan
Department of Chemical Engineering
Indian Institute of Technology, Madras

Lecture – 05
Design of Experiments

So now, let us look at very popular Second Order Statistical Experimental Design, based on the second order model.

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This is the central composite design, you can see that we have a square here; we are talking about 2 factors, 2 variables. So, you have a regular 2^2 design, 2 factors in 2 levels so you have 4 settings, this constitutes the square, low-low and then you also have high-high and so on. So, you have an experimental design involving a 2^2 factorial designs.

In addition to that, you have the axial points you can see this is the first axial point, second axial point, third axial point and fourth axial point. The first second third and fourth are given arbitrary fashion, so you have 4 runs, 2^2 . Then, you have $2k$ axial points where, k is the number of factors, so you have again 4 axial points that makes it the total of 8. But, that is not all you also have the center point which is the geometric center of the design. I said earlier that, the repeats may be performed at the



factorial points or at the centre points. So, you here you can have 4 or 5 repeats for your experiments.

This is a very interesting and very commonly used design for experiments. There is lot of flexibility, where you want to locate you axial point. You may want to locate at either further or close to the factorial design, depending upon your requirements. What those requirements are? And how do you shift them? You will learn when you do a formal course on the Design of Experiments.

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Why are Center Runs required?

- ❖ Repeats at the geometric center of the experimental design (i.e. at $(0,0,...,0)$)
- ❖ Important augmentation to the factorial design
- ❖ Do not contribute to the linear effects and the interaction
- ❖ Detecting curvature if not explicitly estimating them

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So, why do we require the runs at the center? They may represent the repeats okay, rather than repeating the experiments at all the factorial points and this may become very expensive especially, when you have very large number factors, doing experiments at the center of the design space is convenient.

As the center of the designs space, is some kind of representation of the overall design. It also represents an important augmentation to the factorial design and it tells whether there is a curvature in the experimental response okay. Sometimes, there can be only linear variation with the factors but, sometimes if there is interaction between the factors then you have a curvature or a twist in the planar response curve. It may not be a planar or it may not be a simple, but there may be some kind of curvature. So, to detect this curvature you need the center points. How the center points help you to detect curvature is beyond the scope of this introduction lecture.



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Roles played by center points in CCD

- ❖ help in the detection of II order or curvature effects
 $(\beta_{11} + \beta_{22})$ but not in their individual estimation.



So, the center points help to detect the second order or curvature effects. The quadratic terms, it helps to identify whether that beta 11 plus beta 22 is significant or not. But, it doesn't help you to estimate individually beta 11 and beta 22. Where did this beta 11 and beta 22 come from?

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Why so much fuss on II order Models?

II order models are of the form

$$y = \beta_0 + \sum_{i=1}^k \beta_i x_i + \sum_{i < j} \sum_{j=2}^k \beta_{ij} x_i x_j + \sum_{i=1}^k \beta_{ii} x_i^2 + \varepsilon$$

This equations requires estimation of

$$1 + (k) + {}^k C_2 + k = 1 + 2k + k(k-1)/2 \text{ parameters.}$$


Let us go back to the model, if you are having only 2 factors then I said, you will have beta 11 x 1 square plus beta 22 x 2 squared. These represent the quadratic terms and also responsible for the curvature, in addition to the interaction terms okay. So, to find the



beta 11 and beta 22, you require center points. Even though, you may not be able to find out explicitly beta 11 and beta 22 at least we tell you whether beta 11 plus beta 22 is overall significant or insignificant. If beta 11 plus beta 22 is insignificant then, both the beta 11 and beta 22 are not required to be present in the model.

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Axial points of Central Composite Design

- ❖ The axial terms contribute to the estimation of the individual pure quadratic effects significance.
- ❖ If axial points were not present, only the sum of the quadratic terms significance ($\sum_{i=1}^k \beta_{ii}$) could be gauged using the center points

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What is the contribution from the axial points? The axial points contribute to the estimation of the individual pure quadratic effect's significance. If the axial points were not present, only the sum of the quadratic terms significance could be gaged using the center points, this is pretty straight forward.



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Roles played by the axial points of Central Composite Design

- ❖ The axial points do not contribute to the estimation of the interaction effects
- ❖ The center points and the axial points contribute to the flexibility of the CCD.

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The axial points do not contribute to the estimation of the interaction effects. The center points and the axial points contribute to the flexibility of the Central Composite Design.

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Response Surface Methodology

In any experimental work, an important objective could be to identify optimum levels of the various factors which will maximize/minimize a suitable objective for e.g. reaction yield, conversion, process time, energy consumed etc.

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Now, we come into the final topic; Response Surface Methodology. Many industries want to optimize their processes, but did not know where to start and where to end. And, it is not appropriate, especially in the industry to embark on a grand exploratory voyage in the end dimensional space, hoping to sight the promised land sometime or the other.



What is important is, first to do a set of screening experiments where you have a preliminary set of experiments and assess the overall trend. And, the Response Surface Methodology then enables you to identify the direction in which you should proceed. In other words, it points to the direction where you should set your experimental factors, so that you are progressing in the correct direction okay. Suppose, the goal of your process is to optimize then, the Response Surface Methodology will tell you the direction where the process will be increasing or the process response would be increasing in the fastest manner.

This is very useful, it helps you to decide and plan your next level of experiments. So, in any experimental work, an important objective could be to identify optimum levels of the various factors which will maximize, minimize a suitable objective for example, reaction yield, conversion, process time, energy consumed, etcetera. So, your objective function may be either minimum or maximum. So, you have to proceed in such a manner that, you go to the correct desired condition.

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Response Surface Methodology

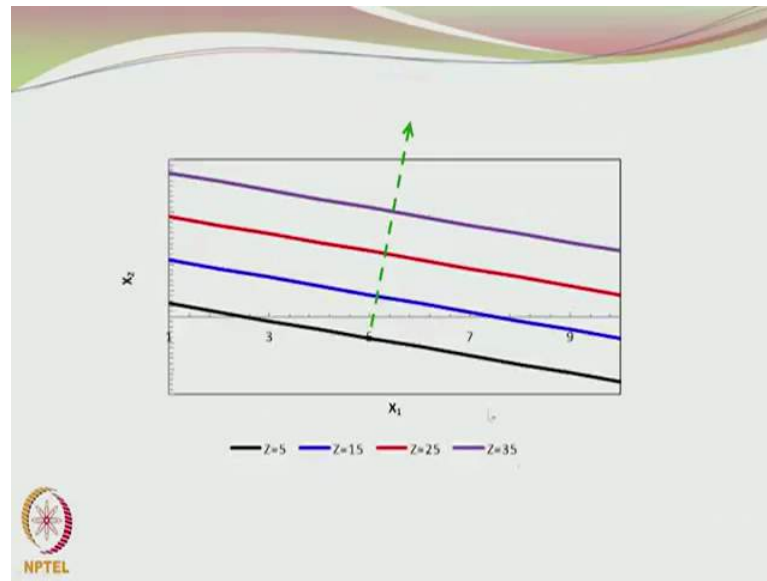
- ❖ The current level of operation may usually be far away from the optimum and we cannot afford to wander in the wilderness of n-dimensional experimental variable space hoping to eventually reach the optimum.
- ❖ Response Surface Methodology deals with identifying optimum settings of the factors in a systematic manner.

The slide features a decorative header with wavy lines in shades of green, yellow, and red. The NPTEL logo is visible in the bottom left corner of the slide content area.

So, the current level of operation may be usually far away from the optimum and we cannot afford to wander in the wilderness of the n-dimensional experimental variable space hoping to eventually reach the optimum. Response Surface Methodology deals with identifying optimum settings of the factors in a systematic manner.



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So, as I said earlier the Response Surface Methodology helps you to find quickly in which direction the processes increasing the fastest. You have to increase in the process responses what you want. So, if you consider case involving 2 variables, x_1 and x_2 . The method of steepest ascent an optimization tool helps you to know the direction in which the responses are increasing in the fastest possible manner and this is directly shown in terms of the green arrow, for this particular case.

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Second Order Model

The second order response may now be expressed as

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_{12} X_1 X_2 + \beta_{11} X_1^2 + \beta_{22} X_2^2 + \epsilon$$

The predicted expression is given below

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_{12} X_1 X_2 + \hat{\beta}_{11} X_1^2 + \hat{\beta}_{22} X_2^2$$



So, when you have 2 factors, this is the form of the equation, the true response involving the random error component and this is the model which is being proposed and you have to estimate the beta naught hat and then the remaining parameters beta hat 1 beta hat 2 beta hat 12 corresponding to the interaction, beta hat 11 and beta hat 22 and the last 2 corresponding to the quadratic terms.

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Location of Stationary Point

The partial derivatives of the predicted response with respect to the variables X_1 and X_2 must be must be set to zero and solving the resulting equations, the coordinates of the stationary point may be identified.

$$\frac{\partial \hat{Y}}{\partial X_k} = 0 \quad k = 1, 2$$

Again, we can use linear algebraic techniques in order to identify the optimum conditions. So, in order to find the stationary points, the so called stationary points, you have to partially differentiate your proposed model with respect to x_1 and x_2 and set them to 0, solve the resulting systems of equations and, or system of equations and identify the stationary conditions.



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Location of Stationary Point in Matrix Notation

Expressing the second order relation in matrix notation as

$$\hat{Y} = \hat{\beta}_0 + \mathbf{X}'\mathbf{b} + \mathbf{X}'\mathbf{B}\mathbf{X}$$

The stationary point is the solution to the equation

$$\frac{\partial \hat{Y}}{\partial \mathbf{X}} = \mathbf{b} + 2\mathbf{B}\mathbf{X} = \mathbf{0}$$


To do this, would be quite tedious especially, when you are having many factors. We may as well use the linear algebra techniques for which several tools are available at present, for example, MATLAB, Scilab and so on, or there is no big deal in writing your own program, if you are having inclination towards that okay. So, coming back to our Response Surface Methodology approach, we can represent the second order model in matrix notation as shown here; $\hat{Y} = \hat{\beta}_0 + \mathbf{X}'\mathbf{b} + \mathbf{X}'\mathbf{B}\mathbf{X}$. And, the stationary point of the solution can be obtained by differentiating this in matrix terms, to get $\mathbf{b} + 2\mathbf{B}\mathbf{X}$ and we get the stationary conditions by equating this to 0. Again, all the bold terms indicates that they are vectors and matrices and not the regular usual scalars.



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Matrix Notation Explained

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ \vdots \\ X_k \end{bmatrix} \quad B = \begin{bmatrix} \hat{\beta}_{11} & \frac{\hat{\beta}_{12}}{2} & \dots & \frac{\hat{\beta}_{1k}}{2} \\ & \hat{\beta}_{22} & \dots & \frac{\hat{\beta}_{2k}}{2} \\ \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \hat{\beta}_{kk} \end{bmatrix} \quad b = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_k \end{bmatrix}$$

So, what is the form of the X vector, the B matrix and the small b vector? We came across all these, in this equation so we have to identify the form for all these. So, you have X as the vector, comprising of the different factors starting from X 1 so on to X k. Then, you have the B matrix, given in terms of these coefficients and what is this beta hat 11? I already told you, what beta hat 11 is. So, this is beta hat 11, beta hat 22. So, you can easily show that this is the form of the model. It might be interesting for you to find out, why some of these terms in the capital B matrix are divided by 2. So, I think it's worth the effort to find out the reason. I will leave it to you. Then you have the b matrix, which again comprises of the parameters to be estimated and it is given in this particular form.



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A presentation slide with a light gray background and a decorative wavy header in shades of green and red. The title "Location of Stationary Point in Matrix Notation" is in bold red text. The text "Solving the above equation we get" is in black. Below it is the equation
$$\mathbf{X}_s = -\frac{1}{2}\mathbf{B}^{-1}\mathbf{b}.$$
 The text "The predicted value of Y at this stationary point is" is in black. Below it is the equation
$$\hat{Y}_s = \hat{\beta}_0 + \frac{1}{2}\mathbf{X}_s'\mathbf{b}$$
 The NPTEL logo is in the bottom left corner.

Location of Stationary Point in Matrix Notation

Solving the above equation we get

$$\mathbf{X}_s = -\frac{1}{2}\mathbf{B}^{-1}\mathbf{b}.$$

The predicted value of Y at this stationary point is

$$\hat{Y}_s = \hat{\beta}_0 + \frac{1}{2}\mathbf{X}_s'\mathbf{b}$$

So, when we want to use this matrix method, we solve this particular equation and then we identify the stationary condition in a very simple way, minus half times the B inverse times the small b vector okay. So, that will give us the stationary conditions and then using those stationary conditions, we can give the estimated predicted value of Y by using this relationship.

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A presentation slide with a light gray background and a decorative wavy header in shades of green and red. The title "Nature of the Optimum Solution" is in bold red text. The text "Check the eigenvalues of B." is in black. Below it is the text "If all the eigenvalues are positive (negative), and the stationary point is within the region of exploration, the stationary point is a minimum (maximum)." The NPTEL logo is in the bottom left corner.

Nature of the Optimum Solution

Check the eigenvalues of $\mathbf{B}$.

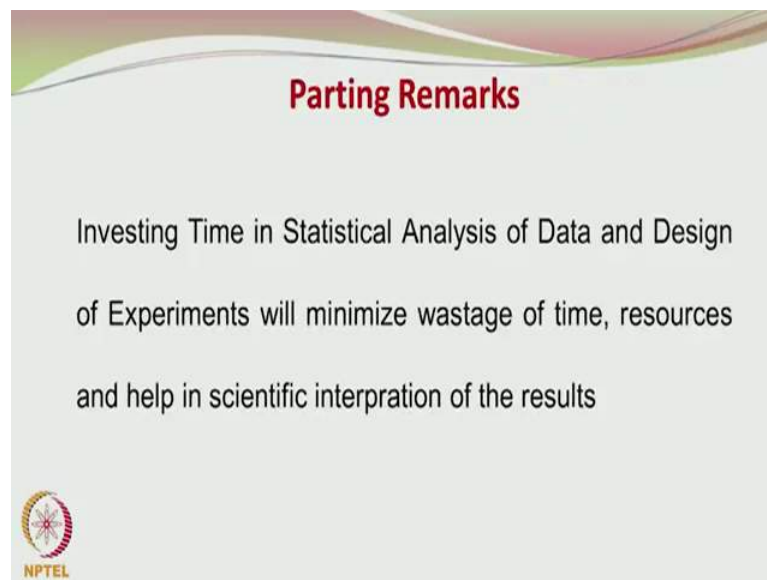
If all the eigenvalues are positive (negative), and the stationary point is within the region of exploration, the stationary point is a minimum (maximum).



So, whether the optimum obtained is maxima or a minima, you can again use or sort to linear algebra tools we can check the eigenvalues of the capital B matrix. So, we find the eigenvalues of this matrix.

So, if the eigenvalues are all positive then, the stationary point in the region of exploration is a minimum, if the eigenvalues are negative then we have hit up on the maximum. Again, there will be some complications when you have 1 eigenvalue and which is positive and another eigenvalue which is negative, I will not get into these complications, but I think there is sufficient unexplored, uncharted territory as far as the student is concerned which he can get into and learn at his own pace.

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So, we are coming to an end of the introductory series of lectures. It has been a real pleasure to talk about the various concepts associated with the Design of Experiments. I didn't only focus on the various experimental designs because I felt to appreciate and understand the experimental design and also to understand, then analyze and present the results from the experimental design concepts, proper introduction into the basics of statistics and probability is also necessary. So, I covered lot of ground talking about normal distribution, the random variable, the sampling distributions of the means, the chi square distribution, the f distribution, the hypothesis testing concepts, believe me all these will come in your Design of Experiments analysis and knowing them would be a good investment, so that you can better appreciate the Design of Experiment concepts.



I have also given NPTEL lectures on this fascinating subject. You are welcome to look at that for getting further information and understanding on this fascinating subject.

Thanks for your attention.



Introduction to Research
Prof. Prathap Haridoss
Prof. Mahesh V Panchagnula
Department of Metallurgical and Materials Engineering
Department of Applied Mechanics
Indian Institute of Technology, Madras

Lecture- 32
Research in Applied Mechanics

Prof. Prathap Haridoss: We are glad to have with us today Dr. Mahesh Panchagnula. He is a professor in the Department of Applied Mechanics. He has a B. Tech in Mechanical Engineering from IIT, Madras and PhD from Purdue University in United States also in the area of Mechanical Engineering. And he has been in the Department of Applied Mechanics here at IIT, Madras. His areas of research include vecting and surface tension and liquid fuel atomization and combustion. And so he is also you know interacted a lot with industry and has a very good perspective of what industry is interested in, what academy has interested in and so on. So, with these words of introduction, we will get started in this discussion. So Mahesh, what are traditional areas of research in your department that incoming students are likely to you know have options to join?

Prof. Mahesh V Panchagnula: Thank you Prathap for having me. Our department is basically kind of structured into 3 groups. We have a group that works on Fluid Mechanics problems, another group that works on Solid Mechanics related problems and a group that works on Bio Mechanics and Bio Medical Engineering related problems. And within these areas we have; these are kind of more granular, it's in other words there is a lot of inter disciplinarily that we are very proud of in our department, we have faculty from 8 different undergraduate trainings all the way from Physicist to Aerospace Engineers, Mechanical Engineers, Civil Engineers are all represented in our department.

So we bring a very melting pot of many different sciences and engineering, and that brings a very different flavor to our department's research. I would strongly encourage all the candidates to go find the faculty profiles on the web and study them, but classically the department has focused on mechanics related problems both in the fields of fluid and solid mechanics and more in more recently we have looked at our faculty had started focusing on fluid structure interaction related problems, cell mechanics which



is a more bio mechanics interfacing with solid mechanics problems, as well as interfacing electrical engineering into mechanical engineering.

Prof. Prathap Haridoss: Of course, I think in any set of research areas even though it may be traditional clearly there are you know boundaries to a research and invariably the research is all about pushing those boundaries. So even though something may be classified as a traditional research you may, you will have you know leading edge part of that research going on. But setting that side, are there areas of research that you would consider you know more recent areas of research that people in your department have started pursuing may be let us say in the last 5, 10 or 15 years which is may be different from what people have been pursuing over the longer time frame?

Prof. Mahesh V Panchagnula: I think these 2 technological advances that I think have enabled the lot of the newer areas of research worldwide. There are extremely fast computational tools that have now become available. And there are extremely sophisticated analysis and visualization tools, electron microscopy etcetera that have now become available. So we have faculty that do experiments and computations using these tools, for example, we have faculty studying tissue properties using nano indentation and atomic force microscopy. We are faculty doing very high in computations on industry scale combustions problems during what is called direct numerical stimulation of a particular kind of a flow.

These are extremely challenging problems and these have been enabled by like I said these tools that have now become available and the insights that have been gained by the use of these tools will soon see the light of day in an industry scale deployment of a solution. I will say these are still emerging research areas, but my vision is that in a few years we will actually see these tools solving industry problems, so these are the newer areas. In the traditional areas also we must point out one difference in the way or we at IIT pursue research in general and our department as well. Using the traditionally developed research tools there are lot of India's specific problems that can be solved, whether it is an organization like DRDO or an ISRO or any of the public sector undertakings or our rural problems. They do not require the extremely sophisticated tools, but they require an India level thinking and India scale thinking that the youth that require may be traditional solutions, and we are not ready to discard those solutions because they are ready for to solve our problems.



(5.16) Prof. Prathap Haridoss: Okay I think that is a very interesting insight into what is driving research these days and how it relates to what is required in our nation. You mentioned that you know in your department you have wide range of different faculty so perceivably you also have a wide range of different students in terms of background.

Prof. Mahesh V Panchagnula: Yes.

Prof. Prathap Haridoss: Who come into your department? So, what sort of issues do they generally face when they come into their MS, PhD program here, are there some general issues that they face as they settled in and if so how would you handle it how they handle it and so on?

Prof. Mahesh V Panchagnula: Sure, I think as a research going from a class room degree program to research program is a big step up in a scholar's journey. In terms of the open-endedness that becomes available to them, that is in front of them. So, until as long as you are in a classroom environment the problems are usually well defined, you know like I say in my lectures given in equation involving 4 variables, given 3 find the fourth. That 's the classical problem solving. Once you get into the research you will still have an equation involving 4 variables, but you are only given 1 variable.

Now, you have to figure out what works for the other 3 variables and justify your own assumptions. That in a very loose definition is kind of where you are getting in to research territory and then beginning to define your own area of research is a second challenge. So very often our faculty allows the scholar the freedom to choose their own area of research, within the boundaries of what they are interested in and what they are equipped to work in. So, this is a second challenge that the scholars usually face and these are good challenges to have because this is where we the scholars grow.

Prof. Prathap Haridoss: Positive challenges.

Prof. Mahesh V Panchagnula: These are very positive challenges and I think the way to approach these challenges is to be open-minded. Is to not think of this as a challenge like you said think of it as a positive challenge and grow in to the solution to that challenge and use the resources available, use your peer group. We have like you said; we have a very diversified peer group in our department. The MS and PhD scholars students population in the department come from many different background, so talk to all of



them and you will see a good prospective emerge of as to what the solution to these 2 questions are for your own training.

Prof. Prathap Haridoss: Ok, Great. See now, when you look at research in an academic setting and then you know there is the industrial setting, where you know people are focused on products that are supposed to come out may be are probably coming out right now or they are coming out in the immediate near future, so sometimes there is always this perception that may be research is doing something which is not directly in line with what the industry is required and may be some of it may even be true, But taking all that into account, are there specific areas of research that are there in your field that you know industry is currently very interested in and are actively looking at?

Prof. Mahesh V Panchagnula: Sure, I think being publicly funded institution we have a mandate to serve Indian industry in some way. I mean it is part of our job description and it is a part of the scholar's job description also when they come into think of what India needs and many of our faculties are involved with like I said DRDO and ISRO very actively both by serving on several expert committees as well as in specific projects. So that's a direct way in which our faculty and students are working with the Indian industry.

A more indirect way, is where we have working on industry related problems, these serve as training rounds for scholars for the industry to absorb later on when they graduate, so many of our faculty of our scholars find jobs in the Indian industry primarily when they graduate whether it is a GE, whether it is a Boeing, whether it is a Mercedes Benz they have traditionally absorbed our research scholars graduating out of our programs. I think large part of it is due to the fact that they have been trained on industry level, industry scale problems during their stint in the department.

Prof. Prathap Haridoss: Okay when you look at, I mean am sure you interact with students you know both belonging to your group, working with you, working with other colleagues of yours and so on and when you look at them going through their you know phases here of I know fresh incoming students, mid way into the research and they getting close to graduation different phases of their you know stay here and then their early part as a professional. So look at this range of time you know frame, how would



you measure you know success in research in this kind of a you know time frame, when they start from nothing and then you know move towards something significant?

Prof. Mahesh V Panchagnula: It's actually a phenomenal growth to watch. And I have had the chance to watch a few of my own students go through the process as well as my colleagues students. When they come in usually find a big change is in their level of confidence in their own abilities and I think that is with that is going to remain with them for rest of their career. The confidence they gain while here at IIT and in the Department of Applied Mechanics. A second a second add-on is their own skills set that they gain while they are here at IIT. In IIT is by any definition of world class institution in term of the infrastructure we provide the students, in terms of the peer group, the faculty group.

I think when they look at this whole this big spread of a multi-course meal in front of them, they end up the research scholars in their limited time they are here on campus end up sampling a lot of the different techniques and they add to their own skills. And at the end of the first few years they are well equipped with these skills. The last few years may be 2 years of that we have spend on campus I am looking at a typical PhD students is where the confidence grows leaps and bounds and I can see this in their committee presentation as they go forward and I think this is a very rewarding progress to watch as a faculty member and I am sure it is for them to reap the benefits of after they leave IIT.

Prof. Prathap Haridoss: Okay you already mentioned something about you know industries that are looking at areas of research here with interest because it is of you know something related to what they are working on. In general when you look at Masters Students and PhD students there is always this perception that they are specialist, which is true they become specialists in a specific area that they working on. And so when they graduate and look for positions there is always this thing that this concept that have to find something that is relevant to what they have become a specialist in. So where do you typically find MS, PhD students graduating from your department from Applied Mechanics what sort of positions do they get and what sort of organizations do they get into?

Prof. Mahesh V Panchagnula: It's a very good question; you know placement at the PhD level is nothing like placement in the undergraduate level. So yes, we are dealing with people with specific skills, but IIT has set up mechanism for industries needing specific



skills to reach out to the candidates who are graduating with those skills. I think by and large a fraction of our candidates of our scholar graduating go into industry positions that build upon the skills that they already have acquired. A similar fraction roughly about, let's say a quarter of our students go into positions where it may not be directly related to the skills they have acquired, but they have that background information necessary to grow into a position that the industry needs them to get into.

But if you look at the problem from the other side from the pull side you know as to where are the jobs for PhD students specifically. The biggest employer in India is the Academia. We now have you know about 16 IITs plus more and more getting added next year. We have several NITs; apart from these we have very good state institutions not to mention the multitude in it's private institutions. They are all looking for good faculties' members and they are all reasonably well paying positions. So a fraction of our faculty of our research scholars do go into these positions, many of them are candidates they already faculty they come gain the training here and go back to become better faculty members where ever they are.

But I will say I will encourage every graduate in PhD students to look at an academic position, because if you want to set this in other 4 as well, but if you want to be a service to the country that is 1 position where India needs you. You look at the demographics and it does not lie, the number of young men and women that are going to come into colleges in the next 10 to 15 years is humongous no other country in the world has this kind of challenge ahead of them and the PhD scholars graduating from institutions like the IITs are the ones that have to solve the training problem there.

Prof. Prathap Haridoss: Okay I think that it's a very important point for young people to look at, because the promise of the nation is also in their hands to deliver. Let me know we have spoken about various issues associated with the PhD process and as it applies to your department and the areas that they work students work in and so on. Let me ask very mundane question I mean during their stay here one of the things, one of the aspects of you know graduate students like is how much they interact with say each other and how much they interact with advisor. In this context I mean in your experience and your view, how often should students be meeting their advisor or guide and how does that know come out come out in play?



Prof. Mahesh V Panchagnula: I think I will say initially you probably want to have a fairly high frequency of meetings, just so you are getting to know each other and then typically I find that every advisor-students pair settles into their own comfort zone of you know how often they meet. Several of our faculty holds regular weekly meetings of the whole group and they may hold more frequent meetings with individual students. We have a few faculty members who, after having gained a certain level of confidence in the scholar may suggest that this scholar go about their own independent research and meet them even less frequently than weeks.

So it depends completely on the advisor and students, but I would look at that in the sense that this is something that you can define with your advisor, it's just two of you sit down you decide level of interaction that is appropriate for you. And we also have a very thriving external program, where the meetings are obviously less frequent but are just as productive in terms of research. So, I will say one last thing as whereas the advisor-student and student-student interaction go, you know we are now at IIT a predominantly PG institution and in the Department of Applied Mechanics is only a PG department.

So, we are all human beings and very often we see conflicts arise, whether it is a student-student conflict or advisor-student conflict, don't be unnerved by that, there are professional mechanism set up to handle conflicts at all levels and I will say that we are also professional beings, so I mean I don't expect to be likeable to every human beings on this planet and if there is a conflict we have a mechanism to professionally resolve it move on and still be friends.

I think, at IIT we take pride in that aspect that often we sometimes have issues with guides and students, but we have a professional mechanism that takes good care of that students graduate, they are very happy when they graduate even after having gone through some issues like this. But I must also point out there is a minuscule fraction of that 2000 odd PhD students that we have on campus that have to encounter anything like this. So, if you have these issues in mind, if you are apprehensive about trying to select a guide I will say don't be fearful go in because if something does not work out there is a

Prof. Prathap Haridoss: There are lot of options available here.

Prof. Mahesh V Panchagnula: I mean just go in with a free mind I am sure things will work out just well.



Prof. Prathap Haridoss: So to wind up I just wanted to get your words on what are the words of advice that you would give to an aspiring student in Applied Mechanics?

Prof. Mahesh V Panchagnula: I will say come in with an open mind, be ready to do interdisciplinary work because that's where world lies today. Many of the problems cannot be put in vertical silos, where I am a mechanical engineer and all I learnt in UG mechanical engineering is enough to solve this problem. So be ready to be more interdisciplinary. And I will say come enjoy the ride, you know to me the third party is to anybody anywhere in the world research is a walk through an uncharted territory, uncharted open area.

So you know you succeeded when your peers outside IIT reward you and say good, we like what you done, whether it is a publication, whether it is a pat on the back at the end of a conference presentation. We all look in the short term that is a very clear success measure, look for those measures and mind you, your peer group is not your immediate 10 students around you, it is somewhere else in the world that's why we publish in international channels. And I think look for those, look for those publications you know, aspire to be to publish in the best channels and I think you will be just fine for the rest of your career.

Prof. Prathap Haridoss: Great. Thank you Dr. Mahesh for joining us, it was a pleasure.



Introduction to Research
Prof. Prathap Haridoss
Prof. Abhijit P. Deshpande
Prof. Renganathan T
Department of Metallurgical and Materials Engineering
Department of Chemical Engineering
Indian Institute of Technology, Madras

Lecture – 33
Research in Chemical Engineering

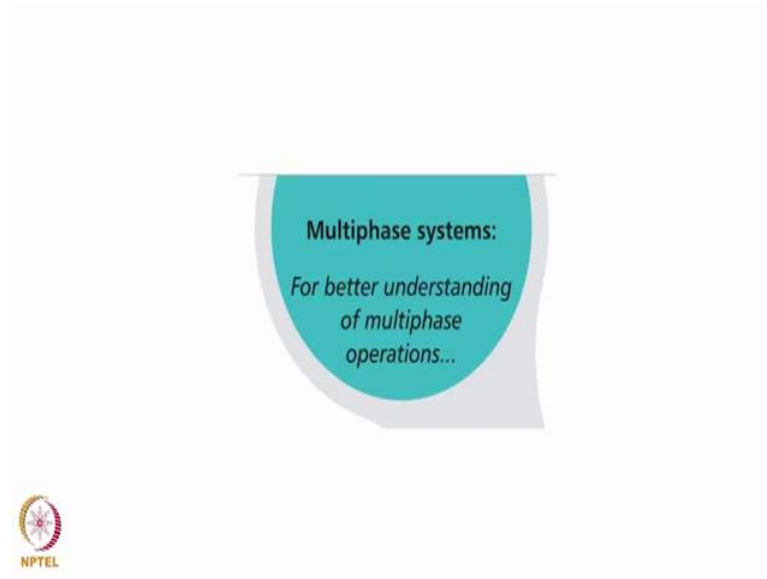
Prof. Prathap Haridoss: Hello, so in this module we discuss Chemical Engineering. So, to discuss Chemical Engineering, we have 2 faculty from the Department of Chemical Engineering at IIT, Madras, who has joined us this morning. We have with us Prof. Abhijit Deshpande, he has a PhD from the University of Washington, he is been a professor and a faculty here at IIT Madras for 20 years now, so he has a lot of experience in teaching and carrying out research in an academic setting, his areas of interest ..research areas of interest include polymers and rheology. We also have a Dr. T Renganathan, he has a PhD in Chemical Engineering from IIT Madras. He is been a faculty here and associate professor here and been with us as a faculty for 6 years and he is also guiding all the incoming PhD students through their initial training processes here, as they get on to carrying out their research in the department.

Between them they have a lot of experience in guiding students, in handling them, leading them through the initial years here and so perhaps they will be able to share their experience with us and tell us a lot about what chemical engineering is about. So, let me begin with this general question, what are you know, of course chemical engineering is one of the old fields of engineering has been around for a long time, what are typically considered as a traditional areas of research which have been there for a very long time in chemical engineering?

Prof. Renganathan T.: Okay. That traditional areas would be based on, let's say fluid flow heat transfer, the courses which are very much familiar to the undergraduate students, Fluid flow when I say applied to systems where you have multiple phases present, simultaneous presence of let say a gas bubble sparged in a liquid column which you called as a bubble column.



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And just like student study fluid mechanics in undergraduate, we study the fluid mechanism in multiphase systems. There has been on traditional area a lot of work was has been carried out across all IIT's, there was also on traditional areas. And then not alone fluid flow, mass transfer are occurring in this equipments the heat transfer and then students to be familiar with the residence and distribution studies. All these are aspects are carried out in detail for these systems.

Prof. Prathap Haridoss: Okay you yourself are an expert in multiphase systems and classification.

Prof. Renganathan T.: Right, right.

Prof. Prathap Haridoss: Okay so, these are, you may mention about traditional areas. So, an incoming graduate students would, if they picked up these kinds of areas you know to work in then there is a lot of pre-existing literature in these areas spread across several years.

Prof. Abhijit P. Deshpande: Also I wanted to add that sometimes these very traditional areas are very essential for a very new topic also. For example, fluid flow in a micro



fluidic channel, so again we have lots of chemical engineering researchers doing work on this, they are actually doing fluid mechanics, but they are doing it for a micro channel where again multiphase may be involved or we have also heat transfer where micro wave heating is being done for a very large scale sterilization.

Prof. Renganathan T.: Fluid flow in a fuel cell, there also flooding phenomenon is involved the flooding phenomenon.

Prof. Prathap Haridoss: Yeah.

Prof. Renganathan T.: Is involved even in a traditional column and absorption column or you are flooding even in enough fuel cell. So, the field of multiphase system are evolved and now applied to newer problems like Fuel Cells and Microfluidics.

Prof. Abhijit P. Deshpande: So, more I mean good way to think about it also may not always be traditional versus novel in terms of area as a whole, but one specific topic may have a more traditional feel to it versus the newer feel to it depending on what precisely you are trying to do in that topic.

Prof. Prathap Haridoss: Okay so, I mean I do understand that you know there is no hard and fast demarcation of.

Prof. Renganathan T.: Similarly, experimentation, for example, like gross measurements where in terms of overall measurement pressure drop and so on. But as it evolved let's say PIV measurements, local measurements were done, so the depth to which measurements were done or even let say even in terms of modeling, reactive level modeling and then CFD level modeling now, let say lattice Boltzmann modeling so, but still applied to a let's say a multiphase system or a fluid flow phenomena.

Prof. Prathap Haridoss: So the difference is in the details.

Prof. Renganathan T.: Details both in terms of experimentation and in terms of modeling.



Prof. Prathap Haridoss: Okay so, I mean I do understand that you know I think across all engineering divisions this is probably similarly true, that you know there is no hard and fast division between old topics versus new topics. But at the same time, I mean are there areas of research that you feel have really come up in the last let's say 5, 10 years which would be, which people I mean consider as a newer areas of research, where may be there is not enough literature from the past and where there lot of new activity going on?

Prof. Abhijit P. Deshpande: Just before we go to the new areas which are there, there is couple of other examples, where it's the old areas with a very new emphasis. And one example is something called Process intensification. Where the idea is that the earlier way of doing chemical processing with multiple operations can be now intensify the process in such a way that we can integrate 2 or 3 processes in 1 operation.

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Prof. Prathap Haridoss: Ok.

Prof. Abhijit P. Deshpande: So, this whole field is called Process intensification where you are doing reaction and separation simultaneously.

Prof. Renganathan T.: Or miniaturization of equipment.



Prof. Abhijit P. Deshpande: Miniaturization of equipment. So, that all of this is actually again you are doing quote and quote traditional work, but actually at a much more efficient way, much more much less energy consumption, may be much more demine in terms of environmental impact. And so these are again examples where processing.

Prof. Renganathan T.: For example, heat transfer as in traditional area inter looking.

Prof. Abhijit P. Deshpande: Yeah.

Prof. Renganathan T.: Pool boiling flow boiling etcetera, but now same thing applied to microwave heating as taken a new view and so on.

Prof. Abhijit P. Deshpande: And, similarly we also have systems and control where again the controls that we can do and the systems we can analyze has with our mathematical computational tools, we can do far more now compared to what we could do earlier and the decision making that we need to do in a chemical plant again we can do it in much more integrated scale, we can borrow ideas from signal processing that electrical engineers have developed for communication and then use them here. So, again this is again border line of novel and traditional area.

Prof. Renganathan T: Earlier it was conventional strategies that have been done, later on evolved to process systems engineering, which involved the whole flow sheet design and then system identification, more on data analysis as they are integration electrical engineering department.

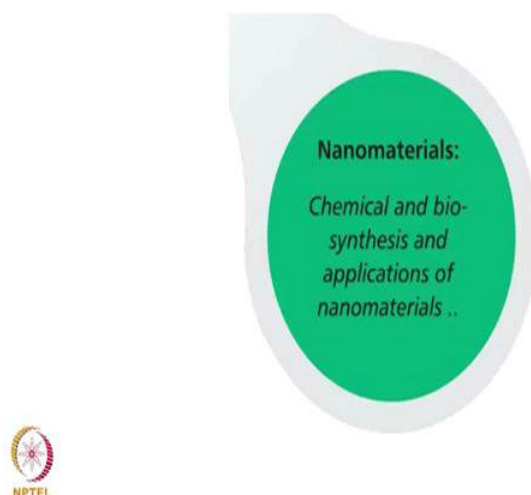
Prof. Prathap Haridoss: Ok.

Prof. Renganathan T.: Now applying the same principle to what distribution it works, fuel cells control of micro fluidic devices, but still the principles are all controlled. But now evolved to more encompassing, more domain; same time apply to modern tools, modern devices and so on.



Prof. Abhijit P. Deshpande: Talking about new areas, we generally in chemical engineering extremely exciting topics are being researched. For example, polymer which we use for insulation, but these days devices are being made from conducting polymers.

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Colloidal systems are being designed for photonic or drug carrier applications. Drop drying, which is a simple fluid mechanical technique is being used to diagnose materials of course with all the emphasis on smart devices, novel electrode materials are being prepared using nano materials. Excitingly also we are using bio processing, so using microbes to actually prepare nano particles. So these are all exciting new areas where chemical engineering is doing more and more research these days.



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As important to do research on these materials; how to process these materials? How to optimize them? Is also equally important and for example, one important area in the last 2 decades or so has been how to process materials for semi conductor applications? We also have more and more foods coming from an industrial setting instead of being done in our kitchen. So, large scale processing of food materials, for example, in micro wave heating thawing of materials is an important research area.

And all of the topics that we have talked about so far have been about information gathering as well as analysis at large scale, but chemical engineers by definition can deal with atoms and molecules and chemical transformations. So it is natural that we work on molecular scales and very powerful simulation techniques are being used at the molecular scale to understand material behavior and their processing.

Prof. Prathap Haridoss: When you look at post graduate degree like MS or a PhD especially in engineering, there is always this perception and partly true that you know we think of the people who complete these degrees as being experts in that area. So, I mean may be that lends itself more naturally to an academic setting. But what degree is industry interested in what the research activities that go on in the general field of chemical engineering?



Prof. Abhijit P. Deshpande: Fine, yeah, if you look at that the best way to look at that is because we have lot of talk about make in India now, and if you look at the sectors that sort of have been identified as key sectors where India should make an impact and lot of investment is being sort.

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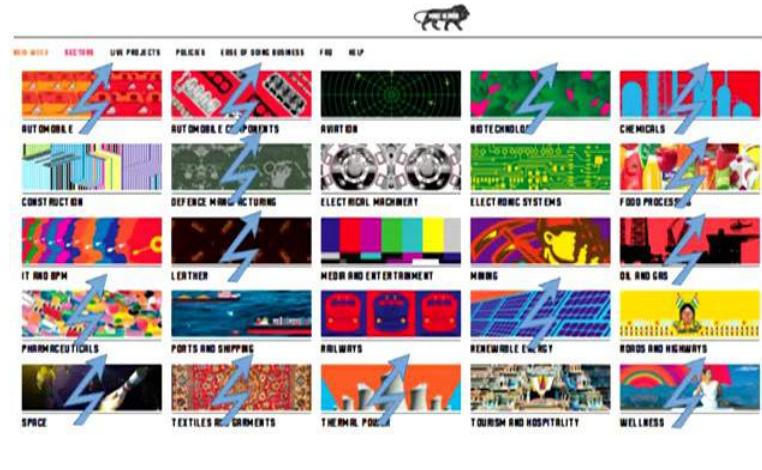


And so, if you look at, there about 25 areas which the Government of India has identified and if you look at the sectors where chemical engineering is directly involved.



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Chemical Engineering directly involved



We have fair bit of, there is oil and gas of course, pharmaceutical, chemical sector itself. But we have also the areas like biotechnology and even wellness where it's let's say research related to the traditional medicines, so in all of these chemical engineering is involved in terms of raw material processing, separations and then the process engineering and then finally getting a material of a desired quality. And even areas such as avionics and electronic system there is significant research that is going on.

Let's say for in our department, where the people are looking at sensors, either for a environmental application which is a very chief sensor or looking at sensor in terms of trying to replace an artificial muscle where it's both sensor as well as actuator. Such applications are being done even in chemical engineering and so all of these are industry relevant topics.

Prof. Renganathan T.: Example related to energy ozone, gasification, biolysis then micro reactors are supplied to pollution treatment devices although then CFD studies applied to various chemical engineering equipments and other equipments as well.



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Prof. Prathap Haridoss: So, I think scale in sense very large chemical industries out there you know in both nationally and internationally.

Prof. Abhijit P. Deshpande: Internationally.

Prof. Prathap Haridoss: So, I think many of the activities you do, I think really help them move to in your levels of sophistication and how they handle it there.

Prof. Abhijit P. Deshpande: Yeah.

Prof. Prathap Haridoss: So, if you look at you know incoming MS, PhD students and you know maybe the things that they are anxious about, the things that they are unfamiliar about and so on. I think may be in school or college, undergraduate college there is clear metric, on you know how well they or how successful they are, there is always an exam, there is a grade and there is a mark and so on.

Now, when they get on to it early to the especially to be early part of their research carrier, I think many things are not very clear. So, in your experience I mean having both done a PhD and now having guided many students, having spent a lot of time in the



academic setting. What could you recommend students to look at, as ways of understanding their progress in research or their success in research? How should they determine? How should they evaluate it and you know how should they you know look at it?

Prof. Renganathan T.: Okay so, when a PhD students or graduates, at least the way in which we look at it, is whether he has a clear understanding what he has done, the bare minimum. And then he is able to explain that to others at different levels, maybe his own friend present before professionals, working his area that something bad. Then next stage maybe to identify the scope of the work he has done, you should know the limitation.

Any research is not complete, so you should know the boundaries with in which he has done his work. That immediately sets him the goals for his future research as far, so can identify research topics or research areas to work on and then of course, he is able to carry out research independently so that he can take up students when we go and takes academic carrier especially. All those are good metrics which you can say that it is a successful researcher at the end of his PhD program.

Prof. Abhijit P. Deshpande: Yeah. And also, of course given that in research is important to communicate our ideas, we have to have publications that come out of research. There has to be a fair representation of his ideas in the open literature by communicating, some papers, having some papers at the end of the PhD.

Prof. Prathap Haridoss: So that I mean sets the standard in terms of, it is a piece of work that they have carried out, that is peer-reviewed, accepted by the peer group. Fine okay so, again as I mentioned earlier typically, students finishing MS and PhD are considered you know experts in that area. So, what sort of positions do they tend to get? What sort of I know placement or positions do they tend to get after an MS or a PhD degree in chemical engineering?

Prof. Abhijit P. Deshpande: So, it varies basically, further research work so, for example, MS going on to PhD or PhD going on to post doctoral work is one set off. Then the other is in terms of academic positions. So, master students can also join a teaching position



and a PhD student can also join an academic research teaching position and then industrial R and D. So, the mix is usually all of these depending on some years for example, in 2005 period my perception was lot of them were joining industrial R and D because there was an expansion of industrial R and D in India.

In the last few years, we have see that there have been more academic positions but again that might change, depending on in our lab for example recently 2, 3 people have joined industry again. So depending on the group the topic that you do, as well as depending on the time when you graduate the mix of these 3 vary; further studies, further research, academic position or industrial R and D.

Prof. Renganathan T.: And few even ideas of entrepreneurial thing, my own student is planning to sort of entrepreneurial, identifying ideas to setup his own.

Prof. Prathap Haridoss: So now, see if you look at the mix of students who come into engineering departments in certainly into chemical engineering and so on. So they come from various backgrounds different types of training that they have had and so on. And they are coming into an academic setting which may be has it is own culture and processes in place. Are there specific issues that students tend to face not just in our department, but in generally in chemical engineering as a whole, when they come in for post graduate studies? Is there something in the undergraduate study that is not know completely preparing them for a post graduate kind of study in the chemical engineering? And I mean what sort of approaches then they would they would need to take to you know overcome those shortcomings?

Prof. Renganathan T.: Okay so the first one can be termed in terms of the course work what they take in their institutes and then in IIT. The number of causes may be less here, but then the rigor the level of analysis that is being done in any course is very detailed. So they will have to like climb up to get that level of attending classes and then analyzing and then submitting assignments and so on. So, rigorousness of course work is one I would say.

Prof. Abhijit P. Deshpande: For the graduate school, it is always much different



compared to an undergraduate.

Prof. Renganathan T.: Yeah, that's right. Then of course, in terms exposure to the computation and techniques not, let's say MATLAB or a CFD tool or a simulator tool and so on. That is almost like integrated part of few courses here. And having known them was useful to just take on, rather than learned that after coming here, that is one. Now, once again the students are introduced to the lab classes in their as part of the UG curriculum, but there even not pay attention to preciseness, accuracy and repeatability and so on. Just say a lab class, but now when they start up MS program or a PhD program where enough detail attention has to be paid to the experiment, the way it is carried out preciseness reputation. They should hear for that as well, which is we do not expect to UG and graduate to get train for that as part of the undergraduate labs. That is one the way in which they look at experiments it should be different.

Prof. Prathap Haridoss: There is something of retuning their mind set approach.

Prof. Renganathan T.: Yeah, yes.

Prof. Abhijit P. Deshpande: One other important retuning that has to be done is at 2 different levels, one is always going back to an undergraduate course as a starting point to think about any set of research area and all that that may not be always very useful. So, like we already said. So, say I want to do in heat transfer may not be a very useful idea because what you have to see is, what are the research area related to heat transfer and they might require very different fields for you to work on that topic. So even though notionally it might seem like there is heat transfer involved in the topic it may not be related to the undergraduate subject that you have done.

So, you need to delink little bit of your aspiration for research with respect to the course work that you have done. It's a good starting point but it's not going to be, because as we always keep on it is in the book then it is not really research then. Something much more than what we have been exposed to as course work will be involved in research. The other extreme of it is to sort of be completely swayed by the new buzz words of a nano technology, again and that is not again is something we should always, we can examine



research topics for their details and things like that and then pursue them.

Rather than get swayed, saying that I am working in nano technology, so by definition whatever I will do will be ground breaking because in the end you will have to, as we said you will have to learn techniques, you will have to understand, you will have to be able to explain, you will have to be able to achieve generate new results and then analyze them and communicate. So all those are very important rather than either just hopping back on one course subject or looking at very hot new area. So, much more holistic, we should be other research, we should be tuned to looking at in a more holistic way rather than pigeon holing our own research topics or our own research areas.

Prof. Prathap Haridoss: Okay so, I been I think in this all the points that you made about you know how the view of a student should change in evolves they become a graduate student. I think one of the things of in a post graduate study, life of a post graduate student is the fact that you know a fair part of your learning process, part of your performance is also got to do with how well you interact with a rest of your group, how well your interact with your advisor and so on. So in this context, in sort of in a mundane way, how often you think students should really meet their advisor? Is there like a guideline that you gave and what are you looking at in this kind of scenario?

Prof. Renganathan T.: When you come to the meeting advisor, the frequency of meeting changes as the progresses along the program. So, let say initially in 1 year he may be meeting very frequently and then for everything he may just go to the guide and so on meet him. Other than that, of course all faculties usually have weekly a formal meeting with the students, other than that of course he can meet him any time whenever required and so on.

So as the years progress and then the number of meetings gradually come down and then also in the student point of view, he will become more independent not to approach the guide for every small thing, which he comes across and one more point of meeting that I would say is he should go on meet his guide when he is not getting result for a prolonged period of time. The results may be wrong that may be interesting may would have happened there. So, the guide based on is such experienced can identify whether it is a



wrong observation or something interesting. Students should not be in a shell that something wrong, he should not be afraid to go and meet the guide. So, these things I would say.

Prof. Abhijit P. Deshpande: And also, the other big situation that we are in as a researcher is actually we will have not just the advisor but a whole lot of other people working around us. So it's a good idea to develop this habit of talking to your own other research scholars because that is a good sounding board also. So, not only should we meet advisors very regularly as Ranga mentioned and every week is probably very good in the beginning and maybe it can become once in 2 weeks if let say you are progressing very well and you are doing very well.

But once in a week is certainly a good idea, but also it's a good idea to every month at least sit down with one of your colleagues and then try to discuss and try to see some broad results, what you have got, are you thinking along in the right direction. If you are planning a major initiative may be just discuss with 2 other students in the lab and see you know this is what I am thinking about, before I meet the advisor I want to check with you. So such exchange of idea is not only with your advisor but with your peers is a very good idea.

Prof. Renganathan T.: And it depends on the guide towards end, instills him more confidence in him that he can do independent research. So that he can do independently, in order at the same time develop some more ideas for his future research and so on.

Prof. Prathap Haridoss: You are recommending that the slowly evolve to becoming independent that something that they should look forward to.

Prof. Renganathan T.: In fact, you are asking how do you judge? One judge is successful he is. Suppose, he knows more than his guide at towards the end that is big mark for us that you are happy that yeah, yeah.

Prof. Prathap Haridoss: So, I think we will yes I would like to conclude by asking you very general question, there are lot of students out there who are looking forward to



careers in research, research in chemical engineering. What are your words of advice for an aspiring chemical engineering post graduate student?

Prof. Renganathan T.: As our Abhijit was mentioning, we work in the recent areas of chemical engineering but the principles involved are still based on our fundamentals of fluid mechanics mass transfer and so on. Our interview process, selection process we do test the fundamentals only. Of course, first we have a short listing and then a written test and then interview process both the written test and interview process test their fundamentals. So, it's good for them to be very good at by the fundamental so that they are successful in that machine process and that is one major. As he said they should know distinguish between the fundamental courses and the research areas what we are we are doing in. It is better to go through a website and then get familiar at least have some glance, what kind of research is going on, so that they know that they should not be looking from a fluid mechanic a master sensor point of view, but look at topics where this principles are involved.

Prof. Abhijit P. Deshpande: Yeah, my general advice is not specifically related to students per say, but generally brought comment about chemical engineering as a discipline saying that you know chemical engineering is one of those few engineering areas where the molecules are still at the center of our considerations. So therefore, going from molecular scale all the way till engineering scale, wherever molecular transformations are involved chemical engineering will be involved.

With that in mind if you approach chemical engineering research you can see boundless sort of opportunities and that is why we showed you how a make in India also out of 25 area, there are 13, 15 areas where chemical engineering is centrally involved. So therefore, if you are sort of looking and considering career in chemical engineering research go for it I would say it is going to be very promising with all the developments that are happening in processors, materials, and energy everywhere chemical engineering will actually play a role. So there will be lot of scope for R and D in chemical engineering in future also.

Prof. Prathap Haridoss: Okay Great, on that positive note; Thank you Abhijit for joining



us. Thank you Ranga for joining us and sharing your insight on us.

Thank you.



Introduction to Research
Prof. Prathap Haridoss
Prof. Ravindra Gettu
Department of Metallurgical and Materials Engineering
Department of Civil Engineering
Indian Institute of Technology, Madras

Lecture - 34
Research in Civil Engineering

Prof. Prathap Haridoss: Hello, So, today we have with us Prof. Gettu Prof. Ravindra Gettu, he is professor in the Department of Civil Engineering here at IIT, Madras. He has a PhD from Northwestern University in Evanston in the United States of America and he works on concrete technology. He is also the associate Dean for Industrial Consultancy and Sponsored Research at IIT, Madras. So, he has a lot of experience working with you know various funding agencies, working with various sponsoring agencies and so on and as a very good in sight into what are the kinds of you know activities that go on as in the institute as a whole and certainly with respect to civil engineering. So first of all welcome.

Prof. Ravindra Gettu: Thank you.

Prof. Prathap Haridoss: To this interview. We just like to start with civil engineering, I mean most of us realize that it is been along for a long time and in fact wherever I go think that comes to my mind is that the first people who arrived here were civil engineers because the road is already there and the building is already there. So what are traditional areas of civil engineering that people do research on, but have been around for a very long time as areas of research?

Prof. Ravindra Gettu: Well, first of all thank you for having me here. It's good that people are interested in civil engineering. Lot of time, we think that people take civil engineering for granted and they don't think any research goes on. Well, at least in our department in most of the places we would have say 4 or 5 major areas of civil engineering. Structural engineering, which deals with the construction of facilities like building, bridges and so on. And then we have Geotechnical engineering which deals with foundation, how these structures are built on? The mechanics of soils and how safe



should the foundation be.

Then we have Environmental engineering, which deals with how people and constructed facilities effect and interact with the environment. We have Water Resources or what can be also called Hydraulic engineering, which deals with facilities like canals, dams and water supply as well as say sewage networks and sanitary networks. Then we have Transportation engineering, which I am sure anybody who has used any road or any system of transportation in our country understands the intricacy and the complications. So, they deal with how to make traffic flow better, viability of different systems and they also look into the future to see how planning should be done.

Then we have a group such as ours, where I come from which deals with construction management and management of projects and to make them more efficient, safer, cost effective and delivered on time. We also deal with materials, materials technologies that are particularly what I do. I work on concrete and other construction materials. Then we also have a group which works on building Physics, like to make sure that our buildings are more comfortable ventilation, thermal, comfort and lighting. So, these are the different aspects. So in each of these areas there is lot of research which goes out. So, I am sure that any one who is interested in research in civil engineering would find an area that is suitable for him.

Prof. Prathap Haridoss: In these range of activities that you are mentioning is there something that's relatively recent that has come on that people have gotten on to in let say, in last 10, 15 years as supposed to something that is been around for a much longer period of time?

Prof. Ravindra Gettu: Well, Civil engineering research like you said in the introduction as being around, there are lot of new things coming in. I works in materials, so the first thing that comes to mind is the new type of materials which are coming in. Polymers are being used extensively in civil engineering. Civil engineering materials where traditionally stone and brick and cement, motor and so on. So but the polymers are coming in very big. Then other technologies, for example, Information technology is affecting a lot, how we model and how we predict anything going from traffic flow to



how a structure behaves and what are the interactions of environmental on a structure. Environmental engineering has become very big now.

This boom in environmental engineering I would say started of long back in other countries, in India may be much later, but in the past decade we have seen so much of research in environmental engineering within the realm of civil engineering. So, there is a lot of lot of studies being done on how to make a water supply safer and how to make sure that the waste that we generate is treated well and disposed off in safe conditions.

Prof. Prathap Haridoss: Ok

Prof. Ravindra Gettu: So, there are a lot of things coming up.

Prof. Prathap Haridoss: When students come to your department for their post graduate degree. So, they all typically would have some kind of a you know under graduate degree on I mean related to civil engineering and they come, most of them I will assume may be you do take students from other backgrounds to. Are some common issues that they face in settling in into a research environment coming from, I mean a course related environment which is what they would have done during under grad or even otherwise just settling into the kind of research activities that go on you know in advanced civil engineering department are there these specific issues that they tend to face?

Prof. Ravindra Gettu: Well generally they do not face any major problems. One thing that comes up often is the way that they have studied in the bachelors or some times even in the masters, is not really amenable to how we expect research to be done. To give an example, lot of students have studied answers for specific questions. And then what we find out is they are not able to analyze new situations or when we have them to do research, they expect or they seem to feel that it would be a series of steps, where the guide tells them to do this and they come back. But when we expect them to analyze the results we find that to be lacking and this comes back to the fact that when they have studied they always being told to study a certain set of answers, for certain set of questions that out of those questions they are examined. So we find out that sometimes that analysis or the ability to handle new type of questions, new type of problems is



lacking. So, we spend say better part of the first year bringing the students up to this level that we required and I would say most of the students who finish say the PhD or their masters now go out with a ability that we expect in a good research.

Prof. Prathap Haridoss: Okay so, in general as things as they leave the institution with the degree in other ways that you feel are the appropriate ways to measure success in their research activity, of course, they are getting a degrees so they have succeeded in some way, but is there something more subtle that you look at and you say this person was a successful researcher?

Prof. Ravindra Gettu: Well, how I would judge researcher say, PhD students still at IIT is, if that person has the ability to analyze and decide, reach decisions based on their results or their research I think they are good researchers.

Prof. Prathap Haridoss: Ok.

Prof. Ravindra Gettu: The proof of the pudding is when they graduate and go out, and they if we follow the career I would expect that say in 5, 10 years they should be one of the top researches in India then we are successful.

Prof. Prathap Haridoss: Ok.

Prof. Ravindra Gettu: In our work in bringing them out as research.

Prof. Prathap Haridoss: So, I think in fact, talking about you know what happens when students live the degree program, I mean in after they graduate, so of course civil engineering is very well connected with the industry I mean just direct link between what you do and may be what very large fraction of industry or that construction industry looks for. So, that link I think most of us understand. Other other industries which pick ups civil engineers where maybe the general public maybe not as aware of it, is there some unique situations like that you can tell us about?



Prof. Ravindra Gettu: Well in our country most of our PhD students go into teaching.

Prof. Prathap Haridoss: Ok.

Prof. Ravindra Gettu: We have, I think I would say about 90 percent or more of our PhD students would go into teaching. Lot of our masters students, those who do MS as well as M. Tech go into the industry. So, they have been people going from chemical manufactures up to structural designers. The whole range in whatever we deal with, there are people who go into the industry and in terms of PhD as I mentioned the openings that they get into are mostly in teaching, but we have had students who go into administration. We have people who go into the IAS, IES; we have students who have gone into the IFS, Indian Forest Service. So, some how civil engineering seems to give them the holistic view, which they hope will be useful them when they go into administration.

Prof. Prathap Haridoss: Okay very nice. So, may be some away bit of a mundane questions now coming of this more philosophical question. Students come in and they become graduate students here for may be say 2 years, 3 years, 5 years depending on their degree program. They go through various phases of learning courses doing research and so on. What do you think is a frequency with which students should meet their advisers and what can you give some input into the kind of interaction that guide and a student should have through the course their PhD?

Prof. Ravindra Gettu: Well it is sort of a difficult question to give a generic answer.

Prof. Prathap Haridoss: Ok.

Prof. Ravindra Gettu: I would say they should meet the guide as often as necessary.

Prof. Prathap Haridoss: Ok.

Prof. Ravindra Gettu: Now, as a thumb rule I think once a week is good enough.



Prof. Prathap Haridoss: Ok.

Prof. Ravindra Gettu: Once a week is good enough. I do not like students to come in every day and ask what they should be doing.

Prof. Prathap Haridoss: Ok.

Prof. Ravindra Gettu: And I think that does not show their independence and I think the guide also should not be doing this. But once a week at least is a good frequency and towards the end of thesis when the thesis has to be corrected, I think more often would be good.

Prof. Prathap Haridoss: Ok.

Prof. Ravindra Gettu: Now this interaction could be in groups. I know research groups like mind, where we have a research meeting of the whole group every week. So the interaction could be in a group or could be one on one.

Prof. Prathap Haridoss: Ok.

Prof. Ravindra Gettu: Depending on the students and I would say the guide.

Prof. Prathap Haridoss: So, may we close with this question, what advice would you give to an aspiring research student in your department?

Prof. Ravindra Gettu: Well, I think the most important thing that I would say and this is something that I ask students say in their comprehensive exam or research proposal. I ask students, how is your thesis going to make the world a better place? This often we do not think how is the thesis going to be important. So, I expect students to believe in themselves, believe in the problem and believe in the solution and know what we are going to solve. So, this is something that sometimes students are caught by surprise.



Prof. Prathap Haridoss: Ok.

Prof. Ravindra Gettu: And this comes back when I was doing my masters. I was working in a problem I thought I was very exciting and when I was applying for a PhD somebody ask me, what is the use of what you are doing? And that made me think I went back to my adviser and said, why I am doing this? Where can I apply this? So, I always ask my students and I tell them, pick up a problem or work on something that is going to have some effect in some way and understand that and work towards it and that I think makes it more pleasant and I think holistically that is the better way.

Prof. Prathap Haridoss: Ok.

Prof. Ravindra Gettu: Certainly.

Prof. Prathap Haridoss: So, thank you Prof. Gettu.

Prof. Ravindra Gettu: Sure.

Prof. Prathap Haridoss: Thank you for joining us.

Prof. Ravindra Gettu: Ok.

Prof. Prathap Haridoss: Pleasure.

Prof. Ravindra Gettu: Thanks Prathap.

Prof. Prathap Haridoss: Thank you.



Introduction to Research
Prof. Prathap Haridoss
Prof. V. Kamakoti
Department of Metallurgical and Materials Engineering
Research in Computer Science & Engineering
Indian Institute of Technology, Madras

Lecture – 35
Introduction to Research

Prof. Prathap Haridoss: Hello, we are very happy to have with us today Prof. Kamakoti from the Department of Computer Science and Engineering, for this department module on Introduction to Research. He has been a faculty in the Department of Computer Science and Engineering for 15 years and has therefore you know interacted a lot with lot of students who have come through the department over the years and also has a lot of interaction with you know researchers around the world and the industry. His areas of research interest include software for VLSI, reconfigurable systems design, compute to architecture and secure systems engineering. So welcome to this class, discussion.

Prof. V. Kamakoti: Thank you.

Prof. Prathap Haridoss: So I would to begin with asking you, you know computer science and engineering has been around I mean for a while and lot of people at least in India, there is a lot of desire to be in computer science and engineering education process. So what are in terms of research though, in going away from under graduate education going into research? What are considered as traditional areas of research in computer science and engineering?

Prof. V. Kamakoti: So Computer science and engineering as you rightly told is a most desired field of many, many people in the country. Reasons are many, if you look at the way man lives today, the actual intervention of computing devices is increasing in a rapid pace. One of the main areas; let us start with the one of the main area which internet of things. Today the entire health, the entire schedule, everything that a man does could be actually monitored, recorded for basically providing some sort of service to the person or to actually for other monitoring purposes using devices.



So, the electronics and basically the compute part of the electronics has become closer to the man kind and that has opened up several you know research areas. The traditional areas if you look they are hardware, software applications and theoretical computer science. In the hardware, basically people were looking at how to make hardware that will work, that can be sold in the market and specifically they were looking at how much power it will consume? How much weight it will be? And how much heat it will be dissipate? To basically see that it is deployed in some environment, these are traditional areas of research when you look at hardware.

The second on the software side, people are looking at how to build, how to actually provide certain infrastructure by which people can develop you know software on the system. Software is of two types, one is the system software which basically comprises the operating system, which comprises the compiler and then you know certain build environments like, for example, certain design kits that we call and lot of a research has gone there which we called traditionally a software engineering of how to make tools that can enable people build large scale software.

The other aspect of software is the applications, where you actually build applications that suits different environments and governing all these three is the theory, which is the foundation. The theoretical computer science basically tells you that, if you want to solve such a problem what is it that you can expect, how much time it will take for a system to execute this particular, to solve this particular problem. So, you get some ideas and based on that ideas you refine your solutions and before actual you start implementing a solution the theory basically gives you lot of insight into how well your final solution could be. So to answer your question in to some of with answer for your question, there are four major areas of research traditionally and it continuous today hardware, software, applications and theory.

Prof. Prathap Haridoss: Ok right, I think nicely summarized in that way about what computer science, in general is in a traditional sense. In the same context, I mean of course, even in traditional areas of computer science like in other form I mean disciplines in engineering. Even in traditional areas I am sure there is you know the latest you know versions of what is happening, and how people work on it. But in addition to that, in



addition to all these areas and the you know cutting edge of those areas, are other areas of research in computer science that are considered new recent areas of research?

Prof. V. Kamakoti: Yeah so, as I started just conversation one of the new areas of researches is internet of things. Internet of things, basically the research there is that I need to monitor something; I need to sense them data for whatever purpose it may be. For example, I need to monitor traffic, I need to monitor pollution level, I need to monitor temperature, I need to monitor say tsunami today. Now, how do I sense that data, how do I take it out and then send it to a global monitoring place, a communicate over the network and then somebody goes and use that data and say there can be a tsunami, the pollution level is high. So there should be somebody who makes a decision based on the data so this entire notion of internet of things as opened up several challenges in the traditional area of research.

For example, sensing is a interdisciplinary topic so we start now interacting with say people who make sensors. Suppose, I am monitoring the structural health of a building, I start interacting with a civil engineer okay, when I want to you have a car which as smart hybrid vehicle system, you start interacting with IC engine person to basically understand. All these are data collection points and that is in now becoming an interdisciplinary research of how I effectively and efficiently and accurately I capture the data. Once the data is taken, then I need to have mechanism by which I do some local processing , many times, so that comes under the hardware design, one of the traditional areas and that challenge is there is that I need to process that data with limited amount of competent. I cannot put a server at every traffic point, suppose I am monitoring the data.

Prof. Prathap Haridoss: Ok.

Prof. V. Kamakoti: Monitoring pollution, I cannot put a server in every signal. So, I need to have some very small devices, which can work probably in a solar type of environment and I need to do some processing, then after that I need to send that data to a server somewhere, that is a problem in computer communication so that is a big problem in communication. Because I may have limited band width and may not even have network connection, say suppose I am doing a remote sensing in high altitude area.



I may not even have network connectivity so how I gather the data and bring it, so that is another challenge that is coming.

Now, IOT is one very, very important challenge, now once you get this data what you do with this is data that becomes a problem of plenty. Now I have to handle this data, I need to visualize it, I need to process it quickly. For example, if there is a monitoring traffic, I need to tell very accurately within the next of few seconds that there is a traffic jam here. If I say 1 hour later, it doesn't make sense so there is some real time requirement for me to quickly accumulate that data, get to some conclusion about data, make a decision and communicate that decision, make an observation and communicate that observation out. This opens up a vast field called data sciences right or data analytics so that is another field, data sensors, data analytics becomes one very important field.

When you look at that field, now there is software by which I need to process the data, I need to write applications to actually extract the data and then this cannot be that just by viewing say millions of bytes of data, for me to come to conclusion is very difficult. I may have to monitor several situations and then come out with some observation and that monitoring mechanism comes under the purview of what you call as machine learning right. So there are machine learning engines, who would have studied this patterns for a period of time and then they will now say this pattern now is coming so this is should be matching on to one of those earlier things, there is a Training phase, machine learning is a another thing that crops up because of this internet of things.

Then covering all these things is now what we call as the information security, for example, monitoring a pace maker of an individual is now done by a device which can be remotely operated. So, it takes your heart parameters and it sends over a network and doctor can monitor. Now, if somebody could penetrate in to this network, a hacker can penetrate in to this network, now you can just go and basically stop the pace maker essentially you can do a cyber murder today.

Prof. Prathap Haridoss: Oh Okay.



Prof. V. Kamakoti: So that opens, as and more your electronics become close to your heart or close to your body and it becomes more penetrations, its penetration into your basic living becomes more and more increasing security becomes a major concern. Now, security is a habit, if you take human being, if that habit is not inculcated at the age of 5, you cannot inculcate it till symmetry, this is saying in Tamil [FL].

“ Anjula Valyadhadhu Ambadhula Valayadhu “

“ Thottil Palakkam Sudugaadu Varay “

Prof. Prathap Haridoss: Yeah, yeah.

Prof. V. Kamakoti: So these are all some statements we will see in Tamil, so the habit that you inculcate in the age of 5, will be thing. But, now computers are in its say around half a century, software development started, big software development started late 1960s even before we both were born, in our previous batch so that is at least now 50 to 60 years old today. Till now people have not been looking at security in a big way and suddenly the security which is like a habit has to be retrofit into this over grown old man.

Prof. Prathap Haridoss: Ok. Yeah, yeah.

Prof. V. Kamakoti: And that becomes a big problem. Today, if I ask you if this mobile phone secure? I do not have an answer. Will my data in this mobile phone be stolen or if they ask you a question, can somebody hack and get some personal details from this mobile phone? I don't have an answer right because this system has so much over grown and I don't know how to basically go and certify that it secure. This information security suddenly with boom of internet and suddenly people realizing that internet can be used for malicious reasons also, information security is growing up as one of the major areas of research today. So whatever you talk this is going to be one information security should be there so everything finally boils down to how secure is the information.

Prof. Prathap Haridoss: Ok.



Prof. V. Kamakoti: Okay the next thing that has happened over a period of time, so we have covered something on application and something on a software and machine learning and all these ends. The other important thing that has come up specifically in large scale computing, what you called as high performance computing is that, we could not increase the frequency of the processor, clock frequency of the processor. As you see we are not seeing any processor that is running more than say 3 to 3.5 giga hertz, the reason is that the power is directly proportional to the frequency and so if I keep increasing the frequency, the power will be so high that it will burn the processor itself, literally burn the processor itself.

Prof. Prathap Haridoss: Ok.

Prof. V. Kamakoti: People have now moved on from a single core model, a single CPU model to multiple CPU so I cannot run give you 10 giga hertz. But I can give you 4 CPU each with 2.5 giga hertz right, so that is possible for me. Now, you cannot actually find a processor CPU today with just 1 core, it is all multi core. The moment I have multi core, then one of the important thing that we need to research upon is, how do I make all the 4 cores work together?

So if your program is a sequential program, namely first step gets executed. So suppose, I want to add 100 numbers, suppose I say add the first number to the second number then to the third number then to the fourth number, this is a sequential program. Now, I need to say take 4 chunks of 25 numbers each, make the first processor add 25, second processor add 25, third processor add 25 and then collect a result and give one answer so is the difference between sequential program and a parallel program. Now, parallelization is now become another major area of interest right so we have to take a core program and see how I can make it into concurrent pieces which can execute at the same point of time and give a result. In nutshell, these are all some of the you know research, new research areas, these are very nascent in stage and there lot of gold mine of open problem for people.

Prof. Prathap Haridoss: Ok, very nice theory, very thorough perspective of; very detailed perspective of what computer science is today, computer science and engineering. Maybe



in this relation, since you both mentioned traditional areas and you know modern areas that are coming up in computer science. If you look at the industry, see they I would like to look at this from say the perspective of the research students who finish an MS or a PhD degree, what sort of activities that go on in computer science are of sort of more immediate interest for the industry, where you know students do research it's also something that they can see immediate applications to?

Prof. V. Kamakoti: So industry today can be broadly classified again in based on the areas of research. For example, traditional hardware design, where there is a lot of stress on today on how to make hardware that is despite very less power, how to make architecture that can suit different environments today. So, one which can do, today if you take a network appliance which for example take some packets and route it, it is also a computer. How do I make device that are very small and that used in millions for the internet of things type of application. There is a set of people, set of industry which are looking at these types of hardware designs of low power, high performance hardware and they are looking at different models of computation.

Today there are notions what you call as approximate computing, stochastic computing, non bunion computing, so different computation models are there and the research labs within these core industry have started looking deeply into this, so this is one line of that. Then the second line is that the systems are becoming very complex, so how do I model a complex system and then how do I verify it is working correctly. There is, how do I make a hardware which lots of complexity so the CPU that was thought for entire semester, in my B. Tech curriculum in 1985, today I teach the same thing within the first week.

Prof. Prathap Haridoss: Ok.

Prof. V. Kamakoti: There is bigger challenge here because history cannot be forgotten and at the same time.

Prof. Prathap Haridoss: Progress is.



Prof. V. Kamakoti: Progress is so rapid that what I learnt for entire semester, I need to teach it within 1 week here. So, essentially the systems are becoming more and more complex and how to model such systems, how to verify such systems, so that is another area. Hardware if you take, there is hard core design and hard core modeling and these two have lot of open problems and lot of job opportunities, so this one set. Now, when we come to the system side, that is the operating system, there are very few groups which work on operating system, there few very individuals in the country who can actually going to a kernel and come out with modifications to the kernel.

There are very few across the globe who can do, few in the sense in terms of percentage it will not even cost 1 percent of the total computer science engineering people who would be graduated, so because it is very, very complex. If people start working on these type issues like kernel level and the compiler basically optimizations and lot of system programming stuff, there is a lot of potential for them in terms of going into a very core research jobs be it a faculty position or an industry position or even a government research establishment versions, so that is also very, very interesting problem in terms of research, in terms of publication, in terms of livelihood everything it's a very good thing.

The next thing is applications, today application wise if you look there are two ends, one is an elephant namely data sciences. Data sciences today is formed between anywhere from, if we have 16 departments in our institution, every department has a data science related problem and that is proved that we have data science laboratory today which has 22 or 27 faculty members across all the 16 departments we have today, that is massive field that is one end.

Another end of this spectrum is, we have this IOT type of devices; in the IOT know the memory will be very small. I cannot write large scale programs, I need to have very effective coding efficient, coding techniques and stuff like that so writing those types of small programs on the devices that's also a very big today, that's a big skill and that we need look into it, so that is from the application side. If you look at hardware, if you look at system software, you look at applications, these are all the big research areas and of course, on the system application side, writing these type of parallel programs also is a skill and very, very few people across the globe has this type of exposure. The very few (



universities even offers such type of parallel architecture and parallel algorithm courses right and so these are something that great research interest here and if somebody aspects to do that, if they take anyone this area and spends 2 to 3 years and demonstrate to the world that they can think new, publish couple of papers write some sort of software I think a they have a great future in front of this.

Prof. Prathap Haridoss: That's I think very good perspective for you know incoming students to consider. Now, over 15 years we have seen several students come in and graduate not may be directly in your group and also around in various groups so on. In your opinion, see there are already some metrics that people use for gazing how you know progress in search is going on and how successful students are in research. In your view what are the parameters that you would look at to a two gaze a success in research that may be student should also consider looking at right?

Prof. V. Kamakoti: If you had asked this question, if you had asked this question in 2001 in 2005, 2010, 2015, my answers have would be different.

Prof. Prathap Haridoss: Ok.

Prof. V. Kamakoti: 2015 the answer which or probably 2020, the answer is that today see those days I could very well say I am a hardware engineer and then forget about everything else and things worked for me, like I could have a career based on hardware engineering just hardware I could go to company which look only a hardware and I could have a career successful career, but that career would have been short stint of say 6 to 7 years, now after that thinks start a things. Today I cannot say I am a hardware engineer everything today is cross layer.

Prof. Prathap Haridoss: Ok.

Prof. V. Kamakoti: So if you look at traditional computer science it should be therefore, any device you take in computing I have something called digital circuits and on top of it I am talking about a digital computer I have a digital circuit there is a micro architecture then an operating system then system software then applications. So, this is tag for me to



even go and tweak some parameter, for example, temperature control at it circuit level I need to know the entire stack I should write a thermal aware compiler, I should have a thermal aware operating system, I need to have thermal aware micro architecture, I need to have a thermal aware digital circuit. So, anything that I want to do anywhere in computing I need to have the entire knowledge of the stack and then only you get, you become a full person and if you don't have that knowledge then it sometimes it becomes a quite you know.

Prof. Prathap Haridoss: Shaky

Prof. V. Kamakoti: Shaky you can be bold over at some point of time. So, my sense of success today is if a student is able to do a research work where he demonstrates that whatever he has found out in whichever layer that 5 layers I told right digital circuit, micro architecture, operating system, system software application, each of there you would have some enhancement, but he should know demonstrate that is well there is a synergy of this and it will effective in all the 5 layers and he is able to argue it and bring it out, I think that is success with respect to and that is also the challenge right.

So, today is the challenge essentially comes normally I give questions in comprehensive viva and typically end sems to design as, they will give a very good answer, but it will take 2 billion dollars to implement. So, that is not engineering. I need to make a hardware or I need to make a system that could be effectively implemented that effectiveness is just not performance, but it is also cost and sell ability in the market. So, might have a very good architecture, but it cost that machine cost 1 billion dollar probably 1 fellow will buy it.

Prof. Prathap Haridoss: Ok.

Prof. V. Kamakoti: So, then it will not be successful. So, to have that a notion of cost performance and have the notion of you know the entire stack and it should be feasibility that is very, very important and that is how we, I feel that a particular research is successful only if it addresses, at least in computer science it addresses this spectrum.



Prof. Prathap Haridoss: Ok.

Prof. V. Kamakoti: Yeah.

Prof. Prathap Haridoss: And now I mean in that context in terms of the context of say successful being successful as a researcher and so on, when I think in general there is always an expectation there is always this feeling that you know when somebody does a MS or a PhD degree they are now specialist in that area and maybe that restricts what kind positions they can go for and that is always a concern that students have when they pick up in an advance degree in any field. In computer science what sort of positions do students who finish MS and PhD degrees tend to get?

Prof. V. Kamakoti: Okay So let me just take, I did PhD in absolute theory I have not done anything on systems but then, I got a job I worked for two postdocs they were more or less theory but then I got a job in the industry which was on processor verification is completely hardware then I came back and I did VLSI here and the today I also consult for a financial bank and I am able to do some you know IT activity there. So if you take essentially we started of the theory, but we are able to do hardware we are able to do application.

Prof. Prathap Haridoss: Ok.

Prof. V. Kamakoti: Just looking at myself that my choice of talking theory that time did not actually restrict me explore all these opportunities.

Prof. Prathap Haridoss: Ok.

Prof. V. Kamakoti: Whether I am successful or not we are some more years to tell that but now, but today I don't feel some handicap because I took theory there so that is reason why I put myself as a case example here.

Prof. Prathap Haridoss: Ok.



Prof. V. Kamakoti: I think computer science is very broad, actually I have seen very good software engineers even from our own institute they are chemical graduates.

Prof. Prathap Haridoss: Ok.

Prof. V. Kamakoti: And interestingly three of big shots I know in that field of software who have been vice president of multi software that goes on to amazon, why an amazon or yahoo or this type of sites, right if you put there within the next few seconds there will be one million hits or something. Whether you are software is working or not will be known within the next few seconds by time you wink your eyes three times your software stability will be not. So these types of projects people are handled and all of them are non computer science people. If non computer scientist could survive in a bigger market, I think somebody was reasonable introduction to computer science irrespective of what they have done databases or whatever they have done I think will certainly have a very good future there. So at least from computer science point of you people need not really worry what area like it.

Prof. Prathap Haridoss: What area like, Ok.

Prof. V. Kamakoti: But there are certain core courses like data structures, algorithms, programming languages, computer architecture, operating systems these core courses are there basically the language of computer science, right. Basically automated theory these courses people either in the undergraduate level or in the post graduate level there should thorough it. What I mean thorough is there are some very standard text books for each one of them and NPTEL has lot of lectures people have to access it and there are very good problems.

I don't know about other fields but one thing very interesting about computer science is, all these books have very interesting problems at the end exercise problems and people have to work out this exercise problems whether you are a teacher or a student, it is a must that you workout all these a exercise work before you start teaching or you start researching. If those things are done then that foundation will be strong and then you can



explore anything. So, that worry that if I take data sciences I may not able to do anything else is all unfound.

Prof. Prathap Haridoss: Ok.

Prof. V. Kamakoti: Which some if you have this foundation anything you can learn and develop within a few years.

Prof. Prathap Haridoss: Great this about the people who we just spoke about the people who are just leaving the department with the post graduate degree. Now, I will step back a bit and look at the people who come in to a department for Masters or a PhD degree, in general do you see then facing any specific kind of difficulty for them to adjust to a research in computer science and engineering and if so, now what are your suggestions or how they are supposed to handle?

Prof. V. Kamakoti: Yeah. My suggestion to all people who are entering computer science department in general which I been coordinating the same research methodology course quite some time before and I insist during the department modules and I tell now, there is one theory like if you don't know the language in which you can communicate you can think, you cannot even think. So, even if I start thinking I think in some language. So if I start if I want to think about computer science concepts, the language of computer science should be there in your mind then you can think anything about computing.

Now, what is this language of computing you just basically formalism, right? Suppose the problem that I see even with students who are about to graduate is that if I ask them to describe something from LKG to plus two. They start with the LKG entrance, then plus two final exam, then some eighth standard half yearly and then third standard quarterly. The organization of ideas is not in a; it's very, very zigzag. What it means to organize that area, suppose I ask you to write a research paper or present something informally the idea should get organized and the way you can do is to basically learn the language of computer science and I strongly suggest that there is a course called Discrete Mathematics. Discrete Mathematics I think there are two offerings in NPTEL, one offering is by Prof. Kamala Keerthivasan and she has done it so nicely that she will write



line by line and repeat every line twice and that type of course and she also has a book, her own book which you can use it and does excellent problems and there are many books in discrete mathematics also.

Take one book, good book and listen these type of NPTEL lecture and get yourself very thorough with Discrete Mathematics there is another book by Knuth, who is the father of computer science and he has written that's called Concrete Mathematics. These type of Discrete Mathematics and Concrete Mathematics if people look in to it just solve at least 20 percent of the problems in some of the standard book there that will give them the organization of ideas. So I want to write a proof. How should I go about? What sort of proof technique should I use? How do I go you know step wise; all these things will come. The moment you have that clarity of thought then I think it becomes very easy see which ever.

So, this is one problem which people face and somehow when we all did our PhD program, we use to listen to our guides. So, my guide was a Prof. Pandurangan he said do all these things we did all that 300, 400 problems in standard book or something like that and we all create. Now, that type of dictate is not working out with our modern students. So one of thing though we do not want to take parental role or a dictator role and say, do it. We will now request them to go and look at these two books and study this; this language of computer science has to come.

Prof. Prathap Haridoss: Ok.

Prof. V. Kamakoti: If it comes in then any subject will become easy because you know how to think, then you can think about anything nothing is rocket science, this is one fundamental problem. Communication skill in this conversation, I would have made say some hundreds of grammatical errors with that mainly we are able to convey hopefully right? So this type of a real a Shakespeare English and all we do not need. We just need them to write the proof clearly, write the concepts clearly, and explain the concept clearly in the language of computer science. These types of training in Discrete Mathematics and Concrete Mathematics would be very helpful. NPTEL has made a great step towards Discrete Mathematics.



Prof. Prathap Haridoss: Ok.

Prof. V. Kamakoti: And Concrete mathematics by Donald Knuth book we should follow this is my general prescription for anybody who wants to come to Computer Science.

Prof. Prathap Haridoss: Great, great, great that's very nice. The in any other engineering discipline I mean especially there is a lot of experimental work. I think the lot of people talk about you know there is the atmosphere in the group and how it impacts the research student and therefore, the meetings that students has with his other group members and the meetings that he or she has with the guide is a very important aspect. Along similar lines in computer science engineering you may be a bit of a mundane question, but how often do you think student should be meeting their guides and you know what sort of an interaction should be there?

Prof. V. Kamakoti: More than meeting the guides, meeting the peer group is very, very important.

Prof. Prathap Haridoss: Ok.

Prof. V. Kamakoti: And then having a year for everything is also very important there should be very keen on observing, for example, I did a course on Distributed Computing with one Proff Mutthu Krishna of our institute, he retired now. But the first quiz, I could not even understand what he was telling when I went and asked the sir, I do not know how to handle your subject though I got reasonable marks but I am not too happy with this thing. Anyway go to Adyar signal and see how their traffic is thick talking about distributor, go to Adyar signal and see how traffic is moving here and there right observe it for 2, 3 days then you will understands more about distributor computing is true.

Actually it was true right. So, how coordination happens, how distributed coordination happens, how clocking, how signals work, what happens if the signal is not obeyed, what happens if there is a delay, what happens signals while so, many things talk about many, many coordination problems in the real Distributor Computer Science. So the observation also is very, very important. So I still believe now, So I have a team of



around 40 students from B. Tech, M. Tech, MS, PhD put together and then we have 5 faculty, 4 faculty in our lab each one have something.

We have around to 100 to 120 people in the lab and that is very, very good. Almost two students review their paper each other and make our life very simple and we also now I have to scaled because we cannot say when we did B. Tech it was 20 to 25 in a class today we has 3 times there are thing or M. Tech programs have grown, our research areas have grown and there is so many projects also. As a faculty the amount of time we can spend on a student as also decreased but how do we compensate for that is to have a larger team and the team members interact and when, when something comes up to you it is after lot of refinement there and I have seen that.

So, when today I need not micro manage M. Tech person, if he is willing to work because the PhD student will actually micro manage him to a large level and no I have to macro manage him. So peer group is very important, going across boundaries and the basically talking to the next team understanding what is happening there computer science is all about applications. Today we exist because you guys have given us problems you guys are solving problems on me if I don't have a user computer science would have gone it is a user grown field.

So people have to start looking at those type of things and that is very good and specifically in our institute with this interdisciplinary stuff that has come actually 4 or 5 interdisciplinary students are there in our lab today and that is creating lot of more you know existing problems. So why I have mentioning here is that you talking to your peers is much more important than talking to your guide. But at least once in a fort-night meeting the guide and also sending some daily reports or weekly reports saying this is what I have done even if you are not done anything I did'nt do anything.

Prof. Prathap Haridoss: Ok.

Prof. V. Kamakoti: So, that is where that will keep (Refer Time: 36:36).



Prof. Prathap Haridoss: So, I would like conclude this discussion with you know, what are your words of advice for anybody who is aspiring to do research, pick up a research degree in you know computer science get into a research degree in computer science?

Prof. V. Kamakoti: My advice is that we need to work hard. Specific advice that is that is for any general.

Prof. Prathap Haridoss: Yes.

Prof. V. Kamakoti: Today as a professor my working schedule is close to 16 hours a day. So, the student I expect him to work double that time. So, go to Sun God and say extend 6 hours and make it 8 hours and make it 32 hours a day. Now, work as hard as possible that is very important today because the competition is very high. If you look at universities across, if you the problems if you are working on problems there are at least today 10 groups which are working on this same problem and this is survival of the fittest and they all send to the top tier conference and that the fittest one goes in and if you miss one dead line there will be three fellows at least who would have published in that area and your one year problem is gone.

So, that race is there always and you to be prepared to be work very hard and to establish yourself and then the next thing is team work is also very, very important that is that should from day one, work in a team to share ideas and be ethical about it because today if you look at many of the major universities across single author paper or student and that author is all gone today. If you take some really seminal papers there are at least 4 to 5 authors right, these all comes out of some large scale collaboration those are exciting area team work is very important and then my traditional advice of getting your formal theory correct your concrete and discrete mathematics very clear that is also very, very important computer science is a rapidly changing field. So, if you think of a problem today within 3 years you do not solve it becomes obsolete at a time you submit your thesis. We have to be on your tradition even I am not too comfortable guiding a PhD beyond 3 years because that original problem, you could have taken they have already become obsolete by 3 years. So, there is big change in field and lot of things especially in computer science is with the industry. So if I get a tool that I am working, this tool is 5



years old they would have tested for 2, 3 years and then release the alpha version to you and so that tool is already 5 years old. So, to keep in trend with the pace of the industry and you know making it irrelevant needs lot of hard work. And so finally, it's all hard work plus team spirit plus good foundation.

Prof. Prathap Haridoss: Thank you Prof. Kamakoti.

Prof. V. Kamakoti: I think this is interesting conversation.

Prof. Prathap Haridoss: Thank you.

Prof. V. Kamakoti: And thank you for selecting me.

Prof. Prathap Haridoss: And not at all. In fact



Introduction of Research
Prof. Prathap Haridoss
Prof. Srikanth Vedantam
Department of Metallurgical and Materials Engineering
Department of Engineering Design
Indian Institute of Technology, Madras

Lecture - 36
Research in Engineering Design

Prof. Prathap Haridoss: Good morning and welcome. It's our pleasure today to have with us Dr. Srikanth Vedantam. He is Professor in the Department of Engineering Design and he is also the Head of the department of the Engineering Design, here at IIT, Madras. He has a B. Tech in Mechanical Engineering from IIT, Madras and PhD from MIT in the United States and he has a lot of industry experience. He has worked with GE Global Research Center in New York and his areas of research include Microstructure evolution, Microfluidics and Wetting of surfaces by Sessile Drops. So, these are you know specific research areas and as he has mentioned his experience in the industry and he has been here as a researcher, as a faculty and as a guide for several years now. So, he has a lot of experience in dealing with research issues, dealing with students, working with students and guiding them through their PhD process. Okay so, with these words of introduction and I am very happy to have you here.

Prof. Srikanth Vedantam: Thank you so much.

Prof. Prathap Haridoss: And, we will go on with this discussion. So, Srikanth, in your department, what would you classify as a traditional areas of research?

Prof. Srikanth Vedantam: Yeah, first with the bio. I have to tell you that our department is little unique. We are a department of engineering design, which there are not many of such departments in the country. So, strictly speaking we don't have really a traditional role as such.

Prof. Prathap Haridoss: Traditional .. Ok. Ok



Prof. Srikanth Vedantam: There are other engineering design department not really department but, definitely focused areas in other countries, in different parts of the world and traditionally, to just answer your question, traditionally there the focus has been on the Design process. So, what we have seen is most traditional departments of Mechanical engineering, Electricals, Civil and so on. At least, in the under graduate curriculum the emphasis is on analysis.

So, somebody has already done the design. The students who come out of the departments are able to do the analysis very well but, the design aspect is not emphasized as much. I mean there are certain definitely certain a like a time, there are times when the people do focus on that, but it is definitely not a majored focus. So, in our department we are being trying to focus on the design aspects. So, to make along like an answer short, in our department the traditional areas have been focused on like materials and then controls aspects for automotive and biomedical educate.

Prof. Prathap Haridoss: Ok ok ok . So, even within this frame work or may be even outside of this frame work are there areas that are more recent and considered you know the modern areas to work in or maybe the cutting edge areas to work in this field?

Prof. Srikanth Vedantam: Yeah, and the answer to this is probably actually, can have a wide ranging answer in the sense that, in order to be in the forefront in our department research areas. You have to be able to cut across disciplines. So, no longer are you if you are a traditional mechanics or a traditional will you able to bring their all everything to bare, to bring better designs forward. So, the ability to be able to transition at least in terms of being able to understand the language of, if you are a mechanical engineer understand the language of electrical engineer and what their constraints are and conversely, if you are an electrical to understand what a mechanical constraints are. The ability to do that is what is the next you know front here in?

Prof. Prathap Haridoss: Ok.

Prof. Srikanth Vedantam: and I think it's true for many other departments as well.



(3.54) Prof. Prathap Haridoss. Ok Ok .So, in your department as you mentioned, you have this multidisciplinary approach. What sort of issues do you know do incoming MS, PhD students face? Let's say, the early part of their you know stay here, what sort of issue do they face you know? And what are typical approaches you would have for handling them that you can expect?

Prof. Srikanth Vedantam: Right. So, the main thing that you know based on the fact that I just mentioned that interdisciplinary and multidisciplinary kind of approach is very important. The one of the main issues is the fact that traditional preparation of incoming graduate students is usually strong in one area. So, the way we try to address this is to have them do courses across many disciplines. So, the first initial portion is spent in some amount of course work, which we believe actually it no, it adds value over the life time anyway. So, some of this is being addressed through this wide ranging course work, where people are not sticking to just one area and trying to focus their efforts in very narrow specialties.

Prof. Prathap Haridoss: Okay from, see when we talk of research, we tend to think of an academic setting and you know students are here in an academic setting and within that and research tends to be a specialization in a particular problem, which has a lot of scientific input to it and so on. In this context at least with respect to your department, where do you see the industry interest? Because, industry typically may have a you know more immediate term focus, they may not necessarily. They may be interested in long term projects but, typically they are focus and thrust is on the immediate future. So, where do you see the you know overlap? Where does industry show interest in activities that you pursue?

Prof. Srikanth Vedantam: Right, it's a good question because, so, in our department we actually are expected, there is a natural acceptance that we would interact much more closely with the industry and try to address problems right from the industry perspective. But, that said, at the end of the day in an academic environment what we view is that, we want people to try to understand the science aspects. Fine because, the really the large jumps that can be made in terms of technology and development can be understood only by the application of the science as if it's well understood.



Otherwise, if we look at it purely from industry development perspective, you can the kind of approaches that even make are only like a sort of small tweaks that will help improve things that will be at a short term. So, in that sense industry does look to our department to try to provide the scientific basis by which you can try to you know have industrial relevant problems but, with the little bit more long term perspective. And, I think that is true, while is true for our department in particular, I think all of I mean all academic departments should be in that mode. Where they are trying to, at least in engineering departments definitely, should be trying to focus on application of science to problems that are just a little bit longer than what the industry perspective is.

Prof. Prathap Haridoss: Yeah you had also mentioned like automotive industry and where I know biomedical industry, from their perspective are there specific you know kinds of projects that they are looking at an academic environment to show the leadership to help them solve?

Prof. Srikanth Vedantam: Yeah.

Prof. Prathap Haridoss: You see.

Prof. Srikanth Vedantam: So, some of our colleagues are you know like an actively participating with the automotive industry in terms of you know new tire designs, handling of vehicles, in terms of the stability, noise vibration aspects and so on. So, when the industry is participating, working on those aspects or problems usually, as I mentioned the approaches they take are only leading to small tweaks which will give incremental advances. So, what they are looking for in our colleagues and the research areas which our students are engaged in, is they can see if they can completely disturb the technology, they can come to probably new.

Prof. Prathap Haridoss: Very nice, yeah, yeah.

Prof. Srikanth Vedantam: So, by application of fundamental science.



Prof. Prathap Haridoss: Ok.

Prof. Srikanth Vedantam: So, in the sense that even though we are a technology oriented department, the science aspects are still important and we focus on them but the application to coming to like the next level, next generation of products.

Prof. Prathap Haridoss: Okay so, the science the basic understanding pushes the boundary much more significantly than you know locally tweaking it.

Prof. Srikanth Vedantam: Exactly, that's the point I want to emphasis really for.

Prof. Prathap Haridoss: That's I think a very valuable insight into the process. So, when, see students come in for Post Graduate degree, a Masters degree and PhD degree they come in with some variety of different back grounds, different colleges and institutions from which they come in and then they go through this process. So, if we look at that process of you know 4 5 years that they are here and then may be even their early carrier. In what ways should they, in what ways do you measure success in their research activity? In what way should they themselves be measuring their success as a researcher?

Prof. Srikanth Vedantam: So, this is a good question because it's not a like a very easy question to answer. There are various measures of success, but at end of the day at least from an academic view point, what I try to you know advice incoming students is that, they should always focus on the process of trying to solve a problem, not necessarily the specific problem itself.

Prof. Prathap Haridoss: Ok.

Prof. Srikanth Vedantam: So, in other words we want people to understand the process by which you are able to take relatively, like you know vaguely stated problem, because not many problems will be stated in a very precise fashion either by the customer in an industry or even like in our research problem in it's initial stages. So, how do you take that vaguely stated problem make it more specific, make very specific assumptions for



that problem, try to develop the tools or use appropriate tools to solve it and then defend what you have, the assumptions you have made and the results conclusions you are drawing in a coherent fashion.

So, this is the process of research which I think spans all like in a discipline and I do emphasize our students and all graduate students that I meet, that they should think about this overall process which they are learning. So, in other words, tomorrow if they are trained right now in one particular application of this process. So, somebody is doing some advance designs in automotives but their understanding the process by which they are going through. Tomorrow, if they have to do it in a slightly different discipline they will be able to carry over this process, the way which they specify make the problem more specific, that how they like obtained the tools to solve them and then how they defend the conclusions they make. So, as a graduate student and academic this is how I measure the success of a graduate student.

Prof. Prathap Haridoss: Ok.

Prof. Srikanth Vedantam: In terms of whether they have learned the process.

Prof. Prathap Haridoss: Learned the process, Okay great.

Prof. Srikanth Vedantam: So.

Prof. Prathap Haridoss: See, when we look at Masters Students and PhD students when they get their degree, I mean the general perception and to some degree it is true. Is that you know they are now specialists, specialist in not just engineering design but they are specialist in particular topic in engineering design and have looked at the topic in great detail and so on. So, when that is the case, what sort of you know what impact does it have you know in terms of their professional growth? Later on, in the sense that, what sort of positions do you normally see students in engineering design, graduating students in engineering design? What sort of positions do they tend to take up once they graduate?



Prof. Srikanth Vedantam: So, so far our department is relatively young. So, we have had a like you know smaller number of people going out after their graduation here, but so far the emphasis has been predominantly into the industry.

Prof. Prathap Haridoss: Ok.

Prof. Srikanth Vedantam: So, we had people of gone into the like research labs of biomedical companies as well as in the automotive R and D sector. So, and but but I see there is a trend where people are trying to go into the academic side as well and have, we are several students who are right now very focusely pursuing their academic carrier as well. So, right now the as assure a story I mean question in a more focus fashion it's mostly industrial.

Prof. Prathap Haridoss: Okay, that's very nice to know I mean I think people tend to think that it is once you do research you have to do, you probably moving away from industry but I guess not. You are saying that in fact, in your discipline lot more people goes I mean the industry interacts much very well with the students graduating from your department. So, a little while earlier you spoke about the process, the process by which a person becomes a researcher and so on. So, to what degree and how much does the interaction between the guide and the student contribute to this process? And, in that context in a more mundane sense, how often do you think students should be meeting their research adviser in during their research process and career?

Prof. Srikanth Vedantam: So, for this my own view in the beginning stages of a research like, I know graduate student trying to start their research program, it should be a lot more frequent kind of meeting may be even once in a week or even at a little smaller time frames. The reason I say that is that, the process of identifying or making a vague statement of our research problem into something more specific that is more tractable is actually one of the more difficult parts of the research methodology.

Prof. Prathap Haridoss: Ok.



Prof. Srikanth Vedantam: So, considering that you have put in the most difficult aspects in the beginning.

Prof. Prathap Haridoss: Beginning.

Prof. Srikanth Vedantam: I think that's where the interaction should be really like you know, more frequent.

Prof. Prathap Haridoss: More regular.

Prof. Srikanth Vedantam: So, then after that there will be a stage where it's just learning tools, and implementing them. That part to the students, researcher could discontinue to do it on their own because first of all the other point of time even the meetings get less frequent may be even a month, once a month. It's better that way because there is no spoon feeding of tools.

Prof. Prathap Haridoss: Ok.

Prof. Srikanth Vedantam: Tools is something that we have to be able to pick up at the end of our research training. We should be able to learn how to read an under graduate text book or even a math's level text book to learn the appropriate tools to solve a problem, that part we have to get that independence. So, the initial part where you are still focusing down the problem may be lot more interactive.

Prof. Prathap Haridoss: Ok, and then.

Prof. Srikanth Vedantam: Once it goes along, then I think it can be as needed like you know how the tools are.

Prof. Prathap Haridoss: Okay great. So, now to sort of wind up and finish this now, close this discussion. What is your advice to students who are aspiring to be you know researchers in the field of engineering design?



Prof. Srikanth Vedantam. So, I would advise them, first of all to try to have much interdisciplinary interests which will make them first of all like much more useful to the eventually industry that they go to or if they go to an academic environment they will still be having like, like they may be bring much more to the table if they do that.

So, first advise is to be much more interdisciplinary and being able to try to like you know, show interest as well as like you know curiosity in subjects that are not immediately related to yours specialized area. Then, other thing is that I think try to have a big picture of what you are trying to solve. In other words, lots of times has when we have focus on our research problem we tend to look at every minute aspects of the problem as they are looking at. But, all the time if you are able to keep a big picture of where this problem is going to fit and how it's not usual that we solve like a really societal need, start to finish but, but each of us will contribute to a small part of it.

Prof. Prathap Haridoss: Small part of it.

Prof. Srikanth Vedantam: That is what we usually have time for and we are able to focus on, but if we understand were this solution that we are providing fix into overall picture of our societal need, that helps us really contribute much more than if you are just focusing on your minor problem.

Prof. Prathap Haridoss: Ok.

Prof. Srikanth Vedantam: Minor problem, that is all.

Prof. Prathap Haridoss: Thank you Srikanth, for joining us. We are really glad that you could join us and share your experiences with us. Thank you for coming.

Prof. Srikanth Vedantam: Thank you.

Prof. Prathap Haridoss: Bye.



Introduction to Research
Prof. Rajesh Kumar
Prof. Prathap Haridoss
Department of Humanities & Social Sciences
Department of Metallurgical and Materials Engineering
Indian Institute of Technology, Madras

Lecture – 37
Research in Humanities & Social Science

Prof. Prathap Haridoss: Hello, today we have with us Dr. Rajesh Kumar. He is faculty in the Department of Humanities here at IIT, Madras and we are going to discuss research in humanities and different aspects of research in humanities, so that if you are aspiring student in this field, it will give you some sense of what to look forward too, when you get into this field. Dr. Rajesh Kumar has a PhD in Linguistics from the University of Illinois at Urbana-Champaign in the United States and he works in the areas of a language in education, social linguistics, linguistic theory and language and cognition. So, he is eminently equipped to discuss these issues in the context of research in the humanities and so, with that brief introduction we will start with discussing this topic with Dr. Rajesh Kumar. So, tell us, what are traditional areas of research in humanities?

Prof. Rajesh Kumar: See, the Department of Humanities in Social Sciences in any IIT is has a unique feature, which is it's intra-disciplinary and multidisciplinary nature. So, in this department at IIT, Madras, we have a traditional areas like economics, literature, linguistics, philosophy, history, sociology and it is several other branches which are is new emerging areas and also, what has been added recently is the developmental studies. Apart from that, Department of Humanities and Social Sciences has 3 more areas which has been added and they are like China studies, European studies that is European union study and also there is a center called Indo-German center for sustainability study. So, most of the traditional areas in that you find in any domain of humanities and social sciences are covered, as research area in the department.

Prof. Prathap Haridoss: Okay and, so do most of the students join your department tend to go towards these, traditional areas of research or is it more towards the more recent areas that you have opened?



Prof. Rajesh Kumar: In the recent times, because of the multidisciplinary in nature of humanities and social sciences in general. You have lot of new emerging areas, for example; in the history one of it's not really new but, one of the new areas that can be counted is history of science. In the developmental studies, the intersections with this with ideals like politics and gender and sociology of religion. In the field of linguistics as well, we have language and technology the use of technologies in teaching methods.

So, they are all kinds of new areas emerging as well. In the traditional area of literature when you divide it into several parts, we cover American literature, African literature, English literature, Indian literature, Indian writing in English and all of these are new emerging areas that have covered in humanities and social sciences. And, most of the students in the recent years who come, they already have one or these, one or the other of these areas in their mind. So, they don't know to come for broad area of studying English or English literature only. And, even within that specific domain of let us say African literature or American literature they have certain specified ideas as well and if not over a period of time those areas emerge in the discussion with their respective faculty, where they go to the specific area in depth.

Prof. Prathap Haridoss: Okay actually, since you mentioned intra-disciplinary areas of research and also the fact that you know many engineering institutions including IIT's have a humanities department. What exactly and since we are been viewed by a general audience, what exactly do you feel are the boundaries of humanities? Where I mean, what are the set of topics that you feel encompass the range of things that people working in humanities focus on, as supposed to any other?

Prof. Rajesh Kumar: Right. In fact, there is when it comes to humanities and social sciences and not almost all IIT's have this department. And, I can speak about the humanities department, here there is no limit in fact and we see the research ranging in health, policy, technology, ethics, religion and these areas do find a space in existing areas as well. So, there is no limit, no boundary that restricts us. In fact, locating this humanities and humanities and social sciences department in IIT gives it, some kind of a expansion where it is easy to transcend to technological domains. And they, faculty and the students both researchers can venture out in to areas where they get data readily



available. For example, someone wants to look at the use of technology in health.

Prof. Prathap Haridoss: Ok.

Prof. Rajesh Kumar: You can easily find someone working in this area in Computer science, Bio technology or Electrical engineering where you see their applications of technology from these areas and then social scientist's do make a research question based on that, what's the impact of these things in assessment of health? For that matter what, how does, for example, how does the entire sector of health does any kind of ethical practice or not.

Prof. Prathap Haridoss: Ok.

Prof. Rajesh Kumar: So, beginning from the technology to ethics to practices locating in an IIT, instead of restricting it form anything else actually opens up more possibilities.

Prof. Prathap Haridoss: More possibilities ok. So, now if you look at it from student perspective, some students who come to join humanities department and may be a humanities department in an IIT, what do you think are major challenges that they may face or they tend to face when they join such a department? Are there such challenges, something uniquely there for them that, they should be aware of?

Prof. Rajesh Kumar: Right, there is one specific thing which becomes technical challenge. That is, it's not a department of a particular discipline. For example, if someone wants to come to study linguistics, there are limited numbers of people in terms of faculty doing research in linguistics and like any other discipline linguistics or history or philosophy or economics they are huge disciplines, they are several sub areas in that. So, the students have to come prepared to be interacting with a limited number of faculty and a staff, in this specific area. And then, for example, linguistics has 4 more core areas like, Syntax which deals with sentences, Morphology which deals with words, Phrenology which deals with sounds and then Social Linguistics that deals with language and society. One person can have limited expertise in all 4 of them, that becomes general linguistic. But, if you want to talk to a specialist in Phrenology or Computational



Linguistics or let's say Morphology, then you have to seek help from outside.

Prof. Prathap Haridoss: Ok.

Prof. Rajesh Kumar: And, that is one of the things I would imagine, applies to all other disciplines that are part of humanities and social sciences in a department like this.

Prof. Prathap Haridoss: Ok.

Prof. Rajesh Kumar: Where you don't have a department of sociology separately, where people study all aspects of sociology or a department of linguistics separately, where people study all aspects of, most of the aspects of linguistics that's one of the limitations that students face. And, in terms of research materials also we there was a time and people face such problems, but now with the emergence of electronic help available in materials being available electronically that is no more a problem. But, it again to go back to the strength part of this take the example of my own discipline linguistics, it's like a virtual discipline, it may not have a department of, it may not look like a department of linguistics, but there it is a virtual department in the sense that, I know at least 4-5 other colleagues working in 4 different departments on several aspects of linguistics.

So, on one side when we do not get to talk to or meet with this specialist on daily basis, on the other side, you have a specialist that you were disposal locally where you can talk to people with a specialties in emerging areas of.

Prof. Prathap Haridoss: Emerging areas.

Prof. Rajesh Kumar: Linguistic. So, it has a combination of a limitations and advantages in being located in IITs

Prof. Prathap Haridoss: Ok.



Prof. Rajesh Kumar: In Technological Institution.

Prof. Prathap Haridoss: So, even in general I mean when you associate were both in IIT's and even outside anywhere else where say humanities departments are there. I mean, I think there is a lot of recognition about the value of humanities in life.

Prof. Rajesh Kumar: Right.

Prof. Prathap Haridoss: And, in general you know social sense. Where do you see, it's utility in say with respect to industry, where in to what degree do you see industry interested in people with a humanities background? What sort of backgrounds are they interested in? And how do a people students graduating with the humanities degree? Where do they see value in, where do they fit in?

Prof. Rajesh Kumar: Very interesting, see one of the things that I use in most of the courses that I teach and it applies to all disciplines and all areas of humanities and social sciences. What we try to prepare students with is, most of the things that you learn in humanities and social sciences either at under graduate level or at a graduate program, are going to be more useful when you go outside the boundaries of IIT or any academic institution for that matter. This is not to underestimate the value of any other discipline, but when you get the disciplinary expertise in Electrical Engineering or a Mechanical Engineering, you demonstrate your learning in a skills of those disciplines only among the peers and in a very restricted environment, where you are suppose to be competing with the other Electrical engineers and Mechanical engineers.

Whatever expertise you deal, you gain out of the courses in humanities and social sciences is going to be useful everywhere that's one broad thing which I wanted to say about that. Now, being a specific to what kind of contributions can humanities in social science graduates with all kinds of expertise in these disciplines make which are useful for industry, that's also numerous. I mean you can simply count. I will tell you the example of one of the things of that I am doing on one of the projects that I am working on. Most of the industries require a cordial relationship with the society in which they are going to be establishing their thing.



Prof. Prathap Haridoss: Immediate community.

Prof. Rajesh Kumar: The Immediate community. So, they need to, you know the complex nature of our country in terms of it is multiculturalism and multilingualism. They need to be in association with the local people. So, to in order to understand the society languages, politics, cultural practices and everything, every industry requires a social scientist to help them navigate through these. I am not sure yet, how industries are hiring people on keeping these issues in mind or not.

Prof. Prathap Haridoss: But you see the value, they have the need for it and they have different insight.

Prof. Rajesh Kumar: Definitely, not only the need it is time that they cannot ignore it any further and we see lot of conflicts emerging in our socio-political condition, situations in this in our country with local issues and industry they are all located in these domains and if they apply the findings of research in humanities and social sciences and they have people equipped with these things then probably they will reduce such difficulties.

Prof. Prathap Haridoss: Okay That is a very I think a very valuable insight.

Prof. Rajesh Kumar: Ok.

Prof. Prathap Haridoss: I think a lot of industries and people from industries who are, I mean may be looking at this kind of information.

Prof. Rajesh Kumar: Absolutely.

Prof. Prathap Haridoss: Possibly find that very useful to apply. So, may be, some mundane questions let me ask you. So, we have in humanities I think the kind of work you do is very different from say, what an engineering activity, yes. There may be I think there is some much more controlled experiment that we do, I am sure some analogues thing may be there in humanities. So, what do you see as, you know the amount of



interaction that must exist between say students coming in and the faculty that they working with, how often you feel they need to meet them, to you know optimally learn something at the same time do something new, how much do you see of this interaction being necessary to what how important to it is how often should it happen?

Prof. Rajesh Kumar: To see as you rightly said there are some areas in humanities and social science also where we need controlled environment for research, but in general and in linguistics and my area in particular, this is again one of the things that I metaphorically use. The laboratory for research on language begins where all laboratories end that is the whole world is laboratory for that.

Prof. Prathap Haridoss: Ok.

Prof. Rajesh Kumar: For that, now, so that's the domain that you have to cover in your research, in humanities and social sciences, given a particular are. How often should the interactions be which is not very different from any other discipline?

Prof. Prathap Haridoss: Ok.

Prof. Rajesh Kumar: I would say based on my own experience in research with, when I was doing my own research with my supervisor or when I do research now with students, I think a meaningful discussion of an hour in a week is sufficient and required. If that meaningful discussion takes place 1 hour of material takes place 3 or 4 hours it depends in individual student and the faculty, but a meaningful contentful discussion where you evaluate what has been done, what has to be done next and where we are today, requires 1 hour.

Prof. Prathap Haridoss: 1 hour.

Prof. Rajesh Kumar: 1 hour a week and then with total honesty research scholar has to translate their discussion into reality, function and work in their own work.



Prof. Prathap Haridoss: Ok.

Prof. Rajesh Kumar: And, whenever they decide their agenda for the next meeting, the ideal and optimal goal for any researcher scholar to be successful in this and to be disciplined in the research, is to come up prepared with at least, I mean they have to give it an effort to be prepared with at least 75 percent of what they discussed their in the previous discussion.

Prof. Prathap Haridoss: In the previous discussion, yeah, yeah.

Prof. Rajesh Kumar: In the next meeting.

Prof. Prathap Haridoss: Ok.

Prof. Rajesh Kumar: So, that kind of discipline in terms of meeting is required in humanities and social sciences in particular and I am sure it is required for all other disciplines as well.

Prof. Prathap Haridoss: I think similar things should be there. Also, when students graduate from your department, even when we talk of MS degree or a PhD degree these are always treated as degrees where somebody as specialized in something very specific and so there is always this feeling amongst people who complete you know under graduate the kind of degrees before they pick up or enroll for a PhD or a MS degree is always this concern that later people may look at us and say we are specialist and there is not an easy fit for us in a position somewhere. Where do you see most of your MS master students or PhD students, students who complete these degrees, where do you see them you know joining for positions that are you know professional positions in locations?

Prof. Rajesh Kumar: Most of the masters students in this department go for their PhD's in higher studies at different places. We have a 5 years integrated program, where students come after their 12th grade. They spend substantial 5 years chunk of their life here and



they learn something and then they must go for some, if they are interested in higher studies to some other institutions to see and verify what they have learnt here is meaningful.

In terms of what they choose to do for master students, we see mostly going to higher studies some of them go to government civil services jobs, some of them preferably join non-government organizations to learn more about this specific areas, take for example, somebody is specialized in some issues in developmental studies like gender, equality or sustainability studies or like a resource management and water resource management or health care management and all that. So, they do find jobs in these areas and there are lot of unconventional areas which were are not available let's say, 10 or 15 years ago are coming up with for hiring of these people. I just interacted last week with 2 of them who are working in the area which I never thought in for, where a student with humanities degree will be working, they were working in marketing.

Prof. Prathap Haridoss: Ok.

Prof. Rajesh Kumar: So, what we were discussing a minute ago certain industries and in particular upcoming industries work kind of startups are realizing the roll of expertise in humanities and social sciences in the areas of marketing, in the areas of, in all kinds of area where you need human interactions.

Prof. Prathap Haridoss: Ok.

Prof. Rajesh Kumar: And, PhD students also have some of them joined Post Doctoral programs and some of them have joined teaching in research programs at various educational institutions. So, these are these are some of the areas where we find then working.

Prof. Prathap Haridoss: Okay Great. Right. So, I think we will close with this question here, which, what advice would you give to an aspiring student who would like to join graduate studies in humanities?



Prof. Rajesh Kumar: Very significant question, this is we face and particularly students face these questions. See, research in any discipline requires certain kind of discipline. So, they should be committed to their discipline to their academic discipline in a disciplined fashion.

Prof. Prathap Haridoss: Ok.

Prof. Rajesh Kumar: That is one. Second, is they should be, it is a full time job. So, 24 hours.

Prof. Prathap Haridoss: Full time activity.

Prof. Rajesh Kumar: Full time activity. And you should, what you are doing it should reflect as your interest, should not be a burden for you. The movement its starts burden means that you are losing your interest, so to keep up your interest in this discipline in any discipline of your choice.

Prof. Prathap Haridoss: You should enjoy doing that.

Prof. Rajesh Kumar: You should enjoy doing that. If you don't enjoy doing that, then that is one of the parameters where you have to realize that you have to find something else to do. Third and very significant that there are lot of things, but at third that I would add from my personal experience is you should be leading something more than you are own discipline as well, so that you are aware of what's happening around you and in other disciplines.

So, if a research program in any discipline, at any academic institution I making it more general is must give you a holistic development and at the end of it if you say, I have done my research in mathematics and I only that also in mathematics I have done it in algebra and I only understand is maths, it does not make you look too good. So, as a mathematician or as an engineer and as a social scientist if you are aware of the fundamentals of other things, you are aware of fundamental of what's happening around



and what's coming up in literature, that keeps you alive and any intern it has huge implication we can we can discuss with at length at some other time. But, what you read outside your own discipline has huge implication for reading and development in your discipline as well.

Prof. Prathap Haridoss: Yeah, I think it opens up lot of new ideas, new thought processors.

Prof. Rajesh Kumar: And the least it does it gives you habit of reading.

Prof. Prathap Haridoss: Yes, of course.

Prof. Rajesh Kumar: So, these are at least 3 things that people should be prepared for, when they take up research in humanities and social sciences in particular and in any other discipline in general that I would think.

Prof. Prathap Haridoss: Okay great. Those are nice words to know, a nice thoughts to keep in mind. So, I would like to thank you for joining us and you know giving us this insight into humanities, which I think a lot of a students across the country consider as an option that they should look at and certainly for you know higher research in these areas, it is nice to have an expert tell us, what is it that they would like to, I mean likely to experience and so I thank you for that.

Prof. Rajesh Kumar: Thank you very much.

Thank you.



Introduction to Research
Prof. Prathap Haridoss
Prof. Babu Viswanath
Department of Metallurgical and Materials Engineering
Department of Mechanical Engineering
Indian Institute of Technology, Madras

Lecture – 38
Research in Mechanical Engineering

Prof. Prathap Haridoss: Welcome to Prof. Babu Viswanath, who is professor in the Department of Mechanical Engineering here at IIT, Madras. He has a very rich background of various things that he has done before. He has a PhD from Ohio State University in the United States; he has worked with Ford, so he has very good industry experience with one of the leading automobile manufacturing companies in the world. Here in IIT, Madras, he is also been our placement adviser so, again a lot of industry interaction. He does a lot of active research in the areas of high performance computing and caustics and propulsion.

Prof. Babu Viswanath: Propulsion.

Prof. Prathap Haridoss: So, these are all areas that he works on. So, he has a very good experience with both industry and research and so we are very glad that he could join us today and to discuss research in the area of Mechanical Engineering and various aspects that researchers might encounter through their stay here, at in their research carrier. So, Prof. Viswanath, we do have you know Mechanical Engineering is a field that is been around for a long time and in fact, I mean I think whenever we talk of Engineering colleges that's the first thing that immediately comes to mind, Mechanical Engineering is there, it is one of the core areas that is there. So, what you think?

Prof. Babu Viswanath: Mechanical, Civil.

Prof. Prathap Haridoss: Sure, sure.

Prof. Babu Viswanath: And Electrical are 3 oldest branches around in the Engineering.



Prof. Prathap Haridoss: 3 holders branches around. So, what do you think what people would think as traditional areas of research in Mechanical Engineering?

Prof. Babu Viswanath: Yeah traditionally, Mechanical Engineering department faculties can be divided into 3 verticals. One is the so called Terminal Engineering and the other one is the so called Design of components and other things, and the third one is the Manufacturing. So, these are the 3 verticals that you can divide Mechanical Engineering research into. The first one deals with Thermodynamic aspects of design and things like that energy conversion, energy conservation and so on.

The second vertical namely Design, works on the material aspects of equipment and devices so, once you have a design on paper that has been let's say, you are talking about an internal combustion engine or jet engine or comprehensive for a refrigerator, once the thermodynamic aspects are being looked at and the requirements have been sized, then comes the task of designing the equipment which actually will achieve those goals. If you let's say, you want to achieve a pressure, certain temperature and so on, we need to select materials which will be used to fabricate the equipment which can deliver these kinds of requirements. So, that is the design aspect of Mechanical Engineering.

The manufacturing aspect logically comes last because, having frozen the design both the thermodynamic aspect as well as the materials aspect. Now, it is necessary for us to figure out, how to actually manufacture this component? So far all these designs are on paper, you know, you say, you select such and such material to make the piston rods or piston itself and so on. But, how do you actually manufacture them? So, that requires that may require special tools, special fixtures, special machines, special machining processors and so on. So, that is the third vertical in Mechanical Engineering. So, if you look at all of 3 of them as a whole, it is end to end design. So, you start from conceptualization, theoretical analysis, energy waste analysis and so on, followed by materials and followed by actual manufacturing details of the components.

Prof. Prathap Haridoss: Okay so, of course, these as you mentioned are, may be what immediately come up as a traditional areas.



Prof. Babu Viswanath: Correct.

Prof. Prathap Haridoss: So, may be even in association with this or independent of these, are there certain areas that are considered modern areas of research which have come up maybe in let say, the last 10 years which are associated with Mechanical Engineering that you think?

Prof. Babu Viswanath: Mechanical Engineering over the years, what has happened is 2 things have happened. One is the even within the 3 verticals that I mentioned earlier. Things have become very, very special. So if, for example, you look at an application like internal combustion engine let say, Diesel engines. The emphasis today as you know is on reducing emissions and pollution and this goal is being pursued in many different funds. One is of course, to look at alternative fuels which are better than hydrocarbon fuels in some sense so, people are looking at fundamental aspects of combustion of alternative fuels. There are also researchers, who are looking at chemical kinetic aspects of combustion of such fuels with the view to reducing the emissions so, these are new areas.

Now, further more there are also researchers who are now, who now have the capability with lasers and high speed cameras and other devices and we now have the capability to actually take real time pictures of what is going on in an engine, then analyzing them to see whether something about the spray or other aspects combustion aspects can be optimized to mere to reduce a emission. So, we now have technology which can take real time images of actual engines. So, these are, within each vertical there have been lots of advancements.

And, the other change that has taken place is many things have become interdisciplinary. Again, if you go back to internal combustion engine example, if you look at an automobile today, everything is electronically controlled. Most of the things are electronically controlled which means, you are working with embedded systems. So, people with backgrounds in Electrical Engineering are coming into Mechanical Engineering and vice versa. So, these 2 things are happening just say, this is becoming more interdisciplinary and there are also developments within the traditional areas of



research itself, things like Mechatronics, things like Nanotechnology which is now being used in Thermal Engineering as well as in Manufacturing also, to get desired finishes, desired quality of machining and things like that. So, these are new things which have come into Mechanical Engineering.

Prof. Prathap Haridoss: Very interesting to know. So, when students come in to the Department of a Mechanical Engineering for PhD or an MS program, they do come there with you know, background from a variety of different places that they have been at for their undergraduate Engineering, which could be small towns, big cities I mean well equipped colleges, may be some are new colleges and so on. What do you see are issues that they tend to face during their initial say initial years here or even through their PhD program is more specifically with respect to your department? Are there any specific things that they initially find that they are you know unprepared for the need to get up to speed on and so on?

Prof. Babu Viswanath: The several problems that students from other colleges face, typically over the years that is what I have seen the biggest problem that they have face probably is the regress academic processes that we follow. Unlike, other institutions where for many different reasons, things will not run in the sequence in the rigid sequence that we follow. Where we have test evaluations the exam papers are discussed in class and some marking schemes are given to students. So, the process runs you know, runs in such a control fashion that students who are coming in for the first time into the IIT classroom, kind of they are surprised by the when the semester suddenly you face in semester exam and the semester is over. So, you take some of while to get adjusted to the academic process that IIT, where things are very well organized and structures and we have very robust academic processes in place.

Now, this is particularly true of a Department like Mechanical Engineering because, it is a probably the biggest department on campus. So, we have about any given time, the department probably has about 1800 students on roll, our department alone. So, it is almost like a mini college.

Prof. Prathap Haridoss: Mini College.



Prof. Babu Viswanath: It's not a college.

Prof. Prathap Haridoss: Not a college.

Prof. Babu Viswanath: So, this actually can also be intermediating to the students who are coming in, because the department is very big and at IIT, Madras particularly the officers of faculty and labs are scattered all over the campus. So, the department office is in one place, other things are in other places. So, it takes a while for students who kind of get used to this geographic diversity of their department.

And, the other thing as I said is the academic process. Being a big department, we actually have, we need to have and we do indeed have very robust academic processes because, that is the only way we can deal with such large numbers. Emphasized processes will lead to lot of confusion, complaints and unfair in acquisitions of unfair treatment and so on. So, we have very, very well structured syllabi, very well structured time lines, number of lectures for each and so on. And, we also in the department as a department we go through a periodic reviews of our curriculum as well as our syllabus to take into account new things that have taken place and also to see whether in a based on a experiences, difficulties that students have, whether some bridge courses are required to make the transition easy for such student. So, we do that also.

Prof. Prathap Haridoss, But, in terms of say let's say preparation for research. Is there something that students you know may not necessarily be even just coming to an IIT, but just their general preparation to face research or to pickup research in an academic setting in the area of Mechanical Engineering. Are there certain things that they are, they feel less prepared for when they come in and when they have to you know learn some thing new in addition to just you know specific course work?

Prof. Babu Viswanath: Yeah, I mean many of the students who come they actually are not very clear about the areas that they want to work in so, that is one major problem for them. So, they come here and then they are exposed to the regress aspects of the area that they have chosen and that might perhaps be little bit unsettling for them because, they thought it was something else and this is actually something else. So, that is one thing



and in that case we try to work with the students to see whether, something can be more interdisciplinary the research topic, can be made more interdisciplinary or whether they can take courses in certain other topics also and what can areas which are suited more to their liking.

So, those are all done on an individual case to case basis between the guide and the student and it is very much possible to pursue interdisciplinary research also. So, we try to take that into account. Understandably, undergraduate students who come here for research many of them, kind of have likes and dislikes but, they are not really sure whether this is something that they want to proceed.

Prof. Prathap Haridoss: To do research on.

Prof. Babu Viswanath: To do research on. So, they find out whether they are actually prepared academically to do research in this topic, is something that they find out, only after they come here so. But, that is understandable as under graduate students.

Prof. Prathap Haridoss: Sure, sure.

Prof. Babu Viswanath: You cannot really expect them to know everything.

Prof. Prathap Haridoss: Absolutely correct.

Prof. Babu Viswanath: So, we are prepared to what with the students. So on is the student is open-minded and willing to make some changes and adjustments and so on, the department is always you know to try to dispatch being so big. The department always has actually accommodated the individual student needs and requirements and you know matured that they have a good experience while they are here.

Prof. Prathap Haridoss: Okay so, maybe since we were talking of processes and procedures and systems in place to for students to have a good experience here. I have bit of mundane question. So, some students join's here for or join's your group for PhD.



What is your, what is your opinion on you know how often a student should you know meet their advisers? Spend time with their adviser? That's an interaction that that's always there inherent in this kind of a setting. What do you think are your views on you know how frequent it should be? How intense it should be and so on?

Prof. Babu Viswanath: This is actually pretty much dependent upon the guide and you know the likes and dislikes of the guide. But, as a general rule what I have tried to do is of course, you know emulate what my PhD adviser did in our research group. What I try to do is, I usually would like to meet with my research scholars on a daily basis.

Prof. Prathap Haridoss: Ok.

Prof. Babu Viswanath: This is not; this does not mean that I expect them to make progress every day. Research is something as someone said, research is something which where you have long periods of frustration, interspersed with brief periods of elation.

Prof. Prathap Haridoss: Ok.

Prof. Babu Viswanath: So, the role of the guide is to prepare the student mentally for those long periods of frustration and teach some that this is a mindset, you know you should not get frustrated. What is more important is the effort that you put in everyday, day in and day out rather than progress or lack they are of. So, that is what research is all about. So, I meet with the students' everyday and most of the days the students will tell me; Sir, I tried that you know it did not work and we have discussions like that. So, I tell them yeah, do not worry about it, why did not you try this or keep trying?

So, in my mind you know this gives students the feeling that the guide is actually sympathetic and understands that you know the progress, the lack of progress is not due to a lack of effort, but because the problem is simply difficult. And, it takes a while you know to figure out it, so otherwise it would not be research that is the difference between a researcher doing research and doing project work. Project work you have the problem, how to solve the problem, there only the details have to be filled in, you do it you get the answer. Whereas, with research many times the problem itself is not well defined and



you don't know how to actually solve the problem, you have handful of options hopefully, that you can try and perhaps one of them will work out. But, until that one works out you have to go through periods of disappointment.

So, many people say frustration, I don't like that word, I prefer to use the word disappointment you try something it doesn't work out you are disappointed, you want to try something else. So, that would be a better way to look at research. How do you handle long periods of these kinds of trials and failures that is the mindset that you need to develop? So, I meet with the students on a daily basis only for that reason that you know.

Prof. Prathap Haridoss: Ok.

Prof. Babu Viswanath: We have an interaction the students feel happy that you know, they have someone to talk to and they can come to me they should not feel that they can come to only to report successes, they should feel comfortable coming to me.

Prof. Prathap Haridoss: And reporting.

Prof. Babu Viswanath: Reporting lack of progress.

Prof. Prathap Haridoss: Lack of progress.

Prof. Babu Viswanath: They would not call it failure. But, I will tell lack of progress

Prof. Prathap Haridoss: Ok.

Prof. Babu Viswanath: They should be, they should feel comfortable to come and say, No, it didn't work. Now, some students are very cryptic, they will say, No, it didn't work. Otherwise, would elaborate a little bit more, but whatever the students feel comfortable with, I would like do that on our daily basis.



Prof. Prathap Haridoss: Very nice, very nice. So, see when we talk of MS and PhD kind of degrees people tend to think that you know these are specializations and in fact, that's the general perception that it is a specialized degree and it is a specialized in a specific problem associated with that general area of Engineering or Science. And, then you come out with the degree. So, what sort of positions in your view that you have seen? What sorts of positions do these people who have done an MS or a PhD in Mechanical Engineering? What sort of positions do they tend to get when once they graduate from the

Prof. Babu Viswanath: See, first of all if you look at the kind of skills you are expected to acquire, I still remember attending a research meeting when I was first year graduate student at United State and an executive from GM, I was talking about, how they make the decision to recruit either an Undergraduate student or a Master student or a PhD student? And, he said something very nice. He said, when we recruit a B. Tech student, we tell the student do this; when we recruit a Master student we tell the student, would you like do this? When we recruit a PhD student we tell them, what should we do?

Prof. Prathap Haridoss: What should we do, ok, very nice.

Prof. Babu Viswanath: So, these are the.

Prof. Prathap Haridoss: These are the way to distinguish the.

Prof. Babu Viswanath: These are the skills that the student should acquire. They should be able to answer these questions you know. So, with PhD or MS which are both degrees based on research you have 2 carrier options, one is to go into research and development in the Indian industry or any other place or R and D is definitely a very good option for them. And, for students with both MS and PhD, these days there is a tremendous requirement for teachers in India today because, the number of colleges being so high, number of students studying Engineering being so high, there is a tremendous requirement. And, many universities and colleges from around the country do come to IIT, Madras, to recruit our students for such positions. So, they can also look to a carrier in teaching and you know they I think at the end of the program here, they should be,



they should equip themselves well to go into one of these 2 carrier options, either carrier in Research and Development or a carrier in Teaching.

Prof. Prathap Haridoss: Okay So Okay so, we now have some students, who are probably considering, you know joining Mechanical Engineering for a higher degree and so on. So, what is your, what are your words of advice for an aspiring student in for higher education in Mechanical Engineering?

Prof. Babu Viswanath: My word of advice to such students would be the following. You look at the options that you have, many of the students actually will probably not be clear whether they want to do higher studies or whether they want to go for a job, may be they have a job offer you know. So, my advice is you know if you trying to decide between the 2. In today's world I would almost always tell you that, if you are able to secure an admission at IIT Madras, for an MS or a PhD you should definitely take it up because it is a very enriching program you have probably the best teachers in the world, and if you look at the rankings of the individual departments in our institute in IIT, Madras, you will see that in QS ranking, they have ranked among the top 100 departments in the world. So, these actually are very, is a very good place to study. And, whatever you did not learn in your undergraduate education for whatever reason, you can probably compensate for that by pursuing a Master's program here.

And, as the you know as the former placement adviser, I can also say that the employment opportunities that you would get at the end of your masters, probably of the best that you get in this country. Best companies come here for selecting, for example, companies like TVS motors, Robot Bosch, General electric, Rolls Royce, Nvidia, Texas instruments, Eaton all these companies come and recruit our Masters, MS students from here. So, you will have your short add, the best possible companies in the country.

Now, so you take this decision that you are grappling with now, whether you want to go for a job or a higher studies, you take that decision. You are not really in a good position to take the decision now okay. Whereas, after you get an MS from IIT, Madras you will be in a much better position to take that decision, you have a very good job that you have at your hand in one of the best companies in the country and now you can decide



whether you want to continue with that or pursue even higher studies like PhD and so on and then go from there. So, my position is you will be in a much better frame of mind both academically as well as personally as well as professionally to take that decision at the end of a Master's program from IIT, Madras. So, my advice is, if you have an admission for MS in IIT Madras, don't even think twice; take it.

Prof. Prathap Haridoss: Great. With those words of advice, thank you very much.

Prof. Babu Viswanath: Thank you.

Prof. Prathap Haridoss: For joining us.

Prof. Babu Viswanath: Thank you Prathap for this opportunity.

Prof. Prathap Haridoss: And, it is very valuable for all the students.

Prof. Babu Viswanath: Thank you for this opportunity.

Prof. Prathap Haridoss: Thank you.

Prof. Babu Viswanath: Thank you.



Introduction to Research
Prof. Prathap Haridoss
Prof. B S Murty
Department of Metallurgical & Materials Engineering
Indian Institute of Technology, Madras

Lecture - 39
Research in Metallurgical & Materials Engineering

Prof. Prathap Haridoss: Hello and welcome to Prof. B. S. Murty, who is joining us here today for discussion on Metallurgical and Materials Engineering. Prof. Murty is the head of our Department of Metallurgical and Materials Engineering, here at IIT, Madras. He is a highly decorated metallurgist, has won many national awards including the Shanthi Swarup Bhatnagar Award, and he is also the fellow of several national academies and also in many international organizations. So, he is imminently qualified to discuss research and what happens in the research setting, what students undergo and what experiences they have in the materials domain and his areas of research included high entropy alloys, bulk metallic glasses and so on. So, with these words of introductions, I welcome you to this interview.

Prof. B. S. Murty: Thank you, Prathap.

Prof. Prathap Haridoss: Yeah, thanks. What would you think our traditional areas of Materials science and Engineering that students are likely to see?

Prof. B. S. Murty: Our department originally started in 59, along with the IIT, Madras as the Department of Metallurgy. So, basically looking at traditional metallurgy, when I said traditional metallurgy, our department particularly concentrated on what is called Casting, Forming, Welding these are the 3 core areas of our department. Not, too much into the extractive metallurgy part of it, though there were few experts in that area too, but that has not been real strength of our department. Particularly, because there are not too many steel plants in and around Chennai okay and that has been mostly the strength of places like IIT, Kharagpur which has been very close to so many steel plants. So, as a result our department traditionally concentrated on as I told Forming, metal forming, how materials deform and then Casting of them because there are quite a number of



foundries in and around Chennai and also what is called Welding okay, there is for example; a Welding Research Institute in Trichy, which used to tie up with the IIT, Madras metallurgy department to a large extent, L&T use to have a lot of tie ups. So, welding which is a part of the manufacturing. So, if you can say the 3 major manufacturing processes which are Casting, Welding and Forming have been the core of our department traditionally for many, many years.

Prof. Prathap Haridoss: Okay so, I think may be the, see, the general audience including aspiring researchers may not be fully aware. So, when you talk of traditional areas like Welding, Casting and Forming, how relevant is research in these seemingly you know old areas relevant today?

Prof. B. S. Murty: For example, I take one example of welding you know, there is enormous research that is going on in welding, on for example, one of the major problems in typical traditional welding which involves let us say, you melt 2 pieces at the joint and then join them, that can give you a lot of I would say weakness at the weld joined because of the melting and the casting that is involved during this solidification process. And, there are a number of applications where you are not allowed to, for the joint to be exposed to very high temperatures. So, there is something called solid state welding that has come up in recent times. So, people are also working on friction welding, friction stir welding. So, where there is no chance of the 2 pieces, weld pieces being you know melted at all and as the result in the solid state they join and they retain a number of properties of these materials, which is very essential for the component to perform in the applications. So, as a result solid state welding has taken up.

Similarly, Joining is no more a simple welding process; people are now joining polymers with metals okay, polymers with ceramics, metals with ceramics. So, there is enormous research area in that okay. Similarly, now if I take forming as an area, there are new forming technologies that have come up, for example, Hydro forming, Sheet metal forming has taken up big you know strides in recent times, for trying to get the whole for example, if I take automotive car body, these people want to have the whole car body made up of 1 single sheet, instead of simply welding and joining a number of pieces.



So, there are technologies that have come up, where there is a lot of research has to go in to develop, what is called Stretchability inside the material and there is a lot of physical metallurgy also goes in, how to develop such stretchability into metals? What kind of new type of alloys have to be developed to have a high stretchability? So, forming also including new area what is called severe plastic deformation that has come in, where people want to generate ultra fine grain sizes inside the material by severely deforming these materials and use that for various applications in terms of enhancement of properties. So, in each of these areas there is a lot of work that is going on, for example; simple thing like Casting. People are now developing what are called Metal foams.

Prof. Prathap Haridoss: Ok.

Prof. B. S. Murty: Which are very useful for; this also a traditional casting process, excepting that you add some foaming agents to develop these foams. And, there the whole technology of how to develop uniformly distributed cells is a big challenge and there people are using newer technology such as X-ray Tomography to actually see the whole foam in 3D and try to design the foam in such a way that you get the proper properties. So, in each of these areas in fact, for example, people are looking at the casting as the solidification takes place. What kind of stresses are developed in the casting using neutron diffraction? I am trying to see whether I can do something to suppress what is called heart tearing? Heart tearing is the major problem in engine blocks and pistons and things like that. When people are developing, there are problems that people encounter during the solidification because of the stresses that are developed. So, people are using modern techniques to study them and trying to identify, how do I control it?

Prof. Prathap Haridoss: Okay So so, I mean the terms for these technologies for these areas of research may look traditional and ancient.

Prof. B. S. Murty: Correct.

Prof. Prathap Haridoss: But, really there is lot of modern activities,



Prof. B. S. Murty: Really, there are lots of modern activities.

Prof. Prathap Haridoss: And lot of challenges in these areas.

Prof. B. S. Murty: Lot of challenges in each of these areas.

Prof. Prathap Haridoss: Okay so, of course, materials is a very large field and then it virtually every field of engineering has materials in it. In recent times, what has been new areas that people have focused on, interesting areas that have come up recently that large groups are working on?

Prof. B. S. Murty: Yes.

Prof. Prathap Haridoss: Ok.

Prof. B. S. Murty: As you rightly side, I always tell people materials is the backbone of everything, without materials you cannot have anything. In fact, I say everything in this world is either spiritual or material. So, to that extent materials are that important. And when we say, as I was telling you we started the journey in our department as a traditional metallurgy and then around 2003, we realize that the non-metallic materials are also equally important, not only the metals. So, we introduced ceramics, polymers, composites everything into our curriculum. In fact, we even changed the name of the department in 2003 to Department of Metallurgical Materials Engineering which is what now the department is named after.

So so, in that connection a lot of newer areas for example, Electronic materials, Polymers, Composites, when I say Composite it would be Polymer matrix composites, Metal matrix composites, Ceramic matrix composites, Ceramics - high temperature ceramics. So, if I now look at the composition of my department, half of the department faculty members work on non-metallic materials areas and as a result materials such as, biomaterials, nanomaterials okay; variety of nanomaterials including carbon nanotubes which you yourself work on. And, materials which are for extremely high temperature



environments, for example, something like nose cones of a missile okay. So, you need materials for which can stand extremely high temperatures.

Similarly, various coatings for high temperature applications people are developing, you just mentioned about high entropy alloys, people are trying to develop them as coating materials on super alloys, so that it gives additional high temperature capabilities for super alloys. Like this a number of newer areas, for example, electronic materials - quite a number of people in our department are working on that. So, after electronic materials, for example, thin films, semi conducting materials and magnetic materials. So, you name every material, now we have people working on these things and they have really taken almost equal importance to traditional metallurgy area that, if I look at the composition of the department is more or less you know, balanced in terms of modern materials and traditional metallurgy.

Prof. Prathap Haridoss: Very nice. So, that gives us also an overview of what sort of you know industries are looking for these kinds of things and where people may be able to you know, take their experience here to some setting where these kinds of things are necessary. So, of course, you are yourself a very successful researcher, very well decorated researcher and so on. And, there are many you know if you look at you do some general reading, there are lot of criteria people used to say that so and so has done well in research and so on. But, forgetting setting aside numbers, in your view in what way would you measure success in research?

Prof. B. S. Murty: Yeah, I would say anyone who can make an impact on either an industry or the future researchers, is what I would consider it as a success. That, the work that you have been doing, has it being really bringing newer technologies, newer process, newer materials which an industry is able to use and make itself as a leading industry in the Global Arena is one way of looking at success or the other way is your research has kind of nucleated a number of other researches okay, which people have taken up a queue from your research and started working on that and taking it forward, this is another way of looking at it. So, basically it all depends on what kind of an impact that you are making either on the industry or on the research.



Prof. Prathap Haridoss: Ok.

Prof. B. S. Murty: Of course, people use numbers as a way of looking at it, but not always numbers are important. Among the numbers, which are the ones which have made more impact on the society of course, it's also important when we talk about in this context. Has it also made an impact on the general you know development of the country in terms of the societal needs?

Prof. Prathap Haridoss: Societal needs.

Prof. B. S. Murty: So, that's also very important. Rural development areas, the number of newer materials that are being developed which are being used for certain applications in even agriculture, so that we grow better. So, these are all the things which are, I would say the matrix people should use.

Prof. Prathap Haridoss: Should used to gauge how.

Prof. B. S. Murty: In my view.

Prof. Prathap Haridoss: Okay very nice. So, if we step back and we have new students who are coming from various backgrounds and who joined research programs in you know various institutions across the country, they are moving away from college education, undergraduate education to postgraduate education and so on. In your view, what are the typical kinds of challenges such student face especially, let's say coming into a materials kind of a department?

Prof. B. S. Murty: Correct. The basic difference between an under graduate education where you are kind of coached okay to study some material or certain reading material and then you go through certain exams and then gets certain grades out of it. To research is the basic difference comes is the, where you try to independently do most of the things yourself. So, PhD is a training ground where we train people, how to plan a certain work and execute it, analyze what comes out of it and then bring out some logical conclusions



out of it. All these things has to be inculcated in a particular student, which most of the students who come from the undergraduate education are not really trained on that. And, also particularly materials research needs handling a number of instruments.

Of course, there is a lot of computational research that also goes on in materials. For example, the new buzzword now is called ICME; Integrated Computational Materials Engineering the people call it as. Starting what people, starting from the atomic level to the macro level can we connect everything, what people call it has multi skill modeling. So, there are people who are trying to do that. There also, many people might have learnt software as a software per se. But, not utilizing the software for solving certain problems, real like problems that posses a lot of challenge when a student actually comes into research. This is one aspect.

Second aspect is in the experimental work, where handling various equipment okay For example, making materials itself is a major challenge and that has to be made in certain condition so that, there is a reproducibility that is essential in any scientific research. So, that is another thing that has to be trained by the faculty and the senior students to the newly joined student. And also, handling various characterizing facilities, just by making a material as simple as I can tell you, anybody can easily make a nanomaterial nowadays. But, to prove to somebody that you have made a nanomaterial is not so easy, it needs various microscope which are extremely expensive starting from anywhere ranging from about 1 crore. It can go up to almost like about 17 crores, we recently have bought a microscope in our Institute which costed us almost like 17 crores, where it can go up to 1 angstrom resolution.

So, you would like to see these materials at that level and that needs a lot of training. So, we have a number of these training programs. For example, we have a course which is called Practical Transmission Electron Microscopy Course. Where, one full semester the student goes through, how to handle this such a sophisticated instrument? Starting from, how to prepare a sample for that instrument to the last day where he will demonstrate good micro structure from such a microscope to be able to you know make himself you know eligible to handle such kind of sophisticated instruments, all these need to be trained. So, that is why the whole 3 or 4 or 5 years, the time that is the students spends



here, for either an MS program or a PhD program is basically towards this kind of training. So, these are the challenges. Of course, they look as challenges when you start with, but there is a lot of enjoyment in learning all these.

Prof. Prathap Haridoss: Lot of opportunities.

Prof. B. S. Murty: Lot of opportunity.

Prof. Prathap Haridoss: Yeah, yeah.

Prof. B. S. Murty: Lot of enjoyment when you do it and when you learn many of these new techniques, they are going to be of a great use when you go for a post doctoral research later where, you are supposed to do a independent research and then also when you yourself become a faculty member to guide the next generation students. All these training will be of a great use.

Prof. Prathap Haridoss: In fact, yes in along those lines. So, what sort of positions did you typically see post graduate students, people completing MS and PhD from our department? What sort of positions did you typically see them going towards?

Prof. B. S. Murty: Typically, the majority of the positions I would say are academic positions, faculty members in various now. In fact, that opportunity has significantly grown, if you look at it, once upon a time there were just 5 IIT's now, there are almost more than 20 IIT's, possibly a number of 23 or so. I do not exactly know the correct number, but something around that.

Prof. Prathap Haridoss: Yes.

Prof. B. S. Murty: And then, NIT's. You have about 1 NIT for each state. I heard that, there are about 30 NIT's and a number of private you know academic institutions. Which are also of reasonably good standard are coming up. In fact, some of them are taking up faculty, who are being called as research faculty, they don't need to even teach, they



simply their job is to keep on doing research, get projects, to publish paper so that their visibility in the research arena is grown. So, this is one spectrum which has grown to a big level that the opportunities for PhD students into those have significantly grown. This is one.

Second is, earlier the industry never used to really take a number of PhD students, they say B. Tech is good enough for us. But, it is now you see quite a number of industries are taking research scholars, I mean people with an MS or PhD background, particularly because they would like to compete globally and become a global leaders. For example, in Chennai itself if you look at it Mahindra and Mahindra have set up a big R&D center, where they are looking for people with the research background. Similarly, a number of Multinational R&D Institutions have come up for example, GE has their own set up in Bangalore, who are taking a large number of materials people with materials background with MS or PhD. Including even Tata steel, every year he is taking a number of people with a PhD background so, for their own R&D. So, the industrial R&D is picking up in a big way.

Prof. Prathap Haridoss: Ok.

Prof. B. S. Murty: Thanks to what we call it as make in India.

Prof. Prathap Haridoss: Make in India movement.

Prof. B. S. Murty: With that movement, a lot of industries are trying to make their products globally competitive and not only globally competitive they would like to be global leaders.

Prof. Prathap Haridoss: Ok.

Prof. B. S. Murty: So, that is where research is very crucial and people with research background are able to fit themselves into such an environment and then grow much better.



Prof. Prathap Haridoss: Okay very nice. Now, let's supposing we step back a bit and look at the you know time that post graduate students spends here as a researcher through his you know degree program, a very mundane question, how often should such a student meet his adviser his or her adviser?

Prof. B. S. Murty: There cannot be a prescription for such a thing.

Prof. Prathap Haridoss: Sure, sure.

Prof. B. S. Murty: Ok, it depends on the student okay how capable he is to handle things on his own. It depends on the background the student comes from and depends on the problem that he is tackling.

Prof. Prathap Haridoss: Okay fine, fine.

Prof. B. S. Murty: If it's a very difficult problem, I would even suggest but, it also depends on whether you are in the beginning of your research or are you moving towards the fag end of your research, PhD degree or so. So, initially it is suggested that you meet almost on a weekly basis okay. So that, you are sure that you are on the right track and once you feel yes, you got a hold on the thing, that things are moving well and then possibly you know I would suggest at least once in a month.

Prof. Prathap Haridoss: At least.

Prof. B. S. Murty: If not frequent, more frequent than that.

Prof. Prathap Haridoss: More frequent than that, Ok.

Prof. B. S. Murty: I usually meet my students once in a fortnight.

Prof. Prathap Haridoss: Ok



Prof. B. S. Murty: So, once in a fortnight, I would an hour I spent with each of the student.

Prof. Prathap Haridoss: Each of the students.

Prof. B. S. Murty: And, I tell them that if you have any other issues, you are free to come and meet me any time.

Prof. Prathap Haridoss: Ok.

Prof. B. S. Murty: But, I would suggest that initial days, it is more important to meet more frequently because the student is not used to you know, first of all planning things. In any research, if you don't plan anything properly ultimately, the result is not going to be as expected.

Prof. Prathap Haridoss: Ok.

Prof. B. S. Murty: So, as a result during the planning stage it is very essential that you are in touch with your guide a more frequently because his guidance becomes very useful in that initial period. Later, you are possibly able to guide yourself on your own.

Prof. Prathap Haridoss: Okay so, I would like to close with this question for you, what would your advice be to an aspiring student in Metallurgical and Materials Engineering?

Prof. B. S. Murty: Okay for that matter, any area if you are doing PhD whether it is metallurgy or not, you should first enjoy this is very important. I would always tell this for a research scholar you need to have 3 qualities okay, you should be able to go for a cup of coffee with your friends and discuss research, not a cricket match that has happen possible the previous day or so. And then, you should be able to go to bed thinking about the problem that you are trying to solve okay that basically means, you are involved in your research, you do not take a research as one of activities that you do on a daily basis. But, it is the activity for you, the most important activity.



Most importantly, I would say the third one is, you should be able to catch an youngster okay just joined in your group are possibly a 12th standard kid and make him excited about what you do. What is very important in research is, to be able to feel that you are possibly the first one to be working in that field, after all you know any research if you simply repeat what somebody else has done, you would not get a degree. So, every PhD student wherever he is, he is working on a problem which possibly nobody else is working on, at least a particular facet of a problem, which he is uniquely trying to develop a solution for that. So, he should feel excited that he is the only one who is working on that and that excitement in research is very essential, if you do not retain that excitement and that fire in you obviously, research is not enjoyable and if you don't enjoy a research there is no point in doing research.

Prof. Prathap Haridoss: So, involvement and enjoyment.

Prof. B. S. Murty: Yes, are very essential.

Prof. Prathap Haridoss: Thank you so much for joining us, they were useful one.

Prof. B. S. Murty: Thank you. Thank you.



Introduction to Research
Prof. Prathap Haridoss
Prof. V. Anantha Subramanian
Department of Metallurgical & Materials Engineering
Department of Ocean Engineering
Indian Institute of Technology, Madras

Lecture – 40
Research in Ocean Engineering

Prof. Prathap Haridoss: Hello, we are very happy to have with us Prof. Anantha Subramaniam. He is the head of Department of Ocean Engineering here at IIT, Madras. He has extensive experience in this field; he has been in this area for over 35 years now. So, lot of industry collaborations, lot of projects, lot of research projects which maybe you know extensions of concepts that we are dealt with through industry collaborations, lots of extensive you know visits and collaborations with international universities and so on. He has very rich experience in research, in dealing with students and dealing with working with industries; you know shaping students' lives as they go about through their you know Masters Degrees, PhD Degrees and so on. So, it is pleasure and privilege that he is able to join us today. His own areas of research are Computer Aided Design and Application of CFD to floating structures including ships and other floating bodies. So, it is a pleasure to have you with us sir.

Prof. V. Anantha Subramanian: Thank you.

Prof. Prathap Haridoss: So, I just wanted to start with this idea that you know in almost any field of engineering there are certain areas that have been what we would consider as traditional areas of research, which have been around for a long time. Of course, even in these areas there are you know newer concepts that people look at, but there are some areas that we would call as traditional areas of research. So, in ocean engineering what would constitute such areas?

Prof. V. Anantha Subramanian: To start with I should say that Ocean Engineering is not a basic discipline by itself. Is an area that has emerged as a discipline, consequent to the oil crisis in the 70's. Otherwise, ocean engineering would cover the major sub set of ships, naval architecture in all its variance. So, offshore structures where then designed to be operated in the challenging environments of the oceans and then of course, there were support results and then of course, ships cannot just go and operate in the deep sea they have to come to the



harbour. So, a whole lot of disciplines have been encompassed in what we now called Ocean Engineering.

So, there is the traditional design and analysis of ships and structures, their interaction with the waves, their propulsion systems, their navigation systems, the manoeuvring, the sea keeping. Now, from there the offshore structures which are huge manmade structures, but not necessarily powered by an engine. They will have their own complexities of being floating bodies or complain bodies or adhered to the ocean bed or articulated in some form. Then you have the coastal structures, ports, harbours, coastal break waters so all these have now come under the Umbrella of Ocean Engineering. And of course, I will be missing out because our department as now forayed into what we called Petroleum Engineering. Although strictly you can deal in petroleum engineering from ocean engineering, but that is what ocean engineering is?

Coming to the traditional areas of research, we can take up the challenges of all associated ship hydro dynamic problems in designing ships, in understanding the structures of ships, in the way the wind and waves interact with the ship and therefore, give rise to stresses, give rise to comfort or discomfort in terms of large motions, the dynamic effects that occur. So, these are the traditional areas of research, where you use modern tools such as of course, finite element analysis, computational fluid dynamics and these help you to therefore, understand better and understand more precisely problems which were thought to be not solvable a few decades ago. So, these are the traditional areas of research.

Now, when you come to the offshore structures, etcetera, there are long terms problems such as fatigue, the failure of certain kinds of joints and the challenges of handling these in the hostile environment in the oceans. So, you inevitably come to interdisciplinary areas, even in this already multidisciplinary area of ocean engineering that would be in terms of control systems etcetera. So these are the new areas of research where it is neither controlled systems experts, such as from let say the basic field of electrical engineering nor is it the naval architect, they have to work together and they have to understand these.

So, these are the new areas of research. Like this we could extent to the area of petroleum engineering because the challenges of exploring below the ocean bed for oil and having to monitor the process of drilling these involve lot of new challenging areas. Again the quest for energy from the oceans, we have the new areas, how to tap energy from the waves? So, we



have a whole lot of devices, some of them have been demonstrated to be technically feasible although still not commercially viable. So, we have structures called ocean oscillating water column, backward bent ducted boil, hydro turbines wherever there is current available, ocean thermal energy conversion. So, all these have given raise to new areas of research.

Prof. Prathap Haridoss: Okay so, lot of traditional areas and several significant new areas that. Of course, I think may be more so then any other discipline I think KLE in ocean engineering there is direct link with you know applications that are visible to the general public.

Prof. V. Anantha Subramanian: Very much.

Prof. Prathap Haridoss: In this context of course, when we talk of research we tend to talk of you know the relaters, the cutting edge and something that may be has a tendency for us to be associated with an academic setting. So, in this context, what aspects of research that go on in ocean engineering or of may be more immediate interest to the industry.

Prof. V. Anantha Subramanian: Yeah, that's an interesting question because traditionally industry looks forward to fast, robust tested solutions rather than deep involved research, which demands handling of it by vertical experts. And, this is something it has been catered to the industry, that we traditionally had the problem of analysis and design because again if I take the example of a ship, is a largest man made mobile object. The challenges of making this structures safe and sea worthy means you have to analyse it, you see today what we have trusted tools like finite element methods, for the structural analysis computational fluid dynamics for analysing the forces that are coming out of it and trying to address the problem of safety and comfort also. Because, you have a lot of dynamic effects occurring such as what we called shipping green water reducing the possibility of water coming on both the deck or trying to minimize the motions.

So, these are solutions which the industry looks forward to, I could add let us say for the example; Indian navy too. That we do offer solutions in these areas by analysing and experimenting with the kind of floating body that they have in their hands, to that extent the department of ocean engineering is well endorsed with the very rare combination of facilities which cannot be thought of under the university umbrella. So, to name them we have the towing tank, we have wave basin, we have the flumes, we have the best of instrumentation, automation that we are able to form, give solutions to the industry by giving them immediate



answers to specific problems. But, long term research continues because as I said earlier, the industry cannot wait for expert solution that requires lot of expertise in handling.

Again, from the industry problems where we get live data and with regard to the ships or with regard to the whatever the floating structure. We take it up further by way of analysis and here comes the interface of academic research, industrial consulting and providing solutions. That is where we have an interesting combination that we are able to use these sources that we get from the industry and try to match it to international levels of understanding, not just offering the solutions sometimes it is understanding the physics of the problem. It may not be that we really form a solution but, we even sometimes know that is certain approach cannot be had and I would say that all these help in contributing towards a research that the department undertakes.

Prof. Prathap Haridoss: Okay great, when I look at ocean engineering and say compare it with the other disciplines that are around including say mechanical engineering, chemical engineering, lot of other disciplines. One of the things I think where the ocean engineering department stands out or stands apart, is the fact that may be there are not enough you know undergraduate programs around in the country or may be even internationally which are focused on in this area. So, by the same token so when MS and PhD students come into your department, may be they are not, I have been I am curious to know to what degree you feel they are adequately prepared for handling what research is in specifically in your field? And if so, I mean or in that context, are there specific difficulties that they face as they come in, challenges that they face and if so, how do you go about addressing?

Prof. V. Anantha Subramanian: Yeah, I think you have pointed out a relevant issue there, because as I have said ocean engineering is not a basic discipline by itself and candidates that we get for our Masters and PhD programs are drawn from many disciplines. Typically, mechanical engineering, civil engineering, some from naval architecture, many from aerospace and of course, because we have the Petroleum Engineering there are some candidates coming from science backgrounds, oceanographic backgrounds, geophysics and so on.

But, let me dwell on the main part of ocean engineering. There is a learning curve that these students have to understand, what the department is about? What are the main courses? We do have a challenge because with a relatively small number of faculty being able to satisfy



the requirements that these students need to go through, we do find that unless they have gone in a little in depth into some of the fundamental aspects such as, hydrodynamic structures and of course, whatever else they want to take it, take up in their research areas. There is a little bit of I would say a limitation that they could do better, if they were geared for that. I must also say that unfortunately research is being made a rat race that there is a tendency to go for numbers, maybe it is the numbers of degrees produced, maybe it is the numbers of visible outputs produced, and that in my personal opinion is counterproductive. If we could address that problem and allow people to learn and experiment and then go about I would say the quality of research would go many notches above.

Prof. Prathap Haridoss: In fact, is in that context I was curious since you have had you know so much experience working with industry and research institutions, internationally as well as in India and so on. In your view, in fact, as exactly as you pointed out you know there are certain matrix that people use and sometimes those are even misused. In your view, what is a good way to look at progress in research from a student prospective or as I know as you interact with the student? How would you feel, when do you feel you know the student is actually making progress in a general sense with respect to research?

Prof. V. Anantha Subramanian: I think the first step would be that the student should not be straight jacketed that from day one the problem is presented and he is asked to work on that. He have to allow the time for a little bit of learning, I know that student will work or for that matter anybody will work only under pressure. So, the pressure has to be applied on them, but this pressure of looking for numbers, if we could take it out, would help them to settle down. The system expects the numbers therefore, they are busy preparing numbers.

So, I think we should be delink this to some extent, without reducing the pressure on them, which means that in first semester when they typically are busy with their courses, the mandatory courses to be taken, we should allow them to interact with other peer group meaning other researchers. Learn the basic tools because I personally believed that the research, whether it is a MS or PhD level is reaching the peak of the pyramid. So, there is a base irrespective of what you are doing, students have to be mandatorily exposed to that and have to get the confidence of that and finally, learn to stand on their feet. If this model could be added, I do not think this is idealistic I have been trying to apply this to my own group and I would say in our humility that we have been successful, maybe the time does'nt meet the stiff deadlines we don't get a PhD in 3 years or something like that. But, I feel this is a much



more hold some way of approaching research and more important is students who come through such a process are easily absorbed in jobs.

Prof. Prathap Haridoss: Ok.

Prof. V. Anantha Subramanian: It is a very satisfying end result of all the work.

Prof. Prathap Haridoss: In fact, in this process by which this student settle and then grows in the research area, in the research setting I think there is a lot of importance to the extent to which the students interact with their peer other students who are present and also with the faculty that they work with, their guide and so on. Maybe a bit of mundane question, but in your view you know what do you see as you know how often students should be meeting their guides? Because, it is very different from a course based kind of learning, there is a lot of I mean influence and importance to this interaction that makes a difference in how they grow. So, what do you think are you know good frequencies with which people should be meeting the guides and how they should be going about it?

Prof. V. Anantha Subramanian: Yeah, that's again a very interesting question because whenever I have addressed the students in their research methodology course, I have tried to put this question as a starter for the course, how many times do you think you should meet the guide? So, I get lots formula on that and of course, many students obediently feel they should meet their guides regularly. I am against that, I would say there would be contact with the guide yes, and the real smart student is one who will get the queue through many casual meeting and also with the meetings of other students.

So, in an ideal sense if you want put a number may be once a week is fine, but more important is the student is able to stand on his feet, put in his efforts because I have always seen that there are so many students who you may be seen around but they come back with results, that is a very important thing. And, they do not get the sense straight, it is not that easy problem, but they have learned to reach out to different sources, put their effort, done their design, done their fabrication, even gone to the market and got the things fabricated it.

Again, I would say maybe I am just putting it in my experience. That I have a group of about 30-40 and we have everybody for everything. So, whether it is a software and induction of the new comers into the learning process for different softwares or whether it is the fabrication or whether it is experimentation. So, I see a harmonious group therefore, I would



say that building a group is very important to help succeed in research and learning from each other rather than, just from the guide. So, I don't think that it is important to meet the guide on a very frequent basis, if they have to meet, most welcome. I do not think that, that is a formula to success.

Prof. Prathap Haridoss: Great, I think that is a very valuable insight into you know what should be going on and at least how the student should approach it in their mindset. So, you were speaking about you know the successful students who complete all their activities here successful and so on. Now, when we look at MS and PhD research type of students, there is always this feeling, in not just in ocean engineering across all disciplines that when you an MS or a PhD you are an expert in an sort of narrow area. As supposed to journalist, who would what now, which is what we would associate with the bachelors degree. Now, of course, being an expert is a good thing but, at the same time that probably narrows down, the kinds of positions that they can get into the jobs that they are getting and so on. So, in your experience, what sort of positions do MS and PhD students completing degrees in ocean engineering, what sort of positions do they go to or they get?

Prof. V. Anantha Subramanian: Yeah, I do believe that acquiring a higher degree in the environment of IIT does not mean achieving proficiency in a particular vertical area alone that is important. But, more important is the problem solving capability, the ability to look at something if they are independently and in depth, the ability to get a team together and work. To me these are the strengths, which give them the job opportunities. Again, I would say I have been very lucky that all my scholars at the time of finishing have been already placed in a job. Of course there are disadvantages, that the student has no time to prepare the visible research outputs, this is the only hassle. Otherwise, I would say that they have been by and large place and I would say that the reason for that is because they have brought these consults, because you do not know where which particular opening you are going to get.

So, essentially you have to take your research here as a learning experience, the ability to solve problems and with this they are able to adapt. Typically, my students have been going to the design organizations, maybe shipyards, may be offshore industry, they have been able join the computational fluid dynamics application software development companies, such as CD-adapco and so many others Eaton and so on. They are there in many industries where, they get compartmentalized and are looking at part of a CFDs solution there, either formulating or developers or testing etcetera. But, by and large I would say whatever they



have done let's say, level confidence which they exclude and that seems to help to meet. That is a very important thing because if you are lucky you manage to work in the same vertical area that you choose for a PhD. But, most of the times you have to be open enough because maybe some other day you will come back to the area or maybe destiny will take you to some other area. So, I think we all have to take it easy, enjoy the research, do what you are doing, don't lose sight of realities, be able to project that you are versatile. For me this is most important.

Prof. Prathap Haridoss: Great. Okay I think maybe you already sort of addressed, what I wanted to ask you next, but in any case I just wanted to know, are there any specific words of advice that you have for students who are aspiring to join an MS and PhD program in ocean engineering?

Prof. V. Anantha Subramanian: Yeah, students who are aspiring to join MS and PhD, first they have to feel convinced that they are not venturing into some unknown area they have to feel comfortable. I do feel that the branch is so broad based and so multidisciplinary and interdisciplinary, the adaptability with which they can go and work in so many wild areas of specialties, that is one thing. The second thing is people need to learn to share and exchange knowledge, it would be very wrong to think that you are working in a narrow area, you would rather individually or you rather work in small groups or even as individual because to me these are much more important in achieving success in a future carrier. The other thing is don't forget to be happy because if you are not happy whatever you do, you cannot get the best of things. Again, I say this from experience because, if a student or a teacher or whoever is not going to enjoy what he does or she does every day, there is no motive because finally, that is a life philosophy that the objective is to be happy everyday and follow that.

Prof. Prathap Haridoss: Okay Great. Very nice words to conclude this discussion with I think. So, it talks about both professional growth and personal growth.

Prof. V. Anantha Subramanian: Both are connected.

Prof. Prathap Haridoss: Great. Thank you Prof. Anantha Subramanian for joining us, it is a pleasure to have you.



Prof. V. Anantha Subramanian: It is my good fortune that you wanted me to say a few things. This will benefit researchers, new students who want to look into IIT and take make the best out of their carrier.

Prof. Prathap Haridoss: Thank you. Thank you so much.



Introduction to Research
Prof. Prathap Haridoss
Prof. Usha Mohan
Department of Metallurgical & Materials Engineering
Department of Management Studies
Indian Institute of Technology, Madras

Lecture – 41
Research in Management Studies

Prof. Prathap Haridoss: We are very happy to have with us today Dr. Usha Mohan from the Department of Management Studies here at IIT Madras and of course, the management is something that you know a lot of people are very interested in and it's you know at least professionally a lot of people pick up degrees in management at some point or the other, especially in an industrial setting or a corporate setting. Of course here, we are taking about research into the areas of management. Dr. Usha Mohan has been here at as a faculty in IIT, Madras for 6 years. Before that she was a faculty in the University of Hyderabad.

Her areas of research, areas of expertise include polyhedral combinatorics, supply chain, risk management, optimization and quantitative models in supply chain management. So, variety of areas that she works on and she also works with the incoming students in the department of management studies to help them get into the research process in a smooth manner. So particularly, she is well qualified and well experienced in addressing the types of aspects that we are going to discuss in this module. So, thank you for joining us.

Prof. Usha Mohan: Welcome.

Prof. Prathap Haridoss: So, I would like to start with this general question we ofcourse, think of management as something that is on the field I mean in a corporate setting and so on. Managing various activities that from going on there and so on. But in terms of research, are there areas of research in your field which are considered you know traditional areas of research, which have been there for a long time where may be there is lot of literature out there that people can refer to?



Prof. Usha Mohan: Yeah, see, I think at this point of time I need to mention that at parent department of this humanities and sciences, social sciences. The department of management studies was born out of this parent department. So, even prior to the research from the department of management studies a lot of our faculty, senior faculty where associated with the humanities department and if you look at the traditional areas of research at that point of time were mainly people who came from an industrial engineering background. We have; we did have finance and HR and this branched out now into 6 areas of functional areas of management once the department of management studies was born.

So, you can put the traditional areas, one into organizational behavior, human resources management, the next is financial management and third would be marketing. The forth is operations and production management, the fifth is systems information systems and then sixth area which you called as integrative management where basically a lot of interdisciplinary within the management studies is addressed to; so this are the traditional areas in 6 functional silos.

Prof. Prathap Haridoss: Ok

Prof. Usha Mohan: Now, if I look at each silo in an organizational behavior in human resources, the typical problems they addressed are motivation, job satisfaction and they look how to come up, they understand organizational citizenship behavior and this is the work-life balance, these are the type of problems they address. In financial, it has been typically investment analysis then they look at commodity price modeling, financial risk analysis etcetera. Marketing, it has been analysis of advertisement and media with market segmentation, consumer behavior. These have been the traditional areas of research.

Operation and production management, I think this owes a lot to the industrial engineering and industrial management background with which the senior faculty came with problems which were mainly focused on resource scheduling, algorithms, coming up with facility planning, transportation and network problems. I am talking about the traditional areas and I think very soon we will be talking about the new areas of research



also. Information systems again, analysis of ERP systems, effectiveness of information systems and integrative; we are trying to come up algorithms with modeling and analysis of virtual organizations, knowledge management processes these have been the traditional areas.

Prof. Prathap Haridoss: Ok, great. That's a very I mean elaborate, detailed layout of what are; what is happening in the general field of management studies in terms of research and so on and as you mentioned you know like with the engineering I presume that even with management there are areas of research that are considered relatively recent in terms of, you know with the way the people have given attention to it and so on.

Prof. Usha Mohan: Yes.

Prof. Prathap Haridoss: Maybe the last 5, 10 years and so on. So what of sort of areas are there in that context?

Prof. Usha Mohan: Yeah, again you can see that as I mentioned that the typically, I will start with the operations. Operations has been more in industrial engineering and production engineering motivated operations management and over the last 5 to 6 years we have seen the shift towards more quantitative modeling and where the methodologies have been more game theoretic methodologies. So, we have both contributions to game theory and we have more game theoretic models like people have been taking about incentive mechanisms and understanding business situation, typically supply chain situations from game theory point of view, game theoretic modeling, how do you model certain situations using game theory? Now, again marketing digital marketing as the new area and we have faculty who have been working in digital marketing as well, sorry and then after sustainability being an important issue. So, we have faculty of working in sustainability, apart from this we have faculty who have been integrating spirituality into psychology of work.

Prof. Prathap Haridoss: Ok.

Prof. Usha Mohan: And all of that so, these are the new areas.



Prof. Prathap Haridoss: New areas that.

Prof. Usha Mohan: Where a lot of faculty have found interest and some very healthy development is lot of people are trying to integrate among areas and we have people who are from two, for example, we have people from HR and analytics coming together to come up with something and say, knowledge management and things like that we do have health care analytics, data analytics and health care operations wise and looking at it from different dimensions, but analytics means the (Refer Time: 06:30).

Prof. Prathap Haridoss: That's very nice to know, The I think again as I mentioned may be and what is probably there in the public perception is of course, a management candidate; a person with the degree in management is often quite well sort after in many various you know settings associated with corporate world, with even you know new entrepreneurial ventures and so on, but in terms of the research that goes on in the area of the management, are there specific areas of research that the industries sort of you know almost immediately interested in? It's quite likely that almost any area of research may be little later they may be interested in out in the future and so on. But are they currently interested in some specific activities that you, in your opinion that you feel is of you know current interest?

Prof. Usha Mohan: Yeah, I think when it comes to the industry though we did have a flagship program which was the MS entrepreneurship program. It was housed in the department of management studies, still I think last year it became an institute wide program and this was very well sort after and if you look at the candidates because we are housed in an Indian Institute of Technology, I think most of our resource scholars and the way we structure our courses itself has an analytical component, no matter which functional area you belong to. So, even if we find the type of jobs the industry seeks from our resource scholars are definitely; they expect them to be in an analytical position and that is where people are trying to go, be it HR analytics or marketing analytics or OM analytics that is so the typical job profiles are more of an analyst type of a profile which people, if they are go into this one, but we have an equal number who get into academics also.



Prof. Prathap Haridoss: Ok. So that also, in fact that was something that I want to touch upon.

Prof. Usha Mohan: Yeah.

Prof. Prathap Haridoss: So those are the kind of positions that people.

Prof. Usha Mohan: Yes we have.

Prof. Prathap Haridoss: Graduate, post graduate students.

Prof. Usha Mohan: In fact, this last batch who graduated last week, 3 of them have gone to the new IIMs; IIM, Kozhikode; IIM, Kashipur and one of the measures

Prof. Prathap Haridoss: Teaching positions.

Prof. Usha Mohan: Teaching positions, yeah.

Prof. Prathap Haridoss: Academic positions.

Prof. Usha Mohan: Academic positions, 4 of them have taken academic positions.

Prof. Prathap Haridoss: Ok ok . And in general.

Prof. Usha Mohan: Yeah.

Prof. Prathap Haridoss: And ok in general and certainly maybe in association with what you see in the set of students that you meet during your you know post graduate program in management studies, how do you see or how will you advise them to measure success in research? I mean they are used to an academic setting or may be, they have done some courses before they came here. So, they have had always had a different metric on how they measure success or their own success.



Prof. Usha Mohan: Yeah.

Prof. Prathap Haridoss: In a research settings, specifically with respect to say management activities, is there anything that you would suggest is a better way for them to measure their success at progress?

Prof. Usha Mohan: Yeah, see since I deal with and interact a lot with new scholars, the first thing we try and tell the scholars is the shift from being a taught course to a research course and once both are MS by a research and PhD, what we try and tell them is ownership is extremely important, ownership to the problem and ownership to the process everything. So, every all your ethics and everything come in there and the way we would like them to define successes rather than, then it should be 'aha' movement, first of all that is something which we expect them feel good about what you are doing and do not chase numbers. The matrices' yes; obviously, publications we strongly encourage this scholars to publish in highly rated journals not just for the numbers and not just for the number of a publication, but the choice of journals is also equally important. So, that is something which we are emphasizing.

So, short term goals might be perhaps publications, good conferences; choose wisely choose conferences, attend good conferences which are rated well by the peers that is something which we strongly just not any conference. So, we try to rather than been counting it should be more by choice, choose wisely that we something which we try and encourage right from day one then after words, there are also minor awards, presentations, best paper awards this is sort of pattern the back and ague to perform better. So, we do have scholars who present both at a national level and international level and regularly obtain awards. So, these are short term. Long term; obviously, would be invitations to chair conference sessions, give plenary talks, but that would be more at a faculty level, but at a scholar level I think the short terms gains, which is to get hold of a good understanding of the research and wide acceptance, wide acceptance could be through conferences or publications.

Prof. Prathap Haridoss: Ok great, I think that that probably applies to a lot of disciplines and so on maybe on a mundane note even you when you when we are talking of a



measuring the search success in research and so on. I think the transformation from you know students life as an undergraduate student or postgraduate student life there is a lot of activity associated with the group that makes a big impact in how successful the student is in their research carrier. So in this context, what do you think is a you know good guideline for; how often students should be meeting their adviser or guide and how does it maybe change with time or what is your opinion on this?

Prof. Usha Mohan: Yeah, I think we can put it as different stages of research. The initial stages anything between meeting the guide once a fortnight to once a month because what we have done in a department now is we have strengthened this course structure. The way we are strengthend the course structure is because department of management studies unlike other core engineering in disciplines, where you get core M. Tech to get into PhD we get a more diverse audience, we get a more diverse group who want to do a PhD management studies.

Now to address this, we have come up with the three prompt course structures in which we have prescribed three sets three courses which form the course research. So, any scholar who is graduating from the department of management studies has to take three courses, earn the credits of which there are three courses which are core. The core is a research methodology which is against, which sort of it is making of a researcher which introduces a researcher to what is expected out of a researcher, how do you deal with problems and all of it's a methodology course.

Of course, we have separate functional areas research methodologies but this is a broad thing the other is data analysis research because we believe that no matter what is your functional area, you should have knowledge of data analytics and the third thing which is academic writing. So, it is making of a researcher, the methodology course and how do you present your research. So, this is what we have termed as a core curriculum. Now, again from management studies it's very important to have a breadth knowledge also, breadth knowledge of at least three of the functional areas against the functional areas are marketing, operations, finance, HR, OV systems and integrative.



So, we have said that student who is from a management study should have breadth knowledge also. So, we have divided into core, breadth and depth and depth is where you dig deeper and you take courses. So this is how we have designed our courses. Any students in the first year we expect them to really focus on this core. So, get yourself prepared to do research. In the first year, we are encouraging scholars to take a lot of course and prepare themselves a research rather than deep dive itself. So, once you gain that maturity which we feel would take at least 6 months to a year, after that your guide starts playing a role. So, first is get hold of your necessary ammunition and then you start deep dive.

Prof. Prathap Haridoss: Ok.

Prof. Usha Mohan: So once that is; that varies between the problem, the guide we have we do have a lot of part-time, external.

Prof. Prathap Haridoss: Ok.

Prof. Usha Mohan: But at that point of time it would be more regular, but the first year is generally getting hold of the course work.

Prof. Prathap Haridoss: Ok.

Prof. Usha Mohan: Preparing you to be a good researcher, we rather put the making of a researcher is the first year is spent for that and then it the frequency keeps increasing. So, at your literature review it would become fortnightly, then it would become to a month then it would be fortnightly to a week and then the thesis writing stage perhaps the thing would become much much more frequent.

Prof. Prathap Haridoss: Great, nice to know these insights. So, maybe I will conclude with this little bit of general question.

Prof. Usha Mohan: Yeah, sure.



Prof. Prathap Haridoss: What are your words of advice for an aspiring student in who wants to do post graduate study in management, an aspiring post graduate student?

Prof. Usha Mohan: Yes

Prof. Prathap Haridoss: For management studies.

Prof. Usha Mohan: The first thing is I think for any researcher, the key quality we look is passion. If a person is not passionate about the research I don't think any researcher would survive because for person with passion it will keep the fire alive, ignite your passion. So, that is something which you have I think one of the key qualities we look for because research is, it's not time bound it can take anything between 3 to 6 years. So, that you should be willing to, we should have both the robust nature and you should have the tenacity to overcome hurdles and look for better things positivity. These are qualities, attributes which we look for and once you have the passion I think everything else falls in place.

How do you develop the passion I think it is a very individualistic goal and that passion and commitment I think these are the two things; commitment to research, these are the two main things which have been and once you have the passion I think everything else will fall on place that's what I would say.

Prof. Prathap Haridoss: But is there any, I think that you feel that is maybe specific to management in this.

Prof. Usha Mohan: See management studies I think it is very, very important that we articulate what the business needs in terms of abstract.

Prof. Prathap Haridoss: Ok Fine.

Prof. Usha Mohan: With because nobody is going to define that for us.



Prof. Prathap Haridoss: Ok.

Prof. Usha Mohan: And the minute that is the idea, there is business requirement; are we going to articulate that into? So, the first transformation happens when you are articulating or translating a business or a decision need; you are abstracting that into an academic environment, you are working in that academic environment. So, all that thing whenever you are working, you need to constantly see that you are not doing it just for the abstract nature, but then after it should lead to something meaningful and again the last stage is to whatever you are developed in the abstract world you will have to get back to the business.

Prof. Prathap Haridoss: Get back for the business, ok.

Prof. Usha Mohan: I think these two things are equally important. How are you translating a business needs to abstract and how are you going to represent in the business world?

Prof. Prathap Haridoss: Ok.

Prof. Usha Mohan: And once that becomes then your research becomes meaningful in a management setup.

Prof. Prathap Haridoss: Okay great. So, thank you Dr. Usha.

Prof. Usha Mohan: Welcome, Yeah.

Prof. Prathap Haridoss: For joining us and sharing so much of insight into a field that you know maybe especially engineers may be not very familiar with, there is a lot of curiosity always about management studies and so on. I thank you very much for joining us.

Prof. Usha Mohan: Thank you.



Prof. Prathap Haridoss: And sharing this.

Prof. Usha Mohan: Thank you.



Introduction to Research
Prof. Prathap Haridoss
Prof. S.R. Chakravarthy
Department of Metallurgical & Materials Engineering
Department of Aerospace Engineering
Indian Institute of Technology, Madras

Lecture - 42
Research in Aerospace Engineering

Prof. Prathap Haridoss: Welcome. We have it's our pleasure to have with us Prof. Satyanarayanan Chakravarthy, who is a professor in the Department of Aerospace Engineering here at IIT, Madras. He is basically a B. Tech from IIT, Madras and he has got a PhD from the Georgia Institute of Technology also called Georgia Tech, I guess. He is been a faculty in IIT, Madras for 20 years. His areas of research are Propulsion and Combustion. In fact, I mean he is recognized expert in both these area. So, he heads the, he is the coordinator for the National Center for Combustion Research and Development, which is housed here in IIT, Madras. He is also the coordinator for the center for propulsion technologies. So, he has a lot of you know experience, worked with a lot of students and so we feel he is ideally suited to discuss research aspects associated with Aerospace Engineering. So, welcome.

Prof. S.R. Chakravarthy: Thanks a lot, Prathap.

Prof. Prathap Haridoss: So, Aerospace Engineering. Generally, so this is the camera that would be easier for you. So, if you look at Aerospace Engineering in general, what you would consider as you know traditional areas of research in Aerospace Engineering which may be you know people may have been working on for a long time. So, there's a lot of literature available in these areas, but still these are areas that you know continued to be looked at.

Prof. S.R. Chakravarthy: Yeah, broadly across the world and most universities, if you look at vehicle Aerospace Engineering Department, there are 3 broad areas. So, the first broadest area is probably aerodynamics, which in many places also includes flight mechanics but, in some places there are specialized faculty researche intense in flight mechanics, so they tried to treat it little separately but normally aerodynamics includes flight mechanics as well.



And then, the second large area is structures, which involves which also a course includes some materials, smart materials, those kinds of areas. But, typically dealing with the shapes in which materials are formed to get air craft, another space craft components and retreading the loads and all those things. And, the third one is propulsion which portents to engines, the once that propel the air craft or the space craft. So, engines would also include things like rockets and so on. So, these are the 3 broad areas we can say that most Aerospace Engineering gets itself slotted into.

Prof. Prathap Haridoss: In addition to these, are there certain areas that are considered you know modern areas of or very recently started areas, initiated areas in Aerospace Engineering.

Prof. S.R. Chakravarthy: Yeah, very much, very much because see for example, as we speak we are now beginning to see emergence of for example, unmanned aerial vehicles, micro air vehicles and so on. So, there is a lot of buzz in the society about these things. So, that's an area that requires a lot of specialized attention and so it is kind of grooming itself into area by itself. But, if you now look at like it is aerodynamics like for example, if you want to mimic insect flight or some such thing, the aerodynamics is horrendously complicated.

So, that gets to be a specialized area there and similarly, the structures of flopping wings and aero-elasticity is like a specialized area that is coming out to be there and of course, these areas have been applied in the past as well, but they are actually getting more specialized in and focus for let say, UAV kind of application, similarly, micro thrusters for propelling these devices. So, in all these traditional areas you are now beginning to see like deeper focus. Similarly, things like Controls, for modern aircraft as well as these newer aircraft that we are talking about, space technology going forward into deep space machines and so on. So, these are all some sort of newer areas that are emerging.

Prof. Prathap Haridoss: OK, and see if you look at say what industry is interested in of course, aerospace industry - I mean yes, directly interested in much of what you are doing. Other industries also which you know, which are may be peripheral not directly aerospace industries are they also interested in some of the kinds of research activities that go on in Aerospace Engineering?



Prof. S.R. Chakravarthy: A lot, actually. So in fact, because aerospace is somewhat exotic, so there is a lot of pioneering research that happens in this field. So, just to give as an aside as an example: For example, the personal computer actually started with the space shuttle, so that was in the 1960's that they were trying to develop and then. So, now it is so ubiquitous. So, likewise lot of things that the automobile people actually adopt like, even things like head-up displays or the driverless cars that we are talking about have been pioneered in aerospace.

So, that way there is a lot of applications that aerospace people can do like, development of new materials, smart structures these things have a lot of applications. On the engine side for example, engines go along with not only propulsion, but also power. So, things like power generation devices electricity generation can be done with things like gas turbines, which are also the same kind of cycles that are adopted in air craft propulsion and there are many such examples that we can give of terrestrial, what we call as terrestrial applications actually.

Prof. Prathap Haridoss: And In fact, in an associated context see we have typically, when we look.

Prof. S.R. Chakravarthy: Wind turbines, I am sorry.

Prof. Prathap Haridoss: Yeah, yeah.

Prof. S.R. Chakravarthy: Wind turbines is something that like, is also because trust is on renewable energy, yeah.

Prof. Prathap Haridoss: Right, in that context if you look at, say again industry expectations of the people that they absorb into the industry generally, there is this impression that if you do a MS or PhD or a specialist and so, there is little narrower area where you may get absorbed. In that context, where do you see you know, let's say the recently graduating MS PhD students, what sort of positions are they getting say in industry or anywhere else? What sort of positions do they end up looking?

Prof. S.R. Chakravarthy: So, I think this answer I would like to actually take a step back and do a little bit broader outlook.

Prof. Prathap Haridoss: Sure.



Prof. S.R. Chakravarthy: So, if somebody is doing research there are about 3 different facets to it. So, one is looking at the phenomena, looking at the process that they are trying to research. The second one is where it is being applied; and the third is what are the skills sets, we are acquiring in order to be able to do the research that we are doing on the phenomena that is applied to something right. So, if you are like a phenomena kind of person, you are probably cut out for academic perceives.

If you are the application kind of person, you probably you are looking at what traditionally we call as core jobs that means like I am. Obviously, when you come to Aerospace Engineering it is very unlikely that you will be actually doing sports medicine in Aerospace Engineering, although it is related okay. So, there is lot of aerodynamics you know and structures of muscles and stuff that there are faculty members in some universities that do those things. So, that is an application area, your core area may be sports medicine or something, but you can always associate a core engineering job, social with applications of what your research is.

Outside of that, that the larger area is the skill sets thing. If you are not this phenomena type of person right, you can now try to leverage the skills that you are learning, right. So, the skills typically broadly again fit into 2 or 3 categories, Experimental skill sets, Computational skill sets and may be Analytical skill sets. So, all these 3 things are there.

So, for example, we do very advanced laser diagnostics which can be applied to let's say, flow pass automotive vehicles in trying to making them more streamline and so on. Or do things like FEM or CFD and analysis which are computational tools that are ubiquitously adopted in lots of different applications. So, the question is which one do you want to leverage? What tickles you? What fascinates you? Are you trying to actually unreliable physics, then you probably want to be an independent researcher all the time in your life and that is one kind of job profile that you will try to attack. If you are applications kind of person you will look for core jobs, but if you are trying to leverage your skills then the world is open for lots of options.

Prof. Prathap Haridoss: Okay so, let me step back a bit now, we spoke about your students are leaving the program and where they may end up going. Let's look at the people who come in, I mean presumably you may have in fact students from different backgrounds coming in.



Prof. S.R. Chakravarthy: Yes, that is correct.

Prof. Prathap Haridoss: And also some coming from Aerospace Engineering, itself from may be different universities and so on. They are also transitioning from you know course based education to a research based education when they go for an MS and PhD degree. Are there specific issues that you see them encountering as they settle into this phase of their educational existence? Is there some other, any things that the challenges that they face or and if so, what should be done for them to you know be prepared?

Prof. S.R. Chakravarthy: This is again a general question that is not necessarily aerospace specific okay. So, I am going to answer it in that way to start with and see where aerospace fits in, if required. But, fundamentally what I see is that, right from kinder garden to let's say undergrad or let's say a course based masters like an M. Tech program, each child to start with and ultimately going to being an adult, is actually programmed. So, we have a time table that is given at the beginning of the semester or a year. So, we know exactly where we need to be sitting in, which room, in which seat, perhaps at which time and so on. So, we are not really trained to think independently on, how we want to spend our time?

And, when you now become a researcher so if you really think about it, even when you are doing course work right, to start with you are doing like lot less number of courses when compared to a typical M. Tech or a B. Tech and lot less when compare to what yourself did when you were a B. Tech or the M. Tech, right. So, you probably did like about 6 courses and then a couple of labs that kept you busy, pretty much all week. With all the home works and assignments to boot. But, now you probably doing like about 2 courses, what you do with the rest of the time?

So, typically I think this hurts the master students more when compared to the others, the PhD's because they have this mindset that I am doing courses so, let me not do any research and that is actually a bad strategy, because they are just wasting a lot of time saying that they doing courses and that chunk of time is like an appreciable part of like let's say, a Masters. So, masters are supposed to be like about 2 to 3 years, but typically everybody thinks it is a 3 year thing rather than 2 year thing which does'nt have to be actually, right. But, they pushed it that way because they had this you know, taking it easy during courses.



I think this we have to inculcate in our students to do a much better time management. They need to understand that there are 24 hours a day, there are 7 days a week, 168 hours a week. If the moment you say like there is 168 hours in a week because somebody they don't even know that because they have not done that little arithmetic, right and they don't know that, how to use those hours and this is something that I find most students are not really cut out managing very well. They get into some kinds of a certain expansion of a lot of freedom where they are not being monitored, they are not being required to be in a particular place doing particular thing, anymore and they are on their own doing things which is very, very hard to sharpen and say at the end of 1 year what have you achieved may be a lot less than, what would have if you have been lot more conscious of your time.

Prof. Prathap Haridoss: Okay great, and also in terms you know preparation for let's say, the kind of detail and regard we have for our course work here and so on. Do you find generally students coming in; are they well prepared enough that once they come through our selection process they are able to handle all what we require of them here? or do you feel they are may be, is there any challenge they face let's say with respect to the math involved in our courses, any other analytical skills are there things they need to be more aware of?

Prof. S.R. Chakravarthy: It's an evolving thing. So, one is maybe they are not really prepared for it may be, but may be they are, so that is a spectrum. But, many of them adapt. So, when they come in they know what is expected and therefore, they try to adapt, they try to pitch at themselves at a higher plane that is required of them and so on. And, progressively I am seeing in the last 20 years that I have been around, our expectations are also getting elevated. So we are, for example, if you see that, there are lots of advance level electives that have being offered with more faculties coming in and with greater levels of specializations and new areas that are being filled in interdisciplinary courses and so on.

So, the level of expertise, expected of the student is increased and if you increase your expectation many times the students rise to the occasion and meet it. So, that the most important thing is actually keep our expectations at the level that is appropriate and not necessarily water it down. Of course, don't make it too higher as well, that gets a bit



unrealistic. But, I think it's important to not let down the expectations and hold it at a certain level and get the students to rise up to attend and they would do it.

Prof. Prathap Haridoss: Okay great.

Prof. S.R. Chakravarthy: Yeah.

(13.47) Prof. Prathap Haridoss: And, okay maybe I have a sort of mundane question related to this kind of you know the life that students have, yes there, doing their research activities here. There is always this idea that you know lot of learning happens when they interact with fellow students, with their faculty, with their guide and so on. What do you feel is a good frequency with which students should be meeting their guide?

Prof. S.R. Chakravarthy: Well, I have been on an average doing like once a week or so and once a week or may be sometimes because of travels and conferences and all that steps may be like once in two weeks. But, I think we need to give some. So, if you are particularly talking about guides versus students that's a different equation, when compare to students meeting fellow students. So, I think that students meeting fellow students has to be happening all the time. I mean they must be in the lab and the lab has to have like a bunch of students working on things, may be some times 1 experiment or whatever it is, it may not be possible to actually be done by 1 student, but it does'nt mean that every student gets a helper to help him something, it dose'nt made sense at all.

So, therefore, like students will have to actually combine the resource and they may be also sharing equipment and so on. So, if so one person is doing his experiment the equipment gets tied to that experiment which means, like it's not available for the next student who is sharing that equipment. So that means, like they have to have a very good helping tendency and they are not really helping for nothing, I mean it is like when I am helping my friend in the lab he is going to help me when I am doing my experiment.

Prof. Prathap Haridoss: So it's a learning experience about.

Prof. S.R. Chakravarthy: So, typically what is happening across the world in many places and you might find this actually, the number of authors is kind of proliferating in many general publications for and there is reason for this. So, many labs are actually getting on a campaign mode. So, they now say OK, now I am going to actually work on a student



axis experiment that means, the entire lab works on student axis experiment on a campaign mode. So that means they finish the experiment in about a week, get that student text to actually process all the data because there is just many times these days we are getting a lot of data, whether it is numerical work or experimental work. It's easy to get the data, lot harder to understand process and there is too much post processing that we need to do, to squeeze physics out of it and all that. So so, let him do the post processing, but he should be available to actually get into the next campaign during the day time or something like that, with the other students and so on. So then, it turns out that many of these people actually get on.

Prof. Prathap Haridoss: Each other's.

Prof. S.R. Chakravarthy: Each other's publications and so on. It's not a bad deal, actually and they learn about each other's experiments, (Refer Time: 16:18) of the problems that they are working on. So, I think that kind of time sharing and time management among students is very important.

As far as the guide is concerned, I think once a week either in individual meeting or least a group meeting is pretty good. Particularly, in the Indian setting I think when compared to let's say Germany or somewhere else where students do not meet the guides for very long time and still on their own because they know how to work with their canes and they know how to craft things and so on. Whereas, I think our students are not very well prepared to do things. So, we have to actually shed our ideas with them and we don't really have a very extensive post doc culture so that means, like all the integrities say something that people gets stuck on and we do end up doing some quite of like a repetitive teaching of these little skills.

So, I think skills development, both moved and for example, teaching and interaction. I think skills development of students has to be focused upon. So, if there is way by which we can actually develop lot of skills for example, things like if somebody wants to learn Fourier transform right, he needs to go to some place and learn it. So that, there could be like some sort of YouTube video, lecture of about an hour with lots of equations or like let's say mat lab programs and so on. That are all available in post free, like an NPTEL things. So, it does not had to be like a formal course for 50 hours or something, can I just learn something quickly, right.



Those kinds of skill development is something that if we have a very good base of it the faculty involvement could be a little lesser, but one of the problems that we have in having to have an involvement is we are not really making a lot of progress, on what our original research goals are in these frequent interactions that seem to be a must right now, simply because we are actually providing through developing skills of each and every student every all over again every time.

Prof. Prathap Haridoss: Okay I think it's a very, very pertinent observation, where we are doing it? How we are approaching things to it? In terms of you know okay, again one of the things you mentioned you know this multiple author publications and so on. And of course, generally we tend to look at publications as one measure of, how progress is happening in the a research scholars' activities? Is there any other way that you feel you know in a more philosophical sense that you feel you need to look at a student or students need to look at himself or herself to understand that they are actually making progress in to research?

Prof. S.R. Chakravarthy: This is a very difficult question and this is sort of highly personnel, as in every researcher and this doesn't have to be a necessarily a faculty member, it also portents to a student researcher. So, every researcher has to actually make up his mind, what is going to make him sleep well that night that he has a sense of accomplishment and of course, as a researcher sometimes I spend sleepless nights thinking about my research so and that doesn't count. So, if I am doing that I am quite excited about my work and so on and that is OK.

But, I think the sense of accomplishment or achievement is highly personal and there are lots of ways by which this can be done. We can measure in terms of matrix like number of publications or the impact factors of the journals, h-index, whatever it is that you want to talk about. And, you may claim that you are bringing in some quality and citation index and all those things into picture in all that stuff. But, I don't know if that's what is going to make you happy, right. So, keep in mind, in the Indian academic context I think most of the academics in India or actually on this job because they wanted to derive satisfaction out of the job that they doing. And, job satisfaction on the whole in most industry is the oxymoron, I mean you either do, you either have a job or you have a satisfaction, so it is one of the, OK.



So, here I think we are trying to do this. So, and we have to actually ask ourselves what satisfies me. So, there are people who want to see what they are doing actually be applied in the industry. There are some people who want actually look the most scholarly and like for example, if you look at what, if you look at G H Hardy's book, an apology of a mathematician, he actually loads his number theory being applied to let's say chess games or something like that. So, there are these pure, experimentalists who do not want any application.

Prof. Prathap Haridoss: Yeah, yeah.

Prof. S.R. Chakravarthy: So, you have to respect them for what they are.

Prof. Prathap Haridoss: You feel you are lowering the.

Prof. S.R. Chakravarthy: Exactly. So, you have to respect them for what they are.

Prof. Prathap Haridoss: Yeah, yeah okay may be to close actually, I just wanted to get your opinion or actually your words of advice, what words of advice would you have for students who are aspiring to join an MS or a PhD program in Aerospace Engineering?

Prof. S.R. Chakravarthy: Simple, 3 words answer is follow your heart.

Prof. Prathap Haridoss: Ok.

Prof. S.R. Chakravarthy: So, I think this is true for anything. So, whatever you want to do, we have a lot of societal pressure unfortunately because we are still a developing country. So, there is like lot of pulls and pressures, family, lots of things. So, I think at the end of the day we need to make up our minds what we want to do and we have to follow our heart. So, if we want to do research and we want to do a particular kind of research, we want to do a particular topic of research, we want to work at the particular guide in a particular department, discipline, whatever it is, just do what you think is a right thing. Don't worry about anything else; everything else will work out for you. I think you can have this attitude like the universe was created for your sake okay, just go on and everybody will follow, not a problem.

Prof. Prathap Haridoss: Great. Thank you very much for joining us, it was a pleasure.

Prof. S.R. Chakravarthy: Thank you.



Prof. Prathap Haridoss: I think very nice insight into what students should look at.



Introduction to Research
Prof. Prathap Haridoss
Prof. Sathyanarayana N Gummadi
Department of Metallurgical & Materials Engineering
Department of Biotechnology
Indian Institute of Technology, Madras

Lecture – 43
Research in Bio Technology

Prof. Prathap Haridoss: Hello, it's our pleasure to have with us Prof. Sathyanarayana Gummadi. He is a Professor in the Department of Biotechnology, here at IIT, Madras. And, he has been with us here for over 15 years now. He has considerable experience in research and a lot of success in research. His areas of research include survival of microbes under extreme conditions and exploiting this aspect for industrial applications. He also looks at molecular and biochemical basis of phospholipid trans location in cellular membranes. So, these are 2 major areas of trust areas of research for Professor Gummadi. He has written several books, several chapters in various books.

He is also NASI Scopus Young Scientist Award winner in the year of 2012 in the field of Biological Sciences and this is given by Elsevier which is you know, body of a journal publication process which you know, which is where lot of scientists publish. He is also in the editorial board of several journals. So, you can see that you know he is very well qualified to discuss research with us in the area of Biotechnology and he has a lot of experience that would be very valuable for us to learn from and listen to. So, thank you for joining us.

Prof. Sathyanarayana N Gummadi: Thank you.

Prof. Prathap Haridoss: So so, let me begin by asking you this general question, in biotechnology are there areas that are considered traditional areas of research which perhaps have been there for a while, where you know if you do a search you will see a lot of literature already existing and so on?

Prof. Sathyanarayana N Gummadi: I think in Biotechnology has evolved from 2 basic disciplines in the country. So, one is from engineering if you say, it came from Chemical



Engineering. Traditional area of research in Chemical Engineering in terms of biotechnology is biochemical engineering, bioprocess engineering to develop process for production of certain metabolites cum industrial valued compounds. Which is replaced by a chemical process; just we want to convert chemical process into non-metallic friendly metabolites. That is one traditional way of approaching, researching from engineering point of view.

And from basic science point view, so biotechnology evolved in M.Sc. programs and B.Sc. programs where they have strong background in terms of Biochemistry, Molecular biology and Genetics. So, these 2 are the traditional areas of research which initially biotech started up in country. Now, it became highly multidisciplinary where it's been working with several fields of engineering and sciences in terms of research.

Prof. Prathap Haridoss: Okay and, in a similar sense I mean, are there you know specific areas that you feel are interesting to highlight as very recent areas of research in biotechnology where you know may be which has sort of gain prominence, might let us say the last 5 years or something like that.

Prof. Sathyanarayana N Gummadi: From industry point of view production of therapeutic proteins, monoclonal antibodies, vaccines, so the drugs for human diseases become a very prominent. So, a big technology has taken over from traditional microbial biotechnology to animal cell cultures and plant cell cultures for production of various drug molecules, so called as the reason term they give it is Biosimilars. So, Biosimilars means the compounds that are similar to what we have in our body.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: For example, for blood deficiency people you give Erythropoietin. So, Erythropoietin is what we have in our body.

Prof. Prathap Haridoss: Have it already in.



Prof. Sathyanarayana N Gummadi: In our body. So, we need to produce certain same compound, with the same structure in Vitro, using memorized cell culture.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: So, that's one of the recently taken over industry, industrial aspects of research where large investments are going on. Secondly, computational biology, people are working on developing new software's in terms of bioinformatics to find structures, structure of proteins, DNA molecules. So, which in turn helpful in making that target in term from bioinformatics. And, recently a very catchy what is coming up is large data analysis.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: Our institute is also very famous in working on large data analysis, big group working on it one. So, in which biotechnology place one of the major roles in large data analysis because there is a huge amount of sequence in data available from human genomes. So, which need to be analyzed and unless to find out, so what could be the probabilities in for disease and all that, so these are the new recent.

Prof. Prathap Haridoss: Very interesting I think, yeah. So, that shows how may be computing.

Prof. Sathyanarayana N Gummadi: Computing.

Prof. Prathap Haridoss: You know facilities are so critical.

Prof. Sathyanarayana N Gummadi: So, critical.

Prof. Prathap Haridoss: For and even you know computational techniques that people specialize in. So, even in biotechnology there's a lot of computation that is inherent to the successes.



Prof. Sathyanarayana N Gummadi: We have big computation group in our department and have very strong interaction between our department and computer science people.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: Doing a lot of projects.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: And, recently we hired an interdisciplinary faculty for this area.

Prof. Prathap Haridoss: OK. Very nice, very nice you know. So, let me also look at it, let us say, looking at it from the student perspective. When students come in here for, specifically for an MS or a PhD degree, if you see you know incoming students based on let's us say, the backgrounds that they have or the range of backgrounds they have and so on. Are there specific areas that you feel they face challenges in where they settle into an MS PhD program? Is the training that is there in traditional colleges that are out there adequate to prepare them for MS, PhD at least as they enter in here in biotechnology? Or you do see some specific you know lack in that they need to address or and if so, you know, how are they going about overcoming such thing.

Prof. Sathyanarayana N Gummadi: I think the new coming student they have lack of lot of hands on experience on newly experiments. So, especially in biotechnology all experimental people, so need to have lot of experience to handle lot of equipment's to do an accuracy. So, that is what lacking in them.

Prof. Prathap Haridoss: Okay and so, it's probably also got to do with, I think in biotechnology may be the requirement for you know say cleanliness and you know, thoroughness in an experiment is probably even more critical because may be our hands can themselves completely pollute sample.



Prof. Sathyanarayana N Gummadi: Sample.

Prof. Prathap Haridoss: So, you may not know what culture you know you are developing there if you not careful may be. So, I think then what you are suggesting is that you know, it's not just may be the ability to start an experiment in an experiment, but the rigger with which that experiment is framed. So, that is something that is people are lacking, as they come in okay. So, are there some you know special courses that you run which particularly to address this. I mean, how do you alert the student that you know their sort of deficient in these? And how do you you know bring them up to the standards that you feel are required?

Prof. Sathyanarayana N Gummadi: So, I think most of us train them in the first one year while they do course work and all that. So, they undergo with some seniors in research scholars and do preliminary experiments. So, we make sure that, they are well equipped to carry out experiments in their own labs. Now, what we expect them to do.

Prof. Prathap Haridoss: Ok ok.

Prof. Sathyanarayana N Gummadi: So, first year majorly we do out time for them to do this.

Prof. Prathap Haridoss: Of course, I mean as you already mentioned that you know there are you know may be specific industries that are looking at what research you do and so on. Of course, at the same time there is some times this impression that an MS degree or a PhD degree has a tendency to come across as being very academic, in the focus because that is how we go about it. Because that level of you know focus and concentration is required to get to carry out research. And, sometimes also a MS and PhD research is on some cutting edge activity which may be the society might benefit a little later from, it may not be like you know is not directly beneficial for them. So, in this scenario, where do you see the industry fit in? How interested are they in what are the current research activities in the field? Or how far you feel they are behind or how far they you feel in line with whatever are going on in the high and research in biotechnology?



Prof. Sathyanarayana N Gummadi: So, to be frank the number of industries biotech available now is much less compared to other fields.

Prof. Prathap Haridoss: Okay Is that just a Indian thing or internationally you also see that.

Prof. Sathyanarayana N Gummadi: Internationally also.

Prof. Prathap Haridoss: Internationally also, OK ok

Prof. Sathyanarayana N Gummadi: Because, it involves high investment to do and the success of the industries also, probability of getting success is also very low.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: Because, most of the thing formal drugs you screen it, do clinical trials and get it to a compound it takes so much investment.

Prof. Prathap Haridoss: So, much investment time and so many regulatory aspects.

Prof. Sathyanarayana N Gummadi: Regulatory aspects. So, especially in those kinds of industries people look for very specific training.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: ; Like Protein expression, how to make potassium, how to make strain improvements, down streaming processing. So, job aspects in those areas are good. For example, in our department students have finished in bio process engineering, so that job placements are better compare to fields from other areas were people have to go for post doctoral training, get more experienced in this one. Then chances of getting good jobs are high. This is so very wide and interdisciplinary, that is the problem. So, people has to, some industry needs a person to be trained on that area.



Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: So, based on that PhD problem and then post doctoral training, might suit to that company.

Prof. Prathap Haridoss: So, do you see students also I mean given that they are doing something totally new and interesting with respect to maybe you know, aspects that can be used in human systems and so on. Are there people opening you know, are there people who are opening their own companies after they graduate, do you see enough of an entrepreneurial you know element in this field or you think again as you said investment is too high and may be therefore, it's not possible what do you.

Prof. Sathyanarayana N Gummadi: I think 3-4 student start up entrepreneurship in our department, in collaborating with the faculty.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: So, they are in initial stages.

Prof. Prathap Haridoss: Ok.

Prof. Sathyanarayana N Gummadi: So, mostly they are making like of sensors, so devices to find out, to investigate some aspects especially in case of disease, how to find out.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: A person is having a disease or not. So, diagnostic kids and more success store in our department is from B. Tech group. So, they started Sea6 Energy, an energy based company.

Prof. Prathap Haridoss: OK.



Prof. Sathyanarayana N Gummadi: From seaweed they make energy.

Prof. Prathap Haridoss: OK ok.

Prof. Sathyanarayana N Gummadi: So, that is more successful and well established company.

Prof. Prathap Haridoss: OK ok.

Prof. Sathyanarayana N Gummadi: In our department now.

Prof. Prathap Haridoss: Ok So, so there is an element also.

Prof. Sathyanarayana N Gummadi: There is an element.

Prof. Prathap Haridoss: And, most of the other students I mean sorry, some amount go to the industry, after that is you are saying that very dependent on their research topic and saying some are opening entrepreneurial ventures. What about you know academia I mean, are there enough positions in academic you know institutions in let's say, in India that MS, PhD students could go to? Because that's normally where many seem to go and that's where they have decided also, many of the other departments, in your case what do you see that?

Prof. Sathyanarayana N Gummadi: There are lots of openings.

Prof. Prathap Haridoss: Lots of openings.

Prof. Sathyanarayana N Gummadi: Lots of openings for academic positions. People who ask for academic position are better to have more publications during the PhD one.

Prof. Prathap Haridoss: OK.



Prof. Sathyanarayana N Gummadi: And, Postdoctoral training abroad is highly preferential to get in good places.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: Then come back and join us as a better way in biological department.

Prof. Prathap Haridoss: Ok ok. So, that's the sequence that you are recommending something.

Prof. Sathyanarayana N Gummadi: Recommending students to come back and (Refer Time: 12:14).

Prof. Prathap Haridoss: Come back and Ok. And, let me ask you let's say something about you know, how you look at progress and research? So, because you have now I mean you have won several awards and so you have been very involved and you are also seeing all these journal from much, much close perspective because you are in the editorial board and so on. Generally, we look at you know the progress of students and as you mentioned you know, the number of publications and good channels and so on. Are there any other ways in which you feel you can gauge the progress of a research student? Because, this research is such a different setting than say course work where it is very clear, there is an exam, there is grade, you get the grade and people say you did well or did not do well. But, in research, what all do you look at as when you see your students and you feel that this person is going as a researcher?

Prof. Sathyanarayana N Gummadi: The first thing that I recognized is that, whether a student is sending me a paper to read.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: Saying that this platform is paper interesting. So, I



can use this one.

Prof. Prathap Haridoss: OK ok.

Prof. Sathyanarayana N Gummadi: That is the first symptoms that he is into this field.

Prof. Prathap Haridoss: Oh Very nice. OK.

Prof. Sathyanarayana N Gummadi: So, he is thinking about the problem and trying to do it making self dependent on his work.

Prof. Prathap Haridoss: And, then make some judgment on the paper that is interesting to read and all that .

Prof. Sathyanarayana N Gummadi: On the paper that is interesting to read in all that. That's the one kind of evaluation I do. Second thing is that, do people come back with entirely new different way of experiments. We have regular set of experiments to do in our research.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: So, people will come; OK, let me try something else.

Prof. Prathap Haridoss: Some different ideas have to probe something.

Prof. Sathyanarayana N Gummadi: Different idea different have to probe this thing, sometime. So, that makes really expand our research area.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: Some of my students did like that. So, mostly from engineering and now I am becoming mostly a biological.



Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: Research in biological sciences is going there. So, that helps in them whether, they are on the right track.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: Progressing themselves and some of them even says that shall we write a grant for this one, I assist you.

Prof. Prathap Haridoss: Ok Very nice. So, they have really thought much more, they are getting confidence.

Prof. Sathyanarayana N Gummadi: Confidence in this one.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: So, there are some people I found then I made a routine in the lab that students when coming to third year, final year should make 1 rough draft of a person and give it to me.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: On his area. Then will see try it correct so, we will submit it. So, that helps them in recommendation later also, he is capable of thinking.

Prof. Prathap Haridoss: OK ok ok .

Prof. Sathyanarayana N Gummadi: I think research apart from publications, so looking at their behavior, how well they maintain the observation book, data.

Prof. Prathap Haridoss: OK.



Prof. Sathyanarayana N Gummadi: So, meeting guide regularly or not. So, some student will regularly meet, everyday or twice in a day and come and show the results, what to do? These are some of the things that we evaluate that.

Prof. Prathap Haridoss: Ok In fact, I mean since you mentioned about student meeting regularly, what you think is a good frequency with which people should be meeting their guides and that is a change with time in their program? What what would you suggest?

Prof. Sathyanarayana N Gummadi: So, in my lab I daily meet them.

Prof. Prathap Haridoss: Meet them daily, fine.

Prof. Sathyanarayana N Gummadi: Daily meet them and see them.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: So, I think in this will, that is better have a daily contact and see you, what's happening and what's going in?

Prof. Prathap Haridoss: Great. So, let me sort of, I mean wind up this discussion with the one more general question for you, what advice would you gave to a student whose is aspiring to join an MS or a PhD program in biotechnology?

Prof. Sathyanarayana N Gummadi: First advice will be not technical this is general advice to any student.

Prof. Prathap Haridoss: Sure.

Prof. Sathyanarayana N Gummadi: First students should try to become a team member in a group, whatever he is trying. That's more important.

Prof. Prathap Haridoss: More important.



Prof. Sathyanarayana N Gummadi: So, that makes his life comfortable here and this is one of the parameter which is asked by all post doctoral positions or in any company, any where do it, whether the person is a team player or not.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: So, that is the first thing that they should learn in this program when they join.

Prof. Prathap Haridoss: Great, great, yeah.

Prof. Sathyanarayana N Gummadi: Second is to identify a problem as quick as possible.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: As quick as possible and make lab record book, observation book very clearly. Whatever they do, even if your experiment is fail that, they should mention as fail.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: When it comes so results. Data handling should be very powerful, that is very important nowadays because people say that this property is wrong, this is wrong, you have to prove them.

Prof. Prathap Haridoss: Yes, yes.

Prof. Sathyanarayana N Gummadi: Or they will say false data or something like that.

Prof. Prathap Haridoss: Yeah, yeah.

Prof. Sathyanarayana N Gummadi: So, the storage of data is very important, raw data's



everything.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: That they have to maintain and they should give a copy to guide and they should also keep reserve that copy for themselves. So that, any issues come it should be maintained up. Third is that they should be very systematic.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: So, systematic in the sense, kind of making a regular timing, coming to lab to work on these things. So, I think this will make them.

Prof. Prathap Haridoss: Okay So, these are things that they should tell themselves that you know these are the kinds of habits I should develop if I want to become a good researcher in I any field, but including biotechnology. Particularly, some of these things you feel are you know a very particular to biotechnology. And, I appreciate the point on being a team player because I think especially when people read for exams may be they have a tendency to read on the own or something and then they have to get use to being a team player and it helps them, it helps the other and in all ways it works very smoothly, yes.

Prof. Sathyanarayana N Gummadi: Also then, don't even talk their own lab people about their problem.

Prof. Prathap Haridoss: Their problem, OK.

Prof. Sathyanarayana N Gummadi: Should talk to that member or other lab members.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: So, more you talk you get quick solution.



Prof. Prathap Haridoss: So, internal communication is as important as the external communication.

Prof. Sathyanarayana N Gummadi: External communication. It is very important and reading up literature is very important.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: Make update on that field. You know, there are many websites where you can click it every week, you get updates what are the papers published on that area.

Prof. Prathap Haridoss: OK.

Prof. Sathyanarayana N Gummadi: Keep reading it up and have broad knowledge on the field is very important. So, that makes them successful in the carrier.

Prof. Prathap Haridoss: OK. Thank you Professor Gummadi, it is a pleasure that we could have you here and I think a lot of insight you have given which is very I think you know, very interesting to look at and I think a certainly students work you know new you are considering who do not know exactly, what they are getting into or what they may be getting into if they get into research. I think a lot of information you have shared will be very useful for them. Thank you so much for joining.

Prof. Sathyanarayana N Gummadi: Thank you for giving me an opportunity.

Prof. Prathap Haridoss: Yeah, sure.

Prof. Sathyanarayana N Gummadi: Thank you for that.



Introduction to Research
Prof. Prathap Haridoss
Prof. B. Viswanathan
Department of Metallurgical & Materials Engineering
Department of Chemistry
Indian Institute of Technology, Madras

Lecture – 44
Research in Chemistry

Prof. Prathap Haridoss: Hello, it is our pleasure to have with us Prof. B. Viswanathan. He is professor emeritus here in IIT, Madras. He was a faculty here at IIT, Madras for almost 35 years, which is a very long time. Much more than you know the age of most post graduate students. He is the author of numerous books. He is considered very important expert, very well accomplished expert in the areas of catalysis and energy amongst other areas in chemistry. He has contributed in many many ways to the development of science in our country. He serves many important committees which make key decisions on you know what projects should be funded and so on.

He has also served in the United Nations development program committee for looking at funding activities in the area of catalysis and so he is both a national as well as an internationally accepted expert. His areas of research include electro chemistry and catalysis so that covers areas such as fuel cells batteries and hydrogen storage, just to name of few and over 35 years. Numerous students have completed MS and PhD under the guidance of Prof. B. Viswanathan and they hold fairly important positions in many industries, many national organizations and so on. So, it is really our privilege pleasure to have Prof. Viswanathan with us and I am sure the views that he has, that he is going to share with us on research in chemistry would definitely be very beneficial for all of us. So, thank you sir for joining us.

Prof. B. Viswanathan: Thank you.

Prof. Prathap Haridoss: So I would like to start by asking you this, like in engineering, I am sure even in chemistry which has been around for as a recognized field of science for a very long time. Are they still areas that are considered you know old traditional an areas of research in chemistry?



Prof. B. Viswanathan: Yes. There are two areas of chemistry which are old, but still it has a relevance to even today, one of them is synthesis organic chemistry, this synthesis has always associated with organic chemistry, but today it is no longer synthesis in organic chemistry, synthesis of chemicals. Okay therefore, the inorganic chemicals or even industrial chemicals, how they can be economically synthesize and they also produced that is the reason. Second thing is science itself, today has changed its phase. Science in the conventional way we were looking at the basic problems. Today the science has to be delivering materials, material means, for example you take a cell phone it is a material called Glylite resonators, this is simple material pervoskite. Therefore, this type of materials have to be designed and fabricated with a very specific properties, dielectric constant has to be specific value. The temperature coefficient of dielectric constant has to be very very low like that.

This is one of the examples, there are many, many materials even for example, you take through solar cells, solar cells today is based upon silicon solar cells, but silicon being a very important material and also costly. So, the design of new solar materials is coming, silver based solar cells and such things are cheaper than this. In the manufacturing process, cost process silver may be costlier than that, but in the solar cell manufacturing thing it can be cheaper than the silicon solar cells. So, therefore, any any area if you take today chemistry is I will say energy materials and environment. These are the three main areas which are evolving in chemistry, but it doesn't mean that this customer area of synthetic organic chemistry or synthetic inorganic chemistry, they are also still important

For example, in synthetic inorganic chemistry, metal organic frame works which is just like a palace, but only thing is it is a palace in chemicals not a brick and mortal. So, this type of metal organic frame works mesoporous materials, they are all the new generation materials, they have a very great advantages and also utility they can be used in the many industries. For example, today if you take the oil industry it is, if even though oil is the main product and their refined oil is the main product, the conversions are taking place only because of these materials. Previously these zeolites were used now mesoporous solids and this MOFs and (Refer Time: 05:17) they are essentially porous solids are being used. Though why we are using this is if now refining is based upon bottom of the barer, we are not taking the crude which is on the top of the layer because it is when you



go to the bottom, the hydro carbons are long thin hydro carbons, therefore they have to be cracked. Previously it was not so long a chain, but previously also long chain, but 30-40 carbon atoms, now 100-200 carbon atoms.

Therefore, the cracking or refining, refining means making the crude oil into those whole products. So, this is requires porous solids of very high porosity, very high pore diameters and their things. These solids have to be fashioned, designed and fabricated. So this is the new generation. In the same way energy conversion and energy storage, these are the two new areas, I will not say new area but newly finding acceptance. Now, the storage is a very important thing, energy conversion is may be possible in where the chemistry, but the storage is a very important problem. The storage must be in such a way that the economy must be equal to the petroleum crude oil or petroleum price. We will simply take fuel price.

Now, if it is not only fuel price, but fuel for example, if you will take a car it takes 5 to 100 liters of petrol. Therefore, if you are storing the fuel, we should store equivalent to that, so that you will be able to travel 500 kilometers or 600 kilometers at a stretch, instead of the intermediates storage. So, this is a main area of research today in materials for energy storage. In chemistry and physics, these are the two immediate areas. The third thing is in designing this solids or materials we have to now functionalize them, the functionalize say is for example, if you take a solid, there are 10 to the power of 15 sites, out of which only 10 to the power of 12 or 10 to the power of 10 sites will be active of all them; how to make those 10 to the power of 10 as active sites, and how to keep them in the active state. This is a very, very important thing because chemists are always using the molar quantities, molar quantities means 10 to the power of 23 molecules.

So, therefore, we can easily analyze them whereas, when you have to analyze 10 to the power of 12 or 10 to the power of 15, your molar concentrations becomes a nano molar, pico molar or even lower than that. So, the analysis and identification and its structure and all those things are very important area. So this is another important area of research in chemistry, I will say general science. The another development that has happened in the last 10 years is how to predict them? How to apriority you have to design a material or you have to say this is the material has to be designed. So, this is called today is a



theoretical chemistry by you can say computational chemistry whichever way you want modelling chemistry whichever way you want to do that, but computation in chemistry today as taken here a new term which is the called the density functional theory.

Density functional theory takes from the basis okay. For example, why hydrogen is present as H_2 , that is because of the one s electron of one hydrogen combines with another one s electron of another hydrogen making a covalent bond. Therefore, we will know, now hydrogen cannot exist in the normal conditions as hydrogen atom, but it can exists only hydrogen molecule because the energy of the hydrogen molecule is lower than that of the energy of the hydrogen atom. So, the same way we can now make this thing for materials.

In the olden days the number of atoms in a molecule was very small may be 10-20, may be 100 let us assume, but today for example, you have metal organic frame work there will be thousands of atoms. So, therefore, we have to minimize the energy and predict them. So, this is done by density functional theory or self consistent fuel theory or calculations, but the recent times density functional theory has over run all these other things because of the time involved in the computation and also comprehension by the chemists.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: Because (Refer Time: 10:19) or self-consistent theory requires a tremendous mathematics. This density functional theory is only grass, but there are other approximations, but if this is the highest approximation that is possible today. So, this theory has taken over the thing and many chemistry laboratories are now facing the problem of how to apply these to their materials. So therefore, what has happened here is, in the chemistry if the research has taken a new turn. It is not enough if you are very good synthetic chemists, if it is not enough you can to make use of the tools to analyze them, but you should also be able to predict them.

Prof. Prathap Haridoss: OK.



Prof. B. Viswanathan: So this is that changing situation in chemistry research today.

Prof. Prathap Haridoss: Ok so that's a very I think a lot of detail you have told us about, both I think traditional areas of research and where you know research is heading in chemistry these days. If I go back a little and let say look at the set of students who probably join a chemistry department for an MS degree or a PhD degree and so on, and you have interacted with a lot of them. They come from a variety of different institutions where they would have done, you know bachelors degree, BSc, Bachelor of Science degree, maybe in chemistry and so on and then before they join here for a masters degree or a PhD degree.

So, in general, in their preparation as they come in do you see that there are any specific significant challenges that these students faced when they try to join? I mean when they get into a masters or a PhD degree, are there certain areas that may be their preparations in generals tends to be a little less for which I mean after they joined their required to pick up more maybe more courses or have some more preparation in those areas? Is there some lacuna in terms of you know either theoretical learning that they have or experimental experience that they have in what you have seen?

Prof. B. Viswanathan: Yeah. That is necessary thing especially in the Indian context because as I told you, analysis has to be at the atomic level today. Previously analysis is on the milli molar or micro molar now it is not. We have single atoms have to be identified, so these types of techniques that are available today for example, XPS or UPS or any other technique, we will not go into all the details that is any the analysis must be the probing system also atomic level, the probed system also at atomic level. Therefore, this means this knowledge is not available in Indian universities.

So, therefore, the students who come for research, but they have to use them because if they have to be competitive in research today and publish in very high impact journals or impact is not a correct thing, in high journals then they have to be using these techniques. There is no other way out that. In the same way as I told you theory also is about lacuna here because computing facilities in many universities are not so high. So, they might not have learnt these things. Even if they have learnt they may not be able to use them in the



actual sense because the programming is a bugback of many of us because we are not professional programming people.

Now, what has happened is the second thing, that is I would not say the students have become, are they a past students are very brilliant, the present students are not brilliant and this. That is not the criterion, the criterion here is today the diversions for the student is very high compare to the diversion that you and I had, that is why the student is when their attention is diverted then they are not able to concentrate.

Prof. Prathap Haridoss: Attention spans are less.

Prof. B. Viswanathan: Yes.

Prof. Prathap Haridoss: These days relatively generally speaking.

Prof. B. Viswanathan: That is what is reflecting.

Prof. B. Viswanathan: Now the students are little behind, but otherwise the capacity of the students remains the same.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: Only thing is, they have because today the computer is there and things, they can use computer for learning instead of that they are using it for playing.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: It is one of them, but in chemistry computing usage is only to the limited extent, that is should be improved.

Prof. Prathap Haridoss: So, generally the people have not been using it as much in the computational sense.



Prof. B. Viswanathan: That is one thing; second thing is even such as a search tool. They can use only Google and all those things. That is not enough now because their literature is so high, because even if I were to tell you, I have been recently writing a book on carbon dioxide to fuels in chemicals because we know very well fuels and chemicals can be used to produce carbon dioxide.

Prof. Prathap Haridoss: Reverse process.

Prof. B. Viswanathan: Reverse process, so that means, we are closing the cycle. This literature in the last 10 years is enormous.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: Ok That is how to convert even though we have not succeeded, still succeeded means in the manufacturing level, but we are succeeded level in converting few molecules.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: Few molecules are CO₂ into chemicals useful chemicals, but this knowledge must be known to them, then only they will be able take up the challenge and do them. This is only one example, another example is as you have already said is fuel cells. In the case of the fuel cells, the problem does not exist is the electrode; the problem exist is the electrode at that the oxygen reduction, not on the fuel combustion. Therefore, the designing proper oxygen reduction electrode is the question here. Up to now we have been using only platinum based electrodes, but now we can use many other materials one of them is hetero atom nitrogen pass plus sulphur boron substituted carbon materials can be easily used for the oxygen reduction reaction.

It can be comparable, not exceeding the platinum, but it is already comparable with platinum today with respect to time it will be improved, then it will be cheaper electrode and it will become a reality. Second thing as I told you before, the storage; storage also is



a very big issue. For example, super capacitors instead of batteries because battery is weighty substance. So, super capacitor, super capacitor also is a possible means of energy conversion divisor, but only thing is the materials in question.

Now, in the material that we are now using is carbon materials are carbon based materials. We will not go into the details of it, so that we have to innovatively bring out as I told you carbon materials with this is hetero atoms and also structurally they should be made more stabled. A carbon materials are layered materials may be layered materials are possible to use a super capacitors, but chalcogenides, dichalcogenides sulphur containing or phosphorous containing all these things will be in the future will be new super capacitors like that we can go on taking about it in the materials in the catalysis or any field. Therefore, what today is the student research, students should be able to focus on these new materials and since they have been conventionally thought they are not able to think the new materials sense.

Prof. Prathap Haridoss: Ok So, lot more reading up is required to catch up on materials that are more recently being looked at and picked up.

Prof. B. Viswanathan: Yes.

Prof. Prathap Haridoss: Okay So, of course, you already mentioned a lot of examples of where you know, may be there is a direct industry application and certainly you know petroleum industries, one that you were taking about. Generally in MS and PhD degrees, we tend to think of research which is, you know significantly ahead of it is time. So, in this context from MS, PhD kind of research activity that takes place are there where do you see industry interest in this kind of you know in a research program kind of activity.

Prof. B. Viswanathan: Okay in industry, since our center is dealing with refining industry. As I told you, we are today refining industry because refining industry is the major industry in India. Nearly 15 refineries are there and all those things. The refining capacity even though increased, the refining material the crude oil is the bottom of the barer. Therefore, the conversion to useful chemicals is a big problem. Not only useful chemicals even the waste chemicals that come out ok for example, that tar or something



other it is not waste, but it is not a high prized commodity, they can used properly if they can. So, all these involve some kind of a refining process.

The refining process means having a crude oil converting them to the size of the molecule that you want and value added molecules. So, this is called can be isomerization, it can be calculation, it can be addition or whatever chemical reaction that we have studied in general organic chemistry, but it has to be done with this molecules which is a very big molecules.

So, therefore, the industry today insists on this. The same way drug, there drug molecules now you drug molecules for example, you have take a hypertension molecular are something like that as a boots synthesis or whatever it is it all requires 5 to 6 steps. These things can be today with these type of materials, porous materials we can make them in single step.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: EBU propene can be synthesized from basic molecule single in step.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: Today it is produced by 5 steps or 6 steps. So, the cost will come down, cost only is not the criterion the drugs will be easily available.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: Therefore, essentially what it means is that, we have to reorient our research; we cannot succeed in one day. Boot synthesis is known for decades now one, if I were to introduced it in one step everybody will laugh at me, but it is possible. Okay But to economically produce it has to be done. So, like this many things can be done. Okay



Prof. Prathap Haridoss: Yes. So, these are the areas that you feel that you know industry itself is directly interested in, a research program that is ongoing.

Prof. B. Viswanathan: Yes.

Prof. Prathap Haridoss: And also, similarly if we had research programs in those areas more industry interaction is likely. So, if you look at the students that graduate, I mean let's say masters and PhD students when they graduate, what sort of positions do you see them getting in industry or any other circumstances?

Prof. B. Viswanathan: OK.

Prof. Prathap Haridoss: Of course, in general in general.

Prof. B. Viswanathan: Our center has a peculiar place, but chemistry in general they find the opportunities in chemical industries especially pharmaceutical industries, if he is an organic chemist. He he is a material chemist means a material development, but today India is in a favorable position because all multi nationals have got their head quarters or semi head quarters in India. So, that for example, the shell or Sabey or any anything GM GE or whatever thing even the VPNG they are all here in India and therefore, they have opportunities there are many more. Previously this type of opportunity was not available for India.

Prof. Prathap Haridoss: Indians.

Prof. B. Viswanathan: India, they have to only look for the research positions abroad or employment abroad.

Prof. Prathap Haridoss: In abroad.



Prof. B. Viswanathan: Yeah. Today equivalent employments are possible in India. So, that is not a problem in other as a matter, but chemists I do not see any problem in the next 20-30 years.

Prof. Prathap Haridoss: Okay that's a very heartening piece of information for that thing lot a students who are in chemistry.

Prof. B. Viswanathan: Yeah. If they have that in a necessary inspiration they can easily fit into any of these things.

Prof. Prathap Haridoss: Okay let's say on a more mundane note, how often do you feel I mean when students come and you now get it to a masters or a PhD program, one of the things they are often told that you know classroom learning is one, but you know interacting with peers, interacting with your adviser, interacting with your guide, these are things that actually contribute a lot to their learning process. So, in your experience what would you recommend as you know how often student should be meeting their guides or how that process should go?

Prof. B. Viswanathan: Okay This is one of the main issues of Indian education itself. I am sorry to tell this, but it is a fact for example, if you our learning in class room, teacher or the professor will come and give lectures and he may not be having time to establish that crossed their barrier of that, it reached to a lecturer to the student. This is the students, also is inhibited to ask questions are raise doubts. So, this is a very big ballot

So, what I suggest is instead of giving lectures and courses, courses should be interactive, more interactive as a matter about questions can be on both sides. The teacher also can ask questions instead of delivering the lecture. See previously what was bothering and lecturing process is covering the syllabus. Now covering the syllabus is not at all the criterion okay. How you have comprehended this syllabus is the question because that is what you are going to make use of in your career. Therefore, the classroom means in your lecturer hour of 60 minutes, there should be 60 questions across.



Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: It can be across between both of them, that is one thing that has to come in India in Indian education. This is not at all.

Prof. Prathap Haridoss: This is a under played, this is quite bit a under played, but what about you know as a student guide interaction I mean.

Prof. B. Viswanathan: The guide should be available on all the time.

Prof. Prathap Haridoss: All the time. OK.

Prof. B. Viswanathan: That is.

Prof. Prathap Haridoss: Easy accessibility of guide is a very important.

Prof. B. Viswanathan: And that also he should treat the students equally.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: Because he should not put him in a fearful situation.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: Or respectful, in both the situation.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: That respect should be only inherent. It should not be shown outside.



Prof. Prathap Haridoss: So, that makes the interaction more effective.

Prof. B. Viswanathan: The students should be able to tell him what is inhibitive to him.

Prof. Prathap Haridoss: Sure.

Prof. B. Viswanathan: That is, in Indian situation, this 'guru-shishya' situation is really that is what is the shishya is capable of asking anything to the guru, guru also is capable of asking the shishya anything that is somehow in between this, disappear.

Prof. Prathap Haridoss: OK. So, that is something that is worth.

Prof. B. Viswanathan: Yes.

Prof. Prathap Haridoss: So also in a similar., sort of in a related sense there are different ways in which people measure, you know how students have performed in, you know research activities in commonly we refer to just publications, over and above publications are there other ways in which you look at students to measure you know how much they have evolved as a researcher.

Prof. B. Viswanathan: Okay this is one thing that in our center, that we are practicing for example for example I will tell you one example then you will and also this example is well known for example, when ammonia synthesis was proposed by Fritz Haber, his boss Ernest, Ernest equation, Ernest put thermodynamics and said it is impossible. So, you cannot work on this, but in spite of it Haber worked on it and got the ammonia synthesis with a BSF, that is not the criteria. If ammonia synthesis were not to be there, world population will be one-half or one-third, because it has provided the food for all the people, that is why world population has increased.

Prof. Prathap Haridoss: OK.



Prof. B. Viswanathan: Now, the question is the same question only we asked, CO₂ to chemicals is not thermodynamically possible.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: So, for example, this is the way we came to this CO₂ problem. If Haber could do ammonia synthesis, why cannot we do carbon dioxide to chemicals?

Prof. Prathap Haridoss: Chemicals. OK.

Prof. B. Viswanathan: So, this type of questions we asked the students and make them interested in the research because these researchers will not provide or give results immediately. So, they should not get discouraged.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: They should be willing to face problems. So, like this today the students should be motivated for this is only one example, many examples are there. For example, super capacitors as I told you, the hetero atom substitution in carbon materials is now known, but when we started 20 years back nobody even thought that nitrogen can be substituted in carbon.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: Because nitrogen is a 5 valence, carbon is 4 valence.

Prof. Prathap Haridoss: valence. OK.

Prof. B. Viswanathan: So, such a solid state material is not possible. Even now many solid state chemistry will not agree with that.

Prof. Prathap Haridoss: OK.



Prof. B. Viswanathan: But those types of things are possible. So, one can unconventionally think.

Prof. Prathap Haridoss: So, you are looking for unconventional thought processors.

Prof. B. Viswanathan: Yes.

Prof. Prathap Haridoss: To

Prof. B. Viswanathan: Or induce those processes.

Prof. Prathap Haridoss: Induce those processes and that gives us the sense that they are actually thinking different from you know being you know put into a stream line and then they are therefore, that is something that they are learning as a research processes that was a very interesting insight on you know how they have to think outside of whatever parameters they are conventionally told about and then that helps them grow as researchers. So, I would like to conclude this discussion by asking you this, I am sure there are lot of students out there, who are considering, you know advance degrees in chemistry masters degree or a PhD degree. So, what are your words of advice to these kinds of aspiring students for aspiring for higher degrees in chemistry?

Prof. B. Viswanathan: Okay Chemistry is generally considered as a substrate design, physics ethics and physics is more basic than chemistry like that is what if the people think, but as I told you, the students should have first thing is that he has analytical skills, have to come to atomic skill, not the molar or milli molar or micro molar it has to come to atomic skill. Now when we it is coming to the atomic scale the probes also have to change for example, electro chemistry it can now do nano skill, but what we want is peco scale or femto scale, 10 to the power of 12 or 10 to the power of minus 15 molar concentrations. Now, the students should have dynamism in taking up that such challenge such thing that is the first thing. Second thing is if the chemists are always afraid of mathematics, now today mathematics is only, even today or everything mathematics is only a language for non mathematicians.



Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: Mathematics is pure mathematics, mathematics for mathematicians, but for all others it is only a language. If I want to say 4, I can say 2 plus 2 is 4. So, therefore, that is how it has to be. So, therefore, we should all.

Prof. Prathap Haridoss: Embrace it.

Prof. B. Viswanathan: Embrace mathematics as a language for the other scientists. If they do that then they will be able to perform. As I told you DFD or HR (Refer Time: 30:56) even self consistent it will become easy, but unfortunately our quantum mechanics is people who teach it for the chemists, they immediately go to a second order differential equation and solution of them and all those things since it of the emphasizing the physical significance of the solutions.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: So therefore, this type of things they should not take it too much. They should look for the analytical solutions of the mathematical expressions.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: Then they will be able to perform well. So, I don't think the chemistry or even physics or even natural sciences or physical science is a challenging job. It is a very entertaining and interesting job.

Prof. Prathap Haridoss: OK.

Prof. B. Viswanathan: So, that students should feel.

Prof. Prathap Haridoss: Comfortable.



Prof. B. Viswanathan: Engineering is one of them, I am not denying that.

Prof. Prathap Haridoss: Sure. Sure.

Prof. B. Viswanathan: Because you are doing it to hands on, the same thing can be done which example of you MOF metal organic frame work is building in the molecules instead of making a building with brick and mortar. So, like that porous solids also is building a material only. So, therefore, the chemistry generally, science can be more interesting than what it is today.

Prof. Prathap Haridoss: Or what it is may be perceived to be, may be people perceive more challenges than the immediately recognize the interested in it.

Prof. B. Viswanathan: Yes.

Prof. Prathap Haridoss: And so an aspiring student, I think if they learnt to distinguish that and you know put away their fear and embrace it in a more you know whole hearted way they probably will be more comfortably fit into masters or a PhD degree. Thank you very much Prof. Vishwanathan.

Prof. B. Viswanathan: Thank you.

Prof. Prathap Haridoss: It is a pleasure, that we could have you. It is a privilege that we will have you in this discussion.

Prof. B. Viswanathan: Thank you very much.



Introduction to Research
Prof. Prathap Haridoss
Prof. Deepa Venkitesh
Department of Metallurgical & Materials Engineering
Department of Electrical Engineering
Indian Institute of Technology, Madras

Lecture – 45
Research in Electrical Engineering

Prof. Prathap Haridoss: Hello. So, it's our pleasure to have with us Prof. Deepa Venkitesh from Department of Electrical Engineering.

Prof. Deepa Venkitesh: Yeah.

Prof. Prathap Haridoss: Here at IIT, Madras. She has been in the department for 7 years now.

Prof. Deepa Venkitesh: 7 years now yeah.

Prof. Prathap Haridoss: And, before that she was as faculty for about 8 years in Mumbai.

Prof. Deepa Venkitesh: Mumbai University.

Prof. Prathap Haridoss: Mumbai University.

Prof. Deepa Venkitesh: Yeah.

Prof. Prathap Haridoss: So, that is where she is been for. So, about 15 years as a faculty, that's a lot of experience as a faculty and she has worked on Photonics, Non-linear optics, Fiber lasers and Optical Signal processing. So, wide range of areas and these are you know technologies that we are we often interact with for lot of our communication processes. So, we are really glad that she could join us. So, we will get an idea of aspects of research associated with Electrical Engineering. So, Deepa in Electrical Engineering it's of course, field of engineering that is being around for a very long time.



Prof. Deepa Venkitesh: Correct.

Prof. Prathap Haridoss: So, are there areas of research that a PhD or an MS student might still encounter, which are considered you know traditional areas of research in Electrical Engineering where there is a lot of literature available that they can refer to and has been around you know, for maybe I don't know 10-15 years, may be much more than 20-30 years, but still considered something that people have to work on and there is a scope there to work on.

Prof. Deepa Venkitesh: So, thank you first of all inviting Prathap. As far as Electrical Engineering is concerned, it's a highly interdisciplinary branch. So, as you rightly pointed out there is lot of material available in tradition. Traditionally Electrical Engineering was to do with power electronics, power systems, you know high voltage engineering and Electrical Engineering or Electronics Engineering as it is called it was in our department it is all both together. It's machines power, electronics, VLSI kind of work. And, research is continuing in that area.

So, for instance in today's context you know, you have lot of solar things coming up and you have local generation of power happening. So, each consumer is now producer of electricity also, so going from the power grid which is a traditional, very traditional area. Now, you have inputs to take from those traditional areas, where earlier it used to be small hydro electric projects which used to feed in power into the system. Now, you would have smaller consumers sitting there and feeding so. Think lot of inputs can come from that area. Similarly, VLSI, microelectronic it is all tradition which is already there, you start from there and then you start growing. Same for photonics, photonics is in area which it's a counter part of electronics as we called, right. Where things in electronics you control electrons and photonics you control photons.

Prof. Prathap Haridoss: Photons.

Prof. Deepa Venkitesh: So, traditionally it is optics, right.

Prof. Prathap Haridoss: OK.



Prof. Deepa Venkitesh: Traditionally you called that it optics and then you start from the principles of optics and then you see how it applies to an engineering perspective. So, yes, there are different traditionally sensitive areas, if you take about for example, Communication, single processing is one of the very, very profound or the very basic subject, which gets applied in multiple different areas, it could be speech processing, it could be image processing, it could be processing for optical communication, it could be a processing for wireless communication.

So, these are traditional areas, definitely they do have a strong hold in Electrical Engineering as on today. But, you know we will see that it will things will change when as far as research is concerned, right. So, in that sense like physics or like chemistry, Electrical Engineering is not that you call it as a chased subject where you keep following one specific area and you keep expanding on that, it's highly multidisciplinary.

Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: So, somebody said if you know mathematics well and if you know the physics of it, then you have pretty much well equipped to do Electrical Engineering.

Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: And, where do you get those mathematics and physics, you got to get your basic.

Prof. Prathap Haridoss: Basics.

Prof. Deepa Venkitesh: Right.

Prof. Prathap Haridoss: Great. So, now I mean in the same token. So, you know like with various engineering disciplines, with a passage of time you know generally new areas come up.



Prof. Deepa Venkitesh: Sure.

Prof. Prathap Haridoss: And, so in Electrical Engineering is there something that is come up much more recently let's say, the last 5 10 years that people focus on a lot.

Prof. Deepa Venkitesh: Absolutely, absolutely. One thing that you see, the solar we were talking about, right. So, now, how do you do this distributed grid? How do you do the smart grid? How do you do distributed generation? How do you feed it into the grid that is in case of power electronics? So, let me also tell you that in general Electrical Engineering we have 5 disciplines, major disciplines we used to classify. One is Communications related area, which is to do with wireless communications that involves lot of hardware to make those antennas, to make the wireless systems and more than hardware a lot of software and mathematics. For example, if you are trying to send a signal from the transmitter to the receiver, the data rates have to be going up so you need to model your channel very well, you need to do an adaptive control of your signal, you need to give a feedback adapter, it is a lot of mathematics probability too that goes in.

So, there is a large influx of you know highly efficient mathematics and signal processing going in that area, that is in the what we call as a Electrical Engineering. One in our, one in our IIT setup, but it's basically communication signal processing and related areas. The second area is Power Systems and Power Electronics. So, as I mentioned earlier the distributed power scenario is one thing, power quality power, when you have distributed system of you know, every consumer starts feeding in power into the system. Now, how does electrical TNEB, for example, ensure the quality of power, right.

Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: So, now there is a software approach to that, there is hardware approach to that, so those are the advances in that area. Same way in renewable power for example, wind solar how you integrate it to the grid, those are issues. And, I think many of you might know the solar DC project.



Prof. Prathap Haridoss: Direct.

Prof. Deepa Venkitesh: You know.

Prof. Prathap Haridoss: Direct DC.

Prof. Deepa Venkitesh: Direct DC consumption where, how do you make your lights? Lights, of course, LED's work on DC; How do you make your fans, your refrigeration? everything on DC so that you don't, you can avoid the ac DC conversion lost, that is a very big project and that is of highly socially relevant project also. So, that's a lot of effort is going in that area. So, that is to do with power systems, power electronics machines. Another important thing in that area is you know, different users will have different requirements, for instance, you are powering your satellite the requirement could be small voltage, but high current application, right. So, how do you do those DC-DC conversions from, you could be drawing the power from the grid or you could be drawing power from solar, but, one DC to another DC of different requirement, how do you do that?

Prof. Prathap Haridoss: All right.

Prof. Deepa Venkitesh: Even robotics, any area that you pick in of engineering you would need that DC-DC converter. So, that's another very important area. You know applications specific DC-DC converter designs. Then the third one, third major area of Electrical Engineering is micro electronics and VLSI. Micro electronics, I don't have to say that is the you know, the granularity has going down, you are following more slog, you are going from nanometer to sub nanometer kind of device dimensions. So, you have all the associated math and you have all the associated physics, along with the fabrication technology for that.

So, you will have. The other important area that is come up there is Microelectronic and you know MEMS what we called as, Micro Electro Mechanical Systems. Now, that is evolved into nano electro mechanical systems, where the device dimensions are very small and you will be able to control or switch certain devices.



Plasmonics is another new area, where metal interfaces can enhance certain properties of light and so that is another area. Silicon photonics is another area, where you know you can just like how you had the electronic integrated circuits you could have photonic integrated circuits where your mother board. So, the biggest problem today in a computer mother board is how fast can you transfer from your CPU to your memory, right. Now, if you can do everything in optics, where your transfer happens in optics, your processing happens in optics, that's the whole area of optical signal processing. You do not have to go into the electronic domain.

Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: Like being the fastest, you can do it at that speed, right. There is lot of development happening around bio medical devices, how do you make simulators for making a surgery efficient or making giving practice to the doctors and things like that? So, definitely yes, all these areas so the controlled an instrumentation and that's where we talked about bio medical engineering. Then of course, we talked about photonics where the other smart city scenario right now people are talking about putting as many sensors as one can and then develop sensor network.

Again, it's completely interdisciplinary you cannot classify whether it is a photonics or a wireless communication because all aspects of it is coming in there, right. So, you know so smart cities you want to put multiple sensors you know, these multiple sensors some of them could be electronic sensors, some of them could be photonic sensors all of them would use a wireless communication link to you know relay the information to 1 central office, then you would do the appropriate process is required. So, all these are coming up organic electronics is another area, where flexible electronics.

Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: Right, you would have seen advertisements from Samsung or somebody who would have made a TV which is flexible, you would have your cell phones which could flexible, you can roll it and put it in your pocket. Question is how efficiently will you be able to develop? How fast will these devices respond? Can I use a



make a use and throw screen for example right ? So, it is completely involving area, right.

Prof. Prathap Haridoss: So, if we now look at let's say a student prospective of you know people coming in and so on. Of course, there is an Electrical Engineering Departments in almost every colleges because it is one of the you know, traditional areas of engineering you have bachelor's degrees in Electrical Engineering being awarded, many institutions around the country. When they come for a masters or a PhD program, are there still certain I mean, are there certain challenges that they face? I mean of course, we are moving from a course based work to a research based work, that itself provides some challenges.

Prof. Deepa Venkitesh: OK.

Prof. Prathap Haridoss: But, other than that specifically for Electrical Engineering in terms of their preparation that you normally see among the student community, that is coming in for higher degree in Electrical Engineering, did you see that they face certain types of you know, handicaps that they need to overcome for which then they are involving certain mechanisms here and if so what?

Prof. Deepa Venkitesh: So, it's a very, very relevant thing that you know, this is something that we keep discussing among our research group at the faculty group and what we have always found is at the end of their undergraduate program they are probably writing a gate exam or you know, some qualifying exam to get into our system. The examination systems are always checking your ability to remember certain things or may be at best in what we call as blooms taxonomy of learning at the base levels. Where they learn something, they are able to reproduce something they are at best able to apply a formula to a given situation.

When they come into the research mold, they also need to research the final goal and the blooms taxonomy is also in creation, right, so you would want them to able to create something. Now, again broadly classifying as in any other engineering or science streams, you could have theoretical work or experimental work. So, let's talk about the



challenges with the theoretical people who try to do simulations or theory. There, what we feel is the mathematics is not strong enough.

Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: Right. There mathematics is not strong enough, as somebody said you know if you get your mathematics, if you get your physics right the rest is mathematics is what somebody say, but that rest is mathematics is not very strong.

Prof. Prathap Haridoss: Yeah, it is very.

Prof. Deepa Venkitesh: So, it is they are able to what we see as a student, if you give a stereo type problem they are able to solve. But, given a situation they are not able to formulate a problem and solve. A part of the research training is towards that I understand but, to correlate from different areas and applying the same problem is something that we are finding, the students are finding it bit.

Prof. Prathap Haridoss: Difficult.

Prof. Deepa Venkitesh: Difficult and challenging. Even in programming people know how to write a c program with same program with or may be a mat lab code as this called these days you know, use 100 lines to write something which is worth may be you know, 15 lines.

Prof. Prathap Haridoss: 15 lines, OK.

Prof. Deepa Venkitesh: So, that's something that smart mathematical thinking is something that we would like the students to develop when they come in. And, as far as experimental skills are concerned, partly because of not their problem, you know, they are mostly not trained in handling equipment's. When I say trained, it means like we are not talking about the research level equipment's we are talking about simple things like



oscilloscopes, right. Again, as I said it's partly not due to their fault, but they even if they get a chance.

Prof. Prathap Haridoss: Exposure lack of exposure.

Prof. Deepa Venkitesh: Yeah, lack of exposure. Most of the colleges probably run 4 or 5 people on 1 equipment's, some of them might be actually using it, and some of them might not using it. So, my only advice to the student is, if they get a chance they have to go and try and run experiments and that experimental skill is something that we, it's not even a experimental skill that experimental sense is something that thinking on feet while running the experiment, those are the things I think our students find it big challenging. Third biggest challenge everyone faces is lack of ownership.

Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: Lack of Ownership.. Lack of ownership for a specific experiment that is done or a specific instrument. This micro electronics and photonics, the most of the instruments are you know huge facilities and I think it is more of ethical culture than anything that can be trained. You, when you are working in a place many of the students feel that it is like a college, where I go in the morning come in the evening.

Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: Close down everything, my job is over there is somebody to take care. But, I think that has to change and they should start getting, they should kind of find a unison with the equipment, only then the results will come out.

Prof. Prathap Haridoss: Okay. Let's look at it from say the industry perspective. So, there is always this perception that when do a masters degree especially in MS or a PhD degree, Research based degree that we, that the person, the student is becoming a specialist in a sort of a narrow field. And, often MS, PhD projects are also may be a little



ahead of their time, in terms of you know because they are really pushing the boundary of some.

Prof. Deepa Venkitesh: Correct.

Prof. Prathap Haridoss: Some areas that they are working.

Prof. Deepa Venkitesh: Otherwise you cannot publish.

Prof. Prathap Haridoss: Yes. So, in that context, to what degree does the industry show interest in what MS PhD students traditionally do in Electrical Engineering? Are there areas that even though they are advanced the industries now ready to absorb also at the same time?

Prof. Deepa Venkitesh: Absolutely, I mean Electrical Engineering by the very nature of the projects that we do is very, there are of course, there are certain areas which industry might not be interested in, let me also say that like example, Quantum Communication for example, right. Or a Quantum Computer for example, it's probably not something where an industry is not.

Prof. Prathap Haridoss: Currently interested in. Yeah, OK.

Prof. Deepa Venkitesh: No, would not spend lot of money on, right.

Prof. Prathap Haridoss: Yeah, yeah.

Prof. Deepa Venkitesh: But, there are certain areas like that. But, many areas that we are talking about for example, these grid, right. I mean, it is industry who wants the result out of this, right.

Prof. Prathap Haridoss: OK.



Prof. Deepa Venkitesh: So, definitely industry is looking forward towards many things that we are doing especially in wireless systems, some of the things that we do evolve as standards.

Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: Right, the kind of work that people do out here, they participate in standards. But, as far as the student is concerned, I think PhD, once you get a PhD I mean, when do you get a PhD I think within a certain reasonable time if you are able to absorb a new topic let's say, industry has a certain requirement, somebody who has a PhD should be able to understand, absorb that topic and provide a solution given a reasonable amount of time. It need not be directly related to what they have actually done.

Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: So, in that sense when they graduate with a PhD, it is probably not necessary that it is directly related to what,

Prof. Prathap Haridoss: What their area of specialization?

Prof. Deepa Venkitesh: Specialization.

Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: So, I think.

Prof. Prathap Haridoss: Yeah, in that context. In fact, I mean taking that forward, what sort of positions do you see MS and PhD students in Electrical Engineering normally speaking up?



Prof. Deepa Venkitesh: Most of the students go for, MS students go for PhD and PhD students go for post doc, but having said that there are many students who are absorbed in industry also.

Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: They do course job. For example, we have from our photonics group 1 of our student have gone into. In fact, my student went to an industry just after PhD, he was observed in Tejas Networks for which is a communication, pure communication company. Whereas, his project was on optical signal process, it has probably 10 years into, I mean I cannot think of an optical signal processor which he had implemented in his PhD getting implemented in the industry for next 10 years, I cannot see that happening.

Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: But, you know, you the kind of expertise that you develop during that, he got placed in Tejas Networks. There is another student who got placed in GE, for example. I mean, there are certain core areas. Again, these are not related to directly to their PhD problem per say. There are of course, also cases where after their Masters or PhD they have their own start ups, people have started for example, Solar electronics, I mean Organic electronics. That are couple of group, group of students who have had their start ups, coming up. So, there are different possibilities one can think of it.

Prof. Prathap Haridoss: Couple of things from let's say faculties view of what is happening with respect to a student. At least, during their student life as a research scholar as a MS or PhD student and may be even their early carrier after that. How other ways in which you measure success in research? Other than you know, I mean traditionally people talk about publications.

Prof. Deepa Venkitesh: Correct.



Prof. Prathap Haridoss: That is certainly something we will still continue to look at. But, is there any other way in which you look at students and feel that you know, in fact, this student you know is growing as a researcher compared to somebody else that you see?

Prof. Deepa Venkitesh: Absolutely, I think that goes without saying every research guide, when do you again get your student ready for graduation and the student is able to give you ideas which the guide is not thought about. Student is independently proving certain things he is coming up with multiple ways of solving the problem. I think that is how you measure success.

Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: And, way to quantify that is of course, your patents, your papers, the number of citations you have had, citation of course, something that comes up over years, but the measure I think the immediate measure which is probably not very quantifiable is coming through discussion between a faculty and a student is when the student is trying to think of or giving us new ideas rather than we giving them the ideas, I think that's the measure of success.

Prof. Prathap Haridoss: In fact, on a more mundane note you know, we always say that students have I mean at least especially as a research scholar, interaction is a very important thing.

Prof. Deepa Venkitesh: Correct.

Prof. Prathap Haridoss: How well they interact with other people in the group? How well and how often they interact with the guide? In your view, you know, how often should students be meeting their guide adviser?

Prof. Deepa Venkitesh: So, there are 2 things. You pointed out 2 things, interaction with the peer group and interaction with the guide. What we have set up in the photonics group is of course, changes from guide to guide but, at least we say once in a week is the



minimum. More meeting, more often than that would mean that the student is dependent on the guide and meeting less often probably student will go out of track. So, we believe once in a week is something, but what we stress on more importantly is their peer interaction. And, one of the things that forces the peer interaction is you know, it's a fairly big group, there are about 6 faculty, 6 of us who are working together in, if it's a research group like that. If people can afford to have that kind of set up.

Best thing is to have them organize peer seminars, it could be on a small topic that faculty is not typically invited for that, it could be discussion of a research paper where you know, peer learning is much more valuable. We feel especially, in certain areas which are highly interdisciplinary. So, I think it's important that the students talk to each other, give presentations to each other, get inputs from the peer the seniors. Then of course, the guide meeting with the guide at least once a week is what I would say.

And, I would also say that the students have to attend other talks, right. Other Talks from related area it could be in different department all together, but if it is having some connection to their research work it is, even if it's not a connections it is good to attend those talks to, I mean that is how they learn and that is how they kind of apply what others are doing to their own work, right. So, they should have that bigger vision or bigger view for, eye for those things.

Prof. Prathap Haridoss: Okay So, in I would say a sort of to conclude, what are you words of advice to any student who is aspiring to you know join an MS or a PhD program in Electrical Engineering?

Prof. Deepa Venkitesh: So, we have covered that mostly, one is their mathematics they need to work on.

Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: Right, whichever area in Electrical Engineering you want to work on. If it is something to do with VLSI or I mean it is your physics and mathematics, basically.



Prof. Prathap Haridoss: OK.

Prof. Deepa Venkitesh: Well work on your physics and mathematics. Then also have this idea that research is not a time bound program, right. The adviser does not have a solution, if the adviser has a solution to the problem he does't need a student, the adviser can work towards every problem. So, do not expect a final answer from an adviser, it is the probably the goal is also not very clear, only the problem is clear, right. So, that will evolve. So, you need to have that patience and that kind of temperament that implicitiveness to go further.

Again, talk to others students and try to understand, what their problems are? How have they solved those kinds of problems? You know, teaching helps so, be a very trying be a proactive teaching assistant if possible, because when you try to teach certain things you will get the connect and your some of your research problems get do gets solved. There is no documentary evidence for that.

Prof. Prathap Haridoss: Yeah, yeah.

Prof. Deepa Venkitesh: But, I feel that teaching helps research and research helps teaching. So, given a chance try and teach certain things when you are in the program and yeah get your maths and physics right.

Prof. Prathap Haridoss: Ok, great on that note. Thank you so much for joining us.

Prof. Deepa Venkitesh: Thank you, thank you very much.

Prof. Prathap Haridoss: It was a pleasure to have you here.

Prof. Deepa Venkitesh: Yeah. Thank you.



Introduction to Research
Prof. Prathap Haridoss
Prof. Arindama Singh
Department of Metallurgical and Materials Engineering
Department of Mathematics
Indian Institute of Technology, Madras

Lecture – 46
Research in Mathematics

Prof. Prathap Haridoss: Hello. We are a pleased to have with us Prof. Arindama Singh from the Department of Mathematics and he will discuss with us you know, research in mathematics and aspects of research associated with you know student life and how it interacts with you know their research at in their department and so on.

Prof. Arindama Singh has a PhD in mathematics from IIT, Kanpur and he has been here at IIT, Madras, as a faculty for more than two decades now, since 1995 and has a lot of you know a rich experience in working with students in actually various capacities because he has also held several positions in the institute which deal with you know critical aspects of a student life here.

And incidentally you should also look him up on the internet. He is a very interesting personality, I had the opportunity of looking up lot of information on a about him. So, if you get a chance please do look him up he has a written a lot of things about his experiences in life in generally in addition to his provisional activities. So, thank you for joining us.

So, I would like to start by looking at this aspect that you know at least in engineering background. We tend to have you know in any engineering division, we have areas of activity which are considered traditional areas of research. In mathematics, is there such a thing, is there such a concept like you know these are the traditional areas of research which are there and then, and which have been there for a long enough time, but there is still research going on it and therefore, there is a lot of literature that people can look at and you know engage themselves against.



Prof. Arindama Singh: Yes, as usual mathematics is in fact, living traditionally so obviously, the old areas never die.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: But new areas being added due to applications and some other programs coming up in daily life. So, like we have the geometry we started from Euclid or so, it is still living.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: Research is going on in geometry in various aspects of it. Similarly we have analysis which is very traditional, algebra. So, these are some of the branches. In fact, these comprise the whole of mathematics. Any branch of mathematics you take it is somewhere related to one of this.

Prof. Prathap Haridoss: One of this.

Prof. Arindama Singh: Or even all of this.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So, they are still living have, however in India sometime back fluid dynamics too care of almost everything.

Prof. Arindama Singh: Any department you go there will be some people working in fluid dynamics.

Prof. Prathap Haridoss: Fluid dynamics.

Prof. Arindama Singh: That is an engineering subject. So, slowly it is dying from mathematics departments.



Prof. Prathap Haridoss: OK ok.

Prof. Arindama Singh: And some pure math, that is being established everywhere.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So, that is the trend now.

Prof. Prathap Haridoss: So, new areas of research of would be.

Prof. Arindama Singh: Yes.

Prof. Prathap Haridoss: I mean in what would be cover? What would constitute like you know areas which have sort of come upon you very recently that well, it may be say let's say the 5-10 years time frame that many may be in their many groups interested in looking at.

Prof. Arindama Singh: So, after this advent of computers, new areas like numerical series.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: Then computer science related mathematics. For example, discrete mathematics, data structures, theory of computation, then image processing and anything related to numerical like - numerical linear algebra, for example. So, these areas have come up recently.

Prof. Prathap Haridoss: So, you yourself in fact, expert in Numerical Analysis.

Prof. Arindama Singh: Yes, I am also related to one of those.



Prof. Prathap Haridoss: Yeah, Computer Science, Theoretical computer science kind of activities. Okay so, these are the newer areas that in mathematics the people work on actually. Okay In the sense in fact, if we look at may be students coming into mathematics department what sort of backgrounds do they come in I mean are they only are they almost uniformly you know bachelor or B. Sc in a mathematics kind of background or you see different, do you see enough engineering students moving to mathematics in the PG programs.

Prof. Arindama Singh: Not many engineers coming to mathematics, but there are one or two.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: In every one or two years, we get one or two engineering students who is so interest in coming to mathematics and they also do good research after that we find, but uniformly it is M. Sc from mathematics.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: That is our intake

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So, in some of the places where mathematics and statistics both are there. So, M. Sc statistics people also come.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: For doing research.

Prof. Prathap Haridoss: OK ok. So, in terms of the students coming in they probably come with an wide range of different institutions from which they have been educated



for their B. Sc and M. Sc and may be as you said you know one or two engineering students and what not. When they come in, in terms of settling into doing a research kind of a program here in mathematics department here, are there a specific issue that you students facing as they settle in it may be the early part of their graduate or post graduate life here in terms of technically adjusting to what is required here and so on.

Prof. Arindama Singh: Yeah. There is in fact, our interns should be able to take care of this issues.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: But it does not because of various reasons. We have to load or narrow down this scope, so that it should be accessible to many people. So, in that sense we keep our interns in such way that it is come on to all almost all the universities in India, which are doing in the B. Sc programs. So, that do we get students from almost all the places and then we have the problem of bringing that standard of, so that they will be almost equal in 1 or 2 years and that's a gigantic task, in one year it's not possible we know, but then we have to give some courses which will be very basic to them. So, that is the idea.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: And then during this course work of period they take the courses which are useful for the research also not only on the general mathematics, but also narrowing down to their research area. So, at least 1 or 2 courses are given from the research area so that they will be going faster.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: This idea of course work is really very important for the research students. So that they will be equalized to you over 1 or 2 years and then they will also be earning some expertise in their narrow down area.



Prof. Prathap Haridoss: In terms of any other preparation in terms you know. So, basically you are saying may be the rigger is what they have to pick up on.

Prof. Arindama Singh: Yes.

Prof. Prathap Haridoss: When they come in and it takes some may be a year or two to settle into this regard rigger, of course, when we think of I know basic sciences you know including physics, chemistry, mathematics and so on, especially with respect to mathematics, The I mean atleast the perception is that you know may be the industry general industry that is out there, which has a lot of people to what degree does that industry I mean see show interest in graduates of a mathematics or in what you know aspects of industry is there you know nice fit between people who are doing an advance degree in mathematics an MS or PhD degree and the what the industry might desire or even say some section of the corporate world may decide.

Prof. Arindama Singh: Yeah. So, in the traditionally people are thinking that industry might take who are doing well in fluid dynamics. But that is not happening in India.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So, there are many industries who use these concepts and foods products are even their process can be optimized by using these methods, but they don't ask for it. And when we go for them they are also reluctant.

Prof. Prathap Haridoss: OK Ok.

Prof. Arindama Singh: Because there is a fear of losing their jobs, or some such.

Prof. Prathap Haridoss: OK Ok.

Prof. Arindama Singh: But, in computer science areas our students are really doing some good job they are going for R&D sections. Recently one person who graduated in a



complexity theory went for this job, for example, in Dell systems. So, there are some areas like this where they are able to go, but it is really again a confine to the R&D sector. It's not going to the productive sector.

Prof. Prathap Haridoss: R&D sector.

Prof. Arindama Singh: In the industries.

Prof. Prathap Haridoss: But what about let's say I mean let's say the world associated with you know all these stock markets and things like that.

Prof. Arindama Singh: Yeah, Finance.

Prof. Prathap Haridoss: Finance.

Prof. Arindama Singh: We are not having any expertise in that.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: There is nobody in actuarial sciences.

Prof. Prathap Haridoss: But.

Prof. Arindama Singh: That is one area where math 27 can be useful. But in abroad they are really.

Prof. Prathap Haridoss: Yeah, they seem to be using that I mean even from what I even say physicists and work not get absorbed by the financial sector because they seem to be doing a lot of analysis or I think the tools may be they learns are suitable for those kinds of positions. So, that is not something that you are saying.

Prof. Arindama Singh: That is not at (Refer Time: 09:21).



Prof. Prathap Haridoss: OK. You see just a there I mean is the just a not aware that there are enough people here with that kind of expertise.

Prof. Arindama Singh: We do not have expertise. In our department, for example, there is no person who is expert in the actuarial sensors.

Prof. Prathap Haridoss: OK, fine. Let us say in our experience which looking at students over all these years. In generally know in the institution we have some metrix of you know how people are progressing, what causes they have done, what grades they have got, what type of publication they made etcetera. But in generally, in a more holistic sense especially for someone who is just you know considering coming in and so on what would you suggest are good ways to measure their progress in research, so that for their own satisfaction also they understand in fact, they are progressing in research especially let's say with respect to mathematics for example, did you think there are ways in which they should think of themselves analyze their situations.

Prof. Arindama Singh: There is one objective factor which is well public essence in good journals.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So, though it looks very objective, there is objective factor in terming which one is good and which one is bad.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So, that is almost settled.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: If you ask any general person whether this journal is good or not he will be able to tell you. Even he publishes in that he may say also it is bad enough.



Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So, there is some set of understanding.

Prof. Prathap Haridoss: Understanding.

Prof. Arindama Singh: So, number of papers that come in good journals that is a.

Prof. Prathap Haridoss: Good metric.

Prof. Arindama Singh: Well.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: But individually when a person discusses with supervisor the supervisor knows how much progress he has taught.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: About that, it's basically not only that area fair he is doing research, but the peripheral areas.

Prof. Prathap Haridoss: OK ok.

Prof. Arindama Singh: How much he has acquired about the null is about the peripherals.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So that after he finishes his PhD, he will be able to really do research on his own.



Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: There are some measures like one publication there is a measure factor they are in the publications. So, one is whether somebody has solved a long standing problem.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: That is a good measure of thing or how far he has progressed towards that. Another is if he has not solved any long standing problem, whether he has done something which will give rise to a long standing problem.

Prof. Prathap Haridoss: Fine, either solved a long standing problem or created a long standing process.

Prof. Arindama Singh: Yes.

Prof. Prathap Haridoss: Problem there is stands for.

Prof. Arindama Singh: So, these are the two things which are very good.

Prof. Prathap Haridoss: OK ok.

Prof. Arindama Singh: And the third one is anyway we have to use always which is the number of the papers in good journals

Prof. Prathap Haridoss: Ok. So, these are some.

Prof. Arindama Singh: Yes.

Prof. Prathap Haridoss: In especially with respect to mathematics on a sort of a mundane thing concept you know when students come in they have come from a you know



academic setting where they do courses and so on, they come and they are now getting into research setting. Generally you know all areas of research we always feel that you know interaction is a very important, technical interaction is a very important way in which people grow as researchers and in that you know working with the people in the group, working with the guide, meeting the guide etcetera plays a very important role. In mathematics where you know there is a lot of thought processes involved, lot of you know it's a very in many ways a very internal thing we are not often not running an experiments sort of how important do you see this role of a student adviser meeting? How often should you think they should meet?

Prof. Arindama Singh: Ideally they should meet every day over coffee.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: Because then mathematics do not have any labs, where they can interact with the things of the work.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: It's really only communicative. It's a linguistic entity.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So, they have to go on communicative.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: By that only they will get experience, there is no other way.

Prof. Prathap Haridoss: OK.



Prof. Arindama Singh: But what we find is usually they are giving some jobs and they come to meet the supervisor after one or two weeks. And then they spend 1 or 2 hours, get some ideas and go away.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: That way it will not happen, mathematics will not happen.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: But sometime you may need that seclusion because if some ideas really stuck you need certain answer time to get it into work.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So, sometimes it may be OK, but not always.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: Ideally it should be everyday one should meet, tell something about what has happened. And then go back.

Prof. Prathap Haridoss: Ok ok and what sort of characteristic you think people should have to be good mathematicians, I mean in the sense how can how can someone you know of course, you have seen one is an inclination you tend to solve some problems and you feel comfortable with idea that I understand, but over and above that is there some visualization capability that you I mean what should a person see in themselves and say okay this I am able to do this, probably I am good at mathematics other than just being able to solve the problem.

Prof. Arindama Singh: Well, if you look at mathematicians there are all sorts of people in that community.



Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: We cannot say that for mathematics one should be like these.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: But certainly they are not fools.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: That is one characteristic that they should be intelligent.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: They should try to find out if there is something to go differ or not.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So that is one characteristic which really characterizes them.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: There are not satisfied just looking at the surface. They would like to go deeper.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: What is the pattern behind it?

Prof. Prathap Haridoss: Right.



Prof. Arindama Singh: See one has that interest. Then probably it can be happened.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: But, if you doesn't have that interest he cannot be a mathematician.

Prof. Prathap Haridoss: So, that thing that you know you should not be a superficial thing.

Prof. Arindama Singh: Yes.

Prof. Prathap Haridoss: Okay much, much deeper into the thought process behind that the solution. What sort of I mean I know you touched upon this little bit, what sort of especially with you know in both certainly in engineering and I am sure in mathematics too when you say an MS or a PhD kind of a degree automatically the thinking is that this person is now an expert in a narrow area. So, there is some kind of a specialization associated with it. So, in that sense what sort of positions are people who create MS or PhD degrees from mathematics department and what sort of positions you see them going to in recent times and where do you think are the possibilities for such.

Prof. Arindama Singh: See at least from IITs when you get one PhD in mathematics, you do not expect him to be very narrow because there are associated with the tutorial classes with the B. Tech process.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So, they are having a lot bigger background than other process.

Prof. Prathap Haridoss: OK.



Prof. Arindama Singh: Where this HTTA concept is missing, probably there the students are also missing.

Prof. Prathap Haridoss: Those students are also missing. Ok, fine they are missing that. So, here it is a nice learning experience.

Prof. Arindama Singh: It is nice learning experience and they get firsthand experience of teaching also.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: Have to communicate mathematics by Vernor communication. So, that is important they can write, but they are not able to speak sometimes.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So, this helps them to speak that way they are better compared to other process, but then that is not everything. So, they have to once they really narrowed down because IITs there are only 17 or 18 IITs and we get a less number of students compared to the other process. So, everywhere it is not possible. So, there, there is an expectation, they are narrowed down to some particular area, but then it should be possible for them to take up when the other thing later, because once that is what one person asked me once that if I come to do M. Sc here what will I gain, I am not going to get any job directly after M. Sc. So, at that time there was no job even in the market for the mathematics students. Now at least there are some employment for doing research or something at that time there is no possible term research also, very few places where having the research compositions. So, my typical answer was whatever job they go they will be able to do it.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: Provided they have the interest.



Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So, these mathematicians pick up this nasty habit that they want to reinvent everything.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: See, if something is done in the book. They will not be satisfied. They would like to do themselves again, though that will be a guideline.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So, this reinvention is also hated in some of the industries.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: Because they have the particular process.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: And if this person doesn't follow the process tries to topple with it or create problems then lot of things gets disturbed.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: So, that is the only thing they have to be conscious, when they go to do any other job outside the academia.

Prof. Prathap Haridoss: OK, but what all.

Prof. Arindama Singh: But usually the position is academic position.



Prof. Prathap Haridoss: Academic position is the most teaching positions.

Prof. Arindama Singh: And then in industries R&D positions.

Prof. Prathap Haridoss: R&D position. These are the.

Prof. Arindama Singh: These are the best ones.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: Best fitted for them really.

Prof. Prathap Haridoss: Okay Fine and okay may be a sort of enclosing I just wanted to get a sense of you know a lot of students I mean in fact, there are students who finish you know even high school who consider mathematics as a something that is very interested and passionate about maybe under grads who consider it in greater you know enthusiasm for it. Is there some, what sort of advise you would give for people who are aspiring to become MS and PhD students to go on to get higher degrees in mathematics, what sort of advise would give you them?

Prof. Arindama Singh: Well, once they do their M. Sc, just to able to find out in which area they are really interested, if they are really interested in some area they should be able to produce some such new things, which may not be very weak thing which expert sees in that area I will say it is nothing, but it should be a deviation from the usual curriculum.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: One such a thing is there I would encourage him to go for doing research in mathematics. If it is not there then it may not be worth doing because I will do the usual things you will for teaching and so on.



Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: But not we will be able to really contribute to mathematics.

Prof. Prathap Haridoss: So, you are saying this would be one nice way for them to gauge whether they are you know in the right process to be going ahead for.

Prof. Arindama Singh: Yes.

Prof. Prathap Haridoss: Higher degrees in mathematics.

Prof. Arindama Singh: Yes.

Prof. Prathap Haridoss: So, they would have to at least for small number of new things.

Prof. Arindama Singh: Yes.

Prof. Prathap Haridoss: They should start dabbling with, so that an expert feels that they are you know comfortable with that.

Prof. Arindama Singh: Yeah. So, you would like to see the playing thing that he really plays with mathematics.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: Not only does the conventional things.

Prof. Prathap Haridoss: OK.

Prof. Arindama Singh: But he is very comfortable with it and he does something there.



Prof. Prathap Haridoss: OK ok. So, that is your advice, before he just jump in to an.

Prof. Arindama Singh: Yes.

Prof. Prathap Haridoss: Master degree or a PhD degree they should first gauge.

Prof. Arindama Singh: Yes.

Prof. Prathap Haridoss: Whether are you know sort of in the right frame of the mind for it and then on that basis proceed. Okay Thank you Dr. Arindama Singh.

Prof. Arindama Singh: Welcome.

Prof. Prathap Haridoss: Enjoyed meeting you and I think a very nice advise for, because mostly people we look at engineering students more often and think a lot of people considered mathematics, but they don't where to take it and I think these are nice words for advise for them to pointer about before they make their decisions and look at their life in graduate school here.

Thank you so much.



Introduction to Research
Prof. Prathap Haridoss
Prof. R. Nirmala
Department of Metallurgical & Materials Engineering
Department of Physics
Indian Institute of Technology, Madras

Lecture – 47
Research in Physics

Prof. Prathap Haridoss: Hello, it's our pleasure to have with us Prof. Nirmala from the Department of Physics here at IIT, Madras. She has been a faculty here for almost a decade now. Before that, I mean she had done her PhD here and after that she has done several post doc positions she has held at IISC, at TIFR, at Ames lab in the US Department of Energy Ames Lab at Ayova. She was also at the SKKU in South Korea so and after all that she is been a faculty here. So, lot of experience in that process working with you know, research groups from across the world and also of course, with the research students here at IIT, Madras. So, it's our pleasure to have her with us today. So, welcome to this interview.

Prof. R. Nirmala: Thanks a lot Prathap, for this chance of talking to in NPTEL.

Prof. Prathap Haridoss: Sure, sure, our pleasure, our pleasure. So, just to start off with, see physics has a department as a field has been around of course, I means it is a one of the fundamental fields and so, I mean for ages people have been doing work in physics. So, if you look at it now, what would you call as traditional areas of research in physics?

Prof. R. Nirmala: Okay so, if you go back in history you will find physics starting from say, Optics, Atomic and Molecular physics and we still continue to work in these areas. And, experimental solid state physics and non-linear dynamics and we have this theoretical and computational physics and this is a very vast area, you can go from condense matters to string theory and etcetera.

Prof. Prathap Haridoss: OK.

Prof. R. Nirmala: These are some traditional areas that physics departments work with.



Prof. Prathap Haridoss: These are areas where there is probably a lot of literature, for somehow, some very long time ago and then you know when somebody starts there is a lot of stuff to look at.

Prof. R. Nirmala: Yeah, yeah.

Prof. Prathap Haridoss: Look at.

Prof. R. Nirmala: Look at.

Prof. Prathap Haridoss: So, also along similar lines if you want to look at you know new areas of research are there new areas that I have come up let's say in the last 10 years or so which lot of people look at?

Prof. R. Nirmala: So, again when I say new areas, some of these areas where were all the time.

Prof. Prathap Haridoss: OK.

Prof. R. Nirmala: But, probably we are concentrating more recently.

Prof. Prathap Haridoss: OK ok.

Prof. R. Nirmala: So, for example, high energy in particle physics it grabbed attention after Higgs boson, for example.

Prof. Prathap Haridoss: Yes, yes.

Prof. R. Nirmala: And, we have gravitation and cosmology. And, we have soft condensed matter on bio physics and quantum computation and information, quantum confinement, quantum interference and the entire set of nanoscience and nanotechnology and this involves low dimensional systems like graphene and similar 2D systems. And, quite a lot



of energy harvesting materials and quantum face transitions that happened close to absolute 0 and strongly correlated electron systems and so on.

Prof. Prathap Haridoss: Okay I mean of course, you see physics I think several of these are also some of these are your areas of expertise. I think your area, you are expert in condense matter, in rarer inter metallic, you also work on strongly correlated electron systems and also on magnetic and transport properties of materials at low temperatures, these are some. So, you are working on really on some of the newer areas.

Prof. R. Nirmala: Newer areas and of course, because it falls into experimental solid state physics I would call that as a traditional as well.

Prof. Prathap Haridoss: Ok, fine.

Prof. R. Nirmala: So there is some link.

Prof. Prathap Haridoss: Now, see if you look at physics. Again, it something that comes from high school, at high school itself students are exposed to a range of topics that get describe to them as physics. So, they are very familiar with what is considered as physics or the thought process that goes into all the things that we look at in physics and so through high school, through you know the process of getting into under graduate programs and also through under graduate programs, they all most all of them have considerable exposure to physics in the engineering field. Given this, when people come in for a Masters program or for a MS program or a PhD program. Do they still face specific challenges in settling into a masters or a PhD program with respect to the physics aspects of it and if so, what do they tend to do to you know handle such challenges?

Prof. R. Nirmala: We do have an M. Tech program and also the conventional M. Sc program under masters and the regular PhD program. So, although the students have exposure to fundamentals of physics, their core physics namely classical mechanics, quantum mechanics, statistical physics, electricity in magnetism and mathematical physics these needs to be strengthened and augmented. So, that is what we do at these



level 2. So, in the initial semesters they undergo a set of these courses, even for a PhD we have a module of foundations in theoretical physics which deals with these core subjects. And, we also have foundations in experimental physics which exposes them to a range of techniques and measurements; they also get some hands on experience doing some basic experiments.

Prof. Prathap Haridoss: OK.

Prof. R. Nirmala: Before they get into the actual program, even if a student registers for a PhD in a theory. He does take this other course as well. So, the person going to do a PhD in experimental physics will also be taking the fundamental theory course.

Prof. Prathap Haridoss: OK.

Prof. R. Nirmala: So, that they become.

Prof. Prathap Haridoss: So, there is some uniform you know setting and they come up to and then from there they are able to. But, any particular thing like for example, is the mathematical skills, do the mathematical skills that they have. Are they sufficient in the general, you know general student community coming in for MS or a PhD or do you feel that is something that they need more work on, when they get in?

Prof. R. Nirmala: Actually, we have variety of curricular across different institutes in a country, which is why we have these courses.

Prof. Prathap Haridoss: OK.

Prof. R. Nirmala: We want to bring them to the same platform, when start from there.

Prof. Prathap Haridoss: OK.

Prof. R. Nirmala: So, if there is any lapse, I think the students get to.



Prof. Prathap Haridoss: Gets to catch up.

Prof. R. Nirmala: Yeah, catch up. Do the things.

Prof. Prathap Haridoss: OK. So, now may be if I shift focus a little bit. See, in certainly in engineering often we tend to see a little greater link between say the industry and what engineering departments do. In the case of, I mean a Science Department like physics there are, I mean there is one aspect that I mean, first of all I mean the industry anyway it looks at you know MS and PhD students as people who are specializing in a field and may be some times the exacts and since they are a specialist, it's always a little narrower field and therefore, unless the industries actually directly working on that field they may find it little less interesting to look at a particular candidate. Now, in the case of physics is this an issue and if not also, are there specific area that typical physics department work on which the industry is you know more interested in, the immediate?

Prof. R. Nirmala: I agree with you that, we work on more fundamentals often. So, this overlap may be restricted to some areas, but I would say that this area by itself is large. For example, this functional and advanced materials, comprising, optical materials, photovoltaics, solar cells and you have this organic electronics, oxide electronics, now the recent nano oxide interfaces, graphene oxide interfaces and you have magnetic memories and spintronics and I think these areas have strong overlap with industry.

Prof. Prathap Haridoss: In industry, OK. And if so, when somebody graduates with an MS or PhD, what sort of positions do they get both industry and otherwise also go?

Prof. R. Nirmala: Yeah. So, the very common place the students, now PhD students get into is Postdoctoral position.

Prof. Prathap Haridoss: OK.

Prof. R. Nirmala: So, some students also get into teaching in universities and colleges as soon as they graduate. And, the students who go for postdoctoral fellowships they come



back and teach at institutes of higher learning like NITs, ICERS, IITs and IISC and so on. So, this is teaching in research. As I said, wherever we have this industry over lab we also have some R&D positions for these students in private and government labs, like defense labs.

Prof. Prathap Haridoss: OK.

Prof. R. Nirmala: And so on so on, yeah.

Prof. Prathap Haridoss: And, you mentioned postdoc positions, in science how important is it to do a postdoc position?

Prof. R. Nirmala: I think it's very important to get this exposure and for the experience. In fact, I think positions are offer to people, who acquire enough postdoctoral experience.

Prof. Prathap Haridoss: OK.

Prof. R. Nirmala: I think it is one of the mandatory norms, if you want to get into these institutes of higher learning I mentioned. I think you should have postdoctoral experience.

Prof. Prathap Haridoss: OK.

Prof. R. Nirmala: It helps you broaden you are skills set, over all values, because during your PhD you would have concentrated on 1 specialized area and now, even in that area you can equip yourselves further that more advanced techniques and so on. And, then come back and contribute.

Prof. Prathap Haridoss: OK.

Prof. R. Nirmala: Yeah.



Prof. Prathap Haridoss: Ok ok. Of course, see you have of course, come up as I mean you have also done a PhD and then you have done several postdoc positions and then helps several postdoc positions and of course, you know guided many students by this time. So, there are certain general perceptions on, how we are supposed to you know measure a progress in research? So, we tend to at least in the more mundane way is to simply look at publications and publications. In your view, are there other ways in which you measure you know progress in research that when you look at students you say yes; in fact this person is you know moving ahead in research?

Prof. R. Nirmala: First, my own personal opinion on measuring this success of research or any work that you do is, whether the end of the day you feel good and satisfied that you have done a good days work. So, if you are going to measure the success of a research student or a research scholar, you should check when he or she graduates whether you have made some independent researcher, whether he or she is able to create a problem and also develop a methodology or a way to solve the problem. And of course, as you said the actual measure will be publications in reputed and in peer reviewed international journals, I think that's a usual way of accessing successful outcome of a PhD program, that must be there to and also this research scholar must be able to present himself or herself before expert audience.

Prof. Prathap Haridoss: OK.

Prof. R. Nirmala: More confidently his or her findings and they should be done in national and international level. One can do it by presenting a seminar or talk in a conference at these levels and also this scholar must be able to interact or by interaction I mean, discussion and collaboration with national and international level experts, so whether they have developed that independence and expertise over the years. And, also I think they should have developed out of the box thinking skills.

Prof. Prathap Haridoss: OK ok.

Prof. R. Nirmala: Their own thinking, the independent thinking and I think one more thing in addition to publication these days is, at the end of the day when they graduate



the students must be able to articulate their problems, the problem that they create.

Prof. Prathap Haridoss: OK ok ok .

Prof. R. Nirmala: In a manner that, it fetches them funding.

Prof. Prathap Haridoss: OK.

Prof. R. Nirmala: They can support themselves.

Prof. Prathap Haridoss: Yeah, yeah.

Prof. R. Nirmala: Yeah, because now we have plenty of opportunities in terms of fellowships and projects.

Prof. Prathap Haridoss: Sure, sure.

Prof. R. Nirmala: Projects, yeah.

Prof. Prathap Haridoss: Okay ok. So, if now this I mean, I guess to some degree we were talking of the students as they you know complete and co ahead and so on. During this stay as MS or PhD students, so one of the things that we often talk about is that, there is a lot of learning that they do by interacting with their peers.

Prof. R. Nirmala: Yeah.

Prof. Prathap Haridoss: And, also of course, interacting with their guide or adviser.

Prof. R. Nirmala: Yeah.

Prof. Prathap Haridoss: In general, in your opinion, how often should students be



meeting their adviser and what do you think it is you know is expected to happen in this process?

Prof. R. Nirmala: I would expect that this should happen every day.

Prof. Prathap Haridoss: Everyday, OK.

Prof. R. Nirmala: Yeah. So, in fact, it can be as often as it can be.

Prof. Prathap Haridoss: OK.

Prof. R. Nirmala: It is not like restricted to few times a week or so. So, I see to it I meet my students every day.

Prof. Prathap Haridoss: OK.

Prof. R. Nirmala: And I think that is good and may be more focus discussion on what has been going on can happen on a weekly basis, you can have weekly group meetings when students comes under reports more formally his or her findings. But otherwise, I think interaction must be on it should be a continuous process.

Prof. Prathap Haridoss: OK ok. So, I think it maybe sort of to conclude, what are your words of you know advice to somebody who is aspiring to do an MS or a PhD degree in physics?

Prof. R. Nirmala: I think it's humbling to be asked this question. But, I think I would say a few things that probably I have learnt from my experience and with my peers and so on. I think there is no replacement for hard work, I think one should develop good set of work ethics and I think one also should continuously update themselves. So, this is about the literature survey that we do with at the beginning of the research. I think it should be a very continues process and students must continuously update in their own research area and they should, now a days we have plenty of avenues for getting these updates



from international journals about the journals contents and so on. So, they must be doing that.

And, they should also connect with their peer group, by peer group I mean contemporaries, also the seniors and juniors. And, they should interact with each other and they should be a part of a discussion forum and they should participate in department level, seminars and colloquia and not only the seminars in their research area, but also in the other areas as well. And, over all they should develop the values of core physics because when you come for a PhD in physics apart from the course, specialized courses that you do in your own research area you must strengthen your basic physics. So, this means classical physics, quantum physics, electricity and magnetism and mathematical physics and statistical physics. So, I think you should get this done, if it was not done before by the time of your PhD program and when you sign up for a PhD I think you have some inner urge.

So, I am sure the scholar themselves will do some self assessment more periodically and keep growing. And, the other thing I find often in our students is this communication skills, very often research is team work not often I think always it is a team work now a days. So, I think they should develop their communication skills as well, both speaking skills as well as writing skills. When I say writing of course, technical writing is also mandatory process. Yeah, I think these are the points that come up.

Prof. Prathap Haridoss: Actually, in fact, I know and I mean you mentioned a point which I thought of very interesting about you know the fact that they should attend colloquies of even other areas and so on. In that context, how much do you think at least, for example, in IIT, Madras, we have a physics department in the midst of a engineering you know environment. Whereas, there are many other places where physics is in a separate you know environment by itself. So, to what degree do you think this has you know assisted may be let's say, the physics students who are in the Physics Department or in a campus like this.

Prof. R. Nirmala: I think there are quite some interdisciplinary programs already running. I am seeing the connections between physics and chemical engineering, physics and



methodology, and physics and e, physics and chemistry and so on. I do find interesting talks happening in the nearby department, I think one should check out and from their point of view and perspective they are presenting their results, whereas you have some view from physics.

Prof. Prathap Haridoss: Yeah, from physics.

Prof. R. Nirmala: So, I have always got some interesting points by attending talks in other departments.

Prof. Prathap Haridoss: OK

Prof. R. Nirmala: Yeah.

Prof. Prathap Haridoss: Ok so, ok great. Thank you very much for joining us, I think you have given a lot of very nice you know I think input and also a lot of advice for the students aspiring to come in. It was a pleasure having you, thank you so much for joining us.

Prof. R. Nirmala: Thanks a lot Prathap. It was nice talking to you.

Prof. Prathap Haridoss: Thank you.

Prof. R. Nirmala: Thanks.



Introduction to Research
Prof. Prathap Haridoss
Prof. S. Sankaran
Department of Metallurgical & Materials Engineering
Indian Institute of Technology, Madras

Lecture - 48
Discussion with Research Scholars

Prof. Prathap Haridoss: Hello, we welcome you to this discussion on research. We have with us a collection of students they are all research scholars here at IIT Madras. There are about 30 of them, spread across about 16 department in the institute. So, we have a broad collection of research scholars, different backgrounds, different departments, different disciplines that they are studying and pursuing research in. And they have typically spent at least 2 years here in campus as research scholars. So, they have gone through all those critical steps that are there as part of research and some of them have been here longer. So, they have very recent experience in what it means to get started as a research scholar.

So I encourage you to look at the department module related to your department which is already put up on this course, as part of this course material, but I would also like you to watch this discussion to see how students, who are you know little bit more senior than you in this research setting and what their experiences have been, what they have learnt and what is it that they would like to share with you. So this is the general context in which will have this discussion. So, I would like to start by asking any of you to, tell me what was your view of research before you came and joined as a research scholar here and if over the time that you have been here if any of these views have changed and if so how they have changed. So, anyone who is willing to volunteer can get started.

Priyanka: Hello I am Priyanka, I am from Electrical Department. I am doing my MS here and I am from Nagpur. So, before coming here I was an under graduate student there in I barely knew what research was but from what idea I had about a researcher was that the someone who you know change the way things have being done and they you know challenge, they accept challenges and just bring about positive change in technology and in society as a whole. After coming here, what I felt was like, I actually realized that what I thought was actually correct and yes there is a lot of, I mean we are creating



impact in the way things were being done and in fact, what has changed in my perspective is that it's not just about you know doing something very exciting, but then at the same time it requires lot of patience, which I realized in the due course of time and I am enjoying this.

Prof. Prathap Haridoss: Okay Great, nice to hear. I think first of all I am very happy that your perspective turned out to be true and that you really do think that we are doing something very useful and I mean, I appreciate your point that there is a lot of patience involved which is I think some of the times, which people don't realize till they get in, that there is a lot of patience involved. May be we will talk of patience also a little bit. Anyone else, please.

Zeeshan: Sir my name is Zeeshan, I am from Department of Biotechnology. I have been a MS scholar here for the past 2 years. Previously I have actually worked outside, so I have some amount of industrial experience. My perspective of research was that it could be more structured; I thought there would be a certain way of doing things. Coming from industry, in industry everything as I said protocol there is SOP for everything, I thought research would be more on those lines something like that, but when I came here I realized that you have to bring the structure into the research it's not all inherent.

So, that part was initially of putting I mean it was strange. I thought it would be a directed approach to you know reach a goal, you would have a goal in mind and you would reach. My perspective has changed in that manner that I have seen that you don't actually start with the fixed goal. Even in research you don't actually start with the fixed goal, your goal actually shows up as you start working hence the process is also not straight forward, you kind of go everywhere and then reach there.

Prof. Prathap Haridoss: So, you tend to meander a bit here and there, move around and then, but let me ask you, in industry what sort of a position where you holding.

Zeeshan: I was a junior research fellow.

Prof. Prathap Haridoss: Okay so, you felt even a research in the industrial setting was



very structured.

Zeeshan: Yes Yes. There we had goals that we had to reach and do. I mean as in every position has it's own, you have your responsibilities, you are expected to fulfill your responsibilities. Lower down in the cadre you do not have big responsibilities so your results are fairly consistent. So, it feels the process is very structured. Generally your team mates tell you that there is this goal that we are trying to reach and we are working towards it. In research even your adviser doesn't tell you what is your goal you kind of have to figure it out as you go along so.

Prof. S. Sankaran: But research you were doing in the job itself right.

Reshan: Sorry.

Prof. S. Sankaran: In the in the previous job you were doing research only right.

Reshan: Yeah it was it was.

Prof. S. Sankaran: But then why did you come to an academic institution to do a research.

Reshan: Well for me it was actually a change of domain. In the industry I was in analytical development, I wanted to shift over to bio process engineering; it's a process engineering stream. So, I did not have the skills that requisite for bio processing engineering and in the industry you don't get to learn as much as do in academy, you have responsibilities you finish them you don't get to learn much.

Prof. Prathap Haridoss: Okay so there is not much flexibility in what you can do. There is a very fixed narrow path you have to walk down that the path. So, research is within that narrow path and you cannot you cannot, so meandering has some use I mean you can go this way and that way helps you learn and grow.



Zeeshan: No, it's just that because in the industry everything is driven economically right, everything you put in money you want some output yeah. So, that's why it is so directed you cannot actually go it's like you optimizing a process you stick with the process, you do not actually start with something entirely new. So, like that.

Prof. Prathap Haridoss: Good very nice to know.

Hemanth: Hello I am Hemanth, I am doing my Ph.D in Chemical Engineering Department. It's been like 4 years or 5 years I have been here, and so when you see when I was in under grade so I thought the research is something you invent a new product and that is the ultimate goal. So when, after coming here, so it's more it's included that you invent a new product on top of that you can develop more scientifically for the existing product also, you can improve the efficiency of the product, and all these basic studies you can do so that the existing product can also be improved, that's that's another prospective of research what I have seen. And when it comes to the way I see the subjects when I was in under grade and when I was a research scholar. So you understand the concepts at the end of the day it is basically you put in the mathematical form of this concepts develops some models for the systems what you have, and then try to understand the systems better with a little experimental activity and also little modeling activity where we tend to do it only experimentally when we are doing a research laboratory in, you don't have research laboratory as such we do in laboratories in under grad you just do a CRE experiment for example, say continuous tank reactor you just do to learn what are the theories that is already established, the same thing you do here for establishing a new theories. So, that's what the main difference we can see from the under grade labs and the labs which you are setup in the IIT's or whatever.

Prathap Haridoss: Okay Great !

Ram: Hi I am Ram; I am doing PhD in the Department of Civil Engineering. So, as an under grade I had bit similar views as others had like you invent a new product or something like kind of an iphone comes up and those are the really the products of our outcomes of the results that happens. But then once I joined here like researcher is a someone who actually works and contributes to the society as a whole. So, because even



I have some amount of industry research experience where it is more as it find out it is more economically like either there is a just to cut down the cost that is happens or how to increase your profits.

But in here we more work on kinds of document how internationally something has been done and for example, the mean things of we even work on something like who works on these stuff like, how do you effect duly managed the waste for example, that something is industry doesn't work at all because something which is there the society is incurring cost in how do you manage it and as a researcher we are actually looking at how do we, an effectively used the waste that the industry is generating. And once we theoretically develop morals which are economical as well and so that you would get something out of it and add value to the society, then the industry and other people start using it. So, we are someone who actually puts the foundation for others to actually arise up in their life, so that is one thing which I have.

Prathap Haridoss: Okay! Very Nice! Very Nice!!

Mahesh Balan: Good evening. So, my name is Mahesh Balan, I am from Management Studies Department. So, I am actually from Kanyakumari. So, my perspective on research was completely altogether different. So, the perspective that I now have on research and the perspective that I had before is completely opposites. So, before I did my engineering then I was working in corporate and they were literally giving me all the stress and all the work. So, my perspective even my dad suggested to go for research, I told - no I am not a geek, I am not a nerd, so I am not fit for research and all those things. So, because I basically had an idea that for doing research we must have that in born interest to do research, that need the hunger to learn more and I basically did not have that in many bachelors.

Then because of work pressure or whatever I opted to do research without having any interest, but now the two thing that I learnt is - one is you need not have that genius thing or intelligence to do good research only thing that we need is interest, patience and hard work, so that I realized that. And, one more thing is in industry you can put a lot of creativity into it and you get recognition for that and all those things, but if we put the



same hard work and creativity in research, but on seeing that being used by someone that small body of knowledge that you add which is new in the complete thing that little happiness that you get and you know the purpose that you are here I think that makes complete difference. So, these two things actually changed my complete view on research before and after that.

Prof. Prathap Haridoss: Very nice, very nice. May be will also, I just wanted to I mean as moving forward I also wanted to get a sense of coming in here and then settling down into a research as a research scholar. What did you feel was the most challenging experience for you that you somehow maybe didn't anticipate and then it took you some effort to get accustomed to and then move on forward as a research as form.

Prof. S. Sankaran: You can also add your anticipation, what was you are anticipation and then how it really shattered or you are happy with whatever.

Sandhya: Hello this is Sandhya, I am from the Department of Management Studies, I am from Hyderabad. I feel personally as research scholars, you get tend to get demotivated very easily because my fellow scholar was also mentioning earlier that it is very unstructured program. So, you would not, there is no hard and fast deadlines as such so, you have to structure you are own program, you need to have your own deadlines and secondly, the output of your research is not very immediate. So, you don't know what is happening, it's as the a process going on; so you tend to get demotivated very easily. So, you need to have that self motivation.

Prof. Prathap Haridoss: Okay self motivation. So, it took you some effort to.

Sandhya: Yes, yes to understand that I need.

Prof. Prathap Haridoss: To go from a point where other people were motivating you to motivate yourself.

Sandhya: Yeah exactly.



Prof. Prathap Haridoss: OK.

Sandhya: True.

Prof. Prathap Haridoss: Ok Fine.

Swathi: Hello my name is Swathi and I am from the Department of Humanities and Social Sciences. So, coming here was a big surprise for me, because precisely because I do my research in humanities and social sciences. So, coming to an IIT which is a science dominated place and doing my research here. I mean I did know that it's going to be different from the universities have been doing my BA and my MA, because they were like you know once that were totally dedicated to humanities in the study of human culture and all that. So, here I don't know the kind of talks that happens and the general culture in, I mean the culture in general is very different from or humanities culture or a culture that is conducive to humanities and social sciences research.

So, initially I did not really like it very much and I thought I might produce a mediocre research if I stay here. So, I thought maybe I should move out from here, but then after a semester or so I thought the kind of challenges this place gives you is to define your discipline, is to have a dialogue across disciplines to make the science people - I mean when I explain what my discipline is to a science person I have more clarity over what I am doing. So, that way you know I do know, I understand my discipline better it gives me better insights into my discipline and you know it's become very broad. Like for example, Terry Eagleton criticizes such that the idea of a university becomes complete only with humanities and social sciences being there, because humanities basically suppose to have a kind of counter intuitive thought, a critical speculative I mean speculative look back at life in general. So, I think if you can integrate this, if you can have a dialogue across disciplines I think we can produce better engineers and better research in humanities.

Prof. Prathap Haridoss: Okay Great. I think it's true actually that I mean even from our discussion with lot of faculty in the institute. So, one of the, I mean aspects about a campus which is multidisciplinary is that in fact, you do have access to lot of other



disciplines easily. So, you can look at for example, in fact, the humanities faculty we interviewed was specifically mentioning that. So, for example, lot of companies now look at the social impact of setting up their industry in a particular location which was not something that they very formally looked at earlier on, previously they just wanted to you know get the regulations, get pass the regulations to set up the industry in some location and they would just set it up and run.

But now they have become much more conscious of the fact that wherever they set up there is you know society near that area, which they have to work with, which they and that society has to actually feel happy with them, they would also you know work with that society get the work force from that society not pollute that location lot of aspects are there where there is a lot human angle in it, which is not not normally recognized or may not have been recognized as much in the past. So so, they are apparently looking at a lot of students with humanities background who would fit into such settings, and such students are probably in a better position to look at develop that aspect of their work when they interact with a lot more engineers who are doing those industrial settings. So Yes. So, that is I think a very nice perspective to have. So, anyone else; yes please.

Swethambul Das: My name is Swethambul Das and I am doing PhD in Physics and I am in my final year of work.

Prof. Prathap Haridoss: And you are from.

Swethambul Das: I am from Bihar, I am from Patna.

Swethambul Das: Okay So, the thing is I hope that I will bring a little different perspective to into the whole discussion because I am from theory side of physics and with the perspective which I share with people from maths department for example is that, the thing is theory theoretical works develop in a little different way than other areas. So, I did cook quite bit of under grade research also before coming here and the thing which I realized doing while doing PhD is that problems unfold itself at every stages. So, there is no set way of going about it and you have to be alert at every stage to tackle the problem in different ways.



So when we see a research paper it look's very nice you know from introduction to conclusion, but it's really messy you have to get your hands dirty to do the whole job. So, that is something I realized while doing you know PhD and also I understood that one has to develop this idea of finding a problem, as far as theories concerned I mean I can't comment on other fields which I have don't have experience in, so solving a problem is a skill and one has to be really smart in doing that, but may be you know a more challenging aspect is to find a particular problem which you can solve and people in theory most of the time do this. So, people who you know young students who wish to come to research and particularly in the theory side they should be dealing to you know get up to this challenge that they should be able to do things independently, should learn at least to do things independently as soon as they can. That's that's what I say.

Prof. Prathap Haridoss: Okay Independent working processes you have to develop and take's a while for a student to do.

Prof. S. Sankaran: Did you come with a definite problem to solve it?

Swethambul Das: Well, such sort of I had a problem in my mind when I came, but my supervisor had a different problem.

Prof. S. Sankaran: OK.

Swethambul Das: So.

Prof. S. Sankaran: How finally, you settled down.

Swethambul Das: Yeah. So, I actually when I joined here one of the things which was asked in my interview is what do you like to do and I told them very specific subject area and problem and the second question was, do you know anyone who works in this field and I knew 3, 4 people but I didn't know whom of them where there in the interview board. So, I picked that random and I said proff V Balakrishnan and everybody was happy and I got. But I am not working with him and I am working with somebody else, but



that's how it all happened.

Prof. Prathap Haridoss: OK.

Ranjith: Hi I am Ranjith, from the Department of Civil Engineering. I am basically from Kerala a place call Trishur. So I would like to share my experience while, some experiences which I faced while settling in to the field of research. So, I am an MS research scholar. So, before this I was doing my under grade. So, we were like a pack of students studying the same subjects writing the same exams from there, here I come to a field and which I am me and my guide and like a small set of people might be doing a similar problem, but the problem to me is specifically pointed out to me and I face difficulties.

So, the thing is we have to from a like a pack culture we are going in to an isolated state where we have to solve problems by ourselves and for that we need a little bit of and not a little bit a real motivation to go on further will face like difficulties our at every stage. So, that self motivation and ability to work independently you have to adapt to it when you get into research. Not just that in the previous studies you will be studying something that is already written, just like my fellow researchers told here will be just accepting that facts, facts and figures in the text books and the papers, but now that complete ideology have change to question everything that you see. So, they have done something in the papers which may be correct and which may have like loopholes also. So, our job is to find the loop holes that is a serious change in attitude that we have to adapt to while you come into the research which I think is really important especially when you are doing the research scientifically. So, something I want to share it with all.

Prof. Prathap Haridoss: Very nice, so getting used to; so Independent working is also a little bit of isolated working is the point you are making so that some takes some while to get used to.

Ranjith: Yes.

Prof. Prathap Haridoss: And also moving away from just accepting whatever you are



shown, to questioning whatever you are shown.

Ranjith: Yes.

Prof. Prathap Haridoss: And making a habit out of it. So, that is nice point.

Bhakti Patel: Hello, I am Bhakti Patel and I am from, my home town is Mumbai. So, while I was doing my under grade I was.

Prof. Prathap Haridoss: Which department?

Bhakti Patel: Which department, Applied Mechanic.

Prof. Prathap Haridoss: OK.

Bhakti Patel: Yeah. So, while I was doing my under grade, I had this idea of research it's like you will be given a problem and then you have to work on it and you have to just go towards that goal, that's all. But while when I joined over here as a research scholar as a PhD scholar, so then it was with my guide that I understood that it's not just like you have to be go in one direction, he gave me a complete freedom as you can do whatever you feel like you have all the facilities which are available over here and you can just manage your own research and be as free as you want and what whenever you face some difficulty you can come over here and then take help from me.

So, that this part it was not, which was something new to me that OK, in my under grade days I was like ok research means just to produce papers and that's all, but over here when I came to became a research scholar, then it was like no we have to get some quality work also done, it's not just the race of getting paper and all. You just have to hang on for some time have some patience and determination and then, just you will spend some time and then you will automatically get some ideas by yourself and then the quality also of your research and all will be increasing gratuity. So, you just have some patience that's all.



Prof. Prathap Haridoss: So, basically you are saying freedom takes a while to get used to.

Bhakti Patel: Yes, yes, yes initially you will be lost completely as in you will be frustrated with yourself and you will be very irritated.

Prof. Prathap Haridoss: So, you wish somebody actually told you no, no don't do.

Bhakti patel: Yeah yeah.

Prof. Prathap Haridoss: I understand.

Prof. S. Sankaran: Finally, it was a happy settling out.

Bhakti patel: No, right still I am struggling.

Prof. S. Sankaran: That is exactly I want to know.

Bakthi patel: Yeah.

Prof. Prathap Haridoss: So, along similar lines may be will go to the opposite end of this spectrum. So, what did you really feel happy about or what was a pleasant surprise for you when you joined which you had not may be as much anticipated.

Vijay Bharadwaj: Hello my name is Vijay Bharadwaj, I hail from Mysore Karnataka. I am a MS research scholar in from Metallurgy and Materials Department. Research is very free in my opinion and particularly in from past one and half year or so, I have found lot of opportunities to develop inter personal skills, soft skills and a very healthy and joyful life style and that was a very present surprise. I did not expect that to happen before I coming here I thought research was all work and no play, but well that is has complete changed out, out of style now. So, that's it.

Rajnath: Hello my name is Rajnath. I am from Applied Mechanics Department, I am



from Hyderabad. Before coming here, during an under grade it was like morely like we study most of them before exams only; like in rest of the time we just pass the time just like that.

Prof. Prathap Haridoss: OK.

Rajnath: Study is not like a major part of the day, but here when we came here everyone this picture like, haan you are from IIT you should be a geek, nerd something like that, but when we came here especially the professors. They are really ground to the earth and very enthusiastic, I have taken some courses they were like really they tend to be like very enthusiastic and you feel more enthusiastic. The way they look at the problem, it's totally different you cannot find that in the text books. What I use to, I am actually in the second semester only. So, whenever we had discussion with some professor I used to search in internet I forgot that one, but I never get anything in internet then I told to that professor sir what is this I am not getting then he said forget everything, you use your brain like that, it's very interesting actually.

Prof. Prathap Haridoss: OK, very nice.

Ashwin: My name is Ashwin, I am from Department of Engineering Design. It's been it's been 3 years from now I have joined IIT and the pleasant surprise is that I got admission in IIT first.

Prof. Prathap Haridoss: We are also very happy to have you here.

Ashwin: Thank you. So, because in my masters I was the you know the last guy who won't study and sitting in last bench and you know say I don't know anything and I am the very you know the 7.5 to 8 CGPA things like that and you know on top of me like everyone will have CGPA of 9 and even my whole family and even my collage mates are very surprised that you got into IIT. So, that is the pleasant surprise.

Prof. Prathap Haridoss: So, you exceeded every bodies expectations including your own.



Okay

Ashwin: I did not expect anything.

Prof. Prathap Haridoss: Yes, please.

Asha Kranthi: I am Asha Kranthi, I am a PhD student in Biotechnology Department and I am basically from Andhra Pradesh. So, one thing that I found surprising okay I came here for a degree I wanted to get a PhD and it was all about research. But right from the third semester onwards I was part of some council or some student body in the institute that challenged me and my own level of confidence, it went down and I picked myself up at a personal level it has been a journey that I think would not have been possible in any of the places that have been till now, so that's the pleasant surprise here.

Prof. Prathap Haridoss: So personal growth is what you experience and growth in personal level, yes.

Revathi: Hi I am Revathi, I am from the Department of Computer Science. I am doing my PhD here. So, the pleasant surprise is that you don't have to be brilliant to do a research. So, the thing is that you can be a mediocre, you just need to have a lot of creativity and imagination and the passion that drives you, you have to feel that okay you are good at this, you can do this. So, that kind of self motivation is what is exactly that is required for a researcher. So, people can wonder I mean when I joined here itself and I started doing my research I used to wonder okay like whether I am going in the right path, whether this is the right thing to do and there is nobody to tell you that okay you are going in the right path or whatever you are threading through is the correct way to approach it.

But the thing is that ultimately whatever especially what my peer group says to me. So it's like you have to explore you have to understand what is happening and whether you are going right or wrong it doesn't matter. It's like, it is continues learning process and whatever you are doing it will be right in one way or another. So, that was the surprise for me because I thought that it was always about doing the right things, but what I



understood what is that it is also OK, to do the wrong things and understand and it's a continues learning process to understand it in a right sense so.

Prof. Prathap Haridoss: Okay Great, and where are you from?

Revathi: I am from Trivandrum, Kerala.

Prof. Prathap Haridoss: OK.

Shruthi: My name is Shruthi, I am from the Department of Metallurgical and Materials Engineering doing my PhD. So, one of the most pleasurable thing was the course work, where in so many concepts were made very clear we have plethora of information from the internet videos now a days, but the way things are explained here by our faculty is so absorbing that I don't think that we would forget it for life, any concept. So, that was a most pleasurable thing. Many times I felt oh that's all, this concept is all about. So, it was a great experience.

Bobu: Good evening everyone my name is Bobu I am doing my PhD in Metallurgy Department, I am from Kochi, Kerala. The pleasant surprise for me was about the amount of freedom that we get in the institute right from the selection of guide to the topic and the amount of time you should take to finish the work. So, that was a pleasant surprise for me which I never had in my under grade or grad school.

Ram: When I started doing my research like before when I was an under grade, seriously when whenever I talked in the family like, there will always be discussion in the family members like what we can do like something like income how do you manage it and all. So, they really don't care what you say at all. So, when you started doing research they think you are a scientist. So, suddenly you become the council for the whole relatives and your friends. So, they everybody will keep calling you what can we do whether my son should study computer science or statistics. Okay so, suddenly you feel a lot of more, you have gained lot of respect and that is it.



Prof. Prathap Haridoss: You suddenly become respectable, to your own pleasant surprise. Okay so, maybe I also wanted to one of the things that we all do as research students and even as you know professional researches is to go for conferences. So that's something that you don't do normally as an under graduate student wherever you are, but it's a inherent aspect of research scholar life. So, I just wanted to wanted you to share with us if you have been to a conference and what was your experience what did you really you know feel about the whole process and how you think it has you know contributed to you as a research.

Madhuri: Hi I am Madhuri, I am a MS scholar in Computer Science Department, I am from Bangalore. So, I attended my first conference after one year after joining MS and I just attended it I did not present any paper. One thing I realized there was, it's not just a show case of the new discoveries that people have made or I working theoretical computer science. So, there are these new theorem people have proved new approaches they have discovered to proof things.

But it is also sort of there is a community feeling that happens in the conference where people are working in different universities across the globe different countries, but they are all come together as a community there is a lot of sharing that happens and it's a very unselfish thing where people discuss come together, do collaborations and conference sort of place a ground for that also to encourage future collaborations and to inspire people to do more. Like teach them on different areas that can be applied, some interdisciplinary things that can be done. So, conferences are really good ground for collaborations and for inspirations.

Prof. Prathap Haridoss: Great, good to know. So, how often have you been going to conferences?

Prof. S. Sankaran: I mean she is too young now.

Prof. Prathap Haridoss: Too young, OK, fine.

Ranjith: Hi my experience with the conference is that basically you get to see the state of work that is happening all over the world like we see in the papers, but the papers will be



at least 6 months to 1 year old at least because it takes a lot of time for it to get published. But then there is lot of work that is going on currently which is fair more advanced than what you see, which is clearly exhibited in that conference. Another thing that I felt in conference was like when you are presenting a paper basically I went in and I made my research. So, I really wanted so much of feedback and we are getting that usually in the institution will get it from lot of I mean peers, our professors, our fellow research scholars etcetera, but we are getting a research from a wider audience and the feedback who is really excellent, on top of that your research visibility increases.

Once I go to a research, I present my work and later I submit a paper, when professor if we are lucky he listened to your presentation or read your proceedings or something like that; he reads or the he knows your work and he reads it he will get a better clarity of your work and give you better reviews. Also when we are applying for post doc or a PhD position, he just known your at least your name or when we writing an email we can say I met you at so and so conference and it is we get a better response from them than the general lot who just send it as a dear sir and something like that.

Prof. Prathap Haridoss: OK.

Ranjith: Visibility is a really good point I see from the conference.

Prof. Prathap Haridoss: Visibility a good level of personal contact.

Ranjith: Yes.

Prof. Prathap Haridoss: Interpersonal activities.

Ranjith: Yeah, it helps in the further collaboration also. We don't know who will be collaborating in our current research itself, we might not have all the facilities, but we can always collaborate with another university we can get the contacts. So, that is a great plus point of a conference.



Prof. Prathap Haridoss: Okay Great, you had something.

Pragyan: Good evening I am Pragyan, I am a MS scholar from department of engineering design. I am doing a Masters here and I am from Orissa particular Bhubaneswar. I have been to two conferences as IIT provides a international and a national conference. So, my first conference was in Bangalore in which I met a lot of DRDO people and my work is related to somewhat defense. So, I saw like they are still like struggling with something which I have the results and they are still struggling with the old techniques. So in that platform I had something to offer to them and also I learnt a lot of things which I didn't know, I taught that it is always all working for me, but they set up some scenario in which my thing was not working. So, in this way we had a lot of collaboration which BEL DRDO and I offered them like you can come to our lab and if you allow we also come and in this way we can help in development of the country rather than spreading across and doing small parts as in why work together and work for development of the country.

Secondly, I went to conference in Poland in which I met lot of people from Russia, Japan, China and US in which I met students I met faculties in which I presented my work I even I told what problem I faced and I invited them to attend my lecture and they also suggested how would I improve my work and what else I should look into which is total unaware of. In this way there is a lot of collaboration and as my friend has already suggested, if you really want to do a PhD abroad, it is better to go and then meet them in person and then ask them to come and see what work you have exactly done. In this way they see you in person and it's a very good way to get a PhD abroad and as I am sending mails across from here. So, IIT provides this good opportunity of going and collaborating and at least interacting with people from national or from international background.

Prof. Prathap Haridoss: Great. So, basically you are saying both you were able to you know give ideas to people and take ideas from them in a collaborative.

Pragyan: Yes, and because everything is not in the internet or because if you see DRDO things is not available anywhere defense.



Prof. Prathap Haridoss: You are actually able to meet people; you would otherwise not be able to access.

Pragyan: Yeah then you see them, what is they are seeing and exactly you can contribute to development the country, because at the end that matters is how a developed to a country is.

Prof. Prathap Haridoss: Great. Yeah, yeah.

Ganesh Govindarajan: Good evening everyone. My name is Ganesh Govindarajan, I am from Motion Engineering Department, PhD scholar from Chennai. Initially I thought that my research is not that much good, my results are not that much good. After that I applied those results to the conference, I presented those results through the conference then I met that the most eminent person in my research area from MIT. He appreciated me, your way of approach is very good you keep on working me, that makes me something like I am going in a good path like that, but everyone something like that is not good like that, but after he is make something feel proudly like that.

Prof. Prathap Haridoss: Okay so, you got a very good feedback on you know and you feel you feel confident about that feedback; lot of people will give you.

Ganesh Govindarajan: The feedback and it is nice to meet with the most eminent persons in from MIT like that, it's nice to get interaction those peoples can.

Prathap Haridoss: Very Nice. Very Nice!

Prof. S. Sankaran: That is one of the primary aim of conducting an international conference to bring all the experts in one place otherwise you don't have an opportunity to meet them. So, we actually in organization of conference that the primary requirement is this, how big people we can bring in one platform and then make available to rest of the audience. So, that is one of the aim for any conference they do, yeah good.



Hemanth: So, I agree with this networking development is the major thing which we get from the conferences. So the suggestion what I would like to give for the new research scholars who are watching this course is plan ahead for your conferences when you are going see the list of invited professors are, list of eminent faculty which are who are coming. Let's say you are in the second year of MS and you are aspiring a PhD position somewhere in the US or Australia whatever. See people who are working in your area and working in the areas which you are interested to pursue your research on PhD, get the emails of those persons and mail them saying I am so and so from Indian Institute of Technology, I saw you or, I came to know, I get to know that you are coming to the conference I am also coming to the conference, please attend my talk and give your feedback and that might help the first thing you need to plan.

So, the other if you are in PhD the same thing applies for a post doctoral fellowship before you go pursue post doctoral fellowship. That way you can get better you know feedback from the professor what who you need to meet and you will get to know them easily than, just going there and dropping say, hi! some professors might not find to talk to at the conference there itself, before giving them information would help you to interact with them in a much more descent manner, just say go for a dinner then talk about your PhD admission or post doctoral fellowship.

Prof. Prathap Haridoss: It is very good for carrier growth, you are able to interact with them and I think that your suggestion is very nice that I think people can plan it out a little better it's almost like an interview for free. So, you go there and you know there.

Hemanth: Yeah they should plan it for a better plan. So, that they can go and meet them and get little more much time then what they get when they meet, on the go.

Prof. Prathap Haridoss: Yeah yeah, Ok.

Swethambul Das: I was in Berlin last week and attended a big international conference. So, first of all it's nice to know that our institute and our society in general feels that, the kind of research that we do can be funded to this extent that we get a lot of money, but in such conferences anywhere in the world. And then first thing I noticed this time, I have



attended other international conferences also within India and elsewhere that we do very good research it's not that India is no longer in the research map any more. We do fantastic research and people from other country come up to us and ask things and discuss things. So, it was a pleasant surprise to learn this fact that we are second to none now, I mean yeah, it takes time to go to the certain extent, but the thing is we are in a in a good hands like professors like you and others in the country and we are doing a good work also that's what I learnt.

Prof. Prathap Haridoss: OK, lovely.

Prasanna: Hello sir, my name is Prasanna, I am from Karnataka. I am third year PhD scholar from Mechanical Department. When I first went to conference in Trivandrum organized by ISRO, VSSC. So, it was very surprise for me because first time I attended the conference, I am able to meet peers in my research areas and also a big big scientist and big big research scholars along with many professors. So, there what I learnt is that in the same group of people when we are going to presenting a paper, so different people are explaining in different perspective based on their application the way their theory they develop in their own mind.

So, this was really give value addition for us to progress in our own field and as well as when people are presenting, the different people presentation skill is very different even though we are working on the same area. So these kind of things will give some kind of value addition to our work and also it will be helpful for us to look for new problem in future and the way we are interacting with them is also very nice for us to further career growth.

Sujith: Hi, this is Sujith. I am doing my PhD; I am in my 4th year I am from Electrical Department.

Prof. Prathap Haridoss: And where are you from?

Sujith: I am from Hyderabad. So, this beginning of this year I went to a conference in my area that is supposed to be the best conference. So, it happens in California as you know



it is surrounded by all the companies that my area of interest. So, one thing I noticed was as people already said you get to meet lot of people, like minded people that are working in your area. But one good thing, one good inspiration also for me was I was asked very frequently when you are completing your PhD.

Prof. Prathap Haridoss: Very nice.

Sujith: By the company people. So, and it became one of the motivations for me to go that conference again, I mean in our area if you have one paper in that conference it is a big thing.

Prof. Prathap Haridoss: Ok.

Sujith: So, it's a very good motivation.

Prof. Prathap Haridoss: Okay very nice. So, may be now we have seen a lot different aspects of your research activities here, were we will look at something more. I was just curious, I mean do you see I mean at least as a undergrad you had lot of other options you could have done gone into management, gone into job, may be some of you have tried some different options before you came here and so on, now you are research student here, research scholar here. Going forward do you see yourself in future continuing to stay in research and in that context may be say 5 or 10 years from now where do you think you see yourself being, I mean what is your expectation or your anticipation that this is what I think I will be doing say 5 10 years from now.

Arjun: I am Arjun, I am doing PhD in Ocean Engineering and I am from Kerala. And I am working in the field of Ship Structures, structural analysis. So, in my field I feel that not much R&Ds happening in India. So, I see that it's a big challenge. So, most of the guys in my field are going for a teaching job.

Prof. Prathap Haridoss: OK.



Arjun: Others, for if I am I opt to go for a research, but I don't know, but I have like to work in industrial research.

Prof. Prathap Haridoss: OK.

Arjun: But not much opportunities are there in India.

Prof. Prathap Haridoss: OK.

Arjun: If I want to work there I want to go abroad or something. So, most probably I would be working as a professor also.

Prof. Prathap Haridoss: So you would either, you will have to do either academic research here or if you want to do an industrial research you feel you have to go abroad somewhere to carry it out.

Unnikrishanan: I am Unnikrishanan, I am doing my MS in Machine Design Section, Mechanical Engineering. I come from Trivandrum. So, in 5 years I see myself working in R&D section in some industry. So, I was part of the placement team last time and I could see that the placement for the research scholars, at least for the MS scholars it was not that good in the industrial perspective, but it has significantly increased over the last 2 years.

So, in the case IIT Madras it has increased from 50 percentage to 71 percentage in the last 2 years. So, now, we can see that the industries are also focusing on research scholars and they are also having close collaboration with lot of research activities that's happening in the campus. So, I feel that there is high scope for research scholars out there in the industry and we should explore it that. So, I would like to work in one industry and see how the things work out there and if I get a good perspective on that then I may stay back over there. Or if I feel that the domain is too vague then I may come back to PhD or I may pursue for some higher education.



Pragyan: I am doing my research in Micro and Electromagnetic Engineering, and I always complain about this there is no company in RF or Microwave in India. So, my professor one day said - stop complaining and make your own company. So, that you at least do something good for the country. Then I realized that see I took a independent problem in which no one worked, in India hardly two groups work on that and I successfully completed it. So, that gives me a confidence you don't need a degree to start a company or something, if you have if you know the subject properly, when you are confident enough that you can carry on then I think in 10 years at least I think I will have my own company in which at least people in my lab won't ever complain you don't have a company in R&D working in microwave. So, that's my that what I see in like after 5 to 10 years

Prof. Prathap Haridoss: Very nice very nice to hear, but you don't have to drop out though, you can finish it.

Ashraf: The first thing when I took a PhD was like I decided my under grade that I wanted to contribute to my field of study. So, that was the reason I took a PhD. So, 5 years from now I want to join the academia and I would like contribute to my field of study.

And one more thing is as I see in our country the research, the academic research and the industry there is a lot of gap in between that. So, I mean if I become a faculty down the line what I would be planning to do is to bridge this gap between the academic research and industry. So, that is one of the aim I have because as you see in the western universities most of the funding comes from the industrial research, industrial sponsored research even the stipends for the scholars, but I mean that has its own disadvantages that our research will become more dependent on the industrial problem statements, but as a country we have to evolve in the closing this gap between the academy and industry. So, that is what I would like to do.

Prof. Prathap Haridoss: Great.

Mukesh: Myself Mukesh, I am from village of Chhattisgarh. I am in Mechanical



Department, I am doing my MS. After 5 year, I want to see myself as a working as a professor, because I wanted to support some research, I mean to say only IITs are doing research basically not all engineering colleges. So, if I go to some normal low level engineering college I want to support like UG levels to go for research also not only get place and go for job. When I came here I thought I will do my MS and I will go for research sorry I will go for placement, but being touch with research I thought we should encourage people for research also that is what I wanted to do.

Prof. Prathap Haridoss: Okay great, we would like to get the next generation of researchers. Yeah, after him.

Ashwin: 5 years before I didn't know that am going to be a researcher first. So, after 5 years may be I wanted to do post doc may be another 5 to 10 years down the line, so that I can explore a different way of doing research in different field for a different applications. So, now I am actually focusing on say nanotechnology. So, how it actually used for a different fields in same in the biotechnology or how to electronics and. So, I would like to go for different labs and to explore different things. So, that I can get a different experience, that's all.

Prof Prathap Haridoss : Okay Great !

Priyanka: My current research topic deals with electric vehicles, now that we see that I mean our world itself is facing a lot of environmental challenges. So, I believe that vehicular technology I mean this electric vehicles itself, electrification and the vehicular technology will revolutionize the way vehicles I mean we see these days. So, 5 years down the line, I would like to see the kind of research I am taking up here those thing a practically being implemented. The commercial viability of those things a lot of research is going on throughout the world and how we can bring those things to India, like in abroad those things are being implemented like we I have heard of Tesla motors and stuff. So, those things are there in US, in Europe, so those parts of the world. So, I would like to bring those things to India and may be you know you just revolutionize the way technology works.



Prof. Prathap Haridoss: Very nice, very nice.

Asha Kranthi: I completed my B Tech in 2000, Electronics and communication engineering and 13 years after my B. Tech I joined my PhD. I am currently in the biotechnology department I always wanted to work on cognitive disabilities, some in the computation neuroscience lab. In the last 4 years I am here, okay in the last 13 years after my B. Tech I worked as a teacher, I worked in the industry doing software programming and hardware integration. So, I have a lot of different skill sets and after coming here in the last 4 years at IITM, I had a broad idea of what are the different areas in which if you take neuroscience, there are people the range of fields or the range of domains in which work is happening with respect neuroscience is very varied. So, I just got an idea of all of those.

When I came here I wanted to complete PhD and go a back as a faculty, but my views have changed over the years. So, 5 years from now I am not really sure exactly this is what I would be doing, but I see that I would be able to figure out something where I can put together my knowledge in electronics as well as my research work here to work out something like one of the colleagues said why don't you set up something.

Prof. Shankaran: Start up.

Asha Kranthi: Why don't you start up something?

Prof. Prathap Haridoss: Okay you want to fully utilize your experience and education.

Asha Kranthi: That is something I have been visualizing that I would be able to put together.

Prof. Prathap Haridoss: Very nice.

Asha Kranthi: Now, and as 13 years ago, I wasn't sure I would be able to work in the neuroscience field given my background of Electronics and Communication. So, today I



feel confident I can figure out a way in that direction.

Prof. Prathap Haridoss: OK, Great.

Hemanth: So, 5 years down the line. So, it is decided when I was joining for PhD I want to become a faculty. So, I still go stick with that, and I will go for an academic profile either in IITs or NITs that's what my hope so. But I would like work for a something that can be done for the rural India, where there is lot of technology and it is not being implemented. So, that I would like to work on projects which uplift the parts of India which are yet to be technolized with this, that is all.

Prof. Prathap Haridoss: Ok. I think we spoke a lot about I just have a couple questions before and we will wind up that. We spoke a lot about the professional aspect of it and how you know may be various aspects of research have been, you became familiar with them you utilize them and so on. I am just curious do you see any change in your personality as a as a person, over the years that you have become been doing the research, are you a different person now then you were in some sense, then you were.

Prof. S. Sankaran: Definitely definitely every one of you.

Prof. Prathap Haridoss: Then you were 2 3 of, 2-3 years ago with before you join here I know that things were much more directed as many of you are saying and now you are suddenly very free, but what what do you think has daily changed in you.

Prof. S. Sankaran: This question is because, not just because you are in IIT or something like that because you tend to meet a different set of people across the country which you might not have had an opportunity before, and also the kind of lifestyle, the kind of curriculum style, everything is you know quite slightly different from the rest of the world. Definitely that will have a some influence I hope mostly is positive influence, but yes, but we would like to hear that what is that influence.

Prof. Prathap Haridoss: What do you feel has changed?



Ashraf : I think you get a sense of self responsibility that is the first thing, where you take research because you have to work hard and you have to be patient enough, so that is one think and one more thing is you start questioning things and theories which you have been.

Prof. Prathap Haridoss: Taking for granted.

Ashraf: Yeah Taking for the granted in your under graduate.

Ashwin: One thing is I can start my own lab now. So, when I joined the IIT. So, my professor's lab he his actually, he is very new to the IITM. So, he just joined just before 2 years back. So, when I entered his lab nothing was there, it is like room is empty. So, and he is a like you need to start developing this lab, so now, I have a confidence that I can develop my own lab in future where ever if is go and join there.

Vijay Bharadwaj: I would like to quote one of one from PhD comics, when I joined for PhD it was like – Oh yeah I will change the world now. Well, now it's not like that, I have grown to be little more realistic and every lab comes with it is own constraints. So, if you choose a problem you learn how to solve it based on the given resources and how much more you can implement it and how much more you can give a change to the lab. That's what you learn, to work under constraints and you are also learned to delete lot of frustration.

Prof. Prathap Haridoss: OK.

Vijay Bharadwaj: It will make you more patient.

Prof. Prathap Haridoss: OK.

Ashraf : Yeah, that was the point I missed. Practicality practicality of any given problem that's what you start assessing.



Prof. Prathap Haridoss: Yeah, very nice yes.

Mahesh Balan: One question I used to ask my professor often is sir why is it a doctor in philosophy and not doctor in management studies or electrical engineering. So, he always says the body of knowledge that you learn in the field is lesser than what you learn about yourself. So, it actually changed myself completely previously when I used to lose my patience often. So, these days I used to go to a restaurant and I don't get my order okay take your own time no problem. So, every aspect of mine is changed. So, since I am from management studies previously like we used to go shopping are discount and go, buy it everything and come and now lot of economic theories come into my mind okay I understand. So, my view and perspective is completely changed and also I would add like as a researcher we also get some humility and simplicity. So, we always have a notion that we are the greatest in the world what we do is awesome, but we understand realities that we just play a small role and there are many good things happening in this (Refer Time: 53:57), so all these things.

Prof. Prathap Haridoss: Very nice, very nice.

Asha Kranthi: I would like to answer this from the perspective of a women, within my family. In the earlier generation there was only lady who is working as a teacher. So, when you aim for PhD the first thing that comes up is why PhD, settle down.

Prof. Prathap Haridoss: Yeah.

Asha Kranthi: So, challenging the notion of settling down working for money and the traditional setup did not come easy. But here, when you see people all around you and when you hear a women faculties share her own experiences of how she pushed through managing both family and a research you get inspired. So, I think you have an environment here that challenges you beyond the traditional setup that is outside. So, that is one thing I would say is what I saw here what helped me a lot.

Prof. Prathap Haridoss: Yeah, very nice.



Swathi: I would like to add to that, now as I said I am doing my research in humanities in social sciences. But IIT is a big targets compared to other humanities universities, you know where real good research happens because I do know the common crowd is more biased with science I guess. But coming to an IIT, I understand that this is a best way to fight patriarchy as in seriously, as in your notion that women cannot do science or women cannot reach somewhere, but then the attitude there is a I mean there is so much of an attitude change when you go back to your hometown or any place.

Like previously I was in English and foreign languages, university and university of Hyderabad, but real good research happens there, but the kind of respect that especially men give to such universities is, I don't know it's very different because humanities is supposed to be a very I don't know feminine subject and sciences is supposed to be masculine and more intelligent. So, when you share that space you get a different respect. So

Prof. Prathap Haridoss: More than your personality other people's personality changes that's what you say.

Swathi: Yeah that changes, you don't have to fight back many a times, and it comes you know.

Prof. Prathap Haridoss: Sure.

Ranjith: Coming into research, we have some of my fellow researchers has told about the freedom that we are getting and with freedom that it comes a lot of responsibility too. So, during this 3 years at IIT Madras, I found that my responsibility as a student, as a may be as an individual improved a lot because I had to handle so much of things which nobody else was like controlling. So, you are completely free to do whatever, but still you have that responsibility this is one thing.

Second thing is like I mean like I am an MS research scholar so this people who have joined with me for M Tech, they might be graduated by now. But you will be still here, I may like we are doing a, in a research kind of stuff it is good like we done a good stuff,



but then going back to my home place there will like okay you just did Masters right, it's more than 2 years are you did you fail or something. They might not know the thing, but now I feel like I am mature enough to understand it's not my problem, but it's their lack of like knowledge in this area.

Prof. Prathap Haridoss: Knowledge in this area.

Ranjith: That is causing and that maturity that is developed during this years at IIT madras.

Prof. Prathap Haridoss: Very nice.

Mukesh: After joining as a research scholar, what I felt myself as a change is like a when previously I went to my home place and small small things I used to argue with my friends. Now after being a research scholar I stopped arguing with them I started observing them what they are saying, I did feel like that. So, the change is like from argument I changed to observation.

Prof. Prathap Haridoss: Yeah, go ahead, go ahead.

Rajnath: Yeah as I joined here I came to meet so many different people like with different mentalities and the way they look at the same situation differently. Earlier we used to be like, yeah what I said should be right, it is right, actually it is right, it should right. But now we start to accept different ideas and different thoughts. In that sense IITM like and also you become more and more enthusiastic everyday and you try to know about everything what is going on, what's actually happening like you be like a very lively person in fact.

Prof. Prathap Haridoss: Very nice.

Zeeshan: My perspective is slightly different; I had the opportunity to go aboard for a semester. So, I was abroad in Germany for 7 months. So, that allowed me to change in



many ways I mean the perspective that you gain going to a foreign land, trying to understand their language, seeing how they do things, it changes you. Here no matter how much time you have spent, the kind of thinking, the way we approach science, the way we approach any problem is very different and how somebody from outside approaches. So, going basically my change came from going aboard from seeing different people's perspective and now the way I think is far more expansive.

Prof. Prathap Haridoss: So, different more cultural influence from different places that make a big difference.

Zeeshan: Changes your perspective in usual time.

Srikanth: Well I think I am last one to introduce myself I am Srikanth, PHD scholar from Electrical Department. I joined MS program here and then I upgraded to PhD.

Prof. Prathap Haridoss: And you are from?

Srikanth: From Warangal, Telangana, so to be frank I joined here as a kid and I think now I have grown up little bit.

Prof. Prathap Haridoss: Very nice, a very nice thing.

Srikanth: And the motto of joining IITM was to get a job. I got an MS program. So, let it be IIT. So, let it be it's an IIT. So, I thought I would get a job after 2 years, but after spending 2 years I thought I am working in a cutting edge some interesting technology like I work in Renewable energy. So, there is a lot of scope. Then I was interested to do PhD. So, then I joined it. Now after while doing my PhD I am thinking like why cannot I commercialize this technology which I am working on. So, the basically I started for a job and then to do a PhD and then it's moving towards starting my own. So, there is a lot of professional change in my carrier. So, such as

Prof. Prathap Haridoss: So you are going from a view where you wanted a job, to



location where may be you will give jobs, so that's good, nice nice to know.

Sujith: My friend said he joined as kid and but he is grown up, but but I mean people have already said that they are getting more freedom, but little bit of maturity. So, I am feeling right now as I am in school again, but with little bit more maturity.

Prof. Prathap Haridoss: Ok.

Sujith: Because we are doing the things that we like, but from the school till B. Tech we have been learning something which we are not aware of, but I mean those things are very alien things to us, but now we are aware of the things and we are doing the things which we like and we have lot of freedom. That's the same thing that we used to do when we are kids right. So, I am feeling like that.

Prof. Prathap Haridoss: Okay very nice, you are having revisiting childhood in a professional way.

Sujith: Exactly.

Prof. Prathap Haridoss: May be I think we will we had lot of nice discussion on various topics I would like to just one last point; if you can comment on. So what are your words of advice or you know any anything that you would like share with a person who is probably watching us and is considering you know taking up research position in and as a MS student or a PhD scholar. Do you have any kind of you know words of wisdom or advice that you would like to share with me?

Sujith: I have a I have only one advice. So, if one person wants to go for research and he has to start as early has like just after the B. Tech, because it's going to take time and you have to think of your family also so.

Prof.Shankaran: Yeah, valid point.



Sujith: Yeah. So, there will be like 21, 22 and 5 years, I mean I mean if they want to do the research I think I would advise them to do soon.

Prof. Prathap Haridoss: Sooner the better.

Sujith: Yeah do a direct PhD and it would like 27 by the end.

Prof. Prathap Haridoss: Very nice sure.

Arun: Hello, I am Arun.from Chemical Engineering Department I am into my second year, I mean third year of MS.

Prof. Prathap Haridoss: And where you from?

Arun: I am from Chennai. I would like to tell people that you know actually perseverance is the most important trait I guess, because like in your under grade and all you just learn from books and everything is just thrown at you, but in research I think you mostly learn by figuring out what doesn't work. Most of the time the experiment don't work so I think, and then again one a good thing is it's not structured. So, it is not like a 9 to 5 job which kinds of 24 x 7, so that is really good.

Prasanna: If anybody wants to perceive in his in the field of research, if he come with proper goal is good. See if most of the research scholar we come here as we got PhD position in IIT Madras. So, if better if we come with proper goal.

Prof. Prathap Haridoss: He may more be focused.

Prasanna: About his carrier growth and future development everything, so later onwards he should not frustrate that while doing PhD. After finishing PhD if he not get job with desired field or whatever it may be according to me.

Prof. Prathap Haridoss: OK, you should think it through a bit and then



Prasanna: Yeah.

Prof. Prathap Haridoss: We have some idea and based on that you should.

Prasanna: You should come yes.

Prof. Prathap Haridoss: Ok.

Madhuri: So, this is a tip for the MS people who are aspiring to join for MS course. You don't have to really know the exact problem you will be working on. So, this is something we can figure out on the ways. So, that was something when I got in and I was worried that I did not really know what problem I wanted to solve, but as we did the courses you can talk to your adviser and figure out what you want to do. So, you do get time to settle in and figure out what to work on and you don't have to jump into it right away.

Prof. Prathap Haridoss: OK. Great Yes.

Vijay Bharadwaj: My piece of advice is mainly intended for undergrad students in their final year. So, I would like to tell that if you are not sure as to whether PhD for me or not, I would like to you push ahead and take the risk of doing masters first. In the masters, in the first year you will undergo lot of different course work at whether MS or M Tech and based on that you will be in a better position to tell okay PhD is for me. No, if not M Tech you are better at a subject and you will be paid more by companies. So, you for undergrads you better take up master before deciding for a PhD.

Ashwin: One thing is we should adapt failures. So, that is the only concern I feel.

Prof. Prathap Haridoss: They should accept.

Ashwin: They should accept failures and they should adapt to the failures.



Prof. Prathap Haridoss: Adapt to the failures. So so, they should become comfortable with the idea. So, so that's something that they should keep in mind as they.

Ashwin: They should not go down even if there are lot of difficulties; get ready for the.

Prof. Prathap Haridoss: Better to prepare to fail. That you can get up stronger get up stronger.

Prof. S. Sankaran: Get up stronger.

Asha Kranthi: I want to add one point to that, learn to recognize success after so many failures instead of thinking this is also a failure.

Prof. Prathap Haridoss: Okay Great, you forget what success is.

Asha Kranthi: Learn to recognize success in the middle of tons of failures.

Prof. Prathap Haridoss: Great.

Shabri Lal: Hello I am Shabrilal, I am doing third year MS in Aerospace, I am from Kerala. Basically the most and foremost thing a research scholar should need is patience, even though he had intelligence and all, it won't work because he should know how to work in a group. If he have enough intelligence and all, if he is a not willing to work in a group because if he has some issues indeed, he should understand how other peoples feel and all.

Prof. Prathap Haridoss: Yes.

Shabri Lal: If he don't know that women culture and all it is difficult to get a positive result from this.

Prof. Prathap Haridoss: Okay so, you are talking about good. So, you are mentioning



team work, team work and also patience that these traits should be there, or at least may be if you are coming you will end up getting those traits I mean, some of it will come. Yeah

Mahesh Balan: So, my advice to people who wanted to do research in PhD is that unlike M. Tech or MBA don't have any expectations. So for the next 4 semesters I am going to put my heart and soul, get good GPA get into a company, that doesn't work for research. So, never have expectations particularly when are you going to graduate and all those things.

Prof. Prathap Haridoss: Yes.

Mahesh Balan: Because in research if you have expectations it doesn't work in that way, if you don't have any expectations you will enjoy the journey and that surprise will make you happier.

Prof. S. Sankaran: Yes fantastic.

Ram: I would suggest that you should really take up a topic which that fascinates you to the core because you are going to spend your hell a lot of time with your research topic. So, something which does not fascinates you stay. So, stay out that's it.

Prof. Prathap Haridoss: Ok.

Zeeshan: Falling upon that point in my opinion I would think that after you do your bachelors it's better that you try to work on something. I mean actually go to the industry try and do something. That will, that way you will figure out what you want to do your research in because that's really critical when you come and don't just say I want to do MS or a PhD, MS in what, PhD in what. So, I think the focus will actually come when you try and do something.

Prof. Prathap Haridoss: Ok.



Srikanth: I just have one suggestion like one important thing is the area in which you are working is very important and whom you are working is important. So, these two things are the basic things before you are joining at research program.

Prof. Prathap Haridoss: So, you apply a thought on both of these so that it goes on smoothly.

Srikanth: Yes and then it is a calculated risk basically.

Prof. Prathap Haridoss: Calculated risk Okay.

Shruthi: Another advice is whichever area you are interested in a topic you could take it, but you shouldn't be worried if your background and the department or the faculty having that say you are from biotech and you want to do research in some materials department or mechanical department that shouldn't come in the way of choosing your research topic. This is out of personal experience and I have seen fellow research scholars, they are still able to make it through as in if they are UG was in biotech M Tech was in nanotech and still they are able to make it and get very good research results.

Prof. Prathap Haridoss: Yeah, I think that's a very important point, that the Indian context because we tend to think that you know especially through our educational process where there is an entrance exam and which puts you into a department and then you some of how think that you are stuck to that particular line of activity. But I think that's a valid point that when you get into research it is much more open in liberating in that sense you can. In fact, almost every department these days says that their work is interdisciplinary. So, in that sense you can always have one foot in one department and still you know seek problems which are more broad based and constrained by that one department.

Prof. S. Sankaran: We had taken a Dentist.

Prof. Prathap Haridoss: Yes.



Prof. S. Sankaran: We have successfully finished PhD in Metallurgical and Materials Engineering Department. So, that it tells.

Prof. Prathap Haridoss: Yeah and who did very well, I mean very well.

Prof. S. Sankaran: Very good excellent no problem, yes.

Asha Kranthi: So, I would like to add one thing B. Tech or M. Tech is more core centric whereas research is where you are trying out things. So, do not wait till you complete your B. Tech to try out something either in the industry or anything. May be right from your second, right from after completing your second year or so in your B, Tech itself start tinkering with something's, start trying out something's beyond the course work.

Prof. Prathap Haridoss: (Refer Time: 69:12) Course work yes, yes, yes.

Asha Kranthi: So, that would also help the students get a clarity as to this aspect of the subject is what I am interested in. So, that would help them get a clarity as to what area of research they want to go in so. Right from the second year or after second year itself start some practical things beyond the course work.

Prasanna: One more point, I would like to add sir, here research scholar, he has to seriously attach to his work and he has to appreciate his work in all the stages failure and success.

Prof. Prathap Haridoss: Okay great! That level of involvement and attachment is very important, something that in aspiring students should be prepared for and have the at least. Okay so, on that note I think we will close this discussion it's been a pleasure to have you all here.

Prof. S. Sankaran: Yes.

Prof. Prathap Haridoss: For us also it's been a nice opportunity to meet so many of you



in this kind of a setting which we don't often have in because of various other scheduling requirements and so on. And very nice discussions, so many of you had so many nice lovely points to make on you know what it is to be a research student here. What's nice about it? What's challenging about it? Where do you think it is going to take you and how much it has impacted you in your life.

So, thank you very much for joining us and I hope other people would also come in to positions that are similar to what you are doing right now, thank you.

Prof. S. Sankaran: Thanks a lot.



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